FOR IIT JEE

Power Coaching

SPEED - II

QUESTION BANK FOR IITJEE

PHYSICS

QUEST TUTORIALS

Power Coaching

QUEST TUTORIALS

TARGET IIT JEE

PHYSICS

KINEMATICS

SHORT QUESTIONS

Q.1 A vector $\vec{a}$ is turned without a change in its length through a small angle $d\theta$. What are $|\Delta\vec{a}|$ and Δa?

Q.2 Does the speedometer of a car measure speed or velocity? Explain.

Q.3 When a particle moves with constant velocity, its average velocity, its instantaneous velocity and its speed are all equal. Comment on this statement.

Q.4 In a given time interval, is the total displacement of a particle equal to the produce of the average velocity and the time interval, even when the velocity is not constant? Explain.

Q.5 Can you have zero displacement and a nonzero average velocity? Can you have a zero displacement and a nonzero velocity? Illustrate your answer on a x-t graph.

Q.6 At which point on its path a projectile has the smallest speed?

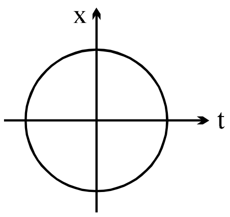
Q.7 A person standing on the edge of a cliff at some height above the ground below throws one ball straight up with initial speed u and then throws another ball straight down with the same initial speed. Which ball, if either, has the larger speed when it hits the ground? Neglect air resistance.

Q.8 An airplane on flood relief mission has to drop a sack of rice exactly in the center of a circle on the ground while flying at a predetermined height and speed. What is so difficult about that? Why doesn't it just drop the sack when it is directly above the circle.

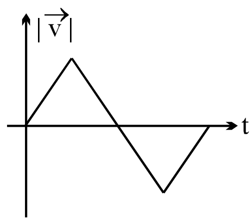
Q.9 Water is collecting in a bucket during a steady downpour. Will the rate at which the bucket is filling change if a steady horizontal wind starts to blow?

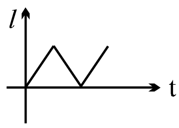
Q.10 Show that, taking the Earth's rotation and revolution into account, a book resting on your table moves faster at night than it does during the daytime. In what reference frame is this statement true?

Q.11 Which of the following graphs cannot possibily represent one dimensional motion of a particle?



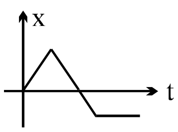
x – displacement



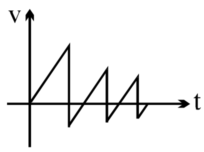


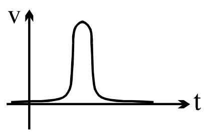
l – length of path

Q.12 Can you suggest a suitable situation from observation around you for each of the follwing?



x – displacement





Q.13 One of the following statements is incorrect.

(a) The car traveled around the track at a constant velocity.

(b) The car traveled around the track at a constant speed. Which statement is incorrect and why?

Q.14 Give an example from your own experience in which the velocity of an object is zero for just an instant of time, but its acceleration is not zero.

Q.15 A ball is dropped from rest from the top of a building and strikes the ground with a speed v_f. From ground level, a second ball is thrown straight upward at the same instant that the first ball is dropped. The initial speed of the second ball is $v_0 = v_f$, the same speed with which the first ball will eventually strike the ground. Ignoring air resistance, decide whether the balls cross paths at half the height of the building, above the halfway point, or below the halfway point. Give your reasoning.

Q.16 The muzzle velocity of a gun is the velocity of the bullet when it leaves the barrel. The muzzle velocity of one rifle with a short barrel is greater than the muzzle velocity of another rifle that has a loner barrel. In which rifle is the acceleration of the bullet larger? Explain your reasoning.

Q.17 On a riverboat cruise, a plastic bottle is accidentally dropped overboard. A passenger on the boat estimate that the boat pulls ahead of the bottle by 5 meters each second. Is it possible to conclude that the boat is moving at 5 m/s with respect to the shore? Account for your answer.

Q.18 A wrench is accidentally dropped from the top of the mast on a sailboat. Will the wrench hit at the same place on the deck whether the sailboat is at rest or moving with a constant velocity? Justify your answer.

Q.19 Is the acceleration of a projectile equal to zero when it reaches the top of its trajectory? If not, why not?

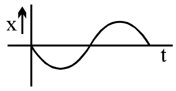
Q.20 A child is playing on the floor of a recreational vehicle (RV) as it moves along the highway at a constant velocity. He has a toy cannon, which shoots a marble at a fixed angle and speed with respect to the floor. The cannon can be aimed toward the front or the rear of the RV. Is the range toward the front the same as, less than, or greater than the range toward the rear? Answer this question (a) from the child's point of view and (b) from the point of view of an observer standing still on the ground. Justify your answers.

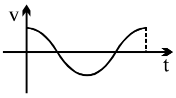
Q.21 Three swimmers can swim equally fast relative to the water. They have a race to see who can swim across a river in the least time. Swimmer A swims perpendicular to the current and lands on the far shore downstream, because the current has swept him in that direction. Swimmer B swims upstream at an angle to the current and lands on the far shore directly opposite the starting point. Swimmer C swims downstream at an angle to the current in an attempt to take advantage of the current. Who crosses the river in the least time? Account for your answer.

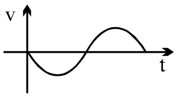
ONLY ONE OPTION IS CORRECT.

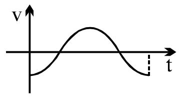
Take approx. 2 minutes for answering each question.

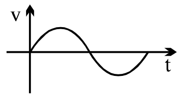
Q.1 If position time graph of a particle is sine curve as shown, what will be its velocity-time graph.



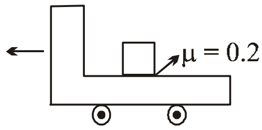
(A) 

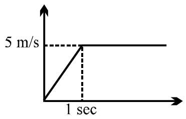
(B) 

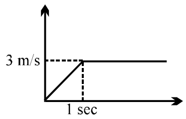
(C) 

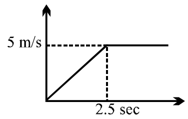
(D) 

Q.2 A truck starting from rest moves with an acceleration of 5 m/s^2 for 1 sec and then moves with constant velocity. The velocity w.r.t ground v/s time graph for block in truck is (Assume that block does not fall off the truck)



(A) 

(B) 

(C) 

(D) None of these

Q.3 If angular velocity of a disc depends an angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration α at $\theta = 1$ rad is :

(A) 8 rad/sec^2 (B) 10 rad/sec^2 (C) 12 rad/sec^2 (D) None

Q.4 If a particle takes t second less and acquires a velocity of v ms^{-1} more in falling through the same distance (starting from rest) on two planets where the accelerations due to gravity are 2 g and 8 g respectively then:

(A) v = 2gt (B) v = 4gt (C) v = 5 gt (D) v = 16 gt

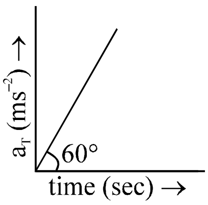
Q.5 It takes one minute for a passenger standing on an escalator to reach the top. If the escalator does not move it takes him 3 minute to walk up . How long will it take for the passenger to arrive at the top if he walks up the moving escalator ?

(A) 30 sec (B) 45 sec (C) 40 sec (D) 35 sec

Q.6 The co-ordinates of a moving particle at a time t, are give by, x = 5 sin 10 t, y = 5 cos 10t. The speed of the particle is :

(A) 25 (B) 50 (C) 10 (D) None

Q.7 Tangential acceleration of a particle moving in a circle of radius 1 m varies with time t as (initial velocity of particle is zero). Time after which total acceleration of particle makes and angle of 30° with radial acceleration is



(A) 4 sec (B) 4/3 sec

(C) $2^{2/3}$ sec (D) $\sqrt{2}$ sec

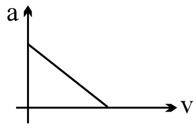
Q.8 A particle is projected from a horizontal plane (x-z plane) such that its velocity vector at time t is given by $\vec{V} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by

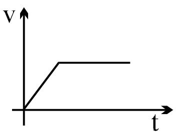
(A) $\frac{ba}{c}$ (B) $\frac{2ba}{c}$ (C) $\frac{3ba}{c}$ (D) None

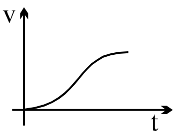
Q.9 A particle moves along a straight line in such a way that it's acceleration is increasing at the rate of 2 m/s^3. It's initial acceleration and velocity were 0, the distance covered by it in t = 3 second is.
(A) 27 m (B) 9 m (C) 3 m (D) 1 m

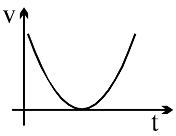
Q.10 A ball is thrown vertically down with velocity of 5m/s. With what velocity should another ball be thrown down after 2 seconds so that it can hit the 1st ball in 2 seconds
(A) 40 m/s (B) 55 m/s (C) 15 m/s (D) 25 m/s

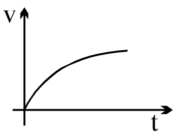
Q.11 Acceleration versus velocity graph of a particle moving in a straight line starting from rest is as shown in figure. The corresponding velocity-time graph would be



(A) 

(B) 

(C) 

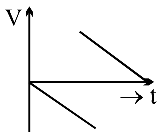
(D) 

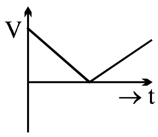
Q.12 A particle is projected vertically upwards from a point A on the ground. It takes t_1 time to reach a point B but it still continues to move up. If it takes further t_2 time to reach the ground from point B then height of point B from the ground is

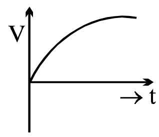
(A) $\frac{1}{2}g(t_1+t_2)^2$ (B) $g\,t_1\,t_2$ (C) $\frac{1}{8}g(t_1+t_2)^2$ (D) $\frac{1}{2}\,g\,t_1\,t_2$

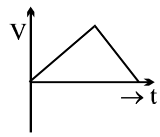
Q.13 Balls are thrown vertically upward in such a way that the next ball is thrown when the previous one is at the maximum height. If the maximum height is 5m, the number of balls thrown per minute will be
(A) 40 (B) 50 (C) 60 (D) 120

Q.14 The velocity- time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?

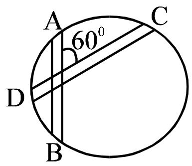
(A) 

(B) 

(C) 

(D) 

Q.15 A disc arranged in a vertical plane has two groves of same length directed along the vertical chord AB and CD as shown in the fig. The same particles slide down along AB and CD. The ratio of the time t_{AB}/t_{CD} is



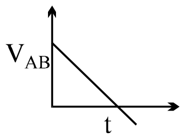
(A) 1 : 2 (B) 1 : $\sqrt{2}$ (C) 2 : 1 (D) $\sqrt{2}$: 1

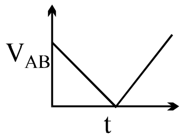
Q.16 The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as

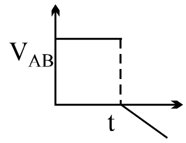
(A) $2a\sin\omega t$ (B) $2a\sin\frac{\omega t}{2}$ (C) $2a\cos\omega t$ (D) $2a\cos\frac{\omega t}{2}$

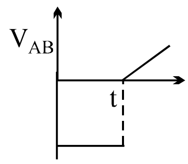
Q.17 A body moves with velocity $v = \ell n\, x$ m/s where x is its position. The net force acting on body is zero at
(A) 0 m (B) $x = e^2$ m (C) x = e m (D) x = 1 m

Q.18 A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of h. Simultaneously another body B is dropped from height h. It strikes the ground and does not rebound. The velocity of A relative to B v/s time graph is best represented by : (upward direction is positive)

(A) 

(B) 

(C) 

(D) 

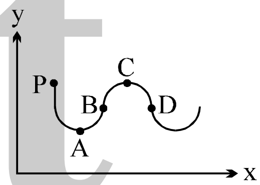
Q.19 A body of mass 1 kg is acted upon by a force $\vec{F} = 2\sin 3\pi t\,\hat{i} + 3\cos 3\pi t\,\hat{j}$ find its position at t = 1 sec if at t = 0 it is at rest at origin.

(A) $\left(\frac{3}{3\pi^2}, \frac{2}{9\pi^2}\right)$ (B) $\left(\frac{2}{3\pi^2}, \frac{2}{3\pi^2}\right)$ (C) $\left(\frac{2}{3\pi}, \frac{2}{3\pi^2}\right)$ (D) none of these

Q.20 A force $F = Be^{-Ct}$ acts on a particle whose mass is m and whose velocity is 0 at t = 0. It's terminal velocity is :

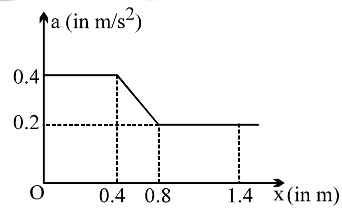
(A) $\frac{C}{mB}$ (B) $\frac{B}{mC}$ (C) $\frac{BC}{m}$ (D) $-\frac{B}{mC}$

Q.21 A man moves in x-y plane along the path shown. At what point is his average velocity vector in the same direction as his instantaneous velocity vector. The man starts from point P.



(A) A (B) B (C) C (D) D

Q.22 The acceleration of a particle which moves along the positive x-axis varies with its position as shown. If the velocity of the particle is 0.8 m/s at x = 0 , the velocity of the particle at x = 1.4 is(in m/s)



(A) 1.6 (B) 1.2
(C) 1.4 (D) none of these

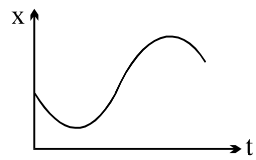
Q.23 A body is thrown up in a lift with a velocity u relative to the lift and the time of flight is found to be t. The acceleration with which the lift is moving up is

(A) $\frac{u - gt}{t}$ (B) $\frac{2u - gt}{t}$ (C) $\frac{u + gt}{t}$ (D) $\frac{2u + gt}{t}$

Q.24 A ball is thrown vertically downwards with velocity $\sqrt{2gh}$ from a height h. After colliding with the ground it just reaches the starting point. Coefficient of restitution is

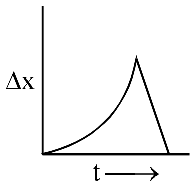
(A) $1/\sqrt{2}$ (B) 1/2 (C) 1 (D) $\sqrt{2}$

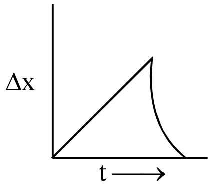
Q.25 The graph of position *x* versus time *t* represents the motion of a particle. If *b* and *c* are both positive constants, which of the following expressions best describes the acceleration a of the particle?

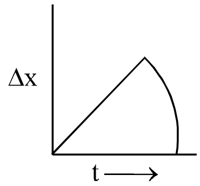


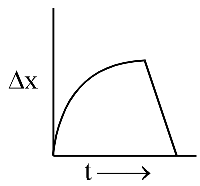
(A) a = b – ct (B) a = + b
(C) a = – c (D) a = b + ct

Q.26 Two stones are thrown up vertically and simultaneously but with different speeds. Which graph correctly represents the time variation of their relative positions Δx. Assume that stones do not bounce after hitting ground.

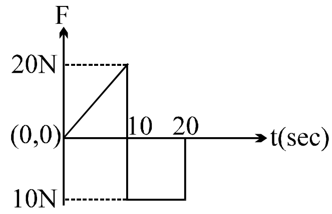
(A) 

(B) 

(C) 

(D) 

Q.27 A particle of mass 1 kg is acted upon by a force 'F' which varies as shown in the figure. If initial velocity of the particle is 10 ms^{-1}, the maximum velocity attained by the particle during the period is



(A) 210 ms^{-1} (B) 110 ms^{-1}
(C) 100 ms^{-1} (D) 90 ms^{-1}

Question No. 28 to 32

Two balls A and B are thrown with same velocity u from the top of a tower. Ball A is thrown vertically upwards and the ball B is thrown vertically downwards.

Q.28 Choose the correct statement
(A) Ball B reaches the ground with greater velocity
(B) Ball A reaches the ground with greater velocity
(C) Both the balls reach the ground with same velocity
(D) Cannot be interpreted

Q.29 If t_A and t_B are the respective times taken by the balls A and B respectively to reach the ground, then identify the correct statement
(A) $t_A > t_B$ (B) $t_A = t_B$ (C) $t_A < t_B$ (D) Cannot be interpreted

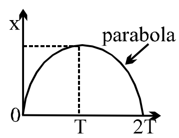
Q.30 If $t_A = 6$ s and $t_B = 2$ s, then the height of the tower is
(A) 80 m (B) 60 m (C) 45 m (D) none of these

Q.31 The velocity u of each ball is
(A) 10 ms^{-1} (B) 15 ms^{-1} (C) 20 ms^{-1} (D) none of these

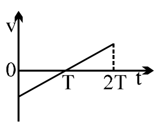
Q.32 If a ball C is thrown with the same velocity but in the horizontal direction from the top of the tower, then it will reach the ground in time t_c equal to
(A) 4 s (B) 3.46 s (C) 4.2 s (D) none of these

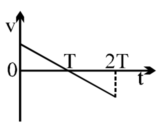
Question No. 33 to 36

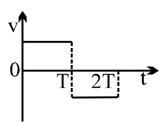
The x-t graph of a particle moving along a straight line is shown in figure

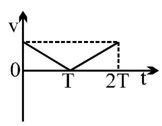


Q.33 The v-t graph of the particle is correctly shown by

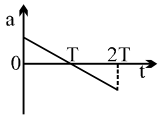
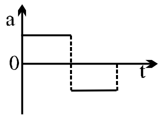
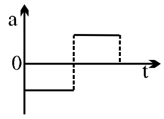
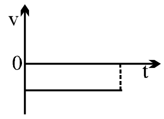
(A) 

(B) 

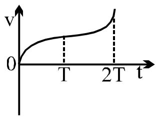
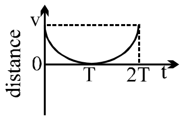
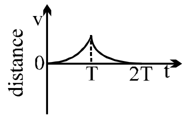
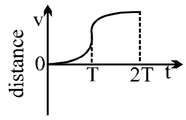
(C) 

(D) 

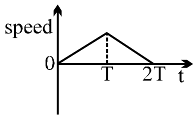
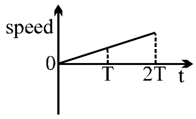
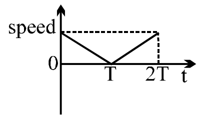
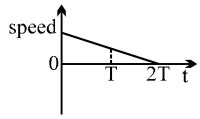
Q.34 The a-t graph of the particle is correctly shown by

(A) 
(B) 
(C) 
(D) 

Q.35 The distance-time graph of the particle is correctly shown by

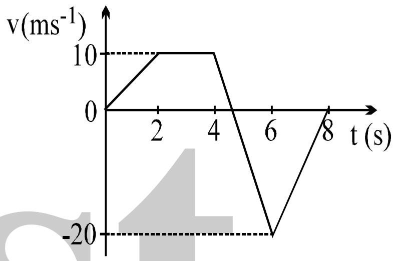
(A) 
(B) 
(C) 
(D) 

Q.36 The speed-time graph of the particle is correctly shown by

(A) 
(B) 
(C) 
(D) 

Question No. 37 to 43

The figure shows a velocity-time graph of a particle moving along a straight line



Q.37 Choose the incorrect statement. The particle comes to rest at
(A) t = 0 s (B) t = 5 s
(C) t = 8 s (D) none of these

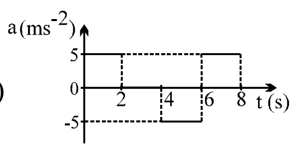
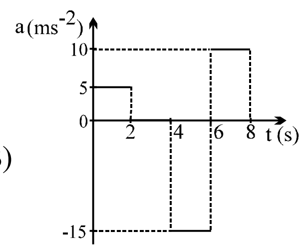
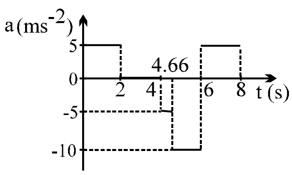
Q.38 Identify the region in which the rate of change of velocity $\left|\frac{\Delta\vec{v}}{\Delta t}\right|$ of the particle is maximum
(A) 0 to 2s (B) 2 to 4s (C) 4 to 6 s (D) 6 to 8 s

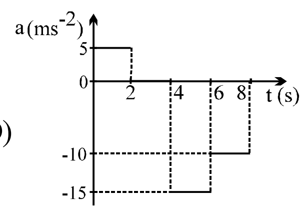
Q.39 If the particle starts from the position x_0 = –15 m, then its position at t = 2s will be
(A) –5 m (B) 5 m (C) 10 m (D) 15 m

Q.40 The maximum displacement of the particle is
(A) 33.3 m (B) 23.3 m (C) 18.3 m (D) zero

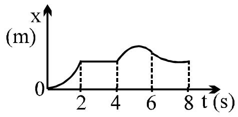
Q.41 The total distance travelled by the particle is
(A) 66.6 m (B) 51.6 m (C) zero (D) 36.6 m

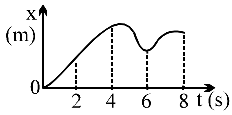
Q.42 The correct acceleration-time graph of the particle is shown as

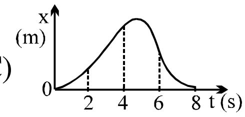
(A) 
(B) 
(C) 

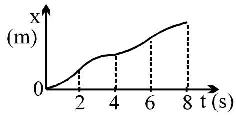
(D) 

Q.43 The correct displacement-time graph of the particle is shown as

(A) 

(B) 

(C) 

(D) 

Q.44 A ball is thrown from a point on ground at some angle of projection. At the same time a bird starts from a point directly above this point of projection at a height h horizontally with speed u. Given that in its flight ball just touches the bird at one point. Find the distance on ground where ball strikes

(A) $2u\sqrt{\frac{h}{g}}$ (B) $u\sqrt{\frac{2h}{g}}$ (C) $2u\sqrt{\frac{2h}{g}}$ (D) $u\sqrt{\frac{h}{g}}$

Q.45 A ball is projected from top of a tower with a velocity of 5 m/s at an angle of 53^0 to horizontal. Its speed when it is at a height of 0.45 m from the point of projection is :
(A) 2 m/s (B) 3 m/s (C) 4 m/s (D) data insufficient.

Q.46 Average velocity of a particle is projectile motion between its starting point and the highest point of its trajectory is : (projection speed = u, angle of projection from horizontal= θ)

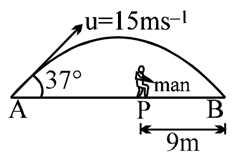
(A) $u\cos\theta$ (B) $\frac{u}{2}\sqrt{1+3\cos^2\theta}$ (C) $\frac{u}{2}\sqrt{2+\cos^2\theta}$ (D) $\frac{u}{2}\sqrt{1+\cos^2\theta}$

Q.47 Find time of flight of projectile thrown horizontally with speed 50 ms^{-1} from a long inclined plane which makes an angle of $\theta = 45°$ from horizontal.
(A) $\sqrt{2}$ sec (B) $2\sqrt{2}$ sec (C) 2 sec (D) none

Q.48 Particle is dropped from the height of 20m from horizontal ground. There is wind blowing due to which horizontal acceleration of the particle becomes 6 ms^{-2}. Find the horizontal displacement of the particle till it reaches ground.
(A) 6 m (B) 10 m (C) 12 m (D) 24 m

Q.49 A ball is dropped from height 5m. The time after which ball stops rebounding if coefficient of restitution between ball and ground e = 1/2, is
(A) 1 sec (B) 2 sec (C) 3 sec (D) infinite

Q.50 A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile.



(A) 3 ms^{-1} (B) 5 ms^{-1}
(C) 9 ms^{-1} (D) 12 ms^{-1}

Q.51 A projectile is fired with a speed u at an angle θ with the horizontal. Its speed when its direction of motion makes an angle 'α' with the horizontal is
(A) $u\sec\theta\cos\alpha$ (B) $u\sec\theta\sin\alpha$ (C) $u\cos\theta\sec\alpha$ (D) $u\sin\theta\sec\alpha$

Q.52 A projectile is fired with a velocity at right angle to the slope which is inclined at an angle θ with the horizontal. The expression for the range R along the incline is

(A) $\frac{2v^2}{g}\sec\theta$ (B) $\frac{2v^2}{g}\tan\theta$ (C) $\frac{2v^2}{g}\tan\theta\sec\theta$ (D) $\frac{v^2}{g}\tan^2\theta$

Q.53 A particle is projected vertically upwards from O with velocity v and a second particle is projected at the same instant from P (at a height h above O) with velocity v at an angle of projection θ. The time when the distance between them is minimum is

(A) $\dfrac{h}{2v\sin\theta}$ (B) $\dfrac{h}{2v\cos\theta}$ (C) h/v (D) h/2v

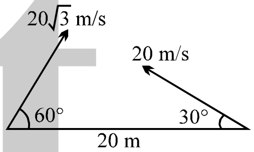
Q.54 A ball is projected from ground with a velocity V at an angle θ to the vertical. On its path it makes an elastic collison with a vertical wall and returns to ground. The total time of flight of the ball is

(A) $\dfrac{2v\sin\theta}{g}$ (B) $\dfrac{2v\cos\theta}{g}$ (C) $\dfrac{v\sin 2\theta}{g}$ (D) $\dfrac{v\cos\theta}{g}$

Q.55 Two particles are moving along two long straight lines, in the same plane, with the same speed = 20 cm/s. The angle between the two lines is 60°, and their intersection point is O. At a certain moment, the two particles are located at distances 3m and 4m from O, and are moving towards O. Subsequently, the shortest distance between them will be

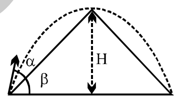
(A) 50 cm (B) $40\sqrt{2}$ cm (C) $50\sqrt{2}$ cm (D) $50\sqrt{3}$ cm

Q.56 In the figure shown, the two projectiles are fired simultaneously. The minimum distance between them during their flight is



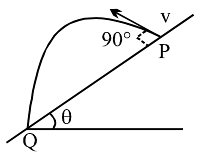
(A) 20 m (B) $10\sqrt{3}$ m (C) 10 m (D) None

Q.57 A shell fired from the base of a mountain just clears it. If α is the angle of projection then the angular elevation of the summit β is



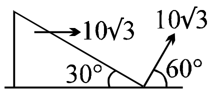
(A) $\dfrac{1}{2}\alpha$ (B) $\tan^{-1}(1/2)$ (C) $\tan^{-1}(1/2\tan\alpha)$ (D) $\tan^{-1}(2\tan\alpha)$

Q.58 If time taken by the projectile to reach Q is T, than PQ =



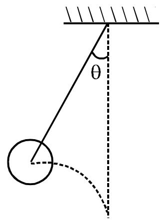
(A) Tv sin θ (B) Tv cos θ (C) Tv sec θ (D) Tv tan θ

Q.59 A particle is projected at angle 60° with speed $10\sqrt{3}$, from the point 'A' as shown in the figure. At the same time the wedge is made to move with speed $10\sqrt{3}$ towards right as shown in the figure. Then the time after which particle will strike with wedge is



(A) 2 sec (B) $2\sqrt{3}$ sec (C) $\dfrac{4}{\sqrt{3}}$ sec (D) None

Q.60 A ball is held in the position shown with string of length 1 m just taut & then projected horizontally with a velocity of 3 m/s. If the string becomes taut again when it is vertical, angle θ is given by

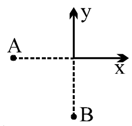


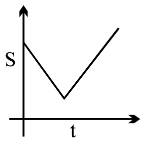
(A) 53° (B) 30° (C) 45° (D) 37°

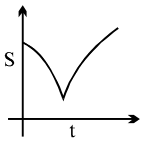
Q.61 Two projectiles A and B are thrown with the same speed such that A makes angle θ with the horizontal and B makes angle θ with the vertical, then

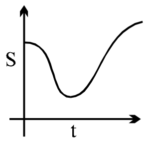
(A) Both must have same time of flight (B) Both must achieve same maximum height
(C) A must have more horizontal range than B (D) Both may have same time of flight

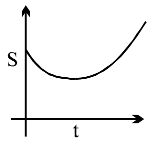
Q.62 Particles A and B are moving with constant velocities along x and y axis respectively, the graph of separation between them with time is



(A) 

(B) 

(C) 

(D) 

Q.63 An aeroplane flying at a constant velocity releases a bomb. As the bomb drops down from the aeroplane,
(A) it will always be vertically below the aeroplane
(B) it will always be vertically below the aeroplane only if the aeroplane is flying horizontally
(C) it will always be vertically below the aeroplane only if the aeroplane is flying at an angle of $45°$ to the horizontal
(D) it will gradually fall behind the aeroplane if the aeroplane is flying horizontally

Q.64 Two particles are projected simultaneously in the same vertical plane, from the same point on ground, but with same speeds but at different angles ($< 90°$) to the horizontal. The path followed by one, as seen by the other, is
(A) a vertical straight line
(B) a straight line making a constant angle with the horizontal
(C) a parabola
(D) a hyperbola

Q.65 Suppose a player hits several baseballs. Which baseball will be in the air for the longest time?
(A) The one with the farthest range.
(B) The one which reaches maximum height.
(C) The one with the greatest initial velocity.
(D) The one leaving the bat at $45°$ with respect to the ground.

Question No. 66 to 69

Two projectiles are thrown simultaneously in the same plane from the same point. If their velocities are v_1 and v_2 at angles θ_1 and θ_2 respectively from the horizontal, then answer the following questions

Q.66 The trajectory of particle 1 with respect to particle 2 will be
(A) a parabola
(B) a straight line
(C) a vertical straight line
(D) a horizontal straight line

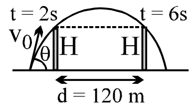
Q.67 If $v_1\cos\theta_1 = v_2\cos\theta_2$, then choose the incorrect statement
(A) one particle will remain exactly below or above the other particle
(B) the trajectory of one with respect to other will be a vertical straight line
(C) both will have the same range
(D) none of these

Q.68 If $v_1\sin\theta_1 = v_2\sin\theta_2$, then choose the incorrect statement
(A) the time of flight of both the particles will be same
(B) the maximum height attained by the particles will be same
(C) the trajectory of one with respect to another will be a horizontal straight line
(D) none of these

Q.69 If $v_1 = v_2$ and $\theta_1 > \theta_2$, then choose the incorrect statement
(A) Particle 2 moves under the particle 1
(B) The slope of the trajectory of particle 2 with respect to 1 is always positive
(C) Both the particle will have the same range if $\theta_1 > 45°$ and $\theta_2 < 45°$ and $\theta_1 + \theta_2 = 90°$
(D) none of these

Question No. 70 to 75

A projectile crosses two walls of equal height H symmetrically as shown



Q.70 The time of flight T is given by

(A) 8 s (B) 9 s (C) 7 s (D) 10 s

Q.71 The height of each wall is

(A) 240 m (B) 120 m (C) 60 m (D) 30 m

Q.72 The maximum height of the projectile is

(A) 120 m (B) 80 m (C) 160 m (D) cannot be obtained

Q.73 If the horizontal distance between the two walls is d = 120 m, then the range of the projectile is

(A) 240 m (B) 160 m (C) 300 m (D) cannot be obtained

Q.74 The angle of projection of the projectile is

(A) $\tan^{-1}(3/4)$ (B) $\tan^{-1}(4/3)$ (C) $\tan^{-1}(4/5)$ (D) $\tan^{-1}(3/5)$

Q.75 The velocity of projection is

(A) 30 ms^{-1} (B) 40 ms^{-1} (C) 50 ms^{-1} (D) none of these

Question No. 76 to 78

A projectile is thrown with a velocity of 50 ms^{-1} at an angle of 53° with the horizontal

Q.76 Choose the incorrect statement

(A) It travels vertically with a velocity of 40 ms^{-1}

(B) It travels horizontally with a velocity of 30 ms^{-1}

(C) The minimum velocity of the projectile is 30 ms^{-1}

(D) None of these

Q.77 Determine the instants at which the projectile is at the same height

(A) t = 1s and t = 7s (B) t = 3s and t = 5s (C) t = 2s and t = 6 s (D) all the above

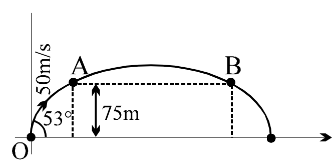
Q.78 The equation of the trajectory is given by

(A) $180y = 240x - x^2$ (B) $180y = x^2 - 240x$

(C) $180y = 135x - x^2$ (D) $180y = x^2 - 135x$

Question No. 79 & 80

At t = 0 a projectile is fired from a point O(taken as origin) on the ground with a speed of 50 m/s at an angle of 53° with the horizontal. It just passes two points A & B each at height 75 m above horizontal as shown.



Q.79 The horizontal separation between the points A and B is

(A) 30 m (B) 60 m (C) 90 m (D) None

Q.80 The distance (in metres) of the particle from origin at t = 2 sec.

(A) $60\sqrt{2}$ (B) 100 (C) 60 (D) 120

Q.81 A particle starts moving rectilinearly at time t = 0 such that its velocity 'v' changes with time 't' according to the equation $v = t^2 - t$ where t is in seconds and v is in m/s. The time interval for which the particle retards is
(A) $t < 1/2$ (B) $1/2 < t < 1$ (C) $t > 1$ (D) $t < 1/2$ and $t > 1$

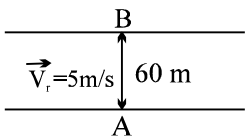
Q.82 A swimmer swims in still water at a speed = 5 km/hr. He enters a 200 m wide river, having river flow speed = 4 km/hr at point A and proceeds to swim at an angle of 127° (sin 37° = 0.6) with the river flow direction. Another point B is located directly across A on the other side. The swimmer lands on the other bank at a point C, from which he walks the distance CB with a speed = 3 km/hr. The total time in which he reachrs from A to B is
(A) 5 minutes (B) 4 minutes (C) 3 minutes (D) None

Q.83 A boat having a speed of 5 km/hr. in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes. The speed of the river in Km/hr.
(A) 1 (B) 3 (C) 4 (D) $\sqrt{41}$

Q.84 Two particles start simultaneously from the same point and move along two straight lines, one with uniform velocity v and other with a uniform acceleration a. If α is the angle between the lines of motion of two particles then the least value of relative velocity will be at time given by
(A) (v/a) sin α (B) (v/a) cos α (C) (v/a) tan α (D) (v/a) cot α

Q.85 A flag is mounted on a car moving due North with velocity of 20 km/hr. Strong winds are blowing due East with velocity of 20 km/hr. The flag will point in direction
(A) East (B) North - East (C) South - East (D) South - West

Q.86 A man is crossing a river flowing with velocity of 5 m/s. He reaches a point directly across at a distance of 60 m in 5 sec. His velocity in still water should be



(A) 12 m/s (B) 13 m/s
(C) 5 m/s (D) 10 m/s

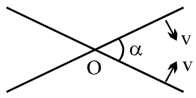
Q.87 A man swimming down stream overcome a float at a point M. After travelling distance D he turned back and passed the float at a distance of D/2 from the point M, then the ratio of speed of swimmer with respect to still water to the speed of the river will be
(A) 2 (B) 3 (C) 4 (D) 2.5

Q.88 A glass wind screen whose inclination with the vertical can be changed is mounted on a car. The car moves horizontally with a speed of 2m/s. At what angle α with the vertical should the wind screen be placed so that the rain drops falling vertically downwards with velocity 6 m/s strike the wind screen perpendicularly.
(A) $\tan^{-1}(3)$ (B) $\tan^{-1}(1/3)$ (C) $\cos^{-1}(3)$ (D) $\sin^{-1}(1/3)$

Q.89 Wind is blowing in the north direction at speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him :
(A) 2 m/s south (B) 2 m/s north (C) 4 m/s west (D) 4 m/s south

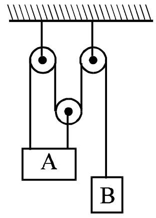
Q.90 Three particles, located initially on the vertices of an equilateral triangle of side L, start moving with a constant tangential acceleration towards each other in a cyclic manner, forming spiral loci that coverage at the centroid of the triangle. The length of one such spiral locus will be
(A) L/3 (B) $2L/\sqrt{3}$ (C) $L/\sqrt{2}$ (D) 2L/3

Q.91 Find the speed of the intersection point O of the two wires if the wires starts moving perpendicular to itself with speed v as shown in figure.



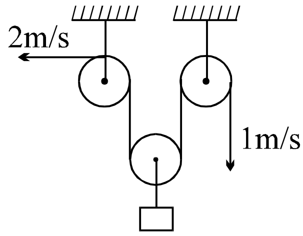
(A) v cosec(α/2) (B) v cosec(α) (C) v cos (α/2) (D) v sec (α/2)

Q.92 At a given instant, A is moving with velocity of 5m/s upwards. What is velocity of B at that time



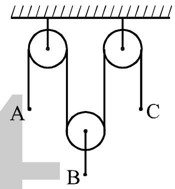
(A) 15 m/s ↓ (B) 15 m/s ↑
(C) 5 m/s ↓ (D) 5 m/s ↑

Q.93 Find the velocity of the hanging block if the velocities of the free ends of the rope are as indicated in the figure.



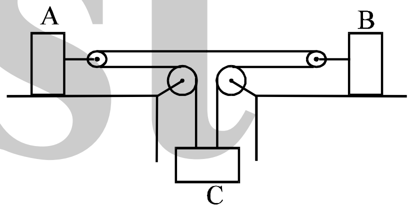
(A) 3/2 m/s ↑
(B) 3/2 m/s ↓
(C) 1/2 m/s ↑
(D) 1/2 m/s ↓

Q.94 The pulleys in the diagram are all smooth and light. The acceleration of A is a upwards and the acceleration of C is f downwards. The acceleration of B is



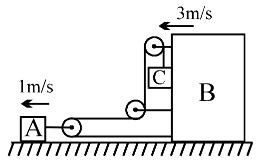
(A) ½(f - a) up (B) ½(a + f) down
(C) ½(a + f) up (D) ½(a - f) up

Q.95 If acceleration of A is 2 m/s^2 to left and acceleration of B is 1m/s^2 to left, then acceleration of C is



(A) 1 m/s^2 upwards (B) 1 m/s^2 downwards
(C) 2 m/s^2 downwards (D) 2 m/s^2 upwards

Q.96 The velocities of A and B are marked in the figure. The velocity of block C is (assume that the pulleys are ideal and string inextensible)



(A) 5 m/s (B) 2 m/s
(C) 3 m/s (D) 4 m/s

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

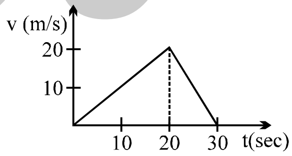
Take approx. 3 minutes for answering each question.

Q.1 A particle is moving along a curve. Then
(A) if its speed is constant it has no acceleration
(B) if its speed is increasing the acceleration of the particle is along its direction of motion
(C) if its speed is constant the magnitude of its acceleration is proportional to its curvature.
(D) the direction of its acceleration cannot be along the tangent.

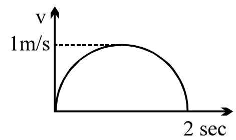
Q.2 A block is thrown with a velocity of 2 ms^{-1} (relative to ground) on a belt, which is moving with velocity 4 ms^{-1} in opposite direction of the initial velocity of block. If the block stops slipping on the belt after 4 sec of the throwing then choose the correct statements (s)
(A) Displacement with respect to ground is zero after 2.66 and magnitude of displacement with respect to ground is 12 m after 4 sec.
(B) Magnitude of displacement with respect to ground in 4 sec is 4 m.
(C) Magnitude of displacement with respect to belt in 4 sec is 12 m.
(D) Displacement with respect to ground is zero in 8/3 sec.

Q.3 A particle has initial velocity 10 m/s. It moves due to constant retarding force along the line of velocity which produces a retardation of 5 m/s^2. Then
(A) the maximum displacement in the direction of initial velocity is 10 m
(B) the distance travelled in first 3 seconds is 7.5 m
(C) the distance travelled in first 3 seconds is 12.5 m
(D) the distance travelled in first 3 seconds is 17.5 m.

Q.4 v-t graph of an object of mass 1 kg is shown
(A) net work done on the object in 30 sec is zero.
(B) the average acceleration of the object is zero.
(C) the average velocity of the object is zero.
(D) the average force on the object is zero.

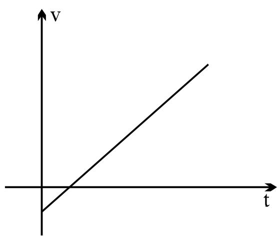


Q.5 Velocity-time graph for a car is semicircle as shown here. Which of the following is correct :
(A) Car must move in circular path.
(B) Acceleration of car is never zero.
(C) Mean speed of the particle is $\pi/4$ m/s.
(D) The car makes a turn once during its motion.



Q.6 Mark the correct statements for a particle going on a straight line
(A) if the velocity is zero at any instant, the acceleration should also be zero at that instant
(B) if the velocity is zero for a time interval, the acceleration is zero at any instant within the time interval
(C) if the velocity and acceleration have opposite sign, the object is slowing down
(D) if the position and velocity have opposite sign, the particle is moving towards the origin

Q.7 From the velocity time garph of a particle moving in straight line decide which of the following is incorrect statement.
(A) the particle crosses its initial position
(B) the speed of the particle increases continuously
(C) the force on the particle is constant
(D) the acceleration of the particle is constant.



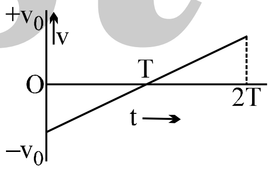
Q.8 A particle initially at rest is subjected to two forces. One is constant, the other is a retarding force proportional to the particle velocity. In the subsequent motion of the particle :
(A) the acceleration will increase from zero to a constant value
(B) the acceleration will decrease from its initial value to zero
(C) the velocity will increase from zero to maximum & then decrease
(D) the velocity will increase from zero to a constant value.

Q.9 A particle moves in a circle of radius R, with a constant speed v. Then, during a time interval [$\pi R/3v$], which of the following is true?
(A) | average acceleration | = $3v^2/\pi R$ (B) | average velocity | = $3v/\pi$
(C) | average acceleration | = $2v^2/\pi R$ (D) average speed = $3v/\pi$

Q.10 An observer moves with a constant speed along the line joining two stationary objects. He will observe that the two objects
(A) have the same speed (B) have the same velocity
(C) move in the same direction (D) move in opposite directions

Q.11 Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one–dimensional motion
(A) $|\vec{v}|$ must decrease when $\vec{a} < 0$
(B) Speed must increase when $\vec{a} > 0$
(C) Speed will increase when both $\vec{v}$ and $\vec{a}$ are < 0
(D) Speed will decrease when $\vec{v} < 0$ and $\vec{a} > 0$

Q.12 The figure shows the velocity (v) of a particle plotted against time (t)



(A) The particle changes its direction of motion at some point
(B) The acceleration of the particle remains constant
(C) The displacement of the particle is zero
(D) The initial and final speeds of the particle are the same

Q.13 Which of the following statements are true for a moving body?
(A) If its speed changes, its velocity must change and it must have some acceleration
(B) If its velocity changes, its speed must change and it must have some acceleration
(C) If its velocity changes, its speed may or may not change, and it must have some acceleration
(D) If its speed changes but direction of motion does not change, its velocity may remain constant

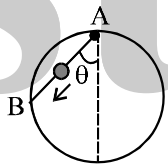
Q.14 Let v and a denote the velocity and acceleration respectively of a body
(A) a can be non zero when $v = 0$
(B) a must be zero when $v = 0$
(C) a may be zero when $v \neq 0$
(D) The direction of a must have some correlation with the direction of v

Q.15 The displacement x of a particle depend on time t as $x = \alpha t^2 - \beta t^3$
(A) particle will return to its starting point after time $\frac{\alpha}{\beta}$.
(B) the particle will come to rest after time $\frac{2\alpha}{3\beta}$
(C) the initial velocity of the particle was zero but its initial acceleration was not zero.
(D) no net force act on the particle at time $\frac{\alpha}{3\beta}$

Q.16 A projectile of mass 1 kg is projected with a velocity of $\sqrt{20}$ m/s such that it strikes on the same level as the point of projection at a distance of $\sqrt{3}$ m. Which of the following options are incorrect:
(A) the maximum height reached by the projectile can be 0.25 m.
(B) the minimum velocity during its motion can be $\sqrt{15}$ m/s
(C) the time taken for the flight can be $\sqrt{\frac{3}{5}}$ sec.
(D) maximum potential energy during its motion can be 6J.

Q.17 Choose the correct alternative (s)
(A) If the greatest height to which a man can throw a stone is h, then the greatest horizontal distance upto which he can throw the stone is 2h.
(B) The angle of projection for a projectile motion whose range R is n times the maximum height is $\tan^{-1}(4/n)$
(C) The time of flight T and the horizontal range R of a projectile are connected by the equation $gT^2 = 2R\tan\theta$ where θ is the angle of projection.
(D) A ball is thrown vertically up. Another ball is thrown at an angle θ with the vertical. Both of them remain in air for the same period of time. Then the ratio of heights attained by the two balls 1 : 1.

Q.18 A bead is free to slide down a smooth wire tightly stretched between points A and B on a vertical circle. If the bead starts from rest at A, the highest point on the circle
(A) its velocity v on arriving at B is proportional to $\cos\theta$
(B) its velocity v on arriving at B is proportional to $\tan\theta$
(C) time to arrive at B is proportional to $\cos\theta$
(D) time to arrive at B is independent of θ



Q.19 If T is the total time of flight, h is the maximum height & R is the range for horizontal motion, the x & y co-ordinates of projectile motion and time t are related as :

(A) $y = 4h\left(\frac{t}{T}\right)\left(1-\frac{t}{T}\right)$ (B) $y = 4h\left(\frac{X}{R}\right)\left(1-\frac{X}{R}\right)$

(C) $y = 4h\left(\frac{T}{t}\right)\left(1-\frac{T}{t}\right)$ (D) $y = 4h\left(\frac{R}{X}\right)\left(1-\frac{R}{X}\right)$

Q.20 A man on a rectilinearly moving cart, facing the direction of motion, throws a ball straight up with respect to himself
(A) The ball will always return to him
(B) The ball will never return to him
(C) The ball will return to him if the cart moves with constant velocity
(D) The ball will fall behind him if the cart moves with some acceleration

Q.21 A particle moves in the xy plane with a constant acceleration 'g' in the negative y-direction. Its equation of motion is $y = ax - bx^2$, where a and b are constants. Which of the following are correct?
(A) The x-component of its velocity is constant.
(B) At the origin, the y-component of its velocity is $a\sqrt{\frac{g}{2b}}$.
(C) At the origin, its velocity makes an angle $\tan^{-1}(a)$ with the x-axis.
(D) The particle moves exactly like a projectile.

Q.22 A ball is rolled off along the edge of a horizontal table with velocity 4 m/s. It hits the ground after time 0.4 s. Which of the following are correct?
(A) The height of the table is 0.8 m
(B) It hits the ground at an angle of 60° with the vertical
(C) It covers a horizontal distance 1.6 m from the table
(D) It hits the ground with vertical velocity 4 m/s

Q.23 A particle is projected from the ground with velocity u at angle θ with horizontal. The horizontal range, maximum height and time of flight are R, H and T respectively. They are given by,

$$R = \frac{u^2 \sin 2\theta}{g}, H = \frac{u^2 \sin^2 \theta}{2g} \text{ and } T = \frac{2u \sin \theta}{g}$$

Now keeping u as fixed, θ is varied from 30° to 60°. Then,
(A) R will first increase then decrease, H will increase and T will decrease
(B) R will first increase then decrease while H and T both will increase
(C) R will decrease while H and T will increase
(D) R will increase while H and T will increase

Q.24 A particle moves with constant speed v along a regular hexagon ABCDEF in the same order. Then the magnitude of the average velocity for its motion from A to
(A) F is v/5 (B) D is v/3 (C) C is $v\sqrt{3}/2$ (D) B is v

ANSWER KEY

ONLY ONE OPTION IS CORRECT.

Q.1	C	Q.2	C	Q.3	C	Q.4	B	Q.5	B	Q.6	B	Q.7	C
Q.8	B	Q.9	B	Q.10	A	Q.11	D	Q.12	D	Q.13	C	Q.14	A
Q.15	B	Q.16	B	Q.17	D	Q.18	C	Q.19	C	Q.20	B	Q.21	C
Q.22	B	Q.23	B	Q.24	A	Q.25	A	Q.26	C	Q.27	B	Q.28	C
Q.29	A	Q.30	B	Q.31	C	Q.32	B	Q.33	B	Q.34	D	Q.35	A
Q.36	C	Q.37	B	Q.38	C	Q.39	A	Q.40	A	Q.41	A	Q.42	B
Q.43	C	Q.44	C	Q.45	C	Q.46	B	Q.47	C	Q.48	C	Q.49	C
Q.50	B	Q.51	C	Q.52	C	Q.53	D	Q.54	B	Q.55	D	Q.56	C
Q.57	C	Q.58	D	Q.59	A	Q.60	D	Q.61	D	Q.62	D	Q.63	A
Q.64	B	Q.65	B	Q.66	B	Q.67	C	Q.68	D	Q.69	B	Q.70	A
Q.71	C	Q.72	B	Q.73	A	Q.74	B	Q.75	C	Q.76	A	Q.77	D
Q.78	A	Q.79	B	Q.80	A	Q.81	B	Q.82	B	Q.83	B	Q.84	B
Q.85	C	Q.86	B	Q.87	B	Q.88	A	Q.89	B	Q.90	D	Q.91	A
Q.92	A	Q.93	A	Q.94	A	Q.95	A	Q.96	A				

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	C,D	Q.2	B,C,D	Q.3	A,C	Q.4	A,B,D
Q.5	C	Q.6	B,C,D	Q.7	B	Q.8	B,D
Q.9	A,B	Q.10	A,B,C	Q.11	C,D	Q.12	A,B,C,D
Q.13	A,C	Q.14	A,C	Q.15	A,B,C,D	Q.16	D
Q.17	A,B,C,D	Q.18	A,D	Q.19	A,B	Q.20	C,D
Q.21	A,B,C,D	Q.22	A,C,D	Q.23	B	Q.24	A,C,D

Power Coaching

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PHYSICS

PARTICLE DYNAMICS

QUESTION FOR SHORT ANSWER

NEWTON LAW & FRICTION

Q.1 In a tug of war, three men pull on a rope to the left at A and three men pull to the right at B with forces of equal magnitude. Now a weight of 5.0 lb is hung vertically from the center of the rope.
(a) Can the men get the rope AB to be horizontal?
(b) If not, explain. If so, determine the magnitude of the forces required at A and B to do this.

Q.2 A massless rope is strung over africtionless pulley. A monkey holds onto one end of the rope and a mirror, having the same weight as the monkey, is attached to the other end of the rope at the mankey's level. Can the monkey get away from his image seen in the mirror.
(a) by climbing up the rope
(b) by climbing down the rope
(c) by releasing the rope?

Q.3 A student standing on the large platform of a spring scale notes his weight. He then takes a step on this platform and noticed that the scale reads less than his weight at the beginning of the step and more than his weight at the end of the step. Explain.

Q.4 An object is placed far away from all the objects that can exert force on it. A frame of reference is constructed by taking the origin and axes fixed in this object. Will the frame be necessarily inertial?

Q.5 The acceleration of a particle is zero as measured from an inertial frame of refrence. Can we conclude that no forces acts on the particle?

Q.6 Two blocks of unequal masses are tied by a spring. The blocks are pulled stretching the spring slightly and the system is released on a frictionless horizontal platform. Are the forces due to the spring on the two blocks equal and opposite? If yes, is it an example of Newton's third law?

Q.7 How could a person who is at rest on completely frictionless ice covering a pond reach shore? Could he do this by walking, rolling, swinging his arms, or kicking his feet? How could a person be placed in such a position in the first place?

Q.8 If you want to stop the car in the shortest distance on an ice road, should you
(a) push hard on the brakes to lock the wheels, (b) push just hard enough to prevent slipping, or (c) "pump" the breakes?

Q.9 How does the earth's rotation affect the apparent weight of a body at the equator?

Q.10 Suppose you need to measure whether a table top in a train is truly horizontal. If you use a spirit level can you determine this when the train is moving down or up a grade? When the train is moving along a curve?

Q.11 A classroom demonstration of Newton's first law is as follows: A glass is covered with a plastic card and a coin is placed on the card. The card is given a quick strike and the coin falls in the glass.
(a) Should the friction coefficient between the card and the coin be small or large?
(b) Should the coin be light or heavy?
(c) Why does the experiment fail if the card is gently pushed?

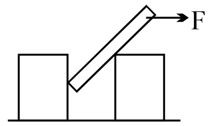
Q.12 Can a sailing boat be propelled by air blown at the sails from a fan attached to the boat?

Q.13 Why is it difficult to walk on sand?

Q.14 Explain why a man getting out of a moving train must run in the same direction for a cerain distance.

Q.15 During a high jump event, it hurts less when an athlete lands on a heap of sand. Explain.

Q.16 A rod not reaching the rough floor is inserted between two identical blocks. A horizontal force F is applied to the upper end of the rod. Which of the blocks will move first?



Q.17 A woman in an elevator lets go of her briefcase but it does not fall to the floor. How is the elevator moving?

Q.18 You take two identical tennis balls and fill one with water. You release both balls simultaneously from the top of a tall building. If air resistance is negligible, which ball strikes the ground first? Explain. What is the answer if air resistance is not negligible?

Q.19 "A ball is thrown from the edge of a high cliff. No matter what the angle at which it is thrown, due to air resistance, the ball will eventually end up moving vertically downward." Justify this statement.

WORK POWER ENERGY

Q.1 How do you explain the fact that when a stone is dropped onto the ground, the change in the momentum of the Earth is equal to that of the stone, while the change in the kinetic energy of the Earth is neglected ?

Q.2 You must have noticed that when you stir a cup of tea, the floating tea leaves collect at the centre of the cup rather than at the outer rim. Can you explain this?

Q.3 Explain why you became physically tired when you push against a wall, fail to move it, and therefore do not work on the wall.

Q.4 A man rowing a boat upstream is at rest with respect to the shore. (a) Is he doing any work? (b) If he stops rowing and moves down with the stream, is any work being done on him?

Q.5 A man bounces on a trampoline, going a little higher with each bounce. Explain how he increases the total mechanical energy.

Q.6 A rope tied to a body is pulled, causing the body to accelerate. But according to Newton's third law, the body pulls back on the rope with an equal and opposite force. Is the total work done then zero? If so, how can the body's kinetic energy change ? Explain.

Q.7 If there is a net force on a moving object that is nonzero and constant in magnitude, is it possible for the total work done on the object to be zero? Explain, with an example that illustrates your answer.

Q.8 Consider yourself sitting in an elevator moving up with a constant velocity. In your reference frame, earth has a large kinetic energy ($\frac{1}{2}M_e v^2$). From where did it gain this kinetic energy?

Q.9 If you hang fuzzy dice from your rear-view mirror and drive through a banked curve, how can you tell whether you are travelling less than, equal to, or greater than the speed used to calculate the banking angle?

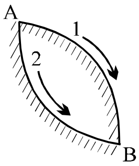
Q.10 If you observer a moving motorcycle moving with velocity v, you observe power expended by engine as $\vec{F}.\vec{v}$. But if you are observing the same motorcycle while moving with velocity $\vec{u}$, you will observe power as $\vec{F}(\vec{v}-\vec{u})$. Still petrol used by engine is same. Explain.

Q.11 A spirit level is tied to a string and whirled rapidly in a horizontal plane. Where will the bubble lie?

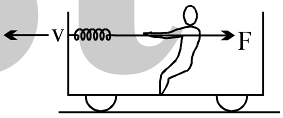
Q.12 A coin is put on the turntable of a record player. The motor is started, but before the final speed of rotation is reached, the coin flies off. Explain.

Q.13 A loose bolt gets detached from the circumference of a big, rotating platform. In what direction will an observer on the platform see it move? In what direction will an observer on the ground see it move?

Q.14 A small body slides along equally rough circular surfaces from A to B via route 1 and 2 of equal radius. In which case will the body's velocity be greater?



Q.15 A man stretches a spring attached to the front wall of railway carriage over a distance l in a uniformly moving train. During this time the train covers a distance L. Does the work done by the man depend on the coordinate system related to the earth or the train? The man moves opposite to the direction of motion of the train as he stretches the spring.



Q.16 Is there any net work done by external forces on a car moving with a constant speed along a straight road?

Q.17 Is a body in circular motion in equilibrium?

Q.18 A wheel rotating at a constant angular velocity ω about a fixed axis is in mechanical equilibrium because no net external force or torque acts on it. However, the particles that make up the wheel undergo a centripetal acceleration $\vec{a}$ directed toward the axis. Since $\vec{a} \neq 0$, how can the wheel be said to be in equilibrium?

Q.19 A car speeds up while the engine delivers constant power. Is the acceleration greater at the beginning of this process or at the end? Explain.

Q.20 A curve in a road has the banking angle calculated for 80 km/h. However, the road is covered with ice, and you plan to creep around the highest lane at 20 km/h. What may happen to your car? Why?

CENTRE OF MASS & MOMENTUM

Q.1 Consider a one-dimensional elastic collision between a given incoming body A and a body B initially rest. How would you choose the mass of B, in comparison to the mass of A, in order that B should recoil with (a) the greatest speed, (b) the greatest momentum, and (c) the greatest kinetic energy?

Q.2 Can the coefficient of restitution ever be greater that 1?

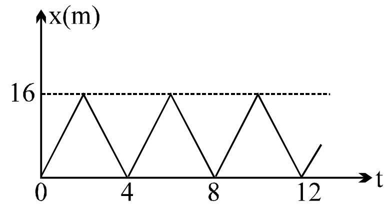
Q.3 A rocket following a parabolic path through the air suddenly explodes into many pieces. What can you say about the motion of this system of pieces?

Q.4 If only an external force can change the momentum of the centre of mass of an object, how can the internal force of the engine accelerate a car?

Q.5 Why does a fielder lower his hand while taking a 'catch'?

Q.6 Why does a gun appear to have a greater 'kick' when fired with the butt held loosely against the shoulder than when held tightly?

Q.7 Figure shows the position-time graph of a particle of mass m =0.5 kg. Suggest a suitable example to fit the curve. What is the interval between ten consecutive impulses? What is the magnitude of each impulse?



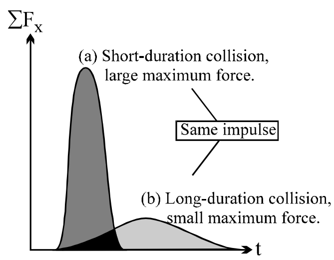
Q.8 Do the cm and the cg coincide for a building? For a lake? Under what conditions does the difference between these two points become insignificant?

Q.9 A car has the same kinetic energy when it is traveling south at 30 m/s as when it is traveling northwest at 30 m/s. Is the momentum of the car the same in both cases? Explain.

Q.10 A truck is accelerating as it speeds down the highway. One inertial frame of reference is attached to the ground with its origin at a fence post. A second frame of reference is attached to a police car that is traveling down the highway at constant velocity. Is the momentum of the truck the same in these two reference frames? Explain. Is the rate of change of the truck's momentum the same in these two frames? Explain.

Q.11 When a large, heavy truck collides with a passenger car, the occupants of the car are more likely to be hurt than the truck driver. Why?

Q.12 A glass dropped on the floor is more likely to break if the floor is concrete than if it is wood.Why?



Q.13 A machine gun is fired at a steel plate. Is the average force on the plate from the bullet impact greater if the bullets bounce off or if they are squashed and stick to the plate? Explain.

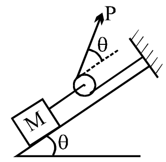
Q.14 A net force with x-component ΣF_x acts on an object from time t_1 to time t_2. The x-component of the momentum of the object is the same at t_1 as it is at t_2, but ΣF_x is not zero at all times between t_1 and t_2. What can you say about the graph of ΣF_x versus t?

Q.15 In a head-on auto collision, passengers not wearing seat belts can be thrown through the windshield. Use Newton's laws of motion to explain why this happens.

ONLY ONE OPTION IS CORRECT.

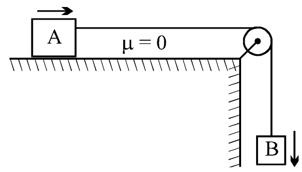
Take approx. 2 minutes for answering each question.

Q.1 What should be the minimum force P to be applied to the string so that block of mass m just begins to move up the frictionless plane.



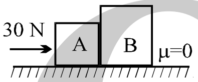
(A) $Mg \tan \theta/2$ (B) $Mg \cot \theta/2$ (C) $\frac{Mg \cos \theta}{1+\sin \theta}$ (D) None

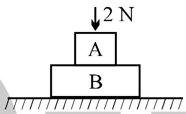
Q.2 Both the blocks shown here are of mass m and are moving with constant velocity in direction shown in a resistive medium which exerts equal constant force on both blocks in direction opposite to the velocity. The tension in the string connecting both of them will be : (Neglect friction)

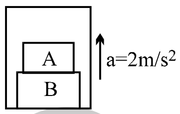


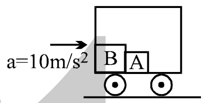
(A) mg (B) mg/2
(C) mg/3 (D) mg/4

Q.3 In which of the following cases is the contact force between A and B maximum ($m_A = m_B = 1$ kg)

(A) 

(B) 

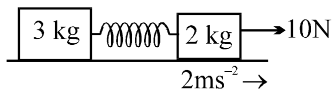
(C) 

(D) 

Q.4 A rope of mass 5 kg is moving vertically in vertical position with an upwards force of 100 N acting at the upper end and a downwards force of 70 N acting at the lower end. The tension at midpoint of the rope is

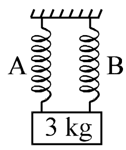
(A) 100 N (B) 85 N (C) 75 N (D) 105 N

Q.5 Find the acceleration of 3 kg mass when acceleration of 2 kg mass is 2 ms^{-2} as shown in figure.



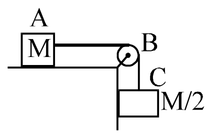
(A) 3 ms^{-2} (B) 2 ms^{-2}
(C) 0.5 ms^{-2} (D) zero

Q.6 Block of 1 kg is initially in equilibrium and is hanging by two identical springs A and B as shown in figures. If spring A is cut from lower point at t=0 then, find acceleration of block in ms^{-2} at t = 0.



(A) 5 (B) 10 (C) 15 (D) 0

Q.7 A block of mass M on a horizontal smooth surface is pulled by a load of mass $M/2$ by means of a rope AB and string BC as shown in the figure. The length & mass of the rope AB are L and $M/2$ respectively. As the block is pulled from AB = L to AB = 0 its acceleration changes from

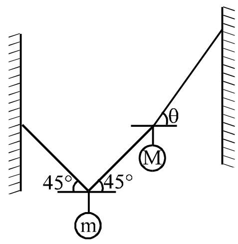


(A) $\frac{3g}{4}$ to g (B) $\frac{g}{4}$ to $\frac{g}{2}$ (C) $\frac{g}{4}$ to g (D) $\frac{3g}{2}$ to 2g

Q.8 A particle of mass m, initially at rest, is acted on by a force $F = F_0 \left\{1-\left(\frac{2t-T}{T}\right)^2\right\}$ during the interval $0 \le t \le T$. The velocity of the particle at the end of the interval is :

(A) $\frac{5F_0T}{6m}$ (B) $\frac{4F_0T}{3m}$ (C) $\frac{2F_0T}{3m}$ (D) $\frac{3F_0T}{2m}$

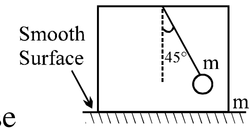
Q.9 Two masses m and M are attached to the strings as shown in the figure. If the system is in equilibrium, then



(A) $\tan\theta = 1 + \frac{2M}{m}$ (B) $\tan\theta = 1 + \frac{2m}{M}$

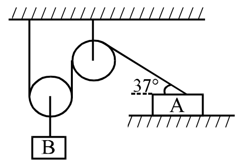
(C) $\cot\theta = 1 + \frac{2M}{m}$ (D) $\cot\theta = 1 + \frac{2m}{M}$

Q.10 A ball connected with string is released at an angle 45° with the vertical as shown in figure. Then the acceleration of the box at this instant will be: [Mass of the box is equal to mass of ball]



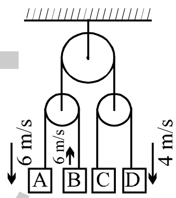
(A) g/4 (B) g/3 (C) g/2 (D) none of these

Q.11 In the figure shown the block B moves down with a velocity 10 m/s. The velocity of A in the position shown is :



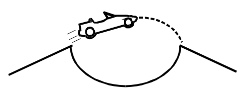
(A) 12.5 m/s (B) 25 m/s
(C) 6.25 m/s (D) none of these

Q.12 In the figure shown the velocity of different blocks is shown. The velocity of C is



(A) 6 m/s
(B) 4 m/s
(C) 0 m/s
(D) none of these

Q.13 A stunt man jumps his car over a crater as shown (neglect air resistance)

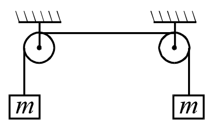


(A) during the whole flight the driver experiences weightlessness
(B) during the whole flight the driver never experiences weightlessness
(C) during the whole flight the driver experiences weightlessness only at the highest point
(D) the apparent weight increases during upward journey

Q.14 A ball of mass m is thrown vertically upwards. Assume the force of air resistance has magnitude proportional to the velocity, and direction opposite to the velocity's. At the highest point, the ball's acceleration is

(A) 0 (B) less than g (C) g (D) greater than g

Q.15 Two identical mass m are connected to a massless string which is hung over two frictionless pulleys as shown in figure. If everything is at rest, what is the tension in the cord?

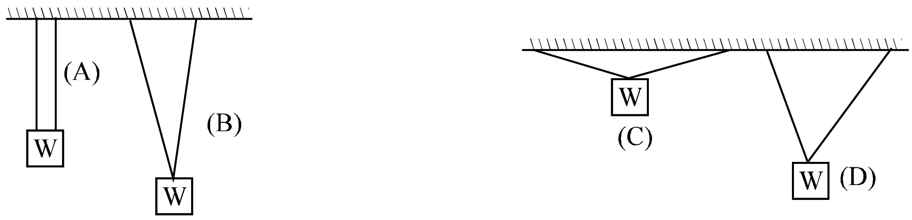


(A) less than mg
(B) exactly mg
(C) more than mg but less than 2mg
(D) exactly 2mg

Q.16 A flexible chain of weight W hangs between two fixed points A & B which are at he same horizontal level. The inclination of the chain with the horizontal at both the points of support is θ. What is the tension of the chain at the mid point?

(A) $\frac{W}{2}$. cosec θ (B) $\frac{W}{2}$. tan θ (C) $\frac{W}{2}$ cot θ (D) none

Q.17 A weight can be hung in any of the following four ways by string of same type. In which case is the string most likely to break?
(A) A (B) B (C) C (D) D



Question No. 18 to 19 (2 questions)
A frictionless pulley is attached to one arm of a balance and a string passed around it carries two masses m_1 and m_2. The pulley is provided with a clamp due to which m_1 and m_2 do not move w.r.t. each other.

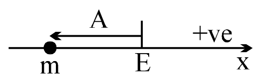
Q.18 On removing the clamp, m_1 and m_2 start moving. How much change in counter mass has to be made to restore balance?

(A) $\frac{(m_1+m_2)^2}{m_1-m_2}$ (B) $\frac{(m_1-m_2)^2}{m_1+m_2}$ (C) $2m_1 - m_2$ (D) $m_1 - m_2$

Q.19 On removing the clamp, if the counter mass restores balance, then acceleration of centre of mass of the masses m_1 and m_2 will have acceleration of magnitude

(A) zero (B) $\left(\frac{m_1-m_2}{m_1+m_2}\right)g$ (C) $\left(\frac{m_1-m_2}{m_1+m_2}\right)^2 g$ (D) g

Question No. 20 to 22 (3 questions)
A particle of mass m is constrained to move on x-axis. A force F acts on the particle. F always points toward the position labeled E. For example, when the particle is to the left of E, F points to the right. The magnitude of F is a constant F except at point E where it is zero.

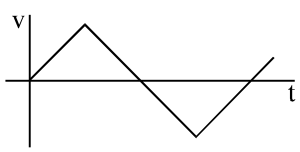


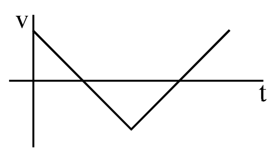
The system is horizontal. F is the net force acting on the particle. The particle is displaced a distance A towards left from the equilibrium position E and released from rest at t = 0.

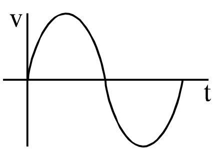
Q.20 What is the period of the motion?

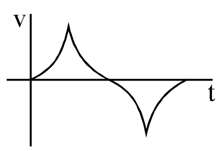
(A) $4\left(\sqrt{\frac{2Am}{F}}\right)$ (B) $2\left(\sqrt{\frac{2Am}{F}}\right)$ (C) $\left(\sqrt{\frac{2Am}{F}}\right)$ (D) None

Q.21 Velocity – time graph of the particle is

(A) 

(B) 

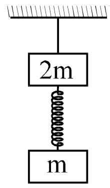
(C) 

(D) 

Q.22 Find minimum time it will take to reach from $x = -\frac{A}{2}$ to 0.

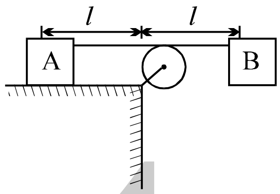
(A) $\frac{3}{2}\sqrt{\frac{mA}{F}}(\sqrt{2}-1)$ (B) $\sqrt{\frac{mA}{F}}(\sqrt{2}-1)$ (C) $2\sqrt{\frac{mA}{F}}(\sqrt{2}-1)$ (D) None

Q.23 Two blocks are connected by a spring. The combination is suspended, at rest, from a string attatched to the ceiling, as shown in the figure. The string breaks suddenly. Immediately after the string breaks, what is the initial downward acceleration of the upper block of mass 2m ?



(A) 0 (B) 3g/2 (C) g (D) 2g

Q.24 Two blocks A and B each of same mass are attached by a thin inextensible string through an ideal pulley. Initially block B is held in position as shown in fig. Now the block B is released. Block A willl slide to right and hit the pulley in time t_A. Block B will swing and hit the surface in time t_B. Assume the surface as frictionless. [Hint: Tension T in the string acting on both blocks is same in magnitude. Acceleration needed for horizontal motion is from T.]



(A) $t_A = t_B$
(B) $t_A < t_B$
(C) $t_A > t_B$
(D) data are not sufficient to get relationship between t_A and t_B.

Q.25 A body is placed on a rough inclined plane of inclination θ. As the angle θ is increased from 0° to 90° the contact force between the block and the plane

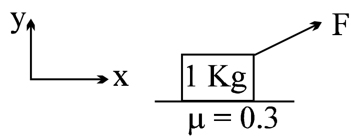
(A) remains constant (B) first remains constant than decreases
(C) first decreases then increases (D) first increases then decreases

Q.26 A block is projected upwards on an inclined plane of inclination 37° along the line of greatest slope of $\mu = 0.5$ with velocity of 5 m/s. The block 1st stops at a distance of __________ from starting point

(A) 1.25 m (B) 2.5 m (C) 10 m (D) 12.5 m

Q.27 A force $\vec{F} = \hat{i} + 4\hat{j}$ acts on block shown. The force of friction acting on the block is :

(A) $-\hat{i}$ (B) $-1.8\,\hat{i}$
(C) $-2.4\,\hat{i}$ (D) $-3\,\hat{i}$



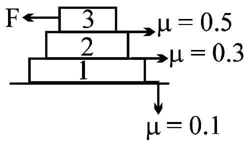
Q.28 A 1.0 kg block of wood sits on top of an identical block of wood, which sits on top of a flat level table made of plastic. The coefficient of static friction between the wood surfaces is μ_1, and the coefficient of static friction between the wood and plastic is μ_2.
A horizontal force F is applied to the top block only, and this force is increased until the top block starts to move. The bottom block will move with the top block if and only if

(A) $\mu_1 < \frac{1}{2}\mu_2$ (B) $\frac{1}{2}\mu_2 < \mu_1 < \mu_2$ (C) $\mu_2 < \mu_1$ (D) $2\mu_2 < \mu_1$

Q.29 A block of mass 2 kg slides down an incline plane of inclination 30°. The coefficient of friction between block and plane is 0.5. The contact force between block and plank is :

(A) 20 Nt (B) $10\sqrt{3}$ Nt (C) $5\sqrt{7}$ Nt (D) $5\sqrt{15}$ Nt

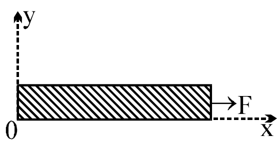
Q.30 If force F is increasing with time and at $t = 0$, $F = 0$ where will slipping first start?



(A) between 3 kg and 2 kg (B) between 2 kg and 1 kg
(C) between 1 kg and ground (D) both (A) and (B)

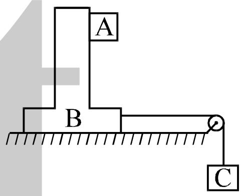
Q.31 A man is standing on a rough ($\mu = 0.5$) horizontal disc rotating with constant angular velocity of 5 rad/sec. At what distance from centre should he stand so that he does not slip on the disc?
(A) $R \le 0.2$m (B) $R > 0.2$ m (C) $R > 0.5$ m (D) $R > 0.3$ m

Q.32 A uniform rod of length L and mass M has been placed on a rough horizontal surface. The horizontal force F applied on the rod is such that the rod is just in the state of rest. If the coefficient of friction varies according to the relation $\mu = Kx$ where K is a +ve constant. Then the tension at mid point of rod is



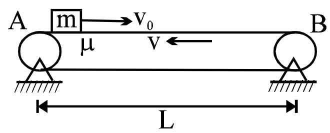
(A) F/2 (B) F/4
(C) F/8 (D) None

Q.33 In the arrangement shown in the figure, mass of the block B and A is 2m and m respectively. Surface between B and floor is smooth. The block B is connected to the block C by means of a string pulley system. If the whole system is released, then find the minimum value of mass of block C so that block A remains stationary w.r.t. B. Coefficient of friction between A and B is μ :



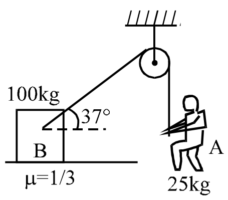
(A) $\dfrac{m}{\mu}$ (B) $\dfrac{2m+1}{\mu+1}$ (C) $\dfrac{3m}{\mu-1}$ (D) $\dfrac{6m}{\mu+1}$

Q.34 With what minimum velocity should block be projected from left end A towards end B such that it reaches the other end B of conveyer belt moving with constant velocity v. Friction coefficient between block and belt is μ.



(A) $\sqrt{\mu g L}$ (B) $\sqrt{2\mu g L}$
(C) $\sqrt{3\mu g L}$ (D) $2\sqrt{\mu g L}$

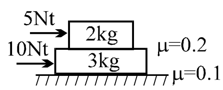
Q.35 Block B of mass 100 kg rests on a rough surface of friction coefficient $\mu = 1/3$. A rope is tied to block B as shown in figure. The maximum acceleration with which boy A of 25 kg can climbs on rope without making block move is :



(A) $\dfrac{4g}{3}$ (B) $\dfrac{g}{3}$ (C) $\dfrac{g}{2}$ (D) $\dfrac{3g}{4}$

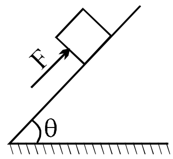
Q.36 A car moves along a circular track of radius R banked at an angle of 30° to the horizontal. The coefficient of static friction between the wheels and the track is μ. The maximum speed with which the car can move without skidding out is
(A) $\left[2gR(1+\mu)/\sqrt{3}\right]^{1/2}$ (B) $\left[gR(1-\mu)/(\mu+\sqrt{3})\right]^{1/2}$
(C) $\left[gR(1+\mu\sqrt{3})/(\mu+\sqrt{3})\right]^{1/2}$ (D) None

Q.37 The system shown in figure is released



(A) $a_1 = 0.35$ m/s^2 ; $a_2 = 4.5$ m/s^2 (B) $a_1 = 3$ m/s^2 ; $a_2 = 0.5$ m/s^2
(C) $a_1 = 2$ m/s^2 ; $a_2 = 2$ m/s^2 (D) $a_1 = 0.5$ m/s^2 ; $a_2 = 3$ m/s^2

Q.38 A block placed on a rough inclined plane of inclination ($\theta=30°$) can just be pushed upwards by applying a force "F" as shown. If the angle of inclination of the inclined plane is increased to ($\theta = 60°$), the same block can just be prevented from sliding down by application of a force of same magnitude. The coefficient of friction between the block and the inclined plane is



(A) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (B) $\frac{2\sqrt{3}-1}{\sqrt{3}+1}$ (C) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (D) None of these

For Q. 39 to Q.43 refer figure-1.(5 questions)

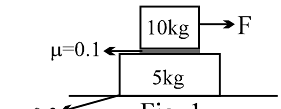


Fig. 1

Q.39 When F = 2N, the frictional force between 5 kg block and ground is
(A) 2N (B) 0
(C) 8 N (D) 10 N

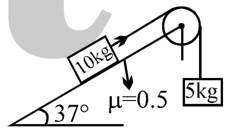
Q.40 When F = 2N, the frictional force between 10 kg block and 5 kg block is
(A) 2N (B) 15 N (C) 10 N (D) None

Q.41 The maximum "F" which will not cause motion of any of the blocks.
(A) 10 N (B) 15 N (C) data insufficient (D) None

Q.42 The maximum acceleration of 5 kg block
(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None

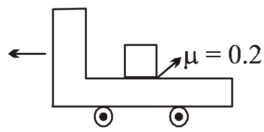
Q.43 The acceleration of 10 kg block when F = 30N
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None

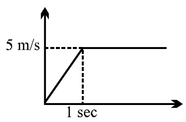
Q.44 The blocks are in equilibrium. The friction force acting on 10 kg block is :

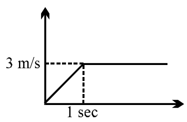


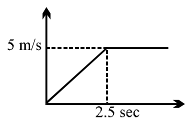
(A) 10 N down the plane (B) 40 N up the plane
(C) 10 N up the plane (D) None

Q.45 A truck starting from rest moves with an acceleration of 5 m/s^2 for 1 sec and then moves with constant velocity. The velocity w.r.t ground v/s time graph for block in truck is (Assume that block does not fall off the truck)



(A) 

(B) 

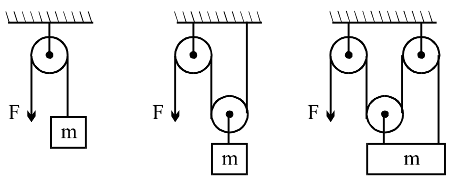
(C) 

(D) None of these

Q.46 A small block of mass m is projected horizontally with speed u where friction coefficient between block and plane is given by $\mu = cx$, where x is displacement of the block on plane. Find maximum distance covered by the block

(A) $\frac{u}{\sqrt{cg}}$ (B) $\frac{u}{\sqrt{2cg}}$ (C) $\frac{2u}{\sqrt{cg}}$ (D) $\frac{u}{2\sqrt{cg}}$

Q.47 Equal force F (> mg) is applied to string in all the 3 cases. Starting from rest, the point of application of force moves a distance of 2 m down in all cases. In which case the block has maximum kinetic energy?



(1) (2) (3)

(A) 1 (B) 2 (C) 3 (D) equal in all 3 cases

Q.48 A body of mass m accelerates uniformly from rest to a speed v_0 in time t_0. The work done on the body till any time t is

(A) $\frac{1}{2} m v_0^2 \left(\frac{t^2}{t_0^2}\right)$ (B) $\frac{1}{2} m v_0^2 \left(\frac{t_0}{t}\right)$ (C) $m v_0^2 \left(\frac{t}{t_0}\right)$ (D) $m v_0^2 \left(\frac{t}{t_0}\right)^3$

Q.49 A man who is running has half the kinetic energy of the boy of half his mass. The man speeds up by 1 m/s and then has the same kinetic energy as the boy. The original speed of the man was

(A) $\sqrt{2}$ m/s (B) $(\sqrt{2} - 1)$ m/s (C) 2 m/s (D) $(\sqrt{2} + 1)$ m/s

Q.50 $F = 2x^2 - 3x - 2$. Choose correct option

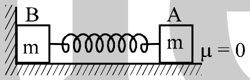
(A) $x = -1/2$ is position of stable equilibrium (B) $x = 2$ is position of stable equilibrium

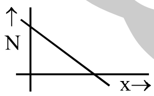
(C) $x = -1/2$ is position of unstable equilibrium (D) $x = 2$ is position of neutral equilibrium

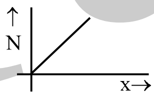
Q.51 A block of mass m is hung vertically from an elastic thread of force constant mg/a. Initially the thread was at its natural length and the block is allowed to fall freely. The kinetic energy of the block when it passes through the equilibrium position will be :

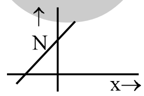
(A) mga (B) mga/2 (C) zero (D) 2mga

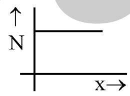
Q.52 The block A is pushed towards the wall by a distance and released. The normal reaction by vertical wall on the block B v/s compression in spring is given by :



(A) 

(B) 

(C) 

(D) 

Q.53 Force acting on a body of mass 1 kg is related to its position x as $F = x^3 - 3x$ N. It is at rest at $x = 1$. Its velocity at $x = 3$ can be :

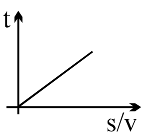
(A) 4 m/s (B) 3 m/s (C) 2 m/s (D) 5 m/s

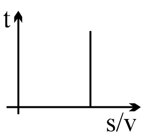
Q.54 Assume the aerodynamic drag force on a car is proportional to its speed. If the power output from the engine is doubled, then the maximum speed of the car.

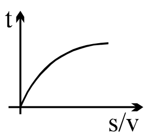
(A) is unchanged (B) increases by a factor of $\sqrt{2}$

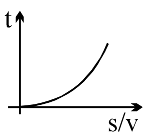
(C) is also doubled (D) increases by a factor of four.

Q.55 A body is moved from rest along a straight line by a machine delivering constant power. The ratio of displacement and velocity (s/v) varies with time t as :

(A) 

(B) 

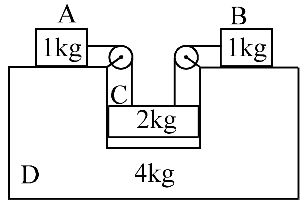
(C) 

(D) 

Q.56 A particle is released from rest at origin. It moves under influence of potential field $U = x^2 - 3x$, kinetic energy at $x = 2$ is

(A) 2J (B) 1 J (C) 1.5 J (D) 0 J

Q.57 In the system shown in the figure there is no friction anywhere. The block C goes down by a distance $x_0 = 10$ cm with respect to wedge D when system is released from rest. The velocity of A with respect to B will be ($g = 10\ m/s^2$) :



(A) zero (B) 1 m/s
(C) 2 m/s (D) None of these

Q.58 Potential energy of a particle is related to x coordinate by equation $x^2 - 2x$. Particle will be in stable equilibrium at
(A) x = 0.5 (B) x = 1 (C) x = 2 (D) x = 4

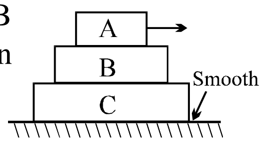
Q.59 A force $\vec{F} = k[y\,\hat{i} + x\,\hat{j}]$ where k is a positive constant acts on a particle moving in x-y plane starting from the point (3,5), the particle is taken along a straight line to (5, 7). The work done by the force is :
(A) zero (B) 35 K (C) 20 K (D) 15 K

Q.60 Water is pumped from a depth of 10 m and delivered through a pipe of cross section $10^{-2} m^2$. If it is needed to deliver a volume of $10^{-1}\ m^3$ per second the power required will be:
(A) 10 kW (B) 9.8 kW (C) 15 kW (D) 4.9 kW

Q.61 A light spring of length 20 cm and force constant 2 kg/cm is placed vertically on a table. A small block of mass 1 kg. falls on it. The length h from the surface of the table at which the ball will have the maximum velocity is
(A) 20 cm (B) 15 cm (C) 10 cm (D) 5 cm

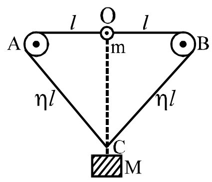
Q.62 The work done in joules in increasing the extension of a spring of stiffness 10 N/cm from 4 cm to 6 cm is:
(A) 1 (B) 10 (C) 50 (D) 100

Q.63 Three blocks A, B and C are kept as shown in the figure. The coefficient of friction between A and B is 0.2, B and C is 0.1 , C and ground is 0.0. The mass of A, B and C are 3 kg, 2 kg and 1 kg respectively. A is given a horizontal velocity 10 m/s. A, B and C always remain in contact i.e. lies as in figure. The total work done by friction will be:



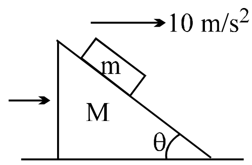
(A) – 75 J (B) 75 J (C) – 150 J (D) – 100 J

Q.64 A loop of light inextensible string passes over smooth small pulleys A and B. Two masses m and M are attached to the points O and C respectively. Then the condition that m and M will cross each other.
[Take AB = 21 and AC = AB = ηl] will be



(A) $\frac{m}{M} > 2\sqrt{\frac{\eta+1}{\eta+3}} - 1$ (B) $\frac{m}{M} > 2\sqrt{\frac{\eta+3}{\eta+1}} - 1$ (C) $\frac{m}{M} > \sqrt{\frac{\eta+1}{\eta+3}} + 1$ (D) none of these

Q.65 In the figure shown all the surfaces are frictionless, and mass of the block, m = 1 kg. The block and wedge are held initially at rest. Now wedge is given a horizontal acceleration of $10\ m/s^2$ by applying a force on the wedge, so that the block does not slip on the wedge. Then work done by the normal force in ground frame on the block in $\sqrt{3}$ seconds is

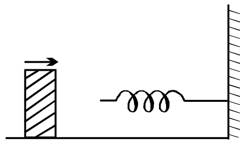


(A) 30 J (B) 60 J
(C) 150 J (D) $100\sqrt{3}$ J

Q.66 When a conservative force does positive work on a body
(A) the potential energy increases (B) the potential energy decreases
(C) total energy increases (D) total energy decreases

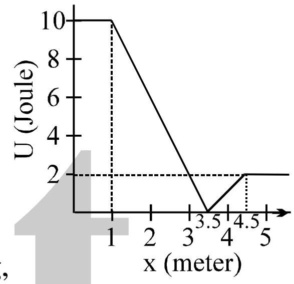
Q.67 The P.E. of a certain spring when stretched from natural length through a distance 0.3 m is10 J. The amount of work in joule that must be done on this spring to stretch it through an additional distance 0.15 m will be
(A) 10 J (B) 20 J (C) 7.5 J (D) 12.5 J

Q.68 A 1.0 kg block collides with a horizontal weightless spring of force constant 2.75 Nm^{-1} as shown in figure. The block compresses the spring 4.0 m from the rest position. If the coefficient of kinetic friction between the block and horizontal surface is 0.25, the speed of the block at the instant of collision is

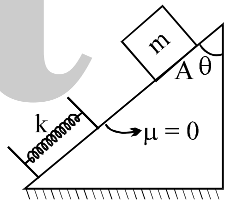


(A) 0.4 ms^{-1} (B) 4 ms^{-1} (C) 0.8 ms^{-1} (D) 8 ms^{-1}

Q.69 A body with mass 2 kg moves in one direction in the presence of a force which is described by the potential energy graph. If the body is released from rest at x = 2m, then its speed when it crosses x = 5 m is
(A) zero (B) 1 ms^{-1}
(C) 2 ms^{-1} (D) 3 ms^{-1}



Q.70 A block of mass 'm' is released from rest at point A. The compression in spring, when the speed of block is maximum

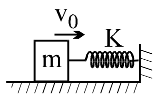


(A) $\frac{mg\sin\theta}{k}$ (B) $\frac{2mg\sin\theta}{k}$

(C) $\frac{mg\cos\theta}{k}$ (D) $\frac{mg}{k}$

Question No. 71 to 76 (6 questions)

A block of mass m moving with a velocity v_0 on a smooth horizontal surface strikes and compresses a spring of stiffness k till mass comes to rest as shown in the figure. This phenomenon is observed by two observers:
A: standing on the horizontal surface **B**: standing on the block



Q.71 To an observer A, the work done by spring force is
(A) negative but nothing can be said about its magnitude (B) $-\frac{1}{2}mv_0^2$

(C) positive but nothing can be said about its magnitude (D) $+\frac{1}{2}mv_0^2$

Q.72 To an observer A, the work done by the normal reaction N between the block and the spring on the block is

(A) zero (B) $-\frac{1}{2}mv_0^2$ (C) $+\frac{1}{2}mv_0^2$ (D) none of these

Q.73 To an observer A, the net work done on the block is

(A) $-mv_0^2$ (B) $+mv_0^2$ (C) $-\frac{1}{2}mv_0^2$ (D) zero

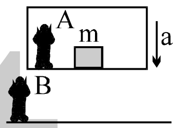
Q.74 According to the observer A
(A) the kinetic energy of the block is converted into the potential energy of the spring
(B) the mechanical energy of the spring-mass system is conserved
(C) the block loses its kinetic energy because of the negative work done by the conservative force of spring
(D) all the above

Q.75 To an observer B, when the block is compressing the spring
(A) velocity of the block is decreasing (B) retardation of the block is increasing
(C) kinetic energy of the block is zero (D) all the above

Q.76 According to observer B, the potential energy of the spring increases
(A) due to the positive work done by pseudo force
(B) due to the positive work done by normal reaction between spring & wall
(C) due to the decrease in the kinetic energy of the block
(D) all the above

Question No. 77 to 80 (4 questions)

A block of mass m is kept in an elevation which starts moving downward with an acceleration a as shown in figure. The block is observed by two observers A and B for a time interval t_0.



Q.77 The observer B finds that the work done by gravity on the block is

(A) $\frac{1}{2}mg^2t_0^2$ (B) $-\frac{1}{2}mg^2t_0^2$ (C) $\frac{1}{2}mgat_0^2$ (D) $-\frac{1}{2}mgat_0^2$

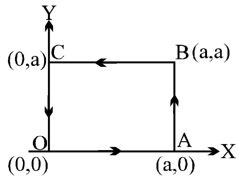
Q.78 The observer B finds that the work done by pseudo force on the block is
(A) zero (B) $-ma^2t_0$ (C) $+ma^2t_0$ (D) $-mgat_0$

Q.79 According to observer B, the net work done on the block is

(A) $-\frac{1}{2}ma^2t_0^2$ (B) $\frac{1}{2}ma^2t_0^2$ (C) $\frac{1}{2}mgat_0^2$ (D) $-\frac{1}{2}mgat_0^2$

Q.80 According to the observer A
(A) the work done by gravity is zero (B) the work done by normal reaction is zero
(C) the work done by pseudo force is zero (D) all the above

Q.81 The work done by the force $\vec{F} = x^2\hat{i} + y^2\hat{j}$ around the path shown in the figure is

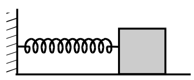


(A) $\frac{2}{3}a^3$ (B) zero (C) a^3 (D) $\frac{4}{3}a^3$

Question No. 82 to 83 (2 questions)

A spring block system is placed on a rough horizontal floor. The block is pulled towards right to give spring an elongation less than $\frac{2\mu mg}{K}$ but more than $\frac{\mu mg}{K}$ and released.

Q.82 Which of the following laws/principles of physics can be applied on the spring block system



(A) conservation of mechanical energy (B) conservation of momentum
(C) work energy principle (D) None

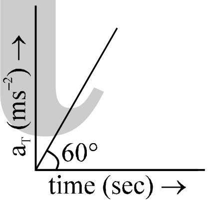
Q.83 The correct statement is
(A) The block will cross the mean position.
(B) The block will come to rest when the forces acting on it are exactly balanced
(C) The block will come to rest when the work done by friction becomes equal to the change in energy stored in spring.
(D) None

Q.84 A particle is rotated in a vertical circle by connecting it to a light rod of length l and keeping the other end of the rod fixed. The minimum speed of the particle when the light rod is horizontal for which the particle will complete the circle is
(A) $\sqrt{gl}$ (B) $\sqrt{2gl}$ (C) $\sqrt{3gl}$ (D) none

Q.85 A body is moving uni-directionally under the influence of a source of constant power. Its displacement in time t is proportional to
(A) $t^{1/2}$ (B) t (C) $t^{3/2}$ (D) t^2

Q.86 If angular velocity of a disc depends an angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration α at $\theta = 1$ rad is :
(A) 8 rad/sec^2 (B) 10 rad/sec^2 (C) 12 rad/sec^2 (D) None

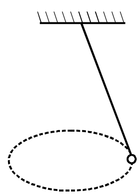
Q.87 Tangential acceleration of a particle moving in a circle of radius 1 m varies with time t as (initial velocity of particle is zero). Time after which total acceleration of particle makes and angle of 30° with radial acceleration is
(A) 4 sec (B) 4/3 sec
(C) $2^{2/3}$ sec (D) $\sqrt{2}$ sec

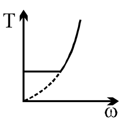


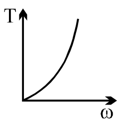
Q.88 The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as
(A) 2 a sinωt (B) 2a sin$\frac{\omega t}{2}$ (C) 2a cos ωt (D) 2a cos $\frac{\omega t}{2}$

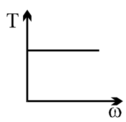
Q.89 A particle originally at rest at the highest point of a smooth vertical circle is slightly displaced. It will leave the circle at a vertical distance h below the highest point, such that
(A) h = R (B) h = R/3 (C) h = R/2 (D) h = 2R

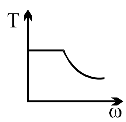
Q.90 In a conical pendulum, the bob is rotated with different angular velocities and tension in the string is calculated for different values of ω. Which of them is correct graph between T & ω.



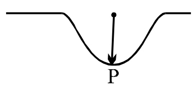
(A) 

(B) 

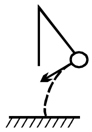
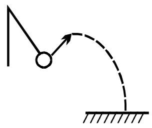
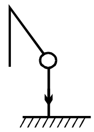
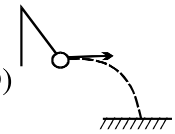
(C) 

(D) 

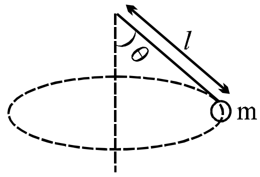
Q.91 A car travelling on a smooth road passes through a curved portion of the road in form of an arc of circle of radius 10 m. If the mass of car is 500 kg, the reaction on car at lowest point P where its speed is 20 m/s is
(A) 35 kN (B) 30 kN (C) 25 kN (D) 20 kN



Q.92 A pendulum bob is swinging in a vertical plane such that its angular amplitude is less than 90^0. At its highest point, the string is cut. Which trajectory is possible for the bob afterwards.

(A)  (B)  (C)  (D) 

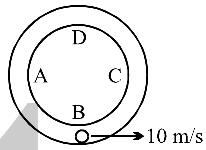
Q.93 A conical pendulum is moving in a circle with angular velocity ω as shown. If tension in the string is T, which of following equations are correct ?



(A) $T = m\omega^2 l$ (B) $T \sin\theta = m\omega^2 l$

(C) $T = mg \cos\theta$ (D) $T = m\omega^2 l \sin\theta$

Q.94 A ball whose size is slightly smaller than width of the tube of radius 2.5 m is projected from bottommost point of a smooth tube fixed in a vertical plane with velocity of 10 m/s. If N_1 and N_2 are the normal reactions exerted by inner side and outer side of the tube on the ball



(A) $N_1 > 0$ for motion in ABC, $N_2 > 0$ for motion in CDA

(B) $N_1 > 0$ for motion in CDA, $N_2 > 0$ for motion in ABC

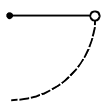
(C) $N_2 > 0$ for motion in ABC & part of CDA

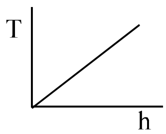
(D) N_1 is always zero.

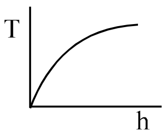
Q.95 A road is banked at an angle of 30° to the horizontal for negotiating a curve of radius $10\sqrt{3}$ m. At what velocity will a car experience no friction while negotiating the curve?

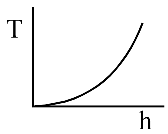
(A) 54 km/hr (B) 72 km/hr (C) 36 km/hr (D) 18 km/hr

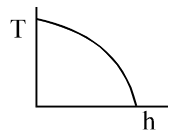
Q.96 A bob attached to a string is held horizontal and released. The tension and vertical distance from point of suspension can be represented by.



(A) 

(B) 

(C) 

(D) 

Q.97 A particle of mass m is tied to one end of a string of length l. The particle is held horizontal with the string taut. It is then projected upward with a velocity u. The tension in the string is $\frac{mg}{2}$ when it is inclined at an angle 30° to the horizontal. The value of u is

(A) $\sqrt{lg}$ (B) $\sqrt{2lg}$ (C) $\sqrt{\frac{lg}{2}}$ (D) $2\sqrt{lg}$

Q.98 The ratio of period of oscillation of the conical pendulum to that of the simple pendulum is : (Assume the strings are of the same length in the two cases and θ is the angle made by the string with the verticla in case of conical pendulum)

(A) $\cos\theta$ (B) $\sqrt{\cos\theta}$ (C) 1 (D) none of these

Q.99 A particle is moving in a circle :

(A) The resultant force on the particle must be towards the centre.

(B) The cross product of the tangential acceleration and the angular velocity will be zero.

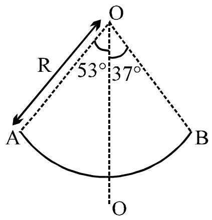
(C) The direction of the angular acceleration and the angular velocity must be the same.

(D) The resultant force may be towards the centre.

Q.100 A particle is moving along the circle $x^2 + y^2 = a^2$ in anti clock wise direction. The x–y plane is a rough horizontal stationary surface. At the point ($a\cos\theta$, $a\sin\theta$), the unit vector in the direction of friction on the particle is:

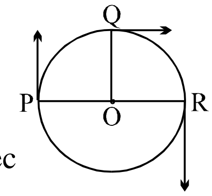
(A) $\cos\theta\,\hat{i} + \sin\theta\,\hat{j}$ (B) $-(\cos\theta\,\hat{i} + \sin\theta\,\hat{j})$ (C) $\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ (D) $\cos\theta\,\hat{i} - \sin\theta\,\hat{j}$

Q.101 A section of fixed smooth circular track of radius R in vertical plane is shown in the figure. A block is released from position A and leaves the track at B. The radius of curvature of its trajectory when it just leaves the track at B is:



(A) R (B) R / 4 (C) R / 2 (D) none of these

Q.102 Three point particles P, Q, R move in circle of radius 'r' with different but constant speeds. They start moving at t = 0 from their initial positions as shown in the figure. The angular velocities (in rad/ sec) of P, Q and R are 5π, 2π & 3π respectively, in the same sense. The time interval after which they are at same angular position.



(A) 2/3 sec (B) 1/6 sec (C) 1/2 sec (D) 3/2 sec

Q.103 In the above question, the number of times P and Q meet in that time interval is:

(A) 4 (B) 1 (C) 3 (D) 9

Q.104 A particle inside the rough surface of a rotating cone about its axis is at rest relative to it at a height of 1m above its vertex. Friction coefficient is $\mu = 0.5$, if half angle of cone is 45°, the maximum angular velocity of revolution of cone can be :

(A) $\sqrt{10}$ rad/s (B) $\sqrt{30}$ rad/s (C) $\frac{\sqrt{40}}{3}$ rad/s (D) $\sqrt{50}$ rad/s

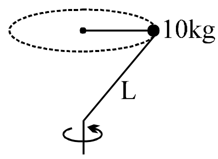
Q.105 A body of mass 1 kg starts moving from rest at t = 0, in a circular path of radius 8 m. Its kinetic energy varies as a function of time as : K.E. = $2t^2$ Joules, where t is in seconds. Then

(A) tangential acceleration = 4 m/s^2 (B) power of all forces at t = 2 sec is 8 watt
(C) first round is completed in 2 sec. (D) tangential force at t = 2 sec is 4 newton.

Q.106 A particle is moving along a circular path of radius R in such a way that at any instant magnitude of radial acceleration & tangential acceleration are equal. If at t = 0 velocity of particle is V_0, the time period of first revolution of the particle is

(A) $\frac{R}{V_0}e^{-2\pi}$ (B) $\frac{R}{V_0}(e^{2\pi}-1)$ (C) $\frac{R}{V_0}$ (D) $\frac{R}{V_0}(1-e^{-2\pi})$

Q.107 A 10 kg ball attached to the end of a rigid massless rod of length 1 m rotates at constant speed in a horizontal circle of radius 0.5 m and period 1.57 sec as in fig. The force exerted by rod on the ball is



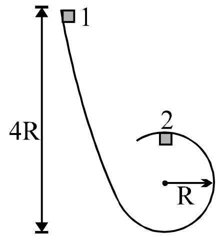
(A) 1.28 N (B) 128 N (C) 10 N (D) 12.8 N

Q.108 Two bodies A & B rotate about an axis, such that angle θ_A (in radians) covered by first body is proportional to square of time, & θ_B (in radians) covered by second body varies linearly. At t = 0, $\theta_A = \theta_B = 0$. If A completes its first revolution in $\sqrt{\pi}$ sec. & B needs 4π sec. to complete half revolution then; angular velocity $\omega_A : \omega_B$ at t = 5 sec. are in the ratio

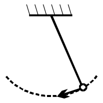
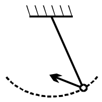
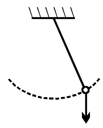
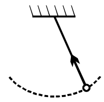
(A) 4 : 1 (B) 20 : 1 (C) 80 : 1 (D) 20 : 4

Q.109 A small cube with mass M starts at rest at point 1 at a height 4R, where R is the radius of the circular part of the track. The cube slides down the frictionless track and around the loop. The force that the track exerts on the cube at point 2 is nearly _____ times the cube's weight Mg.

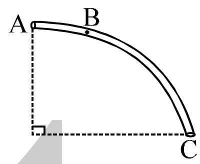
(A) 1 (B) 2 (C) 3 (D) 4



Q.110 Which vector in the figures best represents the acceleration of a pendulum mass at the intermediate point in its swing?

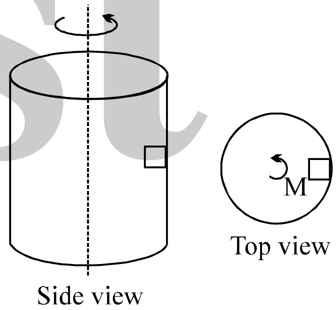
(A)  (B)  (C)  (D) 

Q.111 The tube AC forms a quarter circle in a vertical plane. The ball B has an area of cross–section slightly smaller than that of the tube, and can move without friction through it. B is placed at A and displaced slightly. It will



(A) always be in contact with the inner wall of the tube
(B) always be in contact with the outer wall of the tube
(C) initially be in contact with the inner wall and later with the outer wall
(D) initially be in contact with the outer wall and later with the inner wall

Q.112 A hollow vertical cylinder of radius R is rotated with angular velocity ω about an axis through its center. What is the minimum coefficient of static friction necessary to keep the mass M suspended on the inside of the cylinder as it rotates?



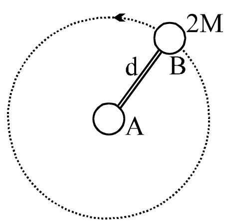
(A) $\mu = \dfrac{gR}{\omega^2}$ (B) $\mu = \dfrac{\omega^2 g}{R}$

(C) $\mu = \dfrac{\omega^2 R}{g}$ (D) $\mu = \dfrac{g}{\omega^2 R}$

Q.113 A horizontal curve on a racing track is banked at a 45° angle. When a vehicle goes around this curve at the curve's safe speed (no friction needed to stay on the track), what is its centripetal acceleration?

(A) g (B) 2g (C) 0.5g (D) none

Q.114 The dumbell is placed on a frictionless horizontal table. Sphere A is attached to a frictionless pivot so that B can be made to rotate about A with constant angular velocity. If B makes one revolution in period P, the tension in the rod is

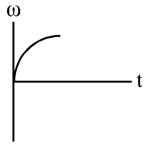


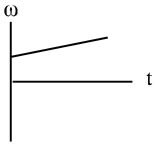
(A) $\dfrac{4\pi^2 Md}{P^2}$ (B) $\dfrac{8\pi^2 Md}{P^2}$ (C) $\dfrac{4\pi^2 Md}{P}$ (D) $\dfrac{2Md}{P}$

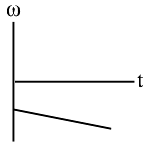
Q.115 Two racing cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 respectively. Their speeds are such that each makes a complete circle in the same time t. The ratio of the angular speeds of the first to the second car is

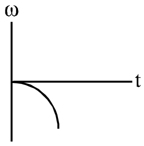
(A) 1 : 1 (B) $m_1 : m_2$ (C) $r_1 : r_2$ (D) $m_1 m_2 : r_1 r_2$

Q.116 The graphs below show angular velocity as a function of time. In which one is the magnitude of the angular acceleration constantly decreasing?

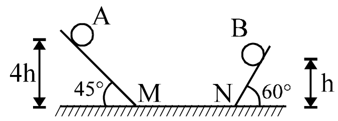
(A) 

(B) 

(C) 

(D) 

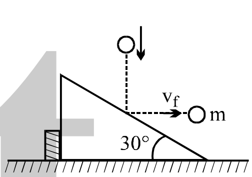
Q.117 Two identical balls A and B are released from the positions shown in figure. They collide elastically on horizontal portion MN. All surfaces are smooth. The ratio of heights attained by A and B after collision will be(Neglect energy loss at M & N)



(A) 1 : 4
(B) 2 : 1
(C) 4 : 13
(D) 2 : 5

Q.118 A ball strikes a smooth horizontal ground at an angle of 45° with the vertical. What **cannot** be the possible angle of its velocity with the vertical after the collision. (Assume $e \le 1$).
(A) 45°
(B) 30°
(C) 53°
(D) 60°

Q.119 As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If $\angle$ of inclined is 30°, the velocity v_f will be

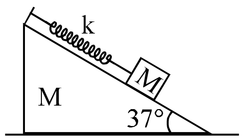


(A) 3 m/s
(B) $\sqrt{3}$ m/s
(C) $1/\sqrt{3}$ m/s
(D) this is not possible

Q.120 An ice block is melting at a constant rate $\left|\frac{dm}{dt}\right| = \mu$. Its initial mass is m_0 and it is moving with velocity on a frictionless horizontal surface. The distance travelled by it till it melts completely is :

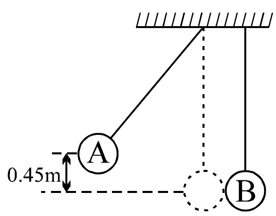
(A) $\frac{2m_0 v}{\mu}$
(B) $\frac{m_0 v}{\mu}$
(C) $\frac{m_0 v}{2\mu}$
(D) can't be said

Q.121 The system of the wedge and the block connected by a massless spring as shown in the figure is released with the spring in its natural length. Friction is absent. maximum elongation in the spring will be



(A) $\frac{3Mg}{5k}$
(B) $\frac{6Mg}{5k}$
(C) $\frac{4Mg}{5k}$
(D) $\frac{8Mg}{5k}$

Q.122 Two massless string of length 5 m hang from the ceiling very near to each other as shown in the figure. Two balls A and B of masses 0.25 kg and 0.5 kg are attached to the string. The ball A is released from rest at a height 0.45 m as shown in the figure. The collision between two balls is completely elastic. Immediately after the collision, the kinetic energy of ball B is 1 J. The velocity of ball A just after the collision is



(A) 5 ms^{-1} to the right
(B) 5 ms^{-1} to the left
(C) 1 ms^{-1} to the right
(D) 1 ms^{-1} to the left

Q.123 A smooth sphere is moving on a horizontal surface with a velocity vector $(2\hat{i} + 2\hat{j})$ m/s immediately before it hit a vertical wall. The wall is parallel to vector $\hat{j}$ and coefficient of restitution between the sphere and the wall is $e = 1/2$. The velocity of the sphere after it hits the wall is

(A) $\hat{i} - \hat{j}$
(B) $-\hat{i} + 2\hat{j}$
(C) $-\hat{i} - \hat{j}$
(D) $2\hat{i} - \hat{j}$

Q.124 A man of mass M stands at one end of a plank of length L which lies at rest on a frictionless surface. The man walks to other end of the plank. If the mass of the plank is $\frac{M}{3}$, then the distance that the man moves relative to ground is :

(A) $\frac{3L}{4}$ (B) $\frac{L}{4}$ (C) $\frac{4L}{5}$ (D) $\frac{L}{3}$

Q.125 Two balls A and B having masses 1 kg and 2 kg, moving with speeds 21 m/s and 4 m/s respectively in opposite direction, collide head on. After collision A moves with a speed of 1 m/s in the same direction, then the coefficient of restitution is

(A) 0.1 (B) 0.2 (C) 0.4 (D) None

Q.126 Two particles of equal mass have velocities $2\hat{i}$ ms^{-1} and $2\hat{j}$ ms^{-1}. First particle has an acceleration $(\hat{i}+\hat{j})$ ms^{-2} while the acceleration of the second particle is zero. The centre of mass of the two particles moves in

(A) circle (B) parabola (C) ellipse (D) straight line

Q.127 A particle of mass 3m is projected from the ground at some angle with horizontal. The horizontal range is R. At the highest point of its path it breaks into two pieces m and 2m. The smaller mass comes to rest and larger mass finally falls at a distance x from the point of projection where x is equal to

(A) $\frac{3R}{4}$ (B) $\frac{3R}{2}$ (C) $\frac{5R}{4}$ (D) 3R

Q.128 A man weighing 80 kg is standing at the centre of a flat boat and he is 20 m from the shore. He walks 8 m on the boat towards the shore and then halts. The boat weight 200 kg. How far is he from the shore at the end of this time ?

(A) 11.2 m (B) 13.8 m (C) 14.3 m (D) 15.4 m

Q.129 From a circle of radius a, an isosceles right angled triangle with the hypotenuse as the diameter of the circle is removed. The distance of the centre of gravity of the remaining position from the centre of the circle is

(A) $3(\pi - 1)a$ (B) $\frac{(\pi-1)a}{6}$ (C) $\frac{a}{3(\pi-1)}$ (D) $\frac{a}{3(\pi+1)}$

Q.130 A sphere strikes a wall and rebounds with coefficient of restitution 1/3. If it rebounds with a velocity of 0.1 m/sec at an angle of 60° to the normal to the wall, the loss of kinetic energy is

(A) 50% (B) $33\frac{1}{3}$% (C) 40% (D) $66\frac{2}{3}$%

Q.131 A truck moving on horizontal road towards east with velocity 20 ms^{-1} collides elastically with a light ball moving with velocity 25 ms^{-1} along west. The velocity of the ball just after collision

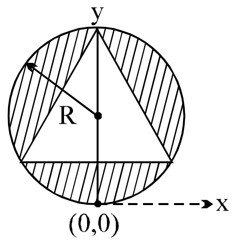
(A) 65 ms^{-1} towards east (B) 25 ms^{-1} towards west

(C) 65 ms^{-1} towards west (D) 20 ms^{-1} towards east

Q.132 A spaceship of speed v_0 travelling along + y axis suddenly shots out one fourth of its part with speed $2v_0$ along + x-axis. xy axes are fixed with respect to ground. The velocity of the remaining part is

(A) $\frac{2}{3}v_0$ (B) $\frac{\sqrt{20}}{3}v_0$ (C) $\frac{\sqrt{5}}{3}v_0$ (D) $\frac{\sqrt{13}}{3}v_0$

Q.133 From a uniform disc of radius R, an equilateral triangle of side $\sqrt{3}\,R$ is cut as shown. The new position of centre of mass is :



(A) (0, 0)　　(B) (0, R)

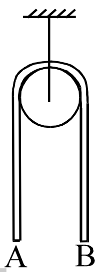
(C) $(0, \frac{\sqrt{3}\,R}{2})$　　(D) none of these

Q.134 If the linear density of a rod of length 3 m varies as $\lambda = 2 + x$, then the position of centre of gravity of the rod is :

(A) 7/3 m　　(B) 12/7 m　　(C) 10/7 m　　(D) 9/7 m

Question No. 135 to 136 (2 questions)

A uniform chain of length 2L is hanging in equilibrium position, if end B is given a slightly downward displacement the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible



Q.135 The acceleration of end B when it has been displaced by distance x, is

(A) $\frac{x}{L}g$　　(B) $\frac{2x}{L}g$　　(C) $\frac{x}{2}g$　　(D) g

Q.136 The velocity v of the string when it slips out of the pulley (height of pulley from floor > 2L)

(A) $\sqrt{\frac{gL}{2}}$　　(B) $\sqrt{2gL}$　　(C) $\sqrt{gL}$　　(D) none of these

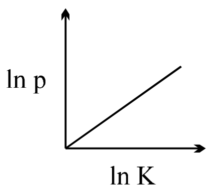
Q.137 A small sphere is moving at a constant speed in a vertical circle. Below is a list of quantities that could be used to describe some aspect of the motion of the sphere.

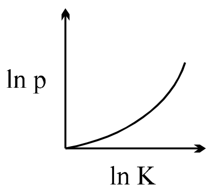
I – kinetic energy
II – gravitational potential energy
III – momentum

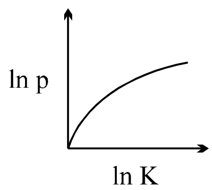
Which of these quantities will change as this sphere moves around the circle?

(A) I and II only　　(B) I and III only　　(C) III only　　(D) II and III only

Q.138 Which of the following graphs represents the graphical relation between momentum (p) and kinetic energy (K) for a body in motion?

(A) 

(B) 

(C) 

(D) none

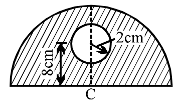
Q.139 When the momentum of a body increases by 100%, its KE increases by

(A) 400%　　(B) 100%　　(C) 300%　　(D) none

Q.140 A small bucket of mass M kg is attached to a long inextensible cord of length L m . The bucket is released from rest when the cord is in a horizontal position. At its lowest position, the bucket scoops up m kg of water and swings up to a height h. The height h in meters is

(A) $\left(\frac{M}{M+m}\right)^2 L$　　(B) $\left(\frac{M}{M+m}\right)L$　　(C) $\left(\frac{M+m}{M}\right)^2 L$　　(D) $\left(\frac{M+m}{M}\right)L$

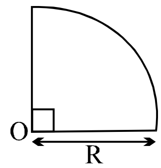
Q.141 In the figure shown a hole of radius 2 cm is made in a semicircular disc of radius 6π at a distance 8 cm from the centre C of the disc. The distance of the centre of mass of this system from point C is :



(A) 4 cm (B) 8 cm (C) 6 cm (D) 12 cm

Q.142 A buggy of mass 100 kg is free to move on a frictionless horizontal track. Two men, each of mass 50 kg, are standing on the buggy, which is initially stationary. The men jump off the buggy with velocity=10m/s relative to the buggy. In one situation, the men jump one after the other. In another situation, the men jump simultaneously. What is the ratio of the recoil velocities of the buggy in two cases?
(A) 5 : 4 (B) 5 : 3 (C) 7 : 6 (D) 7 : 5

Q.143 In the figure one fourth part of a uniform disc of radius R is shown. The distance of the centre of mass of this object from centre 'O' is:

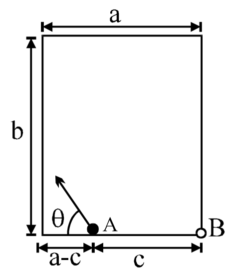


(A) $\frac{4R}{3\pi}$ (B) $\frac{2R}{3\pi}$ (C) $\sqrt{2}\frac{4R}{3\pi}$ (D) $\sqrt{2}\frac{2R}{3\pi}$

Q.144 Two men, of masses 60 kg and 80 kg are sitting at the ends of a boat of mass 60 kg and length 4 m. The boat is stationary. If the men now exchange their positions, then
(A) the centre of mass of the two men shifts by 2 m
(B) the boat moves by 0.4 m
(C) the centre of mass of the two men shifts by 4/7 m.
(D) the boat moves by 0.6 m.

Q.145 On a horizontal smooth surface a disc is placed at rest. Another disc of same mass is coming with impact parameter equal to its own radius. First disc is of radius r. What should be the radius of coming disc so that after collision first disc moves at an angle 45° to the direction of motion of incoming disc :
(A) 2r (B) $r\left(\sqrt{2}-1\right)$ (C) $\frac{r}{\left(\sqrt{2}-1\right)}$ (D) $r\sqrt{2}$

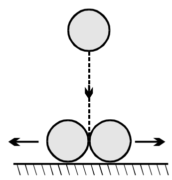
Q.146 A billiard table whose length and width are as shown in the figure. A ball is placed at point A. At what angle 'θ' the ball be projected so that after colliding with two walls, the ball will fall in the pocket B. Assume that all collisions are perfectly elastic (neglect friction)



(A) $\theta = \cot^{-1}\frac{2a-c}{2b}$ (B) $\theta = \tan^{-1}\frac{2a-c}{2b}$

(C) $\theta = \cot^{-1}\frac{c-a}{2b}$ (D) $\theta = \cot^{-1}\frac{c-a}{b}$

Q.147 In the figure shown, the two identical balls of mass M and radius R each, are placed in contact with each other on the frictionless horizontal surface. The third ball of mass M and radius R/2, is coming down vertically and has a velocity = v_0 when it simultaneously hits the two balls and itself comes to rest. Then, each of the two bigger balls will move after collision with a speed equal to



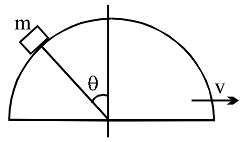
(A) $4v_0/\sqrt{5}$ (B) $2v_0/\sqrt{5}$ (C) $v_0/\sqrt{5}$ (D) None

Q.148 In the above, suppose that the smaller ball does not stop after collision, but continues to move downwards with a speed= $v_0/2$, after the collision. Then, the speed of each bigger ball after collision is
(A) $4v_0/\sqrt{5}$ (B) $2v_0/\sqrt{5}$ (C) $v_0/2\sqrt{5}$ (D) None

Q.149 A body of mass ‘m’ is dropped from a height of ‘h’. Simultaneously another body of mass 2m is thrown up vertically with such a velocity v that they collide at the height h/2. If the collision is perfectly inelastic, the velocity at the time of collision with the ground will be :

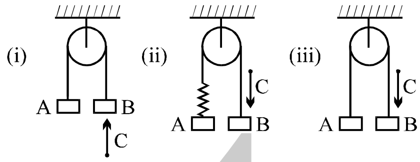
(A) $\sqrt{\frac{5gh}{4}}$ (B) $\sqrt{gh}$ (C) $\sqrt{\frac{gh}{4}}$ (D) $\frac{\sqrt{10gh}}{3}$

Q.150 A hemisphere of mass 3m and radius R is free to slide with its base on a smooth horizontal table. A particle of mass m is placed on the top of the hemisphere. If particle is displaced with a negligible velocity, then find the angular velocity of the particle relative to the centre of the hemisphere at an angular displacement θ, when velocity of hemisphere is v.



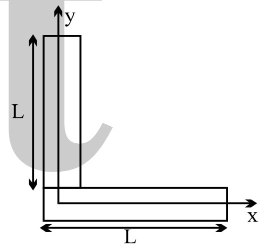
(A) $\frac{4v}{R\cos\theta}$ (B) $\frac{3v}{R\cos\theta}$ (C) $\frac{5v}{R\cos\theta}$ (D) $\frac{2v}{R\cos\theta}$

Q.151 In the figure (i), (ii) & (iii) shown the objects A, B & C are of same mass. String, spring & pulley are massless. C strikes B with velocity ‘u’ in each case and sticks to it. The ratio of velocity of B in case (i) to (ii) to (iii) is



(A) 1 : 1: 1 (B) 3 : 3 : 2
(C) 3 : 2 : 2 (D) none of these

Q.152 Centre of mass of two thin uniform rods of same length but made up of different materials & kept as shown , can be, if the meeting point is the origin of co-ordinates



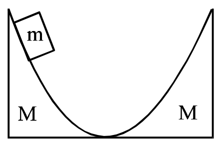
(A) (L/2, L/2) (B) (2L/3, L/2)
(C) (L/3, L/3) (D) (L/3, L/6)

Q.153 A force exerts an impulse I on a particle changing its speed from u to 2u. The applied force and the initial velocity are oppositely directed along the same line. The work done by the force is

(A) $\frac{3}{2}Iu$ (B) $\frac{1}{2}Iu$ (C) I u (D) 2 I u

Q.154 The inclined surfaces of two movable wedges of same mass M are smoothly conjugated with the horizontal plane as shown in figure. A washer of mass m slides down the left wedge from a height h. To what maximum height will the washer rise along the right wedge? Neglect friction.

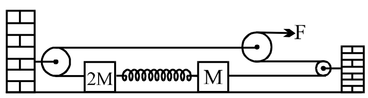
(A) $\frac{h}{(M+m)^2}$ (B) $\frac{hM}{(M+m)^2}$



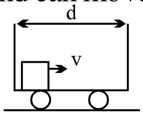
(C) $h\left(\frac{M}{M+m}\right)^2$ (D) $h\left(\frac{M}{M+m}\right)$

Q.155 In the diagram shown, no friction at any contact surface. Initially, the spring has no deformation. What will be the maximum deformation in the spring? Consider all the strings to be sufficiency large. Consider the spring constant to be K.

(A) 4F / 3K (B) 8F / 3K
(C) F / 3K (D) none



Q.156 In a smooth stationary cart of length d, a small block is projected along it's length with velocity v towards front. Coefficient of restitution for each collision is e. The cart rests on a smooth ground and can move freely. The time taken by block to come to rest w.r.t. cart is

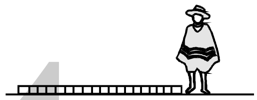


(A) $\frac{ed}{(1-e)v}$ (B) $\frac{ed}{(1+e)v}$ (C) $\frac{d}{e}$ (D) infinite

Q.157 A flexible chain of length 2m and mass 1kg initially held in vertical position such that its lower end just touches a horizontal surface, is released from rest at time t = 0. Assuming that any part of chain which strikes the plane immediately comes to rest and that the portion of chain lying on horizontal surface does not from any heap, the height of its centre of mass above surface at any instant $t = 1/\sqrt{5}$ (before it completely comes to rest) is

(A) 1 m (B) 0.5 m (C) 1.5 m (D) 0.25 m

Q.158 On a smooth horizontal plane, a uniform string of mass M and length is lying in the state of rest. A man of the same mass M is standing next to one end of the string. Now, the man starts collecting the string. Finally the man collects all the string and puts it in his pocket. What is the displacement of the man with respect to earth in the process of collection?



(A) L / 2 (B) L / 4 (C) L / 8 (D) none

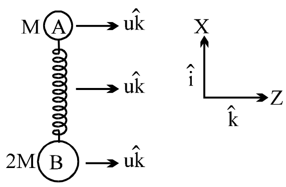
Q.159 An open water tight railway wagon of mass 5×10^3 kg coasts at an initial velocity 1.2 m/s without friction on a railway track. Rain drops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be

(A) 0.5 m/s (B) 2 m/s (C) 1 m/s (D) 1.5 m/s

Q.160 A parallel beam of particles of mass m moving with velocity v impinges on a wall at an angle θ to its normal . The number of particles per unit volume in the beam is n . If the collision of particles with the wall is elastic, then the pressure exerted by this beam on the wall is :

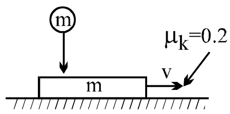
(A) $2\,mnv^2\cos\theta$ (B) $2\,mnv^2\cos^2\theta$ (C) $2\,mnv\cos\theta$ (D) $2\,mnv\cos^2\theta$

Q.161 Two masses A and B of mass M and 2M respectively are connected by a compressed ideal spring. The system is placed on a horizontal frictionless table and given a velocity $u\hat{k}$ in the z-direction as shown in the figure. The spring is then released. In the subsequent motion the line from B to A always points along the $\hat{i}$ unit vector. At some instant of time mass B has a x-component of velocity as $V_x\hat{i}$. The velocity $\vec{V}_A$ of as A at that instant is



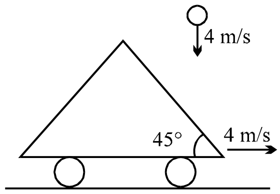
(A) $V_x\hat{i} + u\hat{k}$ (B) $-V_x\hat{i} + u\hat{k}$ (C) $-2V_x\hat{i} + u\hat{k}$ (D) $2V_x\hat{i} + u\hat{k}$

Q.162 A ball of mass m falls vertically from a height h and collides with a block of equal mass m moving horizontally with a velocity v on a surface. The coefficient of kinetic friction between the block and the surface is 0.2, while the coefficient of restitution e between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block decreases by



(A) 0 (B) $0.1\sqrt{2gh}$ (C) $0.3\sqrt{2gh}$ (D) can't be said

Q.163 A small ball falling vertically downward with constant velocity 4m/s strikes elastically a massive inclined cart moving with velocity 4m/s horizontally as shown. The velocity of the rebound of the ball is

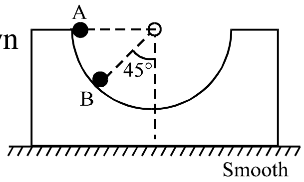


(A) $4\sqrt{2}$ m/s (B) $4\sqrt{3}$ m/s (C) 4m/s (D) $4\sqrt{5}$m/s

Q.164 A rocket of mass 4000 kg is set for vertical firing. How much gas must be ejected per second so that the rocket may have initial upwards acceleration of magnitude 19.6 m/s^2. [Exhaust speed of fuel = 980 m/s.]

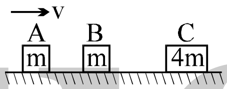
(A) 240 kg s^{-1} (B) 60 kg s^{-1} (C) 120 kg s^{-1} (D) None

Q.165 A ball of mass m is released from A inside a smooth wedge of mass m as shown in the figure. What is the speed of the wedge when the ball reaches point B?



(A) $\left(\frac{gR}{3\sqrt{2}}\right)^{1/2}$ (B) $\sqrt{2gR}$ (C) $\left(\frac{5gR}{2\sqrt{3}}\right)^{1/2}$ (D) $\sqrt{\frac{3}{2}gR}$

Q.166 Three blocks are initially placed as shown in the figure. Block A has mass m and initial velocity v to the right. Block B with mass m and block C with mass 4m are both initially at rest. Neglect friction. All collisions are elastic. The final velocity of block A is



(A) 0.6v to the left (B) 1.4v to the left
(C) v to the left (D) 0.4v to the right

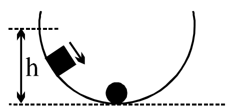
Q.167 Two identical spheres move in opposite directions with speeds v_1 and v_2 and pass behind an opaque screen, where they may either cross without touching (Event 1) or make an elastic head–on collision (Event 2)

(A) We can never make out which event has occurred
(B) We cannot make out which event has occurred only if $v_1 = v_2$
(C) We can always make out which event has occurred
(D) We can make out which event has occurred only if $v_1 = v_2$

Q.168 There are some passengers inside a stationary railway compartment. The track is frictionless. The centre of mass of the compartment itself (without the passengers) is C_1, while the centre of mass of the 'compartment plus passengers' system is C_2. If the passengers move about inside the compartment along the track.

(A) both C_1 and C_2 will move with respect to the ground
(B) neither C_1 nor C_2 will move with respect to the ground
(C) C_1 will move but C_2 will be stationary with respect to the ground
(D) C_2 will move but C_1 will be stationary with respect to the ground

Q.169 A block of mass m starts from rest and slides down a frictionless semi–circular track from a height h as shown. When it reaches the lowest point of the track, it collides with a stationary piece of putty also having mass m. If the block and the putty stick together and continue to slide, the maximum height that the block-putty system could reach is:



(A) h/4 (B) h/2 (C) h (D) independent of h

Q.170 A boy hits a baseball with a bat and imparts an impulse J to the ball. The boy hits the ball again with the same force, except that the ball and the bat are in contact for twice the amount of time as in the first hit. The new impulse equals:

(A) half the original impulse (B) the original impulse
(C) twice the original impulse (D) four times the original impulse

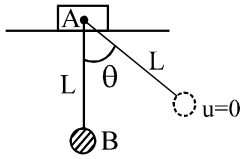
Q.171 Two billiard balls undergo a head-on collision. Ball 1 is twice as heavy as ball 2. Initially, ball 1 moves with a speed v towards ball 2 which is at rest. Immediately after the collision, ball 1 travels at a speed of v/3 in the same direction. What type of collision has occured?

(A) inelastic (B) elastic

(C) completely inelastic (D) Cannot be determined from the information given

Question No. 172 to 175 (4 questions)

A small ball B of mass m is suspended with light inelastic string of length L from a block A of same mass m which can move on smooth horizontal surface as shown in the figure. The ball is displaced by angle θ from equilibrium position & then released.



Q.172 The displacement of block when ball reaches the equilibrium position is

(A) $\frac{L\sin\theta}{2}$ (B) Lsinθ (C) L (D) none of these

Q.173 Tension in string when it is vertical, is

(A) mg (B) mg(2–cosθ) (C) mg (3 – 2cosθ) (D) none of these

Q.174 Maximum velocity of block during subsequent motion of the system after release of ball is

(A) $[gl\,(1\text{-}\cos\theta)]^{1/2}$ (B) $[2gl(1-\cos\theta)]^{1/2}$

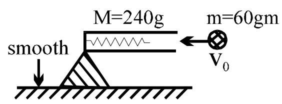
(C) $[gl\cos\theta]^{1/2}$ (D) informations are insufficient to decide

Q.175 The displacement of centre of mass of A + B system till the string becomes vertical is

(A) zero (B) $\frac{L}{2}(1-\cos\theta)$ (C) $\frac{L}{2}(1-\sin\theta)$ (D) none of these

Question No. 176 & 177 (2 questions)

A ball of mass m = 60gm is shot with speed v_0 = 22m/s into the barrel of spring gun of mass M = 240g initially at rest on a frictionless surface. The ball sticks in the barrel at the point of maximum compression of the spring.



Q.176 The speed of the spring gun after the ball stops relative to the barrel, is

(A) 2.2 m/s (B) 4.4 m/s (C) 4.4 cm/s (D) none

Q.177 What fraction of initial kinetic energy of the ball is now stored in the spring?

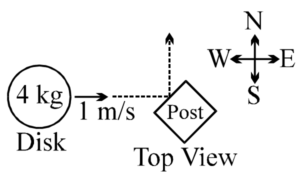
(A) 0.2 (B) 0.8 (C) 0.4 (D) 0.6

Q.178 In an elastic collision of two billiard balls which of the following quantities is not conserved during the short time of collision

(A) momentum (B) total mechanical energy

(C) kinetic energy (D) none

Q.179 A 4-kilogram disk slides over level ice toward the east at a velocity of 1meter per second, as shown. The disk strikes a post and rebounds toward the north at the same speed. The change in the magnitude of the eastward component of the momentum of the disk is



(A) –4 kg·m/s (B) –1 kg·m/s (C) 0 kg·m/s (D) 4 kg·m/s

Q.180 A system of N particles is free from any external forces.

(a) Which of the following is true for the magnitude of the total momentum of the system?

(A) It must be zero

(B) It could be non–zero, but it must be constant

(C) It could be non–zero, and it might not be constant

(D) The answer depends on the nature of the internal forces in the system

(b) Which of the following must be true for the sum of the magnitudes of the momenta of the individual particles in the system?
(A) It must be zero
(B) It could be non–zero, but it must be constant
(C) It could be non–zero, and it might not be constant
(D) It could be zero, even if the magnitude of the total momentum is not zero

Q.181 An isolated rail car of mass M is moving along a straight, frictionless track at an initial speed v_0. The car is passing under a bridge when a crate filled with N bowling balls, each of mass m, is dropped from the bridge into the bed of the rail car. The crate splits open and the bowling balls bounce around inside the rail car, but none of them fall out.

(a) Is the momentum of the rail car + bowling balls system conserved in this collision?
(A) Yes, the momentum is completely conserved.
(B) Only the momentum component in the vertical direction is conserved.
(C) Only the momentum component parallel to the track is conserved.
(D) No components are conserved.

(b) What is the average speed of the rail car + bowling balls system some time after the collision?
(A) $(M + Nm)v_0/M$ (B) $Mv_0/(Nm + M)$ (C) Nmv_0/M
(D) The speed cannot be determined because there is not enough information

Q.182 Consider a one-dimensional collision that involves a body of mass m_1 originally moving in the positive x direction with speed v_0 colliding with a second body of mass m_2 originally at rest. The collision could be completely inelastic, with the two bodies sticking together, completely elastic, or somewhere in between. After the collision, m_1 moves with velocity v_1 while m_2 moves with velocity v_2.

(a) If $m_1 > m_2$, then
(A) $-v_0 < v_1 < 0$ (B) $0 < v_1 < v_0$ (C) $0 < v_1 < 2v_0$ (D) $v_0 < v_1 < 2v_0$

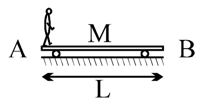
(b) and
(A) $-v_0 < v_2 < 0$ (B) $0 < v_2 < v_0$ (C) $v_0/2 < v_2 < 2v_0$ (D) $v_0 < v_2 < 2v_0$

(c) If $m_1 < m_2$ then
(A) $-v_0 < v_1 < 0$ (B) $-v_0 < v_1 < v_0/2$ (C) $0 < v_1 < v_0/2$ (D) $0 < v_1 < v_0$

(d) and
(A) $-v_0 < v_2 < 0$ (B) $-v_0 < v_2 < v_0/2$ (C) $0 < v_2 < v_0/2$ (D) $0 < v_2 < v_0$

Question No. 183 to 189 (7 questions)

The figure shows a man of mass m standing at the end A of a trolley of mass M placed at rest on a smooth horizontal surface. The man starts moving towards the end B with a velocity u_{rel} with respect to the trolley. The length of the trolley is L.



Q.183 When the man starts moving, then the velocity of the trolley v_2 with respect to ground will be

(A) $\frac{Mu_{rel}}{m+M}$ (B) $\frac{mu_{rel}}{m+M}$ (C) $\frac{m}{M}u_{rel}$ (D) $\frac{M}{m}u_{rel}$

Q.184 The velocity of the man with respect to ground v_1 will be

(A) $\frac{Mu_{rel}}{m+M}$ (B) $\frac{mu_{rel}}{m+M}$ (C) $\frac{m}{M}u_{rel}$ (D) $\frac{M}{m}u_{rel}$

Q.185 The time taken by the man to reach the other end is

(A) $\left(\frac{m+M}{M}\right)\frac{L}{u_{rel}}$ (B) $\left(\frac{m+M}{m}\right)\frac{L}{u_{rel}}$ (C) $\frac{L}{u_{rel}}$ (D) none of these

Q.186 As the man walks on the trolley, the centre of mass of the system (man + trolley)
(A) accelerates towards left (B) accelerates towards right
(C) moves with u_{rel} (D) remains stationary

Q.187 When the man reaches the end B, the distance moved by the trolley with respect to ground is

(A) $\frac{mL}{m+M}$ (B) $\frac{ML}{m+M}$ (C) $\frac{m}{M}L$ (D) $\frac{M}{m}L$

Q.188 The distance moved by the man with respect to ground is

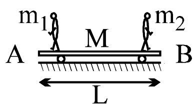
(A) $\frac{mL}{m+M}$ (B) $\frac{ML}{m+M}$ (C) $\frac{m}{M}L$ (D) $\frac{M}{m}L$

Q.189 Choose the correct statement

(A) As the man starts moving the trolley must move backward
(B) The distance moved by the trolley is independent of the speed of the man
(C) The distance moved by the trolley can never exceed L
(D) All the above

Question No. 190 to 194 (5 questions)

Two persons of mass m_1 and m_2 are standing at the two ends A and B respectively, of a trolley of mass M as shown.



Q.190 When the person standing at A jumps from the trolley towards left with u_{rel} with respect to the trolley, then

(A) the trolley moves towards right
(B) the trolley rebounds with velocity $\frac{m_1 u_{rel}}{m_1+m_2+M}$
(C) the centre of mass of the system remains stationary
(D) all the above

Q.191 When only the person standing at B jumps from the trolley towards right while the person at A keeps standing, then

(A) the trolley moves towards left
(B) the trolley moves with velocity $\frac{m_2 u_{rel}}{m_1+m_2+M}$
(C) the centre of mass of the system remains stationary
(D) all the above

Q.192 When both the persons jump simultaneously with same speed then

(A) the centre of mass of the system remains stationary
(B) the trolley remains stationary
(C) the trolley moves toward the end where the person with heavier mass is standing
(D) None of these

Q.193 When both the persons jump simultaneously with u_{rel} with respect to the trolley, then the velocity of the trolley is

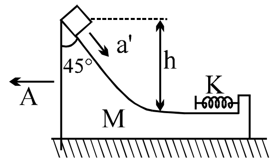
(A) $\frac{|m_1-m_2|u_{rel}}{m_1+m_2+M}$ (B) $\frac{|m_1-m_2|u_{rel}}{M}$ (C) $\left|\frac{m_1 u_{rel}}{m_2+M}-\frac{m_2 u_{rel}}{m_1+M}\right|$ (D) none of these

Q.194 Choose the incorrect statement, if $m_1 = m_2 = m$ and both the persons jump one by one, then

(A) the centre of mass of the system remains stationary
(B) the final velocity of the trolley is in the direction of the person who jumps first
(C) the final velocity of the trolley is $\left(\frac{mu_{rel}}{M+m}-\frac{mu_{rel}}{M+2m}\right)$
(D) none of these

Question No. 195 to 197 (3 questions)

Q.195 A small block of mass m is placed on a wedge of mass M as shown, which is initially at rest. All the surfaces are frictionless . The spring attached to the other end of wedge has force constant k. If a' is the acceleration of m relative to the wedge as it starts coming down and A is the acceleration acquired by the wedge as the block starts coming down, then



(A) $\frac{a'}{\sqrt{2}} < A < a'$ (B) $A < \frac{a'}{\sqrt{2}}$ (C) $A > a'$ (D) None

Q.196 Maximum velocity of M is:

(A) $\sqrt{2gh}$ (B) $\sqrt{\frac{2ghm}{m+M}}$ (C) $\sqrt{\frac{2m^2gh}{mM+M^2}}$ (D) None

Q.197 Maximum retardation of M is:

(A) $\sqrt{\frac{2mghk}{M^2}}$ (B) $\sqrt{\frac{2kgh}{M}}$ (C) $\sqrt{\frac{2kgh}{m}}$ (D) None

Q.198 In a one-dimensional collision, a particle of mass 2m collides with a particle of mass m at rest. If the particles stick together after the collision, what fraction of the initial kinetic energy is lost in the collision?

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) none

Q.199 A ball is dropped from a height h. As it bounces off the floor, its speed is 80 percent of what it was just before it hit the floor. The ball will then rise to a height of most nearly

(A) 0.80 h (B) 0.75 h (C) 0.64 h (D) 0.50 h

Question No.200 to 201(2 questions)

A projectile of mass "m" is projected from ground with a speed of 50 m/s at an angle of 53° with the horizontal. It breaks up into two equal parts at the highest point of the trajectory. One particle coming to rest immediately after the explosion.

Q.200 The ratio of the radii of curvatures of the moving particle just before and just after the explosion are:

(A) 1 : 4 (B) 1 : 3 (C) 2 : 3 (D) 4 : 9

Q.201 The distance between the pieces of the projectile when they reach the ground are:

(A) 240 (B) 360 (C) 120 (D) none

Q.202 A ball is thrown vertically downwards with velocity $\sqrt{2gh}$ from a height h. After colliding with the ground it just reaches the starting point. Coefficient of restitution is

(A) $1/\sqrt{2}$ (B) 1/2 (C) 1 (D) $\sqrt{2}$

Q.203 A ball is dropped from height 5m. The time after which ball stops rebounding if coefficient of restitution between ball and ground e = 1/2, is

(A) 1 sec (B) 2 sec (C) 3 sec (D) infinite

Q.204 A ball is projected from ground with a velocity V at an angle θ to the vertical. On its path it makes an elastic collison with a vertical wall and returns to ground. The total time of flight of the ball is

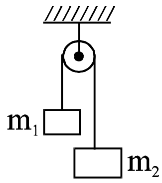
(A) $\frac{2v\sin\theta}{g}$ (B) $\frac{2v\cos\theta}{g}$ (C) $\frac{v\sin 2\theta}{g}$ (D) $\frac{v\cos\theta}{g}$

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

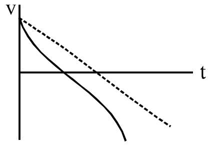
Q.1 A student calculates the acceleration of m_1 in figure shown as

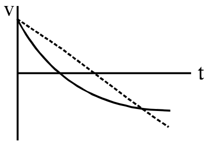
$a_1 = \dfrac{(m_1 - m_2)g}{m_1 + m_2}$. Which assumption is not required to do this calculation.

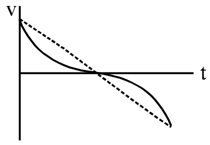


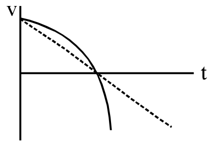
(A) pulley is frictionless
(B) string is massless
(C) pulley is massless
(D) string is inextensible

Q.2 Which graph shows best the velocity-time graph for an object launched vertically into the air when air resistance is given by $|D| = bv$? The dashed line shows the velocity graph if there were no air resistance.

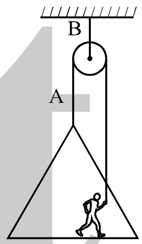
(A) 

(B) 

(C) 

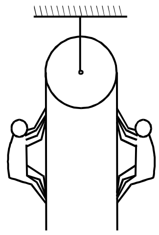
(D) 

Q.3 To paint the side of a building, painter normally hoists himself up by pulling on the rope A as in figure. The painter and platform together weigh 200N. The rope B can withstand 300N. Then



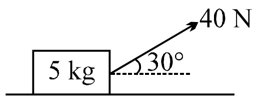
(A) The maximum acceleration that painter can have upwards is $5 m/s^2$.
(B) To hoist himself up, rope B must withstand minimum 400N force.
(C) Rope A will have a tension of 100 N when the painter is at rest.
(D) The painter must exert a force of 200N on the rope A to go downwards slowly.

Q.4 Two men of unequal masses hold on to the two sections of a light rope passing over a smooth light pulley. Which of the following are possible?



(A) The lighter man is stationary while the heavier man slides with some acceleration
(B) The heavier man is stationary while the lighter man climbs with some acceleration
(C) The two men slide with the same acceleration in the same direction
(D) The two men move with accelerations of the same magnitude in opposite directions

Q.5 Adjoining figure shows a force of 40 N acting at 30° to the horizontal on a body of mass 5 kg resting on a smooth horizontal surface. Assuming that the acceleration of free–fall is $10 ms^{-2}$, which of the following statements A, B, C, D, E is (are) correct?

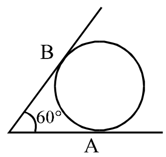


[1] The horizontal force acting on the body is 20 N
[2] The weight of the 5 kg mass acts vertically downwards
[3] The net vertical force acting on the body is 30 N

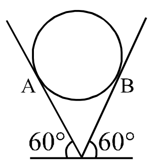
(A) 1, 2, 3
(B) 1, 2
(C) 2 only
(D) 1 only

Q.6 An iron sphere weighing 10 N rests in a V shaped smooth trough whose sides form an angle of 60° as shown in the figure. Then the reaction forces are

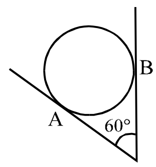
(A) $R_A = 10$ N & $R_B = 0$ in case (i)
(B) $R_A = 10$ N & $R_B = 10$ N in case (ii)
(C) $R_A = \dfrac{20}{\sqrt{3}}$ N & $R_B = \dfrac{10}{\sqrt{3}}$ N in case (iii)
(D) $R_A = 10$ N & $R_B = 10$ N in all the three cases



(i)



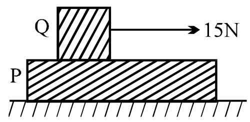
(ii)



(iii)

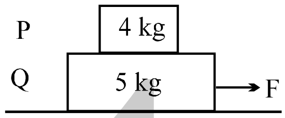
Q.7 For ordinary terrestrial experiments, which of the following observers below are inertial.
(A) a child revolving in a " giant wheel".
(B) a driver in a sports car moving with a constant high speed of 200 km/h on a straight road.
(C) the pilot of an aeroplane which is taking off.
(D) a cyclist negotiating a sharp turn.

Q.8 A long plank P of the mass 5 kg is placed on a smooth floor. On P is placed a block Q of mass 2 kg. The coefficient of friction between P and Q is 0.5. If a horizontal force 15N is applied to Q, as shown, and you may take g as 10N/kg.



(A) The reaction force on Q due to P is 10N
(B) The acceleration of Q relative to P is 2.5 m/s^2
(C) The acceleration of P relative to the Floor is 2.0 m/s^2
(D) The acceleration of centre of mass of P + Q system relative to the floor is $(15/7) m/s^2$

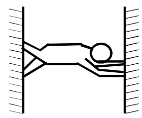
Q.9 The coefficient of friction between 4kg and 5 kg blocks is 0.2 and between 5kg block and ground is 0.1 respectively. Choose the correct statements



(A) Minimum force needed to cause system to move is 17N
(B) When force is 4N static friction at all surfaces is 4N to keep system at rest
(C) Maximum acceleration of 4kg block is $2 m/s^2$
(D) Slipping between 4kg and 5 kg blocks start when F is 17N

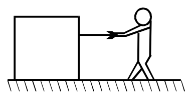
Q.10 In a tug–of–war contest, two men pull on a horizontal rope from opposite sides. The winner will be the man who
(A) exerts greater force on the rope
(B) exerts greater force on the ground
(C) exerts a force on the rope which is greater than the tension in the rope
(D) makes a smaller angle with the vertical

Q.11 A man balances himself in a horizontal position by pushing his hands and feet against two parallel walls. His centre of mass lies midway between the walls. The coefficients of friction at the walls are equal. Which of the following is not correct?



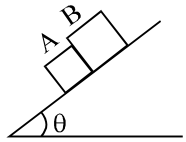
(A) He exerts equal forces on walls
(B) He exerts only horizontal forces on the walls
(C) The forces of friction at the walls are equal
(D) The forces exerted by the walls on him are not horizontal

Q.12 A man pulls a block heavier than himself with a light horizontal rope. The coefficient of friction is the same between the man and the ground, and between the block and the ground



(A) The block will not move unless the man also moves
(B) The man can move even when the block is stationary
(C) If both move, the acceleration of the man is greater than the acceleration of the block
(D) None of the above assertions is correct

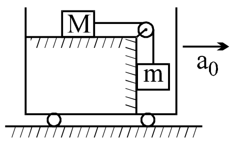
Q.13 The two blocks A and B of equal mass are initially in contact when released from rest on the inclined plane. The coefficients of friction between the inclined plane A and B are μ_1 and μ_2 respectively.



(A) If $\mu_1 > \mu_2$, the blocks will always remain in contact.
(B) If $\mu_1 < \mu_2$, the blocks will slide down with different accelerations. (if blocks slide)
(C) If $\mu_1 > \mu_2$, the blocks will have a common acceleration $\frac{1}{2}(\mu_1+\mu_2)\, g \sin\theta$.
(D) If $\mu_1 < \mu_2$, the blocks will have a common acceleration $\frac{\mu_1\mu_2 g}{\mu_1+\mu_2}\sin\theta$.

Question No. 14 to 16 (3 questions)

Imagine the situation in which the given arrangement is placed inside a trolley that can move only in the horizontal direction, as shown in figure. If the trolley is accelerated horizontally along the positive x-axis with a_0, then



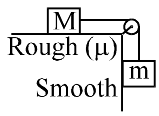
Q.14 Choose the correct statement(s).
(A) There exists a value of $a_0 = \beta$ at which friction force on block M becomes zero
(B) There exists two values of $a_0 = (\beta + \alpha)$ and $(\beta - \alpha)$ at which the magnitudes of friction acting on block M are equal
(C) The maximum value of static friction force acts on the block M at two accelerations a_1 and a_2 such that $a_1 + a_2 = 2\beta$
(D) The maximum value of friction is independent of the acceleration a_0.

Q.15 If a_{min} and a_{max} are the minimum and maximum values of a_0 for which the blocks remain stationary with respect to the surface, then identify the correct statements
(A) If $a_0 < a_{min}$, the block m accelerates downward
(B) If $a_0 > a_{max}$, the block m accelerates upward
(C) The block m does not accelerate up or down when $a_{min} \le a_0 \le a_{max}$
(D) The friction force on the block M becomes zero when $a_0 = \frac{a_{min} + a_{max}}{2}$

Q.16 Identify the correct statement(s) related to the tension T in the string
(A) No value of a_0 exists at which T is equal to zero
(B) There exists a value of a_0 at which T = mg
(C) If T < mg, then it must be more than μMg
(D) If T > mg, then it must be less than μMg

Question No. 17 to 20 (4 questions)

In figure, two blocks M and m are tied together with an inextensible and light string. The mass M is placed on a rough horizontal surface with coefficient of friction μ and the mass m is hanging vertically against a smooth vertical wall.
The pulley is frictionless.

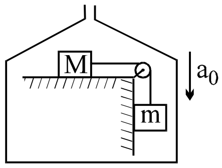


Q.17 Choose the correct statement(s)
(A) The system will accelerate for any value of m
(B) The system will accelerate only when m > M
(C) The system will accelerate only when m > μM
(D) Nothing can be said

Q.18 Choose the correct statement(s) related to the tension T in the string

(A) When $m < \mu M$, $T = mg$
(B) When $m < \mu M$, $T = Mg$
(C) When $m > \mu M$, $\mu Mg < T < mg$
(D) When $m > \mu M$, $mg < T < \mu Mg$

Q.19 Imagine a situation in which the given arrangement is placed inside an elevator that can move only in the vertical direction and compare the situation with the case when it is placed on the ground. When the elevator accelerates downward with $a_0 (< g)$, then



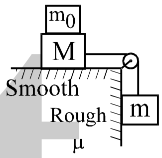
(A) the limiting friction force between the block M and the surface decreases
(B) the system can accelerate with respect to the elevator even when $m < \mu M$
(C) the system does not accelerate with respect to the elevator unless $m > \mu M$
(D) the tension in the string decreases

Q.20 When the downward acceleration of the elevator becomes equal to g, then

(A) both the blocks remain stationary with respect to the elevator
(B) both the blocks accelerate vertically downwards with g with respect to ground
(C) the tension in the string becomes equal to zero
(D) the friction force between the block M and the surface is zero

Question No. 21 to 27 (7 questions)

A block of mass M is placed on a horizontal surface and it is tied with an inextensible string to a block of mass, as shown in figure. A block of mass m_0 is also placed on M



Q.21 If there is no friction between any two surfaces, then

(A) the downward acceleration of the block m is $\dfrac{mg}{m + m_0 + M}$

(B) the acceleration of m_0 is zero
(C) If the tension in the string is T then $Mg < T < mg$
(D) all the above

If a friction force exist between block M and the horizontal surface with the coefficient of friction μ.

Q.22 The minimum value of μ for which the block m remains stationary is

(A) $\dfrac{m}{M}$
(B) $\dfrac{m}{M + m_0}$
(C) $\dfrac{M + m_0}{M}$
(D) $\dfrac{M}{M + m_0}$

Q.23 If $\mu < \mu_{min}$ (the minimum friction required to keep the block m stationary), then the downward acceleration of m is

(A) $\left[\dfrac{m - \mu M}{m + M}\right]g$
(B) $\left[\dfrac{m - \mu(m_0 + M)}{m + m_0 + M}\right]g$

(C) $\left[\dfrac{m - \mu(m_0 + M)}{m + M}\right]g$
(D) $\left[\dfrac{m - \mu M}{m + m_0 + M}\right]g$

Q.24 In previous problem, the tension in the string will be

(A) $\dfrac{mM}{m + M}g$
(B) $\dfrac{m(m_0 + M)}{m + m_0 + M}g$

(C) $\left[\dfrac{m + \mu(m_0 + M)}{m + M}\right]Mg$
(D) $\left[\dfrac{mM + \mu m(m_0 + M)}{m + M}\right]g$

Q.25 If μ_0 be the coefficient of friction between the block M and the horizontal surface then the minimum value of m_0 required to keep the block m stationary is

(A) $\frac{m}{\mu} - M$ (B) $\frac{m - M}{\mu}$ (C) $\frac{m}{\mu} + M$ (D) $\frac{m + M}{\mu}$

Q.26 If friction force exists between the block M and the block m_0 and not between the block M and the horizontal surface, then the minimum value of μ for which the block m remains stationary is

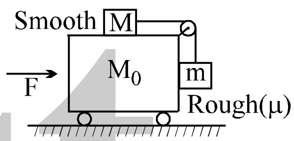
(A) $\frac{m}{m_0}$ (B) $\frac{m}{m_0 + M}$ (C) $\frac{m - m_0}{M}$ (D) none of these

Q.27 The minimum value of μ between the block M and m_0 (taking horizontal surface frictionless) for which all the three blocks move together, is

(A) $\frac{m}{m + m_0 + M}$ (B) $\frac{m}{m + M}$ (C) $\frac{m_0}{m + m_0 + M}$ (D) none of these

Question No. 28 to 31 (4 questions)

Imagine a situation in which the horizontal surface of block M_0 is smooth and its vertical surface is rough with a coefficient of friction μ.



Q.28 Identify the correct statement(s)

(A) If F = 0, the blocks cannot remain stationary

(B) For one unique value of F, the blocks M and m remain stationary with respect to M_0

(C) The limiting friction between m and M_0 is independent of F

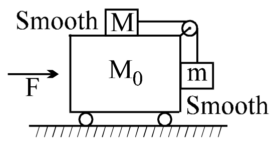
(D) There exist a value of F at which friction force is equal to zero

Q.29 In above problem, choose the correct value(s) of F which the blocks M and m remain stationary with respect to M_0

(A) $(M_0 + M + m)\frac{g}{\mu}$ (B) $\frac{m(M_0 + M + m)g}{M - \mu m}$

(C) $(M_0 + M + m)\frac{mg}{M}$ (D) none of these

Q.30 Consider a special situation in which both the faces of the block M_0 are smooth, as shown in adjoining figure. Mark out the correct statement(s)



(A) If F = 0, the blocks cannot remain stationary

(B) For one unique value of F, the blocks M and m remain stationary with respect to block M_0

(C) There exists a range of F for which blocks M and m remain stationary with respect to block M_0

(D) Since there is no friction, therefore, blocks M and m cannot be in equilibrium with respect to M_0

Q.31 In above problem, the value(s) of F for which M and m are stationary with respect to M_0

(A) $(M_0 + M + m)g$ (B) $(M_0 + M + m)\frac{mg}{M}$ (C) $(M_0 + M + m)\frac{Mg}{m}$ (D) none of these

Q.32 A particle with constant total energy E moves in one dimension in a region where the potential energy is U(x). The speed of the particle is zero where

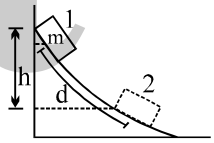
(A) $U(x) = E$ (B) $U(x) = 0$ (C) $\frac{dU(x)}{dx} = 0$ (D) $\frac{d^2U(x)}{dx^2} = 0$

Q.33 A block of mass m slides down a plane inclined at an angle θ. Which of the following will NOT increase the energy lost by the block due to friction?
(A) Increasing the angle of inclination
(B) Increasing the distance that the block travels
(C) Increasing the accelertion due to gravity
(D) Increasing the mass of the block

Q.34 The potential energy in joules of a particle of mass 1 kg moving in a plane is given by $U = 3x + 4y$, the position coordinates of the point being x and y, measured in metres. If the particle is initially at rest at (6,4), then
(A) its acceleration is of magnitude 5 m/s^2
(B) its speed when it crosses the y-axis is 10 m/s
(C) it crosses the y-axis ($x = 0$) at $y = -4$
(D) it moves in a straight line passing through the origin (0,0)

Q.35 The potential energy of a particle of mass 5kg moving in the XY plane is given by $V = -7x+24y$ joules, x and y being in metres. Initially at t=0 the particle is at the origin (0,0) moving with a velocity of $6[\hat{i}(2.4)+\hat{j}(0.7)]$ m/s. Then
(A) the magnitude of velocity of the particle at t = 4 sec is 25m/s
(B) the magnitude of acceleration of the particle is 5m/s^2
(C) the direction of motion of the particle initially at t=0 is at right angles to the direction of acceleration
(D) the path of the particle is a circle.

Q.36 A box of mass m is released from rest at position 1 on the frictionless curved track shown. It slides a distance d along the track in time t to reach position 2, dropping a vertical distance h. Let v and a be the instantaneous speed and instantaneous acceleration, respectively, of the box at position 2. Which of the following equations is valid for this situation?



(A) $h = vt$
(B) $h = (1/2)gt^2$
(C) $d = (1/2)at^2$
(D) $mgh = (1/2)mv^2$

Q.37 A ball of mass m is attached to the lower end of light vertical spring of force constant k. The upper end of the spring is fixed. The ball is released from rest with the spring at its normal (unstretched) length, comes to rest again after descending through a distance x.
(A) $x = mg/k$
(B) $x = 2\, mg/k$
(C) The ball will have no acceleration at the position where it has descended through x/2.
(D) The ball will have an upward acceleration equal to g at its lowermost position.

Q.38 A ball is projected vertically upwards. Air resistance & variation in g may be neglected. The ball rises to its maximum height H in a time T, the height being h after a time t
[1] The graph of kinetic energy E_k of the ball against height h is shown in figure 1
[2] The graph of height h against time t is shown in figure 2
[3] The graph of gravitational energy E_g of the ball against height h is shown in figure 3

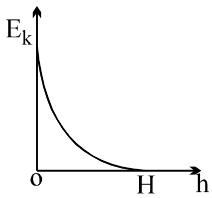


Fig. 1

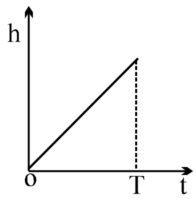


Fig. 2

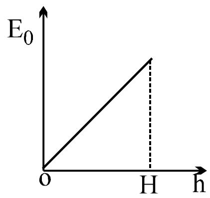
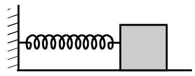


Fig. 3

Which of A, B, C, D, E shows the correct answers?
(A) 3 only
(B) 1, 2
(C) 2, 3
(D) 1 only

Q.39 A spring block system is placed on a rough horizontal floor. The block is pulled towards right to give spring some elongation and released.



(A) The block may stop before the spring attains its mean position.
(B) The block must stop with spring having some compression.
(C) The block may stop with spring having some compression.
(D) It is not possible that the block stops at mean position.

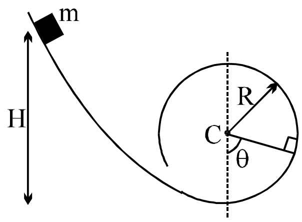
Q.40 In the above situation the block will have maximum velocity when
(A) the spring force becomes zero
(B) the frictional force becomes zero
(C) the net force becomes zero
(D) the acceleration of block becomes zero

Q.41 A particle of mass m is at rest in a train moving with constant velocity with respect to ground. Now the particle is accelerated by a constant force F_0 acting along the direction of motion of train for time t_0. A girl in the train and a boy on the ground measure the work done by this force. Which of the following are **INCORRECT**?
(A) Both will measure the same work
(B) Boy will measure higher value than the girl
(C) Girl will measure higher value than the boy
(D) Data are insufficient for the measurement of work done by the force F_0

Q.42 Two particles move on a circular path (one just inside and the other just outside) with angular velocities ω and 5ω starting from the same point. Then
(A) they cross each other at regular intervals of time $\frac{2\pi}{4\omega}$ when their angular velocities are oppositely directed.
(B) they cross each other at points on the path subtending an angle of 60^o at the centre if their angular velocities are oppositely directed.
(C) they cross at intervals of time $\frac{\pi}{3\omega}$ if their angular velocities are oppositely directed.
(D) they cross each other at points on the path subtending 90^o at the centre if their angular velocities are in the same sense.

Q.43 A cart moves with a constant speed along a horizontal circular path. From the cart, a particle is thrown up vertically with respect to the cart
(A) The particle will land somewhere on the circular path
(B) The particle will land outside the circular path
(C) The particle will follow an elliptical path
(D) The particle will follow a parabolic path

Question No. 44 to 46 (3 questions)
A particle of mass m is released from a height H on a smooth curved surface which ends into a vertical loop of radius R, as shown

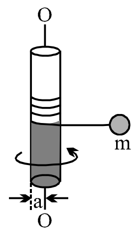


Q.44 Choose the correct alternative(s) if $H = 2R$
(A) The particles reaches the top of the loop with zero velocity
(B) The particle cannot reach the top of the loop
(C) The particle breaks off at a height $H = R$ from the base of the loop
(D) The particle break off at a height $R < H < 2R$

Q.45 If θ is instantaneous angle which the line joining the particle and the centre of the loop makes with the vertical, then identify the correct statement(s) related to the normal reaction N between the block and the surface
(A) The maximum value N occurs at $\theta = 0$
(B) The minimum value of N occurs at $N = \pi$ for $H > 5R/2$
(C) The value of N becomes negative for $\pi/2 < \theta < 3\pi/2$
(D) The value of N becomes zero only when $\theta \geq \pi/2$

Q.46 The minimum value of H required so that the particle makes a complete vertical circle is given by
(A) 5 R (B) 4 R (C) 2.5 R (D) 2 R

Q.47 A small particle of mass m is given an initial high velocity in the horizontal plane and winds its cord around the fixed vertical shaft of radius a. All motion occurs essentially in horizontal plane. If the angular velocity of the cord is ω_0 when the distance from the particle to the tangency point is r_0, then the angular velocity of the cord ω after it has turned through an angle θ is



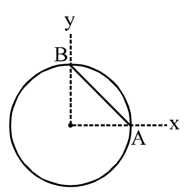
(A) $\omega = \omega_0$ (B) $\omega = \dfrac{a\omega_0}{r_0}$ (C) $\omega = \dfrac{\omega_0}{1 - \dfrac{a\theta}{r_0}}$ (D) $\omega = \omega_0\theta$

Q.48 A particle moving with kinetic energy = 3 joule makes an elastic head on collision with a stationary particle which has twice its mass during the impact.
(A) The minimum kinetic energy of the system is 1 joule.
(B) The maximum elastic potential energy of the system is 2 joule.
(C) Momentum and total kinetic energy of the system are conserved at every instant.
(D) The ratio of kinetic energy to potential energy of the system first decreases and then increases.

Q.49 A ball of mass m collides elastically with an identical ball at rest with some impact parameter.
(A) 100 % energy transfer can never take place
(B) 100 % energy transfer may take place
(C) angle of divergence between the two balls must be 90°
(D) angle of divergence between the two balls depend on impact parameter

Q.50 Two balls A and B having masses 1 kg and 2 kg, moving with speeds 21 m/s and 4 m/s respectively in opposite direction, collide head on. After collision A moves with a speed of 1 m/s in the same direction, then correct statements is :
(A) The velocity of B after collision is 6 m/s opposite to its direction of motion before collision.
(B) The coefficient of restitution is 0.2.
(C) The loss of kinetic energy due to collision is 200 J.
(D) The impulse of the force between the two balls is 40 Ns.

Q.51 An object comprises of a uniform ring of radius R and its uniform chord AB (not necessarily made of the same material) as shown. Which of the following can not be the centre of mass of the object

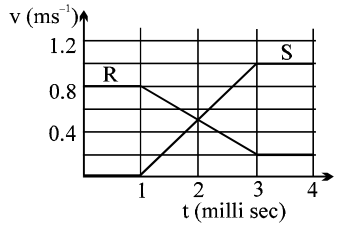


(A) (R/3, R/3) (B) (R/3, R/2)
(C) (R/4, R/4) (D) $(R/\sqrt{2}, R/\sqrt{2})$

Q.52 A ball A collides elastically with another identical ball B initially at rest A is moving with velocity of 10m/s at an angle of 60° from the line joining their centres. Select correct alternative :
(A) velocity of ball A after collision is 5 m/s (B) velocity of ball B after collision is $5\sqrt{3}$ m/s
(C) velocity of ball A after collision is 7.5 m/s (D) velocity of ball B after collision is 5 m/s.

Q.53 Consider following statements
[1] CM of a uniform semicircular disc of radius R = 2R/π from the centre
[2] CM of a uniform semicircular ring of radius R = 4R/3π from the centre
[3] CM of a solid hemisphere of radius R = 4R/3π from the centre
[4] CM of a hemisphere shell of radius R = R/2 from the centre
Which statements are correct?
(A) 1, 2, 4 (B) 1, 3, 4 (C) 4 only (D) 1, 2 only

Q.54 The diagram to the right shows the velocity-time graph for two masses R and S that collided elastically. Which of the following statements is true?



(I) R and S moved in the same direction after the collision.
(II) Kinetic energy of the system (R & S) is minimum at t = 2 milli sec.
(III) The mass of R was greater than mass of S.
(A) I only (B) II only (C) I and II only (D) I, II and III

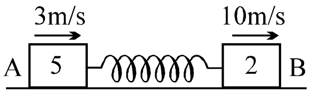
Q.55 In an inelastic collision,
(A) the velocity of both the particles may be same after the collision
(B) kinetic energy is not conserved
(C) linear momentum of the system is conserved.
(D) velocity of separation will be less than velocity of approach.

Q.56 A man of mass 40 kg is standing on a trolley A of mass 140 kg . He pushes another trolley B of same material of mass 60 kg, so that they are set in motion . Then :
(A) speed of trolley A is 3 times that of trolley B immediately after the interaction.
(B) speed of trolley B is 3 times that of trolley A immediately after the interaction.
(C) distance travelled by trolley B is 3 times that of trolley A before they stop.
(D) distance travelled by trolley B is 9 times that of trolley A before they stop.

Q.57 Two identical balls are interconnected with a massless and inextensible thread. The system is in gravity free space with the thread just taut. Each ball is imparted a velocity v, one towards the other ball and the other perpendicular to the first, at t = 0. Then,
(A) the thread will become taut at t = (L/v)
(B) the thread will become taut at some time t < (L/v).
(C) the thread will always remain taut for t > (L/v).
(D) the kinetic energy of the system will always remain mv^2.

Q.58 In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.
(A) The total momentum of the 'A plus B' system is p before and after the impact, and (p-J) during the impact.
(B) During the impact A gives impulse J to B
(C) The coefficient of restitution is $\frac{2J}{p} - 1$
(D) The coefficient of restitution is $\frac{J}{p} + 1$

Q.59 Two blocks A (5kg) and B(2kg) attached to the ends of a spring constant 1120N/m are placed on a smooth horizontal plane with the spring undeformed. Simultaneously velocities of 3m/s and 10m/s along the line of the spring in the same direction are imparted to A and B then



(A) when the extension of the spring is maximum the velocities of A and B are zero.
(B) the maximum extension of the spring is 25cm.
(C) maximum extension and maximum compression occur alternately.
(D) the maximum compression occur for the first time after $\frac{\pi}{56}$ sec.

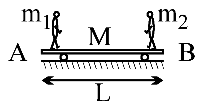
Q.60 In a one–dimensional collision between two particles, their relative velocity is $\vec{v}_1$ before the collision and $\vec{v}_2$ after the collision
(A) $\vec{v}_1 = \vec{v}_2$ if the collision is elastic
(B) $\vec{v}_1 = -\vec{v}_2$ if the collision is elastic
(C) $|\vec{v}_2| = |\vec{v}_1|$ in all cases
(D) $\vec{v}_1 = -k\vec{v}_2$ in all cases, where $k \geq 1$

Q.61 In an elastic collision between disks A and B of equal mass but unequal radii, A moves along the x–axis and B is stationary before impact. Which of the following is possible after impact?
(A) A comes to rest
(B) The velocity of B relative to A remains the same in magnitude but reverses in direction
(C) A and B move with equal speeds, making an angle of 45° each with the x–axis
(D) A and B move with unequal speeds, making angles of 30° and 60° with the x–axis respectively

Q.62 An isolated rail car originally moving with speed v_0 on a straight, frictionles, level track contains a large amount of sand. A release valve on the bottom of the car malfunctions, and sand begins to pour out straight down relative to the rail car.
(a) Is momentum conserved in this process?
(A) The momentum of the rail car alone is conserved
(B) The momentum of the rail car + sand remaining within the car is conserved
(C) The momentum of the rail car + all of the sand, both inside and outside the rail car, is conserved
(D) None of the three previous systems have momentum conservation
(b) What happens to the speed of the rail car as the sand pours out?
(A) The car begins to roll faster
(B) The car maintains the same speed
(C) The car begins to slow down
(D) The problem cannot be solved since momentum is not conserved

Question No. 63 to 66 (4 questions)

Two men of mass m_1 and m_2 are standing at the ends A and B of the trolley, respectively. The mass of the trolley is M and its length is L



The two men can exchange their positions in three different ways:

Case I : m_1 moves towards B with u_{rel} and m_2 remains stationary until m_1 reaches its position; and then m_2 starts moving and reaches the end A.

Case II : m_2 moves towards A with u_{rel} and m_1 remains stationary until m_2 reaches its position, and then m_1 starts moving and reaches the end B.

Case III : Both moves with u_{rel} with respect to trolley towards each other and reach then opposite ends.

Q.63 Choose the correct statement(s) related to **Case I**

(A) As the man m_1 moves, the trolley moves toward left and its velocity becomes maximum when it reaches the end B.

(B) When m_1 reaches the end B, the distance moved by the trolley is $\dfrac{m_1 L}{m_1 + m_2 + M}$

(C) When m_1 and m_2 has exchanged their positions, the displacement of the centre of mass of the system is zero.

(D) When the men have exchanged their positions, the final velocity of the trolley is zero

Q.64 Choose the correct statement(s) related to **Case II**

(A) When the man m_2 reaches the position of m_1, the distance moved by the trolley is $\dfrac{m_2 L}{m_1 + m_2 + M}$

(B) When the man m_1 reaches the position of m_2, the distance moved by the trolley is $\dfrac{m_1 L}{m_1 + m_2 + M}$

(C) When the men have exchanged their positions, the distance moved by the center of mass is

$$\left(\frac{m_1 + m_2}{m_1 + m_2 + M}\right)L$$

(D) When the men have exchanged their position, the displacement of the centre of mass is $\dfrac{(m_1 - m_2)L}{m_1 + m_2 + M}$

Q.65 Choose the correct statement(s) related to **Case III**

(A) As both the men move simultaneously, the velocity of the trolley at any instant is zero

(B) Both men reach their opposite ends simultaneously

(C) The distance travelled by both the men with respect to ground is same

(D) All the above

Q.66 Choose the correct statement(s) related to all the three cases

(A) The centre mass remains stationary at all instants

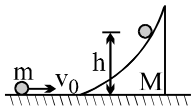
(B) The displacement of the trolley cannot exceed L

(C) The displacement of the trolley is independent of the velocity of each man

(D) The displacement of the trolley in all the three cases is same

Question No. 67 to 73 (7 questions)

A particle of mass m moving horizontally with v_0 strikes a smooth wedge of mass M, as shown in figure. After collision, the ball starts moving up the inclined face of the wedge and rises to a height h.



Q.67 The final velocity of the wedge v_2 is

(A) $\dfrac{mv_0}{M}$ (B) $\dfrac{mv_0}{M+m}$ (C) v_0 (D) insufficient data

Q.68 When the particle has risen to a height h on the wedge, then choose the correct alternative(s)

(A) The particle is stationary with respect to ground

(B) Both are stationary with respect to the centre of mass

(C) The kinetic energy of the centre of mass remians constant

(D) The kinetic energy with respect to centre of mass is converted into potential energy

Q.69 The maximum height h attained by the particle is

(A) $\left(\dfrac{m}{m+M}\right)\dfrac{v_0^2}{2g}$ (B) $\left(\dfrac{m}{M}\right)\dfrac{v_0^2}{2g}$ (C) $\left(\dfrac{M}{m+M}\right)\dfrac{v_0^2}{2g}$ (D) none of these

Q.70 Identify the correct statement(s) related to the situation when the particle starts moving downward.

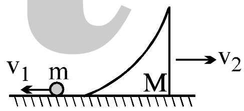
(A) The centre of mass of the system remains stationary

(B) Both the particle and the wedge remain stationary with respect to centre of mass

(C) When the particle reaches the horizontal surface it velocity relative to the wedge is v_0

(D) None of these

Q.71 Suppose the particle when reaches the horizontal surfaces, its velocity with respect to ground is v_1 and that of wedge is v_2. Choose the correct statement(s)



(A) $mv_1 = Mv_2$ (B) $Mv_2 - mv_1 = mv_0$ (C) $v_1 + v_2 = v_0$ (D) $v_1 + v_2 < v_0$

Q.72 Choose the correct statement(s) related to particle m

(A) Its kinetic energy is $K_f = \left(\dfrac{mM}{m+M}\right)gh$ (B) $v_1 = v_0\left(\dfrac{M-m}{M+m}\right)$

(C) The ratio of its final kinetic energy to its initial kinetic energy is $\dfrac{K_f}{K_i} = \left(\dfrac{M}{m+M}\right)^2$

(D) It moves opposite to its initial direction of motion

Q.73 Choose the correct statement related to the wedge M

(A) Its kinetic energy is $K_f = \left(\dfrac{4m^2}{m+M}\right)gh$ (B) $v_2 = \left(\dfrac{2m}{m+M}\right)v_0$

(C) Its gain in kinetic energy is $\Delta K = \left(\dfrac{4mM}{(m+M)^2}\right)\left(\dfrac{1}{2}mv_0^2\right)$

(D) Its velocity is more than the velocity of centre of mass

ANSWER KEY

ONLY ONE OPTION IS CORRECT.

Q.1 A	Q.2 B	Q.3 A	Q.4 B	Q.5 B	Q.6 A	Q.7 B
Q.8 C	Q.9 A	Q.10 B	Q.11 B	Q.12 B	Q.13 A	Q.14 C
Q.15 B	Q.16 C	Q.17 C	Q.18 B	Q.19 C	Q.20 A	Q.21 A
Q.22 B	Q.23 B	Q.24 B	Q.25 B	Q.26 A	Q.27 A	Q.28 D
Q.29 D	Q.30 C	Q.31 A	Q.32 B	Q.33 C	Q.34 B	Q.35 B
Q.36 D	Q.37 C	Q.38 C	Q.39 A	Q.40 A	Q.41 A	Q.42 C
Q.43 A	Q.44 C	Q.45 C	Q.46 A	Q.47 C	Q.48 A	Q.49 D
Q.50 A	Q.51 B	Q.52 B	Q.53 A	Q.54 B	Q.55 A	Q.56 A
Q.57 C	Q.58 B	Q.59 C	Q.60 C	Q.61 B	Q.62 A	Q.63 A
Q.64 A	Q.65 C	Q.66 B	Q.67 D	Q.68 D	Q.69 C	Q.70 C
Q.71 B	Q.72 B	Q.73 C	Q.74 D	Q.75 C	Q.76 B	Q.77 C
Q.78 A	Q.79 B	Q.80 D	Q.81 B	Q.82 C	Q.83 C	Q.84 B
Q.85 C	Q.86 C	Q.87 C	Q.88 B	Q.89 B	Q.90 A	Q.91 C
Q.92 C	Q.93 A	Q.94 C	Q.95 C	Q.96 A	Q.97 B	Q.98 B
Q.99 D	Q.100 C	Q.101 C	Q.102 D	Q.103 C	Q.104 B	Q.105 B
Q.106 D	Q.107 B	Q.108 C	Q.109 C	Q.110 B	Q.111 C	Q.112 D
Q.113 A	Q.114 B	Q.115 A	Q.116 A	Q.117 C	Q.118 B	Q.119 B
Q.120 B	Q.121 B	Q.122 D	Q.123 B	Q.124 B	Q.125 B	Q.126 D
Q.127 C	Q.128 C	Q.129 C	Q.130 D	Q.131 A	Q.132 B	Q.133 B
Q.134 B	Q.135 A	Q.136 C	Q.137 D	Q.138 D	Q.139 C	Q.140 A
Q.141 B	Q.142 C	Q.143 C	Q.144 B	Q.145 C	Q.146 A	Q.147 C
Q.148 C	Q.149 D	Q.150 A	Q.151 B	Q.152 D	Q.153 B	Q.154 C
Q.155 B	Q.156 D	Q.157 D	Q.158 B	Q.159 C	Q.160 B	Q.161 C
Q.162 D	Q.163 D	Q.164 C	Q.165 A	Q.166 A	Q.167 A	Q.168 C
Q.169 A	Q.170 C	Q.171 B	Q.172 A	Q.173 D	Q.174 A	Q.175 B
Q.176 B	Q.177 B	Q.178 C	Q.179 A	Q.180 (a) B ,(b) C		
Q.181 (a) C ,(b) B		Q.182 (a) B, (b) C, (c) B, (d) D			Q.183 B	Q.184 A
Q.185 C	Q.186 D	Q.187 A	Q.188 B	Q.189 D	Q.190 D	Q.191 D
Q.192 A	Q.193 A	Q.194 D	Q.195 B	Q.196 C	Q.197 A	Q.198 B
Q.199 C	Q.200 A	Q.201 A	Q.202 A	Q.203 C	Q.204 B	

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	C	Q.2	B	Q.3	A,C	Q.4	A,B,D
Q.5	C	Q.6	A,B,C	Q.7	B	Q.8	C,D
Q.9	C	Q.10	B	Q.11	B	Q.12	A,B,C
Q.13	A,B	Q.14	A,B,C,D	Q.15	A,B,C,D	Q.16	A,B,C
Q.17	C	Q.18	A,C	Q.19	A,C,D	Q.20	A,B,C,D
Q.21	B	Q.22	B	Q.23	C	Q.24	D
Q.25	A	Q.26	D	Q.27	A	Q.28	A,D
Q.29	B,C	Q.30	A,B	Q.31	B	Q.32	A
Q.33	A	Q.34	A,B,C	Q.35	A,B,C	Q.36	D
Q.37	B,C,D	Q.38	A	Q.39	A,C	Q.40	C,D
Q.41	A,C	Q.42	B,C,D	Q.43	B,D	Q.44	B,D
Q.45	A,B,D	Q.46	C	Q.47	C	Q.48	A,B,D
Q.49	A,C	Q.50	A,B,C	Q.51	B,D	Q.52	D
Q.53	C	Q.54	D	Q.55	A,B,C,D	Q.56	B,D
Q.57	A,C	Q.58	B,C	Q.59	B,C	Q.60	B,D
Q.61	A,B,C,D	Q.62	(a) A,C ;(b) B	Q.63	B,C,D	Q.64	A
Q.65	B	Q.66	A,B,C,D	Q.67	B	Q.68	B,D
Q.69	C	Q.70	C	Q.71	B,C	Q.72	B
Q.73	A,B,C,D						

TARGET IIT JEE

PHYSICS

ROTATIONAL DYNAMICS

QUESTION FOR SHORT ANSWER

Q.1 Should there be any matter at the center of gravity of an object?

Q.2 The torque exerted by a force about some axis depends on the choice of axis. How can the condition, $\sum \tau_{z,ext} = 0$, be satisfied for any choice of axis?

Q.3 What is the path of a particle in a rigid object rotating about a fixed axis?

Q.4 If a rigid object has only translational motion (for example, the body of a car traveling in a straight line on a flat road), are there any points within the object that always have the same velocity as the center of mass? If so, which ones?

Q.5 If a rigid object has only rotational motion about a fixed axis are there any points within the object that always have the same velocity as the center of mass? If so which ones?

Q.6 If a rigid object moves with both transition and rotation about an axis with a fixed orientation (for example, a rolling wheel), are there any points within the object that always have the same velocity as the center of mass? If so, which ones?

Q.7 What is the direction of the angular velocity of a rigid object rotating about a fixed axis? What is the direction of the linear velocity of a particle in a rigid object rotating about a fixed axis?

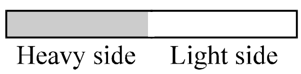
Q.8 A rigid object rotating about a fixed axis has nonzero angular velocity and angular accleration. Particle A in the object is twice as far from the axis of rotation as particle B. What is the ratio of the following quantities for A and B:

(a) the angular speeds
(b) the linear speeds
(c) the magnitudes of the angular accelerations
(d) the tangential components of the accelerations
(e) the radial components of the accelerations
(f) the magnitudes of the linear accelerations?

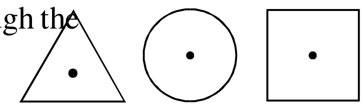
Q.9 Do the angular velocities of the hands of a wall clock point into the wall or out of the wall? At the instant the clock is unplugged, do the angular accelerations of the hands point into the wall or out of the wall?

Q.10 A car is moving forward and slowing down. Is the direction of te angular velocity of the wheels toward the driver's left or right? What is the direction of the angular acceleration of the wheels?

Q.11 One side of a door (figure) is made of material with a larger mass density than the other side. To minimize the moment of inertia about an axis of rotation along the hinges, should the hinges be placed at the heavier side or the lighter side? Explain.



Q.12 Consider three rods made of the same material and with the same length and mass, but with different cross-sectional shapes (figure). Which of the three has the largest moment of inertia about an axis through the center of mass and along the rod's long axis? Which rod has the smallest moment of inertia about that axis?



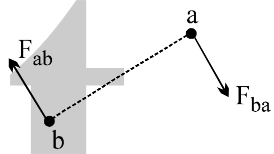
Q.13 Is it possible to find an axis of rotation (call the axis A) about which the moment of inertia for an object is smaller than the moment of inertia about an axis through the center of mass and parallel to A?

Q.14 Suppose you are designing a cart for coasting down a hill. To maximize your coasting speed, should you design the wheels so that their moments of inertia about their rotation axes are large or small, or does it matter? Keeping the moment of inertia of the wheels fixed, will the cart's speed be increased or decreased by increasing the mass of the cart's body? Assume that mechanical energy is conserved.

Q.15 If a particle is in uniform circular motion, is either the direction or the magnitude of the angular momentum about the center of its motion constant? If the particle's speed is changing as it travels in a circle, is either the direction or the magnitude of the angular momentum constant?

Q.16 If the net torque exerted on a particle is in the same direction as the particle's angular momentum, is there a change in the direction of the particle's angular momentum? Is there a change in the magnitude of the particle's angular momentum?

Q.17 Consider an isolated system of two particles a and b that interact with each other such that $F_{ab} = -F_{ba}$ hut the direction of the forces is perpendicular to the line joining the particles, as shown in figure. What happens to this system as time goes on? Is total linear momentum conserved? Is total angular momentum conserved? Is such a system possible? Explain.



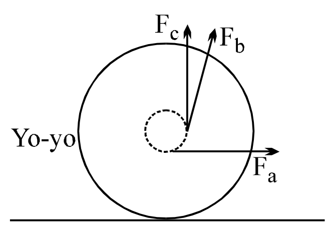
Q.18 When a billiard ball rolls down a slope without sliding, what force is responsible for the torque that causes the angular acceleration about an axis through the center of mass? What force is responsible for the torque that causes the angular acceleration about an axis through the point of contact with the surface?

Q.19 Legend has it that a cat always lands on its feet. High-speed cameras have shown that when a cat begins a fall with its feet up, its tail rotates rapidly and the cat's body also rotates, so that it does, infact, land on its feet. Explain the motion in terms of conservation of angular momentum. Include in your explanation a comparison of the sense of the rotation of the cat's body with that of its tail. How do you think a bobtailed cat might do in a fall that begins with its feet up?

Q.20 A small satellite orbiting the earth has only one window for the astronaut, and the window is facing away from the earth. Explain how the astronaut can rotate the satellite so he can view the earth and not use any rocket fuel in the process.

Q.21 A spinning ice skater rapidly extends his arms. [Neglect friction during the time interval the arms are extended]. Is his kinetic energy conserved? Is his potential energy conserved? Is his mechanical energy conserved? Is his angular momentum conserved? If any of these quantities are not conserved, tell whether they increase or decrease.

Q.22 A yo-yo with half the string wound on its axle is placed on its edge on the floor, as shown in figure. Consider pulling gently on the string in the three different directions indicated by F_a, F_b and F_c in the figure. The force in each case is gentle enough so that the yo-yo does not slide. In which case, if any, does string wind onto the yo-yo, and in which case, if any, does the string wind off the yo-yo?



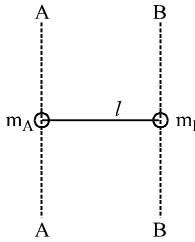
ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

Q.1 Three bodies have equal masses m. Body A is solid cylinder of radius R, body B is a square lamina of side R, and body C is a solid sphere of radius R. Which body has the smallest moment of inertia about an axis passing through their centre of mass and perpendicular to the plane (in case of lamina)

(A) A (B) B (C) C (D) A and C both

Q.2 A point mass m_A is connected to a point mass m_B by a massless rod of length l as shown in the figure. It is observed that the ratio of the moment of inertia of the system about the two axes BB and AA, which is parallel to each other and perpendicular to the rod is $\frac{I_{BB}}{I_{AA}}=3$. The distance of the centre of mass of the system from the mass A is



(A) $(3/4)\, l$ (B) $(2/3)\, l$

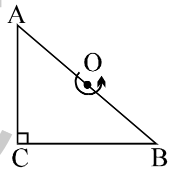
(C) $(1/2)\, l$ (D) $(1/4)\, l$

Q.3 For the same total mass which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body

(A) a disc of radius a (B) a ring of radius a

(C) a square lamina of side 2a (D) four rods forming a square of side 2a

Q.4 Find the moment of inertia of a plate cut in shape of a right angled triangle of mass M, side AC = BC = a about an axis perpendicular to the plane of the plate and passing through the mid point of side AB



(A) $\frac{Ma^2}{12}$ (B) $\frac{Ma^2}{6}$ (C) $\frac{Ma^2}{3}$ (D) $\frac{2Ma^2}{3}$

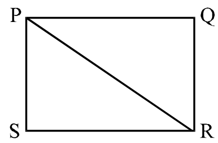
Q.5 Three identical thin rods each of mass m & length l are placed along x, y & z-axis respectively they are placed such that, one end of each rod is at origin 'O'. Then moment of inertia of this system about z-axis is

(A) $\frac{m\ell^2}{3}$ (B) $\frac{2m\ell^2}{3}$ (C) ml^2 (D) $\frac{m\ell^2}{4}$

Q.6 Two rods of equal mass m and length l lie along the x axis and y axis with their centres origin. What is the moment of inertia of both about the line x=y :

(A) $\frac{ml^2}{3}$ (B) $\frac{ml^2}{4}$ (C) $\frac{ml^2}{12}$ (D) $\frac{ml^2}{6}$

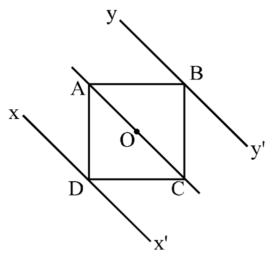
Q.7 Moment of inertia of a rectangular plate about an axis passing through P and perpendicular to the plate is I. Then moment of PQR about an axis perpendicular to the plane of the plate:



(A) about P = I/2 (B) about R = I/2

(C) about P > I/2 (D) about R > I/2

Q.8 Let I_1, I_2 and I_3 be the moment of inertia of a uniform square plate about axes AOC, xDx' and yBy' respectively as shown in the figure. The moments of inertia of the plate $I_1 : I_2 : I_3$ are in the ratio.



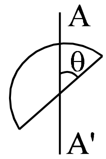
(A) $1 : \frac{1}{7} : \frac{1}{7}$ (B) $1 : \frac{12}{7} : \frac{12}{7}$

(C) $1 : \frac{7}{12} : \frac{7}{12}$ (D) 1 : 7 : 7

Q.9 A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bend in the form of a semicircular arc. Now its moment of inertia through the centre of the semi circular arc and perpendicular to its plane is I_2. The ratio of $I_1 : I_2$ will be ________

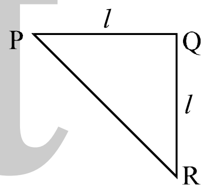
(A) < 1 (B) > 1 (C) = 1 (D) can't be said

Q.10 The moment of inertia of semicircular plate of radius R and mass M about axis AA' in its plane passing through its centre is



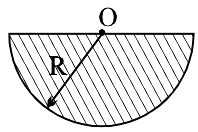
(A) $\frac{MR^2}{2}$ (B) $\frac{MR^2}{4}\cos^2\theta$ (C) $\frac{MR^2}{2}\sin^2\theta$ (D) $\frac{MR^2}{4}$

Q.11 In the triangular sheet given PQ = QR = l. If M is the mass of the sheet. What is the moment of inertial about PR



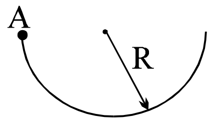
(A) $\frac{Ml^2}{24}$ (B) $\frac{Ml^2}{12}$ (C) $\frac{Ml^2}{6}$ (D) $\frac{Ml^2}{18}$

Q.12 Moment of inertia of a thin semicircular disc (mass = M & radius = R) about an axis through point O and perpendicular to plane of disc, is given by :



(A) $\frac{1}{4}MR^2$ (B) $\frac{1}{2}MR^2$ (C) $\frac{1}{8}MR^2$ (D) MR^2

Q.13 Moment of inertia of a semicircular ring of radius R and mass M ; about an axis passing through A and perpendicular to the plane of the paper is



(A) $\frac{2}{3}MR^2$ (B) MR^2 (C) $\frac{5}{\pi}MR^2$ (D) $2MR^2$

Q.14 A square sheet of edge length L and uniform mass per unit area σ is used to form a hollow cylinder. The moment of inertia of this cylinder about the central axis is

(A) $\frac{2\sigma L^4}{\pi^2}$ (B) $\frac{\sigma L^4}{4\pi^2}$ (C) σL^2 (D) $\frac{\sigma L^4}{3\sqrt{2}\pi^2}$

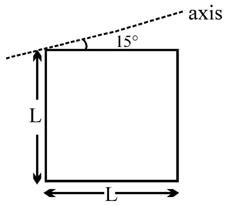
Q.15 A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x, the moment of inertia is given by

$$I = 2x^2 - 12x + 27$$

The x-coordinate of centre of mass is

(A) x = 2 (B) x = 0 (C) x = 1 (D) x = 3

Q.16 A square plate of mass M and edge L is shown in figure. The moment of inertia of the plate about the axis in the plane of plate passing through one of its vertex making an angle 15° from horizontal is.

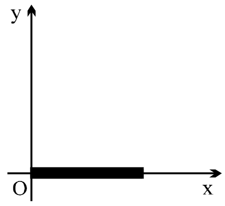


(A) $\frac{ML^2}{12}$ (B) $\frac{11ML^2}{24}$ (C) $\frac{7ML^2}{12}$ (D) none

Q.17 A wire of mass M and length L is bent in the form of a circular ring. The moment of inertia of the ring about its axis is

(A) $\left(\frac{1}{8\pi^2}\right)ML^2$ (B) $(8\pi^2)ML^2$ (C) $\left(\frac{1}{4\pi^2}\right)ML^2$ (D) $(4\pi^2)ML^2$

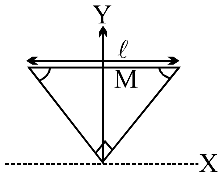
Q.18 The figure shows a uniform rod lying along the x-axis. The locus of all the points lying on the xy-plane, about which the moment of inertia of the rod is same as that about O is



(A) an ellipse (B) a circle (C) a parabola (D) a straight line

Q.19 Consider the following statements
Assertion (A): The moment of inertia of a rigid body reduces to its minimum value as compared to any other parallel axis when the axis of rotation passes through its centre of mass.
Reason (R): The weight of a rigid body always acts through its centre of mass in uniform gravitational field.
Of these statements:
(A) both A and R are true and R is the correct explanation of A
(B) both A and R are true but R is not a correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

Question No. 20 to 23 (4 questions)
The figure shows an isosceles triangular plate of mass M and base L. The angle at the apex is 90°. The apex lies at the origin and the base is parallel to X–axis



Q.20 The moment of inertia of the plate about the z-axis is

(A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{24}$ (C) $\frac{ML^2}{6}$ (D) none of these

Q.21 The moment of inertia of the plate about the x-axis is

(A) $\frac{ML^2}{8}$ (B) $\frac{ML^2}{32}$ (C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{6}$

Q.22 The moment of inertia of the plate about its base parallel to the x-axis is

(A) $\frac{ML^2}{18}$ (B) $\frac{ML^2}{36}$ (C) $\frac{ML^2}{24}$ (D) none of these

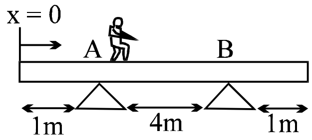
Q.23 The moment of inertia of the plate about the y-axis is

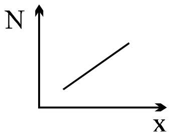
(A) $\frac{ML^2}{6}$ (B) $\frac{ML^2}{8}$ (C) $\frac{ML^2}{24}$ (D) none of these

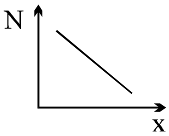
Q.24 A horizontal force F = mg/3 is applied on the upper surface of a uniform cube of mass 'm' and side 'a' which is resting on a rough horizontal surface having $\mu_S = 1/2$. The distance between lines of action of 'mg' and normal reaction 'N' is :

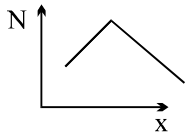
(A) a/2 (B) a/3 (C) a/4 (D) None

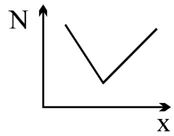
Q.25 A man can move on a horizontal plank supported symmetrically as shown. The variation of normal reaction on support A with distance x of the man from the end of the plank is best represented by :



(A) 

(B) 

(C) 

(D) 

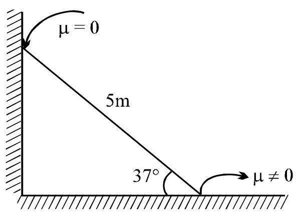
Q.26 A body weighs 6 gms when placed in one pan and 24 gms when placed on the other pan of a false balance. If the beam is horizontal when both the pans are empty, the true weight of the body is :

(A) 13 gm (B) 12 gm (C) 15.5 gm (D) 15 gm

Q.27 A weightless rod is acted on by upward parallel forces of 2N and 4N ends A and B respectively. The total length of the rod AB = 3m. To keep the rod in equilibrium a force of 6N should act in the following manner:

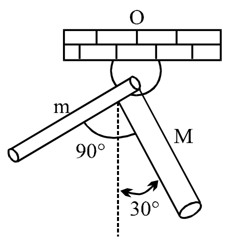
(A) Downwards at any point between A and B.
(B) Downwards at mid point of AB.
(C) Downwards at a point C such that AC = 1m.
(D) Downwards at a point D such that BD = 1m.

Q.28 A 5 m long pole of 3 kg mass is placed against a smooth vertical well as shown in the figure. Under equilibrium condition, if the pole makes an angle of 37° with the horizontal, the frictional force between the pole and horizontal surface is



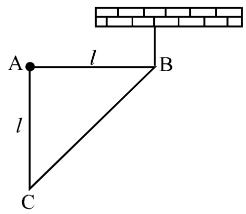
(A) 20 N (B) 30 N
(C) 20 μN (D) 30 μN

Q.29 Two uniform rods of equal length but different masses are rigidly joined to form an L-shaped body, which is then pivoted as shown. If in equilibrium the body is in the shown configuration, ratio M/m will be:



(A) 2 (B) 3 (C) $\sqrt{2}$ (D) $\sqrt{3}$

Q.30 A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support A is:



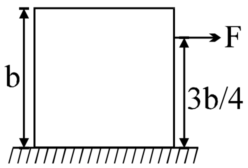
(A) $\dfrac{mg}{3}$ (B) $\dfrac{2mg}{3}$ (C) $\dfrac{mg}{2}$ (D) mg

Q.31 In an experiment with a beam balance on unknown mass m is balanced by two known mass m is balanced by two known masses of 16 kg and 4 kg as shown in figure.

The value of the unknown mass m is

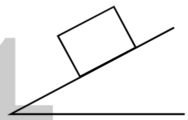
(A) 10 kg (B) 6 kg (C) 8 kg (D) 12 kg

Q.32 A uniform cube of side 'b' and mass M rest on a rough horizontal table. A horizontal force F is applied normal to one of the face at a point, at a height 3b/4 above the base. What should be the coefficient of friction (μ) between cube and table so that is will tip about an edge before it starts slipping?



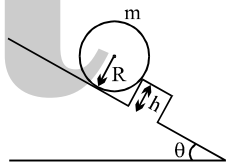
(A) $\mu > \frac{2}{3}$ (B) $\mu > \frac{1}{3}$ (C) $\mu > \frac{3}{2}$ (D) none

Q.33 A homogeneous cubical brick lies motionless on a rough inclined surface. The half of the brick which applies greater pressure on the plane is :



(A) left half (B) right half
(C) both applies equal pressure (D) the answer depend upon coefficient of friction

Q.34 Find minimum height of obstacle so that the sphere can stay in equilibrium.

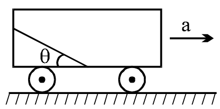


(A) $\frac{R}{1+\cos\theta}$ (B) $\frac{R}{1+\sin\theta}$
(C) R (1– sinθ) (D) R (1 – cosθ)

Q.35 A hallow cone of radius R and height 2R is placed on an inclined plane of inclination θ. If θ is increased gradually, at what value of θ the cone will topple. Assume sufficient friction is present to prevent slipping.

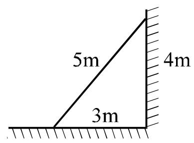
(A) $\tan^{-1}(2/3)$ (B) $\tan^{-1}(3/2)$ (C) $\sin^{-1}(2/3)$ (D) $\cos^{-1}(2/3)$

Q.36 A smooth rod of length ℓ is kept inside a trolley at an angle θ as shown in the figure. What should be the acceleration a of the trolley so that the rod remains in equilibrium with respect to it?



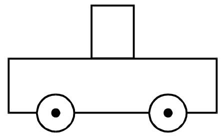
(A) g tanθ (B) g cosθ (C) g sinθ (D) g cotθ

Q.37 A uniform ladder of length 5m is placed against the wall as shown in the figure. If coefficient of friction μ is the same for both the walls, what is the minimum value of μ for it not to slip?



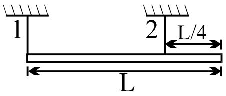
(A) $\mu = \frac{1}{2}$ (B) $\mu = \frac{1}{4}$ (C) $\mu = \frac{1}{3}$ (D) $\mu = \frac{1}{5}$

Q.38 A uniform cylinder rests on a cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, which of the following is the minimum acceleration of the cart needed to cause the cylinder to tip over?



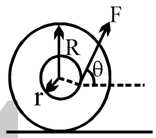
(A) 2 m/s^2
(B) 4 m/s^2
(C) 5 m/s^2
(D) the cylinder would slide before it begins to tip over.

Q.39 A uniform rod of length L and weight W is suspended horizontally by two vertical ropes as shown. The first rope is attached to the left end of the rod while the second rope is attached a distance $L/4$ from the right end. The tension in the second rope is



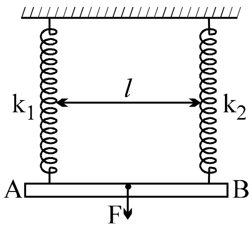
(A) $\frac{W}{2}$ (B) $\frac{W}{4}$ (C) $\frac{W}{3}$ (D) $\frac{2W}{3}$

Q.40 The spool shown in figure is placed on rough horizontal surface and has inner radius r and outer radius R. The angle θ between the applied force and the horizontal can be varied. The critical angle (θ) for which the spool does not roll and remains stationary is given by



(A) $\theta = \cos^{-1}\left(\frac{r}{R}\right)$ (B) $\theta = \cos^{-1}\left(\frac{2r}{R}\right)$ (C) $\theta = \cos^{-1}\sqrt{\frac{r}{R}}$ (D) $\theta = \sin^{-1}\left(\frac{r}{R}\right)$

Q.41 Two light vertical springs with equal natural lengths and spring constants k_1 and k_2 are separated by a distance l. Their upper ends are fixed to the ceiling and their lower ends to the ends A and B of a light horizontal rod AB. A vertical downwards force F is applied at point C on the rod. AB will remain horizontal in equilibrium if the distance AC is



(A) $\frac{l}{2}$ (B) $\frac{l k_1}{k_2 + k_1}$

(C) $\frac{l k_2}{k_1}$ (D) $\frac{l k_2}{k_1 + k_2}$

Q.42 Consider the following statements
Assertion(A) : A cyclist always bends inwards while negotiating a curve
Reason(R) : By bending he lowers his centre of gravity
Of these statements,
(A) both A and R are true and R is the correct explanation of A
(B) both A and R are true but R is not the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

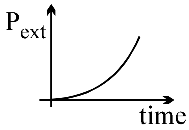
Q.43 A cone of radius r and height h rests on a rough horizontal surface, the coefficient of friction between the cone and the surface being μ. A gradually increasing horizontal force F is applied to the vertex of the cone. The largest value of μ for which the cone will slide before it topples is

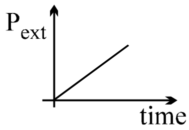
(A) $\mu = \frac{r}{2h}$ (B) $\mu = \frac{2r}{5h}$ (C) $\mu = \frac{r}{h}$ (D) $\mu = \sqrt{\frac{r}{h}}$

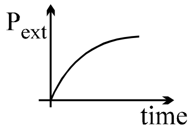
Q.44 A uniform rod of mass m and length l hinged at its end is released from rest when it is in the horizontal position. The normal reaction at the hinge when the rod becomes vertical is :

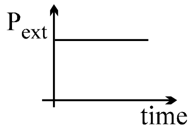
(A) $\frac{Mg}{2}$ (B) $\frac{3Mg}{2}$ (C) $\frac{5Mg}{2}$ (D) 2 Mg

Q.45 A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :

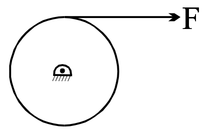
(A) 

(B) 

(C) 

(D) 

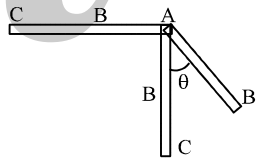
Q.46 A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As the time increases,



(A) its angular velocity increases, but force on hinge remains constant
(B) its angular velocity remains same, but force on hinge increases
(C) its angular velocity increases and force on hinge increases
(D) its angular velocity remains same and force on hinge is constant

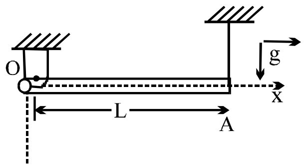
Q.47 The angular momentum of a flywheel having a moment of inertia of 0.4 kg m^2 decreases from 30 to 20 kg m^2/s in a period of 2 second . The average torque acting on the flywheel during this period is :
(A) 10 N . m (B) 2.5 N . m (C) 5 N . m (D) 1.5 N . m

Q.48 A rod hinged at one end is released from the horizontal position as shown in the figure. When it becomes vertical its lower half separates without exerting any reaction at the breaking point. Then the maximum angle 'θ' made by the hinged upper half with the vertical is :



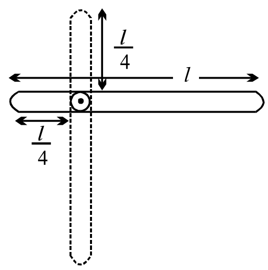
(A) 30° (B) 45° (C) 60° (D) 90°

Q.49 A non uniform rod OA of linear mass density $\lambda = \lambda_0 x$ (λ_0 = const.) is suspended from ceiling with hinge joint O & light string as shown in figure. Find the angular acceleration of rod just after the string is cut.



(A) $\frac{2g}{L}$ (B) $\frac{g}{L}$ (C) $\frac{4g}{3L}$ (D) none of these

Q.50 For the pivoted slender rod of length l as shown in figure, the angular velocity as the bar reaches the vertical position after being released in the horizontal position is

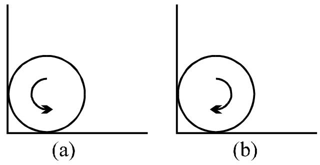


(A) $\sqrt{\frac{g}{l}}$ (B) $\sqrt{\frac{24g}{19l}}$

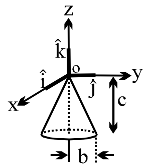
(C) $\sqrt{\frac{24g}{7l}}$ (D) $\sqrt{\frac{4g}{l}}$

Q.51 A sphere is placed rotating with its centre initially at rest in a corner as shown in figure (a) & (b). Coefficient of friction between all surfaces and the sphere is $\frac{1}{3}$. Find the ratio of the frictional force $\frac{f_a}{f_b}$ by ground in situations (a) & (b).

(A) 1
(B) $\frac{9}{10}$
(C) $\frac{10}{9}$
(D) none



Q.52 A solid cone hangs from a frictionless pivot at the origin O, as shown. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors, and a, b, and c are positive constants, which of the following forces F applied to the rim of the cone at a point P results in a torque τ on the cone with a negative component τ_Z?



(A) $F = a\hat{k}$, P is (0,b,–c)
(B) $F = -a\hat{k}$, P is (0,–b,–c)
(C) $F = a\hat{j}$, P is (–b,0,–c)
(D) None

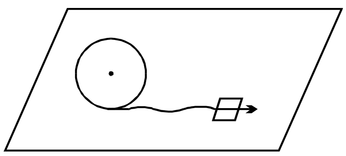
Q.53 A uniform cylinder of mass m can rotate freely about its own axis which is horizontal. A particle of mass m_o hangs from the end of a light string wound round the cylinder which does not slip over it. When the system is allowed to move, the acceleration of the descending mass will be

(A) $\frac{2m_o g}{m + 2m_o}$
(B) $\frac{m_o g}{m + m_o}$
(C) $\frac{2m_o g}{m + m_o}$
(D) $\frac{m_o g}{2m + m_o}$

Q.54 A uniform rod of length l, hinged at the lower end is free to rotate in the vertical plane. If the rod is held vertically in the beginning and then released, the angular acceleration of the rod when it makes an angle of 45° with the horizontal ($I = ml^2/3$)

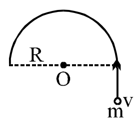
(A) $\frac{3g}{2\sqrt{2}\,l}$
(B) $\frac{6g}{\sqrt{2}\,l}$
(C) $\frac{\sqrt{2}\,g}{l}$
(D) $\frac{2g}{l}$

Q.55 A block of mass m is attached to a pulley disc of equal mass m, radius r by means of a slack string as shown. The pulley is hinged about its centre on a horizontal table and the block is projected with an initial velocity of 5 m/s. Its velocity when the string becomes taut will be



(A) 3 m/s
(B) 2.5 m/s
(C) 5/3 m/s
(D) 10/3 m/s

Q.56 A small bead of mass m moving with velocity v gets threaded on a stationary semicircular ring of mass m and radius R kept on a horizontal table. The ring can freely rotate about its centre. The bead comes to rest relative to the ring. What will be the final angular velocity of the system?



(A) v/R
(B) 2v/R
(C) v/2R
(D) 3v/R

Q.57 A small object is attached to a light string which passes through a hollow tube. The tube is held by one hand and the string by the other. The object is stet into rotation in a circle of radius r_1. The string is then pulled down, shortening the radius of the circle to r_2. The ratio of the new kinetic energy to original kinetic energy is

(A) $\frac{r_1}{r_2}$ (B) 1 (C) $\left(\frac{r_1}{r_2}\right)^2$ (D) $\left(\frac{r_2}{r_1}\right)^2$

Q.58 A man, sitting firmly over a rotating stool has his arms streched. If he folds his arms, the work done by the man is

(A) zero (B) positive

(C) negative (D) may be positive or negative.

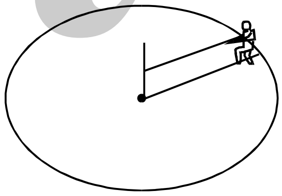
Q.59 A particle of mass 2 kg located at the position $(\hat{i}+\hat{j})$ m has a velocity $2(+\hat{i}-\hat{j}+\hat{k})$ m/s. Its angular momentum about z-axis in kg-m^2/s is:

(A) zero (B) +8 (C) 12 (D) –8

Q.60 A particle is moving in a circular orbit of radius r_1 with an angular velocity ω_1. It jumps to another circular orbit of radius r_2 and attains an angular velocity ω_2. If $r_2 = 0.5\, r_1$ and assuming that no external torque is applied to the system, then the angular velocity ω_2, is given by :

(A) $\omega_2 = 4\,\omega_1$ (B) $\omega_2 = 3\,\omega_1$ (C) $\omega_2 = 2\,\omega_1$ (D) $\omega_2 = \omega_1$

Q.61 A man is sitting in a smooth groove on a horizontal circular table at the edge by holding a rope joined to the centre. The moment of inertia of table is I. Mass of man = M. Man now pulls the rope so that he comes to the centre. The angular velocity of the table :



(A) must increase (B) may increase

(C) must decrease (D) may decrease

Q.62 A particle of mass m is projected with a velocity u making an angle 45° with the horizontal. The magnitude of the torque due to weight of the projectile, when the particle is at its maximum height, about the point of projectile

(A) mu^2 (B) $\frac{3}{4}\,mu^2$ (C) $\frac{1}{4}\,mu^2$ (D) $\frac{1}{2}\,mu^2$

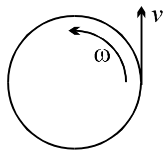
Q.63 In absence of torque the rotational frequency of a body changes from 1 cy/sec to 16 cy/sec, then ratio of radius of gyration in two cases will be :

(A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 4 : 1

Q.64 A particle of mass m is rotating in a plane is a circular path of radius r, its angular momentum is L. The centripital force acting on the particle is :

(A) $\frac{L^2}{mr}$ (B) $\frac{L^2 m}{r}$ (C) $\frac{L^2}{mr^2}$ (D) $\frac{L^2}{mr^3}$

Q.65 A child with mass m is standing at the edge of a disc with moment of inertia I, radius R, and initial angular velocity ω. See figure given below. The child jumps off the edge of the disc with tangential velocity v with respect to the ground. The new angular velocity of the disc is



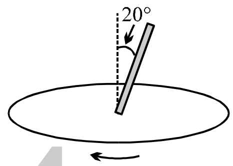
(A) $\sqrt{\dfrac{I\omega^2 - mv^2}{I}}$ (B) $\sqrt{\dfrac{(I+mR^2)\omega^2 - mv^2}{I}}$

(C) $\dfrac{I\omega - mvR}{I}$ (D) $\dfrac{(I+mR^2)\omega - mvR}{I}$

Q.66 A particle of mass 0.5 kg is rotating in a circular path of radius 2m and centrepetal force on it is 9 Newtons. Its angular momentum (in J·sec) is:
(A) 1.5 (B) 3 (C) 6 (D) 18

Question No. 67 & 68 (2 questions)

A uniform rod is fixed to a rotating turntable so that its lower end is on the axis of the turntable and it makes an angle of 20° to the vertical. (The rod is thus rotating with uniform angular velocity about a vertical axis passing through one end.) If the turntable is rotating clockwise as seen from above.



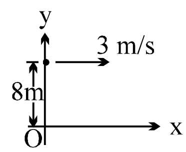
Q.67 What is the direction of the rod's angular momentum vector (calculated about its lower end)?
(A) vertically downwards (B) down at 20° to the horizontal
(C) up at 20° to the horizontal (D) vertically upwards

Q.68 Is there a torque acting on it, and if so in what direction?
(A) yes, vertically (B) yes, horizontally
(C) yes at 20° to the horizontal (D) no

Q.69 A straight rod of length L is released on a frictionless horizontal floor in a vertical position. As it falls + slips, the distance of a point on the rod from the lower end, which follows a quarter circular locus is
(A) L/2 (B) L/4 (C) L/8 (D) None

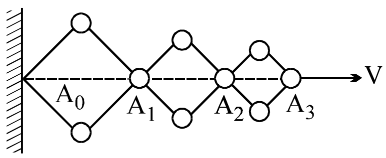
Q.70 Two particles of mass m each are fixed at the opposite ends of a massless rod of length 5m which is oriented vertically on a smooth horizontal surface and released. Find the displacement of the lower mass on the ground when the rod makes an angle of 37° with the vertical.
(A) 1.5 m (B) 2 m (C) 2.5 m (D) 3.5 m

Q.71 A particle starts from the point (0m, 8m) and moves with uniform velocity of $3\hat{i}$ m/s. After 5 seconds, the angular velocity of the particle about the origin will be :



(A) $\dfrac{8}{289}$ rad/s (B) $\dfrac{3}{8}$ rad/s (C) $\dfrac{24}{289}$ rad/s (D) $\dfrac{8}{17}$ rad/s

Q.72 A hinged construction consists of three rhombs with the ratio of sides 5:3:2. Vertex A_3 moves in the horizontal direction at a velocity v. Velocity of A_2 is

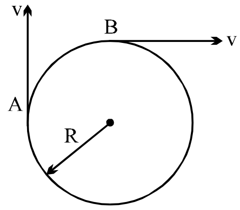


(A) 2.5 V (B) 1.5 V

(C) $\dfrac{2}{3}$ V (D) 0.8 V

Q.73 A wheel of radius r rolling on a straight line, the velocity of its centre being v. At a certain instant the point of contact of the wheel with the grounds is M and N is the highest point on the wheel (diametrically opposite to M). The incorrect statement is:
(A) The velocity of any point P of the wheel is proportional to MP.
(B) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v.
(C) The point of contact M is instantaneously at rest.
(D) The velocities of any two parts of the wheel which are equidistant from centre are equal.

Q.74 Two points of a rigid body are moving as shown. The angular velocity of the body is:



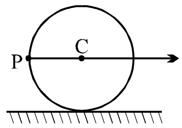
(A) $\frac{\upsilon}{2R}$ (B) $\frac{\upsilon}{R}$ (C) $\frac{2\upsilon}{R}$ (D) $\frac{2\upsilon}{3R}$

Q.75 There is rod of length l. The velocities of its two ends are v_1 and v_2 in opposite directions normal to the rod. The distance of the instantaneous axis of rotation from v_1 is:

(A) zero (B) $\frac{v_2}{v_1+v_2}l$ (C) $\frac{v_1 l}{v_1+v_2}$ (D) $l/2$

Q.76 A thin rod of length L is placed vertically on a frictionless horizontal floor and released with a negligible push to allow it to fall. At any moment, the rod makes an angle θ with the vertical. If the center of mass has acceleration = A, and the rod an angular acceleration = α at initial moment, then
(A) $A = (L\alpha).\sin\theta$ (B) $A/2 = (L\alpha).\sin\theta$ (C) $2A = (L\alpha).\sin\theta$ (D) $A = L\alpha$

Q.77 A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is



(A) zero (B) 45° (C) 135° (D) $\tan^{-1}(1/2)$

Q.78 A ladder of length L is slipping with its ends against a vertical wall and a horizontal floor. At a certain moment, the speed of the end in contact with the horizontal floor is v and the ladder makes an angle $\alpha = 30°$ with the horizontal. Then the speed of the ladder's center must be
(A) $2v/\sqrt{3}$ (B) v/2 (C) v (D) None

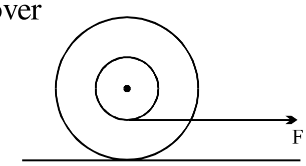
Q.79 In the previous question, if dv/dt = 0, then the angular acceleration of the ladder when $\alpha = 45°$ is
(A) $2v^2/L^2$ (B) $v^2/2L^2$ (C) $\sqrt{2}[v^2/L^2]$ (D) None

Q.80 A ring of radius R rolls without sliding with a constant velocity. The radius of curvature of the path followed by any particle of the ring at the highest point of its path will be
(A) R (B) 2R (C) 4R (D) None

Q.81 Two spheres are rolling with same velocity (for their C. M.) their ratio of kinetic energy is 2 : 1 & radius ratio is 2 : 1, their mass ratio will be :
(A) 2 : 1 (B) 4 : 1 (C) 8 : 1 (D) $2\sqrt{2} : 1$

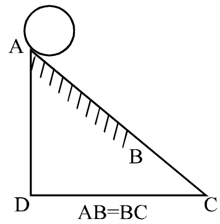
Q.82 Two identical circular loops are moving with same kinetic energy one rolls & other slides. The ratio of their speed is :
(A) 2 : 3 (B) $2 : \sqrt{2}$ (C) $\sqrt{2} : 2$ (D) $\sqrt{5} : \sqrt{3}$

Q.83 Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F as shown in fig. then in case of pure rolling



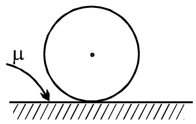
(A) Thread unwinds, spool rotates anticlockwise and friction act leftwards
(B) Thread winds, spool rotates clockwise and friction acts leftwards
(C) Thread winds, spool moves to the right and friction act rightwards
(D) Thread winds, spool moves to the right and friction does not come into existence.

Q.84 Portion AB of the wedge shown in figure is rough and BC is smooth. A solid cylinder rolls without slipping from A to B. The ratio of translational kinetic energy to rotational kinetic energy, when the cylinder reaches point C is :



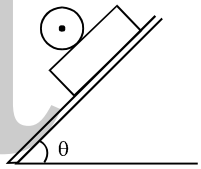
(A) 3/4 (B) 5
(C) 7/5 (D) 8/3

Q.85 A disc of radius r is rotating about its centre with an angular speed ω_0. It is gently placed on a rough horizontal surface. After what time it will be in pure rolling ?



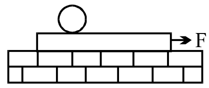
(A) $\frac{\omega_0 r}{2\mu g}$ (B) $\frac{\omega_0 r}{3\mu g}$ (C) $\frac{\omega_0 r}{\mu g}$ (D) $\frac{3}{2}\frac{\omega_0 r}{\mu g}$

Q.86 A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between sphere and plank. If plank and sphere are released from rest, the frictional force on sphere is:

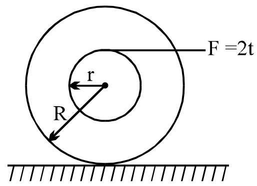


(A) up the plane (B) down the plane (C) horizontal (D) zero

Q.87 A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F. If sphere does not slip over the plank. Which of the following is incorrect.
(A) Acceleration of the centre of sphere is less than that of the plank.
(B) Work done by friction acting on the sphere is equal to its total kinetic energy.
(C) Total kinetic energy of the system is equal to work done by the force F
(D) None of the above

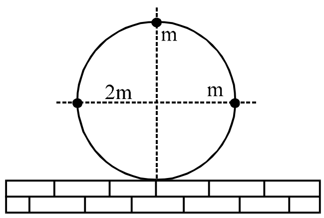


Q.88 A time varying force F = 2t is applied on a spool rolling as shown in figure. The angular momentum of the spool at time t about bottommost point is:



(A) $\frac{r^2t^2}{R}$ (B) $\frac{(R+r)^2}{r}t^2$ (C) $(R+r)t^2$ (D) data is insufficient

Q.89 A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0. The kinetic energy of the system is: (Slipping is absent)

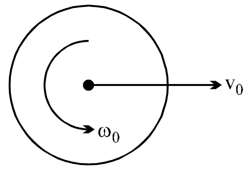


(A) $6\,mv_0^2$ (B) $12\ mv_0^2$
(C) $4\ mv_0^2$ (D) $8\ mv_0^2$

Q.90 The linear speed of a uniform spherical shell after rolling down an inclined plane of vertical height h from rest, is:

(A) $\sqrt{\frac{10gh}{7}}$ (B) $\sqrt{\frac{4gh}{5}}$ (C) $\sqrt{\frac{6gh}{5}}$ (D) $\sqrt{2gh}$

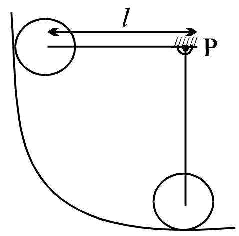
Q.91 A uniform sphere of radius R is placed on a rough horizontal surface and given a linear velocity v_0 angular velocity ω_0 as shown. The sphere comes to rest after moving some distance to the right. It follows that:



(A) $v_0 = \omega_0 R$ (B) $2v_0 = 5\omega_0 R$
(C) $5v_0 = 2\omega_0 R$ (D) $2v_0 = \omega_0 R$

Q.92 A solid uniform disk of mass m rolls without slipping down a fixed inclined plane with an acceleration a. The frictional force on the disk due to surface of the plane is :
(A) 2 ma (B) 3/2 ma (C) ma (D) 1/2 ma

Q.93 A sphere of mass M and radius R is attached by a light rod of length l to a point P. The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. the angular momentum of the system about P when the rod becomes vertical is :



(A) $M\sqrt{\frac{10}{7}gl}\ [l+R]$ (B) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{2}{5}R\right]$

(C) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{7}{5}R\right]$ (D) none of the above

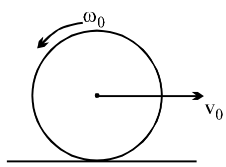
Q.94 A body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body. If the body rolls purely on the surface, its shape can be :
(A) thin pipe (B) uniform cylinder (C) uniform sphere (D) thin spherical shell

Q.95 A slender uniform rod of length ℓ is balanced vertically at a point P on a horizontal surface having some friction. If the top of the rod is displaced slightly to the right, the position of its centre of mass at the time when the rod becomes horizontal :
(A) lies at some point to the right of P (B) lies at some point to the left of P
(C) must be $\ell/2$ to the right of P (D) lies at P

Q.96 A solid sphere with a velocity (of centre of mass) v and angular velocity ω is gently placed on a rough horizontal surface. The frictional force on the sphere:
(A) must be forward (in direction of v) (B) must be backward (opposite to v)
(C) cannot be zero (D) none of the above

Q.97 A cylinder is pure rolling up an incline plane. It stops momentarily and then rolls back. The force of friction
(A) on the cyclinder is zero throughout the journey
(B) is directed opposite to the velocity of the centre of mass throughout the journey
(C) is directed up the plane throughout the journey
(D) is directed down the plane throughout the journey

Q.98 A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is

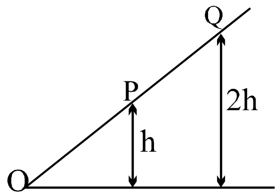


(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

Q.99 On a solid sphere lying on a horizontal surface a force F is applied at a height of R/2 from the centre of mass. The initial acceleration of a point at the top of the sphere is (there is no slipping at any point)

(A) $\frac{15F}{7M}$ (B) $\frac{15F}{14M}$ (C) $\frac{30F}{7M}$ (D) $\frac{F}{M}$

Q.100 A ball rolls down an inclined plane, figure. The ball is first released from rest from P and then later from Q. Which of the following statement is/are correct?



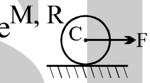
(i) The ball takes twice as much time to roll from Q to O as it does to roll from P to O.
(ii) The acceleration of the ball at Q is twice as large as the acceleration at P.
(iii) The ball has twice as much K.E. at O when rolling from Q as it does when rolling from P.

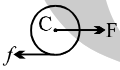
(A) i, ii only (B) ii, iii only (C) i only (D) iii only

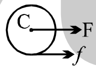
Question No. 101 to 106 (6 questions)

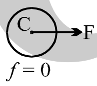
In the following problems, indicate the correct direction of friction force acting on the cylinder, which is pulled on a rough surface by a constant force F.

Q.101 A cylinder of mass M and radius R is pulled horizontally by a force F. The friction force can be given by which of the following diagrams



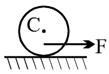
(A) 

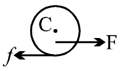
(B) 

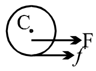
(C) 

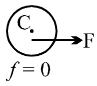
(D) cannot be interpreted

Q.102 A cylinder is pulled horizontally by a force F acting at a point below the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams



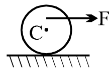
(A) 

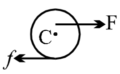
(B) 

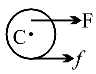
(C) 

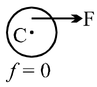
(D) cannot be interpreted

Q.103 A cylinder is pulled horizontally by a force F acting at a point above the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams



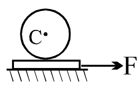
(A) 

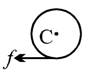
(B) 

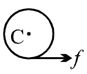
(C) 

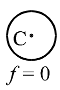
(D) cannot be interpreted

Q.104 A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F. The friction force can be given by which of the following diagrams



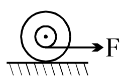
(A) 

(B) 

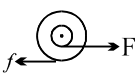
(C) 

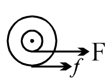
(D) canot be interpreted

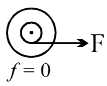
Q.105 A spool is pulled horizontally by a constant force F below the centre of mass.



The friction force can be given by which of the following diagrams

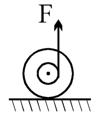
(A) 

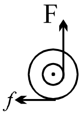
(B) 

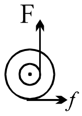
(C) 

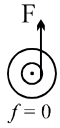
(D) cannot be interpreted

Q.106 A spool is pulled vertically by a constant force F (< Mg) as shown in figure
The friction force can be given by which of the following diagrams



(A) 

(B) 

(C) 

(D) cannot be interpreted

Q.107 One ice skater of mass m moves with speed 2v to the right, while another of the same mass m moves with speed v toward the left, as shown in figure I. Their paths are separated by a distance b. At t = 0, when they are both at x = 0, they grasp a pole of length b and negligible mass. For t > 0, consider the system as a rigid body of two masses m separated by distance b, as shown in figure II. Which of the following is the correct formula for the motion after t = 0 of the skater initially at y = b/2?

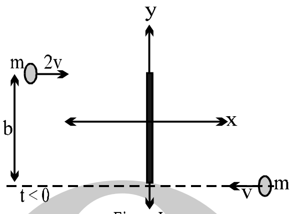


Figure I

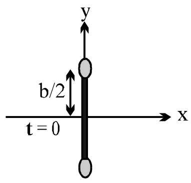


Figure II

(A) $x = 2vt, y = b/2$

(B) $x = vt + 0.5b\sin(3vt/b), y = 0.5b\cos(3vt/b)$

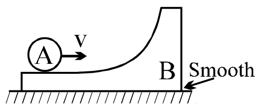
(C) $x = 0.5vt + 0.5b\sin(3vt/b), y = 0.5b\cos(3vt/b)$

(D) $x = 0.5vt + 0.5b\sin(6vt/b), y = 0.5b\cos(6vt/b)$

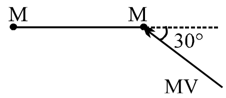
Q.108 Starting from the rest, at the same time, a ring, a coin and a solid ball of same mass roll down an incline without slipping .The ratio of their translational kinetic energies at the bottom will be
(A) 1 : 1 : 1 (B) 10 : 5 : 4 (C) 21 : 28 : 30 (D) None

Q.109 In the figure shown a ring A is initially rolling without sliding with a velocity v on the horizontal surface of the body B (of same mass as A). All surfaces are smooth. B has no initial velocity. What will be the maximum height reached by A on B.

(A) $\frac{3v^2}{4g}$ (B) $\frac{v^2}{4g}$ (C) $\frac{v^2}{2g}$ (D) $\frac{v^2}{3g}$



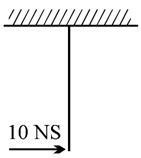
Q.110 Two equal masses each of mass M are joined by a massless rod of length L. Now an impulse MV is given to the mass M making an angle of 30° with the length of the rod. The angular velોctiy of the rod just after imparting the impulse is



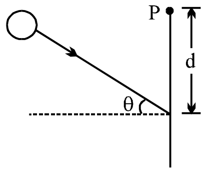
(A) $\frac{v}{L}$ (B) $\frac{2v}{L}$

(C) $\frac{v}{2L}$ (D) none of these.

Q.111 A thin uniform straight rod of mass 2 kg and length 1 m is free to rotate about its upper end when at rest. It receives an impulsive blow of 10 Ns at its lowest point, normal to its length as shown in figure. The kinetic energy of rod just after impact is
(A) 75 J (B) 100 J (C) 200 J (D) none

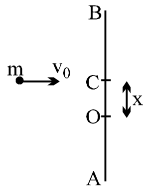


Q.112 A ball of mass m moving with velocity v, collide with the wall elastically as shown in the figure. After impact the change in angular momentum about P is:



(A) 2 *mvd* (B) 2 *mvd* cosθ (C) 2 *mvd* sin θ (D) zero

Q.113 A uniform rod AB of length L and mass M is lying on a smooth table. A small particle of mass m strike the rod with a velocity v_0 at point C at a distance x from the centre O. The particle comes to rest after collision. The value of x, so that point A of the rod remains ststionary just after collision is:

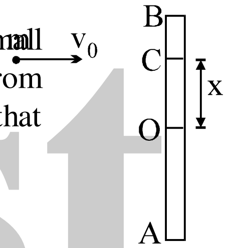


(A) L/3 (B) L/6 (C) L/4 (D) L/12

Q.114 Two particles of equal mass m at A and B are connected by a rigid light rod AB lying on a smooth horizontal table. An impulse J is applied at A in the plane of the table and perpendicular at AB. Then the velocity of particle at A is:

(A) $\frac{J}{2m}$ (B) $\frac{J}{m}$ (C) $\frac{2J}{m}$ (D) zero

Q.115 A uniform rod AB of length L and mass M is lying on a smooth table. A small particle of mass *m* strike the rod with a velocity v_0 at point C a distance x from the centre O. The particle comes to rest after collision. The value of x, so that point A of the rod remains stationary just after collision, is :



(A) L/3 (B) L/6

(C) L/4 (D) L/12

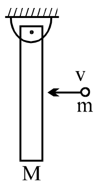
Q.116 A uniform rod of mass M has an impulse applied at right angles to one end. If the other end begins to move with speed V, the magnitude of the impulse is

(A) MV (B) $\frac{MV}{2}$ (C) 2MV (D) $\frac{2MV}{3}$

Q.117 A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse J is applied to the end B, perpendicular to the rod in the horizontal direction. Speed of particle P at a distance $\frac{l}{6}$ from the centre towards A of the rod after time $t = \frac{\pi m l}{12J}$ is

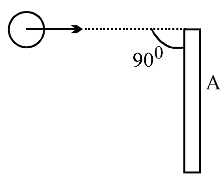
(A) $2\frac{J}{m}$ (B) $\frac{J}{\sqrt{2}m}$ (C) $\frac{J}{m}$ (D) $\sqrt{2}\frac{J}{m}$

Q.118 A uniform rod of mass M is hinged at its upper end. A particle of mass m moving horizontally strikes the rod at its mid point elastically. If the particle comes to rest after collision find the value of M/m =?



(A) 3/4 (B) 4/3 (C) 2/3 (D) none

Q.119 A thin rod of mass M and length L is struck at one end by a ball of clay of mass m, moving with speed v as shown in figure. The ball sticks to the rod. After the collision, the angular momentum of the clay-rod system about A, the midpoint of the rod, is



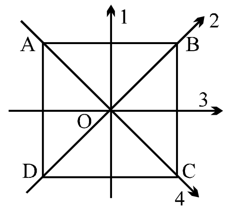
(A) $\left(m+\frac{M}{3}\right)\left(\frac{vL}{2}\right)$ (B) $\left(m+\frac{M}{12}\right)\left(\frac{vL}{2}\right)$

(C) $\frac{mvL}{2}$ (D) mvL

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 ABCD is a square plate with centre O. The moments of inertia of the plate about the perpendicular axis through O is I and about the axes 1, 2, 3 & 4 are I_1, I_2, I_3 & I_4 respectively. It follows that :



(A) $I_2 = I_3$ (B) $I = I_1 + I_4$
(C) $I = I_2 + I_4$ (D) $I_1 = I_3$

Q.2 A rod of weight w is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at a distance x from A.

(A) the normal reaction at A is $\frac{wx}{d}$ (B) the normal reaction at A is $\frac{w(d-x)}{d}$

(C) the normal reaction at B is $\frac{wx}{d}$ (D) the normal reaction at B is $\frac{w(d-x)}{d}$

Q.3 A block with a square base measuring axa and height h, is placed on an inclined plane. The coefficient of friction is m. The angle of inclination (θ) of the plane is gradually increased. The block will

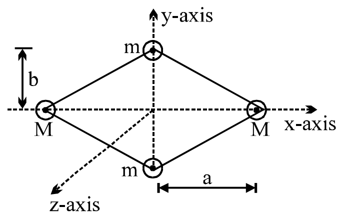
(A) topple before sliding if $\mu > \frac{a}{h}$ (B) topple before sliding if $\mu < \frac{a}{h}$

(C) slide before toppling if $\mu > \frac{a}{h}$ (D) slide before toppling if $\mu < \frac{a}{h}$

Q.4 A body is in equilibrium under the influence of a number of forces. Each force has a different line of action. The minimum number of forces required is
(A) 2, if their lines of action pass through the centre of mass of the body.
(B) 3, if their lines of action are not parallel.
(C) 3, if their lines of action are parallel.
(D) 4, if their lines of action are parallel and all the forces have the same magnitude.

Q.5 If a person sitting on a rotating stool with his hands outstretched, suddenly lowers his hands, then his
(A) Kinetic energy will decrease (B) Moment of inertia will decrease
(C) Angular momentum will increase (D) Angular velocity will remain constant

Q.6 Four point masses are fastened to the corners of a frame of negligible mass lying in the xy plane. Let w be the angular speed of rotation. Then



(A) rotational kinetic energy associated with a given angular speed depends on the axis of rotation.
(B) rotational kinetic energy about y-axis is independent of m and its value is $Ma^2\omega^2$.
(C) rotational kinetic energy about z-axis depends on m and its value is $(Ma^2 + mb^2)\omega^2$.
(D) rotational kinetic energy about z-axis is independent of m and its value is $Mb^2\omega^2$.

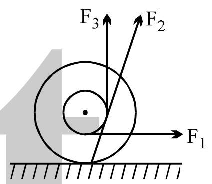
Q.7 A block of mass m moves on a horizontal rough surface with initial velocity v. The height of the centre of mass of the block is h from the surface. Consider a point A on the surface.
(A) angular momentum about A is mvh initially
(B) the velocity of the block decreases at time passes.
(C) torque of the forces acting on block is zero about A
(D) angular mometum is not conserved about A.

Q.8 A paritcle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground.
(A) Angular momentum of the particle about O is increasing .
(B) Torque of the gravitational force on the particle about O is decreasing.
(C) The moment of inertia of the particle about O is decreasing .
(D) The angular velocity of the particle about O is increasing.

Q.9 A man spinning in free space changes the shape of his body, eg. by spreading his arms or curling up. By doing this, he can change his
(A) moment of inertia (B) angular momentum (C) angular velocity (D) rotational kinetic energy

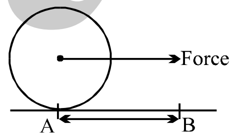
Q.10 A ring rolls without slipping on the ground. Its centre C moves with a constant speed u. P is any point on the ring. The speed of P with respect to the ground is ν.
(A) $0 \le \nu \le 2u$
(B) $\nu = u$, if CP is horizontal
(C) $\nu = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.
(D) $\nu = \sqrt{2}u$, if CP is horizontal

Q.11 A yo-yo is resting on a rough horizontal table. Forces F_1, F_2 and F_3 are applied separately as shown. The correct statement is



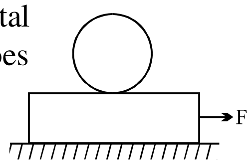
(A) when F_3 is applied the centre of mass will move to the right.
(B) when F_2 is applied the centre of mass will move to the left.
(C) when F_1 is applied the centre of mass will move to the right.
(D) when F_2 is applied the centre of mass will move to the right.

Q.12 A disc of circumference s is at rest at a point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping, and the surface is smooth to the right of B. AB = s. The disc moves from A to B in time T. To the right of B,



(A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged
(B) linear acceleration of the disc will increase
(C) the disc will make one rotation in time T/2
(D) the disc will cover a distance greater than s in further time T.

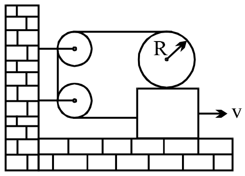
Q.13 A plank with a uniform sphere placed on it, rests on a smooth horizontal plane. Plank is pulled to right by a constant force F. If the sphere does not slip over the plank.



(A) acceleration of centre of sphere is less than that of the plank
(B) acceleration of centre of sphere is greater than the plank because friction acts rightward on the sphere
(C) acceleration of the centre of sphere may be towards left.
(D) acceleration of the centre of sphere relative to plank may be greater than that of the plank relative to floor.

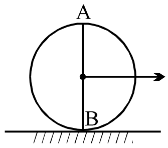
Q.14 A hollow sphere of radius R and mass m is fully filled with water of mass m. It is rolled down a horizontal plane such that its centre of mass moves with a velocity v. If it purely rolls
(A) Kinetic energy of the sphere is $\frac{5}{6}mv^2$ (B) Kinetic energy of the sphere is $\frac{4}{5}mv^2$
(C) Angular momentum of the sphere about a fixed point on ground is $\frac{8}{3}mvR$
(D) Angular momentum of the sphere about a fixed point on ground is $\frac{14}{5}mvR$

Q.15 In the figure shown, the plank is being pulled to the right with a constant speed v. If the cylinder does not slip then:



(A) the speed of the centre of mass of the cylinder is 2v.
(B) the speed of the centre of mass of the cylinder is zero.
(C) the angular velocity of the cylinder is v/R.
(D) the angular velocity of the cylinder is zero.

Q.16 A uniform disc is rolling on a horizontal surface. At a certain instant B is the point of contact and A is at height 2R from ground, where R is radius of disc.



(A) The magnitude of the angular momentum of the disc about B is thrice that about A.
(B) The angular momentum of the disc about A is anticlockwise.
(C) The angular momentum of the disc about B is clockwise
(D) The angular momentum of the disc about A is equal to that about B.

Q.17 If a cylinder is rolling down the incline with sliding.
(A) after some time it may start pure rolling
(B) after sometime it will start pure rolling
(C) it may be possible that it will never start pure rolling
(D) none of these

Q.18 Which of the following statements are correct.
(A) friction acting on a cylinder without sliding on an inclined surface is always upward along the incline irrespective of any external force acting on it.
(B) friction acting on a cylinder without sliding on an inclined surface is may be upward may be downwards depending on the external force acting on it.
(C) friction acting on a cylinder rolling without sliding may be zero depending on the external force acting on it.
(D) nothing can be said exactly about it as it depends on the friction coefficient on inclined plane.

Question No. 19 to 21 (3 questions)

A cylinder and a ring of same mass M and radius R are placed on the top of a rough inclined plane of inclination θ. Both are released simultaneously from the same height h.

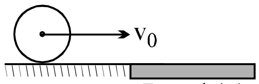
Q.19 Choose the correct statement(s) related to the motion of each body
(A) The friction force acting on each body opposes the motion of its centre of mass
(B) The friction force provides the necessary torque to rotate the body about its centre of mass
(C) Without friction none of the two bodies can roll
(D) The friction force ensures that the point of contact must remain stationary

Q.20 Identify the correct statement(s)
(A) The friction force acting on the cylinder may be more than that acting on the ring
(B) The friction force acting on the ring may be more than that acting on the cylinder
(C) If the friction is sufficient to roll the cylinder then the ring will also roll
(D) If the friction is sufficient to roll the ring then the cylinder will also roll

Q.21 When these bodies roll down to the foot of the inclined plane, then
(A) the mechanical energy of each body is conserved
(B) the velocity of centre of mass of the cylinder is $2\sqrt{\frac{gh}{3}}$
(C) the velocity of centre of mass of the ring is $\sqrt{gh}$
(D) the velocity of centre of mas of each body is $\sqrt{2gh}$

Question No. 22 to 25 (4 questions)

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ, as shown in figure.



Q.22 Choose the correct statement(s)

(A) As the ring enters on the rough surface, the limiting friction force acts on it
(B) The direction of friction is opposite to the direction of motion
(C) The friction force accelerates the ring in the clockwise sense about its centre of mass
(D) As the ring enters on the rough surface it starts rolling

Q.23 Choose the correct statement(s)

(A) The momentum of the ring is conserved
(B) The angular momentum of the ring is conserved about its centre of mass
(C) The angular momentum of the ring conserved about any point on the horizontal surface
(D) The mechanical energy of the ring is conserved

Q.24 Choose the correct statement(s)

(A) The ring starts its rolling motion when the centre of mas stationary
(B) The ring starts rolling motion when the point of contact becomes stationary
(C) The time after which the ring starts rolling is $\frac{v_0}{2\mu g}$
(D) The rolling velocity is $\frac{v_0}{2}$

Q.25 Choose the correct alternative(s)

(A) The linear distance moved by the centre of mass before the ring starts rolling is $\frac{3v_0^2}{8\mu g}$
(B) The net work done by friction force is $-\frac{3}{8}mv_0^2$
(C) The loss is kinetic energy of the ring is $\frac{mv_0^2}{4}$
(D) The gain in rotational kinetic energy is $+\frac{mv_0^2}{8}$

Q.26 Consider a sphere of mass 'm' radius 'R' doing pure rolling motion on a rough surface having velocity $\vec{v}_0$ as shown in the Fig. It makes an elastic impact with the smooth wall and moves back and starts pure rolling after some time again.

(A) Change in angular momentum about 'O' in the entire motion equals $2mv_0R$ in magnitude.
(B) Moment of impulse provided by the wall during impact about O equals $2mv_0R$ in magnitude.
(C) Final velocity of ball will be $\frac{3}{7}\vec{v}_0$
(D) Final velocity of ball will be $-\frac{3}{7}\vec{v}_0$

ANSWER KEY

ONLY ONE OPTION IS CORRECT

Q.1	B	Q.2	D	Q.3	D	Q.4	B	Q.5	B	Q.6	C	Q.7	C
Q.8	D	Q.9	A	Q.10	D	Q.11	B	Q.12	B	Q.13	D	Q.14	B
Q.15	D	Q.16	B	Q.17	C	Q.18	B	Q.19	B	Q.20	C	Q.21	A
Q.22	C	Q.23	C	Q.24	B	Q.25	B	Q.26	B	Q.27	D	Q.28	A
Q.29	D	Q.30	B	Q.31	C	Q.32	A	Q.33	A	Q.34	D	Q.35	B
Q.36	D	Q.37	C	Q.38	B	Q.39	D	Q.40	A	Q.41	D	Q.42	B
Q.43	C	Q.44	C	Q.45	B	Q.46	A	Q.47	C	Q.48	C	Q.49	C
Q.50	C	Q.51	B	Q.52	C	Q.53	A	Q.54	A	Q.55	D	Q.56	C
Q.57	C	Q.58	B	Q.59	D	Q.60	A	Q.61	B	Q.62	D	Q.63	D
Q.64	D	Q.65	D	Q.66	C	Q.67	B	Q.68	B	Q.69	B	Q.70	A
Q.71	C	Q.72	D	Q.73	D	Q.74	B	Q.75	C	Q.76	C	Q.77	B
Q.78	C	Q.79	A	Q.80	C	Q.81	A	Q.82	C	Q.83	B	Q.84	B
Q.85	B	Q.86	D	Q.87	D	Q.88	C	Q.89	A	Q.90	C	Q.91	C
Q.92	D	Q.93	D	Q.94	A	Q.95	A	Q.96	D	Q.97	C	Q.98	A
Q.99	A	Q.100	D	Q.101	A	Q.102	A	Q.103	D	Q.104	B	Q.105	A
Q.106	A	Q.107	C	Q.108	C	Q.109	B	Q.110	C	Q.111	A	Q.112	B
Q.113	B	Q.114	B	Q.115	B	Q.116	B	Q.117	D	Q.118	A	Q.119	C

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	A,B,C,D	Q.2	B,C	Q.3	A,D	Q.4	B,C,D
Q.5	B	Q.6	A,B,C	Q.7	A,B,D	Q.8	A,C,D
Q.9	A,C,D	Q.10	A,C,D	Q.11	C	Q.12	B,C,D
Q.13	A	Q.14	C	Q.15	B,C	Q.16	A,B,C
Q.17	A,C	Q.18	B,C	Q.19	A,B,C,D	Q.20	B,D
Q.21	A,B,C	Q.22	A,B,C	Q.23	C	Q.24	B,C,D
Q.25	A,C,D	Q.26	A,B,D				

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PHYSICS

GRAVITATION

QUESTIONS FOR SHORT ANSWER

Q.1 Two satellites move along a circular orbit in the same direction at a small distance from each other. A container has to be thrown from the first satellite onto the second one. When will the container reach the second satellite faster: if its is thrown in the direction of motion of the first satellite or in the opposite direction ? The velocity of the container with respect to the satellite u is much less than that of the satellite v.

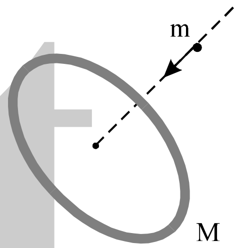
Q.2 After Sputnik I was put into orbit, it was said that it would not return to Earth but would burn up in its descent. Considering the fact that it did not burn up in its ascent, how is this possible ?

Q.3 An artificial satellite is in a circular orbit about the Earth. How will its orbit change if one of its rockets is momentarily fired. (a) towards earth, (b) away from the Earth, (c) in a forward direction, (d) in a backward direction, and (e) at right angles to the plane of the orbit?

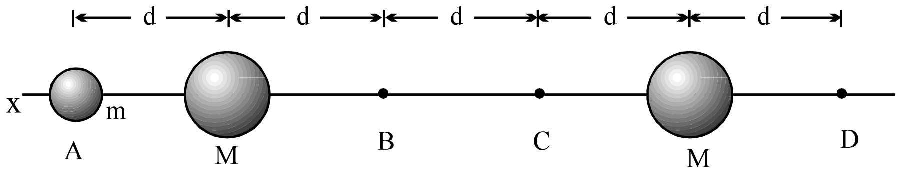
Q.4 A stone is dropped along the centre of a deep vertical mine shaft. Assume no air resistance but consider the Earth's rotation. Will the stone continue along the centre of the shaft ? If not, describe its motion.

Q.5 An iron cube is placed near an iron sphere at a location remote from the Earth's gravity. What can you say about the location of the centre of gravity of the cube? Of the sphere ? In general, does the location of the centre of gravity of an object depend on the nature of the gravitational field in which the object is placed?

Q.6 Figure shows a particle of mass m that is moved from an infinite distance to the centre of a ring of mass M, along the central axis of the ring. For the trip, how does the magnitude of the gravitational force on the particle due to the ring change.



Q.7 In figure, a particle of mass m is initially at point A, at distance d from the centre of one uniform sphere and distance 4d from the centre of another uniform sphere, both of mass M >> m. State whether, if you moved the particle to point D, the following would be positive, negative, or zero:
(a) the change in the gravitational potential energy of the particle,
(b) the work done by the net gravitational force on the particle,
(c) the work done by your force.
(d) What are the answers if, instead, the move were from point B to point C ?



Q.8 Reconsider the situation of above question. Would the work done by you be positive, negative, or zero if you moved the particle (a) from A to B, (b) from A to C, (c) from B to D ? (d) Rank those moves accroding to the absolute value of the work done by your force, greatest first.

ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

Q.1 At what altitude will the acceleration due to gravity be 25% of that at the earth's surface (given radius of earth is R)?

(A) R/4 (B) R (C) 3R/8 (D) R/2

Q.2 At what distance from the centre of the moon is the point at which the strength of the resultant field of earth's and moon's gravitational field is equal to zero. The earth's mass is 81 times that of moon and the distance between centres of these planets is 60R where R is the radius of the earth

(A) 6R (B) 4R (C) 3R (D) 5R

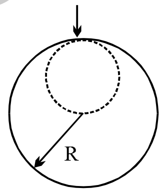
Q.3 Two masses m_1 & m_2 are initially at rest and are separated by a very large distance. If the masses approach each other subsequently, due to gravitational attraction between them, their relative velocity of approach at a separation distance of d is :

(A) $\dfrac{2Gd}{(m_1+m_2)}$ (B) $\dfrac{(m_1+m_2)G}{2d}$ (C) $\left[(m_1+m_2)\dfrac{2G}{d}\right]^{1/2}$ (D) $(m_1+m_2)^{1/2}\,2Gd$

Q.4 Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole (d<<R) The value of d is

(A) $\dfrac{\omega^2R^2}{g}$ (B) $\dfrac{\omega^2R^2}{2g}$ (C) $\dfrac{2\omega^2R^2}{g}$ (D) $\dfrac{\sqrt{Rg}}{g}$

Q.5 A spherical hole of radius R/2 is excavated from the asteroid of mass M as shown in fig. The gravitational acceleration at a point on the surface of the asteroid just above the excavation is



(A) GM/R^2 (B) $GM/2R^2$ (C) $GM/8R^2$ (D) $7GM/8R^2$

Q.6 If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of g the same?

(A) 1/25 (B) 1/5 (C) $1/\sqrt{5}$ (D) 5

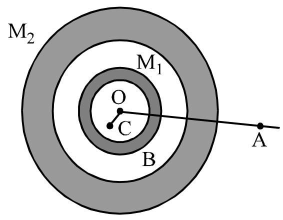
Q.7 A man of mass m starts falling towards a planet of mass M and radius R. As he reaches near to the surface, he realizes that he will pass through a small hole in the planet. As he enters the hole, he sees that the planet is really made of two pieces a spherical shell of negligible thickness of mass $\dfrac{2M}{3}$ and a point mass $\dfrac{M}{3}$ at the centre. Change in the force of gravity experienced by the man is

(A) $\dfrac{2}{3}\dfrac{GMm}{R^2}$ (B) 0 (C) $\dfrac{1}{3}\dfrac{GMm}{R^2}$ (D) $\dfrac{4}{3}\dfrac{GMm}{R^2}$

Q.8 An infinite number of masses, each of one kg are placed on the +ve X axis at 1m, 2m, 4m, ______ from the origin. The magnitude of the gravitational field at origin due to this distribution of masses is:

(A) 2G (B) $\dfrac{4G}{3}$ (C) $\dfrac{3G}{4}$ (D) ∞

Q.9 Two concentric shells of uniform density of mass M_1 and M_2 are situated as shown in the figure. The forces experienced by a particle of mass m when placed at positions A, B and C respectively are (given OA = p, OB = q and OC = r)



(A) zero, $G\frac{M_1 m}{q^2}$ and $G\frac{(M_1+M_2)m}{p^2}$

(B) $G\frac{(M_1+M_2)m}{p^2}$, $G\frac{(M_1+M_2)m}{q^2}$ and $G\frac{M_1 m}{r^2}$

(C) $G\frac{M_1 m}{q^2}$, $G\frac{(M_1+M_2)m}{p^2}$ and zero

(D) $G\frac{(M_1+M_2)m}{p^2}$, $G\frac{M_1 m}{q^2}$ and zero

Q.10 A satellite of the earth is revolving in circular orbit with a uniform velocity V. If the gravitational force suddenly disappears, the satellite will
(A) continue to move with the same velocity in the same orbit.
(B) move tangentially to the original orbit with velocity V.
(C) fall down with increasing velocity.
(D) come to a stop somewhere in its original orbit.

Q.11 A newly discovered planet has a density eight times the density of the earth and a radius twice the radius of the earth. The time taken by 2 kg mass to fall freely through a distance S near the surface of the earth is 1 second. Then the time taken for a 4 kg mass to fall freely through the same distance S near the surface of the new planet is
(A) 0.25 sec. (B) 0.5 sec (C) 1 sec. (D) 4 sec.

Q.12 At what height above the earth's surface does the acceleration due to gravity fall to 1% of its value at the earth's surface?
(A) 9R (B) 10R (C) 99R (D) 100R

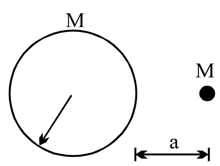
Q.13 The radius of a planet is R. A satellite revolves around it in a circle of radius r with angular velocity ω_0. The acceleration due to the gravity on planet's surface is

(A) $\frac{r^3\omega_0}{R}$ (B) $\frac{r^3\omega_0^3}{R^2}$ (C) $\frac{r^3\omega_0^2}{R}$ (D) $\frac{r^3\omega_0^2}{R^2}$

Q.14 The mass and diameter of a planet are twice those of earth. What will be the period of oscillation of a pendulum on this planet if it is a seconds pendulum on earth?

(A) $\sqrt{2}$ second (B) $2\sqrt{2}$ seconds (C) $\frac{1}{\sqrt{2}}$ second (D) $\frac{1}{2\sqrt{2}}$ second

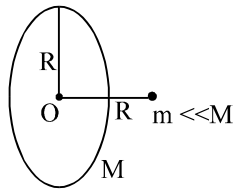
Q.15 A particle of mass M is at a distance a from surface of a thin spherical shell of equal mass and having radius a.



(A) Gravitational field and potential both are zero at centre of the shell.
(B) Gravitational field is zero not only inside the shell but at a point outside the shell also.
(C) Inside the shell, gravitational field alone is zero.
(D) Neither gravitational field nor gravitational potential is zero inside the shell.

Q.16 A satellite revolves in the geostationary orbit but in a direction east to west. The time interval between its successive passing about a point on the equator is :
(A) 48 hrs (B) 24 hrs (C) 12 hrs (D) never

Q.17 A particle starts from rest at a distance R from the centre and along the axis of a fixed ring of radius R & mass M. Its velocity at the centre of the ring is:



(A) $\sqrt{\frac{\sqrt{2}\,GM}{R}}$ (B) $\sqrt{\frac{2GM}{R}}$
(C) $\sqrt{\left(1-\frac{1}{\sqrt{2}}\right)\frac{GM}{R}}$ (D) $\sqrt{\left(2-\sqrt{2}\right)\frac{GM}{R}}$

Q.18 A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V. Due to the rotation of planet about its axis the acceleration due to gravity g at equator is 1/2 of g at poles. The escape velocity of a particle on the planet in terms of V.
(A) $V_e = 2V$ (B) $V_e = V$ (C) $V_e = V/2$ (D) $V_e = \sqrt{3}\,V$

Q.19 Two point masses of mass 4*m* and *m* respectively separated by *d* distance are revolving under mutual force of attraction. Ratio of their kinetic energies will be :
(A) 1 : 4 (B) 1 : 5 (C) 1 : 1 (D) 1 : 2

Q.20 Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of the escape velocity $\frac{V_A}{V_B}$ is
(A) 2 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) $1/2$

Q.21 A ball 'A' of mass m falls to the surface of the earth from infinity. Another ball 'B' of mass 2m falls to the earth from the height equal to six times radius of the earth then ratio of velocities of 'A' and 'B' on reaching the earth is
(A) $\sqrt{(6/5)}$ (B) $\sqrt{(5/6)}$ (C) 1 (D) $\sqrt{(7/6)}$

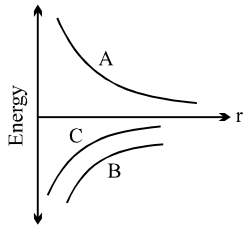
Q.22 The ratio of gravitational acceleration at height 3R to that at height 4R from the surface of the earth is : (where R is the radius of the earth)
(A) 9/16 (B) 25/16 (C) 16/25 (D) 16/9

Q.23 A rocket is launched straight up from the surface of the earth. When its altitude is one fourth of the radius of the earth, its fuel runs out and therefore it coasts. The minimum velocity which the rocket must have when it starts to coast if it is to escape from the gravitational pull of the earth is [escape velocity on surface of earth is 11.2km/s]
(A) 1km/s (B) 5km/s (C) 10km/s (D) 15km/s

Q.24 Select the correct choice(s):
(A) The gravitational field inside a spherical cavity, within a spherical planet must be non zero and uniform.
(B) When a body is projected horizontally at an appreciable large height above the earth, with a velocity less than for a circular orbit, it will fall to the earth along a parabolic path.
(C) A body of zero total mechanical energy placed in a gravitational field will escape the field
(D) Earth's satellite must be in equatorial plane .

Q.25 A satellite of mass m, initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is
(A) $\frac{\sqrt{3}}{4}$ mgR (B) $\frac{1}{2}$ mgR (C) $\frac{1}{4}$ mgR (D) $\frac{3}{4}$ mgR

Q.26 The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



(A) A shows the kinetic energy, B the total energy and C the potential energy of the system.
(B) C shows the total energy, B the kinetic energy and A the potential energy of the system.
(C) C and A are kinetic and potential energies respectively and B is the total energy of the system.
(D) A and B are kinetic and potential energies and C is the total energy of the system.

Q.27 A particle is projected from the mid-point of the line joining two fixed particles each of mass m. If the distance of separation between the fixed particles is l, the minimum velocity of projection of the particle so as to escape is equal to

(A) $\sqrt{\frac{GM}{l}}$ (B) $\sqrt{\frac{GM}{2l}}$ (C) $\sqrt{\frac{2GM}{l}}$ (D) $2\sqrt{\frac{2GM}{l}}$

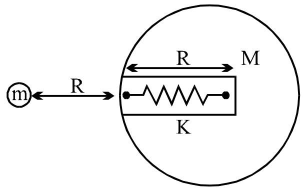
Q.28 The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(A) v_e (B) $\frac{v_e}{\sqrt{2}}$ (C) $\frac{v_e}{2}$ (D) zero

Q.29 A small ball of mass 'm' is released at a height 'R' above the earth surface, as shown in the figure above. If the maximum depth of the ball to which it goes is R/2 inside the earth through a narrow grove before coming to rest momentarily. The grove, contain an ideal spring of spring constant K and natural length R, find the value of K if R is radius of earth and M mass of earth

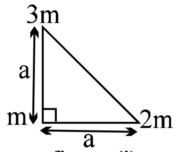
(A) $\frac{3GMm}{R^3}$ (B) $\frac{6GMm}{R^3}$

(C) $\frac{9GMm}{R^3}$ (D) $\frac{7GMm}{R^3}$

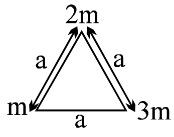


Q.30 A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre
(A) increases
(B) decreases
(C) remains same
(D) during the compression increases then returns at the previous value.

Q.31 Consider two configurations of a system of three particles of masses m, 2m and 3m. The work done by external agent in changing the configuration of the system from figure (i) to figure (ii) is



figure(i)



figure(ii)

(A) zero (B) $-\frac{6Gm^2}{a}\left(1+\frac{1}{\sqrt{2}}\right)$

(C) $-\frac{6Gm^2}{a}\left(1-\frac{1}{\sqrt{2}}\right)$ (D) $-\frac{6Gm^2}{a}\left(2-\frac{1}{\sqrt{2}}\right)$

Q.32 A uniform spherical planet (Radius R) has acceleration due to gravity at its surface g. Points P and Q located inside and outside the planet have acceleration due to gravity $\frac{g}{4}$. Maximum possible separation between P and Q is

(A) $\frac{7R}{4}$ (B) $\frac{3R}{2}$ (C) $\frac{9R}{4}$ (D) none

Q.33 A particle is dropped on Earth from height R (radius of Earth) and it bounces back to a height R/2 the coefficient of restitution for collision is (ignore air resistance and rotation of Earth)

(A) $\frac{2}{3}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\sqrt{\frac{1}{3}}$ (D) $\sqrt{\frac{1}{2}}$

Q.34 A body of mass m is lifted up from the surface of the earth to a height three times the radius of the earth. The change in potential energy of the body is

(A) 3mgR (B) 3/4 mgR (C) 1/3 mgR (D) 2/3 mgR

where g is acceleration due to gravity at the surface of earth.

Q.35 When a satellite moves around the earth in a certain orbit, the quantity which remains constant is :

(A) angular velocity (B) kinetic energy (C) aerial velocity (D) potential energy

Q.36 A satellite is launched into a circular orbit of radius R around the earth. A second satellite is launched into an orbit of radius 1.02R. The period of second satellite is larger than the first one by approximately

(A) 1.5% (B) 3% (C) 1% (D) 2%

Q.37 A satellite of mass 5M orbits the earth in a circular orbit. At one point in its orbit, the satellite explodes into two pieces, one of mass M and the other of mass 4M. After the explosion the mass M ends up travelling in the same circular orbit, but in opposite direction. After explosion the mass 4M is in

(A) bound orbit
(B) unbound orbit
(C) partially bound orbit
(D) data is insufficient to determine the nature of the orbit.

Q.38 A satellite can be in a geostationary orbit around a planet at a distance r from the centre of the planet. If the angular velocity of the planet about its axis doubles, a satellite can now be in a geostationary orbit around the planet if its distance from the centre of the planet is

(A) $\frac{r}{2}$ (B) $\frac{r}{2\sqrt{2}}$ (C) $\frac{r}{(4)^{1/3}}$ (D) $\frac{r}{(2)^{1/3}}$

Q.39 A planet of mass m is in an elliptical orbit about the sun ($m << M_{sun}$) with an orbital period T. If A be the area of orbit, then its angular momentum would be:

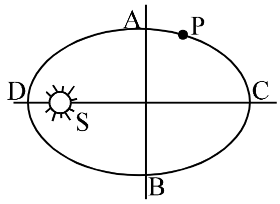
(A) $\frac{2mA}{T}$ (B) mAT (C) $\frac{mA}{2T}$ (D) 2mAT

Q.40 Suppose the gravitational force varies inversely as the n^{th} power of distance. Then the time period of a planet in circular orbit of radius R around the sun will be proportional to

(A) $R^{\left(\frac{n+1}{2}\right)}$ (B) $R^{\left(\frac{n-1}{2}\right)}$ (C) R^n (D) $R^{\left(\frac{n-2}{2}\right)}$

Question No. 41 to 42 (2 questions)

Figure shows the orbit of a planet P round the sun S. AB and CD are the minor and major axes of the ellipse.



Q.41 If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then

(A) $t_1 = t_2$ (B) $t_1 > t_2$ (C) $t_1 < t_2$ (D) nothing can be concluded

Q.42 If U is the potential energy and K kinetic energy then $|U| > |K|$ at

(A) Only D (B) Only C

(C) both D & C (D) neither D nor C

Q.43 If a tunnel is cut at any orientation through earth, then a ball released from one end will reach the other end in time(neglect earth rotation)

(A) 84.6 minutes (B) 42.3 minutes (C) 8 minutes (D) depends on orientation

Questions 44 to 48 (5 questions)

Two stars bound together by gravity orbit each other because of their mutual attraction. Such a pair of stars is referred to as a binary star system. One type of binary system is that of a black hole and a companion star. The black hole is a star that has collapsed on itself and is so massive that not even light rays can escape its gravitational pull. Therefore, when describing the relative motion of a black hole and a companion star, the motion of the black hole can be assumed negligible compared to that of the companion.

The orbit of the companion star is either elliptical with the black hole at one of the foci or circular with the black hole at the centre. The gravitational potential energy is given by $U = -GmM/r$, where G is the universal gravitational constant, m is the mass of the companion star, M is the mass of the black hole, and *r* is the distance between the centre of the companion star and the centre of the black hole. Since the gravitational force is conservative, the companion star's total mechanical energy is a constant of the motion. Because of the periodic nature of the orbit, there is a simple relation between the average kinetic energy $\langle K \rangle$ of the companion star and its average potential energy $\langle U \rangle$. In particular, $\langle K \rangle = -\langle U/2 \rangle$

Two special points along the orbit are singled out by astronomers. *Perigee* is the point at which the companion star is closest to the black hole, and *apogee* is the point at which it is furthest from the black hole.

Q.44 At which point in the elliptical orbit does the companion star attain its maximum kinetic energy?

(A) Apogee (B) Perigee (C) The point midway from apogee to perigee

(D) All points in the orbit, since the kinetic energy is a constant of the motion.

Q.45 For circular orbits, the potential energy of the companion star is constant throughout the orbit. If the radius of the orbit doubles, what is the new value of the velocity of the companion star?

(A) It is 1/2 of the old value (B) It is $1/\sqrt{2}$ of the old value

(C) It is the same as the old value. (D) It is double the old value

Q.46 The work done on the companion star in one complete orbit by the gravitational force of the black hole equals

(A) the difference in the kinetic energy of the companion star between apogee and perigee.

(B) the total mechanical energy of the companion star

(C) zero

(D) the gravitational force on the companion star times the distance that it travels in one orbit.

Q.47 For a circular orbit, which of the following gives the correct expression for the total energy?

(A) $-(1/2)\,mv^2$ (B) mv^2 (C) $-(GmM)/r$ (D) $(GmM)/2r$

Q.48 What is the ratio of the acceleration of the black hole to that of the companion star?

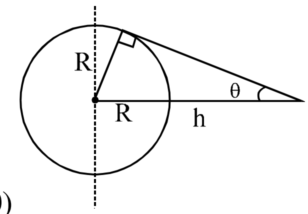
(A) M / m (B) m / M (C) mM / r (D) 1 / 1

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 Assuming the earth to be a sphere of uniform density the acceleration due to gravity
(A) at a point outside the earth is inversely proportional to the square of its distance from the centre
(B) at a point outside the earth is inversely proportional to its distance from the centre
(C) at a point inside is zero
(D) at a point inside is proportional to its distance from the centre.

Q.2 A geostationary satellite is at a height h above the surface of earth. If earth radius is R
(A) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1}(R/R+h)$.
(B) The maximum colatitudes on earth upto which the satellite can be used for communication is $\sin^{-1}(R/R+h)$.



(C) The area on earth escaped from this satellite is given as $2\pi R^2(1+\sin\theta)$
(D) The area on earth escaped from this satellite is given as $2\pi R^2(1+\cos\theta)$

Q.3 Gravitational potential at the centre of curvature of a hemispherical bowl of radius R and mass M is V.
(A) gravitational potential at the centre of curvature of a thin uniform wire of mass M, bent into a semicircle of radius R, is also equal to V.
(B) In part (A) if the same wire is bent into a quarter of a circle then also the gravitational potential at the centre of curvature will be V.
(C) In part (A) if the same wire mass is nonuniformly distributed along its length and it is bent into a semicircle of radius R, gravitational potential at the centre is V.
(D) none of these

Q.4 When a satellite in a circular orbit around the earth enters the atmospheric region, it encounters small air resistance to its motion. Then
(A) its kinetic energy increases
(B) its kinetic energy decreases
(C) its angular momentum about the earth decreases
(D) its period of revolution around the earth increases

Q.5 A communications Earth satellite
(A) goes round the earth from east to west
(B) can be in the equatorial plane only
(C) can be vertically above any place on the earth
(D) goes round the earth from west to east

Q.6 An earth satellite is moved from one stable circular orbit to another larger and stable circular orbit. The following quantities increase for the satellite as a result of this change
(A) gravitational potential energy (B) angular vleocity
(C) linear orbital velocity (D) centripetal acceleration

Q.7 Two satellites s_1 & s_2 of equal masses revolve in the same sense around a heavy planet in coplanar circular orbit of radii R & 4R
(A) the ratio of period of revolution s_1 & s_2 is 1 : 8.
(B) their velocities are in the ratio 2 : 1
(C) their angular momentum about the planet are in the ratio 2 : 1
(D) the ratio of angular velocities of s_2 w.r.t. s_1 when all three are in the same line is 9 : 5.

Q.8 A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of the earth

(A) the acceleration of S is always directed towards the centre of the earth

(B) the angular momentum of S about the centre of the earth changes in direction, but its magnitude remains constant

(C) the total mechanical energy of S varies periodically with time

(D) the linear momentum of S remains constant in magnitude

Q.9 If a satellite orbits as close to the earth's surface as possible,

(A) its speed is maximum

(B) time period of its rotation is minimum

(C) the total energy of the 'earth plus satellite' system is minimum

(D) the total energy of the 'earth plus satellite'system is maximum

Q.10 For a satellite to orbit around the earth, which of the following must be true?

(A) It must be above the equator at some time

(B) It cannot pass over the poles at any time

(C) Its height above the surface cannot exceed 36,000 km

(D) Its period of rotation must be > $2\pi\sqrt{R/g}$ where R is radius of earth

Answer Key

ONLY ONE OPTION IS CORRECT.

Q.1	B	Q.2	A	Q.3	C	Q.4	A	Q.5	B	Q.6	B	Q.7	A
Q.8	B	Q.9	D	Q.10	B	Q.11	A	Q.12	A	Q.13	D	Q.14	B
Q.15	D	Q.16	C	Q.17	D	Q.18	A	Q.19	A	Q.20	A	Q.21	D
Q.22	B	Q.23	C	Q.24	C	Q.25	D	Q.26	D	Q.27	D	Q.28	B
Q.29	D	Q.30	B	Q.31	C	Q.32	C	Q.33	B	Q.34	B	Q.35	C
Q.36	B	Q.37	B	Q.38	C	Q.39	A	Q.40	A	Q.41	B	Q.42	C
Q.43	B	Q.44	B	Q.45	B	Q.46	C	Q.47	A	Q.48	B		

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	A,D	Q.2	A,C	Q.3	A,C	Q.4	A,C
Q.5	B,D	Q.6	A	Q.7	A,B,D	Q.8	A
Q.9	A,B,C	Q.10	A,D				

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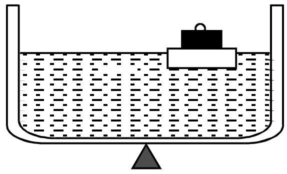
PHYSICS

FLUID MECHANICS

QUESTION FOR SHORT ANSWER

Q.1 Liquid is flowing inside a horizontal pipe which has a constriction along its length. Vertical tube manometers are attached at both the wide portion and the narrow portion of the pipe. If a stopcock at the exit end is closed, will the liquid in the manometer tubes rise or fall? Explain.

Q.2 A vessel filled with water is supported on a knife-edge. Will the equilibrium be disturbed if a small board carrying a weight is placed on the surface of the water?



Q.3 A wooden cylinder floats in water in a vessel with its axis vertical. How will the level of water in the vessel change if the cylinder floats with its axis horizontal?

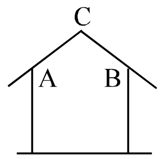
Q.4 Mercury is poured into a vertical U-tube, and water is poured in above it. The level of water is the same in both arms. Will the level of the water and the mercury be the same if a piece of wood is dropped into one arm and some water equal in weight to this piece is added to the other?

Q.5 A tank containing water is placed on a spring balance. A stone of weight w is hung and lowered into the water without touching the sides and the bottom of the tank. Explain how the reading will change.

Q.6 An open bucket of water is on a smooth inclined plane, forming an angle α with the horizontal. How will be the level relative to the plane when the bucket is alloowed to slide down the plane?

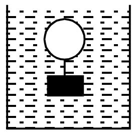
Q.7 Does the difference in pressure between the lower and upper surfaces of an airplane wing depend on the altitude of the moving plane? Explain.

Q.8 During storms, the strong winds tear off the roofs of thatched houses along the ridge C if the roof is fastened more firmly at the edges A and B than at the ridge. On the other hand, if the roof is secured more firmly at the ridge C than at the edges, the wind will first lift the roof up and then carry it away. Explain.



Q.9 Liquids leak through a pin hole at the bottom of a vessel. Will kerosene leak faster than water? Give reasons for your answer (assume same velocity).

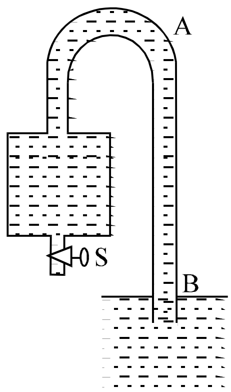
Q.10 A balloon filled with air is weighed so that it barely floats in water as shown in the figure. Explain why it sinks to the bottom when it is submerged more



Q.11 Explain why one has to blow over a piece of paper rather than under it, to keep it horizontal.

Q.12 Why do water jets taper when the tap is almost closed?

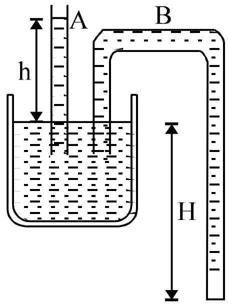
Q.13 The vessel shown in figure is entirely filled with water. What will happen if the tap S is opened?



Q.14 A sailor found a small hole in the hold of his vessel, through which water was entering into it. He tried to stop the flow with a plank but the stream of water pushed the plank away. He managed to bring the plank close to the hole with the aid of another sailor, and then found that he could hold the plank alone. Explain why the pressure on the plank is different in the two cases.

Q.15 Water can rise to a height h in a certain capillary tube. Suppose that this tube is immersed in water so that only a height h/2 is above the surface. Will there be a fountain? Explain.

Q.16 Two capillary tubes A and B are immersed in water – one is straight and the other is in the form of a rectangular U-tube. The tube A is sufficiently long. The lower end of the bent tube is at a depth H. What form will the meniscus take and will there be any flow of the water? Consider the following five cases:



(a) H > h
(b) H = h
(c) 0 < H < h
(d) H = 0
(e) H < 0

Q.17 Two ships happen to sail parallel and adjacent to each other : and if they are not far away. They experience a pull to bring them together. Explain with diagram.

Q.18 Explain why two glass plates with a thin film of water between them are difficult to separate by a direct pull but can easily be separated by sliding?

Q.19 Explain these observation: (a) water forms globules on a greasy plate but not on a clean one; (b) small bubbles on the surface of water cluster together.

Q.20 When paddling a canoe, one can attain a certain critical speed with relatively little effort, and then a much greater effort is required to make the canoe go even a little faster. Why?

Q.21 Why do jet airplanes usually fly at altitudes above 30,000 feet, through it takes a lot of fuel to climb that high ?

ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

Q.1 The vertical limbs of a U shaped tube are filled with a liquid of density ρ upto a height h on each side. The horizontal portion of the U tube having length 2h contains a liquid of density 2ρ. The U tube is moved horizontally with an accelerator g/2 parallel to the horizontal arm. The difference in heights in liquid levels in the two vertical limbs, at steady state will be

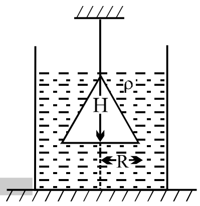
(A) $\frac{2h}{7}$ (B) $\frac{8h}{7}$ (C) $\frac{4h}{7}$ (D) None of these

Q.2 A bucket contains water filled upto a height = 15 cm. The bucket is tied to a rope which is passed over a frictionless light pulley and the other end of the rope is tied to a weight of mass which is half of that of the (bucket + water). The water pressure above atmosphere pressure at the bottom is

(A) 0.5 kPa (B) 1 kPa

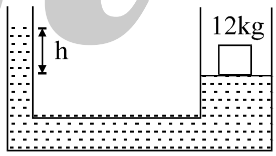
(C) 5 kPa (D) None of these

Q.3 A cone of radius R and height H, is hanging inside a liquid of density ρ by means of a string as shown in the figure. The force, due to the liquid acting on the slant surface of the cone is



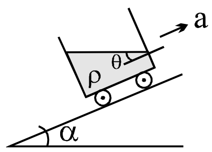
(A) $\rho\pi gHR^2$ (B) $\pi\rho HR^2$ (C) $\frac{4}{3}\pi\rho gHR^2$ (D) $\frac{2}{3}\pi\rho gHR^2$

Q.4 The area of cross-section of the wider tube shown in figure is $800\ cm^2$. If a mass of 12 kg is placed on the massless piston, the difference in heights h in the level of water in the two tubes is :



(A) 10 cm (B) 6 cm (C) 15 cm (D) 2 cm

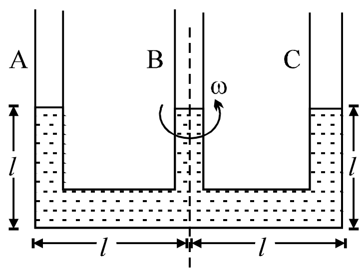
Q.5 A fluid container is containing a liquid of density ρ is accelerating upward with acceleration a along the inclined place of inclination α as shown. Then the angle of inclination θ of free surface is :



(A) $\tan^{-1}\left[\frac{a}{g\cos\alpha}\right]$ (B) $\tan^{-1}\left[\frac{a+g\sin\alpha}{g\cos\alpha}\right]$

(C) $\tan^{-1}\left[\frac{a-g\sin\alpha}{g(1+\cos\alpha)}\right]$ (D) $\tan^{-1}\left[\frac{a-g\sin\alpha}{g(1-\cos\alpha)}\right]$

Q.6 Figure shows a three arm tube in which a liquid is filled upto levels of height l. It is now rotated at an angular frequency ω about an axis passing through arm B. The angular frequency ω at which level of liquid in arm B becomes zero.



(A) $\sqrt{\frac{2g}{3l}}$ (B) $\sqrt{\frac{g}{l}}$ (C) $\sqrt{\frac{3g}{l}}$ (D) $\sqrt{\frac{3g}{2l}}$

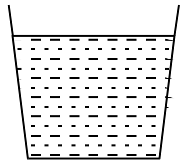
Q.7 An open cubical tank was initially fully filled with water. When the tank was accelerated on a horizontal plane along one of its side it was found that one third of volume of water spilled out. The acceleration was

(A) g/3 (B) 2g/3 (C) 3g/2 (D) None

Q.8 Some liquid is filled in a cylindrical vessel of radius R. Let F_1 be the force applied by the liquid on the bottom of the cylinder. Now the same liquid is poured into a vessel of uniform square crss-section of side R. Let F_2 be the force applied by the liquid on the bottom of this new vessel. Then:

(A) $F_1 = \pi F_2$ (B) $F_1 = \frac{F_2}{\pi}$ (C) $F_1 = \sqrt{\pi} F_2$ (D) $F_1 = F_2$

Q.9 A liquid of mass 1 kg is filled in a flask as shown in figure. The force exerted by the flask on the liquid is ($g = 10\ m/s^2$)[Neglect atmospheric pressure]:

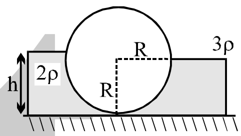


(A) 10 N (B) greater than 10N (C) less than 10N (D) zero

Q.10 A U – tube having horizontal arm of length 20 cm, has uniform cross-sectional area = $1 cm^2$. It is filled with water of volume 60 cc. What volume of a liquid of density 4 g/cc should be poured from one side into the U – tube so that no water is left in the horizontal arm of the tube?

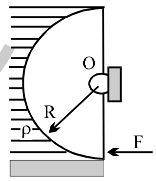
(A) 60 cc (B) 45 cc (C) 50 cc (D) 35 cc

Q.11 In the figure shown, the heavy cylinder (radius R) resting on a smooth surface separates two liquids of densities 2ρ and 3ρ. The height 'h' for the equilibrium of cylinder must be



(A) 3R/2 (B) $R\sqrt{\frac{3}{2}}$ (C) $R\sqrt{2}$ (D) None

Q.12 A light semi cylindrical gate of radius R is piovted at its mid point O, of the diameter as shown in the figure holding liquid of density ρ. The force F required to prevent the rotation of the gate is equal to

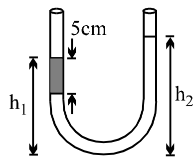


(A) $2\pi R^3 \rho g$ (B) $2\rho g R^3 l$ (C) $\frac{2R^2 l \rho g}{3}$ (D) none of these

Q.13 The pressure at the bottom of a tank of water is 3P where P is the atmospheric pressure . If the water is drawn out till the level of water is lowered by one fifth., the pressure at the bottom of the tank will now be

(A) 2P (B) (13/5) P (C) (8/5) P (D) (4/5)P

Q.14 An open-ended U-tube of uniform cross-sectional area contains water (density 1.0 gram/centimeter^3) standing initially 20 centimeters from the bottom in each arm. An immiscible liquid of density 4.0 grams/ centimeter^3 is added to one arm until a layer 5 centimeters high forms, as shown in the figure above. What is the ratio h_2/h_1 of the heights of the liquid in the two arms?



(A) 3/1 (B) 5/2 (C) 2/1 (D) 3/2

Q.15 A heavy hollow cone of radius R and height h is placed on a horizontal table surface, with its flat base on the table. The whole volume inside the cone is filled with water of density ρ. The circular rim of the cone's base has a watertight seal with the table's surface and the top apex of the cone has a small hole. Neglecting atmospheric pressure find the total upward force exerted by water on the cone is

(A) $(2/3)\pi R^2 h\rho g$ (B) $(1/3)\pi R^2 h\rho g$ (C) $\pi R^2 h\rho g$ (D) None

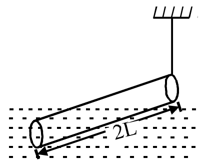
Q.16 Two cubes of size 1.0 m sides, one of relative density 0.60 and another of relative density = 1.15 are connected by weightless wire and placed in a large tank of water. Under equilibrium the lighter cube will project above the water surface to a height of

(A) 50 cm (B) 25 cm (C) 10 cm (D) zero

Q.17 A cuboidal piece of wood has dimensions a, b and c. Its relative density is d. It is floating in a large body of water such that side a is vertical. It is pushed down a bit and released. The time period of SHM executed by it is :

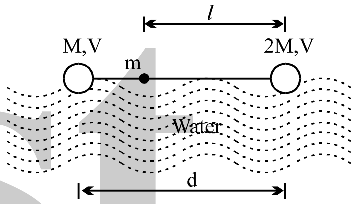
(A) $2\pi\sqrt{\frac{abc}{g}}$ (B) $2\pi\sqrt{\frac{g}{da}}$ (C) $2\pi\sqrt{\frac{bc}{dg}}$ (D) $2\pi\sqrt{\frac{da}{g}}$

Q.18 A slender homogeneous rod of length $2L$ floats partly immersed in water, being supported by a string fastened to one of its ends, as shown. The specific gravity of the rod is 0.75. The length of rod that extends out of water is :



(A) L (B) $\frac{1}{2}$ L (C) $\frac{1}{4}$ L (D) 3 L

Q.19 A dumbbell is placed in water of density ρ. It is observed that by attaching a mass m to the rod, the dumbbell floats with the rod horizontal on the surface of water and each sphere exactly half submerged as shown in the figure. The volume of the mass m is negligible. The value of length l is



(A) $\frac{d(V_\rho - 3M)}{2(V_\rho - 2M)}$ (B) $\frac{d(V_\rho - 2M)}{2(V_\rho - 3M)}$

(C) $\frac{d(V_\rho + 2M)}{2(V_\rho - 3M)}$ (D) $\frac{d(V_\rho - 2M)}{2(V_\rho + 3M)}$

Q.20 Two bodies having volumes V and 2V are suspended from the two arms of a common balance and they are found to balance each other. If larger body is immersed in oil (density $d_1 = 0.9$ gm/cm^3) and the smaller body is immersed in an unknown liquid, then the balance remain in equilibrium. The density of unknown liquid is given by :

(A) 2.4 gm/cm^3 (B) 1.8 gm/cm^3 (C) 0.45 gm/cm^3 (D) 2.7 gm/cm^3

Q.21 A container of large surface area is filled with liquid of density ρ. A cubical block of side edge a and mass M is floating in it with four-fifth of its volume submerged. If a coin of mass m is placed gently on the top surface of the block is just submerged. M is

(A) 4m/5 (B) m/5 (C) 4m (D) 5m

Q.22 A boy carries a fish in one hand and a bucket(not full) of water in the other hand . If he places the fish in the bucket , the weight now carried by him (assume that water does not spill) :

(A) is less than before (B) is more than before
(C) is the same as before (D) depends upon his speed

Q.23 A cork of density 0.5gcm^{-3} floats on a calm swimming pool. The fraction of the cork's volume which is under water is

(A) 0% (B) 25% (C) 10% (D) 50%

Q.24 Two cylinders of same cross-section and length L but made of two material of densities d_1 and d_2 are cemented together to form a cylinder of length 2L. The combination floats in a liquid of density d with a length L/2 above the surface of the liquid. If $d_1 > d_2$ then:

(A) $d_1 > \frac{3}{4}d$ (B) $\frac{d}{2} > d_1$ (C) $\frac{d}{4} > d_1$ (D) $d < d_1$

Q.25 A piece of steel has a weight W in air, W_1 when completely immersed in water and W_2 when completely immersed in an unknown liquid. The relative density (specific gravity) of liquid is:

(A) $\frac{W - W_1}{W - W_2}$ (B) $\frac{W - W_2}{W - W_1}$ (C) $\frac{W_1 - W_2}{W - W_1}$ (D) $\frac{W_1 - W_2}{W - W_2}$

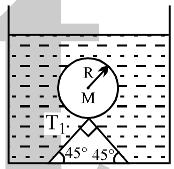
Q.26 A ball of relative density 0.8 falls into water from a height of 2m. The depth to which the ball will sink is (neglect viscous forces):

(A) 8 m (B) 2 m (C) 6 m (D) 4 m

Q.27 A small wooden ball of density ρ is immersed in water of density σ to depth h and then released. The height H above the surface of water up to which the ball will jump out of water is

(A) $\frac{\sigma h}{\rho}$ (B) $\left(\frac{\sigma}{\rho} - 1\right)h$ (C) h (D) zero

Q.28 A hollow sphere of mass M and radius r is immersed in a tank of water (density ρ_w). The sphere would float if it were set free. The sphere is tied to the bottom of the tank by two wires which makes angle 45° with the horizontal as shown in the figure. The tension T_1 in the wire is :



(A) $\frac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{\sqrt{2}}$ (B) $\frac{2}{3}\pi R^3 \rho_w g - Mg$

(C) $\frac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{2}$ (D) $\frac{4}{3}\pi R^3 \rho_w g + Mg$

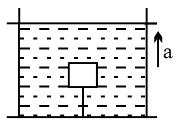
Q.29 A metal ball of density 7800 kg/m^3 is suspected to have a large number of cavities . It weighs 9.8 kg when weighed directly on a balance and 1.5 kg less when immersed in water . The fraction by volume of the cavities in the metal ball is approximately :

(A) 20 % (B) 30 % (C) 16 % (D) 11 %

Q.30 A sphere of radius R and made of material of relative density σ has a concentric cavity of radius r. It just floats when placed in a tank full of water. The value of the ratio R/r will be

(A) $\left(\frac{\sigma}{\sigma - 1}\right)^{1/3}$ (B) $\left(\frac{\sigma - 1}{\sigma}\right)^{1/3}$ (C) $\left(\frac{\sigma + 1}{\sigma}\right)^{1/3}$ (D) $\left(\frac{\sigma - 1}{\sigma + 1}\right)^{1/3}$

Q.31 A body having volume V and density ρ is attached to the bottom of a container as shown. Density of the liquid is d(>ρ). Container has a constant upward acceleration a. Tension in the string is



(A) V[Dg – ρ(g+a)] (B) V(g+a) (d – ρ) (C) V (d – ρ) g (D) none

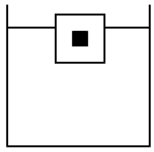
Q.32 A hollow cone floats with its axis vertical upto one-third of its height in a liquid of relative density 0.8 and with its vertex submerged. When another liquid of relative density ρ is filled in it upto one-third of its height, the cone floats upto half its vertical height. The height of the cone is 0.10 m and the radius of the circular base is 0.05 m. The specific gravity ρ is given by

(A) 1.0 (B) 1.5 (C) 2.1 (D) 1.9

Q.33 A beaker containing water is placed on the platform of a spring balance. The balance reads 1.5 kg. A stone of mass 0.5 kg and density 500 kg/m^3 is immersed in water without touching the walls of beaker. What will be the balance reading now?

(A) 2 kg (B) 2.5 kg (C) 1 kg (D) 3 kg

Q.34 There is a metal cube inside a block of ice which is floating on the surface of water. The ice melts completely and metal falls in the water. Water level in the container

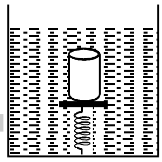


(A) Rises (B) Falls
(C) Remains same (D) Nothing can be concluded

Q.35 A uniform solid cylinder of density 0.8 g/cm^3 floats in equilibrium in a combination of two non-mixing liquid A and B with its axis vertical. The densities of liquid A and B are 0.7 g/cm^3 and 1.2 gm/cm^3. The height of liquid A is h_A = 1.2 cm and the length of the part of cylinder immersed in liquid B is h_B = 0.8 cm. Then the length part of the cylinder in air is

(A) 0.21 cm (B) 0.25 cm (C) 0.35 cm (D) 0.4 cm

Q.36 A cylindrical block of area of cross–section A and of material of density ρ is placed in a liquid of density one–third of density of block. The block compresses a spring and compression in the spring is one–third of the length of the block. If acceleration due to gravity is g, the spring constant of the spring is:



(A) ρAg (B) 2ρAg (C) 2ρAg/3 (D) ρAg/3

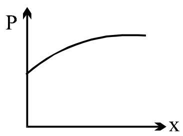
Q.37 A rectangular tank is placed on a horizontal ground and is filled with water to a height *H* above the base. A small hole is made on one vertical side at a depth *D* below the level of the water in the tank. The distance x from the bottom of the tank at which the water jet from the tank will hit the ground is

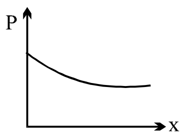
(A) $2\sqrt{D(H-D)}$ (B) $2\sqrt{DH}$ (C) $2\sqrt{D(H+D)}$ (D) $\frac{1}{2}\sqrt{DH}$

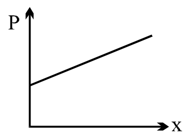
Q.38 A jet of water with cross section of 6 cm^2 strikes a wall at an angle of 60° to the normal and rebounds elastically from the wall without losing energy. If the velocity of the water in the jet is 12 m/s, the force acting on the wall is

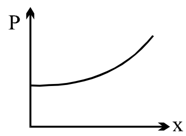
(A) 0.864 Nt (B) 86.4 Nt (C) 72 Nt (D) 7.2 Nt

Q.39 The cross sectional area of a horizontal tube increases along its length linearly, as we move in the direction of flow. The variation of pressure, as we move along its length in the direction of flow (x-direction), is best depicted by which of the following graphs

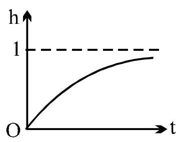
(A) 

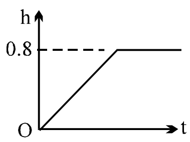
(B) 

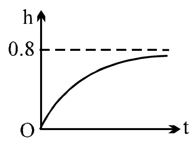
(C) 

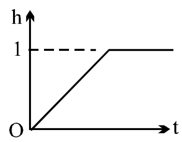
(D) 

Q.40 A cylindrical tank of height 1 m and cross section area A = 4000 cm^2 is initially empty when it is kept under a tap of cross sectional area 1 cm^2. Water starts flowing from the tap at t = 0, with a speed = 2 m/s. There is a small hole in the base of the tank of cross-sectional area 0.5 cm^2. The variation of height of water in tank (in meters) with time t is best depicted by

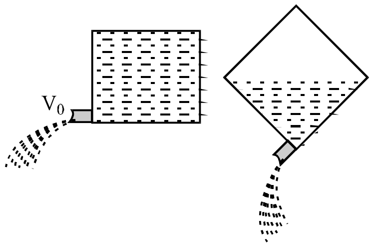
(A) 

(B) 

(C) 

(D) 

Q.41 A cubical box of wine has a small spout located in one of the bottom corners. When the box is full and placed on a level surface, opening the spout results in a flow of wine with a initial speed of v_0 (see figure). When the box is half empty, someone tilts it at 45° so that the spout is at the lowest point (see figure). When the spout is opened the wine will flow out with a speed of

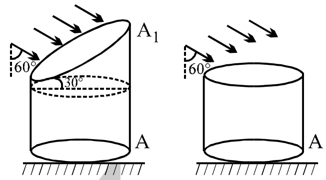


(A) v_0 (B) $v_0/2$ (C) $v_0/\sqrt{2}$ (D) $v_0/\sqrt[4]{2}$

Q.42 Water is flowing steadily through a horizontal tube of non uniform cross-section. If the pressure of water is 4×10^4 N/m^2 at a point where cross-section is 0.02 m^2 and velocity of flow is 2 m/s, what is pressure at a point where cross-section reduces to 0.01 m^2.

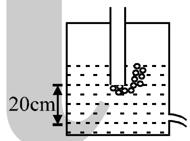
(A) 1.4×10^4 N/m^2 (B) 3.4×10^4 N/m^2 (C) 2.4×10^{-4} N/m^2 (D) none of these

Q.43 A vertical cylindrical container of base area A and upper cross-section area A_1 making an angle 30° with the horizontal is placed in an open rainy field as shown near another cylindrical container having same base area A. The ratio of rates of collection of water in the two containers will be



(A) $2/\sqrt{3}$ (B) $4/\sqrt{3}$ (C) 2 (D) None

Q.44 A tube is attached as shown in closed vessel containing water. The velocity of water coming out from a small hole is :



(A) $\sqrt{2}$ m/s (B) 2 m/s
(C) depends on pressure of air inside vessel (D) None of these

Q.45 A large tank is filled with water to a height H. A small hole is made at the base of the tank. It takes T_1 time to decrease the height of water to H/η, ($\eta > 1$) and it takes T_2 time to take out the rest of water. If $T_1 = T_2$, then the value of η is :

(A) 2 (B) 3 (C) 4 (D) $2\sqrt{2}$

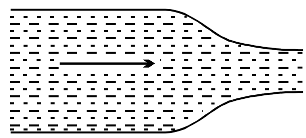
Q.46 In the case of a fluid, Bernoulli's theorem expresses the application of the principle of conservation of :

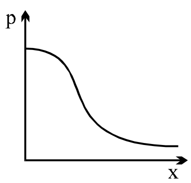
(A) linear momentum (B) energy (C) mass (D) angular momentum

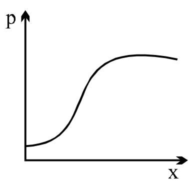
Q.47 Fountains usually seen in gardens are generated by a wide pipe with an enclosure at one end having many small holes. Consider one such fountain which is produced by a pipe of internal diameter 2 cm in which water flows at a rate 3 ms^{-1}. The enclosure has 100 holes each of diameter 0.05 cm. The velocity of water coming out of the holes ids (in ms^{-1}):

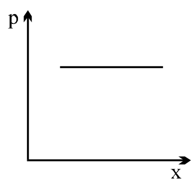
(A) 0.48 (B) 96 (C) 24 (D) 48

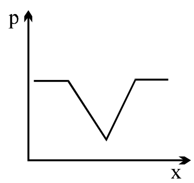
Q.48 Water flows through a frictionless duct with a cross-section varying as shown in figure. Pressure p at points along the axis is represented by



(A)


(B)


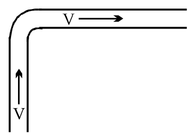
(C)


(D)


Q.49 A cylindrical vessel filled with water upto the height H becomes empty in time t_0 due to a small hole at the bottom of the vessel. If water is filled to a height 4H it will flow out in time
(A) t_0 (B) $4t_0$ (C) $8t_0$ (D) $2t_0$

Q.50 A cylindrical vessel open at the top is 20cm high and 10cm in diameter. A circular hole whose cross-sectional area 1 cm^2 is cut at the centre of the bottom of the vessel. Water flows from a tube above it into the vessel at the rate 100 cm^3s^{-1}. The height of water in the vessel under steady state is (Take g=1000 cms^{-2})
(A) 20 cm (B) 15 cm (C) 10 cm (D) 5 cm

Q.51 A fire hydrant delivers water of density ρ at a volume rate L. The water travels vertically upward through the hydrant and then does 90° turn to emerge horizontally at speed V. The pipe and nozzle have uniform crosssection throughout. The force exerted by the water on the corner of the hydrant is



(A) ρVL (B) zero (C) $2\rho VL$ (D) $\sqrt{2}\rho VL$

Q.52 A vertical tank, open at the top, is filled with a liquid and rests on a smooth horizontal surface. A small hole is opened at the centre of one side of the tank. The area of cross-section of the tank is N times the area of the hole, where N is a large number. Neglect mass of the tank itself. The initial acceleration of the tank is

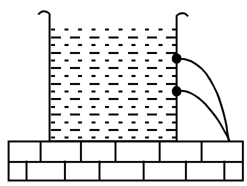
(A) $\frac{g}{2N}$ (B) $\frac{g}{\sqrt{2N}}$ (C) $\frac{g}{N}$ (D) $\frac{g}{2\sqrt{N}}$

Q.53 Two water pipes P and Q having diameters 2×10^{-2}m and 4×10^{-2}m, respectively, are joined in series with the main supply line of water. The velocity of water flowing in pipe P is
(A) 4 times that of Q (B) 2 times that of Q
(C) 1/2 times of that of Q (D) 1/4 times that of Q

Q.54 Water flows into a cylindrical vessel of large cross-sectional area at a rate of 10^{-4} m^3/s. It flows out from a hole of area 10^{-4} m^2, which has been punched through the base. How high does the water rise in the vessel?
(A) 0.075 m (B) 0.051 m (C) 0.031 m (D) 0.025 m

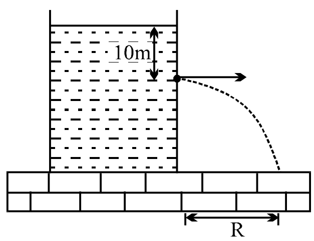
Q.55 A tank is filled up to a height 2H with a liquid and is placedon a platform of height H from the ground. The distance x from the ground where a small hole is punched to get the maximum range R is:
(A) H (B) 1.25 H (C) 1.5 H (D) 2 H

Q.56 In a cylindrical vessel containing liquid of density ρ, there are two holes in the side walls at heights of h_1 and h_2 respectively such that the range of efflux at the bottom of the vessel is same. The height of a hole, for which the range of efflux would be maximum, will be



(A) $h_2 - h_1$ (B) $h_2 + h_1$
(C) $\frac{h_2 - h_1}{2}$ (D) $\frac{h_2 + h_1}{2}$

Q.57 A large tank is filled with water (density = 10^3 kg/m^3). A small hole is made at a depth 10 m below water surface. The range of water issuing out of the hole is R on ground. What extra pressure must be applied on the water surface so that the range becomes 2R (take 1 atm = 10^5 Pa and g = 10 m/s^2):



(A) 9 atm (B) 4 atm
(C) 5 atm (D) 3 atm

Q.58 A water barrel stands on a table of height h. If a small hole is punched in the side of the barrel at its base, it is found that the resultant stream of water strikes the ground at a horizonatl distance R from the barrel. The depth of water in the barrel is

(A) $\frac{R}{2}$ (B) $\frac{R^2}{4h}$ (C) $\frac{R^2}{h}$ (D) $\frac{h}{2}$

Q.59 A cyclindrical vessel of cross-sectional area 1000 cm^2, is fitted with a frictionless piston of mass 10 kg, and filled with water completely. A small hole of cross-sectional area 10 mm^2 is opened at a point 50 cm deep from the lower surface of the piston. The velocity of efflux from the hole will be

(A) 10.5 m/s (B) 3.4 m/s (C) 0.8 m/s (D) 0.2 m/s

Q.60 A laminar stream is flowing vertically down from a tap of cross-section area 1 cm^2. At a distance 10 cm below the tap, the cross-section area of the stream has reduced to 1/2 cm^2. The volumetric flow rate of water from the tap must be about

(A) 2.2 litre/min (B) 4.9 litre/min (C) 0.5 litre/min (D) 7.6 litre/min

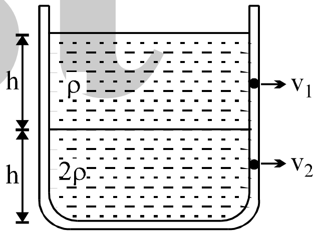
Q.61 A horizontal right angle pipe bend has crosssectional area = 10 cm^2 and water flows through it at speed = 20 m/s. The force on the pipe bend due to the turning of water is :

(A) 565.7 N (B) 400 N (C) 20 N (D) 282.8 N

Q.62 A jet of water having velocity = 10 m/s and stream cross-section = 2 cm^2 hits a flat plate perpendicularly, with the water splashing out parallel to plate. The plate experiences a force of

(A) 40 N (B) 20 N (C) 8 N (D) 10 N

Q.63 Equal volumes of two immiscible liquids of densities ρ and 2ρ are filled in a vessel as shown in figure. Two small holes are punched at depth h/2 and 3h/2 from the surface of lighter liquid. If v_1 and v_2 are the velocities of a flux at these two holes, then v_1/v_2 is :



(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{\sqrt{2}}$

Q.64 A horizontal pipe line carries water in a streamline flow. At a point along the tube where the cross-sectional area is 10^{-2} m^2, the water velocity is 2 ms^{-1} and the pressure is 8000 Pa. The pressure of water at another point where the cross-sectional area is 0.5×10^{-2} m^2 is :

(A) 4000 Pa (B) 1000 Pa (C) 2000 Pa (D) 3000 Pa

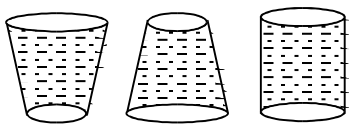
Q.65 A cylindrical vessel open at the top is 20 cm high and 10 cm in diameter. A circular hole whose cross-sectional area 1 cm^2 is cut at the centre of the bottom of the vessel. Water flows from a tube above it into the vessel at the rate 100 cm^3 s^{-1}. The height of water in the vessel under steady state is (Take g = 1000 cm s^{-2})

(A) 20 cm (B) 15 cm (C) 10 cm (D) 5 cm

Q.66 Water is pumped from a depth of 10 m and delivered through a pipe of cross section $10^{-2}$$m^2$. If it is needed to deliver a volume of 10^{-1} m^3 per second the power required will be:

(A) 10 kW (B) 9.8 kW (C) 15 kW (D) 4.9 kW

Q.67 The three water filled tanks shown have the same volume and height. If small identical holes are punched near this bottom, which one will be the first to get empty.



(i) (ii) (iii)

(A) (i) (B) (ii)
(C) (iii) (D) All will take same time

Q.68 A cylindrical vessel filled with water upto height of H stands on a horizontal plane. The side wall of the vessel has a plugged circular hole touching the bottom. The coefficient of friction between the bottom of vessel and plane is μ and total mass of water plus vessel is M. What should be minimum diameter of hole so that the vessel begins to move on the floor if plug is removed (here density of water is ρ)

(A) $\sqrt{\frac{2\mu M}{\pi\rho H}}$ (B) $\sqrt{\frac{\mu M}{2\pi\rho H}}$ (C) $\sqrt{\frac{\mu M}{\rho H}}$ (D) none

Q.69 Which of the following is not an assumption for an ideal fluid flow for which Bernoulli's principle is valid
(A) Steady flow (B) Incompressible (C) Viscous (D) Irrotational

Q.70 A body of density ρ' is dropped from rest at a height h into a lake of density ρ, where $\rho > \rho'$. Neglecting all dissipative forces, calculate the maximum depth to which the body sinks before returning to float on the surface.

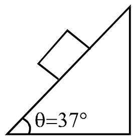
(A) $\frac{h}{\rho-\rho'}$ (B) $\frac{h\rho'}{\rho}$ (C) $\frac{h\rho'}{\rho-\rho'}$ (D) $\frac{h\rho}{\rho-\rho'}$

Q.71 A Newtonian fluid fills the clearance between a shaft and a sleeve. When a force of 800N is applied to the shaft, parallel to the sleeve, the shaft attains a speed of 1.5 cm/sec. If a force of 2.4 kN is applied instead, the shaft would move with a speed of
(A) 1.5 cm/sec (B) 13.5 cm/sec (C) 4.5 cm/sec (D) None

Q.72 A solid metallic sphere of radius r is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross-sectional area and to the square of the velocity, then the terminal velocity of the sphere is proportional to which of the following?
(A) r^2 (B) r (C) $r^{3/2}$ (D) $r^{1/2}$

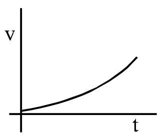
Q.73 Two drops of same radius are falling through air with steady velocity of v cm/s. If the two drops coalesce, what would be the terminal velocity?
(A) 4 v (B) $(4)^{1/3}$ v (C) 2 v (D) 64 v

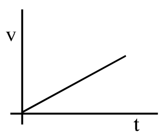
Q.74 A cubical block of side 'a' and density 'ρ' slides over a fixed inclined plane with constant velocity 'v'. There is a thin film of viscous fluid of thickness 't' between the plane and the block. Then the coefficient of viscosity of the thin film will be:

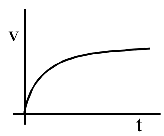


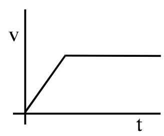
(A) $\frac{3\rho a g t}{5v}$ (B) $\frac{4\rho a g t}{5v}$ (C) $\frac{\rho a g t}{v}$ (D) none of these

Q.75 Which of the following graphs best represents the motion of a raindrop?

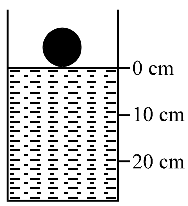
(A) 

(B) 

(C) 

(D) 

Q.76 A spherical ball of density ρ and radius 0.003m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 N.m^{-2} and its density $\rho_L = \rho/2$ = 1260 kg.m^{-3}. Assume the ball reaches a terminal speed by the 10 cm mark. The time taken by the ball to traverse the distance between the 10 cm and 20 cm mark is



(A) 500 μs (B) 50 ms (C) 0.5 s (D) 5 s
(g = acceleration due to gravity = 10 ms^{-2})

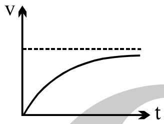
Q.77 A sphere is dropped under gravity through a fluid of viscosity η. If the average acceleration is half of the initial acceleration, the time to attain the terminal velocity is (ρ = density of sphere ; r = radius)

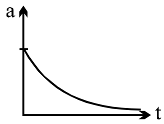
(A) $\frac{4\rho r^2}{9\eta}$ (B) $\frac{9\rho r^2}{4\eta}$ (C) $\frac{4\rho r}{9\eta}$ (D) $\frac{9\rho r}{4\eta}$

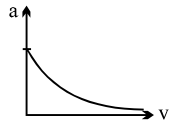
Q.78 A ball of mass m and radius r is gently released in a viscous liquid. The mass of the liquid displaced by it is m' such that m > m'. The terminal velocity is proportional to

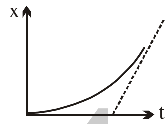
(A) $\frac{m-m'}{r}$ (B) $\frac{m+m'}{r}$ (C) $\frac{(m+m')}{r^2}$ (D) $(m-m')\,r^2$

Q.79 Which of the following is the ***incorrect*** graph for a sphere falling in a viscous liquid?
(Given at t = 0, velocity v = 0 and displacement x = 0.)

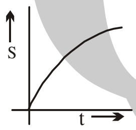
(A) 

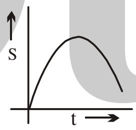
(B) 

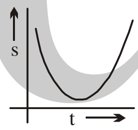
(C) 

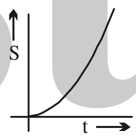
(D) 

Q.80 The displacement of a ball falling from rest in a viscous medium is platted against time. Choose a possible option

(A) 

(B) 

(C) 

(D) 

Q.81 There is a 1mm thick layer of glycerine between a flat plate of area 100 cm^2 & a big fixed plate. If the coefficient of viscosity of glycerine is 1.0 kg/m-s then how much force is required to move the plate with a velocity of 7 cm/s?
(A) 3.5 N (B) 0.7 N (C) 1.4 N (D) None

Q.82 There is a horizontal film of soap solution. On it a thread is placed in the form of a loop. The film is pierced inside the loop and the thread becomes a circular loop of radius R. If the surface tension of the loop be T, then what will be the tension in the thread?
(A) $\pi R^2/T$ (b) $\pi R^2 T$ (C) $2\pi RT$ (D) 2RT

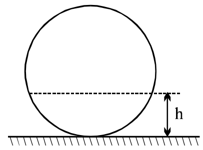
Q.83 A container, whose bottom has round holes with diameter 0.1 mm is filled with water. The maximum height in cm upto which water can be filled without leakage will be what?
Surface tension = 75×10^{-3} N/m and g = 10 m/s^2:
(A) 20 cm (B) 40 cm (C) 30 cm (D) 60 cm

Q.84 If two soap bubbles of different radii are connected by a tube,
(A) air flows from the bigger bubble to the smaller bubble till the sizes become equal
(B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged
(C) air flows from the smaller bubble to the bigger
(D) there is no flow of air.

Q.85 Two soap bubbles with radii r and ($r_1 > r_2$) come in contact. Their common surface has a radius of curvature r.

(A) $r = \dfrac{r_1 + r_2}{2}$ (B) $r = \dfrac{r_1 r_2}{r_1 - r_2}$ (C) $r = \dfrac{r_1 r_2}{r_1 + r_2}$ (D) $r = \sqrt{r_1 r_2}$

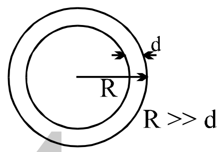
Q.86 A liquid is filled in a spherical container of radius R till a height h. At this positions the liquid surface at the edges is also horizontal. The contact angle is



(A) 0 (B) $\cos^{-1}\left(\dfrac{R-h}{R}\right)$

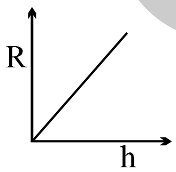
(C) $\cos^{-1}\left(\dfrac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\dfrac{R-h}{R}\right)$

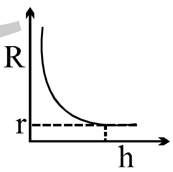
Q.87 A soap bubble has radius R and thickness d (<< R) as shown. It colapses into a spherical drop. The ratio of excess pressure in the drop to the excess pressure inside the bubble is

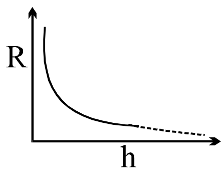


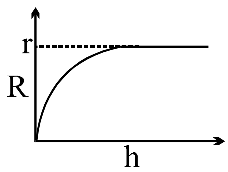
(A) $\left(\dfrac{R}{3d}\right)^{\frac{1}{3}}$ (B) $\left(\dfrac{R}{6d}\right)^{\frac{1}{3}}$ (C) $\left(\dfrac{R}{24d}\right)^{\frac{1}{3}}$ (D) None

Q.88 A long capillary tyube of radius 'r' is initially just vertically completely imerged inside a liquid of angle of contact 0^0. If the tube is slowly raised then relation between radius of curvature of of miniscus inside the capillary tube and displacement (h) of tube can be represented by

(A)


(B)


(C)


(D)


ONE OR MORE THAN ONE OPTION MAY BE CORRECT

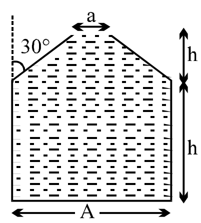
Take approx. 3 minutes for answering each question.

Q.1 A beaker is filled in with water is accelerated a m/s^2 in +x direction. The surface of water shall make on angle

(A) $\tan^{-1}(a/g)$ backwards (B) $\tan^{-1}(a/g)$ forwards

(C) $\cot^{-1}(g/a)$ backwards (D) $\cot^{-1}(g/a)$ forwards

Q.2 The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h. The upper part is a conical vessel of height h with base area 'A' and top area 'a' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height 2h. Neglecting atmospheric pressure.



(A) The force F exerted by the liquid on the base of the vessel is $2h\rho g\,\dfrac{(A+a)}{2}$

(B) the pressure P at the base of the vessel is $2h\rho g\,\dfrac{A}{a}$

(C) the weight of the liquid W is greater than the force exerted by the liquid on the base

(D) the walls of the vessel exert a downward force (F–W) on the liquid.

Q.3 The weight of an empty balloon on a spring balance is w_1. The weight becomes w_2 when the balloon is filled with air. Let the weight of the air itself be w .Neglect the thickness of the balloon when it is filled with air. Also neglect the difference in the densities of air inside & outside the balloon. Then :

(A) $w_2 = w_1$ (B) $w_2 = w_1 + w$ (C) $w_2 < w_1 + w$ (D) $w_2 > w_1$

Q.4 A cubical block of wood of edge 10cm and mass 0.92kg floats on a tank of water with oil of rel. density 0.6 to a depth of 4cm above water. When the block attains equilibrium with four of its sides edges vertical

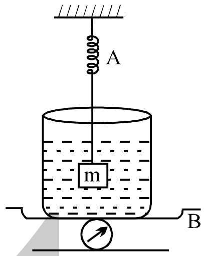
(A) 1cm of it will be above the free surface of oil.

(B) 5cm of it will be under water.

(C) 2cm of it will be above the common surface of oil and water.

(D) 8cm of it will be under water.

Q.5 The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in the figure in this situation:



(A) the balance A will read more than 2 kg

(B) the balance B will read more than 5 kg

(C) the balance A will read less than 2 kg and B will read more than 5 kg

(D) the balances A and B will read 2 kg and 5 kg respectively.

Q.6 **Assertion** : A helium filled balloon does not rise indefinately in air but halts after a certain height.

Reason : Viscosity opposes the motion of balloon.

Choose any one of the following four responses:

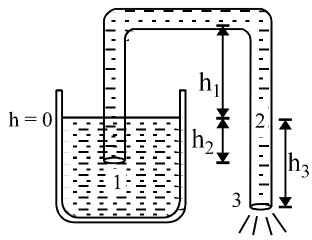
(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

Q.7 Figure shows a siphon. Choose the wrong statement:



(A) Siphon works when $h_3 > 0$

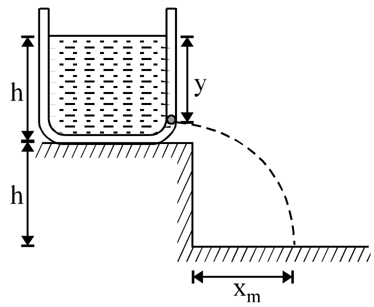
(B) Pressure at point 2 is $P_2 = P_0 - \rho g h_3$

(C) Pressure at point 3 is P_0

(D) None of the above

(P_0 = atmospheric pressure)

Q.8 A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then



(A) $x_m = 2h$

(B) $x_m = 1.5$ h

(C) y = h

(D) y = 0.75 h

Q.9 Water coming out of a horizontal tube at a speed ν strikes normally a vertically wall close to the mouth of the tube and falls down vertically after impact. When the speed of water is increased to 2ν.

(A) the thrust exerted by the water on the wall will be doubled

(B) the thrust exerted by the water on the wall will be four times

(C) the energy lost per second by water strikeup the wall will also be four times

(D) the energy lost per second by water striking the wall be increased eight times.

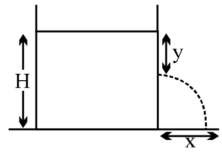
Q.10 A cylindrical vessel is filled with a liquid up to height H. A small hole is made in the vessel at a distance y below the liquid surface as shown in figure. The liquid emerging from the hole strike the ground at distance x

(A) if y is increased from zero to H, x will decrease and then increase

(B) x is maximum for $y = \frac{H}{2}$

(C) the maximum value of x is $\frac{H}{2}$

(D) the maximum value of x increases with the increases in density of the liquid



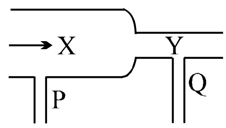
Q.11 A steady flow of water passes along a horizontal tube from a wide section X to the narrower section Y, see figure. Manometers are placed at P and Q at the sections. Which of the statements A, B, C, D, E is most correct?

(A) water velocity at X is greater than at Y

(B) the manometer at P shows lower pressure than at Q

(C) kinetic energy per m^3 of water at X = kinetic energy per m^3 at Y

(D) the manometer at P shows greater pressure than at Y



Q.12 When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble

(A) is greater than the pressure outside it
(B) is less than the pressure outside it
(C) increases as the bubble moves up
(D) decreases as the bubble moves up

ANSWER KEY

ONLY ONE OPTION IS CORRECT

Q.1	B	Q.2	B	Q.3	D	Q.4	C	Q.5	B	Q.6	C	Q.7	B
Q.8	D	Q.9	A	Q.10	D	Q.11	B	Q.12	D	Q.13	B	Q.14	C
Q.15	A	Q.16	B	Q.17	D	Q.18	A	Q.19	B	Q.20	B	Q.21	C
Q.22	C	Q.23	D	Q.24	A	Q.25	B	Q.26	A	Q.27	B	Q.28	A
Q.29	C	Q.30	A	Q.31	B	Q.32	D	Q.33	B	Q.34	B	Q.35	B
Q.36	B	Q.37	A	Q.38	B	Q.39	A	Q.40	C	Q.41	D	Q.42	B
Q.43	C	Q.44	B	Q.45	C	Q.46	B	Q.47	D	Q.48	A	Q.49	D
Q.50	D	Q.51	D	Q.52	C	Q.53	A	Q.54	B	Q.55	C	Q.56	D
Q.57	D	Q.58	B	Q.59	B	Q.60	B	Q.61	A	Q.62	B	Q.63	D
Q.64	C	Q.65	D	Q.66	C	Q.67	A	Q.68	A	Q.69	C	Q.70	C
Q.71	C	Q.72	D	Q.73	B	Q.74	A	Q.75	C	Q.76	D	Q.77	A
Q.78	A	Q.79	C	Q.80	D	Q.81	B	Q.82	D	Q.83	C	Q.84	C
Q.85	B	Q.86	B	Q.87	C	Q.88	B						

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	A,C	Q.2	D	Q.3	A,C	Q.4	C,D
Q.5	B,C	Q.6	B	Q.7	D	Q.8	A,C
Q.9	B,D	Q.10	B	Q.11	D	Q.12	A,D

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PHYSICS

CALORIMETRY AND HEAT TRANSFER

QUESTION FOR SHORT ANSWER

Q.1 Can one object be hotter than another if they are at the same temperature? Explain.

Q.2 What qualities make a particular thermometric property suitable for use in a practical thermometer?

Q.3 You put two uncovered pails of water, one containing hot water and one containing cold water, outside in below-freezing weather. The pail with the hot water will usually begin to freeze first. Why? What would happen if you covered the pails?

Q.4 Can a temperature be assigned to a vacuum?

Q.5 What are the dimensions of α, the coefficient of linear expansion? Does the value of α depend on the unit of length used? When Fahrenheit degrees are used instead of Celsius degrees as the unit of temperature change, does the numerical value of α change? If so, how? If not, prove it.

Q.6 A metal ball can pass through a metal ring. When the ball is heated, however, it gets stuck in the ring. What would happen if the ring, rather than the ball, were heated?

Q.7 Two strips, one of iron and one of zinc, are riveted together side by side to form a straight bar that curves when heated. Why is the iron on the inside of the curve?

Q.8 Explain how the period of a pendulum clock can be kept constant with temperature by attaching vertical tubes of mercury to the bottom of the pendulum.

Q.9 What causes water pipes to burst in the winter?

Q.10 Do the pressure and volume of air in a house change when the furnace raises the temperature significantly? If not, is the ideal gas law violated?

Q.11 If two systems are in thermal equilibrium, they have the same temperature. Is the converse true? That is, if two systems have the same temperature, are they in thermal equilibrium? What can you say about two systems that have different temperatures?

Q.12 As a practical matter, there is always a temperature difference between a system and some part of its environment, however remote. Must there always be some heat transferred because of that temperature difference? Explain.

ONLY ONE OPTION IS CORRECT.

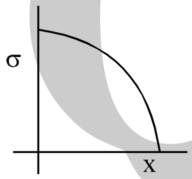
Take approx. 2 minutes for answering each question.

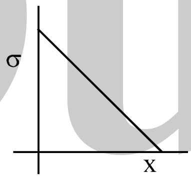
Q.1 Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is ($Y = 2.0 \times 10^{11}\ Nm^{-2}$)
(A) $3.2 \times 10^{9}\ Nm^{-2}$ (B) $3.2 \times 10^{7}\ Nm^{-2}$ (C) $3.6 \times 10^{9}\ Nm^{-2}$
(D) $3.6 \times 10^{7}\ Nm^{-2}$ (E) $4.08 \times 10^{3}\ Nm^{-3}$

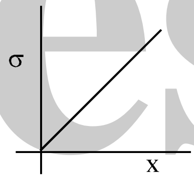
Q.2 A solid sphere of radius R made of of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is
(A) mg/AK (B) mg/3AK (C) mg/A (D) mg/3AR

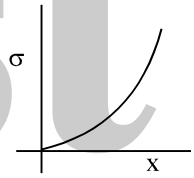
Q.3 A cylindrical wire of radius 1 mm, length 1 m, Young's modulus = $2 \times 10^{11}\ N/m^2$, poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become
(A) 0.99998 mm (B) 0.99999 mm (C) 0.99997 mm (D) 0.99995 mm

Q.4 A uniform rod rotating in gravity free region with certain constant angular velocity. The variation of tensile stress with distance x from axis of rotation is best represented by which of the following graphs.

(A) 

(B) 

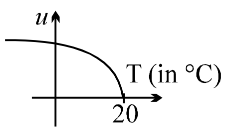
(C) 

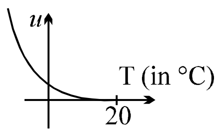
(D) 

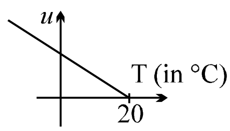
Q.5 A metallic wire of length L is fixed between two rigid supports. If the wire is cooled through a temperature difference ΔT (Y = young's modulus, ρ = density, α = coefficient of linear expansion) then the frequency of transverse vibration is proportional to :

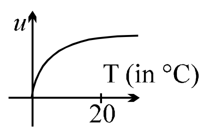
(A) $\frac{\alpha}{\sqrt{\rho Y}}$ (B) $\sqrt{\frac{Y\alpha}{\rho}}$ (C) $\frac{\rho}{\sqrt{Y\alpha}}$ (D) $\sqrt{\frac{\rho\alpha}{Y}}$

Q.6 A metal wire is clamped between two vertical walls. At 20 °C the unstrained length of the wire is exactly equal to the separation between walls. If the temperature of the wire is decreased the graph between elastic energy density (u) and temperature (T) of the wire is

(A) 

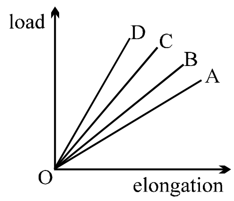
(B) 

(C) 

(D) 

Q.7 A rod of length 2m rests on smooth horizontal floor. If the rod is heated from 0°C to 20°C. Find the longitudinal strain developed? ($\alpha = 5 \times 10^{-5}/°C$)
(A) 10^{-3} (B) 2×10^{-3} (C) Zero (D) None

Q.8 The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line
(A) OB (B) OA (C) OD (D) OC



Q.9 A steel tape gives correct measurement at 20°C. A piece of wood is being measured with the steel tape at 0°C. The reading is 25 cm on the tape, the real length of the given piece of wood must be:
(A) 25 cm (B) <25 cm (C) >25 cm (D) can not say

Q.10 A rod of length 20 cm is made of metal. It expands by 0.075 cm when its temperature is raised from 0°C to 100°C. Another rod of a different metal B having the same length expands by 0.045 cm for the same change in temperature, a third rod of the same length is composed of two parts one of metal A and the other of metal B. Thus rod expand by 0.06 cm.for the same change in temperature. The portion made of metal A has the length :
(A) 20 cm (B) 10 cm (C) 15 cm (D) 18 cm

Q.11 A sphere of diameter 7 cm and mass 266.5 gm floats in a bath of a liquid. As the temperature is raised, the sphere just begins to sink at a temperature 35°C. If the density of a liquid at 0°C is 1.527 gm/cc, then neglecting the expansion of the sphere, the coefficient of cubical expansion of the liquid is f :
(A) 8.486×10^{-4} per °C (B) 8.486×10^{-5} per °C
(C) 8.486×10^{-6} per °C (D) 8.486×10^{-3} per °C

Q.12 The volume of the bulb of a mercury thermometer at 0°C is V_0 and cross section of the capillary is A_0. The coefficient of linear expansion of glass is a_g per °C and the cubical expansion of mercury γ_m per °C. If the mercury just fills the bulb at 0°C, what is the length of mercury column in capillary at T°C.
(A) $\frac{V_0T(\gamma_m+3a_g)}{A_0(1+2a_gT)}$ (B) $\frac{V_0T(\gamma_m-3a_g)}{A_0(1+2a_gT)}$ (C) $\frac{V_0T(\gamma_m+2a_g)}{A_0(1+3a_gT)}$ (D) $\frac{V_0T(\gamma_m-2a_g)}{A_0(1+3a_gT)}$

Q.13 A metallic rod l cm long with a square cross-section is heated through 1°C. If Young's modulus of elasticity of the metal is E and the mean coefficient of linear expansion is α per degree Celsius, then the compressional force required to prevent the rod from expanding along its length is :(Neglect the change of cross-sectional area)
(A) EAαt (B) EAαt/(1 + αt) (C) EAαt/(1–αt) (D) E/αt

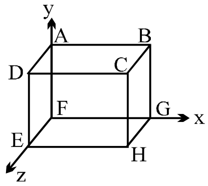
Q.14 The loss in weight of a solid when immersed in a liquid at 0°C is W_0 and at t°C is W. If cubical coefficient of expansion of the solid and the liquid by γ_s and γ_1 respectively, then W is equal to :
(A) $W_0 [1 + (\gamma_s - \gamma_l) t]$ (B) $W_0 [1 - (\gamma_s - \gamma_l)t]$
(C) $W_0 [(\gamma_s - \gamma_l) t]$ (D) $W_0t/(\gamma_s - \gamma_l)$

Q.15 A thin walled cylindrical metal vessel of linear coefficient of expansion 10^{-3} °C^{-1} contains benzenr of volume expansion coefficient 10^{-3}°C^{-1}. If the vessel and its contents are now heated by 10°C, the pressure due to the liquid at the bottom.
(A) increases by 2% (B) decreases by 1% (C) decreases by 2% (D) remains unchanged

Q.16 A rod of length 2m at 0°C and having expansion coefficient $\alpha = (3x + 2) \times 10^{-6}$ °C^{-1} where x is the distance (in cm) from one end of rod. The length of rod at 20°C is :
(A) 2.124 m (B) 3.24 m (C) 2.0120 m (D) 3.124 m

Q.17 A copper ring has a diameter of exactly 25 mm at its temperature of 0°C. An aluminium sphere has a diameter of exactly 25.05 mm at its temperature of 100°C. The sphere is placed on top of the ring and two are allowed to come to thermal equilibrium, no heat being lost to the surrounding. The sphere just passes through the ring at the equilibrium temperature. The ratio of the mass of the sphere & ring is :
(given : $\alpha_{Cu} = 17 \times 10^{-6}$ /°C, $\alpha_{Al} = 2.3 \times 10^{-5}$ /°C, specific heat of Cu = 0.0923 Cal/g°C and specific heat of Al = 0.215 cal/g°C)
(A) 1/5 (B) 23/108 (C) 23/54 (D) 216/23

Q.18 A cuboid ABCDEFGH is anisotropic with $\alpha_x = 1 \times 10^{-5}$ /°C, $\alpha_y = 2 \times 10^{-5}$ /°C, $\alpha_z = 3 \times 10^{-5}$ /°C. Coefficient of superficial expansion of faces can be



(A) $\beta_{ABCD} = 5 \times 10^{-5}$ /°C (B) $\beta_{BCGH} = 4 \times 10^{-5}$ /°C
(C) $\beta_{CDEH} = 3 \times 10^{-5}$ /°C (D) $\beta_{EFGH} = 2 \times 10^{-5}$ /°C

Q.19 An open vessel is filled completely with oil which has same coefficient of volume expansion as that of the vessel. On heating both oil and vessel,
(A) the vessel can contain more volume and more mass of oil
(B) the vessel can contain same volume and same mass of oil
(C) the vessel can contain same volume but more mass of oil
(D) the vessel can contain more volume but same mass of oil

Q.20 A metal ball immersed in Alcohol weights W_1 at 0°C and W_2 at 50°C. The coefficient of cubical expansion of the metal $(\gamma)_m$ is less than that of alcohol $(\gamma)_{Al}$. Assuming that density of metal is large compared to that of alcohol, it can be shown that
(A) $W_1 > W_2$ (B) $W_1 = W_2$ (C) $W_1 < W_2$ (D) any of (A) , (B) or (C)

Q.21 A solid ball is completely immersed in a liquid. The coefficients of volume expansion of the ball and liquid are 3×10^{-6} and 8×10^{-6} per °C respectively. The percentage change in upthrust when the temperature is increased by 100 °C is
(A) 0.5 % (B) 0.11 % (C) 1.1 % (D) 0.05 %

Q.22 A thin copper wire of length L increase in length by 1% when heated from temperature T_1 to T_2. What is the percentage change in area when a thin copper plate having dimensions $2L \times L$ is heated from T_1 to T_2?
(A) 1% (B) 2% (C) 3% (D) 4%

Q.23 If two rods of length L and 2L having coefficients of linear expansion α and 2α respectively are connected so that total length becomes 3L, the average coefficient of linear expansion of the composition rod equals:
(A) $\frac{3}{2}\alpha$ (B) $\frac{5}{2}\alpha$ (C) $\frac{5}{3}\alpha$ (D) none of these

Q.24 The bulk modulus of copper is 1.4×10^{11} Pa and the coefficient of linear expansion is 1.7×10^{-5} $(C°)^{-1}$. What hydrostatic pressure is necessary to prevent a copper block from expanding when its temperature is increased from 20°C to 30°C?
(A) 6.0×10^5 Pa (B) 7.1×10^7 Pa (C) 5.2×10^6 Pa (D) 40 atm

Q.25 The coefficients of thermal expansion of steel and a metal X are respectively 12×10^{-6} and 2×10^{-6} per°C. At 40°C, the side of a cube of metal X was measured using a steel vernier callipers. The reading was 100 mm. Assuming that the calibration of the vernier was done at 0°C, then the actual length of the side of the cube at 0°C will be
(A) > 100 mm (B) < 100 mm (C) = 100 mm (D) data insufficient to conclude

Q.26 A glass flask contains some mercury at room temperature. It is found that at different temperatures the volume of air inside the flask remains the same. If the volume of mercury in the flask is 300 cm^3, then volume of the flask is (given that coefficient of volume expansion of mercury and coefficient of linear expansion of glass are 1.8×10^{-4} $(°C)^{-1}$ and 9×10^{-6} $(°C)^{-1}$ respectively)
(A) 4500 cm^3 (B) 450 cm^3 (C) 2000 cm^3 (D) 6000 cm^3

Question No. 27 to 31 (5 question)

Solids and liquids both expand on heating. The density of substance decreases on expanding according to the relation

$$\rho_2 = \frac{\rho_1}{1+\gamma(T_2 - T_1)}$$

where, $\rho_1 \longrightarrow$ density at T_1

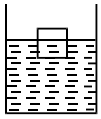
$\rho_2 \longrightarrow$ density at T_2

$\gamma \longrightarrow$ coeff. of volume expansion of substances

when a solid is submerged in a liquid, liquid exerts an upward force on solid which is equal to the weight of liquid displaced by submerged part of solid.

Solid will float or sink depends on relative densities of solid and liquid.

A cubical block of solid floats in a liquid with half of its volume submerged in liquid as shown in figure (at temperature T)



$\alpha_S \longrightarrow$ coeff. of linear expansion of solid

$\gamma_L \longrightarrow$ coeff. of volume expansion of liquid

$\rho_S \longrightarrow$ density of solid at temp. T

$\rho_L \longrightarrow$ density of liquid at temp. T

Q.27 The relation between densities of solid and liquid at temperature T is

(A) $\rho_S = 2\rho_L$ (B) $\rho_S = (1/2)\rho_L$ (C) $\rho_S = \rho_L$ (D) $\rho_S = (1/4)\rho_L$

Q.28 If temperature of system increases, then fraction of solid submerged in liquid

(A) increases (B) decreases (C) remains the same (D) inadequate information

Q.29 Imagine fraction submerged does not change on increasing temperature the relation between γ_L and α_S is

(A) $\gamma_L = 3\alpha_S$ (B) $\gamma_L = 2\alpha_S$ (C) $\gamma_L = 4\alpha_S$ (D) $\gamma_L = (3/2)\alpha_S$

Q.30 Imagine the depth of the block submerged in the liquid does not change on increasing temperature then

(A) $\gamma_L = 2\alpha$ (B) $\gamma_L = 3\alpha$ (C) $\gamma_L = (3/2)\alpha$ (D) $\gamma_L = (4/3)\alpha$

Q.31 Assume block does not expand on heating. The temperature at which the block just begins to sink in liquid is

(A) $T + 1/\gamma_L$ (B) $T + 1/(2\gamma_L)$ (C) $T + 2/\gamma_L$ (D) $T + \gamma_L/2$

Q.32 The coefficient of apparent expansion of a liquid in a copper vessel is C and in a silver vessel is S. The coefficient of volume expansion of copper is γ_c. What is the coefficient of linear expansion of silver?

(A) $\frac{(C+\gamma_c+S)}{3}$ (B) $\frac{(C-\gamma_c+S)}{3}$ (C) $\frac{(C+\gamma_c-S)}{3}$ (D) $\frac{(C-\gamma_c-S)}{3}$

Q.33 An aluminium container of mass 100 gm contains 200 gm of ice at – 20°C. Heat is added to the system at the rate of 100 cal/s. The temperature of the system after 4 minutes will be (specific heat of ice = 0.5 and L = 80 cal/gm, specific heat of A*l* = 0.2 cal/gm/°C)

(A) 40.5°C (B) 25.5°C (C) 30.3°C (D) 35.0°C

Q.34 A thermally insulated vessel contains some water at 0^0C. The vessel is connected to a vacuum pump to pump out water vapour. This results in some water getting frozen. It is given Latent heat of vaporization of water at 0°C $=21 \times 10^5$ J/kg and latent heat of freezing of water $= 3.36 \times 10^5$ J/kg. The maximum percentage amount of water that will be solidified in this manner will be
(A) 86.2% (B) 33.6% (C) 21% (D) 24.36%

Q.35 A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body = 0.1 Cal/gm°C.
(A) 1 kg (B) 1.5 kg (C) 2 kg (D) 2.5 kg

Q.36 10 gm of ice at 0°C is kept in a calorimeter of water equivalent 10 gm. How much heat should be supplied to the apparatus to evaporate the water thus formed? (Neglect loss of heat)
(A) 6200 cal (B) 7200 cal (C) 13600 cal (D) 8200 cal

Q.37 Heat is being supplied at a constant rate to a sphere of ice which is melting at the rate of 0.1 gm/sec. It melts completely in 100 sec. The rate of rise of temperature thereafter will be
(Assume no loss of heat.)
(A) 0.8 °C/sec (B) 5.4 °C/sec (C) 3.6 °C/sec (D) will change with time

Q.38 1 kg of ice at – 10°C is mixed with 4.4 kg of water at 30°C. The final temperature of mixture is :
(specific heat of ice is 2100 J/kg/k)
(A) 2.3°C (B) 4.4°C (C) 5.3°C (D) 8.7°C

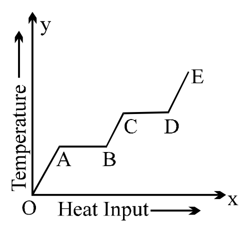
Q.39 Steam at 100°C is added slowly to 1400 gm of water at 16°C until the temperature of water is raised to 80°C. The mass of steam required to do this is ($L_V = 540$ cal/gm) :
(A) 160 gm (B) 125 mg (C) 250 gm (D) 320 gm

Q.40 A 2100 W continuous flow geyser (instant geyser) has water inlet temperature = 10°C while the water flows out at the rate of 20 g/sec. The outlet temperature of water must be about
(A) 20°C (B) 30°C (C) 35°C (D) 40°C

Q.41 A continuous flow water heater (geyser) has an electrical power rating = 2 kW and efficiency of conversion of electrical power into heat = 80%. If water is flowing through the device at the rate of 100 cc/sec, and the inlet temperature is 10°C, the outlet temperature will be
(A) 12.2°C (B) 13.8°C (C) 20°C (D) 16.5°C

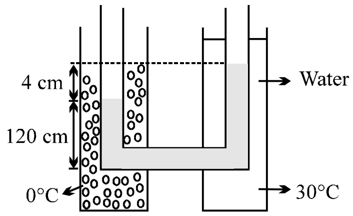
Q.42 Ice at 0°C is added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and has all melted the temperature of the flask and contents is 40°C. When a further 80g of ice has been added and has all metled, the temperature of the whole is 10°C. Calculate the specific latent heat of fusion of ice.[Take S_w =1 cal /gm °C.]
(A) 3.8×10^5 J/ kg (B) 1.2×10^5 J/ kg (C) 2.4×10^5 J/ kg (D) 3.0×10^5 J/ kg

Q.43 A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represent.



(A) latent heat of liquid
(B) latent heat of vapour
(C) heat capacity of vapour
(D) inverse of heat capacity of vapour

Q.44 Two vertical glass tubes filled with a liquid are connected by a capillary tube as shown in the figure. The tube on the left is put in an ice bath at 0°C while the tube on the right is kept at 30°C in a water bath. The difference in the levels of the liquid in the two tubes is 4 cm while the height of the liquid column at 0°C is 120 cm. The coefficient of volume expansion of liquid is (Ignore expansion of glass tube)



(A) 22×10^{-4}/°C (B) 1.1×10^{-4}/°C
(C) 11×10^{-4}/°C (D) 2.2×10^{-4}/°C

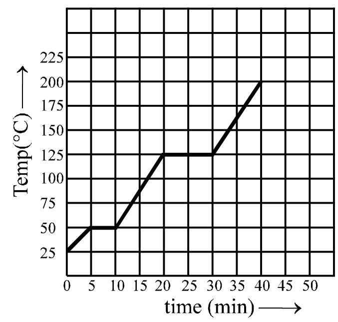
Q.45 A system S receives heat continuously from an electrical heater of power 10W. The temperature of S becomes constant at 50°C when the surrounding temperature is 20°C. After the heater is switched off, S cools from 35.1°C to 34.9°C in 1 minute. The heat capacity of S is
(A) 100J/°C (B) 300J/°C (C) 750J/°C (D) 1500J/°C

Q.46 A block of ice with mass *m* falls into a lake. After impact, a mass of ice *m*/5 melts. Both the block of ice and the lake have a temperature of 0°C. If L represents the heat of fusion, the minimum distance the ice fell before striking the surface is
(A) $\frac{L}{5g}$ (B) $\frac{5L}{g}$ (C) $\frac{gL}{5m}$ (D) $\frac{mL}{5g}$

Q.47 Pure water super cooled to −15°C is contained in a thermally insulated flask. Small amount of ice is thrown into the flask. The fraction of water frozen into ice is :
(A) 3/35 (B) 6/35 (C) 6/29 (D) 2/35

Q.48 The specific heat of a metal at low temperatures varies according to $S = aT^3$ where a is a constant and T is the absolute temperature. The heat energy needed to raise unit mass of the metal from T = 1 K to T = 2 K is
(A) 3 a (B) $\frac{15a}{4}$ (C) $\frac{2a}{3}$ (D) $\frac{12a}{5}$

Q.49 The graph shown in the figure represent change in the temperature of 5 kg of a substance as it abosrbs heat at a constant rate of 42 kJ min^{-1}. The latent heat of vapourazation of the substance is :



(A) 630 kJ kg^{-1}
(B) 126 kJ kg^{-1}
(C) 84 kJ kg^{-1}
(D) 12.6 kJ kg^{-1}

Q.50 The density of a material A is 1500 kg/m^3 and that of another material B is 2000 kg/m^3. It is found that the heat capacity of 8 volumes of A is equal to heat capacity of 12 volumes of B. The ratio of specific heats of A and B will be
(A) 1 : 2 (B) 3 : 1 (C) 3 : 2 (D) 2 : 1

Q.51 Find the amount of heat supplied to decrease the volume of an ice water mixture by 1 cm^3 without any change in temperature. ($\rho_{ice} = 0.9\ \rho_{water}$, $L_{ice} = 80$ cal/gm).
(A) 360 cal (B) 500 cal (C) 720 cal (D) none of these

Q.52 Some steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C so that the temperature of the calorimeter and its contents rises to 80°C. What is the mass of steam condensing. (in kg)
(A) 0.130 (B) 0.065 (C) 0.260 (D) 0.135

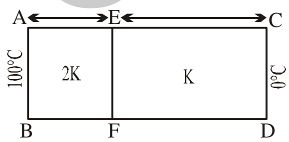
Q.53 One end of a 2.35m long and 2.0cm radius aluminium rod ($K = 235\ W.m^{-1}K^{-1}$) is held at 20^0C. The other end of the rod is in contact with a block of ice at its melting point. The rate in $kg.s^{-1}$ at which ice melts is
(A) $48\pi \times 10^{-6}$ (B) $24\pi \times 10^{-6}$ (C) $2.4\pi \times 10^{-6}$ (D) $4.8\pi \times 10^{-6}$

[Take latent heat of fusion for ice as $\frac{10}{3} \times 10^5 J.kg^{-1}$]

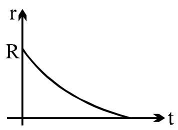
Q.54 Four rods of same material with different radii r and length l are used to connect two reservoirs of heat at different temperatures. Which one will conduct most heat ?
(A) r =2cm, l=0.5m (B) r = 2cm, l = 2m (C) r =0.5cm, l =0.5m (D) r = 1 cm, l = 1 m

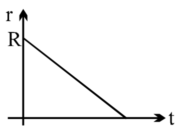
Q.55 A cylinder of radius R made of a material of thermal conductivity k_1 is surrounded by a cylindrical shell of inner radius R and outer radius 2R made of a material of thermal conductivity k_2. The two ends of the combined system are maintained at different temperatures. There is no loss of heat from the cylindrical surface and the system is in steady state. The effective thermal conductivity of the system is
(A) $k_1 + k_2$ (B) $\frac{k_1k_2}{k_1+k_2}$ (C) $\frac{1}{4}(k_1 + 3k_2)$ (D) $\frac{1}{4}(3k + k_2)$

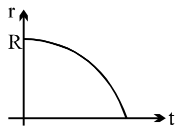
Q.56 Heat is conducted across a composite block of two slabs of thickness d and 2d. Their thermal conductivities are 2k and k respectively. All the heat entering the face AB leaves from the face CD. The temperature in °C of the junction EF of the two slabs is :
(A) 20 (B) 50 (C) 60 (D)80

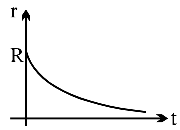


Q.57 A sphere of ice at 0°C having initial radius R is placed in an environment having ambient temperature > 0°C. The ice melts uniformly, such that shape remains spherical. After a time 't' the radius of the sphere has reduced to r. Assuming the rate of heat absorption is proportional to the surface area of the sphere at any moment, which graph best depicts r (t).

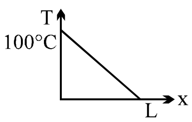
(A) 

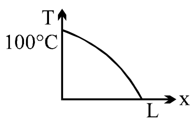
(B) 

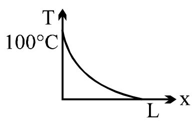
(C) 

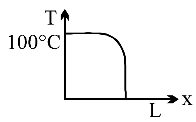
(D) 

Q.58 A rod of length L and uniform cross-sectional area has varying thermal conductivity which changes linearly from 2K at end A to K at the other end B. The ends A and B of the rod are maintained at constant temperature 100°C and 0°C, respectively. At steady state, the graph of temperature : T = T(x) where x = distance from end A will be

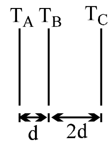
(A) 

(B) 

(C) 

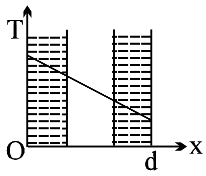
(D) 

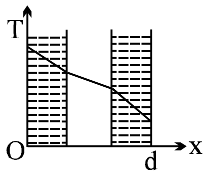
Q.59 Two sheets of thickness d and 2 d and same area are touching each other on their face. Temperature T_A, T_B, T_C shown are in geometric progression with common ratio r = 2. Then ratio of thermal conductivity of thinner and thicker sheet are

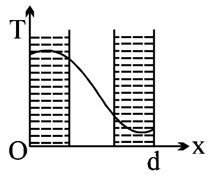


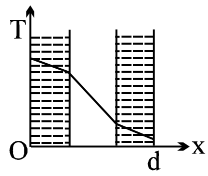
(A) 1 (B) 2 (C) 3 (D) 4

Q.60 The wall with a cavity consists of two layers of brick separated by a layer of air. All three layers have the same thickness and the thermal conductivity of the brick is much greater than that of air. The left layer is at a higher temperature than the right layer and steady state condition exists. Which of the following graphs predicts correctly the variation of temperature T with distance d inside the cavity?

(A) 

(B) 

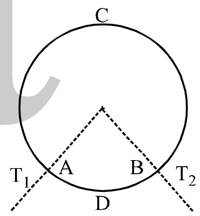
(C) 

(D) 

Q.61 A wall has two layer A and B each made of different material, both the layers have the same thickness. The thermal conductivity of the material A is twice that of B. Under thermal equilibrium the temperature difference across the wall B is 36°C. The temperature difference across the wall A is

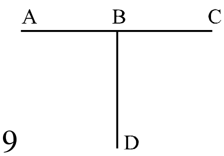
(A) 6°C (B) 12°C (C) 18°C (D) 72°C

Q.62 A ring consisting of two parts ADB and ACB of same conductivity k carries an amount of heat H. The ADB part is now replaced with another metal keeping the temperatures T_1 and T_2 constant. The heat carried increases to 2H. What should be the conductivity of the new ADB part? Given $\dfrac{ACB}{ADB} = 3$:



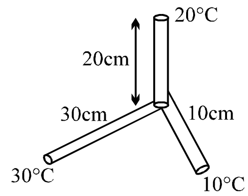
(A) $\dfrac{7}{3}$ k (B) 2 k (C) $\dfrac{5}{2}$ k (D) 3 k

Q.63 Three conducting rods of same material and cross-section are shown in figure. Temperatures of A, D and C are maintained at 20°C, 90°C and 0°C. The ratio of lengths of BD and BC if there is no heat flow in AB is:



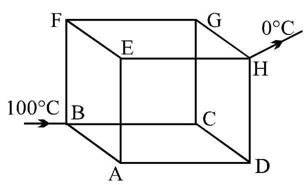
(A) 2 / 7 (B) 7 / 2 (C) 9 / 2 (D) 2 / 9

Q.64 Three rods made of the same material and having same cross-sectional area but different lengths 10cm, 20 cm and 30 cm are joined as shown. The temperature of the joint is:



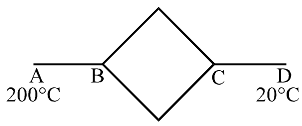
(A) 20°C (B) 23.7°C (C) 16.4°C (D) 18.2°C

Q.65 Twelve conducting rods form the riders of a uniform cube of side 'l'. If in steady state, B and H ends of the rod are at 100°C and 0°C. Find the temperature of the junction 'A'.



(A) 80°C (B) 60°C (C) 40°C (D) 70°C

Q.66 Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperature of 200°C and 20°C respectively. The temperature of junction B will be:

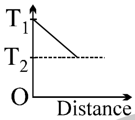


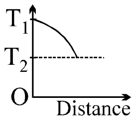
(A) 120°C (B) 100°C (C) 140°C (D) 80°C

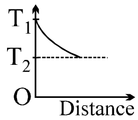
Q.67 A metallic rod of cross-sectional area 9.0 cm^2 and length 0.54 m, with the surface insulated to prevent heat loss, has one end immersed in boiling water and the other in ice-water mixture. The heat conducted through the rod melts the ice at the rate of 1 gm for every 33 sec. The thermal conductivity of the rod is
(A) 330 $Wm^{-1}\ K^{-1}$ (B) 60 $Wm^{-1}\ K^{-1}$ (C) 600 $Wm^{-1}\ K^{-1}$ (D) 33 $Wm^{-1}\ K^{-1}$

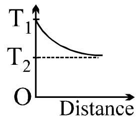
Q.68 A hollow sphere of inner radius R and outer radius 2R is made of a material of thermal conductivity K. It is surrounded by another hollow sphere of inner radius 2R and outer radius 3R made of same material of thermal conductivity K. The inside of smaller sphere is maintained at 0°C and the outside of bigger sphere at 100°C. The system is in steady state. The temperature of the interface will be :
(A) 50°C (B) 70°C (C) 75°C (D) 45°C

Q.69 The ends of a metal bas of constant cross-sectional area are maintained at temperatures T_1 and T_2 which are both higher than the temperature of the surroundings. If the bar is unlagged, which one of the following sketches best represents the variation of temperature with distance along the bar?

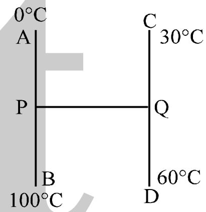
(A) 

(B) 

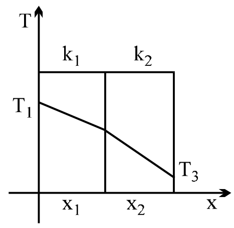
(C) 

(D) 

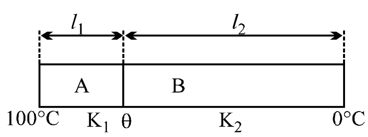
Q.70 Three identical rods AB, CD and PQ are joined as shown. P and Q are mid points of AB and CD respectively. Ends A, B, C and D are maintained at 0°C, 100°C, 30°C and 60°C respectively. The direction of heat flow in PQ is
(A) from P to Q (B) from Q to P
(C) heat does not flow in PQ (D) data not sufficient



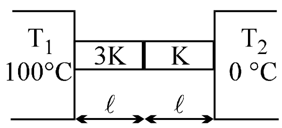
Q.71 The temperature drop through each layer of a two layer furnace wall is shown in figure. Assume that the external temperature T_1 and T_3 are maintained constant and $T_1 > T_3$. If the thickness of the layers x_1 and x_2 are the same, which of the following statements are correct.
(A) $k_1 > k_2$
(B) $k_1 < k_2$
(C) $k_1 = k_2$ but heat flow through material (1) is larger then through (2)
(D) $k_1 = k_2$ but heat flow through material (1) is less than that through (2)



Q.72 Two rods A and B of different materials but same cross section are joined as in figure. The free end of A is maintained at 100°C and the free end of B is maintained at 0°C. If $l_2 = 2l_1$, $K_1 = 2K_2$ and rods are thermally insulated from sides to prevent heat losses then the temperature θ of the junction of the two rods is
(A) 80°C (B) 60°C (C) 40°C (D) 20°C



Question No. 73 to 75 (3 questions)
Two rods A and B of same cross-sectional are A and length l connected in series between a source (T_1 = 100°C) and a sink (T_2 = 0°C) as shown in figure. The rod is laterally insulated



Q.73 The ratio of the thermal resistance of the rod is

(A) $\frac{R_A}{R_B} = \frac{1}{3}$ (B) $\frac{R_A}{R_B} = 3$ (C) $\frac{R_A}{R_B} = \frac{3}{4}$ (D) $\frac{4}{3}$

Q.74 If T_A and T_B are the temperature drops across the rod A and B, then

(A) $\frac{T_A}{T_B} = \frac{3}{1}$ (B) $\frac{T_A}{T_B} = \frac{1}{3}$ (C) $\frac{T_A}{T_B} = \frac{3}{4}$ (D) $\frac{T_A}{T_B} = \frac{4}{3}$

Q.75 If G_A and G_B are the temperature gradients across the rod A and B, then

(A) $\frac{G_A}{G_B} = \frac{3}{1}$ (B) $\frac{G_A}{G_B} = \frac{1}{3}$ (C) $\frac{G_A}{G_B} = \frac{3}{4}$ (D) $\frac{G_A}{G_B} = \frac{4}{3}$

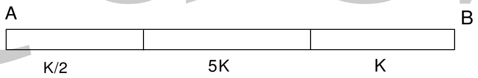
Q.76 Two sheets of thickness d and 3d, are touching each other. The temperature just outside the thinner sheet side is A, and on the side of the thicker sheet is C. The interface temperature is B. A, B and C are in arithmetic progressing, the ratio of thermal conductivity of thinner sheet and thicker sheet is

(A) 1 : 3 (B) 3 : 1 (C) 2 : 3 (D) 1 : 9

Q.77 A cylindrical rod with one end in a steam chamber and the outer end in ice results in melting of 0.1 gm of ice per second. If the rod is replaced by another with half the length and double the radius of the first and if the thermal conductivity of material of second rod is 1/4 that of first, the rate at which ice melts is gm/sec will be

(A) 3.2 (B) 1.6 (C) 0.2 (D) 0.1

Q.78 A composite rod made of three rods of equal length and cross-section as shown in the fig. The thermal conductivities of the materials of the rods are K/2, 5K and K respectively. The end A and end B are at constant temperatures. All heat entering the face A goes out of the end B there being no loss of heat from the sides of the bar. The effective thermal conductivity of the bar is



(A) 15K/16 (B) 6K/13 (C) 5K/16 (D) 2K/13.

Q.79 A rod of length L with sides fully insulated is of a material whose thermal conductivity varies with temperature as $K=\frac{\alpha}{T}$, where α is a constant. The ends of the rod are kept at temperature T_1 and T_2. The temperature T at x, where x is the distance from the end whose temperature is T_1 is

(A) $T_1\left(\frac{T_2}{T_1}\right)^{\frac{x}{L}}$ (B) $\frac{x}{L}\ln\frac{T_2}{T_1}$ (C) $T_1 e^{\frac{T_2 x}{T_1 L}}$ (D) $T_1 + \frac{T_2 - T_1}{L}x$

Q.80 The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $3/4\lambda_0$, the power radiated by it will increase by a factor of

(A) 4/3 (B) 16/9 (C) 64/27 (D) 256/81

Q.81 A black metal foil is warmed by radiation from a small sphere at temperature 'T' and at a distance 'd'. It is found that the power received by the foil is P. If both the temperature and distance are doubled, the power received by the foil will be :

(A) 16 P (B) 4 P (C) 2 P (D) P

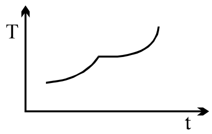
Q.82 Star S_1 emits maximum radiation of wavelength 420 nm and the star S_2 emits maximum radiation of wavelength 560 nm, what is the ratio of the temperature of S_1 and S_2 :

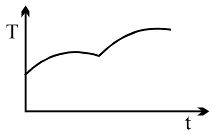
(A) 4/3 (B) $(4/3)^{1/4}$ (C) 3/4 (D) $(3/4)^{1/2}$

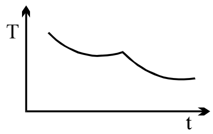
Q.83 Spheres P and Q are uniformly constructed from the same material which is a good conductor of heat and the radius of Q is thrice the radius of P. The rate of fall of temperature of P is x times that of Q when both are at the same surface temperature. The value of x is :

(A) 1/4 (B) 1/3 (C) 3 (D) 4

Q.84 An ice cube at temperature – 20°C is kept in a room at temperature 20°C. The variation of temperature of the body with time is given by

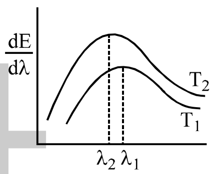
(A) 

(B) 

(C) 

(D) 

Q.85 The spectral emissive power E_λ for a body at temperature T_1 is plotted against the wavelength and area under the curve is found to be A. At a different temperature T_2 the area is found to be 9A. Then λ_1/λ_2 =



(A) 3 (B) 1/3 (C) $1/\sqrt{3}$ (D) $\sqrt{3}$

Q.86 The intensity of radiation emitted by the Sun has its maximum value at a wavelength of 510 nm and that emitted by the North Star has the maximum value at 350 nm. If these stars behave like black bodies then the ratio of the surface temperature of the Sun and the North Star is

(A) 1.46 (B) 0.69 (C) 1.21 (D) 0.83

Q.87 Two bodies P and Q have thermal emissivities of ε_P and ε_Q respectively. Surface areas of these bodies are same and the total radiant power is also emitted at the same rate. If temperature of P is θ_P kelvin then temperature of Q i.e. θ_Q is

(A) $\left(\frac{\varepsilon_Q}{\varepsilon_P}\right)^{1/4}\theta_P$ (B) $\left(\frac{\varepsilon_P}{\varepsilon_Q}\right)^{1/4}\theta_P$ (C) $\left(\frac{\varepsilon_Q}{\varepsilon_P}\right)^{1/4}\times\frac{1}{\theta_P}$ (D) $\left(\frac{\varepsilon_Q}{\varepsilon_P}\right)^{4}\theta_P$

Q.88 A black body calorimeter filled with hot water cools from 60°C to 50°C in 4 min and 40°C to 30°C in 8 min. The approximate temperature of surrounding is :

(A) 10°C (B) 15°C (C) 20°C (D) 25°C

Q.89 The rate of emission of radiation of a black body at 273°C is E, then the rate of emission of radiation of this body at 0°C will be

(A) $\frac{E}{16}$ (B) $\frac{E}{4}$ (C) $\frac{E}{8}$ (D) 0

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 Four rods A, B, C, D of same length and material but of different radii r, $r\sqrt{2}$, $r\sqrt{3}$ and 2r respectively are held between two rigid walls. The temperature of all rods is increased by same amount. If the rods donot bend, then

(A) the stress in the rods are in the ratio 1 : 2 : 3 : 4.

(B) the force on the rod exerted by the wall are in the ratio 1 : 2 : 3 : 4.

(C) the energy stored in the rods due to elasticity are in the ratio 1 : 2 : 3 : 4.

(D) the strains produced in the rods are in the ratio 1 : 2 : 3 : 4.

Q.2 A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is l.

(A) Loss in gravitational potential energy of M is Mgl

(B) The elastic potential energy stored in the wire is Mgl

(C) The elastic potential energy stored in the wire is 1/2 Mgl

(D) Heat produced is 1/2 Mgl.

Q.3 When the temperature of a copper coin is raised by 80°C, its diameter increases by 0.2%.

(A) Percentage rise in the area of a face is 0.4 %

(B) Percentage rise in the thickness is 0.4 %

(C) Percentage rise in the volume is 0.6 %

(D) Coefficient of linear expansion of copper is 0.25×10^{-4} $C°^{-1}$.

Q.4 An experiment is perfomed to measure the specific heat of copper. A lump of copper is heated in an oven, then dropped into a beaker of water. To calculate the specific heat of copper, the experimenter must know or measure the value of all of the quantities below EXCEPT the

(A) heat capacity of water and beaker

(B) original temperature of the copper and the water

(C) final (equilibrium) temperature of the copper and the water

(D) time taken to achieve equilibrium after the copper is dropped into the water

Q.5 One end of a conducting rod is maintained at temperature 50°C and at the other end, ice is melting at 0°C. The rate of melting of ice is doubled if:

(A) the temperature is made 200°C and the area of cross-section of the rod is doubled

(B) the temperature is made 100°C and length of rod is made four times

(C) area of cross-section of rod is halved and length is doubled

(D) the temperature is made 100°C and the area of cross-section of rod and length both are doubled.

Q.6 Two metallic sphere A and B are made of same material and have got identical surface finish. The mass of sphere A is four times that of B. Both the spheres are heated to the same temperature and placed in a room having lower temperature but thermally insulated from each other.

(A) The ratio of heat loss of A to that of B is $2^{4/3}$.

(B) The ratio of heat loss of A to that of B is $2^{2/3}$.

(C) The ratio of the initial rate of cooling of A to that of B is $2^{-2/3}$.

(D) The ratio of the initial rate of cooling of A to that of B is $2^{-4/3}$.

Q.7 Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies radiate energy at the same rate. The wavelength λ_B, corresponding to the maximum spectral radiancy in the radiation from B, is shifted from the wavelength corresponding to the maximum spectral radiancy in the radiation from A by 1.00 μm. If the temperature of A is 5802 K,

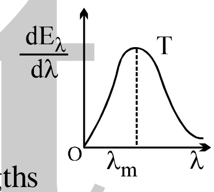
(A) the temperature of B is 1934 K
(B) $\lambda_B = 1.5$ μm
(C) the temperature of B is 11604 K
(D) the temperature of B is 2901 K

Q.8 Three bodies A, B and C have equal surface area and thermal emissivities in the ratio $e_A : e_B : e_C = 1 : \frac{1}{2} : \frac{1}{4}$. All the three bodies are radiating at same rate. Their wavelengths corresponding to maximum intensity are λ_A, λ_B and λ_C respectively and their temperatures are T_A, T_B and T_C on kelvin scale, then select the **incorrect** statement.

(A) $\sqrt{T_A T_C} = T_B$
(B) $\sqrt{\lambda_A \lambda_C} = \lambda_B$
(C) $\sqrt{e_A T_A}\ \sqrt{e_C T_C} = e_B T_B$
(D) $\sqrt{e_A \lambda_A T_A \cdot e_B \lambda_B T_B} = e_C \lambda_C T_C$

Question No. 9 to 11 (3 questions)

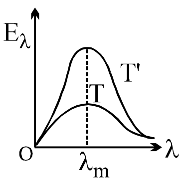
The figure shows a radiant energy spectrum graph for a black body at a temperature T.

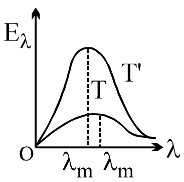


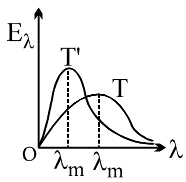
Q.9 Choose the correct statement(s)
(A) The radiant energy is not equally distributed among all the possible wavelengths
(B) For a particular wavelength the spectral intensity is maximum
(C) The area under the curve is equal to the total rate at which heat is radiated by the body at that temperature
(D) None of these

Q.10 If the temperature of the body is raised to a higher temperature T', then choose the correct statement(s)
(A) The intensity of radiation for every wavelength increases
(B) The maximum intensity occurs at a shorter wavelength
(C) The area under the graph increases
(D) The area under the graph is proportional to the fourth power of temperature

Q.11 Identify the graph which correctly represents the spectral intensity versus wavelength graph at two temperatures T' and T (T < T')

(A)


(B)


(C)


(D) none of these

Answer Key

ONLY ONE OPTION IS CORRECT.

Q.1	E	Q.2	B	Q.3	D	Q.4	A	Q.5	B	Q.6	B	Q.7	C
Q.8	C	Q.9	B	Q.10	B	Q.11	A	Q.12	B	Q.13	B	Q.14	A
Q.15	C	Q.16	C	Q.17	C	Q.18	C	Q.19	D	Q.20	C	Q.21	D
Q.22	B	Q.23	C	Q.24	B	Q.25	A	Q.26	C	Q.27	B	Q.28	A
Q.29	A	Q.30	A	Q.31	A	Q.32	C	Q.33	B	Q.34	A	Q.35	B
Q.36	D	Q.37	A	Q.38	D	Q.39	A	Q.40	C	Q.41	B	Q.42	A
Q.43	D	Q.44	C	Q.45	D	Q.46	A	Q.47	B	Q.48	B	Q.49	C
Q.50	D	Q.51	C	Q.52	A	Q.53	C	Q.54	A	Q.55	C	Q.56	D
Q.57	B	Q.58	B	Q.59	A	Q.60	D	Q.61	C	Q.62	A	Q.63	B
Q.64	C	Q.65	B	Q.66	C	Q.67	B	Q.68	C	Q.69	C	Q.70	A
Q.71	A	Q.72	A	Q.73	A	Q.74	B	Q.75	B	Q.76	A	Q.77	C
Q.78	A	Q.79	A	Q.80	D	Q.81	B	Q.82	A	Q.83	C	Q.84	B
Q.85	D	Q.86	B	Q.87	B	Q.88	B	Q.89	A				

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	B,C	Q.2	A,CD	Q.3	A,C,D	Q.4	D
Q.5	D	Q.6	A,C	Q.7	A,B	Q.8	D
Q.9	A,B	Q.10	A,B,C,D	Q.11	B		

Power Coaching

QUEST TUTORIALS

TARGET IIT JEE

PHYSICS

KTG & THERMODYNAMICS

QUESTION FOR SHORT ANSWER

Q.1 How is the speed of sound related to the gas variables in the kinetic theory model?

Q.2 How can you best use a spoon to cool a cup of coffee? Stirring – which involves doing work– would seem to heat the coffee rather than cool it.

Q.3 How does a layer of snow protect plants during cold weather? During freezing spells, citrus growers in Florida often spray their fruit with water, hoping it will freeze. How does that help?

Q.4 Explain why the latent heat of vaporization of a substance might be expected to be considerably greater than its latent heat of fusion.

Q.5 A thermos bottle contains coffee. The thermos bottle is vigorously shaken. Consider the coffee as the system.

(a) Does its temperature rise? (b) Has heat been added to it?

(c) Has work been done on it? (d) Has its internal energy changed?

Q.6 Consider that heat can be transferred by convection and radiation, as well as by conduction, and explain why a thermos bottle is doubled-walled, evacuated, and silvered.

Q.7 The outside surface of a metallic container of a gas is buffed vigorously by a polishing wheel. Is the energy transferred to the gas inside the container called heat or work? Explain.

Q.8 Can you "heat up" a bowl of soup without adding heat? Must the temperature of a system change if heat is added?

Q.9 An ice cube is placed in a well-insulated beaker of lukewarm water. Take the system to consist of the ice cube and the water. The ice melts, and the final state of this system is liquid water at some final lower temperatures. Is this process adiabatic? What provided the energy to melt the ice?

Q.10 In a cycle the final state of the system is the same as the initial state. Given the heat added in a cycle, can you determine how much work is done by the system? Explain.

Q.11 In a pure gas, all molecules are identical and have the same mass. Is the average translational kinetic energy still given by (3/2)kT for a mixture of gases such as air? Explain.

Q.12 The walls of a container of gas also consist of molecules. Thus a gas molecule colliding with a wall having the same temperature as the gas actually collides with one or more molecules. How can we justify, in an average sense, our assumption that the perpendicular component of velocity of a molecule merely reverses on a collision with the wall?

Q.13 Can the gravitational potential energy of interaction of the molecules of a gas with the earth be neglected in comparison with the kinetic energy of the molecules? What about the gravitational potential energy of interaction of the molecules with each other? Explain.

Q.14 Consider the air in a basketball during a game. Is (the vector) $(\vec{v}) = 0$ for the air? Explain.

Q.15 How can the temperature of a gas change during an adiabatic process, since no heat is exchanged with the surroundings? (Hint: Consider the change in velocity of a molecule colliding with a moving wall.)

Q.16 Suppose that an opposite pair of walls of a container of gas are maintained at different temperatures. By what mechanism involving molecular collisions is heat conducted through this gas? Note that the gas is not at a uniform temperature.

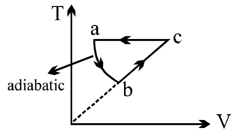
Q.17 Why is it improper to speak of the temperature of a molecule? Of a system of 100 molecules? How many molecules must a system have before we can speak meaningfully of the temperature of that system?

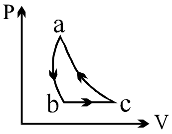
Q.18 The speed of sound in He is greater than the speed of sound in air at the same temperature and pressure. How, can this be understood at the molecular level?

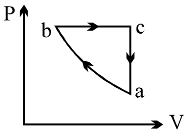
<u>***ONLY ONE OPTION IS CORRECT.***</u>

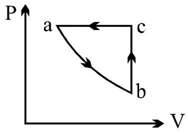
Take approx. 2 minutes for answering each question.

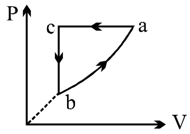
Q.1 PV curve for the process whose VT curve is



(A) 

(B) 

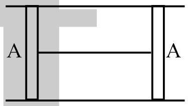
(C) 

(D) 

Q.2 Find the approx. number of molecules contained in a vessel of volume 7 litres at 0°C at 1.3×10^5 pascal

(A) 2.4×10^{23} (B) 3×10^{23} (C) 6×10^{23} (D) 4.8×10^{23}

Q.3 A cylindrical tube of cross-sectional area A has two air tight frictionless pistons at its two ends. The pistons are tied with a straight two ends. The pistons are tied with a straight piece of metallic wire. The tube contains a gas at atmospheric pressure P_0 and temperature T_0. If temperature of the gas is doubled then the tension in the wire is



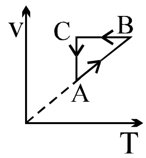
(A) $4\ P_0\ A$ (B) $P_0A/2$

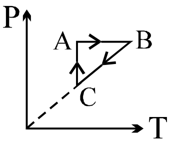
(C) $P_0\ A$ (D) $2\ P_0\ A$

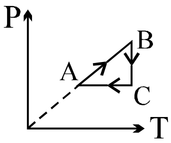
Q.4 An ideal gas mixture filled inside a balloon expands according to the relation $PV^{2/3}$ = constant. The temperature inside the balloon is

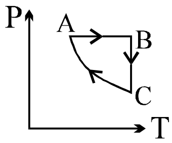
(A) increasing (B) decreasing (C) constant (D) can't be said

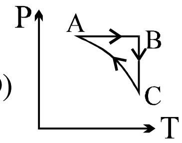
Q.5 An ideal gas undergoes a thermodynamics cycle as shown in figure. Which of the following graphs represents the same cycle?



(A) 

(B) 

(C) 

(D) 

Q.6 A rigid tank contains 35 kg of nitrogen at 6 atm. Sufficient quantity of oxygen is supplied to increase the pressure to 9 atm, while the temperature remains constant. Amount of oxygen supplied to the tank is :

(A) 5 kg (B) 10 kg (C) 20 kg (D) 40 kg

Q.7 A perfect gas of a given mass is heated first in a small vessel and then in a large vessel, such that their volumes remain unchanged. The P-T curves are

(A) parabolic with same curvature (B) parabolic with different curvature

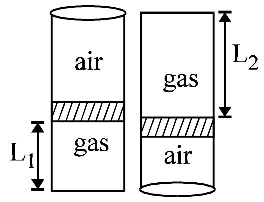
(C) linear with same slopes (D) linear with different slopes

Q.8 At a temperature T K, the pressure of 4.0g argon in a bulb is p. The bulb is put in a bath having temperature higher by 50K than the first one. 0.8g of argon gas had to be removed to maintained original pressure. The temperature T is equal to

(A) 510 K (B) 200 K (C) 100 K (D) 73 K

Q.9 When 2 gms of a gas are introduced into an evacuated flask kept at 25^0C the pressure is found to be one atmosphere. If 3 gms of another gas added to the same flask the pressure becomes 1.5 atmospheres. The ratio of the molecular weights of these gases will be
(A) 1 : 3 (B) 3 : 1 (C) 2 : 3 (D) 3 : 2

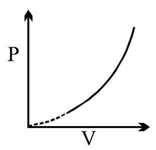
Q.10 An ideal gas is trapped inside a test tube of cross-sectional area 20×10^{-6} m^2 as shown in the figure. The gas occupies a height L_1 at the bottom of the tube and is separated from air at atmospheric pressure by a mercury column of mass 0.002 kg. If the tube is quickly turned isothermally, upside down so that mercury column encloses the gas from below. The gas now occupies height L_1 in the tube. The ratio $\frac{L_2}{L_1}$ is [Take atmospheric pressure = 10^5 Nm^{-2}]
(A) $\frac{102}{101}$ (B) $\frac{101}{99}$ (C) $\frac{99}{100}$ (D) $\frac{100}{99}$

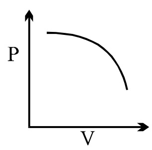


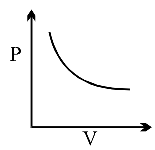
Q.11 An open and wide glass tube is immersed vertically in mercury in such a way that length 0.05 m extends above mercury level. The open end of the tube is closed and the tube is raised further by 0.43 m. The length of air column above mercury level in the tube will be : Take P_{atm} = 76 cm of mercury
(A) 0.215 m (B) 0.2 m (C) 0.1 m (D) 0.4 m

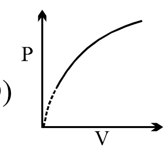
Q.12 A vessel of volume 0.02 m^3 contains a mixture of hydrogen and helium at 20°C and 2 atmospheric pressure. The mass of mixture is 5 gms. Find the ratio of mass of hydrogen to that of helium in the mixture.
(A) 1 : 2 (B) 1 : 3 (C) 2 : 3 (D) 3 : 2

Q.13 An ideal gas follows a process PT = constant. The correct graph between pressure & volume is

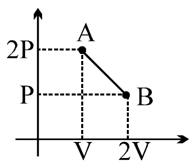
(A) 

(B) 

(C) 

(D) 

Q.14 The process AB is shown in the diagram. As the gas is taken from A to B, its temperature

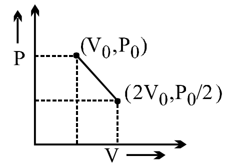


(A) initially increases then decreases (B) initially decreases then increases
(C) remains constant (D) variation depends on type of gas

Q.15 During an experiment an ideal gas oberys an addition equation of state P^2V = constant. The initial temperature and pressure of gas are T and V respectively. When it expands to volume 2V, then its temperature will be :
(A) T (B) $\sqrt{2}$ T (C) 2 T (D) $2\sqrt{2}$ T

Q.16 A barometer tube, containing mercury, is lowered in a vessel containing mercury until only 50 cm of the tube is above the level of mercury in the vessel. If the atmospheric pressure is 75 cm of mercury, what is the pressure at the top of the tube?
(A) 33.3 kPa (B) 66.7 kPa (C) 3.33 MPa (D) 6.67 MPa

Q.17 One mole of a gas expands obeying the relation as shown in the P/V diagram. The maximum temperature in this process is equal to



(A) $\frac{P_0V_0}{R}$ (B) $\frac{3P_0V_0}{R}$ (C) $\frac{9P_0V_0}{8R}$ (D) None of these

Q.18 A vessel with open mouth contains air at 60°C. When the vessel is heated upto temperature T, one fourth of the air goes out. The value of T is
(A) 80°C (B) 171°C (C) 333°C (D) 444°C

Q.19 28 gm of N_2 gas is contained in a flask at a pressure of 10 atm and at a temperature of 57°. It is found that due to leakage in the flask, the pressure is reduced to half and the temperature reduced to 27°C. The quantity of N_2 gas that leaked out is
(A) 11/20 gm (B) 20/11 gm (C) 5/63 gm (D) 63/5 gm

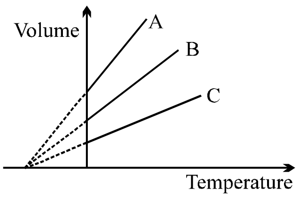
Q.20 If a mixture of 28 g of Nitrogen, 4 g of Hydrogen and 8 gm of Helium is contained in a vessel at temperature 400 K and pressure 8.3×10^5 Pa, the density of the mixture will be :
(A) 3 kg/m^3 (B) 0.2 kg/m^3 (C) 2 g/litre (D) 1.5 g/litre

Q.21 The temperature of a gas is doubled (i) on absolute scale (ii) on centigrade scale. The increase in root mean square velocity of gas will be
(A) More in case (i) (B) More in case (ii) (C) Same in both case (D) Information not sufficient

Q.22 A cylinder containing gas at 27°C is divided into two parts of equal volume each 100cc and at equal pressure by a piston of cross sectional area 10.85 cm^2. The gas in one part is raised in temperature to 100°C while the other maintained at original temperature. The piston and wall are perfect insulators. How far will the piston move during the change in temperature?
(A) 1 cm (B) 2 cm (C) 0.5 cm (D) 1.5 cm

Q.23 12gms of gas occupy a volume of 4×10^{-3} m^3 at a temperature of 7°C. After the gas is heated at constant pressure its density becomes 6×10^{-4}gm/cc. What is the temperature to which the gas was heated.
(A) 1000K (B) 1400K (C) 1200K (D) 800K

Q.24 The expansion of an ideal gas of mass m at a constant pressure P is given by the straight line B. Then the expansion of the same ideal gas of mass 2 m at a pressure 2P is given by the straight line



(A) C (B) A (C) B (D) none

Q.25 A vessel contains 1 mole of O_2 gas (molar mass 32) at a temperature T. The pressure of the gas is P. An identical vessel containing one mole of He gas (molar mass 4) at a temperature 2T has a pressure of
(A) P/8 (B) P (C) 2P (D) 8P

Q.26 A container X has volume double that of contianer Y and both are connected by a thin tube. Both contains same ideal gas. The temperature of X is 200K and that of Y is 400K. If mass of gas in X is m then in Y it will be:
(A) m/8 (B) m/6 (C) m/4 (D) m/2

Q.27 An ideal gas of Molar mass M is contained in a vertical tube of height H, closed at both ends. The tube is accelerating vertically upwards with acceleration g. Then, the ratio of pressure at the bottom and the mid point of the tube will be
(A) exp[2MgH/RT] (B) exp[–2MgH/RT] (C) exp[MgH/RT] (D) MgH/RT

Q.28 The ratio of average translational kinetic energy to rotational kinetic energy of a diatomic molecule at temperature T is

(A) 3 (B) 7/5 (C) 5/3 (D) 3/2

Q.29 One mole of an ideal gas at STP is heated in an insulated closed container until the average speed of its molecules is doubled. Its pressure would therefore increase by factor.

(A) 1.5 (B) $\sqrt{2}$ (C) 2 (D) 4

Q.30 Three particles have speeds of 2u , 10u and 11u. Which of the following statements is correct?

(A) The r.m.s. speed exceeds the mean speed by about u.

(B) The mean speed exceeds the r.m.s. speed by about u.

(C) The r.m.s. speed equals the mean speed.

(D) The r.m.s. speed exceeds the mean speed by more than 2u.

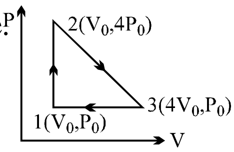
Q.31 The ratio of specific heats of a gas is $\frac{9}{7}$, then the number of degrees of freedom of the gas molecules for translational motion is :

(A) 7 (B) 3 (C) 6 (D) none

Q.32 A diatomic gas of molecules weight 30 gm/mole is filled in a container at 27°C. It is moving at a velocity 100 m/s. If it is suddenly stopped, the rise in temperature of gas is :

(A) $\frac{60}{R}$ (B) $\frac{600}{R}$ (C) $\frac{6\times10^4}{R}$ (D) $\frac{6\times10^5}{R}$

Q.33 One mole of an ideal diatomic gas is taken through the cycle as shown in the figure.



$1 \rightarrow 2$: isochoric process $2 \rightarrow 3$: straight line on P-V diagram

$3 \rightarrow 1$: isobaric process

The average molecular speed of the gas in the states 1, 2 and 3 are in the ratio

(A) 1 : 2 : 2 (B) 1 : $\sqrt{2}$: $\sqrt{2}$ (C) 1 : 1 : 1 (D) 1 : 2 : 4

Question No. 34 to 37 (4 questions)

Five moles of helium are mixed with two moles of hydrogen to form a mixture. Take molar mass of helium $M_1 = 4g$ and that of hydrogen $M_2 = 2g$

Q.34 The equivalent molar mass of the mixture is

(A) 6g (B) $\frac{13g}{7}$ (C) $\frac{18g}{7}$ (D) none

Q.35 The equivalent degree of freedom f of the mixture is

(A) 3.57 (B) 1.14 (C) 4.4 (D) none

Q.36 The equivalent value of γ is

(A) 1.59 (B) 1.53 (C) 1.56 (D) none

Q.37 If the internal energy of He sample of 100J and that of the hydrogen sample is 200J, then the internal energy of the mixture is

(A) 900 J (B) 128.5 J (C) 171.4 J (D) 300 J

Q.38 N(<100) molecules of a gas have velocities 1, 2, 3........ N/km/s respectively. Then
(A) rms speed and average speed of molecules is same.
(B) ratio of rms speed to average speed is $\sqrt{(2N+1)(N+1)/6N}$
(C) ratio of rms speed to average speed is $\sqrt{(2N+1)(N+1)/6}$
(D) ratio of rms speed to average speed of a molecule is $2/\sqrt{6} \times \sqrt{(2N+1)/(N+1)}$

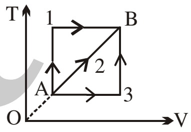
Q.39 Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. The temperature of the mixture will be :

(A) $\frac{T_1 + T_2}{n_1 + n_2}$ (B) $\frac{T_1}{n_1} + \frac{T_2}{n_2}$ (C) $\frac{n_2T_1 + n_1T_2}{n_1 + n_2}$ (D) $\frac{n_1T_1 + n_2T_2}{n_1 + n_2}$

Q.40 At temperature T, N molecules of gas A each having mass m and at the same temperature 2N molecules of gas B each having mass 2m are filled in a container. The mean sqaure velocity of molecules of gas B is v^2 and mean square of x component of velocity of molecules of gas A is w^2. The ratio of w^2/v^2 is :
(A) 1 (B) 2 (C) 1/3 (D) 2/3

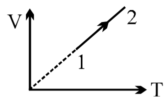
Q.41 Five particles have speeds 1, 2, 3, 4, 5 m/s. the average velocity of the particles is (in m/s)
(A) 3 (B) 0 (C) 2.5 (D) cannot be calculated.

Q.42 A given mass of a gas expands from a state A to the state B by three paths 1, 2 and 3 as shown in T-V indicator diagram. If W_1, W_2 and W_3 respectively be the work done by the gas along the three paths, then



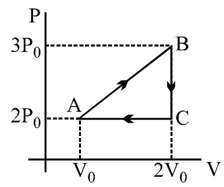
(A) $W_1 > W_2 > W_3$ (B) $W_1 < W_2 < W_3$
(C) $W_1 = W_2 = W_3$ (D) $W_1 < W_2$, $W_1 > W_3$

Q.43 An ideal gas undergoes the process $1 \rightarrow 2$ as shown in the figure, the heat supplied and work done in the process is ΔQ and ΔW respectively. The ratio $\Delta Q : \Delta W$ is



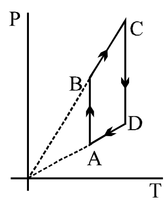
(A) $\gamma : \gamma - 1$ (B) γ
(C) $\gamma - 1$ (D) $\gamma - 1/\gamma$

Q.44 In the above thermodynamic process, the correct statement is



(A) Heat given in the complete cycle ABCA is zero
(B) Work done in the complete cycle ABCA is zero
(C) Work done in the complete cycle ABCA is $(1/2\, P_0V_0)$
(D) None

Q.45 Pressure versus temperature graph of an ideal gas is shown in figure



(A) During the process AB work done by the gas in positive
(B) during the process CD work done by the gas is negative
(C) during the process BC internal energy of the gas is increasing
(D) None

Q.46 A reversible adiabatic path on a P-V diagram for an ideal gas passes through state A where $P=0.7\times10^5$ N/ m^{-2} and $v = 0.0049\ m^3$. The ratio of specific heat of the gas is 1.4. The slope of path at A is :
(A) $2.0 \times 10^7\ Nm^{-5}$ (B) $1.0 \times 10^7\ Nm^{-5}$ (C) $-2.0 \times 10^7\ Nm^{-5}$ (D) $-1.0 \times 10^7\ Nm^{-5}$

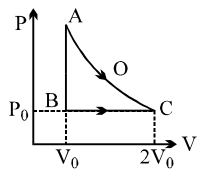
Q.47 An ideal gas at pressure P and volume V is expanded to volume 2V. Column I represents the thermodynamic processes used during expansion. Column II represents the work during these processes in the random order.

Column I	**Column II**
(p) isobaric	(x) $\frac{PV(1-2^{1-\gamma})}{\gamma-1}$
(q) isothermal	(y) PV
(r) adiabatic	(z) PV $ln\,2$

The correct matching of column I and column II is given by :

(A) p-y, q-z, r-x (B) p-y, q-x, r-z (C) p-x, q-y, r-z (D) p-z, q-y, r-x

Q.48 An ideal gas is taken from point A to point C on P-V diagram through two process AOC and ABC as shown in the figure. Process AOC is isothermal

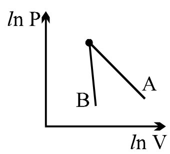


(A) Process AOC requires more heat than process ABC.
(B) Process ABC requires more heat than process AOC.
(C) Both process AOC & ABC require same amount of heat.
(D) Data is insufficient for comparison of heat requirement for the two processes.

Q.49 One mole of an ideal gas is contained with in a cylinder by a frictionless piston and is initially at temperature T. The pressure of the gas is kept constant while it is heated and its volume doubles. If R is molar gas constant, the work done by the gas in increasing its volume is

(A) RT $ln2$ (B) 1/2 RT (C) RT (D) 3/2 RT

Q.50 The figure, shows the graph of logarithmic reading of pressure and volume for two ideal gases A and B undergoing adiabatic process. From figure it can be concluded that



(A) gas B is diatomic (B) gas A and B both are diatomic
(C) gas A is monoatomic (D) gas B is monoatomic & gas A is diatomic

Q.51 A thermodynamic cycle takes in heat energy at a high temperature and rejects energy at a lower temperature. If the amount of energy rejected at the low temperature is 3 times the amount of work done by the cycle, the efficiency of the cycle is

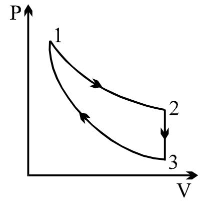
(A) 0.25 (B) 0.33 (C) 0.67 (D) 0.9

Q.52 Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume becomes half the initial volume.
(A) If the compression is adiabatic then monoatomic gas will have maximum final pressure.
(B) If the compression is adiabatic then triatomic gas will have maximum final pressure.
(C) If the compression is adiabatic then their final pressure will be same.
(D) If the compression is isothermal then their final pressure will be different.

Q.53 If heat is added at constant volume, 6300J of heat are required to raise the temperature of an ideal gas by 150K. If instead, heat is added at constant pressure, 8800 joules are required for the same temperature change. When the temperature of the gas changes by 300K, the internal energy of the gas changes by

(A) 5000J (B) 12600J (C) 17600J (D) 22600J

Q.54 Three processes form a thermodynamic cycle as shown on P-V diagram for an ideal gas. Process $1 \rightarrow 2$ takes place at constant temperature (300K). Process $2 \rightarrow 3$ takes place at constant volume. During this process 40J of heat leaves the system. Process $3 \rightarrow 1$ is adiabatic and temperature T_3 is 275K. Work done by the gas during the process $3 \rightarrow 1$ is



(A) -40J (B) -20J
(C) +40J (D) +20J

Q.55 When unit mass of water boils to become steam at 100^0C, it absorbs Q amount of heat. The densities of water and steam at 100^0C are ρ_1 and ρ_2 respectively and the atmospheric pressure is p_0. The increase in internal energy of the water is

(A) Q (B) $Q + p_0\left(\frac{1}{\rho_1} - \frac{1}{\rho_2}\right)$ (C) $Q + p_0\left(\frac{1}{\rho_2} - \frac{1}{\rho_1}\right)$ (D) $Q - p_0\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)$

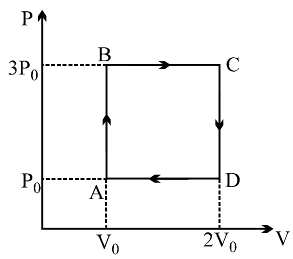
Q.56 A polyatomic gas with six degrees of freedom does 25J of work when it is expanded at constant pressure. The heat given to the gas is
(A) 100J (B) 150J (C) 200J (D) 250J

Q.57 An ideal gas expands from volume V_1 to V_2. This may be achieved by either of the three processes: isobaric, isothermal and adiabatic. Let ΔU be the change in internal energy of the gas, Q be the quantity of heat added to the system and W be the work done by the system on the gas. Identify which of the following statements is false for ΔU?
(A) ΔU is least under adiabatic process.
(B) ΔU is greatest under adiabatic process.
(C) ΔU is greatest under the isobaric process.
(D) ΔU in isothermal process lies in-between the values obtained under isobaric and adiabatic processes.

Q.58 In an isobaric expansion of an ideal gas, which of the following is zero?
(A) work done (B) ΔQ (C) ΔU (D) d^2V/dT^2

Q.59 A perfect gas is found to obey the relation $PV^{3/2}$ = constant, during an adiabatic process. If such a gas, initially at a temperature T, is compressed adiabatically to half its initial volume, then its final temperature will be
(A) 2T (B) 4T (C) $\sqrt{2}$T (D) $2\sqrt{2}$T

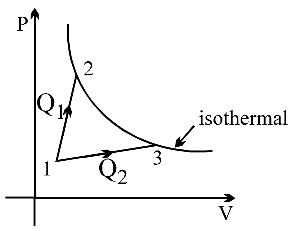
Q.60 A ideal monoatomic gas is carried around the cycle ABCDA as shown in the fig. The efficiency of the gas cycle is



(A) $\frac{4}{21}$ (B) $\frac{2}{21}$

(C) $\frac{4}{31}$ (D) $\frac{2}{31}$

Q.61 A gas takes part in two processes in which it is heated from the same initial state 1 to the same final temperature. The processes are shown on the P-V diagram by the straight line 1-2 and 1-3. 2 and 3 are the points on the same isothermal curve. Q_1 and Q_2 are the heat transfer along the two processes. Then



(A) $Q_1 = Q_2$ (B) $Q_1 < Q_2$
(C) $Q_1 > Q_2$ (D) insufficient data

Q.62 In thermodynamic process pressure of a fixed mass of gas is changed in such a manner that the gas releases 30 joule of heat and 18 joule of work was done on the gas. If the initial internal energy of the gas was 60 joule, then, the final internal energy will be :

(A) 32 joule (B) 48 joule (C) 72 joule (D) 96 joule

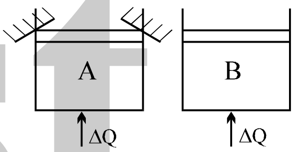
Q.63 A cylinder made of perfectly non conducting material closed at both ends is divided into two equal parts by a heat proof piston. Both parts of the cylinder contain the same masses of a gas at a temperature $t_0 = 27°$ and pressure $P_0 = 1$ atm. Now if the gas in one of the parts is slowly heated to $t = 57°C$ while the temperature of first part is maintained at t_0 the distance moved by the piston from the middle of the cylinder will be (length of the cylinder = 84 cm)

(A) 3 cm (B) 5 cm (C) 2 cm (D) 1 cm

Q.64 1 gm water at 100°C and 10^5Pa pressure converts into 1841cm^3 of steam at constant temperature and pressure. If latent heat of vapourization of water is 2250 J/gm. The change in internal energy of water in this process is

(A) zero (B) 2250 J (C) 2066 J (D) none

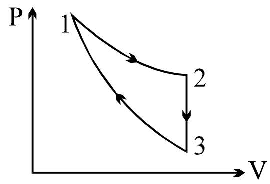
Q.65 Two identical vessels A & B contain equal amount of ideal monoatomic gas. The piston of A is fixed but that of B is free. Same amount of heat is absorbed by A & B. If B's internal energy increases by 100 J the change in internal energy of A is



(A) 100 J (B) $\frac{500}{3}$ J

(C) 250 J (D) none of these

Q.66 Three processes compose a thermodynamics cycle shown in the PV diagram. Process 1→2 takes place at constant temperature. Process 2→3 takes place at constant volume, and process 3→1 is adiabatic. During the complete cycle, the total amount of work done is 10 J. During process 2→3, the internal energy decrease by 20J and during process 3→1, 20 J of work is done on the system. How much heat is added to the system during process 1→2?



(A) 0 (B) 10 J

(C) 20 J (D) 30 J

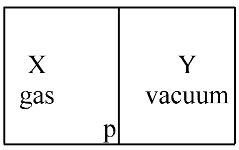
Q.67 An ideal gas undergoes an adiabatic process obeying the relation $PV^{4/3}$ = constant. If its initial temperature is 300 K and then its pressure is increased upto four times its initial value, then the final temperature is (in Kelvin):

(A) $300\sqrt{2}$ (B) $300\sqrt[3]{2}$ (C) 600 (D) 1200

Q.68 The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is

(A) 0 Nm^{-2} (B) 1 Nm^{-2} (C) 1.4×10^4 Nm^{-2} (D) 1.4×10^5 Nm^{-2}

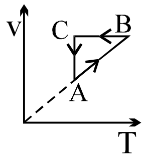
Q.69 A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y. Which of the following statements is correct?

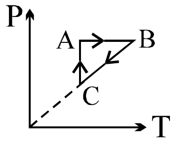
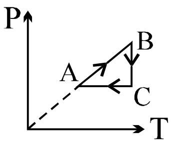
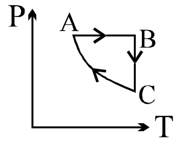
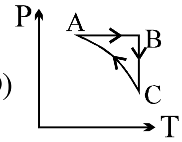


(A) No work is done by X (B) X decreases in temperature

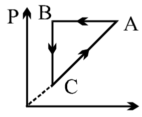
(C) X increases in internal energy (D) X doubles in pressure

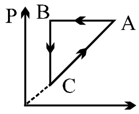
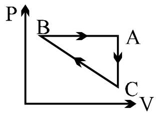
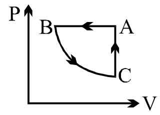
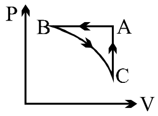
Q.70 An ideal gas undergoes a thermodynamics cycle as shown in figure. Which of the following graphs represents the same cycle?



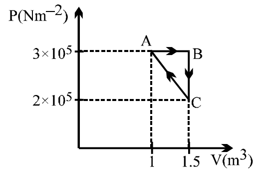
(A)  (B)  (C)  (D) 

Q.71 A cyclic process ABCA is shown in PT diagram. When presented on PV, it would



(A)  (B)  (C)  (D) 

Q.72 Considere the thermodynamics cycle shown on PV diagram. The process $A \rightarrow B$ is isobaric, $B \rightarrow C$ is isochoric and $C \rightarrow A$ is a straight line process. The following internal energy and heat are given :



$$\Delta U_{A \rightarrow B} = +400 \text{ kJ and } Q_{B \rightarrow C} = -500 \text{ kJ}$$

The heat flow in the process $Q_{C \rightarrow A}$ is :

(A) – 20 kJ (B) + 25 kJ (C) – 25 kJ (D) Data are insufficient

Q.73 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :

(A) 32 kJ (B) 5 kJ (C) – 4 kJ (D) 13 kJ

Q.74 A vessel contains an ideal monoatomic gas which expands at constant pressure, when heat Q is given to it. Then the work done in expansion is :

(A) Q (B) $\frac{3}{5}Q$ (C) $\frac{2}{5}Q$ (D) $\frac{2}{3}Q$

Q.75 One mole of an ideal monoatomic gas at temperature T_0 expands slowly according to the law P/V = constant. If the final temperature is $2\,T_0$, heat supplied to the gas is :

(A) $2\,R\,T_0$ (B) $\frac{3}{2}\,R\,T_0$ (C) $R\,T_0$ (D) $\frac{1}{2}\,R\,T_0$

Q.76 A diatomic gas follows equation PV^m = constant, during a process. What should be the value of m such that its molar heat capacity during process = R

(A) 2/3 (B) 1 (C) 1.5 (D) 5/3

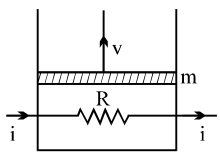
Q.77 One mole of an ideal gas at temperature T_1 expends according to the law $\frac{P}{V^2}$ = a (constant). The work done by the gas till temperature of gas becomes T_2 is :

(A) $\frac{1}{2}R(T_2 - T_1)$ (B) $\frac{1}{3}R(T_2 - T_1)$ (C) $\frac{1}{4}R(T_2 - T_1)$ (D) $\frac{1}{5}R(T_2 - T_1)$

Q.78 2 moles of a diatomic gas undergoes the process : PT^2/V = constant. Then, the molar heat capacity of the gas during the process will be equal to

(A) 5R/2 (B) 9R/2 (C) 3R (D) 4R

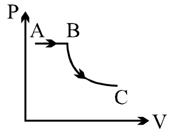
Q.79 A resistance coil connected to an external battery is placed inside an adiabatic cylinder fitted with a frictionless pistn and containing an ideal gas. A current i flows through the coil which has a resistance R. At what speed must the piston move upward in order that the temperature of the gas remains uchanged? Neglect atmospheric pressure.



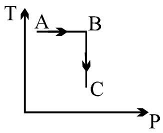
(A) $\frac{i^2 m}{Rg}$ (B) $\frac{Rmg}{i^2}$ (C) $\frac{mg}{i^2}$ (D) $\frac{i^2 R}{mg}$

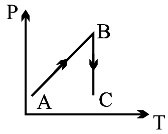
ONE OR MORE THAN ONE OPTION MAY BE CORRECT

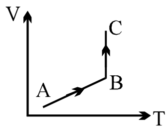
Take approx. 3 minutes for answering each question.

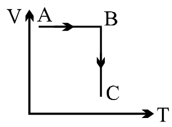


Q.1 A process is shown in the diagram. Which of the following curves may represent the same process ?

(A) 

(B) 

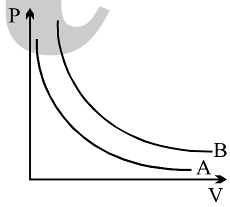
(C) 

(D) 

Q.2 An ideal gas expands in such a way that PV^2 = constant throughout the process.
(A) The graph of the process of T-V diagram is a parabola.
(B) The graph of the process of T-V diagram is a straight line.
(C) Such an expansion is possible only with heating.
(D) Such an expansion is possible only with cooling.

Q.3 A gas expands such that its initial and final temperature are equal. Also, the process followed by the gas traces a straight line on the P-V diagram :
(A) The temperature of the gas remains constant throughout.
(B) The temperature of the gas first increases and then decreases
(C) The temperature of the gas first decreases and then increases
(D) The stright line has a negative slope.

Q.4 Figure shows the pressure P versus volume V graphs for two different gas sample at a given temperature. M_A and M_B are masses of two samples, n_A and n_B are numbers of moles. Which of the following ***must be incorect.***



(A) $M_A > M_B$ (B) $M_A < M_B$
(C) $n_A > n_B$ (D) $n_A < n_B$

Question No. 5 to 7 (3 questions)

A very tall vertical cylinder is filled with a gas of molar mass M under isothermal conditions at temperature T. The density and pressure of the gas at the base of the container is ρ_0 and p_0, respectively

Q.5 Choose the correct statement(s)
(A) Pressure decreases with height
(B) The rate of decreases of pressure with height is a constant.
(C) $\frac{dp}{dh} = -\rho g$ where ρ is density of the gas at a height h.
(D) $p = \rho \frac{RT}{M}$

Q.6 Choose the correct statement(s) if gravity is assumed to be constant throughout the container
(A) Both pressure and density decreases exponentially with height.
(B) The variation of pressure is $p = p_0 e^{-\frac{Mgh}{RT}}$
(C) The variation of density $\rho = \rho_0 e^{-\frac{Mgh}{RT}}$
(D) The molecular density decreases as one moves upwards.

Q.7 Choose the correct statement(s)
(A) The density of gas cannot be uniform throughout the cylinder.
(B) The density of gas cannot be uniform throughout the cylinder under isothermal conditions.
(C) The density of gas is constant if $\left|\frac{dT}{dh}\right| = \frac{R}{Mg}$
(D) The density of gas is uniform if $\left|\frac{dT}{dh}\right| = \frac{Mg}{R}$

Q.8 During an experiment, an ideal gas is found to obey a condition VP^2 = constant. The gas is initially at a temperature T, pressure P and volume V. The gas expands to volume 4V.
(A) The pressure of gas changes to $\frac{P}{2}$
(B) The temperature of gas changes to 4T
(C) The graph of above process on the P-T diagram is parabola
(D) The graph of above process on the P-T diagram is hyperbola.

Q.9 During an experiment, an ideal gas is found to obey a condition $\frac{P^2}{\rho}$ = constant [ρ = density of the gas]. The gas is initially at temperature T, pressure P and density ρ. The gas expands such that density changes to $\rho/2$.
(A) The pressure of the gas changes to $\sqrt{2}$ P
(B) The temperature of the gas changes to $\sqrt{2}$ T
(C) The graph of above process on the P–T diagram is parabola
(D) The graph of the above process on the P–T diagram is hyperbola.

Q.10 According to kinetic theory of gases, which of following statement will be true.
(A) Ideal gases can not be liquified
(B) The molecules of ideal gas do not obey newtons laws of motion.
(C) Pressure of gas is always inversely proportional to its volume
(D) molecules of gas never move in straight line.

Q.11 According to kinetic theory of gases,
(A) The velocity of molecules decreases for each collision
(B) The pressure exerted by a diatomic gas is proportional to the mean velocity of the molecule.
(C) The K.E. of the gas decreases on expansion at constant temperature.
(D) The mean translational K.E. of a diatomic gas increases with increase in absolute temperature.

Q.12 Two vessels of the same volume contain the same gas at same temperature. If the pressure in the vessels be in the ratio of 1 : 2, then
(A) the ratio of the average kinetic energy is 1 : 2
(B) the ratio of the root mean square velocity is 1 : 1
(C) the ratio of the average velocity is 1 : 2
(D) the ratio of number of molecules is 1 : 2

Q.13 The total kinetic energy of translatory motion of all the molecules of 5 litres of nitrogen exerting a pressure P is 3000 J.
(A) the total k.e. of 10 litres of N_2 at a pressure of 2P is 3000 J
(B) the total k.e. of 10 litres of He at a pressure of 2P is 3000 J
(C) the total k.e. of 10 litres of O_2 at a pressure of 2P is 20000 J
(D) the total k.e. of 10 litres of Ne at a pressure of 2P is 12000 J

Q.14 A vertical cylinder with heat-conducting walls is closed at the bottom and is fitted with a smooth light piston. It contains one mole of an ideal gas. The temperature of the gas is always equal to the surrounding's temperature, T_0. The piston is moved up slowly to increase the volume of the gas to η times. Which of the following is incorrect?
(A) Work done by the gas is $RT_0 \ln \eta$.
(B) Work done against the atmosphere is $RT_0(\eta - 1)$.
(C) There is no change in the internal energy of the gas.
(D) The final pressure of the gas is $\frac{1}{(\eta - 1)}$ times its initial pressure.

Q.15 A closed vessel contains a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A contained in the vessel is 2 times that of B. The following statements are given
(i) Average kinetic energy per molecule of A is equal to that of B.
(ii) Root mean square value of translational velocity of B is four times that of A.
(iii) Pressure exerted by B is eight times of that exerted by A.
(iv) Number of molecules of B in the cylinder is eight time that of A.
(A) (i), (ii) and (iii) are true
(B) (ii), (iii) and (iv) are true
(C) (i), (ii) and (iv) are true
(D) All are true

Q.16 A mixture of ideal gases 7 kg of nitrogen and 11 kg of CO_2. Then
(A) equivalent molecular weight of the mixture is 36.
(B) equivalent molecular weight of the mixture is 18.
(C) γ for the mixture is 5/2
(D) γ for the mixture is 47/35.
(Take γ for nitrogen and CO_2 as 1.4 and 1.3 respectively)

Q.17 What is/are the same for O_2 and NH_3 in gaseous state
(A) ratio of specific heats
(B) average velocity
(C) maximum no. of vibrational degree of freedom
(D) None of these

Q.18 A piston is slowly pushed into a metal cylinder containing an ideal gas. Which of the following statements is/are incorrect?
(A) The pressure of the gas increases
(B) The number of the molecules per unit volume increases
(C) The average speed of gas molecules increases
(D) The frequency of collision of the gas molecules with the piston increases.

Q.19 Which of the following statements is/are not an assumption of the kinetic theory for an ideal gas?
(A) The duration of a collision is negligible as compared to the time between successive collisions
(B) The molecules have negligible attraction for each other
(C) The molecules have negligible momentum change on collision with the container walls
(D) There is no total kinetic energy change of the molecules on colliding with each other or with the walls of the container.

Q.20 Select the incorrect statement about ideal gas.
(A) Molecules of a gas are in incessant random motion colliding against one another and with the walls of the container.
(B) The gas is not isotropic and the constant (1/3) in equation $P = (1/3)\rho\, v_{rms}^2$ is result of this property
(C) The time during which a collision lasts is negligible compared to the time of free path between collisions.
(D) There is no force of interaction between molecules among themselves or between molecules and the wall except during collision.

Q.21 Select the **incorrect** statement(s)
(A) RMS speed of 8 gm oxygen gas in container at 27° C is approximately 484 m/s
(B) RMS speed of 8 gm oxygen in container at 27° C is approximately 968 m/s
(C) For number of molecules greater than one, RMS speed is greater than average speed
(D) A gas behaves more closely as an ideal gas at low pressures and high temperatures

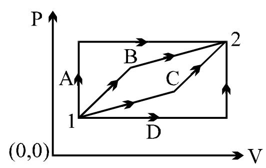
Q.22 Hydrogen gas and oxygen gas have volume 1 cm^3 each at N.T.P.
(A) Number of molecules is same in both the gases.
(B) The rms velocity of molecules of both the gases is the same.
(C) The internal energy of each gases is the same
(D) The average velocity of molecules of each gas is the same.

Q.23 A gas is enclosed in a vessel at a constant temperature at a pressure of 5 atmosphere and volume 4 litre. Due to a leakage in the vessel, after some time, the pressure is reduced to 4 atmosphere. As a result, the
(A) volume of the gas decreased by 20%
(B) average K.E. of gas molecule decreases by 20%
(C) 20% of the gas escaped due to the leakage
(D) 25% of the gas escaped due to the leakage

Q.24 In case of hydrogen and oxygen at N.T.P., which of the following quantities is / are the same?
(A) average momentum per molecule
(B) average kinetic energy per molecule
(C) kinetic energy per unit volume
(D) kinetic energy per unit mass

Q.25 A container holds 10^{26} molecules/m^3, each of mass 3×10^{-27} kg. Assume that 1/6 of the molecules move with velocity 2000 m/s directly towards one wall of the container while the remaining 5/6 of the molecules move either away from the wall or in perpendicular direction, and all collisions of the molecules with the wall are elastic
(A) number of molecules hitting 1 m^2 of the wall every second is 3.33×10^{28}.
(B) number of molecules hitting 1 m^2 of the wall every second is 2×10^{29}.
(C) pressure exerted on the wall by molecules is 24×10^5 Pa.
(D) pressure exerted on the wall by molecules is 4×10^5 Pa.

Q.26 An ideal gas is taken from state 1 to state 2 through optional path A, B, C & D as shown in P-V diagram. Let Q, W and U represent the heat supplied, work done & internal energy of the gas respectively. Then



(A) $Q_B - W_B > Q_C - W_C$
(B) $Q_A - Q_D = W_A - W_D$
(C) $W_A < W_B < W_C < W_D$
(D) $Q_A > Q_B > Q_C > Q_D$

Q.27 A student records ΔQ, ΔU & ΔW for a thermodynamic cycle $A \rightarrow B \rightarrow C \rightarrow A$. Certain entries are missing. Find correct entry in following options.

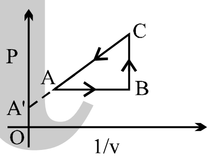
	AB	BC	CA
ΔW	40J		30J
ΔU		50J	
ΔQ	150J	10J	

(A) $W_{BC} = -70$ J
(B) $\Delta Q_{CA} = 130$ J
(C) $\Delta U_{AB} = 190$ J
(D) $\Delta U_{CA} = -160$ J

Q.28 Two moles of monoatomic gas is expanded from (P_0, V_0) to $(P_0, 2V_0)$ under isobaric condition. Let ΔQ_1, be the heat given to the gas, ΔW_1 the work done by the gas and ΔU_1 the change in internal energy. Now the monoatomic gas is replaced by a diatomic gas. Other conditions remaining the same. The corresponding values in this case are ΔQ_2, ΔW_2, ΔU_2 respectively, then
(A) $\Delta Q_1 - \Delta Q_2 = \Delta U_1 - \Delta U_2$
(B) $\Delta U_2 + \Delta W_2 > \Delta U_1 + \Delta W_1$
(C) $\Delta U_2 > \Delta U_1$
(D) All of these

Q.29 For an ideal gas
(A) The change in internal energy in a constant pressure process from temperature T_1 to T_2 is equal to $nC_v(T_2 - T_1)$ where C_v is the molar specific heat at constant volume and n is the number of the moles of the gas.
(B) The change in internal energy of the gas and the work done by the gas are equal in magnitude in an adiabatic proces.
(C) The internal energy does not change in an isothermal process.
(D) A, B and C

Q.30 An enclosed ideal gas is taken through a cycle as shown in the figure. Then



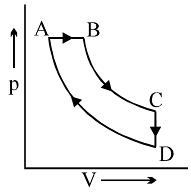
(A) Along AB, temperature decreases while along BC temperature increases
(B) Along AB, temperature increases while along BC the temperature decreases.
(C) Along CA work is done by the gas and the internal energy remains constant.
(D) Along CA work is done on the gas and internal energy of the gas increases.

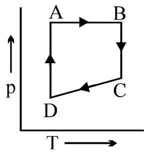
Q.31 Two gases have the same initial pressure, volume and temperatue. They expand to the same final volume, one adiabatically and the other isothermally
(A) The final temperature is greater for the isothermal process
(B) The final pressure is greater for the isothermal process
(C) The work done by the gas is greater for the isothermal process
(D) All the above options are incorrect

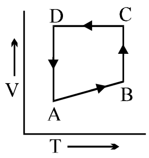
Q.32 In the previous question, if the two gases are compressed to the same final volume
(A) the final temperature is greater for the adiabatic process
(B) the final pressure is greater for the adiabatic process
(C) the work done on the gas is greater for the adiabatic process
(D) all the above options are incorrect

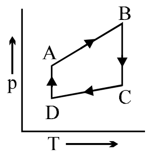
Q.33 The first law of thermodynamics can be written as $\Delta U = \Delta Q + \Delta W$ for an ideal gas. Which of the following statements is correct?
(A) ΔU is always zero when no heat enters or leaves the gas
(B) ΔW is the work done by the gas in this written law.
(C) ΔU is zero when heat is supplied and the temperature stays constant
(D) $\Delta Q = -\Delta W$ when the temperature increases very slowly.

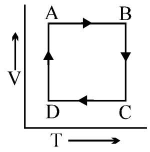
Q.34 A cyclic process ABCD is shown in the p-V diagram. Which of the following curves represents the same process if BC & DA are isothermal processes



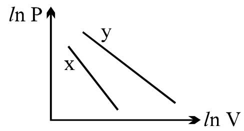
(A) 

(B) 

(C) 

(D) 

Q.35 For two different gases X and Y, having degrees of freedom f_1 and f_2 and molar heat capacities at constant volume C_{V_1} and C_{V_2} respectively, the ln P versus ln V graph is plotted for adiabatic process, as shown



(A) $f_1 > f_2$ (B) $f_2 > f_1$
(C) $C_{V_2} > C_{V_1}$ (D) $C_{V_1} > C_{V_2}$

Q.36 2 moles of a monoatomic gas are expanded to double its initial volume, through a process P/V = constant. If its initial temperature is 300 K, then which of the following is not true.
(A) $\Delta T = 900$ K (B) $\Delta Q = 3200$ R (C) $\Delta Q = 3600$ R (D) W = 900 R

Answer Key

ONLY ONE OPTION IS CORRECT.

Q.1	A	Q.2	A	Q.3	C	Q.4	A	Q.5	A	Q.6	C	Q.7	D
Q.8	B	Q.9	A	Q.10	B	Q.11	C	Q.12	A	Q.13	C	Q.14	A
Q.15	B	Q.16	A	Q.17	C	Q.18	B	Q.19	D	Q.20	C	Q.21	A
Q.22	A	Q.23	B	Q.24	C	Q.25	C	Q.26	C	Q.27	C	Q.28	D
Q.29	D	Q.30	A	Q.31	B	Q.32	A	Q.33	A	Q.34	D	Q.35	A
Q.36	C	Q.37	D	Q.38	D	Q.39	D	Q.40	D	Q.41	D	Q.42	A
Q.43	A	Q.44	C	Q.45	C	Q.46	C	Q.47	A	Q.48	A	Q.49	C
Q.50	D	Q.51	A	Q.52	A	Q.53	B	Q.54	A	Q.55	B	Q.56	A
Q.57	B	Q.58	D	Q.59	C	Q.60	A	Q.61	B	Q.62	B	Q.63	C
Q.64	C	Q.65	B	Q.66	D	Q.67	A	Q.68	D	Q.69	A	Q.70	A
Q.71	C	Q.72	C	Q.73	D	Q.74	C	Q.75	A	Q.76	D	Q.77	B
Q.78	D	Q.79	D										

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	C	Q.2	D	Q.3	B,D	Q.4	C
Q.5	A,C,D	Q.6	A,B,C,D	Q.7	A,B,D	Q.8	A,D
Q.9	B,D	Q.10	A	Q.11	D	Q.12	B,D
Q.13	C,D	Q.14	D	Q.15	D	Q.16	A,D
Q.17	B	Q.18	C	Q.19	C	Q.20	B
Q.21	B	Q.22	A,C,D	Q.23	C	Q.24	A,B,C
Q.25	A,D	Q.26	B,D	Q.27	D	Q.28	D
Q.29	D	Q.30	A	Q.31	A,B,C	Q.32	A,B,C
Q.33	C	Q.34	A,B	Q.35	B,C	Q.36	B

Power Coaching

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PHYSICS

MECHANICAL WAVES

QUESTION FOR SHORT ANSWER

Q.1 When the two waves interfere, does one alter the progress of the other?

Q.2 If two waves differ only in amplitude and are propagated in opposite directions through a medium, will they produce standing waves? Is energy transported? Are there any nodes?

Q.3 Energy can be transferred along a string by wave motion. However, in a standing wave on a string, no energy can ever be transferred past at node. Why not ?

Q.4 Can a standing wave be produced on a string by superposing two waves travelling in opposite directions with the same frequency but different amplitudes ? Why or Why not ? Can a standing wave be produced by superposing two waves travelling its opposite directions with different frequencies but the same amplitude? Why or why not?

Q.5 A sound source and a listener are both at rest on the earth, but a strong wind is blowing from the source toward the listener. Is there a Doppler effect ? Why or why not?

Q.6 Can a great singer cause a glass object to shatter by this singing? Explain with reason.

Q.7 The wave functions $y = A\sqrt{(x - vt)}$, $y = (x + vt)^3$, $y = A(x - vt)$ or $y = A\log(x + vt)$ though are of the form $y = f(x \pm vt)$, but not useful in wave motion. Explain why?

Q.8 The source of energy of sun is fusion of hydrogen which provides energy in the form of heat, light and sound. Explain why sound from sun does not reach earth while heat and light do.

Q.9 Explain why:
(a) Velocity of sound is generally greater in solids than in gases.
(b) The velocity of sound in oxygen is lesser than in hydrogen.

Q.10 Why is it sometimes difficult to recognise the speaker's voice over the telephone?

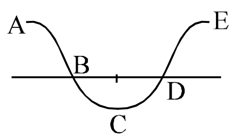
Q.11 Transverse waves are possible in solids but not in fluids Why?

Q.12 Snakes have no ears. Yet they can hear our footsteps. How is it possible?

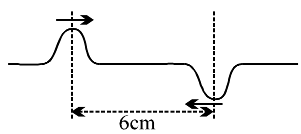
Q.13 Whys is sound wave of intensity 10^{-12} W m^{-2} and frequency 1000 Hz taken as the standard for expressing the intensity level of all other sound waves?

Q.14 If two sound waves of frequencies 500 Hz and 550 Hz superimpose, will they produce beats? Would you hear the beats?

Q.15 Which parts of the curve in the figure represent compression and refraction for a longitudinal wave?



Q.16 Two pulses are travelling along a string in opposite directions as shown in the figure here. If the wave velocity is 2 cm s^{-1} and the pulses are 6 cm apart, sketch the pattern after 1.5 s. What happens to the energy at this instant?



Q.17 Why does the faint sound from a tuning fork become quite loud when the stem is placed upon a table?

Q.18 On a hot day, the voice of a man cannot be heard from a great distance in an open field. Explain.

Q.19 A plane wave of sound travelling in air is incident upon a plane water surface. The angle of incidence is 60°. Assuming Snell's law to be valid for sound waves, it follows that the wave will be refracted into the water away from the normal. Is this true or false?

Q.20 Why do you see lightning before you hear the thunder? A familiar rule of thumb is to start counting slowly, once per second, when you see the lightning; when you hear the thunder, divide the number you have reached by 3 to obtain your distance from the lightning in kilometers (or by 5 to obatin your distance in miles). Why does this work, or does it?

Q.21 Childern make toy telephones by sticking each end of a long string through a hole in the bottom of a paper cup and knotting it so it will not pull out. When the spring is pulled taut, sound can be transmitted from one cup to the other. How does this work? Why is the transmitted sound louder than the sound travelling through air for the same distance?

Q.22 Justify the following statement: "A wave transfers energy fastest where the particles of the medium are moving fastest." In particular, explain how can this can be true for a transverse wave, in which the particles of the medium move in a direction perpendicualr to that of wave propagation.

Q.23 A musical interval of an octave corresponds to a factor of two in frequency. By what factor must the tension in a guitar or violin string be increased to raise its pitch one octave? To raise it two octaves? Explain your reasoning. Is there any danger in attempting these changes in pitch?

Q.24 By touching a string lightly at its center while bowing, a violinist can produce a note exactly one octave above the note to which the string is tuned, that is a note with exactly twice the frequency. Why is this possible?

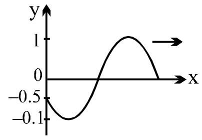
ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

Q.1 A wave is propagating along x-axis. The displacement of particles of the medium in z-direction at $t = 0$ is given by: $z = \exp[-(x+2)^2]$, where 'x' is in meters. At $t = 1s$, the same wave disturbance is given by: $z = \exp[-(2-x)^2]$. Then, the wave propagation velocity is

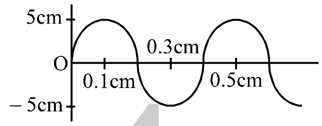
(A) 4 m/s in + x direction (B) 4 m/s in –x direction
(C) 2 m/s in + x direction (D) 2 m/s in – x direction

Q.2 The equation of a wave travelling along the positive x-axis, as shown in figure at t = 0 is given by



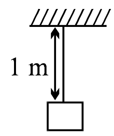
(A) $\sin\left(kx - \omega t + \dfrac{\pi}{6}\right)$ (B) $\sin\left(kx - \omega t - \dfrac{\pi}{6}\right)$ (C) $\sin\left(\omega t - kx + \dfrac{\pi}{6}\right)$ (D) $\sin\left(\omega t - kx - \dfrac{\pi}{6}\right)$

Q.3 Figure shown the shape of part of a long string in which transverse waves are produced by attaching one end of the string to tuning fork of frequency 250 Hz. What is the velocity of the waves?



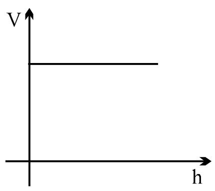
(A) 1.0 ms^{-1} (B) 1.5 ms^{-1}
(C) 2.0 ms^{-1} (D) 2.5 ms^{-1}

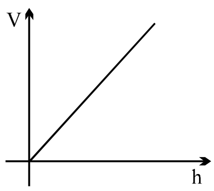
Q.4 A block of mass 1 kg is hanging vertically from a string of length 1 m and mass/length = 0.001 Kg/m. A small pulse is generated at its lower end. The pulse reaches the top end in approximately

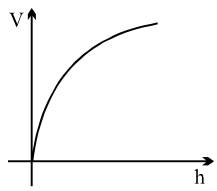


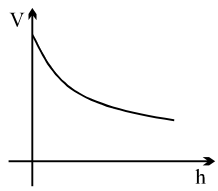
(A) 0.2 sec (B) 0.1 sec (C) 0.02 sec (D) 0.01 sec

Q.5 A uniform rope having some mass hanges vertically from a rigid support. A transverse wave pulse is produced at the lower end. The speed (v) of the wave pulse varies with height (h) from the lower end as:

(A)


(B)


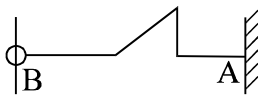
(C)


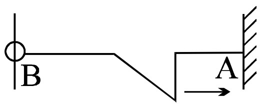
(D)


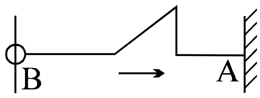
Q.6 A wire of 10^{-2} kgm^{-1} passes over a frictionless light pulley fixed on the top of a frictionless inclined plane which makes an angle of 30° with the horizontal. Masses m and M are tied at two ends of wire such that m rests on the plane and M hangs freely vertically downwards. The entire system is in equilibrium and a transverse wave propagates along the wire with a velocity of 100 ms^{-1}.

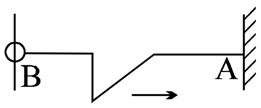
(A) M = 5 kg (B) $\dfrac{m}{M} = \dfrac{1}{4}$ (C) m = 20 kg (D) $\dfrac{m}{M} = 4$

Q.7 A pulse shown here is reflected from the rigid wall A and then from free end B. The shape of the string after these 2 reflection will be



(A)


(B)


(C)


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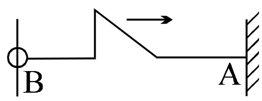


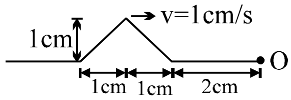
Q.8 A composition string is made up by joining two strings of different masses per unit length $\rightarrow \mu$ and 4μ. The composite string is under the same tension. A transverse wave pulse : $Y = (6\text{ mm})\sin(5t + 40x)$, where 't' is in seconds and 'x' in meters, is sent along the lighter string towards the joint. The joint is at $x = 0$. The equation of the wave pulse reflected from the joint is

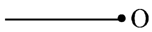
(A) $(2\text{ mm})\sin(5t - 40x)$
(B) $(4\text{ mm})\sin(40x - 5t)$
(C) $-(2\text{ mm})\sin(5t - 40x)$
(D) $(2\text{ mm})\sin(5t - 10x)$

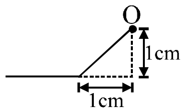
Q.9 In the previous question, the percentage of power transmitted to the heavier string through the joint is approximately

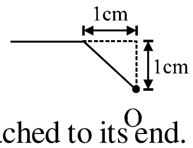
(A) 33%
(B) 89%
(C) 67%
(D) 75%

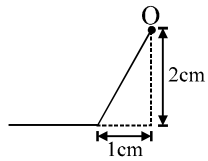
Q.10 A wave pulse on a string has the dimension shown in figure. The waves speed is v = 1 cm/s. If point O is a free end. The shape of wave at time t = 3 s is :



(A) 

(B) 

(C) 

(D) 

Q.11 A string 1m long is drawn by a 300Hz vibrator attached to its end. The string vibrates in 3 segments. The speed of transverse waves in the string is equal to

(A) 100 m/s
(B) 200 m/s
(C) 300 m/s
(D) 400 m/s

Q.12 A wave is represented by the equation $y = 10 \sin 2\pi(100t - 0.02x) + 10 \sin 2\pi(100t + 0.02x)$. The maximum amplitude and loop length are respectively

(A) 20 units and 30 units
(B) 20 units and 25 units
(C) 30 units and 20 units
(D) 25 units and 20 units

Q.13 The resultant amplitude due to superposition of two waves $y_1 = 5\sin(wt - kx)$ and $y_2 = -5\cos(wt - kx - 150°)$

(A) 5
(B) $5\sqrt{3}$
(C) $5\sqrt{2-\sqrt{3}}$
(D) $5\sqrt{2+\sqrt{3}}$

Q.14 A wave represented by the equation $y = A\cos(kx - \omega t)$ is superimposed with another wave to form a statioary wave such that the point x =0 is a node. The equation of the other wave is:

(A) $-A\sin(kx + \omega t)$
(B) $-A\cos(kx + \omega t)$
(C) $A\sin(kx + \omega t)$
(D) $A\cos(kx + \omega t)$

Q.15 A taut string at both ends vibrates in its n^{th} overtone. The distance between adjacent Node and Antinode is found to be 'd'. If the length of the string is L, then

(A) $L = 2d(n + 1)$
(B) $L = d(n + 1)$
(C) $L = 2dn$
(D) $L = 2d(n - 1)$

Q.16 A metallic wire of length L is fixed between two rigid supports. If the wire is cooled through a temperature difference ΔT (Y = young's modulus, ρ = density, α = coefficient of linear expansion) then the frequency of transverse vibration is proportional to :

(A) $\dfrac{\alpha}{\sqrt{\rho Y}}$
(B) $\sqrt{\dfrac{Y\alpha}{\rho}}$
(C) $\dfrac{\rho}{\sqrt{Y\alpha}}$
(D) $\sqrt{\dfrac{\rho\alpha}{Y}}$

Q.17 A standing wave $y = A\sin\left(\dfrac{20}{3}\pi x\right)\cos(1000\pi t)$ is maintained in a taut string where y and x are expressed in meters. The distance between the successive points oscillating with the amplitude A/2 across a node is equal to

(A) 2.5cm
(B) 25cm
(C) 5cm
(D) 10cm

Q.18 A string of length 1m and linear mass density $0.01 kgm^{-1}$ is stretched to a tension of 100N. When both ends of the string are fixed, the three lowest frequencies for standing wave are f_1, f_2 and f_3. When only one end of the string is fixed, the three lowest frequencies for standing wave are n_1, n_2 and n3. Then

(A) $n_3 = 5n_1 = f_3 = 125$ Hz (B) $f_3 = 5f_1 = n_2 = 125$ Hz

(C) $f_3 = n_2 = 3f_1 = 150$ Hz (D) $n_2 = \frac{f_1 + f_2}{2} = 75$ Hz

Q.19 The frequency of a sonometer wire is f, but when the weights producing the tensions are completely immersed in water the frequency becomes f/2 and on immersing the weights in a certain liquid the frequency becomes f/3. The specific gravity of the liquid is:

(A) $\frac{4}{3}$ (B) $\frac{16}{9}$ (C) $\frac{15}{12}$ (D) $\frac{32}{27}$

Q.20 A firecracker exploding on the surface of a lake is heard as two sounds a time interval *t* apart by a man on a boat close to water surface. Sound travels with a speed *u* in water and a speed *v* in air. The distance from the exploding firecracker to the boat is

(A) $\frac{uvt}{u+v}$ (B) $\frac{t(u+v)}{uv}$ (C) $\frac{t(u-v)}{uv}$ (D) $\frac{uvt}{u-v}$

Q.21 A wave travels uniformly in all directions from a point source in an isotropic medium. The displacement of the medium at any point at a distance r from the source may be represented by
(A is a constant representing strength of source)

(A) $[A/\sqrt{r}] \sin(kr - \omega t)$ (B) $[A/r] \sin(kr - \omega t)$

(C) $[Ar] \sin(kr - \omega t)$ (D) $[A/r^2] \sin(kr - \omega t)$

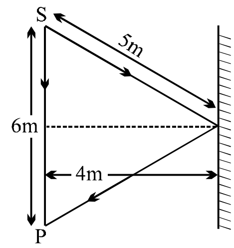
Q.22 How many times more intense is 90 dB sound than 40 dB sound?

(A) 5 (B) 50 (C) 500 (D) 10^5

Q.23 Three coherent waves of equal frequencies having amplitude 10 μm, 4μm and 7 μm respectively, arrive at a given point with successive phase difference of $\pi/2$. The amplitude of the resulting wave in mm is given by

(A) 5 (B) 6 (C) 3 (D) 4

Q.24 A person standing at a distance of 6 m from a source of sound receives sound wave in two ways, one directly from the source and other after reflection from a rigid boundary as shown in the figure. The maximum wavelength for which, the person will receive maximum sound intensity, is



(A) 4 m (B) $\frac{16}{3}$ m (C) 2 m (D) $\frac{8}{3}$ m

Q.25 The ratio of intensities between two coherent soud sources is 4 : 1. The differenmce of loudness in dB between maximum and minimum intensities when they interfere in space is:

(A) 10 log 2 (B) 20 log 3 (C) 10 log 3 (D) 20 log 2

Q.26 In Quincke's tube a detector detects minimum intensity. Now one of the tube is displaced by 5 cm. During displacement detector detects maximum intensity 10 times, then finally a minimum intensity (when displacement is complete). The wavelength of sound is:

(A) 10/9 cm (B) 1 cm (C) 1/2 cm (D) 5/9 cm

Q.27 The ratio of maximum to minimum intensity due to superposition of two waves is $\frac{49}{9}$. Then the ratio of the intensity of component waves is

(A) $\frac{25}{4}$ (B) $\frac{16}{25}$ (C) $\frac{4}{49}$ (D) $\frac{9}{49}$

Q.28 Two waves of sound having intensities I and 4I interfere to produce interference pattern. The phase difference between the waves is $\pi/2$ at point A and π at point B. Then the difference between the resultant intensities at A and B is

(A) 2I (B) 4I (C) 5I (D) 7I

Q.29 Sound waves of frequency 660 Hz fall normally on a perfectly reflecting wall. The shortest distance from the wall at which the air particle has maximum amplitude of vibration is (velocity of sound in air is 330 m/s)

(A) 0.125 m (B) 0.5 m (C) 0.25 m (D) 2 m

Q.30 An open organ pipe of length L vibrates in second harmonic mode. The pressure vibration is maximum

(A) at the two ends (B) at a distance L/4 from either end inside the tube
(C) at the mid-point of the tube (D) none of these

Q.31 An open organ pipe of length l is sounded together with another organ pipe of length l + x in their fundamental tones (x << l). The beat frequency heard will be (speed of sound is v) :

(A) $\frac{vx}{4l^2}$ (B) $\frac{vl^2}{2x}$ (C) $\frac{vx}{2l^2}$ (D) $\frac{vx^2}{2l}$

Q.32 A sufficiently long close organ pipe has a small hole at its bottom. Initially the pipe is empty. Water is poured into the pipe at a constant rate. The fundamental frequency of the air column in the pipe

(A) continuously increasing (B) first increases and them becomes constant
(C) continuously decreases (D) first decreases and them become constant

Q.33 A tuning fork of frequency 340 Hz is vibrated just above a cylindrical tube of length 120 cm. Water is slowly poured in the tube. If the speed of sound is 340 ms^{-1} then the minimum height of water required for resonance is:

(A) 95 cm (B) 75 cm (C) 45 cm (D) 25 cm

Q.34 An organ pipe P_1 closed at one end vibrating in its first overtone. Another pipe P_2 open at both ends is vibrating in its third overtone. They are in a resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is :

(A) 8/3 (B) 3/8 (C) 1/2 (D) 1/3

Q.35 In a closed end pipe of length 105 cm, standing waves are set up corresponding to the third overtone. What distance from the closed end, amongst the following, is a pressure Node?

(A) 20 cm (B) 60 cm (C) 85 cm (D) 45 cm

Q.36 A pipe's lower end is immersed in water such that the length of air column from the top open end has a certain length 25 cm. The speed of sound in air is 350 m/s. The air column is found to resonate with a tuning fork of frequency 1750 Hz. By what minimum distance should the pipe be raised in order to make the air column resonate again with the same tuning fork?

(A) 7 cm (B) 5 cm (C) 35 cm (D) 10 cm

Q.37 In case of closed organ pipe which harmonic the pth overtone will be

(A) 2p + 1 (B) 2p – 1 (C) p + 1 (D) p – 1

Q.38 A closed organ pipe of radius r_1 and an open organ pipe of radius r_2 and having same length 'L' resonate when excited with a given tunning fork. Closed organ pipe resonates in its fundamental mode where as open organ pipe resonates in its first overtone, then

(A) $r_2 - r_1 = L$ (B) $r_2 - r_1 = L/2$ (C) $r_2 - 2r_1 = 2.5\ L$ (D) $2r_2 - r_1 = 2.5\ L$

Q.39 First overtone frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. Further nth harmonic of closed organ pipe is also equal to the mth harmonic of open pipe, where n and m are:

(A) 5, 4 (B) 7, 5 (C) 9, 6 (D) 7, 3

Q.40 If l_1 and l_2 are the lengths of air column for the first and second resonance when a tuning fork of frequency n is sounded on a resonance tube, then the distance of the displacement antinode from the top end of the resonance tube is:

(A) $2(l_2 - l_1)$ (B) $\frac{1}{2}(2l_1 - l_2)$ (C) $\frac{l_2 - 3l_1}{2}$ (D) $\frac{l_2 - l_1}{2}$

Q.41 A closed orgain pipe has length 'l'. The air in it is vibrating in 3rd overtone with maximum displacement amplitude 'a'. The displacement amplitude at distance $l/7$ from closed end of the pipe is:

(A) 0 (B) a (C) a / 2 (D) none of these

Q.42 The first resonance length of a resonance tube is 40 cm and the second resonance length is 122 cm. The third resonance length of the tube will be

(A) 200 cm (B) 202 cm (C) 203 cm (D) 204 cm

Q.43 A tuning fork of frequency 280 Hz produces 10 beats per sec when sounded with a vibrating sonometer string. When the tension in the string increases slightly, it produces 11 beats per sec. The original frequency of the vibrating sonometer string is :

(A) 269 Hz (B) 291 Hz (C) 270 Hz (D) 290 Hz

Q.44 Two tuning forks A & B produce notes of frequencies 256 Hz & 262 Hz respectively. An unknown note sounded at the same time as A produces beats . When the same note is sounded with B, beat frequency is twice as large . The unknown frequency could be :

(A) 268 Hz (B) 260 Hz (C) 250 Hz (D) 242 Hz

Q.45 A closed organ pipe and an open pipe of same length produce 4 beats when they are set into vibrations simultaneously. If the length of each of them were twice their initial lengths, the number of beats produced will be

(A) 2 (B) 4 (C) 1 (D) 8

Q.46 The speed of sound in a gas, in which two waves of wavelength 1.0 m and 1.02 m produce 6 beats per second, is approximately:

(A) 350 m/s (B) 300 m/s (C) 380 m/s (D) 410 m/s

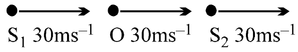
Q.47 In a test of subsonic Jet flies over head at an altitude of 100 m. The sound intensity on the ground as the Jet passes overhead is 160 dB. At what altitude should the plane fly so that the ground noise is not greater than 120 dB.

(A) above 10 km from ground (B) above 1 km from ground
(C) above 5 km from ground (D) above 8 km from ground

Q.48 The frequency changes by 10% as a sound source approaches a stationary observer with constant speed v_s. What would be the percentage change in frequency as the source recedes the observer with the same speed. Given that $v_s < v$. (v = speed of sound in air)

(A) 14.3% (B) 20% (C) 10.0% (D) 8.5%

Q.49 Consider two sound sources S_1 and S_2 having same frequency 100Hz and the observer O located between them as shown in the fig. All the three are moving with same velocity in same direction. The beat frequency of the observer is

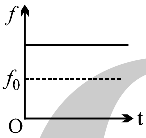


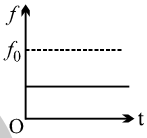
(A) 50Hz (B) 5 Hz (C) zero (D) 2.5 Hz

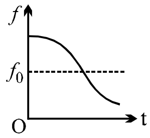
Q.50 An engine whistling at a constant frequency n_0 and moving with a constant velocity goes past a stationary observer. As the engine crosses him, the frequency of the sound heard by him changes by a factor f. The actual difference in the frequencies of the sound heard by him before and after the engine crosses him is

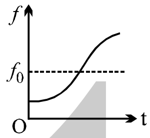
(A) $\frac{1}{2}n_0(1-f^2)$ (B) $\frac{1}{2}n_0\left(\frac{1-f^2}{f}\right)$ (C) $n_0\left(\frac{1-f}{1+f}\right)$ (D) $\frac{1}{2}n_0\left(\frac{1-f}{1+f}\right)$

Q.51 Source and observer both start moving simultaneously from origin, one along x-axis and the other along y-axis with speed of source = twice the speed of observer. The graph between the apparent frequency observed by observer f and time t would approximately be :

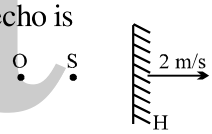
(A) 

(B) 

(C) 

(D) 

Q.52 A stationary sound source 's' of frequency 334 Hz and a stationary observer 'O' are placed near a reflecting surface moving away from the source with velocity 2 m/sec as shown in the figure. If the velocity of the sound waves is air is V = 330 m/sec, the apparent frequency of the echo is

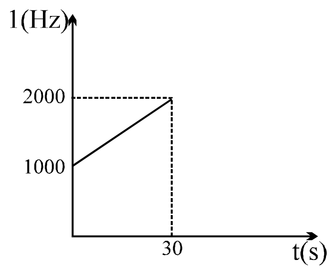


(A) 332 Hz (B) 326 Hz
(C) 334 Hz (D) 330 Hz

Q.53 A source S of frequency f_0 and an observer O, moving with speeds v_1 and v_2 respectively, are movinng away from each other. When they are separated by distance a (t =0), a pulse is emitted by the source. This pulse is received by O at time t_1 then t_1, is equal to

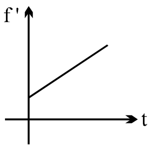
(A) $\frac{a}{v_s + v_2}$ (B) $\frac{a}{v_1 + v_s}$ (C) $\frac{a}{v_s - v_2}$ (D) $\frac{a}{v_1 + v_2 + v_s}$

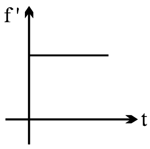
Q.54 A detector is released from rest over a source of sound of frequency $f_0 = 10^3$ Hz. The frequency observed by the detector at time t is plotted in the graph. The speed of sound in air is (g = 10 m/s^2)

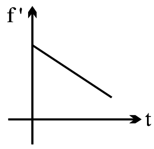


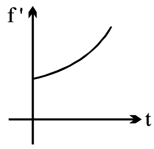
(A) 330 m/s (B) 350 m/s
(C) 300 m/s (D) 310 m/s

Q.55 An observer starts moving with uniform acceleration 'a' towards a stationary sound source of frequency f. As the observer approaches the source, the apparent frequency f' heard by the observer varies with time t as:

(A) 

(B) 

(C) 

(D) 

Q.56 A sounding body of negligible dimension emitting a frequency of 150 Hz is dropped from a height. During its fall under gravity it passes near a balloon moving up with a constant velocity of 2m/s one second after it started to fall.The difference in the frequency observed by the man in balloon just before and just after crossing the body will be : (Given that -velocity of sound = 300m/s; g = 10m/s^2)

(A) 12 (B) 6 (C) 8 (D) 4

Q.57 A source of sound S having frequency f. Wind is blowing from source to observer O with velocity u. If speed of sound with respect to air is C, the wavelength of sound detected by O is:

(A) $\frac{C+u}{f}$ (B) $\frac{C-u}{f}$ (C) $\frac{C(C+u)}{(C-u)f}$ (D) $\frac{C}{f}$

<u>ONE OR MORE THAN ONE OPTION MAY BE CORRECT</u>

Take approx. 3 minutes for answering each question.

Q.1 A sinusoidal progressive wave is generated in a string. It's equation is given by $y = (2\text{ mm}) \sin (2\pi x - 100\pi t + \pi/3)$. The time when particle at x = 4 m first passes through mean position, will be

(A) $\frac{1}{150}$ sec (B) $\frac{1}{12}$ sec (C) $\frac{1}{300}$ sec (D) $\frac{1}{100}$ sec

Q.2 A transverse wave is described by the equation $y = A \sin [2\pi (ft - x/\lambda)]$.The maximum particle velocity is equal to four times the wave velocity if:

(A) $\lambda = \pi A/4$ (B) $\lambda = \pi A/2$ (C) $\lambda = \pi A$ (D) $\lambda = 2\pi A$

Q.3 A wave equation is given as $y = \cos(500t - 70x)$, where y is in mm, x in m adn t is in sec.

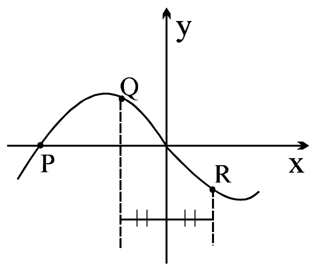
(A) the wave must be a transverse porpagating wave.

(B) The speed of the wave is 50/7 m/s

(C) The frequency of oscillations 1000π Hz

(D) Two closest points which are in same phase have separation $20\pi/7$ cm.

Q.4 At a certain moment, the photograph of a string on which a harmonic wave is travelling to the right is shown. Then, which of the following is true regarding the velocities of the points P, Q and R on the string.

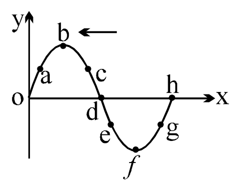


(A) v_P is upwards (B) $v_Q = -v_R$

(C) $|v_P| > |v_Q| = |v_R|$ (D) $v_Q = v_R$

Question No. 5 to 8 (4 questions)

The figure represents the instantaneous picture of a transverse harmonic wave traveling along the negative x-axis. Choose the correct alternative(s) related to the movement of the nine points shown in the figure.



Q.5 The points moving upward is/are

(A) a (B) c (C) f (D) g

Q.6 The points moving downwards is/are

(A) o (B) b (C) d (D) h

Q.7 The stationary points is/are

(A) o (B) b (C) f (D) h

Q.8 The points moving with maximum velocity is/are

(A) b (B) c (C) d (D) h

Q.9 A perfectly elastic uniform string is suspended vertically with its upper end fixed to the ceiling and the lower end loaded with the weight. If a transverse wave is imparted to the lower end of the string, the pulse will

(A) not travel along the length of the string
(B) travel upwards with increasing speed
(C) travel upwards with decreasing speed
(D) travelled upwards with constant acceleration

Q.10 One end of a string of length L is tied to the ceiling of a lift accelerating upwards with an acceleration 2g. The other end of the string is free. The linear mass density of the string varies linearly from 0 to λ from bottom to top.

(A) The velocity of the wave in the string will be 0.
(B) The acceleration of the wave on the string will be 3g/4 every where.
(C) The time taken by a pulse to reach from bottom to top will be $\sqrt{8L/3g}$.
(D) The time taken by a pulse to reach from bottom to top will be $\sqrt{4L/3g}$.

Q.11 A plane wave y=A sin $\omega\left(t-\frac{x}{v}\right)$ undergo a normal incidence on a plane boundary separating medium M_1 and M_2 and splits into a reflected and transmitted wave having speeds v_1 and v_2 then

(A) for all values of v_1 and v_2 the phase of transmitted wave is same as that of incident wave
(B) for all values of v_1 and v_2 the phase of reflected wave is same as that of incident wave
(C) the phase of transmitted wave depends upon v_1 and v_2
(D) the phase of reflected wave depends upon v_1 and v_2

Q.12 The vibration of a string fixed at both ends are described by Y= 2 sin(πx) sin(100πt) where Y is in mm, x is in cm, t in sec then

(A) Maximum displacement of the particle at x = 1/6 cm would be 1 mm.
(B) velocity of the particle at x = 1/6 cm at time t = 1/600 sec will be 157 $\sqrt{3}$ mm/s
(C) If the length of the string be 10 cm, number of loop in it would be 5
(D) None of these

Q.13 In a standing wave on a string.

(A) In one time period all the particles are simultaneously at rest twice.
(B) All the particles must be at their positive extremes simultaneously once in one time period.
(C) All the particles may be at their positive extremes simultaneouslyonce in a time period.
(D) All the particles are never at rest simultaneously.

Q.14 A standing wave pattern of amplitude A in a string of length L shows 2 nodes (plus those at two ends). If one end of the string corresponds to the origin and v is the speed of progressive wave, the disturbance in the string, could be represented (with appropriate phase) as:

(A) $y(x,t) = A\sin\left(\frac{2\pi x}{L}\right)\cos\left(\frac{2\pi vt}{L}\right)$
(B) $y(x,t) = A\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{3\pi vt}{L}\right)$
(C) $y(x,t) = A\cos\left(\frac{4\pi x}{L}\right)\cos\left(\frac{4\pi vt}{L}\right)$
(D) $y(x,t) = A\sin\left(\frac{3\pi x}{L}\right)\cos\left(\frac{3\pi vt}{L}\right)$

Q.15 The length, tension, diameter and density of a wire B are double than the corresponding quantities for another stretched wire A. Then.

(A) Fundamental frequency of B is $\frac{1}{2\sqrt{2}}$ times that of A.

(B) The velocity of wave in B is $\frac{1}{\sqrt{2}}$ times that of velocity in A.

(C) The fundamental frequency of A is equal to the third overtone of B.

(D) The velocity of wave in B is half that of velocity in A.

Q.16 A string is fixed at both ends vibrates in a resonant mode with a separation 2.0 cm between the consecutive nodes. For the next higher resonant frequency, this separation is reduced to 1.6 cm. The length of the string is

(A) 4.0 cm (B) 8.0 cm (C) 12.0 cm (D) 16.0 cm

Q.17 A clamped string is oscillating in nth harmonic, then

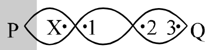
(A) total energy of oscillations will be n^2 times that of fundamental frequency

(B) total energy of oscillations will be $(n-1)^2$ times that of fundamental frequency

(C) average kinetic energy of the string over a complete oscillations is half of that of the total energy of the string.

(D) none of these

Q.18 Figure, shows a stationary wave between two fixed points P and Q. Which point(s) of 1, 2 and 3 are in phase with the point X?



(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 3 only

Q.19 Which of the following statements are wrong about the velocity of sound in air:

(A) decreases with increases in temperature (B) increases with decrease in temperature

(C) decreases as humidity increases (D) independent of density of air.

Q.20 The particle displacement of a travelling longitudinal wave is represented by $\xi = \xi\,(x, t)$. The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

(A) $|\partial\xi / \partial x|_C = |\partial\xi / \partial x|_R$

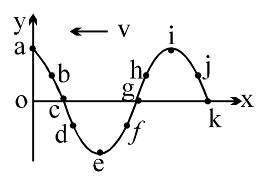
(B) $|\partial\xi / \partial t|_C = |\partial\xi / \partial t|_R = 0$

(C) $(\text{pressure})_C - (\text{pressure})_R = 2\,|\partial\xi / \partial x|_C$ x Bulk modulus of air.

(D) Particles of air are stationary mid-way between 'C' and 'R'.

Question No. 21 to 26 (6 questions)

The figure represents the instantaneous picture of a longitudinal harmonic wave travelling along the negative x-axis. Identify the correct statement(s) related to the movement of the points shown in the figure.



Q.21 The points moving in the direction of wave are

(A) b (B) c (C) f (D) i

Q.22 The points moving opposite to the direction of propagation are

(A) a (B) d (C) f (D) j

Q.23 The stationary points are

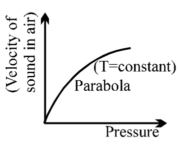
(A) a (B) c (C) g (D) k

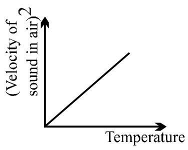
Q.24 The maximum displaced points are
(A) a (B) e (C) g (D) i

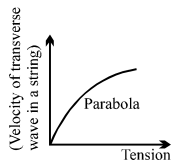
Q.25 The points of maximum compression are
(A) c (B) g (C) e (D) k

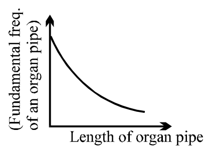
Q.26 The points of maximum rarefaction are
(A) a (B) e (C) g (D) i

Q.27 Which of the following graphs is/are correct.

(A) 

(B) 

(C) 

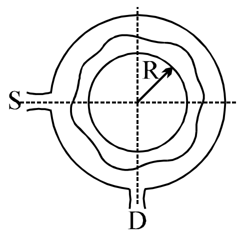
(D) 

Q.28 Two waves are propagating along a taut string that coincides with the x-axis. The first wave has the wave function y_1 = Acos [k(x – vt)] and the second has the wave function y = A cos [k(x + vt) + ϕ].
(A) For constructive interference at x = 0, $\phi = \pi$.
(B) For constructive interference at x = 0, $\phi = 3\pi$.
(C) For destructive interference at x = 0, $\phi = \pi$.
(D) For destructive interference at x = 0, $\phi = 2\pi$.

Q.29 Two interfering waves have the same wavelength, frequency, and amplitude. They are traveling in the same direction but are 90° out of phase. Compared to the individual waves, the resultant wave will have the same.
(A) amplitude and velocity but different wavelength
(B) amplitude and wavelength but different velocity
(C) wavelength and velocity but different amplitude
(D) amplitude and frequency but different velocity

Question No. 30 to 34 (5 questions)

A narrow tube is bent in the form of a circle of radius R, as shown in the figure. Two small holes S and D are made in the tube at the positions right angle to each other. A source placed at S generated a wave of intensity I_0 which is equally divided into two parts : One part travels along the longer path, while the other travels along the shorter path. Both the part waves meet at the point D where a detector is placed



Q.30 If a maxima is formed at the detector then, the magnitude of wavelength λ of the wave produced is given by

(A) πR (B) $\frac{\pi R}{2}$ (C) $\frac{\pi R}{4}$ (D) $\frac{2\pi R}{3}$

Q.31 If the minima is formed at the detector then, the magnitude of wavelength λ of the wave produced is given by

(A) $2\pi R$ (B) $\frac{3\pi R}{2}$ (C) $\frac{2\pi R}{3}$ (D) $\frac{2\pi R}{5}$

Q.32 The maximum intensity produced at D is given by
(A) $4I_0$ (B) $2I_0$ (C) I_0 (D) $3I_0$

Q.33 The maximum value of λ to produce a maxima at D is given by

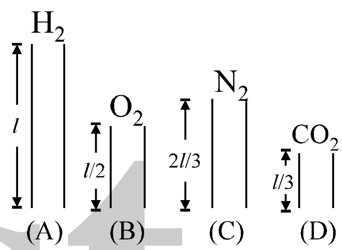
(A) πR (B) $2\pi R$ (C) $\frac{\pi R}{2}$ (D) $\frac{3\pi R}{2}$

Q.34 The maximum value of λ to produce a minima at D is given by

(A) πR (B) $2\pi R$ (C) $\frac{\pi R}{2}$ (D) $\frac{3\pi R}{2}$

Q.35 The second overtone of an open organ pipe A and a closed pipe B have the same frequency at a given temperature. It follows that the ratio of the
(A) length of A and B is 4 : 3
(B) fundamental frequencies of A & B is 5 : 6
(C) lengths of B to that of A is 5 : 6
(D) frequencies of first overtone of A & B is 10 : 9

Q.36 Four open organ pipes of different lengths and different gases at same temperature as shown in figure. Let f_A, f_B, f_C and f_D be their fundamental frequencies then :[Take $\gamma_{CO_2} = 7/5$]



(A) $f_A/f_B = \sqrt{2}$ (B) $f_B/f_C = \sqrt{72/28}$
(C) $f_C/f_D = \sqrt{11/28}$ (D) $f_D/f_A = \sqrt{76/11}$

Q.37 A gas is filled in an organ pipe and it is sounded with an organ pipe in fundamental mode. Choose the correct statement(s) : (T = constant)
(A) If gas is changed from H_2 to O_2, the resonant frequency will increase
(B) If gas is changed from O_2 to N_2, the resonant frequency will increase
(C) If gas is changed from N_2 to He, the resonant frequency will decrease
(D) If gas is changed from He to CH_4, the resonant frequency will decrease

Q.38 A closed organ pipe of length 1.2 m vibrates in its first overtone mode. The pressure variation is maximum at:
(A) 0.8 m from the open end (B) 0.4 m from the open end
(C) at the open end (D) 1.0 m from the open end

Q.39 The equation of a wave disturbance is given as : $y = 0.02\cos\left(\frac{\pi}{2} + 50\pi t\right)\cos(10\pi x)$, where x and y are in meters and t in seconds. Choose the wrong statement:
(A) Antinode occurs at x = 0.3 m (B) The wavelength is 0.2 m
(C) The speed of the constituent waves is 4 m/s (D) Node occurs at x = 0.15 m

Q.40 For a certain organ pipe three successive resonance frequencies are observed at 425 Hz, 595 Hz and 765 Hz respectively. If the speed of sound in air is 340 m/s, then the length of the pipe is:
(A) 2.0 m (B) 0.4 m (C) 1.0 m (D) 0.2 m

Q.41 In an organ pipe whose one end is at x = 0, the pressure is expressed by

$$p = p_0 \cos\frac{3\pi x}{2}\sin 300\pi t$$ where x is in meter and t in sec. The organ pipe can be

(A) closed at one end, open at another with length = 0.5m
(B) open at both ends, length = 1m
(C) closed at both ends, length = 2m
(D) closed at one end, open at another with length = $\frac{2}{3}$ m

Q.42 Two whistles A and B each have a frequency of 500Hz. A is stationary and B is moving towards the right (away from A) at a speed of 50 m/s. An observer is between the two whistles moving towards the right with a speed of 25 m/s. The velocity of sound in air is 350 m/s. Assume there is no wind. Then which of the following statements are true:
(A) The apparent frequency of whistle B as heard by A is 444Hz approximately
(B) The apparent frequency of whistle B as heard by the observer is 469Hz approximately
(C) The difference in the apparent frequencies of A and B as heard by the observer is 4.5 Hz.
(D) The apparent frequencies of the whistles of each other as heard by A and Bare the same.

Q.43 A source of sound moves towards an observer
(A) the frequency of the source is increased.
(B) the velocity of sound in the medium is increased.
(C) the wavelength of sound in the medium towards the observer is decreased.
(D) the amplitude of vibration of the particles is increased.

Q.44 A car moves towards a hill with speed v_c. It blows a horn of frequency f which is heared by an observer following the car with speed v_0. The speed of sound in air is v.
(A) the wavelength of sound reaching the hill is $\frac{v}{f}$
(B) the wavelength of sound reaching the hill is $\frac{v - v_c}{f}$
(C) the beat frequency observed by the observer is $\left(\frac{v + v_o}{v - v_c}\right)f$
(D) the beat frequency observed by the observer is $\frac{2v_c(v + v_o)f}{v^2 - v_c^2}$

ANSWER KEY

ONLY ONE OPTION IS CORRECT.

Q.1	A	Q.2	D	Q.3	A	Q.4	D	Q.5	C	Q.6	C	Q.7	A
Q.8	C	Q.9	B	Q.10	D	Q.11	B	Q.12	B	Q.13	A	Q.14	B
Q.15	A	Q.16	B	Q.17	C	Q.18	D	Q.19	D	Q.20	D	Q.21	B
Q.22	D	Q.23	A	Q.24	A	Q.25	B	Q.26	B	Q.27	A	Q.28	B
Q.29	A	Q.30	B	Q.31	C	Q.32	B	Q.33	C	Q.34	B	Q.35	D
Q.36	D	Q.37	A	Q.38	C	Q.39	C	Q.40	C	Q.41	B	Q.42	D
Q.43	D	Q.44	C	Q.45	A	Q.46	B	Q.47	A	Q.48	D	Q.49	C
Q.50	B	Q.51	B	Q.52	D	Q.53	C	Q.54	C	Q.55	A	Q.56	A
Q.57	A												

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	C	Q.2	B	Q.3	A,B,D	Q.4	C,D
Q.5	A,D	Q.6	C	Q.7	B,C	Q.8	C,D
Q.9	B,D	Q.10	B,C	Q.11	A,D	Q.12	A,B
Q.13	A,C	Q.14	D	Q.15	C,D	Q.16	B
Q.17	A,C	Q.18	D	Q.19	A,B,C,D	Q.20	A,C,D
Q.21	B	Q.22	C	Q.23	A	Q.24	A,B,D
Q.25	A,D	Q.26	C	Q.27	B,C	Q.28	C
Q.29	C	Q.30	A,B,C	Q.31	A,C,D	Q.32	B
Q.33	A	Q.34	B	Q.35	C,D	Q.36	C
Q.37	B,D	Q.38	B	Q.39	C	Q.40	C
Q.41	C	Q.42	B,C	Q.43	C	Q.44	B,D

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QUEST TUTORIALS

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PHYSICS

SIMPLE HARMONIC MOTION

QUESTION FOR SHORT ANSWER

Q.1 Suppose we have a block of unknown mass and a spring of unknown force constant. Show how we can predict the period of oscillation of this block-spring system simply by measuring the extension of the spring produced by attaching the block to it.

Q.2 How are each of the following properties of a simple harmonic oscillator affected by doubling the amplitude: period, force constant, total mechanical energy, maximum velocity, maximum acceleration?

Q.3 A person stands on a bathroom-type scale, which rests on a platform suspended by a large spring. The whole system executes simple harmonic motion in a vertical direction. Describe the variation in scale reading during a period of motion.

Q.4 Predict by qualitative arguments whether a pendulum oscillating with large amplitude will have a period longer or shorter than the period for oscillations with small amplitude. (Consider extreme cases.)

Q.5 If taken to the Moon, will there be any change in the frequency of oscillation of a torsional pendulum? A simple pendulum? A spring-block oscillator? A physical pendulum?

Q.6 How can a pendulum be used to trace out a sinusoidal curve?

Q.7 During SHM, are the displacement and velocity ever in the same direction? The velocity and acceleration? The displacement and the acceleration?

ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

Q.1 A simple harmonic motion having an amplitude A and time period T is represented by the equation : $y = 5 \sin \pi(t + 4)$ m
Then the values of A (in m) and T (in sec) are :
(A) A = 5; T = 2 (B) A = 10 ; T = 1 (C) A = 5 ; T = 1 (D) A = 10 ; T = 2

Q.2 The displacement of a body executing SHM is given by $x = A \sin (2\pi t + \pi/3)$. The first time from t = 0 when the velocity is maximum is
(A) 0.33 sec (B) 0.16 sec (C) 0.25 sec (D) 0.5 sec

Q.3 The maximum acceleration of a particle in SHM is made two times keeping the maximum speed to be constant. It is possible when
(A) amplitude of oscillation is doubled while frequency remains constant
(B) amplitude is doubled while frequency is halved
(C) frequency is doubled while amplitude is halved
(D) frequency is doubled while amplitude remains constant

Q.4 The potential energy of a simple harmonic oscillator of mass 2 kg in its mean position is 5 J. If its total energy is 9J and its amplitude is 0.01 m, its time period would be
(A) $\pi/10$ sec (B) $\pi/20$ sec (C) $\pi/50$ sec (D) $\pi/100$ sec

Q.5 A plank with a small block on top of it is under going vertical SHM. Its period is 2 sec. The minimum amplitude at which the block will separate from plank is :
(A) $\frac{10}{\pi^2}$ (B) $\frac{\pi^2}{10}$ (C) $\frac{20}{\pi^2}$ (D) $\frac{\pi}{10}$

Q.6 Time period of a particle executing SHM is 8 sec. At t = 0 it is at the mean position. The ratio of the distance covered by the particle in the 1st second to the 2nd second is :
(A) $\frac{1}{\sqrt{2}+1}$ (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}+1$

Q.7 The angular frequency of motion whose equation is $4\frac{d^2y}{dt^2} + 9y = 0$ is (y = displacement and t = time)
(A) $\frac{9}{4}$ (B) $\frac{4}{9}$ (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

Q.8 Two particles are in SHM in a straight line. Amplitude A and time period T of both the particles are equal. At time t=0, one particle is at displacement $y_1 = +A$ and the other at $y_2 = -A/2$, and they are approaching towards each other. After what time they cross each other ?
(A) T/3 (B) T/4 (C) 5T/6 (D) T/6

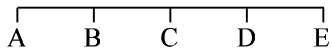
Q.9 Two particles execute SHM of same amplitude of 20 cm with same period along the same line about the same equilibrium position. The maximum distance between the two is 20 cm. Their phase difference in radians is
(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q.10 Two particles P and Q describe simple harmonic motions of same period, same amplitude, along the same line about the same equilibrium position O. When P and Q are on opposite sides of O at the same distance from O they have the same speed of 1.2 m/s in the same direction, when their displacements are the same they have the same speed of 1.6 m/s in opposite directions. The maximum velocity in m/s of either particle is
(A) 2.8 (B) 2.5 (C) 2.4 (D) 2

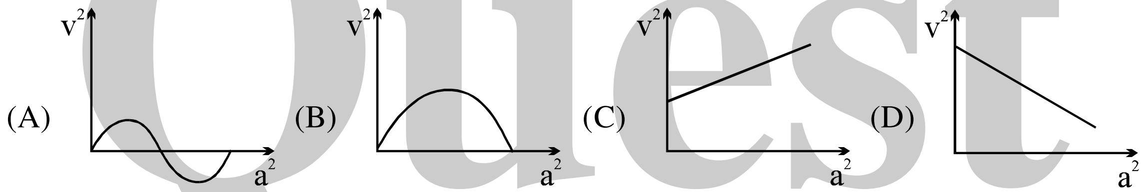
Q.11 A particle performs SHM with a period T and amplitude a. The mean velocity of the particle over the time interval during which it travels a distance a/2 from the extreme position is
(A) a/T (B) 2a/T (C) 3a/T (D) a/2T

Q.12 A body performs simple harmonic oscillations along the straight line ABCDE with C as the midpoint of AE. Its kinetic energies at B and D are each one fourth of its maximum value. If AE = 2R, the distance between B and D is

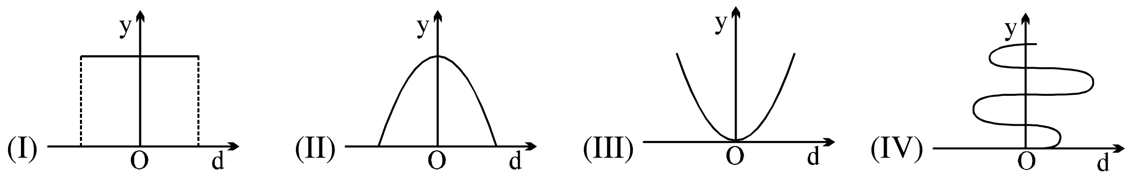
(A) $\frac{\sqrt{3}R}{2}$ (B) $\frac{R}{\sqrt{2}}$ (C) $\sqrt{3}$ R (D) $\sqrt{2}$ R



Q.13 A graph of the square of the velocity against the square of the acceleration of a given simple harmonic motion is



Question No. 14 to 16 (3 questions)
The graphs in figure show that a quantity y varies with displacement d in a system undergoing simple harmonic motion.



Which graphs best represents the relationship obtained when y is

Q.14 The total energy of the system
(A) I (B) II (C) III (D) IV

Q.15 The time
(A) I (B) II (C) III (D) IV

Q.16 The unbalanced force acting on the system.
(A) I (B) II (C) III (D) IV

Q.17 A small mass executes linear SHM about O with amplitude a and period T. Its displacement from O at time T/8 after passing through O is:
(A) a/8 (B) $a/2\sqrt{2}$ (C) a/2 (D) $a/\sqrt{2}$

Q.18 A particle executes SHM of period 1.2 sec and amplitude 8 cm. Find the time it takes to travel 3cm from the positive extremity of its oscillation.
(A) 0.28 sec (B) 0.32 sec (C) 0.17 sec (D) 0.42 sec

Q.19 A particle executes SHM on a straight line path. The amplitude of oscillation is 2 cm. When the displacement of the particle from the mean position is 1 cm, the numerical value of magnitude of acceleration is equal to the numerical value of magnitude of velocity. The frequency of SHM (in second^{-1}) is:

(A) $2\pi\sqrt{3}$ (B) $\frac{2\pi}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2\pi}$ (D) $\frac{1}{2\pi\sqrt{3}}$

Q.20 A stone is swinging in a horizontal circle 0.8 m in diameter at 30 rev / min. A distant horizontal light beam causes a shadow of the stone to be formed on a nearly vertical wall. The amplitude and period of the simple harmonic motion for the shadow of the stone are

(A) 0.4 m, 4 s (B) 0.2 m. 2 s (C) 0.4 m, 2 s (D) 0.8 m, 2 s

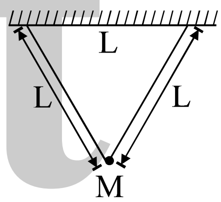
Q.21 Find the ratio of time periods of two identical springs if they are first joined in series & then in parallel & a mass m is suspended from them :

(A) 4 (B) 2 (C) 1 (D) 3

Q.22 In an elevator, a spring clock of time period T_S (mass attached to a spring) and a pendulum clock of time period T_P are kept. If the elevator accelerates upwards

(A) T_S well as T_P increases (B) T_S remain same, T_P increases
(C) T_S remains same, T_P decreases (D) T_S as well as T_P decreases

Q.23 A man is swinging on a swing made of 2 ropes of equal length L and in direction perpendicular to the plane of paper. The time period of the small oscillations about the mean position is



(A) $2\pi\sqrt{\frac{L}{2g}}$ (B) $2\pi\sqrt{\frac{\sqrt{3}L}{2g}}$

(C) $2\pi\sqrt{\frac{L}{2\sqrt{3}g}}$ (D) $\pi\sqrt{\frac{L}{g}}$

Q.24 Two bodies P & Q of equal mass are suspended from two separate massless springs of force constants k_1 & k_2 respectively. If the maximum velocity of them are equal during their motion, the ratio of amplitude of P to Q is :

(A) $\frac{k_1}{k_2}$ (B) $\sqrt{\frac{k_2}{k_1}}$ (C) $\frac{k_2}{k_1}$ (D) $\sqrt{\frac{k_1}{k_2}}$

Q.25 Vertical displacement of a plank with a body of mass 'm' on it is varying according to law $y = \sin\omega t + \sqrt{3}\cos\omega t$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after t = 0 are given by: (y is positive vertically upwards)

(A) $\sqrt{\frac{g}{2}}, \frac{\sqrt{2}}{6}\frac{\pi}{\sqrt{g}}$ (B) $\frac{g}{\sqrt{2}}, \frac{2}{3}\sqrt{\frac{\pi}{g}}$ (C) $\sqrt{\frac{g}{2}}, \frac{\pi}{3}\sqrt{\frac{2}{g}}$ (D) $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

Q.26 A ring of diameter 2m oscillates as a compound pendulum about a horizontal axis passing through a point at its rim. It oscillates such that its centre move in a plane which is perpendicular to the plane of the ring. The equivalent length of the simple pendulum is

(A) 2m (B) 4m (C) 1.5m (D) 3m

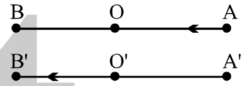
Q.27 Two pendulums have time periods T and 5T/4. They start SHM at the same time from the mean position. After how many oscillations of the smaller pendulum they will be again in the same phase:
(A) 5 (B) 4 (C) 11 (D) 9

Q.28 A particle of mass *m* moves in a one-dimensional potential energy $U(x) = -ax^2 + bx^4$, where 'a' and 'b' are positive constants. The angular frequency of small oscillations about the minima of the potential energy is equal to

(A) $\pi\sqrt{\frac{a}{2b}}$ (B) $2\sqrt{\frac{a}{m}}$ (C) $\sqrt{\frac{2a}{m}}$ (D) $\sqrt{\frac{a}{2m}}$

Q.29 A tunnel is dug in the earth across one of its diameter. Two masses 'm' & '2m' are dropped from the ends of the tunnel. The masses collide and stick to each other and perform S.H.M. Then amplitude of S.H.M. will be: [R = radius of the earth]
(A) R (B) R / 2 (C) R / 3 (D) 2R / 3

Q.30 Two particles undergo SHM along parallel lines with the same time period (T) and equal amplitudes. At a particular instant, one particle is at its extreme position while the other is at its mean position. They move in the same direction. They will cross each other after a further time

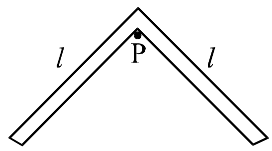


(A) T/8 (B) 3T/8 (C) T/6 (D) 4T/3

Q.31 A heavy brass sphere is hung from a light spring and is set in vertical small oscillation with a period T. The sphere is now immersed in a non-viscous liquid with a density 1/10th the density of the sphere. If the system is now set in vertical S.H.M., its period will be
(A) (9/10)T (B) $(9/10)^2$T (C) (10/9) T (D) T

Q.32 A simple pendulum is oscillating in a lift. If the lift is going down with constant velocity, the time period of the simple pendulum is T_1. If the lift is going down with some retardation its time period is T_2, then
(A) $T_1 > T_2$
(B) $T_1 < T_2$
(C) $T_1 = T_2$
(D) depends upon the mass of the pendulum bob

Q.33 A system of two identical rods (L-shaped) of mass *m* and length *l* are resting on a peg P as shown in the figure. If the system is displaced in its plane by a small angle θ, find the period of oscillations:



(A) $2\pi\sqrt{\frac{\sqrt{2}l}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{2}l}{3g}}$ (C) $2\pi\sqrt{\frac{2l}{3g}}$ (D) $3\pi\sqrt{\frac{l}{3g}}$

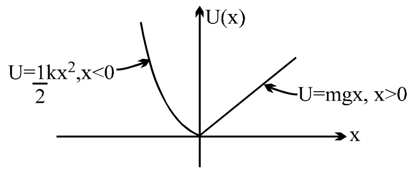
Q.34 A particle starts oscillating simple harmonically from its equilibrium position then, the ratio of kinetic energy and potential energy of the particle at the time T/12 is : (T = time period)
(A) 2 : 1 (B) 3 : 1 (C) 4 :1 (D) 1 : 4

Q.35 A spring mass system preforms S.H.M. If the mass is doubled keeping amplitude same, then the total energy of S.H.M. will become :
(A) double (B) half (C) unchanged (D) 4 times

Q.36 A mass at the end of a spring executes harmonic motion about an equilibrium position with an amplitude A. Its speed as it passes through the equilibrium position is V. If extended 2A and released, the speed of the mass passing through the equilibrium position will be

(A) 2V (B) 4V (C) $\frac{V}{2}$ (D) $\frac{V}{4}$

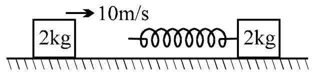
Q.37 A particle of mass m moves in the potential energy U shown above. The period of the motion when the particle has total energy E is



(A) $2\pi\sqrt{m/k}+4\sqrt{2E/mg^2}$ (B) $2\pi\sqrt{m/k}$

(C) $\pi\sqrt{m/k}+2\sqrt{2E/mg^2}$ (D) $2\sqrt{2E/mg^2}$

Q.38 A 2 Kg block moving with 10 m/s strikes a spring of constant π^2 N/m attached to 2 Kg block at rest kept on a smooth floor. The time for which rear moving block remain in contact with spring will be



(A) $\sqrt{2}$ sec (B) $\frac{1}{\sqrt{2}}$ sec

(C) 1 sec (D) $\frac{1}{2}$ sec

Q.39 In the **above question**, the velocity of the rear 2 kg block after it separates from the spring will be :
(A) 0 m/s (B) 5 m/s (C) 10 m/s (D) 7.5 m/s

Q.40 A particle is subjected to two mutually perpendicular simple harmonic motions such that its x and y coordinates are given by

$$x = 2\sin\omega t \ ; \ y = 2\sin\left(\omega t+\frac{\pi}{4}\right)$$

The path of the particle will be :
(A) an ellipse (B) a straight line (C) a parabola (D) a circle

Q.41 The amplitude of the vibrating particle due to superposition of two SHMs,

$$y_1 = \sin\left(\omega t+\frac{\pi}{3}\right) \text{ and } y_2 = \sin\omega t \text{ is :}$$

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Q.42 Two simple harmonic motions $y_1 = A\sin\omega t$ and $y_2 = A\cos\omega t$ are superimposed on a particle of mass m. The total mechanical energy of the particle is:

(A) $\frac{1}{2}m\omega^2A^2$ (B) $m\omega^2A^2$ (C) $\frac{1}{4}m\omega^2A^2$ (D) zero

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 Two particles are in SHM on same straight line with amplitude A and 2A and with same angular frequency ω. It is observed that when first particle is at a distance $A/\sqrt{2}$ from origin and going toward mean position, other particle is at extreme position on other side of mean position. Find phase difference between the two particles

(A) 45° (B) 90° (C) 135° (D) 180°

Q.2 A particle is executing SHM of amplitude A, about the mean position x = 0. Which of the following cannot be a possible phase difference between the positions of the particle at x = + $A/2$ and x = – $A/\sqrt{2}$.

(A) 75° (B) 165° (C) 135° (D) 195°

Q.3 Speed v of a particle moving along a straight line, when it is at a distance x from a fixed point on the line is given by $v^2 = 108 - 9x^2$ (all quantities in S.I. unit). Then

(A) The motion is uniformly accelerated along the straight line

(B) The magnitude of the acceleration at a distance 3 cm from the fixed point is 0.27 m/s^2.

(C) The motion is simple harmonic about x = $\sqrt{12}$ m.

(D) The maximum displacement from the fixed point is 4 cm.

Q.4 A block is placed on a horizontal plank. The plank is performing SHM along a vertical line with amplitude of 40cm. The block just loses contact with the plank when the plank is momentarily at rest. Then:

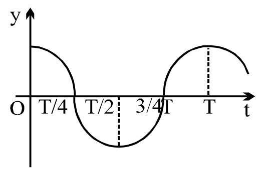
(A) the period of its oscillations is 2π/5 sec.

(B) the block weights on the plank double its weight, when the plank is at one of the positions of momentary rest.

(C) the block weights 1.5 times its weight on the plank halfway down from the mean position.

(D) the block weights its true weight on the plank, when velocity of the plank is maximum.

Q.5 The displacement-time graph of a particle executing SHM is shown. Which of the following statements is/are true?



(A) The velocity is maximum at t = T/2

(B) The acceleration is maximum at t = T

(C) The force is zero at t = 3T/4

(D) The potential energy equals the oscillation energy at t = T/2.

Q.6 The potential energy of a particle of mass 0.1kg, moving along x-axis, is given by U = 5x(x-4)J where x is in metres. It can be concluded that

(A) the particle is acted upon by a constant force.

(B) the speed of the particle is maximum at x = 2 m

(C) the particle executes simple harmonic motion

(D) the period of oscillation of the particle is π/5 s.

Q.7 A particle is executing SHM with amplitude A, time period T, maximum acceleration a_o and maximum velocity v_0. Its starts from mean position at t=0 and at time t , it has the displacement A/2, acceleration a and velocity v then

(A) t=T/12 (B) $a=a_o/2$ (C) $v=v_o/2$ (D) t=T/8

Q.8 The amplitude of a particle executing SHM about O is 10 cm. Then:
(A) When the K.E. is 0.64 of its max. K.E. its displacement is 6cm from O.
(B) When the displacement is 5 cm from O its K.E. is 0.75 of its max.P.E.
(C) Its total energy at any point is equal to its maximum K.E.
(D) Its velocity is half the maximum velocity when its displacement is half the maximum displacement.

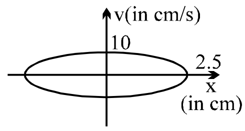
Q.9 The displacement of a particle varies according to the relation $x = 3\sin 100t + 8\cos^2 50t$. Which of the following is/are correct about this motion.
(A) the motion of the particle is not S.H.M.
(B) theamplitude of the S.H.M. of the particle is 5 units
(C) the amplitude of the resultant S.H. M. is $\sqrt{73}$ units
(D) the maximum displacement of the particle from the origin is 9 units.

Q.10 In SHM, acceleration versus displacement (from mean position) graph:
(A) is always a straight line passing through origin and slope –1
(B) is always a straight line passing through origin and slope +1
(C) is a straight line not necessarily passing through origin
(D) none of the above

Q.11 A particle moves in xy plane according to the law $x = a\sin\omega t$ and $y = a(1\text{-}\cos\omega t)$ where a and ω are constants. The particle traces
(A) a parabola
(B) a straight line equallyinclined to x and y axes
(C) a circle
(D) a distance proportional to time.

Q.12 For a particle executing S.H.M., x = displacement from equilibrium position, v = velocity at any instant and a = acceleration at any instant, then
(A) v-x graph is a circle
(B) v-x graph is an ellipse
(C) a-x graph is a straight line
(D) a-v graph is an ellipse

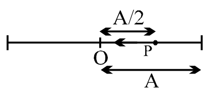
Q.13 The figure shows a graph between velocity and displacement (from mean position) of a particle performing SHM:



(A) the time period of the particle is 1.57s
(B) the maximum acceleration will be $40 cm/s^2$
(C) the velocity of particle is $2\sqrt{21}$ cm/s when it is at a distance 1 cm from the mean position.
(D) none of these

Q.14 Equations $y = 2A\cos^2 \omega t$ and $y = A(\sin\omega t + \sqrt{3}\cos\omega t)$ represent the motion of two particles.
(A) Only one of these is S.H.M.
(B) Ratio of maximum speeds is 2 : 1
(C) Ratio of maximum speeds is 1 : 1
(D) Ratio of maximum accelerations is 1 : 4

Q.15 A particle starts from a point P at a distance of A/2 from the mean position O & travels towards left as shown in the figure. If the time period of SHM, executed about O is T and amplitude A then the equation of motion of particle is :



(A) $x = A\sin\left(\frac{2\pi}{T}t + \frac{\pi}{6}\right)$
(B) $x = A\sin\left(\frac{2\pi}{T}t + \frac{5\pi}{6}\right)$
(C) $x = A\cos\left(\frac{2\pi}{T}t + \frac{\pi}{6}\right)$
(D) $x = A\cos\left(\frac{2\pi}{T}t + \frac{\pi}{3}\right)$

Q.16 A mass of 0.2kg is attached to the lower end of a massless spring of force-constant 200 N/m, the upper end of which is fixed to a rigid support. Which of the following statements is/are true?
(A) In equilibrium, the spring will be stretched by 1cm.
(B) If the mass is raised till the spring is unstretched state and then released, it will go down by 2cm before moving upwards.
(C) The frequency of oscillation will be nearly 5 Hz.
(D) If the system is taken to the moon, the frequency of oscillation will be the same as on the earth.

Q.17 The angular frequency of a spring block system is ω_0. This system is suspended from the ceiling of an elevator moving downwards with a constant speed v_0. The block is at rest relative to the elevator. Lift is suddenly stopped. Assuming the downwards as a positive direction, choose the wrong statement:

(A) The amplitude of the block is $\frac{v_0}{\omega_0}$

(B) The initial phase of the block is π.

(C) The equation of motion for the block is $\frac{v_0}{\omega_0} \sin \omega_0 t$.

(D) The maximum speed of the block is v_0.

Q.18 A cylindrical block of density ρ is partially immersed in a liquid of density 3ρ. The plane surface of the block remains parallel to the surface of the liquid. The height of the block is 60 cm. The block performs SHM when displaced from its mean position. [Use $g = 9.8$ m/s^2]
(A) the maximum amplitude is 20 cm.
(B) the maximum amplitude is 40 cm
(C) the time period will be $2\pi/7$ seconds.
(D) none

Q.19 A system is oscillating with undamped simple harmonic motion. Then the
(A) average total energy per cycle of the motion is its maximum kinetic energy.

(B) average total energy per cycle of the motion is $\frac{1}{\sqrt{2}}$ times its maximum kinetic energy.

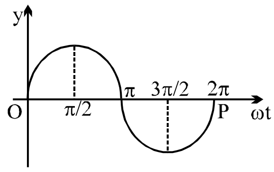
(C) root mean square velocity is $\frac{1}{\sqrt{2}}$ times its maximum velocity

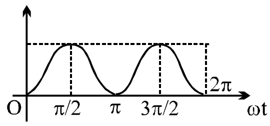
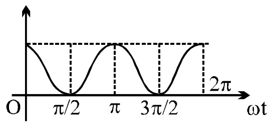
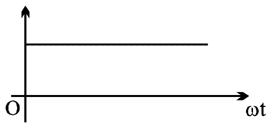
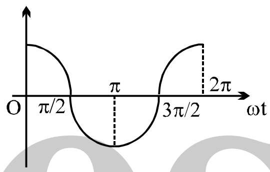
(D) mean velocity is 1/2 of maximum velocity.

Q.20 A particle of mass m performs SHM along a straight line with frequency f and amplitude A.
(A) The average kinetic energy of the particle is zero.
(B) The average potential energy is $m\,\pi^2 f^2 A^2$.
(C) The frequency of ocillation of kinetic energy is 2f.
(D) Velocity function leads acceleration by $\pi/2$.

Q.21 A linear harmonic oscillator of force constant 2×10^6 Nm^{-1} and amplitude 0.01 m has a total mechanical energy of 160 J. Its
(A) maximum potential energy is 100 J
(B) maximum kinetic energy is 100 J
(C) maximum potential energy is 160 J
(D) minimum potential energy is zero.

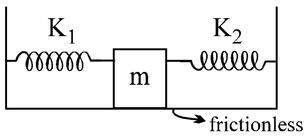
Q.22 The graph plotted between phase angle (ϕ) and displacement of a particle from equilibrium position (y) is a sinusoidal curve as shown below. Then the best matching is



Column A		Column B	
(a)	K.E. versus phase angle curve	(i)	
(b)	P.E. versus phase angle curve	(ii)	
(c)	T.E. versus phase angle curve	(iii)	
(d)	Velocity versus phase angle curve	(iv)	

(A) (a)-(i), (b)-(ii), (c)-(iii) & (d)-(iv)
(B) (a)-(ii), (b)-(i), (c)-(iii) & (d)-(iv)
(C) (a)-(ii), (b)-(i), (c)-(iv) & (d)-(iii)
(D) (a)-(ii), (b)-(iii), (c)-(iv) & (d)-(i)

Q.23 Two springs with negligible masses and force constant of $K_1 = 200\ Nm^{-1}$ and $K_2 = 160\ Nm^{-1}$ are attached to the block of mass m = 10 kg as shown in the figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time t = 0, a sharp impulse of 50 Ns is given to the block with a hammer.



(A) Period of oscillations for the mass m is $\frac{\pi}{3}$ s.
(B) Maximum velocity of the mass m during its oscillation is 5 ms^{-1}.
(C) Data are insufficient to determine maximum velocity.
(D) Amplitude of oscillation is 0.42 m.

Answer Key

ONLY ONE OPTION IS CORRECT.

Q.1	A	Q.2	A	Q.3	C	Q.4	D	Q.5	A
Q.6	D	Q.7	C	Q.8	D	Q.9	C	Q.10	D
Q.11	C	Q.12	C	Q.13	D	Q.14	A	Q.15	D
Q.16	D	Q.17	D	Q.18	C	Q.19	C	Q.20	C
Q.21	B	Q.22	C	Q.23	B	Q.24	B	Q.25	A
Q.26	C	Q.27	A	Q.28	B	Q.29	C	Q.30	B
Q.31	D	Q.32	A	Q.33	B	Q.34	B	Q.35	C
Q.36	A	Q.37	C	Q.38	C	Q.39	A	Q.40	A
Q.41	C	Q.42	B						

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	C	Q.2	C	Q.3	B	Q.4	A, B, C, D
Q.5	B, C, D	Q.6	B, C, D	Q.7	A, B	Q.8	A, B, C
Q.9	B, D	Q.10	D	Q.11	C, D	Q.12	B, C, D
Q.13	A, B, C	Q.14	C	Q.15	D	Q.16	A, B, C, D
Q.17	B	Q.18	A, C	Q.19	A, C	Q.20	B, C
Q.21	B, C	Q.22	B	Q.23	A, B		

Power Coaching

QUEST TUTORIALS

TARGET IIT JEE

PHYSICS

ELECTROSTATICS

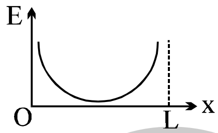
ONLY ONE OPTION IS CORRECT.

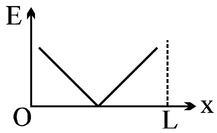
Take approx. 2 minutes for answering each question.

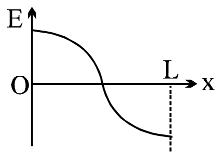
Q.1 Two identical conducting spheres, having charges of opposite sign, attract each other with a force of 0.108 N when separated by 0.5 m. The spheres are connected by a conducting wire, which is then removed, and thereafter, they repel each other with a force of 0.036 N. The initial charges on the spheres are

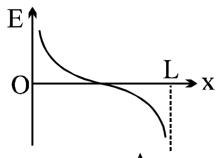
(A) $\pm 5 \times 10^{-6}$ C and $\mp$ 15×10^{-6} C
(B) $\pm 1.0 \times 10^{-6}$ C and $\mp$ 3.0×10^{-6} C
(C) $\pm 2.0 \times 10^{-6}$ C and $\mp$ 6.0×10^{-6} C
(D) $\pm 0.5 \times 10^{-6}$ C and $\mp$ 1.5×10^{-6} C

Q.2 Two identical point charges are placed at a separation of l. P is a point on the line joining the charges, at a distance x from any one charge. The field at P is E. E is plotted against x for values of x from close to zero to slightly less than l. Which of the following best represents the resulting curve?

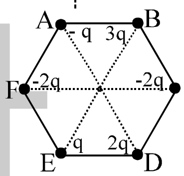
(A) 

(B) 

(C) 

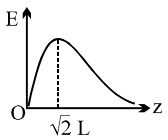
(D) 

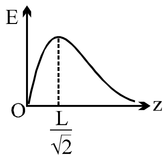
Q.3 Six charges are placed at the corner of a regular hexagon as shown. If an electron is placed at its centre O, force on it will be:

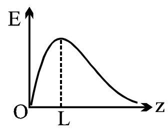


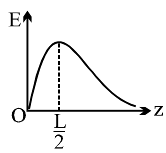
(A) Zero
(B) Along OF
(C) Along OC
(D) None of these

Q.4 Four equal positive charges are fixed at the vertices of a square of side L. Z-axis is perpendicular to the plane of the square. The point z = 0 is the point where the diagonals of the square intersect each other. The plot of electric field due to the four charges, as one moves on the z-axis.

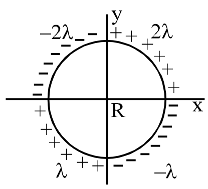
(A) 

(B) 

(C) 

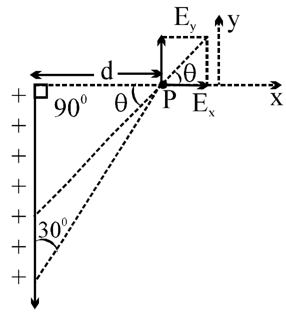
(D) 

Q.5 The charge per unit length of the four quadrant of the ring is 2λ, -2λ, λ and $-\lambda$ respectively. The electric field at the centre is



(A) $-\dfrac{\lambda}{2\pi\varepsilon_0 R}\hat{i}$
(B) $\dfrac{\lambda}{2\pi\varepsilon_0 R}\hat{j}$
(C) $\dfrac{\sqrt{2}\,\lambda}{4\pi\varepsilon_0 R}\hat{i}$
(D) None

Q.6 The direction (θ) of $\vec{E}$ at point P due to uniformly charged finite rod will be



(A) at angle 30^0 from x-axis
(B) 45^0 from x - axis
(C) 60^0 from x-axis
(D) none of these

Q.7 Two equal negative charges are fixed at the points [0, a] and [0, –a] on the y-axis. A positive charge Q is released from rest at the points [2a, 0] on the x-axis . The charge Q will

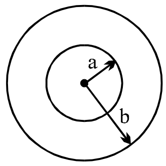
(A) execute simple harmonic motion about the origin
(B) move to the origin and remain at rest
(C) move to infinity
(D) execute oscillatory but not simple harmonic motion.

Q.8 Which of the following is a volt :
(A) Erg per cm
(B) Joule per coulomb
(C) Erg per ampere
(D) Newton / (coulomb x m^2)

Q.9 n small drops of same size are charged to V volts each. If they coalesce to form a signal large drop, then its potential will be
(A) V/n
(B) Vn
(C) $Vn^{1/3}$
(D) $Vn^{2/3}$

Q.10 Potential difference between centre & the surface of sphere of radius R and uniform volume charge density ρ within it will be :
(A) $\frac{\rho R^2}{6\epsilon_0}$
(B) $\frac{\rho R^2}{4\epsilon_0}$
(C) 0
(D) $\frac{\rho R^2}{2\epsilon_0}$

Q.11 If the electric potential of the inner metal sphere is 10 volt & that of the outer shell is 5 volt, then the potential at the centre will be :



(A) 10 volt
(B) 5 volt
(C) 15 volt
(D) 0

Q.12 Three concentric metallic spherical shell A, B and C or radii a, b and c (a < b < c) have surface charge densities $-\sigma$, $+\sigma$, and $-\sigma$ respectively. The potential of shell A is :
(A) $(\sigma/\varepsilon_0)[a+b-c]$
(B) $(\sigma/\varepsilon_0)[a-b+c]$
(C) $(\sigma/\varepsilon_0)[b-a-c]$
(D) none

Q.13 A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R. The force of gravity acts downwards. The equilibrium of the particle will be stable
(A) for all values of H
(B) only if $H > \frac{R}{\sqrt{2}}$
(C) only if $H < \frac{R}{\sqrt{2}}$
(D) only if $H = \frac{R}{\sqrt{2}}$

Q.14 A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre?
(A) R
(B) R/2
(C) R/3
(D) 2R

Q.15 An infinite nonconducting sheet of charge has a surface charge density of 10^{-7} C/m^2. The separation between two equipotential surfaces near the sheet whose potential differ by 5V is
(A) 0.88 cm
(B) 0.88 mm
(C) 0.88 m
(D) 5×10^{-7} m

Q.16 Two identical thin rings, each of radius R meter are coaxially placed at distance R meter apart. If Q_1 and Q_2 coulomb are respectively the charges uniformly spread on the two rings, the work done in moving a charge *q* from the centre of one ring to that of the other is
(A) zero
(B) $q(Q_1-Q_2)(\sqrt{2}-1)/(\sqrt{2}.4\pi\varepsilon_0 R)$
(C) $q\sqrt{2}(Q_1+Q_2)/4\pi\varepsilon_0 R$
(D) $q(Q_1-Q_2)(\sqrt{2}+1)/(\sqrt{2}.4\pi\varepsilon_0 R)$

Q.17 Two positively charged particles X and Y are initially far away from each other and at rest. X begins to move towards Y with some initial velocity. The total momentum and energy of the system are p and E.
(A) If Y is fixed, both p and E are conserved.
(B) If Y is fixed, E is conserved, but not p.
(C) If both are free to move, p is conserved but not E.
(D) If both are free, E is conserved, but not p.

Q.18 Two particles X and Y, of equal mass and with unequal positive charges, are free to move and are initially far away from each other. With Y at rest, X begins to move towards it with initial velocity u. After a long time, finally

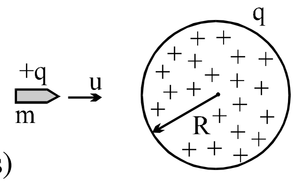
(A) X will stop, Y will move with velocity u.

(B) X and Y will both move with velocities u/2 each.

(C) X will stop, Y will move with velocity < u.

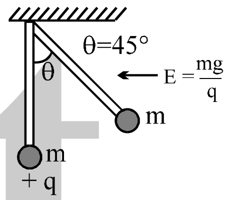
(D) both will move with velocities < u/2.

Q.19 A bullet of mass m and charge q is fired towards a solid uniformly charged sphere of radius R and total charge + q. If it strikes the surface of sphere with speed u, find the minimum speed u so that it can penetrate through the sphere. (Neglect all resistance forces or friction acting on bullet except electrostatic forces)



(A) $\frac{q}{\sqrt{2\pi\varepsilon_0 mR}}$ (B) $\frac{q}{\sqrt{4\pi\varepsilon_0 mR}}$ (C) $\frac{q}{\sqrt{8\pi\varepsilon_0 mR}}$ (D) $\frac{\sqrt{3}q}{\sqrt{4\pi\varepsilon_0 mR}}$

Q.20 In space of horizontal EF (E = (mg)/q) exist as shown in figure and a mass m attached at the end of a light rod. If mass m is released from the position shown in figure find the angular velocity of the rod when it passes through the bottom most position

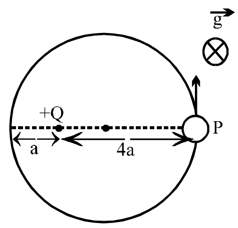


(A) $\sqrt{\frac{g}{l}}$ (B) $\sqrt{\frac{2g}{l}}$ (C) $\sqrt{\frac{3g}{l}}$ (D) $\sqrt{\frac{5g}{l}}$

Q.21 Two identical particles of mass m carry a charge Q each. Initially one is at rest on a smooth horizontal plane and the other is projected along the plane directly towards first particle from a large distance with speed ν. The closest distance of approach be

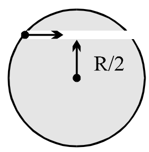
(A) $\frac{1}{4\pi\varepsilon_0}\frac{Q^2}{m\nu}$ (B) $\frac{1}{4\pi\varepsilon_0}\frac{4Q^2}{m\nu^2}$ (C) $\frac{1}{4\pi\varepsilon_0}\frac{2Q^2}{m\nu^2}$ (D) $\frac{1}{4\pi\varepsilon_0}\frac{3Q^2}{m\nu^2}$

Q.22 The diagram shows a small bead of mass m carrying charge q. The bead can freely move on the smooth fixed ring placed on a smooth horizontal plane. In the same plane a charge +Q has also been fixed as shown. The potential atthe point P due to +Q is V. The velocity with which the bead should projected from the point P so that it can complete a circle should be greater than



(A) $\sqrt{\frac{6qV}{m}}$ (B) $\sqrt{\frac{qV}{m}}$ (C) $\sqrt{\frac{3qV}{m}}$ (D) none

Q.23 A unit positive point charge of mass m is projected with a velocity V inside the tunnel as shown. The tunnel has been made inside a uniformly charged non conducting sphere. The minimum velocity with which the point charge should be projected such it can it reach the opposite end of the tunnel, is equal to



(A) $[\rho R^2/4m\varepsilon_0]^{1/2}$

(B) $[\rho R^2/24m\varepsilon_0]^{1/2}$

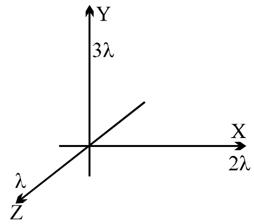
(C) $[\rho R^2/6m\varepsilon_0]^{1/2}$

(D) zero because the initial and the final points are at same potential.

Q.24 A conducting sphere of radius a has charge Q on it. It is enclosed by a neutral conducting concentric spherical shell having inner radius 2a and outer radius 3a. Find electrostatic energy of system.

(A) $\frac{5}{12}\frac{kQ^2}{a}$ (B) $\frac{11}{12}\frac{kQ^2}{a}$ (C) $\frac{kQ^2}{2a}$ (D) none

Q.25 The diagram shows three infinitely long uniform line charges placed on the X, Y and Z axis. The work done in moving a unit positive charge from (1, 1, 1) to (0, 1, 1) is equal to



(A) $(\lambda\, ln\, 2) / 2\pi\varepsilon_0$ (B) $(\lambda\, ln\, 2) / \pi\varepsilon_0$
(C) $(3\lambda\, ln\, 2) / 2\pi\varepsilon_0$ (D) None

Q.26 A charged particle of charge Q is held fixed and another charged particle of mass *m* and charge *q* (of the same sign) is released from a distance r. The impulse of the force exerted by the external agent on the fixed charge by the time distance between Q and q becomes 2r is

(A) $\sqrt{\frac{Qq}{4\pi\epsilon_0\, mr}}$ (B) $\sqrt{\frac{Qqm}{4\pi\epsilon_0\, r}}$ (C) $\sqrt{\frac{Qqm}{\pi\epsilon_0\, r}}$ (D) $\sqrt{\frac{Qqm}{2\pi\epsilon_0\, r}}$

Q.27 In a uniform electric field, the potential is 10V at the origin of coordinates, and 8V at each of the points (1, 0, 0), (0, 1, 0) and (0, 0, 1). The potential at the point (1, 1, 1) will be
(A) 0 (B) 4 V (C) 8 V (D) 10 V

Q.28 In a regular polygon of n sides, each corner is at a distance r from the centre. Identical charges are placed at (n – 1) corners. At the centre, the intensity is E and the potential is V. The ratio V/E has magnitude.
(A) r n (B) r (n – 1) (C) (n – 1)/r (D) r(n – 1)/n

Q.29 The equation of an equipotential line in an electric field is y = 2x, then the electric field strength vector at (1, 2) may be

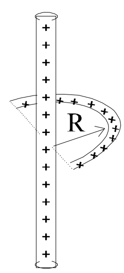
(A) $4\hat{i}+3\hat{j}$ (B) $4\hat{i}+8\hat{j}$ (C) $8\hat{i}+4\hat{j}$ (D) $-8\hat{i}+4\hat{j}$

Q.30 A charge 3 coulomb experiences a force 3000 N when placed in a uniform electric field. The potential difference between two points separated by a distance of 1 cm along the field lines is
(A) 10 V (B) 90 V (C) 1000 V (D) 9000V

Q.31 In a certain region of space, the potential is given by : $V = k[2x^2 - y^2 + z^2]$. The electric field at the point (1, 1, 1) has magnitude =

(A) $k\sqrt{6}$ (B) $2k\sqrt{6}$ (C) $2k\sqrt{3}$ (D) $4k\sqrt{3}$

Q.32 Find the force experienced by the semicircular rod charged with a charge q, placed as shown in figure. Radius of the wire is R and the line of charge with linear charge density λ is passing through its centre and perpendicular to the plane of wire.

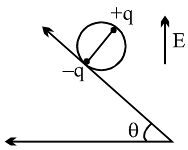


(A) $\frac{\lambda q}{2\pi^2\varepsilon_0 R}$ (B) $\frac{\lambda q}{\pi^2\varepsilon_0 R}$ (C) $\frac{\lambda q}{4\pi^2\varepsilon_0 R}$ (D) $\frac{\lambda q}{4\pi\varepsilon_0 R}$

Q.33 Uniform electric field of magnitude 100 V/m in space is directed along the line y = 3 + x. Find the potential difference between point A (3, 1) & B (1, 3)

(A) 100 V (B) $200\sqrt{2}$ V (C) 200 V (D) 0

Q.34 A wheel having mass m has charges +q and –q on diametrically opposite points. It remains in equilibrium on a rough inclined plane in the presence of uniform vertical electric field E =



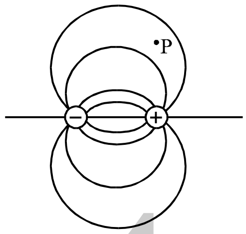
(A) $\frac{mg}{q}$ (B) $\frac{mg}{2q}$ (C) $\frac{mg\tan\theta}{2q}$ (D) none

Q.35 A and B are two points on the axis and the perpendicular bisector respectively of an electric dipole. A and B are far away from the dipole and at equal distance from it. The field at A and B are $\vec{E}_A$ and $\vec{E}_B$.

(A) $\vec{E}_A = \vec{E}_B$ (B) $\vec{E}_A = 2\vec{E}_B$

(C) $\vec{E}_A = -2\vec{E}_B$ (D) $|E_B| = \frac{1}{2}|E_A|$, and $\vec{E}_B$ is perpendicular to $\vec{E}_A$

Q.36 Figure shows the electric field lines around an electric dipole. Which of the arrows best represents the electric field at point P ?



(A) $\uparrow$ (B) $\nwarrow$

(C) $\nearrow$ (D) $\swarrow$

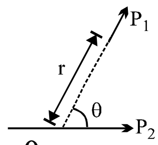
Q.37 The dipole moment of a system of charge +q distributed uniformly on an arc of radius R subtending an angle $\pi/2$ at its centre where another charge -q is placed is :

(A) $\frac{2\sqrt{2}qR}{\pi}$ (B) $\frac{\sqrt{2}qR}{\pi}$ (C) $\frac{qR}{\pi}$ (D) $\frac{2qR}{\pi}$

Q.38 An electric dipole is kept on the axis of a uniformly charged ring at distance $R/\sqrt{2}$ from the centre of the ring. The direction of the dipole moment is along the axis. The dipole moment is P, charge of the ring is Q and radius of the ring is R. The force on the dipole is nearly

(A) $\frac{4kPQ}{3\sqrt{3}R^2}$ (B) $\frac{4kPQ}{3\sqrt{3}R^3}$ (C) $\frac{2kPQ}{3\sqrt{3}R^3}$ (D) zero

Q.39 Two short electric dipoles are placed as shown. The energy of electric interaction between these dipoles will be

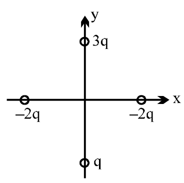


(A) $\frac{2kP_1P_2\cos\theta}{r^3}$ (B) $\frac{-2kP_1P_2\cos\theta}{r^3}$ (C) $\frac{-2kP_1P_2\sin\theta}{r^3}$ (D) $\frac{-4kP_1P_2\cos\theta}{r^3}$

Q.40 Point P lies on the axis of a dipole. If the dipole is rotated by 90° anticlock wise, the electric field vector $\vec{E}$ at P will rotate by

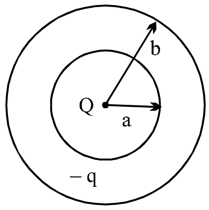
(A) 90° clock wise (B) 180° clock wise (C) 90° anti clock wise (D) none

Q.41 4 charges are placed each at a distance 'a' from origin. The dipole moment of configuration is



(A) $2qa\hat{j}$ (B) $3qa\hat{j}$ (C) $2aq[\hat{i}+\hat{j}]$ (D) none

Q.42 Both question (a) and (b) refer to the system of charges as shown in the figure. A spherical shell with an inner radius 'a' and an outer radius 'b' is made of conducting material. A point charge +Q is placed at the centre of the spherical shell and a total charge – q is placed on the shell.



(a) Charge – q is distributed on the surfaces as

(A) – Q on the inner surface, – q on outer surface

(B) – Q on the inner surface, – q + Q on the outer surface

(C) +Q on the inner surface, –q – Q on the outer surface

(D) The charge –q is spread uniformly between the inner and outer surface.

(b) Assume that the electrostatic potential is zero at an infinite distance from the spherical shell. The electrostatic potential at a distance R (a < R < b) from the centre of the shell is

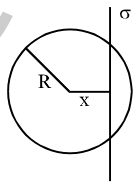
(A) 0 (B) $\frac{KQ}{a}$ (B) $K\frac{Q-q}{R}$ (D) $K\frac{Q-q}{b}$

(where $K = \frac{1}{4\pi\varepsilon_0}$)

Q.43 Electric flux through a surface of area 100 m^2 lying in the xy plane is (in V-m) if $\vec{E} = \hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k}$

(A) 100 (B) 141.4 (C) 173.2 (D) 200

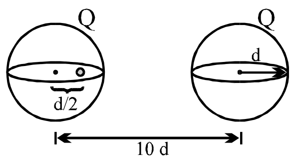
Q.44 An infinite, uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux Φ through the Gaussian surface is



(A) $\frac{\pi R^2 \sigma}{\varepsilon_0}$ (B) $\frac{2\pi(R^2 - x^2)\sigma}{\varepsilon_0}$

(C) $\frac{\pi(R-x)^2\sigma}{\varepsilon_0}$ (D) $\frac{\pi(R^2 - x^2)\sigma}{\varepsilon_0}$

Q.45 Two spherical, nonconducting, and very thin shells of uniformly distributed positive charge Q and radius d are located a distance 10d from each other. A positive point charge q is placed inside one of the shells at a distance d/2 from the center, on the line connecting the centers of the two shells, as shown in the figure. What is the net force on the charge q?



(A) $\frac{qQ}{361\pi\varepsilon_0 d^2}$ to the left (B) $\frac{qQ}{361\pi\varepsilon_0 d^2}$ to the right

(C) $\frac{362qQ}{361\pi\varepsilon_0 d^2}$ to the left (D) $\frac{360qQ}{361\pi\varepsilon_0 d^2}$ to the right

Q.46 A positive charge q is placed in a spherical cavity made in a positively charged sphere. The centres of sphere and cavity are displaced by a small distance $\vec{l}$. Force on charge q is :

(A) in the direction parallel to vector $\vec{l}$

(B) in radial direction

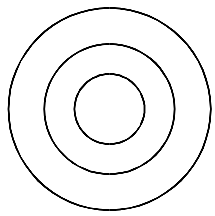
(C) in a direction which depends on the magnitude of charge density in sphere

(D) direction can not be determined.

Q.47 There are four concentric shells A, B, C and D of radii a, 2a, 3a and 4a respectively. Shells B and D are given charges +q and –q respectively. Shell C is now earthed. The potential difference $V_A - V_C$ is :

(A) $\frac{Kq}{2a}$ (B) $\frac{Kq}{3a}$ (C) $\frac{Kq}{4a}$ (D) $\frac{Kq}{6a}$

Q.48 A metal ball of radius R is placed concentrically inside a hollow metal sphere of inner radius 2R and outer radius 3R. The ball is given a charge +2Q and the hollow sphere a total charge – Q. The electrostatic potential energy of this system is :

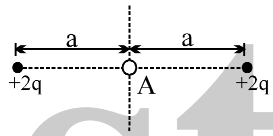


(A) $\frac{7Q^2}{24\pi\varepsilon_0 R}$ (B) $\frac{5Q^2}{16\pi\varepsilon_0 R}$ (C) $\frac{5Q^2}{8\pi\varepsilon_0 R}$ (D) None

<u>ONE OR MORE THAN ONE OPTION MAY BE CORRECT</u>

Take approx. 3 minutes for answering each question.

Q.1 A negative point charge placed at the point A is

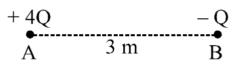


(A) in stable equilibrium along x-axis
(B) in unstable equilibrium along y-axis
(C) in stable equilibrium along y-axis
(D) in unstable equilibrium along x-axis

Q.2 Five balls numbered 1 to 5 are suspended using separate threads. Pairs (1,2), (2,4) and (4,1) show electrostatic attraction while pairs (2,3) and (4,5) show repulsion. Therefore ball 1 must be
(A) positively charged (B) negatively charged (C) neutral (D) made of metal

Q.3 Four charges of 1 μC, 2 μC, 3 μC, and – 6μC are placed one at each corner of the square of side 1m. The square lies in the x-y plane with its centre at the origin.
(A) The electric potential is zero at the origin.
(B) The electric potential is zero everywhere along the x-axis only of the sides of the square are parallel to x and y axis.
(C) The electric potential is zero everywhere along the z-axis for any orientation of the square in the x-y plane.
(D) The electric potential is not zero along the z-axis except at the origin.

Q.4 Two fixed charges 4Q (positive) and Q (negative) are located at A and B, the distance AB being 3 m.

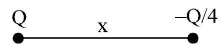


(A) The point P where the resultant field due to both is zero is on AB outside AB.
(B) The point P where the resultant field due to both is zero is on AB inside AB.
(C) If a positive charge is placed at P and displaced slightly along AB it will execute oscillations.
(D) If a negative charge is placed at P and displaced slightly along AB it will execute oscillations.

Q.5 Two identical charges +Q are kept fixed some distance apart. A small particle P with charge q is placed midway between them. If P is given a small displacement Δ, it will undergo simple harmonic motion if
(A) q is positive and Δ is along the line joining the charges.
(B) q is positive and Δ is perpendicular to the line joining the charges.
(C) q is negative and Δ is perpendicular to the line joining the charges.
(D) q is negative and Δ is along the line joining the charges.

Q.6 Select the correct statement : (Only force on a particle is due to electric field)
(A) A charged particle always moves along the electric line of force.
(B) A charged particle may move along the line of force
(C) A charge particle never moves along the line of force
(D) A charged particle moves along the line of force only if released from rest.

Q.7 Two point charges Q and – Q/4 are separated by a distance x. Then

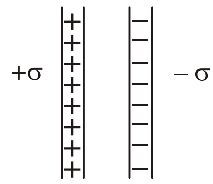


(A) potential is zero at a point on the axis which is x/3 on the right side of the charge – Q/4
(B) potential is zero at a point on the axis which is x/5 on the left side of the charge – Q/4
(C) electric field is zero at a point on the axis which is at a distance x on the right side of the charge – Q/4
(D) there exist two points on the axis where electric field is zero.

Q.8 An electric charge 10^{-8} C is placed at the point (4m, 7m, 2m). At the point (1m, 3m, 2m), the electric
(A) potential will be 18 V
(B) field has no Y-component
(C) field will be along Z-axis
(D) potential will be 1.8 V

Q.9 Three point charges Q, 4Q and 16Q are placed on a straight line 9 cm long. Charges are placed in such a way that the system has minimum potential energy. Then
(A) 4Q and 16Q must be at the ends and Q at a distance of 3 cm from the 16Q.
(B) 4Q and 16Q must be at the ends and Q at a distance of 6 cm from the 16Q.
(C) Electric field at the position of Q is zero.
(D) Electric field at the position of Q is $\dfrac{Q}{4\pi\varepsilon_0}$.

Q.10 Two infinite sheets of uniform charge density +σ and –σ are parallel to each other as shown in the figure. Electric field at the
(A) points to the left or to the right of the sheets is zero.
(B) midpoint between the sheets is zero.
(C) midpoint of the sheets is σ / ε_0 and is directed towards right.
(D) midpoint of the sheet is $2\sigma / \varepsilon_0$ and is directed towards right.



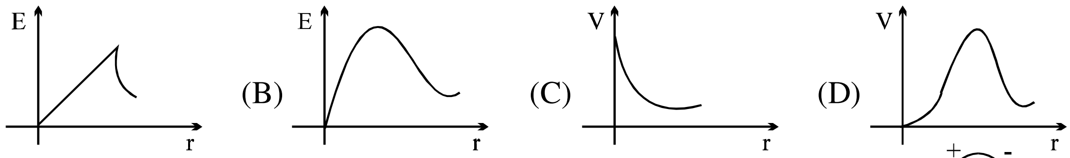
Q.11 Potential at a point A is 3 volt and at a point B is 7 volt , an electron is moving towards A from B.
(A) It must have some K.E. at B to reach A
(B) It need not have any K.E. at B to reach A
(C) to reach A it must have more than or equal to 4 eV K. E. at B.
(D) when it will reach A, it will have K.E. more then or at least equal to 4 eV if it was released from rest at B.

Q.12 A conducting sphere of radius r has a charge. Then
(A) The charge is uniformly distributed over its surface, if there is an external electric field.
(B) Distribution of charge over its surface will be non uniform if no external electric field exist in space.
(C) Electric field strength inside the sphere will be equal to zero only when no external electric field exists
(D) Potential at every point of the sphere must be same

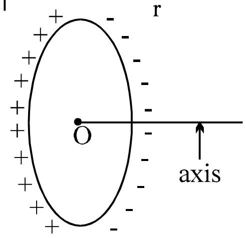
Q.13 For a spherical shell
(A) If potential inside it is zero then it necessarily electrically neutral
(B) electric field in a charged conducting spherical shell can be zero only when the charge is uniformly distributed.
(C) electric potential due to induced charges at a point inside it will always be zero
(D) none of these

Q.14 A circular ring carries a uniformly distributed positive charge. The electric field (E) and potential (V) varies with distance (r) from the centre of the ring along its axis as

(A) (B) (C) (D)



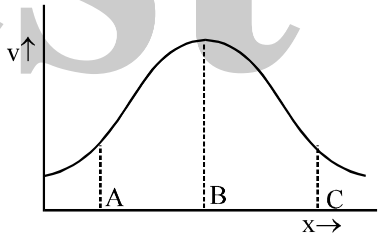
Q.15 The figure shows a nonconducting ring which has positive and negative charge non uniformly distributed on it such that the total charge is zero. Which of the following statements is true?



(A) The potential at all the points on the axis will be zero.
(B) The electric field at all the points on the axis will be zero.
(C) The direction of electric field at all points on the axis will be along the axis.
(D) If the ring is placed inside a uniform external electric field then net torque and force acting on the ring would be zero.

Q.16 At distance of 5cm and 10cm outwards from the surface of a uniformly charged solid sphere, the potentials are 100V and 75V respectively . Then
(A) potential at its surface is 150V.
(B) the charge on the sphere is $(5/3) \times 10^{-10}$C.
(C) the electric field on the surface is 1500 V/m.
(D) the electric potential at its centre is 225V.

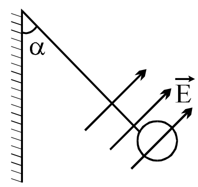
Q.17 Variation of electrostatic potential along x-direction is shown in the graph. The correct statement about electric field is



(A) x component at point B is maximum
(B) x component at point A is towards positive x-axis.
(C) x component at point C is along negative x-axis
(D) x component at point C is along positive x-axis

Q.18 A particle of charge 1 μC & mass 1 gm moving with a velocity of 4 m/s is subjected to a uniform electric field of magnitude 300 V/m for 10 sec. Then it's final speed cannot be:
(A) 0.5 m/s (B) 4 m/s (C) 3 m/s (D) 6 m/s

Q.19 A charged cork of mass m suspended by a light string is placed in uniform electric filed of strength $E = (\hat{i} + \hat{j}) \times 10^5\ NC^{-1}$ as shown in the fig. If in equilibrium position tension in the string is $\frac{2mg}{(1+\sqrt{3})}$ then angle 'α' with the vertical is



(A) 60° (B) 30° (C) 45° (D) 18°

Q.20 A proton and a deuteron are initially at rest and are accelerated through the same potential difference. Which of the following is false concerning the final properties of the two particles ?
(A) They have different speeds
(B) They have same momentum
(C) They have same kinetic energy
(D) They have been subjected to same force

Q.21 Charge Q is distributed non-uniformly over a ring of radius R, P is a point on the axis of ring at a distance $\sqrt{3}R$ from its centre. Which of the following is a wrong statement.

(A) Potential at P is $\dfrac{KQ}{2R}$

(B) Magnitude of electric field at P may be greater than $\dfrac{\sqrt{3}KQ}{8R^2}$

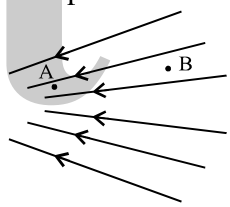
(C) Magnitude of electric field at P must be equal to $\dfrac{\sqrt{3}KQ}{8R^2}$

(D) Magnitude of electric field at P cannot be less than $\dfrac{\sqrt{3}KQ}{8R^2}$

Q.22 An electric dipole moment $\vec{p} = (2.0\hat{i} + 3.0\hat{j})\ \mu C.\ m$ is placed in a uniform electric field $\vec{E} = (3.0\hat{i} + 2.0\hat{k}) \times 10^5\ N\ C^{-1}$.

(A) The torque that $\vec{E}$ exerts on $\vec{p}$ is $(0.6\hat{i} - 0.4\hat{j} - 0.9\hat{k})$ Nm.
(B) The potential energy of the dipole is –0.6 J.
(C) The potential energy of the dipole is 0.6 J.
(D) If the dipole is rotated in the electric field, the maximum potential energy of the dipole is 1.3 J.

Q.23 Which of the following is true for the figure showing electric lines of force?
(E is electrical field, V is potential)



(A) $E_A > E_B$ (B) $E_B > E_A$
(C) $V_A > V_B$ (D) $V_B > V_A$

Q.24 If we use permittivity ε, resistance R, gravitational constant G and voltage V as fundamental physical quantities, then
(A) [angular displacement] = $\varepsilon^0R^0G^0V^0$ (B) [Velocity] = $\varepsilon^{-1}R^{-1}G^0V^0$
(C) [dipole moment] = $\varepsilon^1R^0G^0V^1$ (D) [force] = $\varepsilon^1R^0G^0V^2$

Q.25 Units of electric flux are

(A) $\dfrac{N-m^2}{Coul^2}$ (B) $\dfrac{N}{Coul^2-m^2}$ (C) volt-m (D) Volt-m^3

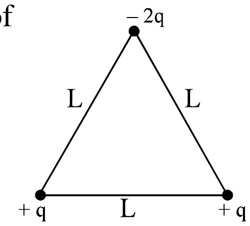
Q.26 A thin-walled, spherical conducting shell S of radius R is given charge Q. The same amount of charge is also placed at its centre C. Which of the following statements are correct?

(A) On the outer surface of S, the charge density is $\dfrac{Q}{2\pi R^2}$.

(B) The electric field is zero at all points inside S.
(C) At a point just outside S, the electric field is double the field at a point just inside S.
(D) At any point inside S, the electric field is inversely proportional to the square of its distance from C.

Q.27 A hollow closed conductor of irregular shape is given some charge. Which of the following statements are correct?
(A) The entire charge will appear on its outer surface.
(B) All points on the conductor will have the same potential.
(C) All points on its surface will have the same charge density.
(D) All points near its surface and outside it will have the same electric intensity.

Q.28 Three points charges are placed at the corners of an equilateral triangle of side L as shown in the figure.

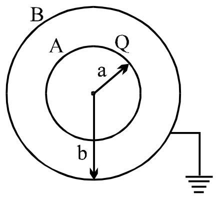


(A) The potential at the centroid of the triangle is zero.
(B) The electric field at the centroid of the triangle is zero.
(C) The dipole moment of the system is $\sqrt{2}\,qL$
(D) The dipole moment of the system is $\sqrt{3}\,qL$.

Q.29 An electric dipole is placed at the centre of a sphere. Mark the correct answer
(A) the flux of the electric field through the sphere is zero
(B) the electric field is zero at every point of the sphere.
(C) the electric potential is zero everywhere on the sphere.
(D) the electric potential is zero on a circle on the surface.

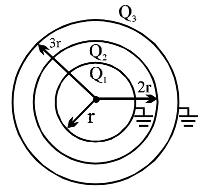
Q.30 An electric field converges at the origin whose magnitude is given by the expression E = 100*r* Nt/Coul, where *r* is the distance measured from the origin.
(A) total charge contained in any spherical volume with its centre at origin is negative.
(B) total charge contained at any spherical volume, irrespective of the location of its centre, is negative.
(C) total charge contained in a spherical volume of radius 3 cm with its centre at origin has magnitude 3×10^{-13}C.
(D) total charge contained in a spherical volume of radius 3 cm with its centre at origin has magnitude 3×10^{-9} Coul.

Q.31 A conducting sphere A of radius a, with charge Q, is placed concentrically inside a conducting shell B of radius b. B is earthed. C is the common centre of the A and B.



(A) The field is a distance r from C, where $a \le r \le b$ is $\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$.
(B) The potential at a distance r from C, where $a \le r \le b$, is $\frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$.
(C) The potential difference between A and B is $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{a}-\frac{1}{b}\right)$
(D) The potential at a distance r from C, where $a \le r \le b$, $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{r}-\frac{1}{b}\right)$.

Q.32 Three concentric conducting spherical shells have radius r, 2r and 3r and Q_1, Q_2 and Q_3 are final charges respectively. Innermost and outermost shells are already earthed as shown in figure. Choose the wrong statement.



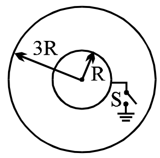
(A) $Q_1 + Q_3 = - Q_2$

(B) $Q_1 = \frac{-Q_2}{4}$

(C) $\frac{Q_3}{Q_1} = 3$

(D) $\frac{Q_3}{Q_2} = \frac{-1}{3}$

Q.33 Two thin conducting shells of radii R and 3R are shown in the figure. The outer shell carries a charge + Q and the inner shell is neutral. The inner shell is earthed with the help of a switch S.



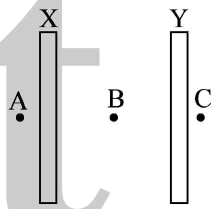
(A) With the switch S open, the potential of the inner sphere is equal to that of the outer.

(B) When the switch S is closed, the potential of the inner sphere becomes zero.

(C) With the switch S closed, the charge attained by the inner sphere is – q/3.

(D) By closing the switch the capacitance of the system increases.

Q.34 X and Y are large, parallel conducting plates closed to each other. Each face has an area A. X is given a charge Q. Y is without any charge. Points A, B and C are as shown in figure.



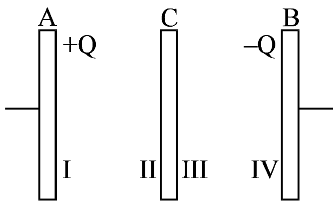
(A) The field at B is $\frac{Q}{2\varepsilon_0 A}$

(B) The field at B is $\frac{Q}{\varepsilon_0 A}$

(C) The fields at A, B and C are of the same magnitude.

(D) The field at A and C are of the same magnitude, but in opposite directions.

Q.35 Plates A and B constitute an isolated, charge parallel-plate capacitor. The inner surfaces (I and IV) of A and B have charges +Q and –Q respectively. A third plate C with charge +Q is now introduced midway between A and B. Which of the following statements is not correct?



(A) The surfaces I and II will have equal and opposite charges.

(B) The surfaces III and IV will have equal and opposite charges.

(C) the charge on surface III will be greater than Q.

(D) The potential difference between A and C will be equal to the potential difference between C and B.

Q.36 A particle of mass m and charge q is thrown in a region where uniform gravitational field and electric field are present. The path of particle

(A) may be a straight line

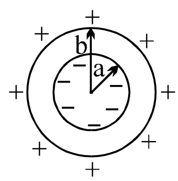
(B) may be a circle

(C) may be a parabola

(D) may be a hyperbola

Question No. 37 to 42 (6 questions)

An empty thick conducting shell of inner radius a and outer radius b is shown in figure. If it is observed that the inner face of the shell carries a uniform charge density $-\sigma$ and the surface carries a uniform charge density 'σ'



Q.37 If a point charge q_A is placed at the center of the shell, then choose the correct statement(s)

(A) The charge must be positive

(B) The charge must be negative

(C) The magnitude of charge must be $4\pi\sigma a^2$

(D) The magnitude of charge must be $4\pi\sigma(b^2 - a^2)$

Q.38 If another point charge q_B is also placed at a distance c (>b) the center of shell, then choose the correct statements

(A) force experienced by charge A is $\frac{\sigma q_A b^2}{\varepsilon_0 c^2}$

(B) force experienced by charge A is zero

(C) The force experienced by charge B is $\frac{\sigma q_B b^2}{\varepsilon_0 c^2}$

(D) The force experienced by charge B is $\frac{k q_A q_B}{c^2}$

Q.39 If the charge q_A is slowly moved inside the shell, then choose the statement(s)
(A) Charge distribution on the inner and outer face of the shell changes
(B) The force acting on the charge B charges
(C) The charge B also starts moving slowly
(D) None of these

Q.40 Choose the correct statement related to the potential of the shell in absence of q_B
(A) Potential of the outer surface is more than that of the inner surface because it is positively charged
(B) Potential of the outer surface is more than that of the inner surface because it carries more charge
(C) Both the surfaces have equal potential

(D) The potential of the outer surface is $\frac{\sigma b}{\varepsilon_0}$

Q.41 If the outer surface of the shell is earthed, then identify the correct statement(s)
(A) Only the potential of outer surface becomes zero
(B) Charge on the outer surface also becomes zero
(C) The outer surface attains negative charge
(D) Negative charge on the inner surface decreases

Q.42 If the inner surface of the shell is earthed, then identify the correct statement(s)
(A) The potential of both the inner and outer surface of the shell becomes zero
(B) Charge on the outer surface becomes zero
(C) Charge on the inner surface decreases
(D) Positive charge flows from the shell to the earth

ANSWER KEY

ONLY ONE OPTION IS CORRECT

Q.1	B	Q.2	D	Q.3	D	Q.4	D	Q.5	A	Q.6	A	Q.7	D
Q.8	B	Q.9	D	Q.10	A	Q.11	A	Q.12	C	Q.13	B	Q.14	C
Q.15	B	Q.16	B	Q.17	B	Q.18	A	Q.19	B	Q.20	B	Q.21	B
Q.22	A	Q.23	A	Q.24	A	Q.25	B	Q.26	B	Q.27	B	Q.28	B
Q.29	D	Q.30	A	Q.31	B	Q.32	B	Q.33	D	Q.34	B	Q.35	C
Q.36	B	Q.37	A	Q.38	D	Q.39	B	Q.40	A	Q.41	A		
Q.42	(a) B (b) D			Q.43	C	Q.44	D	Q.45	A	Q.46	A	Q.47	D
Q.48	A												

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	C,D	Q.2	C,D	Q.3	A,C	Q.4	A,D
Q.5	A,C	Q.6	B	Q.7	A,B,C	Q.8	A
Q.9	B,C	Q.10	A,C	Q.11	A,C	Q.12	D
Q.13	D	Q.14	B	Q.15	A	Q.16	A,C,D
Q.17	D	Q.18	A	Q.19	A,B	Q.20	B
Q.21	C	Q.22	A,B,D	Q.23	A,D	Q.24	A,B,D
Q.25	C	Q.26	A,C,D	Q.27	A,B	Q.28	A,D
Q.29	A,D	Q.30	A,B,C	Q.31	A,C,D	Q.32	D
Q.33	A,B,C,D	Q.34	A,C,D	Q.35	D	Q.36	A,C
Q.37	A,C	Q.38	B	Q.39	D	Q.40	C,D
Q.41	B	Q.42	A,B,D				

Power Coaching

QUEST TUTORIALS

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PHYSICS

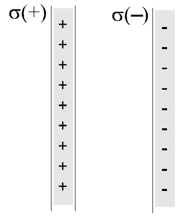
CAPACITANCE

QUESTION FOR SHORT ANSWER

Q.1 The electric strength of air is about 30, 000 V/cm. By this we mean that when the electric field intensity exceeds this value, a spark will jump through the air. We say that "electric breakdown" has occurred. Using this value, estimate the potential difference between two objects where a spark jumps. A typical situation might be the spark that jumps between your body and a metal door handle after you have walked on a deep carpet or slid across a plastic car seat in very dry weather.

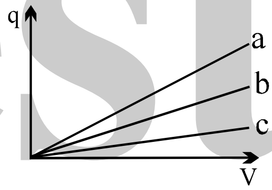
Q.2 If you grasp the two wires leading from the two plates of a charged capacitor, you may feel a shock. The effect is much greater for a 2–μF capacitor than for a 0.02μF capacitor, even though both are are charged to the same potential difference. Why?

Q.3 Three infinite nonconducting sheets, with uniform surface charge densities σ, 2σ and 3σ are arranged to be parallel like the two sheets in Fig. What is their order, from left to right, if the electric field $\vec{E}$ produced by the arrangement has magnitude $E = 0$ in one region and $E = 2\sigma/\varepsilon_0$ in another region?



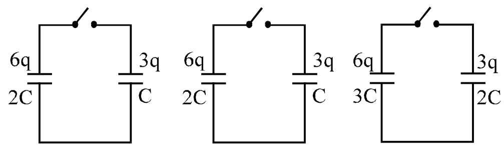
Q.4 As shown in the figure plots of charge versus potential difference for three parallel plate capacitors, which have the plate areas and separations given in the table. Which of the plots goes with which of the capacitors?

Capacitor	Area	Separation
1	A	d
2	2A	d
3	A	2d

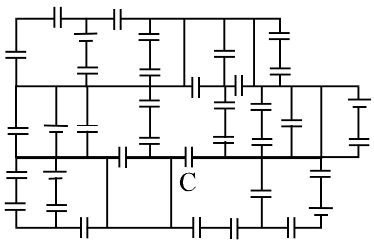


Q.5 Initially, a single capacitance C_1 is wired to a battery. Then capacitance C_2 is added in parallel. Are (a) the potential difference across C_1 and (b) the charge q_1 on C_1 now more than, less than, or the same as previously? (c) Is the equivalent capacitance C_{12} of C_1 and C_2 more than, less than, or equal to C_1? (d) Is the total charge stored on C_1 and C_2 together more than, less than, or equal to the charge stored previously on C_1?

Q.6 As shown in the figure three circuits, each consisting of a switch and two capacitors, initially charged as indicated. After the switches have been closed, in which circuit (if any) will the charge on the left–hand capacitor (a) increase, (b) decrease and (c) remain the same?



Q.7 Cap-monster maze. In the Figure all the capacitors have a capacitance of 6.0 μF, and all the batteries have an emf of 10V. What is the charge on capacitor C? (If you can find the proper loop through this maze, you can answer the question with a few seconds of mental calculation.)



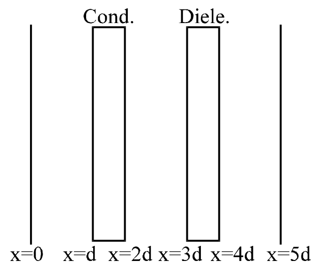
Q.8 An oil filled capacitor has been designed to have a capacitance C and to operate safely at or below a certain maximum potential difference V_m without arcing over. However, the designer did not do a good job and the capacitor occasionally arcs over. What can be done to redesign the capacitor, keeping C and V_m unchanged and using the same dielectric?

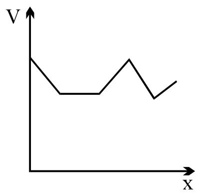
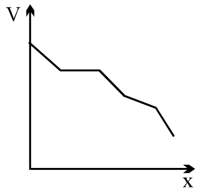
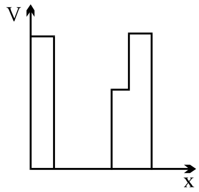
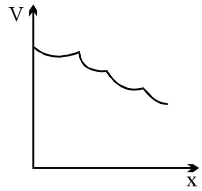
Q.9 One of the plates of a capacitor connected to battery is earthed. Will the potential diffrence between the plates change if the earthing wire is removed?

ONLY ONE OPTION IS CORRECT.

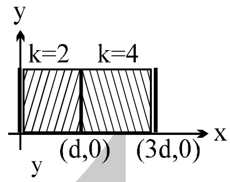
Take approx. 2 minutes for answering each question.

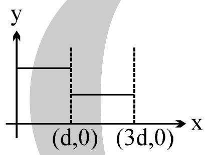
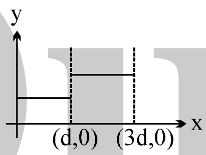
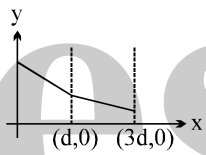
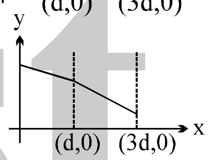
Q.1 The distance between plates of a parallel plate capacitor is 5d. Let the positively charged plate is at x=0 and negatively charged plate is at x=5d. Two slabs one of conductor and other of a dielectric of equal thickness d are inserted between the plates as shown in figure. Potential versus distance graph will look like :



(A) 
(B) 
(C) 
(D) 

Q.2 A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate.

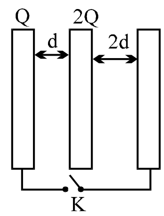


(A) 
(B) 
(C) 
(D) 

Q.3 The distance between the plates of a charged parallel plate capacitor is 5 cm and electric field inside the plates is 200 Vcm^{-1}. An uncharged metal bar of width 2 cm is fully immersed into the capacitor. The length of the metal bar is same as that of plate of capacitor. The voltage across capacitor after the immersion of the bar is

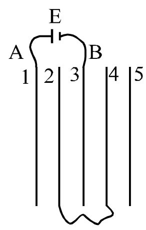
(A) zero (B) 400 V (C) 600 V (D) 100 V

Q.4 Three large plates are arranged as shown. How much charge will flow through the key k if it is closed?



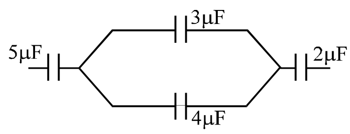
(A) $\frac{5Q}{6}$ (B) $\frac{4Q}{3}$ (C) $\frac{3Q}{2}$ (D) none

Q.5 Five conducting parallel plates having area A and separation between them d, are placed as shown in the figure. Plate number 2 and 4 are connected wire and between point A and B, a cell of emf E is connected. The charge flown through the cell is



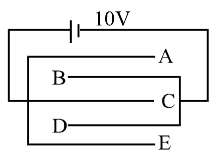
(A) $\frac{3}{4}\frac{\varepsilon_0 AE}{d}$ (B) $\frac{2}{3}\frac{\varepsilon_0 AE}{d}$ (C) $\frac{4\varepsilon_0 AE}{d}$ (D) $\frac{\varepsilon_0 AE}{2d}$

Q.6 If charge on left plane of the 5µF capacitor in the circuit segment shown in the figure is –20µC, the charge on the right plate of 3µF capacitor is



(A) +8.57 µC (B) –8.57 µC (C) +11.42 µC (D) –11.42 µC

Q.7 Five identical capacitor plates are arranged such that they make capacitors each of 2 µF. The plates are connected to a source of emf 10 V. The charge on plate C is

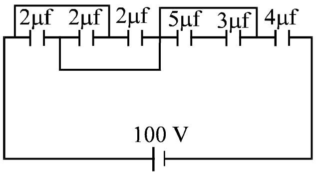


(A) + 20 µC (B) + 40 µC (C) + 60 µC (D) + 80 µC

Q.8 A capacitor of capacitance C is charged to a potential difference V from a cell and then disconnected from it. A charge +Q is now given to its positive plate. The potential difference across the capacitor is now

(A) V (B) $V + \frac{Q}{C}$ (C) $V + \frac{Q}{2C}$ (D) $V - \frac{Q}{C}$, if $V < CV$

Q.9 In the circuit shown in figure charge stored in the capacitor of capacity 5 μf is

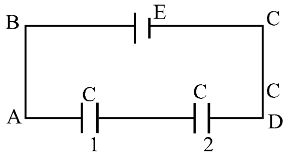


(A) 60 μC (B) 20 μC
(C) 30 μC (D) zero

Q.10 A conducting body 1 has some initial charge Q, and its capacitance is C. There are two other conducting bodies, 2 and 3, having capacitances : $C_2 = 2C$ and $C_3 \to \infty$. Bodies 2 and 3 are initially uncharged. "Body 2 is touched with body 1. Then, body 2 is removed from body 1 and touched with body 3, and then removed." This process is repeated N times. Then, the charge on body 1 at the end must be
(A) $Q/3^N$ (B) $Q/3^{N-1}$ (C) Q/N^3 (D) None

Q.11 Condenser A has a capacity of 15 μF when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity 1 μF with air between the plates. Both are charged separately by a battery of 100V . After charging, both are connected in parallel without the battery and the dielectric material being removed. The common potential now is
(A) 400V (B) 800V (C) 1200V (D) 1600V

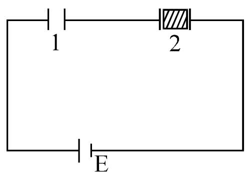
Q.12 In the adjoining figure, capacitor (1) and (2) have a capacitance 'C' each. When the dielectric of dielectric consatnt K is inserted between the plates of one of the capacitor, the total charge flowing through battery is



(A) $\frac{KCE}{K+1}$ from B to C (B) $\frac{KCE}{K+1}$ from C to B

(C) $\frac{(K-1)CE}{2(K+1)}$ from B to C (D) $\frac{(K-1)CE}{2(K+1)}$ from C to B

Q.13 Two identical capacitors 1 and 2 are connected in series to a battery as shown in figure. Capacitor 2 contains a dielectric slab of dielectric constant k as shown. Q_1 and Q_2 are the charges stored in the capacitors. Now the dielectric slab is removed and the corresponding charges are Q'_1 and Q'_2. Then



(A) $\frac{Q'_1}{Q_1} = \frac{k+1}{k}$ (B) $\frac{Q'_2}{Q_2} = \frac{k+1}{2}$ (C) $\frac{Q'_2}{Q_2} = \frac{k+1}{2k}$ (D) $\frac{Q'_1}{Q_1} = \frac{k}{2}$

Q.14 The area of the plates of a parallel plate capacitor is A and the gap between them is d. The gap is filled with a non-homogeneous dielectric whose dielectric constant varies with the distance 'y' from one plate as : $K = \lambda\sec(\pi y/2d)$, where λ is a dimensionless constant. The capacitance of this capacitor is
(A) $\pi\varepsilon_0\lambda A/2d$ (B) $\pi\varepsilon_0\lambda A/d$ (C) $2\pi\varepsilon_0\lambda A/d$ (D) none

Q.15 A capacitor stores 60μC charge when connected across a battery. When the gap between the plates is filled with a dielectric , a charge of 120μC flows through the battery. The dielectric constant of the material inserted is :
(A) 1 (B) 2 (C) 3 (D) none

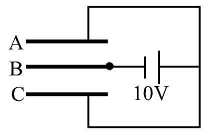
Q.16 In the **above question**, if the initial capacitance of the capacitor was 2μF, the amount of heat produced when the dielectric is inserted.

(A) 3600μJ (B) 2700μJ (C) 1800μJ (D) none

Q.17 A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of 2V with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be

(A) 1.75 (B) 2.25 (C) 2.5 (D) 1/2

Q.18 Three plates A, B and C each of area 0.1 m^2 are separated by 0.885 mm from each other as shown in the figure. A 10 V battery is used to charge the system. The energy stored in the system is



(A) 1 μJ (B) 10^{-1} μJ (C) 10^{-2} μJ (D) 10^{-3} μJ

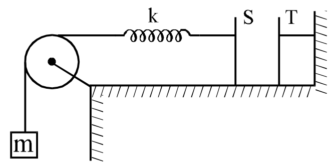
Q.19 A parallel plate capacitor of capacitance C is connected to a battery and is charged to a potential difference V. Another capacitor of capacitance 2C is similarly charged to a potential difference 2V. The charging battery is now disconnected and the capacitors are connect in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is

(A) zero (B) $\frac{3}{2}CV^2$ (C) $\frac{25}{6}CV^2$ (D) $\frac{9}{2}CV^2$

Q.20 A 2 μF capacitor is charged to a potential = 10V. Another 4 μF capacitor is charged to a potential = 20V. The two capacitors are then connected in a single loop, with the positive plate of one connected with negative plate of the other. What heat is evolved in the circuit?

(A) 300 μJ (B) 600 μJ (C) 900 μJ (D) 450 μJ

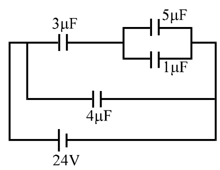
Q.21 The plates S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are insulating and massless. The magnitude of charge on one of the capacitor plates is: [Area of plates = A]



(A) $\sqrt{2mgA\epsilon_0}$ (B) $\sqrt{\frac{4mgA\epsilon_0}{k}}$

(C) $\sqrt{mgA\epsilon_0}$ (D) $\sqrt{\frac{2mgA\epsilon_0}{k}}$

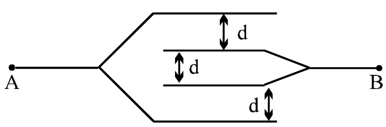
Q.22 In the circuit shown, the energy stored in 1μF capacitor is



(A) 40 μJ (B) 64 μJ

(C) 32 μJ (D) none

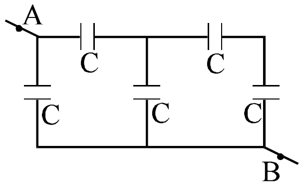
Q.23 Four metallic plates arearranged as shown in the figure. If the distance between each plate then capacitance of the given system between points A and B is (Given d << A)



(A) $\frac{\varepsilon_0 A}{d}$ (B) $\frac{2\varepsilon_0 A}{d}$

(C) $\frac{3\varepsilon_0 A}{d}$ (D) $\frac{4\varepsilon_0 A}{d}$

Q.24 What is the equivalent capacitance of the system of capacitors between A & B



(A) $\frac{7}{6}C$ (B) 1.6 C (C) C (D) None

Q.25 From a supply of identical capacitors rated 8 μF, 250 V, the minimum number of capacitors required to form a composite 16 μF, 1000 V is :
(A) 2 (B) 4 (C) 16 (D) 32

Q.26 The minimum number of capacitors each of 3 μF required to make a circuit with an equivalent capacitance 2.25 μF is
(A) 3 (B) 4 (C) 5 (D) 6

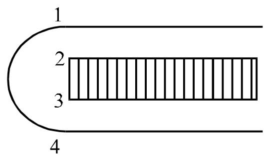
Q.27 The capacitance (C) for an isolated conducting sphere of radius (a) is given by $4\pi\varepsilon_0 a$. If the sphere is enclosed with an earthed concentric sphere. The ratio of the radii of the spheres being $\frac{n}{(n-1)}$ then the capacitance of such a sphere will be increased by a factor

(A) n (B) $\frac{n}{(n-1)}$ (C) $\frac{(n-1)}{n}$ (D) a . n

Q.28 Two capacitor having capacitances 8 μF and 16 μF have breaking voltages 20 V and 80 V. They are combined in series. The maximum charge they can store individually in the combination is
(A) 160 μC (B) 200 μC (C) 1280 μC (D) none of these

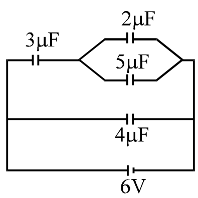
Q.29 A capacitor of capacitance 1 μF withstands the maximum voltage 6 kV while a capacitor of 2 μF withstands the maximum voltage 4 kV. What maximum voltage will the system of these two capacitor withstands if they are connected in series?
(A) 10 kV (B)12 kV (C) 8 kV (D) 9 kV

Q.30 Four identical plates 1, 2, 3 and 4 are placed parallel to each other at equal distance as shown in the figure. Plates 1 and 4 are joined together and the space between 2 and 3 is filled with a dielectric of dielectric constant k = 2. The capacitance of the system between 1 and 3 & 2 and 4 are C_1 and C_2 respectively. The ratio $\frac{C_1}{C_2}$ is :



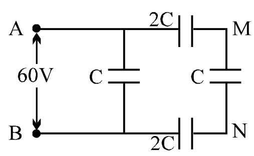
(A) $\frac{5}{3}$ (B) 1 (C) $\frac{3}{5}$ (D) $\frac{5}{7}$

Q.31 In the circuit shown in figure, the ratio of charges on 5μF and 4μF capacitor is :



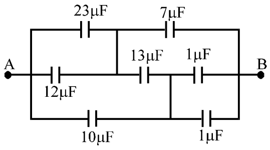
(A) 4/5 (B) 3/5
(C) 3/8 (D) 1/2

Q.32 In the circuit shown, a potential difference of 60V is applied across AB. The potential difference between the point M and N is



(A) 10 V (B) 15 V
(C) 20 V (D) 30 V

Q.33 Find the equivalent capacitance across A & B



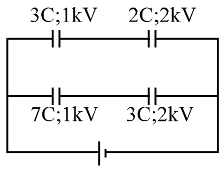
(A) $\frac{28}{3}$ μf (B) $\frac{15}{2}$ μF

(C) 15 μF (D) none

Q.34 A capacitor of capacitance 1 μF with stands the maximum voltages 6 KV while a capacitor of capacitance 2.0 μF with stands the maximum voltage = 4KV. if the two capacitors are connected in series, then the two capacitors combined can take up a maximum voltage of

(A) 2.4 KV (B) 5 KV (C) 9 KV (D) 10 KV

Q.35 The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?[Hint: First of all find out the break down voltages of each branch. After that compare them.]

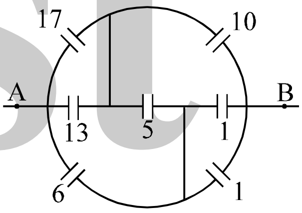


(A) 2.5 kV (B) 10 / 3kV (C) 3 kV (D) 1 kV

Q.36 Three capacitors 2 μF, 3 μF and 5 μF can withstand voltages to 3V, 2V and 1V respectively. Their series combination can withstand a maximum voltage equal to

(A) 5 Volts (B) (31/6) Volts (C) (26/5) Volts (D) None

Q.37 Find equivalent capacitance across AB (all capacitances in μF)



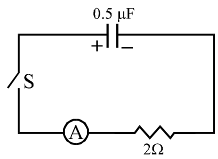
(A) $\frac{20}{3}$ μF (B) 9μF

(C) 48 μF (D) None

Q.38 Three long concentric conducting cylindrical shells have radii R, 2R and $2\sqrt{2}$ R. Inner and outer shells are connected to each other. The capacitance across middle and inner shells per unit length is:

(A) $\frac{\frac{1}{3}\in_0}{ln\,2}$ (B) $\frac{6\pi\in_0}{ln\,2}$ (C) $\frac{\pi\in_0}{2ln\,2}$ (D) None

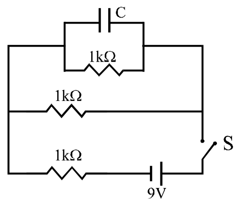
Q.39 A charged capacitor is allowed to discharge through a resistance 2Ω by closing the switch S at the instant t = 0. At time t = $ln\,2$ μs, the reading of the ammeter falls half of its initial value. The resistance of the ammeter equal to



(A) 0 (B) 2Ω

(C) ∞ (D) 2MΩ

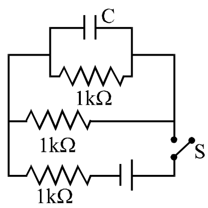
Q.40 A capacitor C = 100 μF is connected to three resistor each of resistance 1 kΩ and a battery of emf 9V. The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant



(A) 33 ms (B) 5 ms

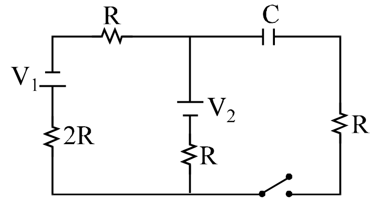
(C) 3.3 ms (D) 50 ms

Q.41 A capacitor C = 100 μF is connected to three resistors each of resistance 1 kW and a battery of emf 9V. The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant.



(A) 33 ms (B) 5 ms (C) 3.3 ms (D) 50 ms

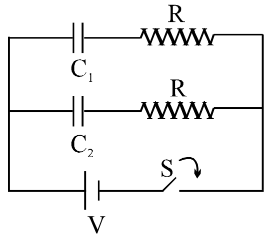
Q.42 In the transient shown the time constant of the circuit is :



(A) $\frac{5}{3}RC$ (B) $\frac{5}{2}RC$

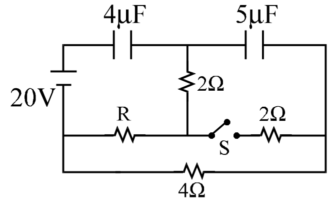
(C) $\frac{7}{4}RC$ (D) $\frac{7}{3}RC$

Q.43 In the circuit shown in figure C_1=2C_2. Switch S is closed at time t=0. Let i_1 and i_2 be the currents flowing through C_1 and C_2 at any time t, then the ratio i_1/i_2



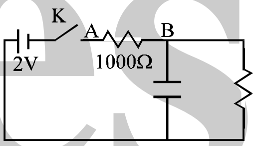
(A) is constant
(B) increases with increase in time t
(C) decreases with increase in time t
(D) first increases then decreases

Q.44 Find heat produced in the capacitors on closing the switch S



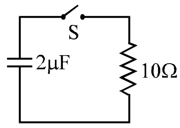
(A) 0.0002 J (B) 0.0005 J
(C) 0.00075 (D) zero

Q.45 In the circuit shown, when the key k is pressed at time t = 0, which of the following statements about current I in the resistor AB is true



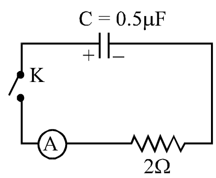
(A) I = 2mA at all t
(B) I oscillates between 1 mA and 2mA
(C) I = 1 mA at all t
(D) At t = 0, I = 2mA and with time it goes to 1 mA

Q.46 In the R – C circuit shown in the figure the total energy of 3.6×10^{-3} J is dissipated in the 10 Ω resistor when the switch S is closed. The initial charge on the capacitor is



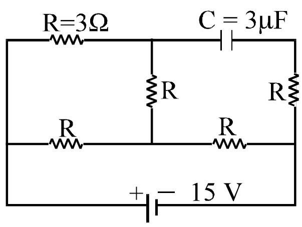
(A) 60 μC (B) 120 μC (C) $60\sqrt{2}$ μC (D) $\frac{60}{\sqrt{2}}$ μC

Q.47 A charged capacitor is allowed to discharge through a resistor by closing the key at the instant t =0. At the instant t = (ln 4) μs, the reading of the ammeter falls half the initial value. The resistance of the ammeter is equal to



(A) 1 MΩ (B) 1Ω (C) 2Ω (D) 2MΩ

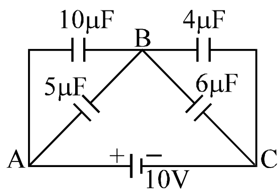
Q.48 In the circuit shown, the cell is ideal, with emf = 15 V. Each resistance is of 3Ω. The potential difference across the capacitor is



(A) zero (B) 9 V
(C) 12 V (D) 15 V

Question No. 49 to 52 (4 questions)

In the circuit shown in figure, four capacitors are connected to a battery.



Q.49 The equivalent capacitance of the circuit is
(A) 25 μF (B) 6 μF (C) 8.4 μF (D) none

Q.50 The charge on the 5 μF capacitor is
(A) 60 μC (B) 24 μC (C) 12 μC (D) 20 μC

Q.51 The potential difference across the 6 μF capacitor is
(A) 6V (B) 4V (C) 5V (D) none

Q.52 The maximum energy is stored in the capacitor of
(A) 10 μF (B) 6 μF (C) 5 μF (D) 4 μF

Q.53 A parallel plate capacitor has an electric field of 10^5V/m between the plates. If the charge on the capacitor plate is 1μC, then the force on each capacitor plate is
(A) 0.1Nt (B) 0.05Nt (C) 0.02Nt (D) 0.01Nt

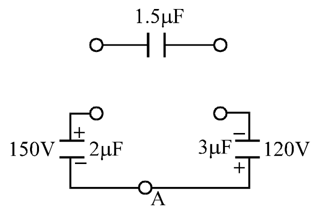
Q.54 A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved
(A) remains the same (B) becomes eight times
(C) becomes four times (D) becomes two times

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

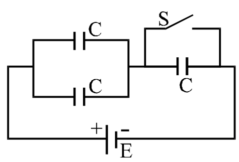
Q.1 A parallel plate capacitor A is filled with a dielectric whose dielectric constant varies with applied voltage as K = V. An identical capacitor B of capacitance C_0 with air as dielectric is connected to voltage source V_0 = 30V and then connected to the first capacitor after disconnecting the voltage source. The charge and voltage on capacitor.
(A) A are $25C_0$ and 25V (B) A are $25C_0$ and 5 V
(C) B are $5c_0$ and 5V (D) B are $5C_0$ and 25 V

Q.2 Two capacitors of 2 μF and 3 μF are charged to 150 volt and 120 volt respectively. The plates of capacitor are connected as shown in the figure. A discharged capacitor of capacity 1.5 μF falls to the free ends of the wire. Then



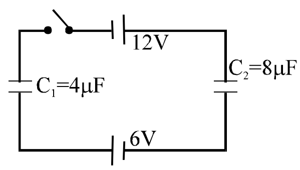
(A) charge on the 1.5 μF capacitors is 180 μC
(B) charge on the 2μF capacitor is 120 μC
(C) charge flows through A from right to left.
(D) charge flows through A from left to right.

Q.3 In the circuit shown, each capacitor has a capacitance C. The emf of the cell is E. If the switch S is closed



(A) positive charge will flow out of the positive terminal of the cell
(B) positive charge will enter the positive terminal of the cell
(C) the amount of charge flowing through the cell will be CE.
(D) the amount of charge flowing through the cell will be 4/3 CE.

Q.4 In the circuit shown initially C_1, C_2 are uncharged. After closing the switch

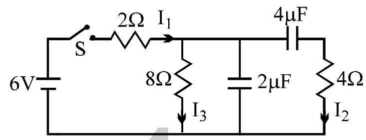


(A) The charge on C_2 is greater that on C_1
(B) The charge on C_1 and C_2 are the same
(C) The potential drops across C_1 and C_2 are the same
(D) The potential drops across C_2 is greater than that across C_1

Q.5 A parallel plate air-core capacitor is connected across a source of constant potential difference. When a dielectric plate is introduced between the two plates then :
(A) some charge from the capacitor will flow back into the source.
(B) some extra charge from the source will flow back into the capacitor.
(C) the electric field intensity between the two plate does not change.
(D) the electric field intensity between the two plates will decrease.

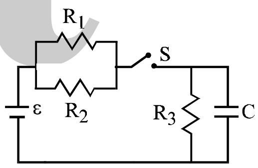
Q.6 A parallel plate capacitor has a parallel sheet of copper inserted between and parallel to the two plates, without touching the plates. The capacity of the capacitor after the introduction of the copper sheet is :
(A) minimum when the copper sheet touches one of the plates.
(B) maximum when the copper sheet touches one of the plates.
(C) invariant for all positions of the sheet between the plates.
(D) greater than that before introducing the sheet.

Q.7 In the circuit shown in the figure, the switch S is initially open and the capacitor is initially uncharged. I_1, I_2 and I_3 represent the current in the resistance 2Ω, 4Ω and 8Ω respectively.



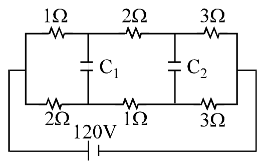
(A) Just after the switch S is closed, $I_1 = 3A$, $I_2 = 3A$ and $I_3 = 0$
(B) Just after the switch S is closed, $I_1 = 3A$, $I_2 = 0$ and $I_3 = 0$
(C) long time after the switch S is closed, $I_1 = 0.6$ A, $I_2 = 0$ and $I_3 = 0$
(D) long after the switch S is closed, $I_1 = I_2 = I_3 = 0.6$ A.

Q.8 The circuit shown in the figure consists of a battery of emf ε = 10 V ; a capacitor of capacitance C = 1.0 μF and three resistor of values $R_1 = 2\Omega$, $R_2 = 2\Omega$ and $R_3 = 1\Omega$. Initially the capacitor is completely uncharged and the switch S is open. The switch S is closed at t = 0.



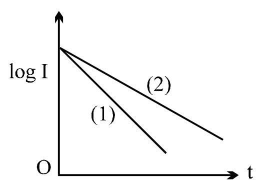
(A) The current through resistor R_3 at the moment the switch closed is zero.
(B) The current through resistor R_3 a long time after the switch closed is 5A.
(C) The ratio of current through R_1 and R_2 is always constant.
(D) The maximum charge on the capacitor during the operation is 5μC.

Q.9 In the circuit shown in figure $C_1 = C_2 = 2\mu F$. Then charge stored in



(A) capacitor C_1 is zero (B) capacitor C_2 is zero
(C) both capacitor is zero (D) capacitor C_1 is 40 μC

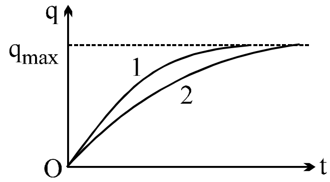
Q.10 A capacitor of capacity C is charged to a steady potential difference V and connected in series with an open key and a pure resistor 'R'. At time t = 0, the key is closed. If I = current at time t, a plot of log I against 't' is as shown in (1) in the graph. Later one of the parameters i.e. V, R or C is changed keeping the other two constant, and the graph (2) is recorded. Then



(A) C is reduced (B) C is increased
(C) R is reduced (D) R is increased

Question No.11 to 12 (2 questions)

The charge across the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure.



Q.11 Choose the correct statement(s) related to the two circuits.

(A) Both the capacitors are charged to the same charge.

(B) The emf's of cells in both the circuit are equal.

(C) The emf's of the cells may be different.

(D) The emf E_1 is more than E_2

Q.12 Identify the correct statement(s) related to the R_1, R_2, C_1 and C_2 of the two RC circuits.

(A) $R_1 > R_2$ if $E_1 = E_2$ (B) $C_1 < C_2$ if $E_1 = E_2$

(C) $R_1C_1 > R_2C_2$ (D) $\dfrac{R_1}{R_2} < \dfrac{C_2}{C_1}$

Q.13 A parallel plate capacitor is charged by connecting it to a battery. The battery is disconnected and the plates of the capacitor are pulled apart to make the separation between the plates twice. Again the capacitor is connected to the battery (with same polarity) then

(A) Charge from the battery flows into the capacitor after reconnection

(B) Charge from capacitor flows into the battery after reconnection.

(C) The potential difference between the plates increases when the plates are pulled apart.

(D) After reconnection of battery potential difference between the plate will immediately becomes half of the initial potential difference. (Just after disconnecting the battery)

Q.14 The plates of a parallel plate capacitor with no dielectric are connected to a voltage source. Now a dielectric of dielectric constant K is inserted to fill the whole space between the plates with voltage source remaining connected to the capacitor.

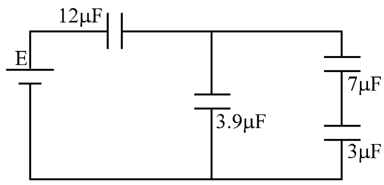
(A) the energy stored in the capacitor will become K–times

(B) the electric field inside the capacitor will decrease to K–times

(C) the force of attraction between the plates will increase to K^2–times

(D) the charge on the capacitor will increase to K–times

Q.15 Four capacitors and a battery are connected as shown. The potential drop across the 7 μF capacitor is 6 V. Then the :



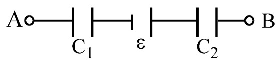
(A) potential difference across the 3 μF capacitor is 10 V

(B) charge on the 3 μF capacitor is 42 μC

(C) e.m.f. of the battery is 30 V

(D) potential difference across the 12 μF capacitor is 10 V.

Q.16 A circuit shown in the figure consists of a battery of emf 10 V and two capacitance C_1 and C_2 of capacitances 1.0 μF and 2.0 μF respectively. The potential difference $V_A - V_B$ is 5V



(A) charge on capacitor C_1 is equal to charge on capacitor C_2

(B) Voltage across capacitor C_1 is 5V.

(C) Voltage across capacitor C_2 is 10 V

(D) Energy stored in capacitor C_1 is two times the energy stored in capacitor C_2.

Q.17 A capacitor C is charged to a potential difference V and battery is disconnected. Now if the capacitor plates are brought close slowly by some distance :

(A) some +ve work is done by external agent (B) energy of capacitor will decrease

(C) energy of capacitor will increase (D) none of the above

Q.18 The capacitance of a parallel plate capacitor is C when the region between the plate has air. This region is now filled with a dielectric slab of dielectric constant k. The capacitor is connected to a cell of emf E, and the slab is taken out

(A) charge CE(k – 1) flows through the cell
(B) energy $E^2C(k-1)$ is absorbed by the cell.
(C) the energy stored in the capacitor is reduced by $E^2C(k-1)$
(D) the external agent has to do $\frac{1}{2}E^2C(k-1)$ amount of work to take the slab out.

Q.19 Two capacitors of capacitances 1μF and 3μF are charged to the same voltages 5V. They are connected in parallel with oppositely charged plates connected together. Then:

(A) Final common voltage will be 5V.
(B) Final common voltage will be 2.5 V
(C) Heat produced in the circuit will be zero.
(D) Heat produced in the circuit will be 37.5 μJ

Q.20 The two plates X and Y of a parallel plate capacitor of capacitance C are given a charge of amount Q each. X is now joined to the positive terminal and Y to the negative terminal of a cell of emf E = Q/C.

(A) Charge of amount Q will flow from the negative terminal to the positive terminal of the cell inside it.
(B) The total charge on the plate X will be 2Q.
(C) The total charge on the plate Y will be zero.
(D) The cell will supply CE^2 amount of energy.

Q.21 A dielectric slab is inserted between the plates of an isolated charged capacitor. Which of the following quantities will remain the same?

(A) the electric field in the capacitor
(B) the charge on the capacitor
(C) the potential difference between the plates
(D) the stored energy in the capacitor.

Q.22 The separation between the plates of a isolated charged parallel plate capacitor is increased. Which of the following quantities will change?

(A) charge on the capacitor
(B) potential difference across the capacitor
(C) energy of the capacitor
(D) energy density between the plates.

Q.23 Each plate of a parallel plate capacitor has a charge q on it. The capacitor is now connected to a battery. Now,

(A) the facing surfaces of the capacitor have equal and opposite charges.
(B) the two plates of the capacitor have equal and opposite charges.
(C) the battery supplies equal and opposite charges to the two plates.
(D) the outer surfaces of the plates have equal charges.

Q.24 Following operations can be performed on a capacitor :

X – connect the capacitor to a battery of emf E.
Y – disconnect the battery
Z – reconnect the battery with polarity reversed.
W – insert a dielectric slab in the capacitor

(A) In XYZ (perform X, then Y, then Z) the stored electric energy remains unchanged and no thermal energy is developed.
(B) The charge appearing on the capacitor is greater after the action XWY than after the action XYW.
(C) The electric energy stored in the capacitor is greater after the action WXY than after the action XYW.
(D) The electric field in the capacitor after the action XW is the same as that after WX.

Q.25 A parallel plate capacitor is charged and then disconnected from the source of potential difference. If the plates of the condenser are then moved farther apart by the use of insulated handle, which one of the following is true?

(A) the charge on the capacitor increases
(B) the charge on the capacitor decreases
(C) the capacitance of the capacitor increases
(D) the potential difference across the plate increases

Q.26 A parallel plate capacitor is charged and then disconnected from the source steady E.M.F. The plates are then drawn apart farther. Again it is connected to the same source. Then :
(A) the potential difference across the plate increases, while the plates are being drawn apart.
(B) the charge from the capacitor flows into the source, when the capacitor is reconnected.
(C) more charge is drawn to the capacitor from the source, during the reconnection.
(D) the electric intensity between the plates remains constant during the drawing apart of plates.

Q.27 When a parallel plates capacitor is connected to a source of constant potential difference,
(A) all the charge drawn from the source is stored in the capacitor.
(B) all the energy drawn from the source is stored in the capacitor.
(C) the potential difference across the capacitor grows very rapidly initially and this rate decreases to zero eventually.
(D) the capacity of the capacitor increases with the increase of the charge in the capacitor.

Q.28 When two identical capacitors are charged individually to different potentials and connected parallel to each other, after disconnecting them from the source :
(A) net charge on connected plates is less than the sum of initial individual charges.
(B) net charge on connected plates equals the sum of initial charges.
(C) the net potential difference across them is different from the sum of the individual initial potential differences.
(D) the net energy stored in the two capacitors is less than the sum of the initial individual energies.

Q.29 A parallel plate capacitor of plate area A and plate seperation d is charged to potential difference V and then the battery is disconnected. A slab of dielectric constant K is then inserted between the plates of the capacitor so as to fill the space between the plates. If Q, E and W denote respectively, the magnitude of charge on each plate, the electric field between the plates (after the slab is inserted) and the work done on the system, in question, in the process of inserting the slab, then

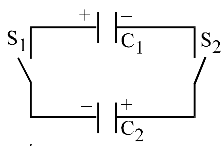
(A) $Q = \dfrac{\varepsilon_0 AV}{d}$ (B) $Q = \dfrac{\varepsilon_0 KAV}{d}$ (C) $E = \dfrac{V}{Kd}$ (D) $W = -\dfrac{\varepsilon_0 AV^2}{2d}\left(1-\dfrac{1}{K}\right)$

Q.30 A parallel plate capacitor is connected to a battery. The quantities charge, voltage, electric field and energy associated with the capacitor are given by Q_0, V_0, E_0 and U_0 respectively. A dielectric slab is introduced between plates of capacitor but battery is still in connection. The corresponding quantities now given by Q, V, E and U related to previous ones are
(A) $Q > Q_0$ (B) $V > V_0$ (C) $E > E_0$ (D) $U < U_0$

Q.31 A parallel-plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges +Q and –Q respectively. A third plate C, identical to A and B, with charge +Q, is now introduced midway between A and B, parallel to them. Which of the following are correct?

(A) The charge on the inner face of B is now $-\dfrac{3Q}{2}$
(B) There is no change in the potential difference between A and B.
(C) The potential difference between A and C is one-third of the potential difference between B and C.
(D) The charge on the inner face of A is now $Q/2$.

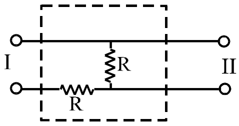
Q.32 Two capacitors C_1 = 4 μF and C_2 = 2μF are charged to same potential V = 500 Volt, but with opposite polarity as shown in the figure. The switches S_1 and S_2 are closed.

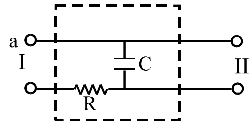


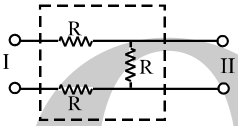
(A) The potential difference across the two capacitors are same and is given by $500/3$ V
(B) The potential difference across the two capacitors are same and is given by $1000/3$ V
(C) The ratio of final energy to initial energy of the system is 1/9.
(D) The ratio of final energy to initial energy of the system is 4/9.

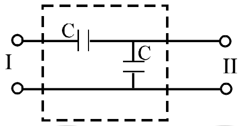
Q.33 A parallel plate capacitor is charged to a certain potential and the charging battery is then disconnected. Now, if the plates of the capacitor are moved apart then:
(A) The stored energy of the capacitor increases
(B) Charge on the capacitor increases
(C) Voltage of the capacitor decreases
(D) The capacitance increases

Q.34 If a battery of voltage V is connected across terminals I of the block box shown in figure, an ideal voltmeter connected to terminals II gives a reading of V/2, while if the battery is connected to terminals II, a voltmeter across terminals
I reads V. The black box may contain

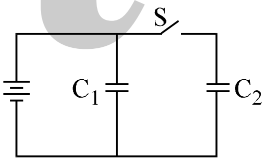
(A) 

(B) 

(C) 

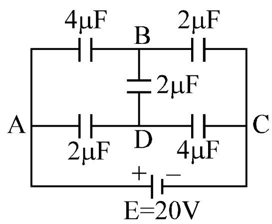
(D) 

Q.35 Two capacitors of equal capacitance ($C_1 = C_2$) are shown in the figure. Initially, while the switch S is open, one of the capacitors is uncharged and the other carries charge Q_0. The energy stored in the charged capacitor is U_0. Sometimes after the switch is closed, the capacitors C_1 and C_2 carry charges Q_1 and Q_2, respectively; the voltages across the capacitors are V_1 and V_2; and the energies stored in the capacitors are U_1 and U_2. Which of the following statements is INCORRECT ?



(A) $Q_0 = \frac{1}{2}(Q_1 + Q_2)$ (B) $Q_1 = Q_2$
(C) $V_1 = V_2$ (D) $U_1 = U_2$
(E) $U_0 = U_1 + U_2$

Question No.36 to 39 (4 questions)
The figure shows a diagonal symmetric arrangement of capacitors and a battery

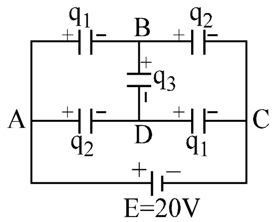


Q.36 Identify the correct statements.
(A) Both the 4μF capacitors carry equal charges in opposite sense.
(B) Both the 4μF capacitors carry equal charges in same sense.
(C) $V_B - V_D > 0$
(D) $V_D - V_B > 0$

Q.37 If the potential of C is zero, then
(A) $V_A = +20V$ (B) $4(V_A - V_B) + 2(V_D - V_B) = 2V_B$
(C) $2(V_A - V_D) + 2(V_B - V_D) = 4V_D$ (D) $V_A = V_B + V_D$

Q.38 The potential of the point B and D are
(A) $V_B = 8V$ (B) $V_B = 12V$ (C) $V_D = 8V$ (D) $V_D = 12V$

Q.39 The value of charge q_1, q_2 and q_3 as shown in the figure are



(A) $q_1 = 32\ \mu C$; $q_2 = 24\ \mu C$; $q_3 = -8\ \mu C$
(B) $q_1 = 48\ \mu C$; $q_2 = 16\ \mu C$; $q_3 = +8\ \mu C$
(C) $q_1 = 32\ \mu C$; $q_2 = 24\ \mu C$; $q_3 = +8\ \mu C$
(D) $q_1 = 3\ \mu C$; $q_2 = 4\ \mu C$; $q_3 = +2\ \mu C$

Q.40 If Q is the charge on the plates of a capacitor of capacitance C, V the potential difference between the plates, A the area of each plate and d the distance between the plates, the force of attraction between the plates is

(A) $\frac{1}{2}\left(\frac{Q^2}{\varepsilon_0 A}\right)$ (B) $\frac{1}{2}\left(\frac{CV^2}{d}\right)$ (C) $\frac{1}{2}\left(\frac{CV^2}{A\varepsilon_0}\right)$ (D) $\frac{1}{4}\left(\frac{Q^2}{\pi\varepsilon_0 d^2}\right)$

ANSWER KEY

ONLY ONE OPTION IS CORRECT

Q.1	B	Q.2	A	Q.3	C	Q.4	A	Q.5	B	Q.6	A
Q.7	B	Q.8	C	Q.9	D	Q.10	A	Q.11	B	Q.12	D
Q.13	C	Q.14	A	Q.15	C	Q.16	C	Q.17	B	Q.18	B
Q.19	B	Q.20	B	Q.21	A	Q.22	C	Q.23	B	Q.24	B
Q.25	D	Q.26	B	Q.27	A	Q.28	A	Q.29	D	Q.30	B
Q.31	C	Q.32	D	Q.33	B	Q.34	C	Q.35	A	Q.36	B
Q.37	B	Q.38	B	Q.39	A	Q.40	D	Q.41	D	Q.42	C
Q.43	B	Q.44	D	Q.45	D	Q.46	B	Q.47	C	Q.48	C
Q.49	B	Q.50	D	Q.51	B	Q.52	B	Q.53	B	Q.54	C

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	B,C	Q.2	A,B,C	Q.3	A,D	Q.4	B
Q.5	B,C	Q.6	C,D	Q.7	B	Q.8	A,B,C,D
Q.9	B,D	Q.10	B	Q.11	A,C	Q.12	D
Q.13	B,C	Q.14	A,C,D	Q.15	B,C,D	Q.16	A,D
Q.17	B	Q.18	A,B,D	Q.19	B,D	Q.20	A,B,C,D
Q.21	B	Q.22	B,C	Q.23	A,C,D	Q.24	B,C,D
Q.25	D	Q.26	A,B,D	Q.27	A,C	Q.28	B,C,D
Q.29	A,C,D	Q.30	A	Q.31	A,B,C,D	Q.32	A,C
Q.33	A	Q.34	D	Q.35	E	Q.36	B,C
Q.37	A,B,C,D	Q.38	B,C	Q.39	C	Q.40	A,B

TARGET IIT JEE

PHYSICS

CURRENT ELECTRICITY

QUESTION FOR SHORT ANSWER

Q.1 Fluorescent light bulbs are usually more efficient light emitters than incandescent bulbs. That is, for the same input energy, the fluorescent bulb gives off more light than the incandescent bulb. Carefully touch a fluorescent bulb and in an incandescent one after each has been lit for a few minutes. Explain why the incandesent bulb is a less efficient light emitter.

Q.2 Birds perch on high tension wires all the time. Why are they not electrocuted, even when they perch on a part of the wire where the insulation has worn off?

Q.3 Explain why touching an exposed circuit wire when you are in a damp basement is much more dangerous than touching the same wire when you are on the second floor.

Q.4 Initiallly, a single resistor R_1 is wired to a battery. Then resistor R_2 is added in parallel. Are (a) the potential difference across R_1 and (b) the current i_1 through R_1 now more than, less than, or the same as previously? (c) Is the equivalent resistance R_{12} of R_1 and R_2 more than, less than, or equal to R_1? (d) Is the total current through R_1 and R_2 together more than, less than, or equal to the current through R previously?

Q.5 A current enters the top of a copper sphere of radius R and leaves through the diametrically opposite point. are all parts equally effective in dissipating joule heat?

Q.6 How can an electric heater designed for 220 V be adopted for 110 V without changing the length of the coil and also without a change in the consumed power?

Q.7 The brilliance of lamps in a room noticeably drops as soon as a high power electric iron is switched on and after a short interval, the bulbs regain their original brilliance.Explain.

Q.8 Consider a circuit containing an ideal battery connected to a resistor. Do "work done by the battery" and " the thermal energy developed" represent two names of the same physical quantity?

Q.9 A current is passed through a steel wire which gets heated to a dull red. then half the wire is immersed in cold water. The portion out of the water becomes brighter. Why?

Q.10 A non ideal battery is connected to a resistor. Is work done by the battery equal to the thermal energy developed in the resistor? Does your answer change if the battery is ideal?

Q.11 For manual control of the current of a circuit, two rheostats in parallel are preferable to a single rheostat. Why?

Q.12 The drift velocity of electrons is quite small. How then does a bulb light up as soon as the switch is turned on, although the bulb may be quite far from the switch?

Q.13 Some times it is said that "heat is developed" in a resistance when there is an electric current in it. Recall that heat is defined as the energy being transferred due to the temperature difference. Is the statement under quotes technically correct?

Q.14 Does emf have electrostatic origin?

Q.15 The resistance of the human body is about 10 kΩ. If the resistance of our body is so large, why does one experience a strong shock from a live wire of 220 V supply ?

Q.16 Would you prefer a voltmeter or a potentiometer to measure the emf of a battery?

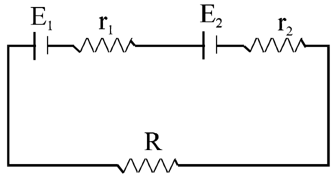
Q.17 Can the potential difference across a battery be greater than its emf?

ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

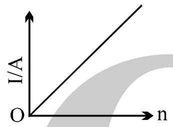
Q.1 A storage battery is connected to a charger for charging with a voltage of 12.5Volts. The internal resistance of the storage battery is 1Ω. When the charging current is 0.5 A, the emf of the storage battery is:
(A) 13 Volts (B) 12.5 Volts (C) 12 Volts (D) 11.5 Volts

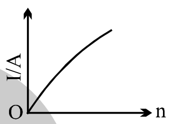
Q.2 Under what condition current passing through the resistance R can be increased by short circuiting the battery of emf E_2. The internal resistances of the two batteries are r_1 and r_2 respectively.

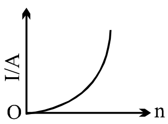


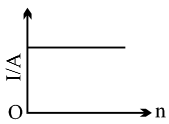
(A) $E_2r_1 > E_1 (R + r_2)$ (B) $E_1r_2 > E_2 (R + r_1)$
(C) $E_2r_2 > E_1 (R + r_2)$ (D) $E_1r_1 > E_2 (R + r_1)$

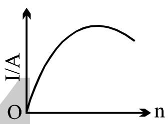
Q.3 A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured.
Which one of the graph below shows the relationship between I and n?

(A) 

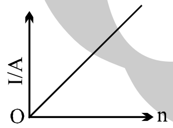
(B) 

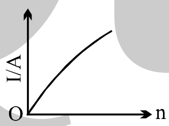
(C) 

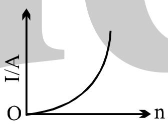
(D) 

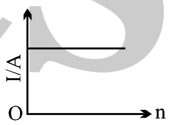
(E) 

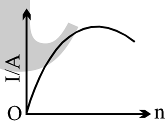
Q.4 In previous problem, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n?

(A) 

(B) 

(C) 

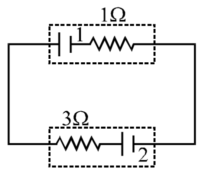
(D) 

(E) 

Q.5 n identical cells are joined in series with its two cells A and B in the loop with reversed polarities. EMF of each shell is E and internal resistance r. Potential difference across cell A or B is (here n>4)

(A) $\frac{2E}{n}$ (B) $2E\left(1-\frac{1}{n}\right)$ (C) $\frac{4E}{n}$ (D) $2E\left(1-\frac{2}{n}\right)$

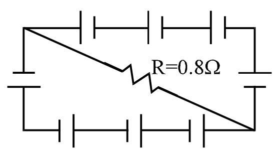
Q.6 In the figure shown, battery 1 has emf = 6 V and internal resistance = 1 Ω. Battery 2 has emf = 2V and internal resistance = 3 Ω. The wires have negligible resistance. What is the potential difference across the terminals of battery 2 ?



(A) 4 V (B) 1.5 V
(C) 5 V (D) 0.5 V

Q.7 The terminal voltage across a battery of emf E can be
(A) 0 (B) > E (C) < E (D) all of above

Q.8 A circuit is comprised of eight identical batteries and a resistor R = 0.8Ω. Each battery has an emf of 1.0 V and internal resistance of 0.2Ω. The voltage difference across any of the battery is

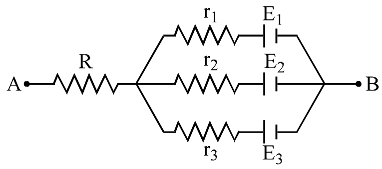


(A) 0.5V (B) 1.0V
(C) 0 V (D) 2 V

Q.9 In order to determine the e.m.f. of a storage battery it was connected in series with a standard cell in a certain circuit and a current I_1 was obtained. When the battery is connected to the same circuit opposite to the standard cell a current I_2 flow in the external circuit from the positive pole of the storage battery was obtained. What is the e.m.f. ε_1 of the storage battery? The e.m.f. of the standard cell is ε_2.

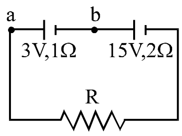
(A) $\varepsilon_1 = \dfrac{I_1 + I_2}{I_1 - I_2}\varepsilon_2$ (B) $\varepsilon_1 = \dfrac{I_1 + I_2}{I_2 - I_1}\varepsilon_2$ (C) $\varepsilon_1 = \dfrac{I_1 - I_2}{I_1 + I_2}\varepsilon_2$ (D) $\varepsilon_1 = \dfrac{I_2 - I_1}{I_1 + I_2}\varepsilon_2$

Q.10 In the network shown the potential difference between A and B is ($R = r_1 = r_2 = r_3 = 1\ \Omega$, $E_1 = 3$ V, $E_2 = 2$ V, $E_3 = 1$ V)



(A) 1 V (B) 2 V
(C) 3 V (D) 4 V

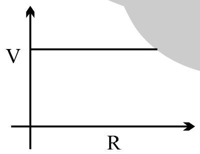
Q.11 Two batteries one of the emf 3V, internal resistance 1 ohm and the other of emf 15 V, internal resistance 2 ohm are connected in series with a resistance R as shown. If the potential difference between a and b is zero the resistance of R in ohm is

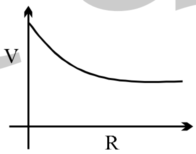


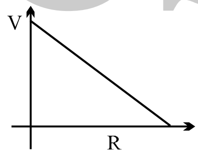
(A) 5 (B) 7 (C) 3 (D) 1

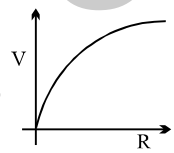
Q.12 A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in time t. N number of similar cells is now connected in series with a wire of the same material and cross section but of length 2L. The temperature of the wire is raised by the same amount ΔT in the same time t. The value of N is :
(A) 4 (B) 6 (C) 8 (D) 9

Q.13 A cell of emf E has an internal resistance r & is connected to rheostat. When resistance R of rheostat is changed correct graph of potential difference across it is

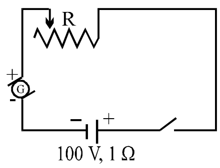
(A) 

(B) 

(C) 

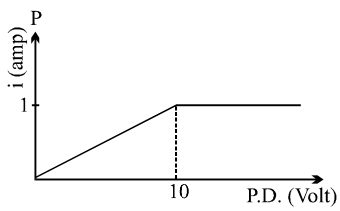
(D) 

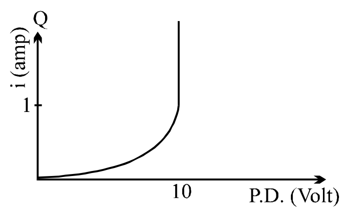
Q.14 The battery in the diagram is to be charged by the generator G. The generator has a terminal voltage of 120 volts when the charging current is10 amperes. The battery has an emf of 100 volts and an internal resistance of 1 ohm. In order to charge the battery at 10 amperes charging current, the resistance R should be set at



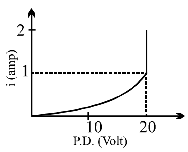
(A) 0.1 Ω (B) 0.5 Ω
(C) 1.0 Ω (D) 5.0 Ω

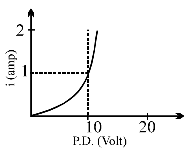
Q.15 Two current elements P and Q have current voltage characteristics as shown below :

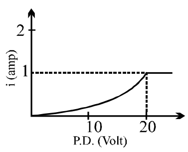


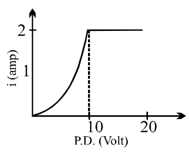


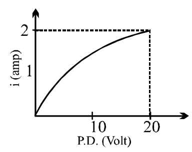
Which of the graphs given below represents current voltage characteristics when P and Q are in series.

(A) 

(B) 

(C) 

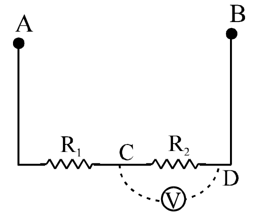
(D) 

(E) 

Q.16 A wire of cross-section area A, length L_1, resistivity ρ_1 and temperature coefficient of resistivity α_1 is connected to a second wire of length L_2, resistivity ρ_2, temperature coefficient of resistivity α_2 and the same area A, so that wire carries same current. Total resistance R is independent of temperature for small temperature change if (Thermal expansion effect is negligible)

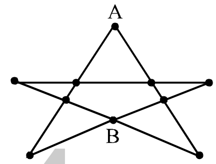
(A) $\alpha_1 = -\alpha_2$ (B) $\rho_1 L_1 \alpha_1 + \rho_2 L_2 \alpha_2 = 0$
(C) $L_1 \alpha_1 + L_2 \alpha_2 = 0$ (D) None

Q.17 Resistances R_1 and R_2 each 60Ω are connected in series as shown in figure. The Potential difference between A and B is kept 120 volt. Then what will be the reading of voltmeter connected between the point C & D if resistance of voltmeter is 120Ω.



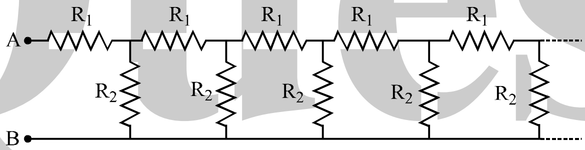
(A) 48 V (B) 24 V
(C) 40V (D) None

Q.18 The resistance of all the wires between any two adjacent dots is R. Then equivalent resistance between A and B as shown in figure is :



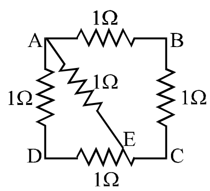
(A) 7/3 R (B) 7/6 R
(C) 14/8 R (D) None of these

Q.19 Consider an infinte ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage is halved after each section.



(A) $R_1/R_2 = 1$ (B) $R_1/R_2 = 1/2$ (C) $R_1/R_2 = 2$ (D) $R_1/R_2 = 3$

Q.20 ABCD is a square where each side is a uniform wire of resistance 1Ω. A point E lies on CD such that if a uniform wire of resistance 1Ω is connected across AE and constant potential difference is applied across A and C then B and E are equipotential.

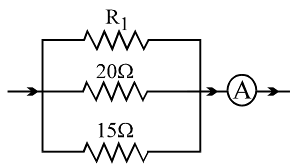


(A) $\frac{CE}{ED} = 1$ (B) $\frac{CE}{ED} = 2$ (C) $\frac{CE}{ED} = \frac{1}{\sqrt{2}}$ (D) $\frac{CE}{ED} = \sqrt{2}$

Q.21 In order to increase the resistance of a given wire of uniform cross section to four times its value, a fraction of its length is stretched uniformly till the full length of the wire becomes $\frac{3}{2}$ times the original length what is the value of this fraction?

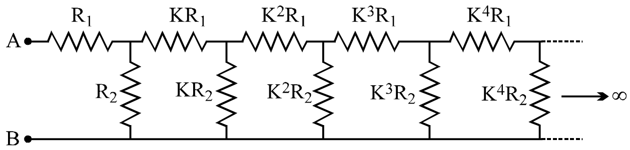
(A) $\frac{1}{4}$ (B) $\frac{1}{8}$ (C) $\frac{1}{16}$ (D) $\frac{1}{6}$

Q.22 In the given circuit the current flowing through the resisitance 20 ohms is 0.3 ampere while the ammetre reads 0.8 ampere. What is the value of R_1?



(A) 30 ohms (B) 40 ohms (C) 50 ohms (D) 60 ohms

Q.23 The circuit diagram shown consists of a large number of element (each element has two resistors R_1 and R_2). The resistance of the resistors in each subsequent element differs by a factor of $K = 1/2$ from the resistance of the resistors in the previous elements. The equivalent reistance between A and B shown in figure is :



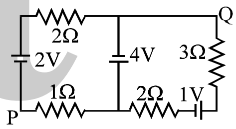
(A) $\dfrac{R_1 - R_2}{2}$ (B) $\dfrac{(R_1 - R_2) + \sqrt{6R_1R_2}}{2}$

(C) $\dfrac{(R_1 - R_2) + \sqrt{R_1^2 + R_2^2 + 6R_1R_2}}{2}$ (D) None of these

Q.24 A brass disc and a carbon disc of same radius are assembled alternatively to make a cylindrical conductor. The resistance of the cylinder is independent of the temperature. The ratio of thickness of the brass disc to that of the carbon disc is [α is temperature coefficient of resistance & Neglect linear expansion]

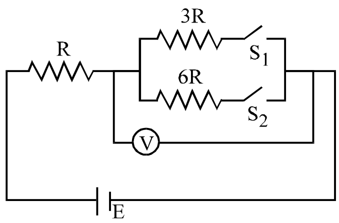
(A) $\left|\dfrac{\alpha_C \rho_C}{\alpha_B \rho_B}\right|$ (B) $\left|\dfrac{\alpha_C \rho_B}{\alpha_B \rho_C}\right|$ (C) $\left|\dfrac{\alpha_B \rho_C}{\alpha_C \rho_B}\right|$ (D) $\left|\dfrac{\alpha_B \rho_B}{\alpha_C \rho_C}\right|$

Q.25 In the circuit shown, what is the potential difference V_{PQ}?



(A) + 3V (B) + 2V (C) –2V (D) none

Q.26 In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed then

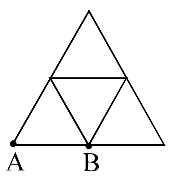


(A) $V_2 > V_1 > V_3$ (B) $V_3 > V_2 > V_1$
(C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

Q.27 One end of a Nichrome wire of length 2L and cross-sectional area A is attatched to an end of another Nichrome wire of length L and cross-sectional area 2A. If the free end of the longer wire is at an electric potential of 8.0 volts, and the free end of the shorter wire is at an electric potential of 1.0 volt, the potential at the junction of the two wires is equal to

(A) 2.4 V (B) 3.2 V (C) 4.5 V (D) 5.6 V

Q.28 In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is



(A) $\dfrac{11R}{7}$ (B) $\dfrac{18R}{11}$ (C) $\dfrac{7R}{11}$ (D) $\dfrac{11R}{18}$

Q.29 Power generated across a uniform wire connected across a supply is H. If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

(A) $\dfrac{H}{n^2}$ (B) n^2H (C) nH (D) $\dfrac{H}{n}$

Q.30 A constant voltage is applied between the two ends of a uniform metallic wire. Some heat is developed in it. The heat developed is doubled if
(A) both the length and the radius of the wire are halved.
(B) both the length and the radius of the wire are doubled
(C) the radius of the wire is doubled
(D) the length of the wire is doubled

Q.31 When electric bulbs of same power, but different marked voltage are connected in series across the power line, their brightness will be :
(A) proportional to their marked voltage
(B) inversely proportional to their marked voltage
(C) proportional to the square of their marked voltage
(D) inversely proportional to the square of their marked voltage
(E) the same for all of them

Q.32 Two bulbs rated (25 W – 220V) and (100W – 220V) are connected in series to a 440 V line. Which one is likely to fuse?
(A) 25 W bulb (B) 100 W bulb (C) both bulbs (D) none

Q.33 Rate of dissipation of Joule's heat in resistance per unit volume is (symbols have usual meaning)
(A) σE (B) σJ (C) $J E$ (D) None

Q.34 The charge flowing through a resistance R varies with time as $Q = 2t - 8t^2$. The total heat produced in the resistance is (for $0 \le t \le \frac{1}{8}$)
(A) $\frac{R}{6}$ joules (B) $\frac{R}{3}$ joules (C) $\frac{R}{2}$ joules (D) R joules

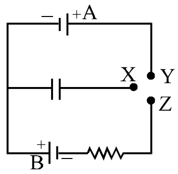
Q.35 A total charge Q flows across a resistor R during a time interval = T in such a way that the current vs. time graph for $0 \to T$ is like the loop of a sin curve in the range $0 \to \pi$. The total heat generated in the resistor is
(A) $Q^2\pi^2R / 8T$ (B) $2Q^2\pi^2R / T$ (C) $2Q^2\pi R / T$ (D) $Q^2\pi^2R / 2T$

Q.36 If the length of the filament of a heater is reduced by 10%, the power of the heater will
(A) increase by about 9% (B) increase by about 11%
(C) increase by about 19% (D) decrease by about 10%

Q.37 A heater A gives out 300 W of heat when connected to a 200 V d.c. supply. A second heater B gives out 600 W when connected to a 200 v d.c. supply. If a series combination of the two heaters is connected to a 200 V d.c. supply the heat output will be
(A) 100 W (B) 450 W (C) 300 W (D) 200 W

Q.38 Two bulbs one of 200 volts, 60 watts & the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be
(A) 37.5 watt (B) 160 watt (C) 62.5 watt (D) 110 watt

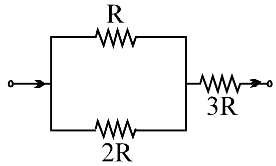
Q.39 In the circuit shown the cells are ideal and of equal emfs, the capacitance of the capacitor is C and the resistance of the resistor is R. X is first joined to Y and then to Z. After a long time, the total heat produced in the resistor will be



(A) equal to the energy finally stored in the capacitor
(B) half of the energy finally stored in the capacitor
(C) twice the energy finally stored in the capacitor
(D) 4 times the energy finally stored in the capacitor

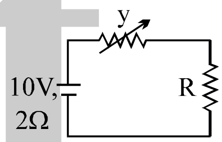
Q.40 Three 60 W light bulbs are mistakenly wired in series and connected to a 120 V power supply. Assume the light bulbs are rated for single connection to 120 V. With the mistaken connection, the power dissipated by each bulb is:
(A) 6.7 W (B) 13.3 W (C) 20 W (D) 40 W

Q.41 The ratio of powers dissipatted respectively in R and 3R, as shown is:



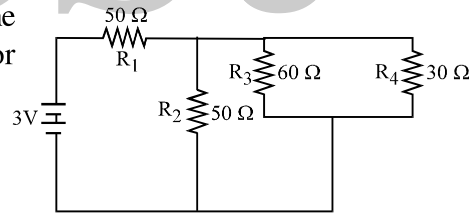
(A) 9 (B) 27/4 (C) 4/9 (D) 4/27

Q.42 In the figure shown the power generated in y is maximum when y = 5Ω.
Then R is



(A) 2 Ω (B) 6 Ω
(C) 5 Ω (D) 3 Ω

Q.43 In the circuit shown, the resistances are given in ohms and the battery is assumed ideal with emf equal to 3.0 volts. The resistor that dissipates the most power is



(A) R_1
(B) R_2
(C) R_3
(D) R_4

Q.44 What amount of heat will be generated in a coil of resistance R due to a charge q passing through it if the current in the coil decreases to zero uniformly during a time interval Δt

(A) $\frac{4}{3}\frac{q^2R}{\Delta t}$ (B) $ln\frac{q^2R}{2\Delta t}$ (C) $\frac{2q^2R}{3\Delta t}$ (D) $\ln\frac{(2\Delta t)}{q^2R}$

Q.45 The variation of current (I) and voltage (V) is as shown in figure A. The variation of power P with current I is best shown by which of the following graph

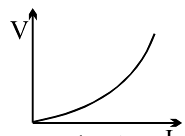
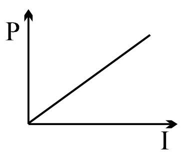
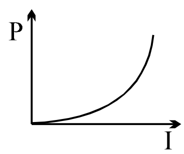
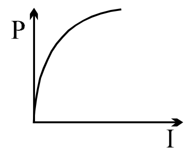
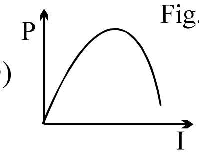


Fig. A

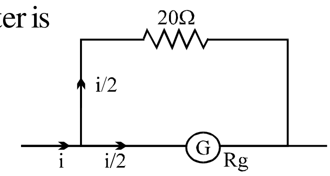
(A) 

(B) 

(C) 

(D) 

Q.46 In a galvanometer, the deflection becomes one half when the galvanometer is shunted by a 20Ω resistor. The galvanometer resistance is



(A) 5Ω (B) 10Ω
(C) 40Ω (D) 20Ω

Q.47 When a galvanometer is shunted with a 4Ω resistance, the deflection is reduced to one-fifth. If the galvanometer is further shunted with 2Ω wire, the further reduction in the deflection will be (the main current remains same)

(A) $\frac{8}{13}$ of the deflection when shunted with 4Ω only

(B) $\frac{5}{13}$ of the deflection when shunted with 4Ω only

(C) $\frac{3}{4}$ of the deflection when shunted with 4Ω only

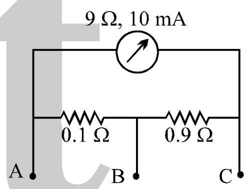
(D) $\frac{3}{13}$ of the deflection when shunted with 4Ω only

Q.48 A galvanometer has a resistance of 20Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

(A) 0.01Ω resistor connected across it (B) 0.02Ω resistor connected across it

(C) 200Ω resistor connected in series with it (D) 2000 Ω resistor connected in series with it

Q.49 A milliammeter of range 10 mA and resistance 9 Ω is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B (C is left isolated). The value of I is

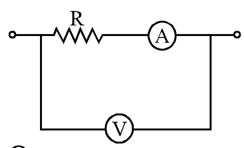


(A) 100 mA (B) 900 mA (C) 1 A (D) 1.1 A

Q.50 A galvanometer coil has a resistance 90 Ω and full scale deflection current 10 mA. A 910Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1V, the number of divisions on its scale is

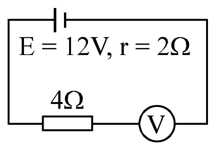
(A) 90 (B) 91 (C) 100 (D) none

Q.51 In the circuit shown the resistance of voltmeter is 10,000 ohm and that of ammeter is 20 ohm. The ammeter reading is 0.10 Amp and voltmeter reading is 12 volt. Then R is equal to



(A) 122 Ω (B) 140 Ω (C) 116 Ω (D)100 Ω

Q.52 By error, a student places moving-coil voltmeter V (nearly ideal) in series with the resistance in a circuit in order to read the current, as shown. The voltmeter reading will be



(A) 0 (B) 4V (C) 6V (D) 12V

Q.53 In a balanced wheat stone bridge, current in the galvanometer is zero. It remains zero when:

[1] battery emf is increased

[2] all resistances are increased by 10 ohms

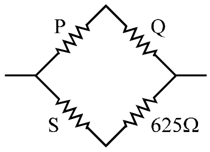
[3] all resistances are made five times

[4] the battery and the galvanometer are interchanged

(A) only [1] is correct (B) [1], [2] and [3] are correct

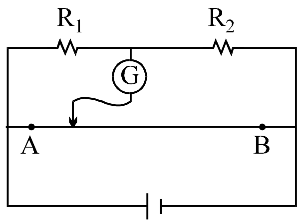
(C) [1], [3] and [4] are correct (D) [1] and [3] are correct

Q.54 A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51Ω to secure balance. The unknown resistance in the fourth arm is



(A) 625 Ω (B) 650 Ω (C) 676 Ω (D) 600 Ω

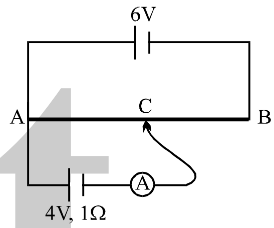
Q.55 In the figure shown for gives values of R_1 and R_2 the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of 10 Ω, balance shifts to 50 cm. R_1 and R_2 are (AB = 1 m):



(A) $\frac{10}{3}$ Ω, 5 Ω (B) 20 Ω, 30 Ω

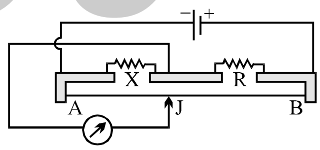
(C) 10 Ω, 15 Ω (D) 5 Ω, $\frac{15}{2}$ Ω

Q.56 A 6 V battery of negligible internal resistance is connected across a uniform wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance 1 Ω is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. The AC is equal to



(A) 2/3 m (B) 1/3 m
(C) 3/5 m (D) 1/2 m

Q.57 The figure shows a metre-bridge circuit, with AB = 100 cm, X = 12Ω and R = 18Ω, and the jockey J in the position of balance.
If R is now made 8Ω, through what distance will J have to be moved to obtain balance?



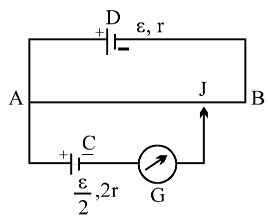
(A) 10 cm (B) 20 cm
(C) 30 cm (D) 40 cm

Q.58 A potentiometer wire has length 10 m and resistance 10Ω. It is connected to a battery of EMF 11 volt and internal resistance 1 Ω, then the potential gradient in the wire is
(A) 10 V/m (B) 1 V/m (C) 0.1 V/m (D) none

Q.59 The length of a potentiometer wire is l. A cell of emf E is balanced at a length $l/3$ from the positive end of the wire. If the length of the wire is increased by $l/2$. At what distance will the same cell give a balance point.

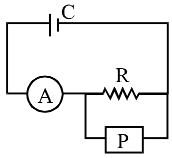
(A) $\frac{2l}{3}$ (B) $\frac{l}{2}$ (C) $\frac{l}{6}$ (D) $\frac{4l}{3}$

Q.60 In the figure, the potentiometer wire AB of length L and resistance 9r is joined to the cell D of emf ε and internal resistance r. The cell C's emf is ε/2 and its internal resistance is 2r. The galvanometer G will show no deflection when the length AJ is



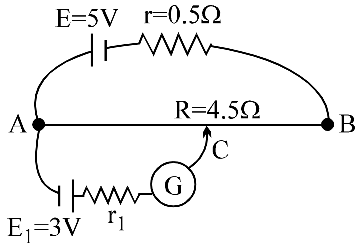
(A) $\frac{4L}{9}$ (B) $\frac{5L}{9}$ (C) $\frac{7L}{18}$ (D) $\frac{11L}{18}$

Q.61 An ammeter A of finite resistance, and a resistor R are joined in series to an ideal cell C. A potentiometer P is joined in parallel to R. The ammeter reading is I_0 and the potentiometer reading is V_0. P is now replaced by a voltmeter of finite resistance. The ammeter reading now is I and the voltmeter reading is V.



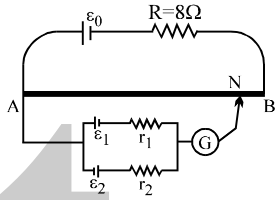
(A) $I > I_0$, $V < V_0$ (B) $I > I_0$, $V = V_0$ (C) $I = I_0$, $V < V_0$ (D) $I < I_0$, $V = V_0$

Q.62 In the given potentiometer circuit length of the wire AB is 3 m and resistance is $R = 4.5\ \Omega$. The length AC for no deflection in galvanometer is



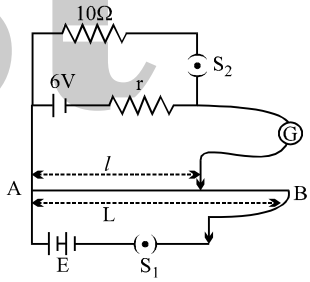
(A) 2 m (B) 1.8 m
(C) dependent on r_1 (D) none of these

Q.63 A battery of emf $E_0 = 12$ V is connected across a 4m long uniform wire having resistance 4Ω/m. The cells of small emfs $\varepsilon_1 = 2$V and $\varepsilon_2 = 4$V having internal resistance 2Ω and 6Ω respectively, are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to



(A) $\frac{1}{6}$ m (B) $\frac{1}{3}$ m (C) 25 cm (D) 50 cm

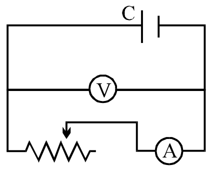
Q.64 In the arrangement shown in figure when the switch S_2 is open, the galvanometer shows no deflection for $l = L/2$. When the switch S_2 is closed, the galvanometer shows no deflection for $l = 5L/12$. The internal resistance (r) of 6 V cell, and the emf E of the other battery are respectively

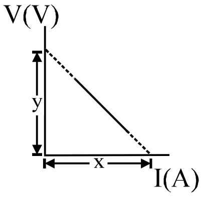


(A) 3Ω, 8V (B) 2Ω, 12V
(C) 2Ω, 24V (D) 3Ω, 12V

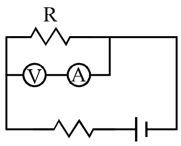
Q.65 The diagram besides shows a circuit used in an experiment to determine the emf and internal resistance of the cell C. A graph was plotted of the potential difference V between the terminals of the cell against the current I, which was varied by adjusting the rheostat. The graph is shown on the right ; x and y are the intercepts of the graph with the axes as shown. What is the internal resistance of the cell ?

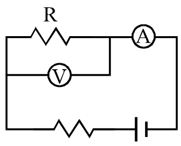
(A) x (B) y
(C) x/y (D) y/x

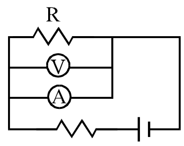


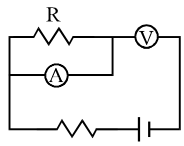


Q.66 Which of the following wiring diagrams could be used to experimentally determine R using ohm's law? Assume an ideal voltmeter and an ideal ammeter.

(A) 

(B) 

(C) 

(D) 

Q.67 A current of (2.5 ± 0.05) A flows through a wire and develops a potential difference of (10 ± 0.1) volt. Resistance of the wire in ohm, is

(A) 4 ± 0.12 (B) 4 ± 0.04 (C) 4 ± 0.08 (D) 4 ± 0.02

Q.68 Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1 : 4, the drift velocity of electrons in the two wires will be in the ratio:
(A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

Q.69 In a wire of cross-section radius r, free electrons travel with drift velocity v when a current I flows through the wire. What is the current in another wire of half the radius and of the same material when the drift velocity is 2v?
(A) 2I (B) I (C) I/2 (D) I/4

Q.70 Read the following statements carefully :
Y : The resistivity of a semiconductor decreases with increases of temperature.
Z : In a conducting solid, the rate of collision between free electrons and ions increases with increase of temperature.
Select the correct statement from the following :
(A) Y is true but Z is false (B) Y is false but Z is true.
(C) Both Y and Z are true. (D) Y is true and Z is the correct reason for Y.

Q.71 A piece of copper and another of germanium are cooled from room temperature to 80 K. The resistance of :
(A) each of them increases
(B) each of them decreases
(C) copper increases and germanium decreases
(D) copper decreases and germanium increases.

Q.72 An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions$\rightarrow$ their charges being –e and +2e. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v (–ve ion) and v/4 (+ve ion). Both ions have the same number per unit volume = n. The current flowing through the pipe is
(A) nev A/2 (B) nev A/4 (C) 5nev A/2 (D) 3nev A/2

Q.73 As the temperature of a conductor increases, its resistivity and conductivity change. The ratio of resistivity to conductivity
(A) increases
(B) decreases
(C) remains constant
(D) may increase or decrease depending on the actual temperature.

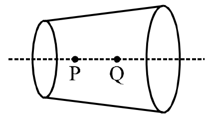
Q.74 Current density in a cylindrical wire of radius R is given as $J = \begin{cases} J_0\left(\frac{x}{R}-1\right) \text{for} 0 \le x < \frac{R}{2} \\ J_0\frac{x}{R} \text{for} \frac{R}{2} \le x \le R \end{cases}$.

The current flowing in the wire is:

(A) $\frac{7}{24}\pi J_0R^2$ (B) $\frac{1}{6}\pi J_0R^2$ (C) $\frac{7}{12}\pi J_0R^2$ (D) $\frac{5}{12}\pi J_0R^2$

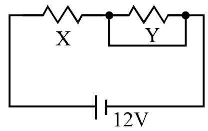
Q.75 A current I flows through a uniform wire of diameter d when the mean electron drift velocity is V. The same current will flow through a wire of diameter d/2 made of the same material if the mean drift velocity of the electron is :
(A) v/4 (B) v/2 (C) 2v (D) 4v

Q.76 A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and q is v_P and v_Q.



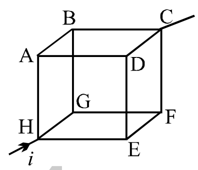
(A) $v_P = v_Q$ (B) $v_P < v_Q$
(C) $v_P > v_Q$ (D) Data insufficient

Q.77 When an ammeter of negligible internal resistance is inserted in series with circuit it reads 1A. When the voltmeter of very large resistance is connected across X it reads 1V. When the point A and B are shorted by a conducting wire, the voltmeter measures 10 V across the battery. The internal resistance of the battery is equal to



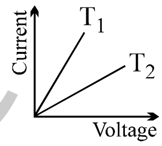
(A) zero
(B) 0.5 Ω
(C) 0.2 Ω
(D) 0.1 Ω

Q.78 In the box shown current i enters at H and leaves at C. If $i_{AB} = \frac{i}{6}$, $i_{DC} = \frac{2i}{3}$, $i_{HA} = \frac{i}{2}$, $i_{GF} = \frac{i}{6}$, $i_{HE} = \frac{i}{6}$, choose the branch in which current is zero



(A) BG (B) FC (C) ED (D) none

Q.79 The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2. Which is correct

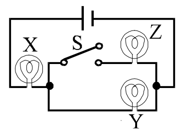


(A) $T_1 > T_2$ (B) $T_1 < T_2$
(C) $T_1 = T_2$ (D) none

Q.80 A uniform copper wire carries a current i amperes and has p carriers per metre^3. The length of the wire is ℓ metres and its cross-section area is s metre^2. If the charge on a carrier is q coulombs, the drift velocity in ms^{-1} is given by

(A) $i/\ell sq$ (B) i/psq (C) psq/i (D) $i/ps\ell q$

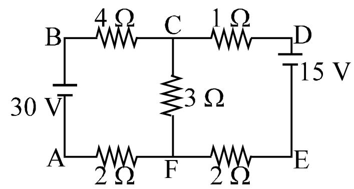
Q.81 If X, Y and Z in figure are identical lamps, which of the following changes to the brightnesses of the lamps occur when switch S is closed?



(A) X stays the same, Y decreases (B) X increases, Y decreases
(C) X increases, Y stays the same (D) X decreases, Y increases

Question No. 82 to 85 (4 questions)

The figure shows a network of five resistances and two batteries



Q.82 The current through the 30V battery is
(A) 3A (B) 1A (C) 2A (D) none

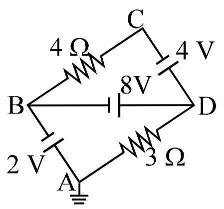
Q.83 The current through the 15V battery is
(A) zero (B) 1A (C) 3A (D) none

Q.84 Which of the batteries is getting charged.
(A) 30V (B) 15V (C) both (D) none

Q.85 The total electrical power consumed by the circuit is
(A) 15W (B) 75W (C) 105W (D) 90W

Question No. 86 to 88 (3 questions)

The figure shows a network of resistances in which the point A is earthed.



Q.86 The point which has the least potential is

(A) A (B) B (C) C (D) D

Q.87 The current through the 3Ω resistor is

(A) 2A from D to A (B) 2A from A to D

(C) 3.33A from A to D (D) 3.33A from D to A

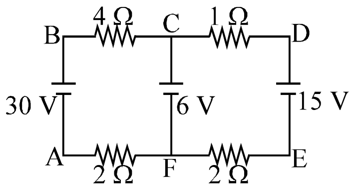
Q.88 The current through the 4Ω resistor is

(A) 0.5 A from B to C (B) 0.5A from C to B

(C) 1A from C to B (D) 1A from B to C

Question No. 89 to 93 (5 questions)

The figure shows a network of four resistances and three batteries



Q.89 Choose the correct alternative

(A) The potential difference $V_C - V_F = 6V$.

(B) No current flows in the branch CF.

(C) Current flows in the branch from F to C.

(D) Both (A) and (C)

Q.90 Mark the incorrect statement.

(A) The current flowing in the left loop is independent of the right loop.

(B) The current flowing in the right loop is independent of the left loop.

(C) Both 30V and 15V batteries do not produce current in the branch CF

(D) both (A) and (B)

Q.91 Which of the battery is getting charged.

(A) Only 6V (B) both 6V and 15V (C) Only 15V (D) None

Q.92 The current through the branch CF is

(A) 4A (B) 3A (C) 7A (D) 1A

Q.93 The electrical power dissipated as heat is

(A) 207 W (B) 123 W (C) 165 W (D) none

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 A battery is of emf E is being charged from a charger such that positive terminal of the battery is connected to terminal A of charger and negative terminal of the battery is connected to terminal B of charger. The internal resistance of the battery is r.

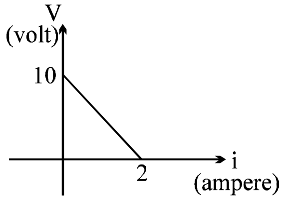
(A) Potential difference across points A and B must be more than E.

(B) A must be at higher potential than B

(C) In battery, current flows from positive terminal to the negative terminal

(D) No current flows through battery

Q.2 A battery of emf E and internal resistance r is connected across a resistance R. Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative(s).



(A) internal resistance of battery is 5Ω

(B) emf of the battery is 20V

(C) maximum current which can be taken from the battery is 4A

(D) V- i graph can never be a straight line as shown in figure.

Q.3 The equivalent resistance of a group of resistances is R. If another resistance is connected in parallel to the group, its new equivalent becomes R_1 & if it is connected in series to the group, its new equivalent becomes R_2 we have :

(A) $R_1 > R$ (B) $R_1 < R$ (C) $R_2 > R$ (D) $R_2 < R$

Q.4 Two identical fuses are rated at 10A. If they are joined

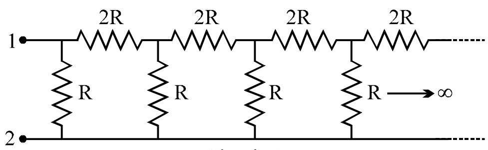
(A) in parallel, the combination acts as a fuse of rating 20A

(B) in parallel, the combination acts as a fuse of rating 5A

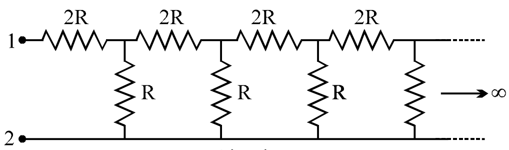
(C) in series, the combination acts as a fuse of rating 10A.

(D) in series, the combination acts as a fuse of rating 20A.

Q.5 Two circuits (shown below) are called 'Circuit A' and 'Circuit B'. The equivalent resistance of 'Circuit a' is x and that of 'Circuit B' is y between 1 and 2.



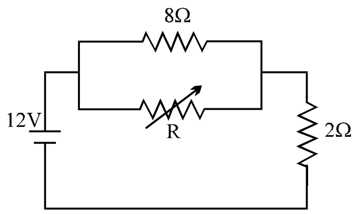
Circuit A



Circuit B

(A) $y > x$ (B) $y = (\sqrt{3} + 1)R$ (C) $xy = 2R^2$ (D) $x - y = 2R$

Q.6 The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition



(A) R = 0

(B) R = 8Ω

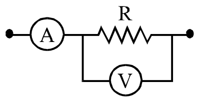
(C) power dissipated in the 2 Ω resistor is 72 W.

(D) power dissipated in the 2 Ω resistor is 8 W.

Q.7 A galvanometer may be converted into ammeter or voltmeter. In which of the following cases the resistance of the device will be the largest ? (Asssume maximum range of galvanometer = 1 mA)
(A) an ammeter of range 10A (B) a voltmeter of range 5 V
(C) an ammeter of range 5 A (D) a voltmeter of range 10 V.

Q.8 Mark out the correct options.
(A) An ammeter should have small resistance.
(B) An ammeter should have large resistance.
(C) A voltmeter should have small resistance.
(D) A voltmeter should have large resistance.

Q.9 In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non ideal, then R is :



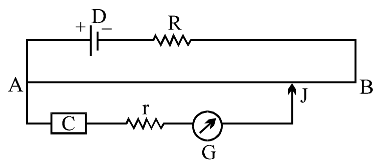
(A) 5Ω (B) less than 5Ω
(C) greater than 5Ω (D) between 4Ω & 5Ω

Q.10 A micrometer has a resistance of 100Ω and a full scale range of 50μA. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).
(A) 50 V range with 10 kΩ resistance in series. (B) 10 V range with 200 kΩ resistance in series.
(C) 5 mA range with 1 Ω resistance in parallel. (D) 10 mA range with 1 kΩ resistance in parallel.

Q.11 In a potentiometer arrangement. E_1 is the cell establishing current in primary circuit. E_2 is the cell to be measured. AB is the potentiometer wire and G is a galvanometer. Which of the following are the essential condition for balance to be obtained.
(A) The emf of E_1 must be greater than the emf of E_2.
(B) Either the positive terminals of both E_1 and E_2 or the negative terminals of both E_1 and E_2 must be joined to one end of potentiometer wire.
(C) The positive terminals of E_1 and E_2 must be joined to one end of potentiometer wire.
(D) The resistance of G must be less than the resistance of AB.

Q.12 In a potentiometer wire experiment the emf of a battery in the primary circuit is 20V and its internal resistance is 5Ω. There is a resistance box in series with the battery and the potentiometer wire, whose resistance can be varied from 120Ω to 170Ω. Resistance of the potentiometer wire is 75Ω. The following potential differences can be measured using this potentiometer.
(A) 5V (B) 6V (C) 7V (D) 8V

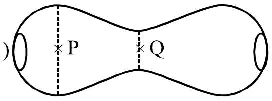
Q.13 In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0. C is a cell of internal resistance r. The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this?



(A) $r > R_0$ (B) $R >> R_0$
(C) emf of C > emf of D (D) The negative terminal of C is connected to A.

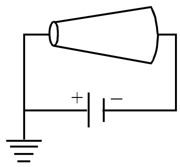
Q.14 Which of the following quantities do not change when a resistor connected to a battery is heated due to the current?
(A) drift speed (B) resistivity (C) resistance (D) number of free electrons

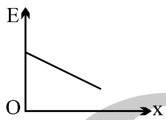
Q.15 A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :

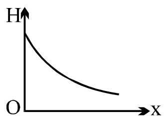


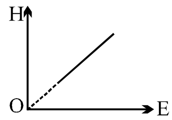
(A) the current at the cross-section P equals the current at the cross-section Q
(B) the electric field intensity at P is less than that at Q.
(C) the rate of heat generated per unit time at Q is greater than that at P
(D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.

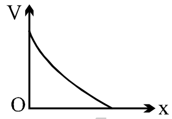
Q.16 A conductor is made of an isotropic material and has shape of a truncated cone. A battery of constant emf is connected across it and its left end is earthed as shown in figure. If at a section distant x from left end, electric field intensity, potential and the rate of generation of heat per unit length are E, V and H respectively, which of the following graphs is/are correct?



(A) 

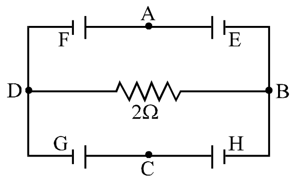
(B) 

(C) 

(D) 

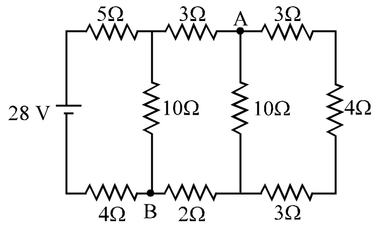
Q.17 A simple circuit contains an ideal battery and a resistance R. If a second resistor is placed in parallel with the first,
(A) the potential across R will decrease
(B) the current through R will decreased
(C) the current delivered by the battery will increase
(D) the power dissipated by R will increased.

Q.18 In the circuit shown E, F, G and H are cells of e.m.f. 2V, 1V, 3V and 1V respectively and their internal resistances are 2Ω, 1Ω, 3Ω and 1Ω respectively.



(A) $V_D - V_B = -2/13$ V
(B) $V_D - V_B = 2/13$ V
(C) $V_G = 21/13$ V = potential difference across G.
(D) $V_H = 19/13$ V = potential difference across H.

Q.19 Consider the circuit shown in the figure

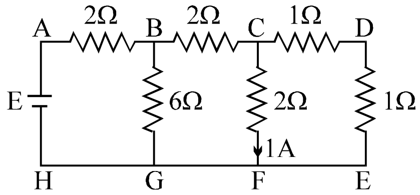


(A) the current in the 5 Ω resistor is 2 A
(B) the current in the 5 Ω resistor is 1 A
(C) the potential difference $V_A - V_B$ is 10 V
(D) the potential difference $V_A - V_B$ is 5 V

Q.20 A current passes through a wire of nonuniform cross section. Which of the following quantities are independent of the cross-section?
(A) the charge crossing in a given time interval.
(B) drift speed
(C) current density
(D) free-electron density.

Question No. 21 to 24 (4 questions)

The figure shows a network of resistors and a battery. If 1A current flows through the branch CF, then answer the following questions



Q.21 The current through

(A) branch DE is 1A
(B) branch BC is 2A
(C) branch BG is 4A
(D) branch HG is 6A

Q.22 The emf E of the battery is

(A) 24 V
(B) 12 V
(C) 18V
(D) 6V

If a zero resistance wire is connected in parallel to branch CF

Q.23 The current through

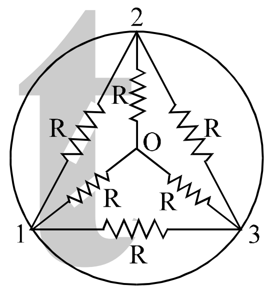
(A) branch DE is zero
(B) branch BC is zero
(C) branch BG is 0.5A
(D) branch AB is 1.5A

Q.24 The emf E of the battery is

(A) 9V
(B) 6.6V
(C) 5.25V
(D) 10.5V
(E) 12V

Question No. 25 to 27 (3 questions)

Inside a super conducting ring six identical resistors each of resistance R are connected as shown in figure.



Q.25 The equivalent resistance(s)

(A) between 1 & 3 is zero.
(B) between 1 & 3 is R/2
(C) between 1 & 2, 2 & 3, 3 & 1 are all equal.
(D) between 1 & 3 is two times that between 1 & 2.

Q.26 The equivalent resistance(s)

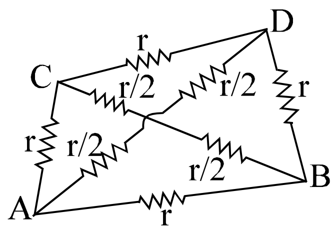
(A) between 0 & 1 is R.
(B) between 0 & 1 is R/3
(C) between 0 & 1 is zero.
(D) between 0 & 1, 0 & 2 and 0 & 3 are all equal.

Q.27 Imagine a battery of emf E between the point 0 and 1, with its positive terminal connected with O.

(A) The current entering at O is equally divided into three resistances.
(B) the current in the other three resistances R_{12}, R_{13}, R_{23} is zero.
(C) The resistances R_{02} and R_{03} have equal magnitudes of current while the resistance R_{01} have different current.
(D) Potential $V_2 = V_3 > V_1$.

Question No. 28 to 30 (3 questions)

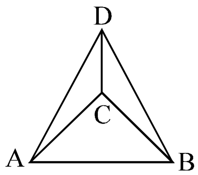
The figure shows a tetrahedron, each side of which has a resistance r

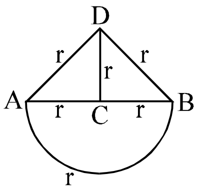


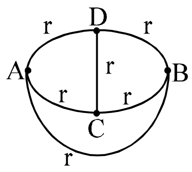
Q.28 Choose the correct statement(s) related to the resistance between any two points.

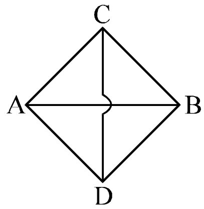
(A) $R_{AB} = R_{BD} = R_{BC} = R_{CD} = R_{CA} = R_{AD}$
(B) $R_{AB} = R_{AC} = R_{AD} = R_{BD} = R_{BC} \neq R_{CD}$
(C) R_{CD} is the least
(D) $R_{AB} = R_{AC} = R_{BC}$ and $R_{CD} = R_{AD} = R_{BD}$

Q.29 Choose the correct diagram(s), which show two-dimensional equivalent of the tetrahedron.

(A) 

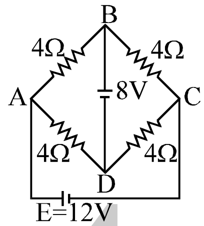
(B) 

(C) 

(D) 

Q.30 If a battery is connected between any two points of the tetrahedron, then identify the correct statement(s).
(A) The potentials of the other two points are always equal.
(B) There always exists a branch through which no current flows.
(C) The current coming out of the battery in each case is same.
(D) None of these

Question No. 31 to 33 (3 questions)
The given figure shows a network of resistances and a battery.



Q.31 Identify the correct statement(s)
(A) The circuit satisfies the condition of a balanced Wheatstone bridge.
(B) $V_B - V_D = 0$
(C) $V_B - V_D = 8$
(D) no current flows in the branch BD

Q.32 Which of the two batteries is getting charged?
(A) 8V battery (B) 12 V battery (C) none (D) can't be said

Q.33 Choose the correct statement(s).
(A) The current coming out of the 8V battery is 2A
(B) The current coming out of the 12V battery is 3A
(C) The current flowing in all the 4Ω branches is same.
(D) The current flowing in the diagonally opposite branches is same

ANSWER KEY

ONLY ONE OPTION IS CORRECT

Q.1	C	Q.2	B	Q.3	D	Q.4	A	Q.5	D	Q.6	C	Q.7	D
Q.8	C	Q.9	A	Q.10	B	Q.11	C	Q.12	B	Q.13	D	Q.14	C
Q.15	C	Q.16	B	Q.17	A	Q.18	B	Q.19	B	Q.20	D	Q.21	B
Q.22	D	Q.23	C	Q.24	A	Q.25	B	Q.26	A	Q.27	A	Q.28	D
Q.29	B	Q.30	B	Q.31	C	Q.32	A	Q.33	C	Q.34	A	Q.35	A
Q.36	B	Q.37	D	Q.38	A	Q.39	D	Q.40	A	Q.41	D	Q.42	D
Q.43	A	Q.44	A	Q.45	B	Q.46	D	Q.47	A	Q.48	B	Q.49	C
Q.50	C	Q.51	D	Q.52	D	Q.53	C	Q.54	B	Q.55	A	Q.56	A
Q.57	B	Q.58	B	Q.59	B	Q.60	B	Q.61	A	Q.62	D	Q.63	C
Q.64	B	Q.65	D	Q.66	B	Q.67	A	Q.68	C	Q.69	C	Q.70	C
Q.71	D	Q.72	D	Q.73	A	Q.74	D	Q.75	D	Q.76	C	Q.77	C
Q.78	B	Q.79	B	Q.80	B	Q.81	B	Q.82	A	Q.83	B	Q.84	D
Q.85	C	Q.86	B	Q.87	A	Q.88	C	Q.89	A	Q.90	C	Q.91	A
Q.92	C	Q.93	B										

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	A,B,C	Q.2	A	Q.3	B,C	Q.4	A,C
Q.5	A,B,C	Q.6	A,C	Q.7	D	Q.8	A,D
Q.9	C	Q.10	B,C	Q.11	A,B	Q.12	A,B,C
Q.13	A	Q.14	D	Q.15	A,B,C,D	Q.16	B,C
Q.17	C	Q.18	A,C,D	Q.19	A	Q.20	A,D
Q.21	A,B	Q.22	B	Q.23	A	Q.24	E
Q.25	A,C,D	Q.26	B,D	Q.27	A,B	Q.28	A,D
Q.29	A,B,C,D	Q.30	A,B,C	Q.31	C	Q.32	C
Q.33	A,B,D						

TARGET IIT JEE

PHYSICS

MAGNETIC EFFECT OF CURRENT

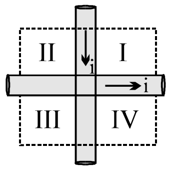
QUESTION FOR SHORT ANSWER

Q.1 Consider a magnetic field line. Is the magnitude of $\vec{B}$ constant or variable along such a line? Can you give an example of each case?

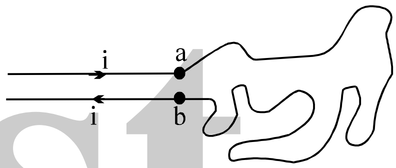
Q.2 A current is sent through a vertical spring from whose lower end a weight is hanging. What will happen?

Q.3 $B = \mu_0 i / 2\pi d$ suggets that a strong magnetic field is set up at points near a long wire carrying a current. Since there is a current i and magnetic field $\vec{B}$, why is there not a force on the wire in accord with the equation $\vec{F}_B = i\vec{L} \times \vec{B}$?

Q.4 Two fixed wires cross each other perpendicularly so that they do not actually touch but are close to each other, as shown in figure. Equal currents i exist in each wire in the directions indicated. In what region(s) will there be some points of zero net magnetic field?

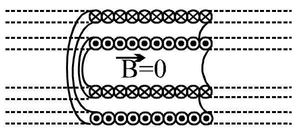


Q.5 A messy loop of limp wire is placed on a frictionless table and anchored at points a and b as shown in figure. If a current i is now passed through the wire, will it try to form a circular loop or will it try to bunch up further?



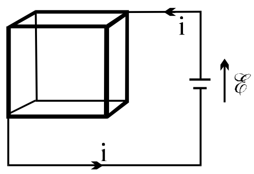
Q.6 A very long conductor has a square cross section and contains a coaxial cavity also with a square cross section. Current is distributed uniformly over the material cross section of the conductor. Is the magnetic field in the cavity equal to zero? Justify you answer.

Q.7 Two long solenoids are nested on the same axis, as in figure. They carry identical currents but in opposite directions. If there is no magnetic field inside the inner solenoid, what can you say about n, the number of turns per unit length, for the two solenoids? Which one, if either, has the larger value?



Q.8 The magnetic field at the center of a circular current loop has the value $B = \mu_0 i / 2R$. However, the electric field at the center of a ring of charge is zero. Why this difference?

Q.9 A steady current is set up in a cubical network of resistive wires, as in figure. Use symmetry arguments to show that the magnetic field at the center of the cube is zero



Q.10 A copper pipe filled with an electrolyte. When a voltage is applied, the current in the electrolyte is constituted by the movement of positive and negative ions in opposite directions. Will such a pipe experience a force when placed in a magnetic field perpendicular to the current.

Q.11 Magnetic moments arise due to charges. Can a system have magnetic moments even though it has no charge.

Q.12 Imagine that the room in which you are seated is fillled with a uniform magnetic field with B pointing vertically upward. A circular loop of wire has its plane horizontal . For what direction of current in the loop, as viewed from above, will the loop be in stable eqiulibrium with respect to forces & torques of magnetic origin ?

Q.13 Two current-carrying wires may attract each other. In absence of other forces, the wires will move towards each other increasing the kinetic energy. From where does this energy come?

Q.14 In order to have a current in a long wire, it should be connected to a battery or some such device. Can we obtain the magnetic field due to a straight, long wire by using Ampere's law without mentioning this other part of the circuit.

Q.15 A uniform magnetic field fills a certian cubical region of space. Can an electron be fired into this cube from the outside in such a way that it will travel in a closed circular path inside the cube?

Q.16 In Ampere's law $\oint \vec{B}.d\vec{l} = \mu_0 i$ the current outside the curve is not included on the right hand side. Does it mean that the magnetic field B calculated by using Ampere's law, gives the contribution of only the currents crossing the area bounded by the curve ?

Q.17 A magnetic field that varies in magnitude form point to point, but has constant direction (East to West) is set up in a chamber . A charged particle enters the chamber and travels undeflected along a straight path with constant speed . What can you say about the initial velocity of the particle?

Q.18 A charged particle enters an environment of a strong & non-uniform magnetic field varying from point to point both in magnitude and direction and comes out of it following a complicated trajectory. Would its final speed equal the initial speed , if it suffered no collisions with the environment.

Q.19 A straight wire carrying on electric current is placed along the axis of a uniformly charged ring. Will there be a magnetic force on the wire if the ring starts rotating about the wire ? If yes, in which direction ?

Q.20 An electron travelling West to East enters a chamber having a uniform electrostatic field in North to South direction . Specify the direction in which a uniform magnetic field should be set up to prevent the electron from deflecting from its straight line path .

Q.21 The magnetic field inside a tightly wound, long solenoid is $B = \mu_0 ni$. It suggests that the field does not depend on the total length of the solenoid, and hence if we add more loops at the ends of a solenoid the field should not increase. Explain qualitatively why the extra-added loops do not have a considerable effect on the field inside the solenoid.

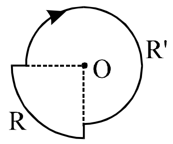
Q.22 A lightening conductor is connected to the earth by a circular copper pipe. After lightning strikes, it is discovered that the pipe has turned into a circular rod. Explain the cause of this phenomenon.

Q.23 We know that the work required to turn a current loop end for end in an external magnetic field is $2\mu B$. Does this hold no matter what the original orientaion of the loop was ?

ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

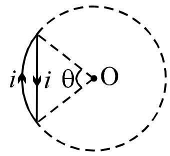
Q.1 A current of i ampere is flowing through each of the bent wires as shown the magnitude and direction of magnetic field at 0 is



(A) $\frac{\mu_0 i}{4}\left(\frac{1}{R}+\frac{2}{R'}\right)$ (B) $\frac{\mu_0 i}{4}\left(\frac{1}{R}+\frac{3}{R'}\right)$

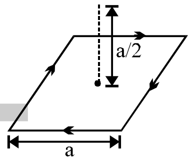
(C) $\frac{\mu_0 i}{8}\left(\frac{1}{R}+\frac{3}{2R'}\right)$ (D) $\frac{\mu_0 i}{8}\left(\frac{1}{R}+\frac{3}{R'}\right)$

Q.2 Net magnetic field at the centre of the circle O due to a current carrying loop as shown in figure is ($\theta < 180°$)



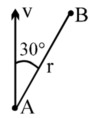
(A) zero
(B) perpendicular to paper inwards
(C) perpendicular to paper outwards
(D) is perpendicular to paper inwards if $\theta \le 90°$ and perpendicular to paper outwards if $90° \le \theta < 180°$

Q.3 The magnetic field due to a current carrying square loop of side a at a point located symmetrically at a distance of a/2 from its centre (as shown is)



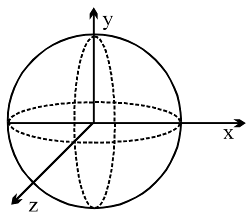
(A) $\frac{\sqrt{2}\,\mu_0 i}{\sqrt{3}\,\pi a}$ (B) $\frac{\mu_0\, i}{\sqrt{6}\,\pi a}$ (C) $\frac{2\mu_0 i}{\sqrt{3}\,\pi a}$ (D) zero

Q.4 A charge particle A of charge q = 2 C has velocity v = 100 m/s. When it passes through point A and has velocity in the direction shown. The strength of magnetic field at point B due to this moving charge is (r = 2 m).



(A) 2.5 μT (B) 5.0 μT (C) 2.0 μT (D) None

Q.5 Three rings, each having equal radius R, are placed mutually perpendicular to each other and each having its centre at the origin of co-ordinate system. If current I is flowing thriugh each ring then the magnitude of the magnetic field at the common centre is



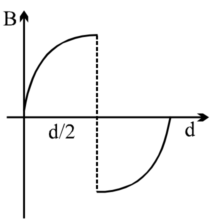
(A) $\sqrt{3}\frac{\mu_0 I}{2R}$ (B) zero (C) $\left(\sqrt{2}-1\right)\frac{\mu_0 I}{2R}$ (D) $\left(\sqrt{3}-\sqrt{2}\right)\frac{\mu_0 I}{2R}$

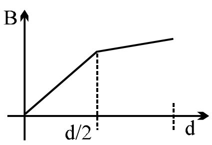
Q.6 Two concentric coils X and Y of radii 16 cm and 10 cm lie in the same vertical plane containing N-S direction. X has 20 turns and carries 16 A. Y has 25 turns & carries 18A. X has current in anticlockwise direction and Y has current in clockwise direction for an observer, looking at the coils facing the west. The magnitude of net magnetic field at their common centre is

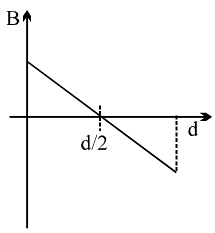
(A) $5\pi \times 10^{-4}$ T towards west (B) $13\pi \times 10^{-4}$ T towards east

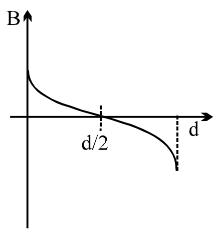
(C) $13\pi \times 10^{-4}$ T towards west (D) $5\pi \times 10^{-4}$ T towards east

Q.7 A uniform beam of positively charged particles is moving with a constant velocity parallel to another beam of negatively charged particles moving with the same velocity in opposite direction separated by a distance d. The variation of magnetic field B along a perpendicular line draw between the two beams is best represented by

(A) 

(B) 

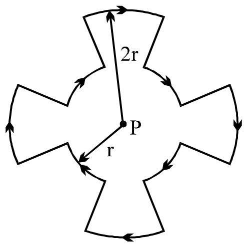
(C) 

(D) 

Q.8 The dimension of $\sqrt{\frac{\mu}{\in}}$ where μ is permeability & ε is permittivity is same as :

(A) Resistance (B) Inductance (C) Capacitance (D) None of these

Q.9 A current I flows around a closed path in the horizontal plane of the circle as shown in the figure. The path consists of eight arcs with alternating radii r and 2r. Each segment of arc subtends equal angle at the common centre P. The magnetic field produced by current path at point P is



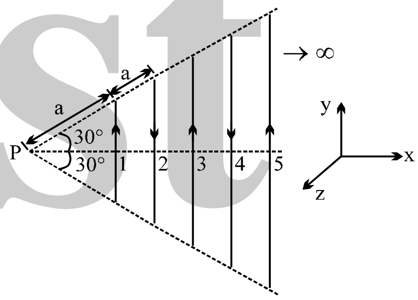
(A) $\frac{3}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed inward.

(B) $\frac{3}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed outward.

(C) $\frac{1}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed inward.

(D) $\frac{1}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed outward..

Q.10 Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is



(A) $\frac{\mu_0 I}{4\pi}\frac{\ln 2}{\sqrt{3}\,a}\hat{k}$ (B) $\frac{\mu_0 I}{4\pi}\frac{\ln 4}{\sqrt{3}\,a}\hat{k}$

(C) $\frac{\mu_0 I}{4\pi}\frac{\ln 4}{\sqrt{3}\,a}(-\hat{k})$ (D) Zero

Q.11 A direct current is passing through a wire. It is bent to form a coil of one turn. Now it is further bent to form a coil of two turns but at smaller radius. The ratio of the magnetic induction at the centre of this coil and at the centre of the coil of one turn is

(A) 1 : 4 (B) 4 : 1 (C) 2 : 1 (D) 1 : 1

Q.12 Two mutually perpendicular conductors carrying currents I_1 and I_2 lie in one plane. Locus of the point at which the magnetic induction is zero, is a

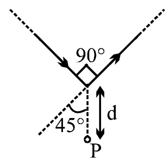
(A) circle with centre as the point of intersection of the conductor.

(B) parabola with vertex as the point of intersection of the conductors

(C) straight line passing through the point of intersection of the conductors.

(D) rectangular hyperbola

Q.13 Find the magnetic field at P due to the arrangement shown

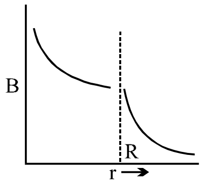


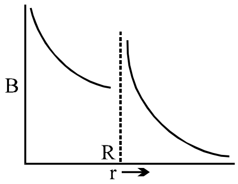
(A) $\frac{\mu_0 i}{\sqrt{2}\,\pi d}\left(1-\frac{1}{\sqrt{2}}\right)\otimes$ (B) $\frac{2\mu_0 i}{\sqrt{2}\,\pi d}\otimes$ (C) $\frac{\mu_0 i}{\sqrt{2}\,\pi d}\otimes$ (D) $\frac{\mu_0 i}{\sqrt{2}\,\pi d}\left(1+\frac{1}{\sqrt{2}}\right)\otimes$

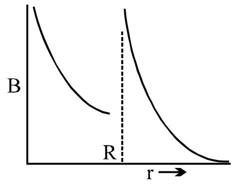
Q.14 Equal current *i* is flowing in three infinitely long wires along positive x, y and z directions. The magnitude field at a point (0, 0, –a) would be:

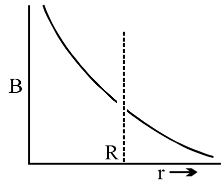
(A) $\frac{\mu_0 i}{2\pi a}(\hat{j}-\hat{i})$ (B) $\frac{\mu_0 i}{2\pi a}(\hat{i}+\hat{j})$ (C) $\frac{\mu_0 i}{2\pi a}(\hat{i}-\hat{j})$ (D) $\frac{\mu_0 i}{2\pi a}(\hat{i}+\hat{j}+\hat{k})$

Q.15 A thin, straight conductor lies along the axis of a hollow conductor of radius R. The two carry equal currents in the same direction. The magnetic field B is plotted against the distance r from the axis. Which of the following best represents the resulting curve?

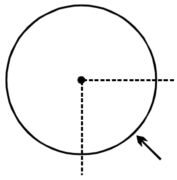
(A) 

(B) 

(C) 

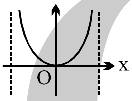
(D) 

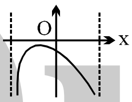
Q.16 A long thin walled pipe of radius R carries a current I along its length. The current density is uniform over the circumference of the pipe. The magnetic field at the center of the pipe due to quarter portion of the pipe shown, is

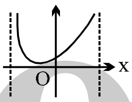


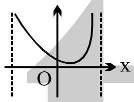
(A) $\frac{\mu_0 I\sqrt{2}}{4\pi^2 R}$ (B) $\frac{\mu_0 I}{\pi^2 R}$ (C) $\frac{2\mu_0 I\sqrt{2}}{\pi^2 R}$ (D) None

Q.17 Two very long straight parallel wires, parallel to y-axis, carry currents 4I and I, along +y direction and –y direction, respectively. The wires are passes through the x-axis at the points (d, 0, 0) and (– d, 0, 0) respectively. The graph of magnetic field z-component as one moves along the x-axis from x = – d to x = +d, is best given by

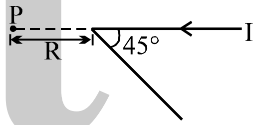
(A) 

(B) 

(C) 

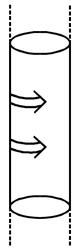
(D) 

Q.18 A long straight wire, carrying current I, is bent at its midpoint to from an angle of 45°. Induction of magnetic field at point P, distant R from point of bending is equal to :



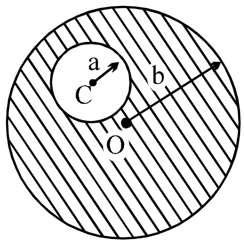
(A) $\frac{(\sqrt{2}-1)\mu_0 I}{4\pi R}$ (B) $\frac{(\sqrt{2}+1)\mu_0 I}{4\pi R}$ (C) $\frac{(\sqrt{2}-1)\mu_0 I}{4\sqrt{2}\pi R}$ (D) $\frac{(\sqrt{2}+1)\mu_0 I}{4\sqrt{2}\pi R}$

Q.19 A hollow cylinder having infinite length and carrying uniform current per unit length λ along the circumference as shown. Magnetic field inside the cylinder is



(A) $\frac{\mu_0 \lambda}{2}$ (B) $\mu_0\lambda$ (C) $2\mu_0\lambda$ (D) none

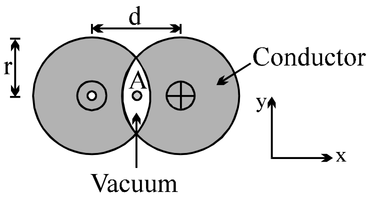
Q.20 A long straight metal rod has a very long hole of radius 'a' drilled parallel to the rod axis as shown in the figure. If the rod carries a current 'i' find
the value of magnetic induction on the axis of the hole, where OC = c



(A) $\frac{\mu_0 ic}{\pi(b^2 - a^2)}$ (B) $\frac{\mu_0 ic}{2\pi(b^2 - a^2)}$

(C) $\frac{\mu_0 i(b^2 - a^2)}{2\pi c}$ (D) $\frac{\mu_0 ic}{2\pi a^2 b^2}$

Q.21 Two long conductors are arranged as shown above to form overlapping cylinders, each of raidus r, whose centers are separated by a distance d. Current of density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point A?



(A) $(\mu_0/2\pi)\pi dJ$, in the +y-direction (B) $(\mu_0/2\pi)d^2/r$, in the +y-direction
(C) $(\mu_0/2\pi)4d^2J/r$, in the –y-direction (D) $(\mu_0/2\pi)Jr^2/d$, in the –y-direction
(E) There is no magnetic field at A.

Q.22 An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force of electron is zero
(A) positive z- axis (B) negative z-axis (C) positive y-axis (D) negative y-axis

Q.23 A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{i}$ with velocity $\vec{v} = v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time
(A) $t = \frac{2mv_0}{qE}$ (B) $t = \frac{2Bq}{mv_0}$ (C) $t = \frac{\sqrt{3}\,Bq}{mv_0}$ (D) $t = \frac{\sqrt{3}\,mv_0}{qE}$

Q.24 An electron is projected with velocity v_0 in a uniform electric field E perpendicular to the field. Again it is projetced with velocity v_0 perpendicular to a uniform magnetic field B/ If r_1 is initial radius of curvature just after entering in the electric field and r_2 is initial radius of curvature just after entering in magnetic field then the ratio r_1/r_2 is equal to
(A) $\frac{Bv_0^2}{E}$ (B) $\frac{B}{E}$ (C) $\frac{Ev_0}{B}$ (D) $\frac{Bv_0}{E}$

Q.25 A uniform magnetic field $\vec{B} = B_0\,\hat{j}$ exists in a space. A particle of mass m and charge q is projected towards negative x-axis with speed v from the a point (d, 0, 0). The maximum value v for which the particle does not hit y-z plane is
(A) $\frac{2Bq}{dm}$ (B) $\frac{Bqd}{m}$ (C) $\frac{Bq}{2dm}$ (D) $\frac{Bqd}{2m}$

Q.26 Two protons move parallel to each other, keeping distance r between them, both moving with same velocity $\vec{V}$. Then the ratio of the electric and magnetic force of interaction between them is
(A) c^2/V^2 (B) $2c^2/V^2$ (C) $c^2/2V^2$ (D) None

Q.27 A charged particle of specific charge α is released from origin at time t = 0 with velocity $\vec{V} = V_o\hat{i} + V_o\hat{j}$ in magnetic field $\vec{B} = B_o\hat{i}$. The coordinates of the particle at time $t = \frac{\pi}{B_o\alpha}$ are (specific charge α= q/m)
(A) $\left(\frac{V_o}{2B_o\alpha}, \frac{\sqrt{2}V_o}{\alpha B_o}, \frac{-V_o}{B_o\alpha}\right)$ (B) $\left(\frac{-V_o}{2B_o\alpha}, 0, 0\right)$
(C) $\left(0, \frac{2V_o}{B_o\alpha}, \frac{V_o\pi}{2B_o\alpha}\right)$ (D) $\left(\frac{V_o\pi}{B_o\alpha}, 0, -\frac{2V_o}{B_o\alpha},\right)$

Q.28 Three ions H^+, He^+ and O^{+2} having same kinetic energy pass through a region in which there is a uniform magnetic field perpendicular to their velocity, then :
(A) H^+ will be least deflected. (B) He^+ and O^{+2} will be deflected equally.
(C) O^{+2} will be deflected most. (D) all will be deflected equally.

Q.29 An electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction $\vec{B}$. If kinetic energy is doubled and magnetic induction tripled, the radius will become
(A) $\frac{3R}{2}$ (B) $\sqrt{\frac{3}{2}}\,R$ (C) $\sqrt{\frac{2}{9}}\,R$ (D) $\sqrt{\frac{4}{3}}\,R$

Q.30 An electron (mass $= 9.1 \times 10^{-31}$; charge $= -1.6 \times 10^{-19}$C) experiences no deflection if subjected to an electric field of 3.2×10^5 V/m and a magnetic field of 2.0×10^{-3} Wb/m^2. Both the fields are normal to the path of electron and to each other . If the electric field is removed, then the electron will revolve in an orbit of radius :
(A) 45 m (B) 4.5 m (C) 0.45 m (D) 0.045 m

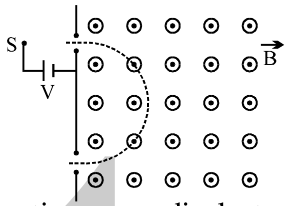
Q.31 A charged particle moves in a magnetic field $\vec{B} = 10\hat{i}$ with initial velocity $\vec{u} = 5\hat{i} + 4\hat{j}$. The path of the particle will be

(A) straight line (B) circle (C) helical (D) none

Q.32 A electron experiences a force $(4.0\hat{i} + 3.0\hat{j}) \times 10^{-13}$ N in a uniform magnetic field when its velocity is $2.5\hat{k} \times 10^{7}$ ms^{-1}. When the velocity is redirected and becomes $(1.5\hat{i} - 2.0\hat{j}) \times 10^{7}$ ms^{-1}, the magnetic force of the electron is zero. The magnetic field vector $\vec{B}$ is :

(A) $-0.075\hat{i} + 0.1\hat{j}$ (B) $0.1\hat{i} + 0.075\hat{j}$ (C) $0.075\hat{i} - 0.1\hat{j} + \hat{k}$ (D) $0.075\hat{i} - 0.1\hat{j}$

Q.33 A mass spectrometer is a device which select particle of equal mass. An iron with electric charge $q > 0$ and mass m starts at rest from a source S and is accelerated through a potential difference V. It passes through a hole into a region of constant magnetic field $\vec{B}$ perpendicular to the plane of the paper as shown in the figure. The particle is deflected by the magnetic field and emerges through the bottom hole at a distance d from the top hole. The mass of the particle is



(A) $\dfrac{qBd}{mV}$ (B) $\dfrac{qB^2d^2}{4V}$ (C) $\dfrac{qB^2d^2}{8V}$ (D) $\dfrac{qBd}{2mV}$

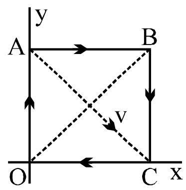
Q.34 Electrons moving with different speeds enter a uniform magnetic field in a direction perpendicular to the field. They will move along circular paths.

(A) of same radius
(B) with larger radii for the faster electrons
(C) with smaller radii for the faster electrons
(D) either (B) or (C) depending on the magnitude of the magnetic field

Q.35 In the previous question, time periods of rotation will be :

(A) same for all electrons
(B) greater for the faster electrons
(C) smaller for the faster electrons
(D) either (B) or (C) depending on the magnitude of the magnetic field

Q.36 OABC is a current carrying square loop an electron is projected from the centre of loop along its diagonal AC as shown. Unit vector in the direction of initial acceleration will be



(A) $\hat{k}$ (B) $-\left(\dfrac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(C) $-\hat{k}$ (D) $\dfrac{\hat{i} + \hat{j}}{\sqrt{2}}$

Q.37 A particle having charge of 1 C, mass 1 kg and speed 1 m/s enters a uniform magnetic field, having magnetic induction of 1 T, at an angle $\theta = 30°$ between velocity vector and magnetic induction. The pitch of its helical path is (in meters)

(A) $\dfrac{\sqrt{3}\pi}{2}$ (B) $\sqrt{3}\pi$ (C) $\dfrac{\pi}{2}$ (D) π

Q.38 A charged particle is released from rest in a region of uniform electric and magnetic fields, which are parallel to each other. The locus of the particle will be

(A) helix of constant pitch (B) straight line
(C) helix of varying pitch (D) cycloid

Q.39 A particle of specific charge (charge/mass) α starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{k}$. Its velocity at $(x_0, y_0, 0)$ is $(4\hat{i} + 3\hat{j})$. The value of x_0 is:

(A) $\frac{13}{2}\frac{\alpha E_0}{B_0}$ (B) $\frac{16\alpha B_0}{E_0}$ (C) $\frac{25}{2\alpha E_0}$ (D) $\frac{5\alpha}{2B_0}$

Q.40 A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity [$u\mathbf{i} - v\mathbf{j}$]. Uniform electric magnetic fields exist in the region along the +y direction, of magnitude E and B. The particle will definitely return to the origin once if

(A) $[vB/2\pi E]$ is an integer (B) $(u^2 + v^2)^{1/2}$ $[B/\pi E]$ is an integer

(C) $[vB/\pi E]$ in an integer (D) $[uB/\pi E]$ is an integer

Q.41 An electron moving with a velocity $\vec{V}_1 = 2\hat{i}$ m/s at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}$N. If the electron is moving with a velocity $\vec{V}_2 = 2\hat{j}$ m/s at the same point, it experiences a force $\vec{F}_2 = +2\hat{i}$ N. The force the electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}$ m/s at the same point is

(A) zero (B) $2\hat{k}$N (C) $-2\hat{k}$N (D) information is insufficient

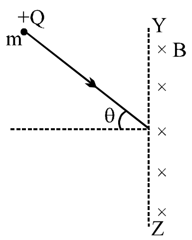
Q.42 Two particles of charges +Q and –Q are projected from the same point with a velocity v in a region of uniform magnetic field B such that the velocity vector makes an angle q with the magnetic field. Their masses are M and 2M, respectively. Then, they will meet again for the first time at a point whose distance from the point of projection is

(A) $2\pi Mv\cos\theta/QB$ (B) $8\pi Mv\cos\theta/QB$ (C) $\pi Mv\cos\theta/QB$ (D) $4\pi Mv\cos\theta/QB$

Q.43 A particle of charge Q and mass M moves in a circular path of radius R in a uniform magnetic field of magnitude B. The same particle now moves with the same speed in a circular path of same radius R in the space between the cylindrical electrodes of the cylindrical capacitor. The radius of the inner electrode is R/2 while that of the outer electrode is 3R/2. Then the potential difference between the capacitor electrodes must be

(A) $QBR(ln3)/M$ (B) $QB^2R^2(ln3)/2M$ (C) $QB^2R^2(ln3)/M$ (D) None

Q.44 A particle with charge +Q and mass m enters a magnetic field of magnitude B, existing only to the right of the boundary YZ. The direction of the motion of the particle is perpendicular to the direction of B. Let $T = 2\pi\frac{m}{QB}$. The time spent by the particle in the field will be

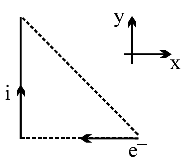


(A) $T\theta$ (B) $2T\theta$ (C) $T\left(\frac{\pi+2\theta}{2\pi}\right)$ (D) $T\left(\frac{\pi-2\theta}{2\pi}\right)$

Q.45 In the previous question, if the particle has –Q charge, the time spend by the particle in the field will be

(A) $T\theta$ (B) $2T\theta$ (C) $T\left(\frac{\pi+2\theta}{2\pi}\right)$ (D) $T\left(\frac{\pi-2\theta}{2\pi}\right)$

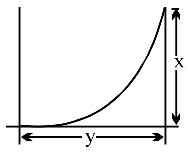
Q.46 The direction of magnetic force on the electron as shown in the diagram is along



(A) y-axis (B) –y-axis

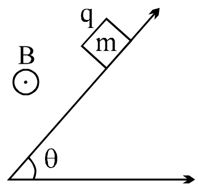
(C) z-axis (D) –z-axis

Q.47 A particle having charge q enters a region of uniform magnetic field $\vec{B}$ (directed inwards) and is deflected a distance x after travelling a distance y. The magnitude of the momentum of the particle is:



(A) $\frac{qBy}{2}$ (B) $\frac{qBy}{x}$ (C) $\frac{qB}{2}\left(\frac{y^2}{x}+x\right)$ (D) $\frac{qBy^2}{2x}$

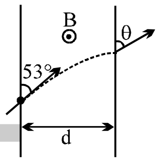
Q.48 A block of mass m & charge q is released on a long smooth inclined plane magnetic field B is constant, uniform, horizontal and parallel to surface as shown. Find the time from start when block loses contact with the surface.



(A) $\frac{m\cos\theta}{qB}$ (B) $\frac{m\,\mathrm{cosec}\,\theta}{qB}$

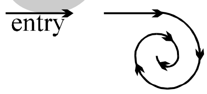
(C) $\frac{m\cot\theta}{qB}$ (D) none

Q.49 A particle moving with velocity v having specific charge (q/m) enters a region of magnetic field B having width $d=\frac{3mv}{5qB}$ at angle 53° to the boundary of magnetic field. Find the angle θ in the diagram.



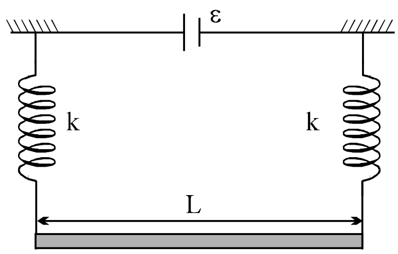
(A) 37° (B) 60° (C) 90° (D) none

Q.50 A charged particle enters a uniform magnetic field perpendicular to its initial direction travelling in air. The path of the particle is seen to follow the path in figure. Which of statements 1–3 is/are correct?
[1] The magnetic field strength may have been increased while the particle was travelling in air
[2] The particle lost energy by ionising the air
[3] The particle lost charge by ionising the air



(A) 1, 2, 3 are correct (B) 1, 2 only are correct
(C) 2, 3 only are correct (D) 1 only

Q.51 A straight rod of mass m and length L is suspended from the identical spring as shown in the figure. The spring stretched by a distance of x_0 due to the weight of the wire. The circuit has total resistance RΩ. When the magnetic field perpendicular to the plane of the paper is switched on, springs are observed to extend further by the same distance. The magnetic field strength is



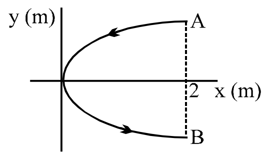
(A) $\frac{mgR}{\varepsilon L}$; directed outward from the plane of the paper

(B) $\frac{mgR}{2\varepsilon x_0}$; directed outward from the plane of the paper

(C) $\frac{mgR}{\varepsilon L}$; directed into the plane of the paper

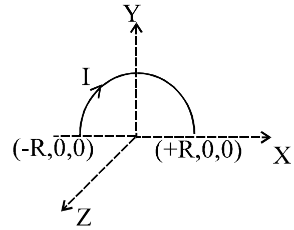
(D) $\frac{mgR}{\varepsilon x_0}$; directed into the plane of the paper

Q.52 A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current $i = 2$ A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ Tesla. The magnetic force on the wire is (in newton)



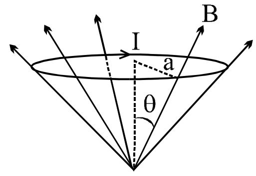
(A) $-16\hat{i}$ (B) $32\hat{i}$ (C) $-32\hat{i}$ (D) $16\hat{i}$

Q.53 A semi circular current carrying wire having radius R is placed in x-y plane with its centre at origin 'O'. There is non-uniform magnetic field $\vec{B} = \frac{B_o x}{2R}\hat{k}$ (here B_o is +ve constant) is existing in the region. The magnetic force acting on semi circular wire will be along



(A) – x-axis
(B) + y-axis
(C) – y-axis
(D) + x-axis

Q.54 A circular current loop of radius a is placed in a radial field B as shown. The net force acting on the loop is

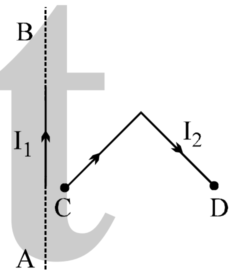


(A) zero
(B) $2\pi B a I\cos\theta$
(C) $2\pi a IB\sin\theta$
(D) None

Q.55 A conductor of length l and mass m is placed along the east-west line on a table. Suddenly a certain amount of charge is passed through it and it is found to jump to a height h. The earth's magnetic induction is B. The charge passed through the conductor is:

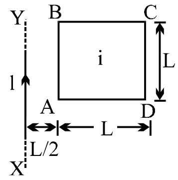
(A) $\frac{1}{Bmgh}$
(B) $\frac{\sqrt{2gh}}{Blm}$
(C) $\frac{gh}{Blm}$
(D) $\frac{m\sqrt{2gh}}{Bl}$

Q.56 In the figure shown a current I_1 is established in the long straight wire AB. Another wire CD carrying current I_2 is placed in the plane of the paper. The line joining the ends of this wire is perpendicular to the wire AB. The force on the wire CD is:



(A) zero
(B) towards left
(C) directed upwards
(D) none of these

Q.57 A square loop ABCD, carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be

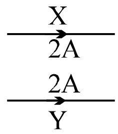


(A) $\frac{2\mu_0 Ii}{3\pi}$
(B) $\frac{\mu_0 Ii}{2\pi}$
(C) $\frac{2\mu_0 Iil}{3\pi}$
(D) $\frac{\mu_0 Iil}{2\pi}$

Q.58 A metal ring of radius r = 0.5 m with its plane normal to a uniform magnetic field B of induction 0.2 T carries a current I = 100 A. The tension in newtons developed in the ring is:

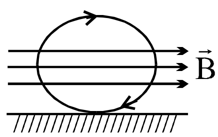
(A) 100
(B) 50
(C) 25
(D) 10

Q.59 In given figure, X and Y are two long straight parallel conductors each carrying a current of 2 A. The force on each conductor is F newtons. When the current in each is changed to 1 A and reversed in direction, the force on each is now



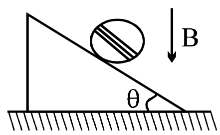
(A) F/4 and unchanged in direction
(B) F/2 and reversed in direction
(C) F/2 and unchanged in direction
(D) F/4 and reversed in direction

Q.60 A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current i = 4A. A horizontal magnetic field B = 10T is switched on at time t = 0 as shown in figure. The initial angular acceleration of the ring will be



(A) 40 π rad/s²
(B) 20 π rad/s²
(C) 5 π rad/s²
(D) 15 π rad/s²

Q.61 In the figure shown a coil of single turn is wound on a sphere of radius R and mass m. The plane of the coil is parallel to the plane and lies in the equatorial plane of the sphere. Current in the coil is i. The value of B if the sphere is in equilibrium is



(A) $\frac{mg\cos\theta}{\pi iR}$
(B) $\frac{mg}{\pi iR}$
(C) $\frac{mg\tan\theta}{\pi iR}$
(D) $\frac{mg\sin\theta}{\pi iR}$

Q.62 The magnetic moment of a circular orbit of radius 'r' carrying a charge 'q' and rotating with velocity v is given by

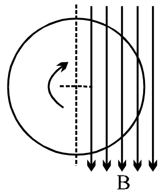
(A) $\frac{qvr}{2\pi}$ (B) $\frac{qvr}{2}$ (C) $qv\pi r$ (D) $qv\pi r^2$

Q.63 The dimensional formula for the physical quantity $\frac{E^2\mu_0\varepsilon_0}{B^2}$ is

(E = electric field and B = magnetic field)

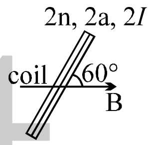
(A) $L^0M^0T^0$ (B) $L^1M^0T^{-1}$ (C) $L^{-1}M^0T^1$ (D) $L^{1/2}M^0T^{-1/2}$

Q.64 A thin non conducting disc of radius R is rotating clockwise (see figure) with an angular velocity w about its central axis, which is perpendicular to its plane. Both its surfaces carry +ve charges of uniform surface density. Half the disc is in a region of a uniform, unidirectional magnetic field B parallel to the plane of the disc, as shown. Then,



(A) The net torque on the disc is zero.
(B) The net torque vector on the disc is directed leftwards.
(C) The net torque vector on the disc is directed rightwards.
(D) The net torque vector on the disc is parallel to B.

Q.65 A rectangular coil PQ has 2n turns, an area 2a and carries a current $2I$, (refer figure). The plane of the coil is at 60° to a horizontal uniform magnetic field of flux density B. The torque on the coil due to magnetic force is

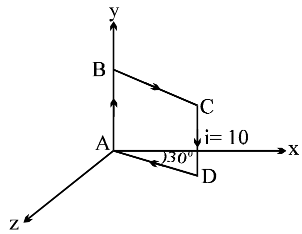


(A) $BnaI \sin60°$ (B) $8BnaI \cos60°$ (C) $4naI B\sin60°$ (D) none

Q.66 A straight current carrying conductor is placed in such a way that the current in the conductor flows in the direction out of the plane of the paper. The conductor is placed between two poles of two magnets, as shown. The conductor will experience a force in the direction towards

(A) P (B) Q (C) R (D) S

Q.67 Figure shows a square current carrying loop ABCD of side 10 cm and current i = 10A. The magnetic moment $\vec{M}$ of the loop is



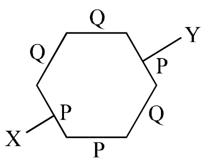
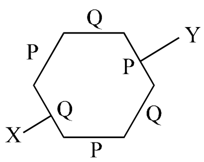
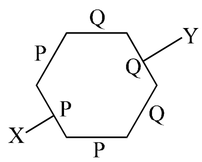
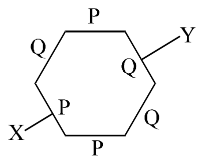
(A) $(0.05)\left(\hat{i}-\sqrt{3}\hat{k}\right)A-m^2$ (B) $(0.05)\left(\hat{j}+\hat{k}\right)A-m^2$

(C) $(0.05)\left(\sqrt{3}\hat{i}+\hat{k}\right)A-m^2$ (D) $\left(\hat{i}+\hat{k}\right)A-m^2$

<u>ONE OR MORE THAN ONE OPTION MAY BE CORRECT</u>

Take approx. 3 minutes for answering each question.

Q.1 In the following hexagons, made up of two different material P and Q, current enters and leaves from points X and Y respectively. In which case the magnetic field at its centre is not zero.

(A)  (B)  (C)  (D) 

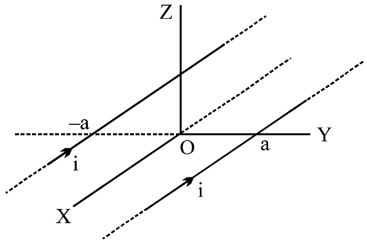
Q.2 Consider the magnetic field produced by a finitely long current carrying wire.

(A) the lines of field will be concentric circles with centres on the wire.
(B) There can be two points in the same plane where magnetic fields are same.
(C) There can be large number of points where the magnetic field is same.
(D) The magnetic field at a point is inversally proportional to the distance of the point from the wire.

Q.3 Consider three quantities $x = E/B$, $y = \sqrt{1/\mu_0\varepsilon_0}$ and $z = \dfrac{l}{CR}$. Here, l is the length of a wire, C is a capacitance and R is a resistance. All other symbols have standard meanings.

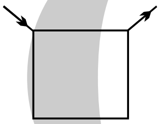
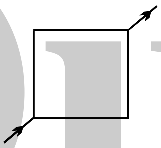
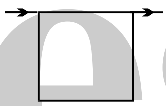
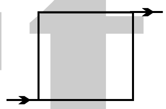
(A) x, y have the same dimensions
(B) y, z have the same dimensions
(C) z, x have the same dimensions
(D) none of the three pairs have the same dimensions.

Q.4 Two long thin, parallel conductors carrying equal currents in the same direction are fixed parallel to the x-axis, one passing through $y = a$ and the other through $y = -a$. The resultant magnetic field due to the two conductors at any point is B. Which of the following are correct?

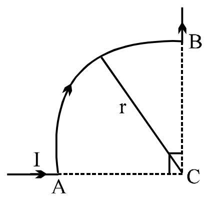


(A) B = 0 for all points on the x-axis
(B) At all points on the y-axis, excluding the origin, B has only a z-component.
(C) At all points on the z-axis, excluding the origin, B has only a y-component.
(D) B cannot have an x-component.

Q.5 Current flows through uniform, square frames as shown. In which case is the magnetic field at the centre of the frame not zero?

(A)  (B)  (C)  (D) 

Q.6 A wire carrying I is shaped as shown. Section AB is a quarter circle of radius r. The magnetic field at C is directed



(A) along the bisector of the angle ACB, away from AB
(B) along the bisector of the angle ACB, towards AB
(C) perpendicular to the plane of the paper, directed into the paper
(D) at an angle $\pi/4$ to the plane of the paper

Q.7 A long straight wire carries a current along the x-axis. Consider the points A(0, 1, 0), B(0, 1, 1), C(1, 0, 1) and D(1, 1, 1). Which of the following pairs of points will have magnetic fields of the same magnitude?

(A) A and B
(B) A and C
(C) B and C
(D) B and D

Q.8 In the previous question, if the current is i and the magnetic field at D has magnitude B,

(A) $B = \dfrac{\mu_0 i}{2\sqrt{2}\pi}$
(B) $B = \dfrac{\mu_0 i}{2\sqrt{3}\pi}$
(C) B is parallel to the x-axis
(D) B makes an angle of 45° with the xy plane

Q.9 Which of the following statement is correct :

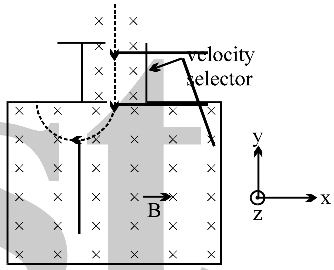
(A) A charged particle enters a region of uniform magnetic field at an angle 85^0 to magnetic lines of force. The path of the particle is a circle.
(B) An electron and proton are moving with the same kinetic energy along the same direction. When they pass through uniform magnetic field perpendicular to their direction of motion, they describe circular path.
(C) There is no change in the energy of a charged particle moving in a magnetic field although magnetic force acts on it.
(D) Two electrons enter with the same speed but in opposite direction in a uniform transverse magnetic field. Then the two describe circle of the same radius and these move in the same direction.

Q.10 Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a, b and c be the ratio of their time periods, radii and pitches of the helical paths than
(A) abc = 1 (B) abc > 1 (C) abc < 1 (D) a = bc

Q.11 Consider the following statements regarding a charged particle in a magnetic field . Which of the statements are true :
(A) Starting with zero velocity, it accelerates in a direction perpendicular to the magnetic field.
(B) While deflecting in magnetic field its energy gradually increases .
(C) Only the component of magnetic field perpendicular to the direction of motion of the charged particle is effective in deflecting it.
(D) Direction of deflecting force on the moving charged particle is perpendicular to its velocity.

Q.12 A particle of charge q and velocity v passes undeflected through a space with non-zero electric field E and magnetic field B. The undeflecting conditions will hold if.
(A) signs of both q and E are reversed.
(B) signs of both q and B are reversed.
(C) both B and E are changed in magnitude, but keeping the product of |B| and |E| fixed.
(D) both B and E are doubled in magnitude.

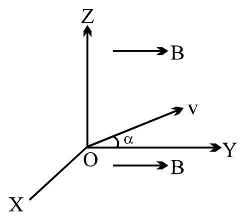
Q.13 Two charged particle A and B each of charge +e and masses 12 amu and 13 amu respectively follow a circular trajectory in chamber X after the velocity selector as shown in the figure. Both particles enter the velocity selector with speed 1.5×10^6 ms^{-1}. A uniform magnetic field of strength 1.0 T is maintained within the chamber X and in the velocity selector.



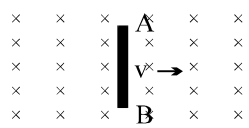
(A) Electric field across the conducting plate of the velocity selector is -10^6 NC^{-1} $\hat{i}$.
(B) Electric field across the conducting plate of the velocity selector is 10^6 NC^{-1} $\hat{i}$.
(C) The ratio r_A/r_B of the radii of the circular paths for the two particles is $12/13$.
(D) The ratio r_A/r_B of the radii of the circular paths for the two particles is $13/12$.

Q.14 An electron is moving along the positive X-axis. You want to apply a magnetic field for a short time so that the electron may reverse its direction and move parallel to the negative X–axis. This can be done by applying the magnetic field along
(A) Y-axis (B) Z-axis (C) Y-axis only (D) Z-axis only

Q.15 In a region of space, a uniform magnetic field B exists in the y-direction. A proton is fired from the origin, with its initial velocity v making a small angle α with the y-direction in the yz plane. In the subsequent motion of the proton,
(A) its x-coordinate can never be positive
(B) its x- and z-coordinates cannot both be zero at the same time
(C) its z-coordinate can never be negative
(D) its y-coordinate will be proportional to the square of its time of flight



Q.16 A rod AB moves with a uniform velocity v in a uniform magnetic field as shown in figure.
(A) The rod becomes electrically charged.
(B) The end A becomes positively charged.
(C) The end B becomes positively charged.
(D) The rod becomes hot because of Joule heating.

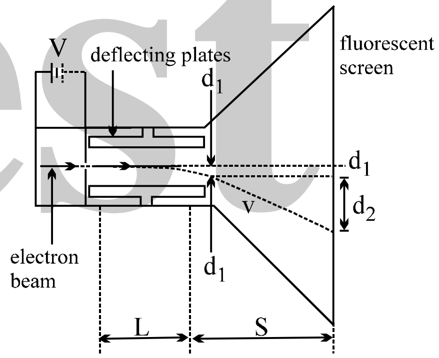


Question No. 17 to 21 (5 questions)

The following experiment was performed by J.J.Thomson in order to measure the ratio of the charge e to the mass m of an electron. Figure shows a modern version of Thomson's apparatus. Electrons emitted from a hot filament are accelerated by a potential difference V. As the electrons pass through the deflector plates, they encounter both electric and magnetic fields. When the electrons leave the plates they enter a field-free region that extends to the fluorescent screen. The beam of electrons can be observed as a spot of light on the screen. The entire region in which the electrons travel is evacuated with a vacuum pump.

Thomson's procedure was to first set both the electric and magnetic fields to zero, note the position of the undeflected electron beam on the screen, then turn on only the electric field and measure the resulting deflection. The deflection of an electron in an electric field of magnitude E is given by $d_1 = eEL^2/2mv^2$, where L is the length of the deflecting plates, and v is the speed of the electron. The deflection d_1 can also be calculated from the total deflection of the spot on the screen, $d_1 + d_2$ and the geometry of the apparatus. In the second part of the experiment, Thomson adjusted the magnetic field so as to exactly cancel the force applied by the electric field, leaving the electron beam undeflected. This gives eE = evB. By combining this relation with the expression for d_1, one can calculate the charge to mass ratio of the electron as a function of the known quantities. The result is:

$$\frac{e}{m} = \frac{2d_1E}{B^2L^2}$$



Q.17 Why was it important for Thomson to evacuate the air from the apparatus?

(A) Electrons travel faster in a vacuum, making the deflection d_1 smaller.

(B) Electromagnetic waves propagate in a vacuum.

(C) The electron collisions with the air molecules cause them to be scattered, and a focused beam will not be produced.

(D) It was not important and could have been avoided.

Q.18 One might have considered a different experiment in which no magnetic field is needed. The ratio e/m can then be calculated directly from the expression for d_1. Why might Thomson have introduced the magnetic field B in his experiment?

(A) To verify the correctness of the equation for the magnetic force.

(B) To avoid having to measure the electron speed v.

(C) To cancel unwanted effects of the electric field E.

(D) To make sure that the electric field does not exert a force on the electron.

Q.19 If the electron speed were doubled by increasing the potential difference V, which of the following would have to be true in order to correctly measure e/m?

(A) The magnetic field would have to be cut in half in order to cancel the force applied by the electric field.

(B) The magnetic field would have to be doubled in order to cancel the force applied by the electric field.

(C) The length of the plates, L, would have to be doubled to keep the deflection, d_1, from changing.

(D) Nothing needs to be changed.

Q.20 The potential difference V, which accelerates the electrons, also creates an electric field. Why did Thomson NOT consider the deflection caused this electric field in his experiment?

(A) This electric field is much weaker than the one between the deflecting plates and can be neglected.

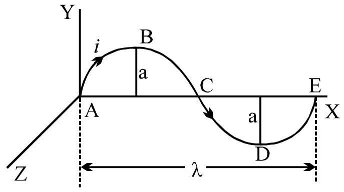
(B) Only the deflection, $d_1 + d_2$ caused by the deflecting plates is measured in the experiment.

(C) There is no deflection from this electric field

(D) The magnetic field cancels the force caused by this electric field.

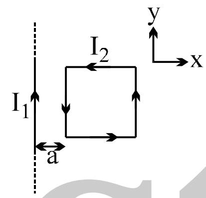
Q.21 If the electron is deflected downward when only the electric field is turned on (as shown in figure) then in what directions do the electric and magnetic fields point in the second part of the experiment?
(A) The electric field points to the bottom, while the magnetic field points into the page.
(B) The electric field points to the bottom, while the magnetic field points out of the page.
(C) The electric field points to the top, while the magnetic field points into the page.
(D) The electric field points to the top, while the magnetic field points out of the page.

Q.22 A conductor ABCDE, shaped as shown, carries a current i. It is placed in the xy plane with the ends A and E on the x-axis. A uniform magnetic field of magnitude B exists in the region. The force acting on it will be



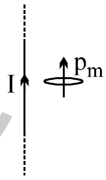
(A) zero, if B is in the x-direction
(B) $\lambda B i$ in the z-direction, if B is in the y-direction
(C) $\lambda B i$ in the negative y-direction, if B is in the z-direction
(D) $2aBi$, if B is in the x-direction

Q.23 A square loop of side ℓ is placed in the neighbourhood of an infinitely long straight wire carrying a current I_1. The loop carries a current I_2 as shown in figure



(A) The magnetic moment of the loop is $\vec{p}_m = l^2 I_2 \hat{k}$
(B) The magnetic moment of the loop is $\vec{p}_m = -l^2 I_2 \hat{k}$
(C) The potential energy of the loop is minimum
(D) The torque experienced by the loop is maximum

Q.24 The magnetic dipole $\vec{p}_m$ is placed parallel to an infinitely long straight wire as shown in figure



(A) the potential energy of the dipole is minimum
(B) the torque acting on the dipole is zero
(C) the force acting on the dipole is zero
(D) none of these

ANSWER KEY

ONLY ONE OPTION IS CORRECT.

Q.1	D	Q.2	C	Q.3	C	Q.4	A	Q.5	A	Q.6	A	Q.7	D
Q.8	A	Q.9	A	Q.10	B	Q.11	B	Q.12	C	Q.13	A	Q.14	A
Q.15	B	Q.16	A	Q.17	C	Q.18	A	Q.19	B	Q.20	B	Q.21	A
Q.22	B	Q.23	D	Q.24	D	Q.25	B	Q.26	A	Q.27	D	Q.28	B
Q.29	C	Q.30	C	Q.31	C	Q.32	A	Q.33	C	Q.34	B	Q.35	A
Q.36	B	Q.37	B	Q.38	B	Q.39	C	Q.40	C	Q.41	A	Q.42	D
Q.43	C	Q.44	C	Q.45	D	Q.46	A	Q.47	C	Q.48	C	Q.49	C
Q.50	B	Q.51	A	Q.52	B	Q.53	A	Q.54	C	Q.55	D	Q.56	D
Q.57	A	Q.58	D	Q.59	A	Q.60	A	Q.61	B	Q.62	B	Q.63	A
Q.64	B	Q.65	B	Q.66	B	Q.67	A						

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	A	Q.2	A,B,C	Q.3	A,B,C	Q.4	A,B,C,D
Q.5	C	Q.6	C	Q.7	B,D	Q.8	A,D
Q.9	B,C	Q.10	A,D	Q.11	C,D	Q.12	D
Q.13	C	Q.14	A,B	Q.15	A	Q.16	B
Q.17	C	Q.18	B	Q.19	A	Q.20	C
Q.21	D	Q.22	A,B,C	Q.23	A	Q.24	C

TARGET IIT JEE

PHYSICS

ELECTROMAGNETIC INDUCTION
&
ALTERNATING CURRENT

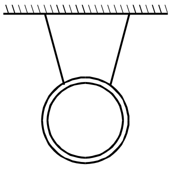
QUESTION FOR SHORT ANSWER

Q.1 Are induced emfs and currents different in any way from emfs and currents provided by a battery connected to a conducting loop?

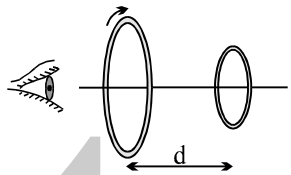
Q.2 Can a charged particle at rest be set in motion by the action of a magnetic field? If not, why not? If so, how? Consider both static and time-varying fields.

Q.3 In Faraday's law of induction, does the induced emf depend on the resistance of the circuit? If so, how?

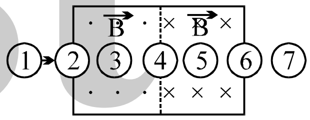
Q.4 Figure shows a copper ring that is hung from a ceiling by two threads. Describe in detail how you might most effectively use a bar magnet to get this ring to swing back and forth.



Q.5 Two conducting loops face each other a distance d apart, as shown in figure. An observer sights along their common axis from left to right. A clockwise current i is suddenly established in the larger loop by a battery not shown. (a) What is the direction of the induced current in the smaller loop? (b) What is the direction of the force (if any) that acts on the smaller loop?



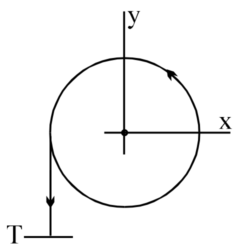
Q.6 A circular loop moves with constant velocity through regions where uniform magnetic fields of the same magnitude are directed into or out of the plane of the page, as indicated in figure. At which of the seven indicated positions will the induced current be (a) clockwise, (b) counterclockwise, and (c) zero?



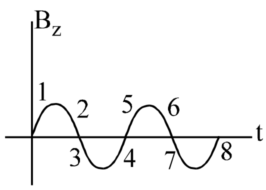
Q.7 Can an induced current ever establish a magnetic field $\vec{B}$ that is in the same direction as the magnetic field inducing the current? Justify your answer.

Q.8 A plane closed loop is placed in a uniform magnetic field. In what ways can the loop be moved without inducing an emf? Consider motions both of translation and rotation.

Q.9 Figure (a) shows a top view of the electron orbit in a betatron. Electrons are accelerated in a circular orbit in the xy plane and then withdrawn to strike the target T. The magnetic field $\vec{B}$ is along the z axis (the positive z axis is out of the page). The magnetic field B_z along this axis varies sinusoidally as shown in figure (b). Recall that the magnetic field must (i) guide the electrons in their circular path and (ii) generate the electric field that accelerates the electrons. Which quarter cycle(s) in figure are suitable (a) according to (i), (b) according to (ii), and (c) for operation of the betatron?



(a)



(b)

Q.10

(i) A piece of metal and a piece of non-metallic stone are dropped from the same height near the surface of the earth. Which one will reach the ground earlier?

(ii) A metallic loop is placed in a nonuniform magnetic field. will an emf be induced in the loop ?

(iii) A wire loop is held with its plane horizontal. A magnet with its north pole downward is allowed to fall through it from some height. Will the magnet fall with constant acceleration? What will happen if the poles are reversed?

(iv) A magnet is dropped down into long vertically copper tube . Show that, even neglecting air resistance the magnet will reach a constant terminal velocity .

(v) A magnet is dropped from the ceiling along the axis of a copper loop lying flat on the floor. If the falling magnet is photographed with a time sequence camera, what differences, if any will be noted if,
(i) the loop is at room temperature (ii) the loop is packed in dry ice ?

Q.11 A copper ring is suspended in a vertical plane by a thread. A steel bar is passed through the ring in the horizontal direction which is perpendicular to the plane of the loop. Then a magnet is similarly passed through the loop. Will the motion of the magnet and the bar affect the position of the ring?

Q.12 If the magnetic field outside a copper box is suddenly changed, what happens to the magnetic field inside the box ? Such low-resistivity metals are used to form enclosures which shield objects inside them against varying magnetic fields.

Q.13 Metallic (nonferromagnetic) and nonmetallic particles in a solid waste may be separated as follows. The waste is allowed to slide down an incline over permanent magnets. The metallic particles slow down as compared to the nonmetallic ones and hence are separated. Discuss the role of eddy currents in the process.

Q.14 A jet plane is flying due north . A potential difference is produced between he wing tips of the plane. Will a passenger sitting inside the plane also expect some emf between the wing tips? Will a tiny bulb connected to the wing tips glow?

Q.15 Is the inductance per unit length for a solenoid near its centre ;
(a) the same as (b) less than or (c) greater than
the inductance per unit length near its ends ?

Q.16 Two solenoids A & B have the same diameter & length & contain only one layer of windings, with adjacent turns touching, insulation thickness being negligible . Solenoid A contains many turns of fine wire & solenoid B contains fewer turns of heavier wire.
(i) which solenoid has the larger inductance ?
(i) which solenoid has the larger inductive time constant ? (material is same)

Q.17 If the flux passing through each turn of a coil is the same, the inductance of the coil may be computed from $L = \frac{N\phi_B}{i}$. How might one compute L for a coil for which this assumption is not valid .

Q.18 If a current in a source of emf is in the direction of the emf, the energy of the source decreases, if a current is in a direction opposite to the emf (as in charging a battery), the energy of the source increases . Do these statements apply to the inductor .

Q.19 Does the time required for the current in particular LR circuit to build up to any given fraction of its equilibrium value depend on the value of the applied emf .

Q.20 A steady current is set up in a coil with a very large inductive time constant . When the current is interupted with a switch a heavy arc tends to appear at the switch blades . Explain? [Note : interrupting currents in highly inductive circuits can be dangerous]

Q.21 What is the advantage of placing the two electric wires carrying ac close together?

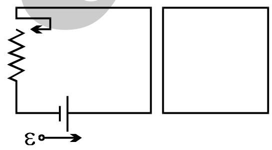
Q.22 In an LR series circuit the self induced emf is a maximum at the instant the switch is closed. How can this be since there is no current in the inductance at this instant .

Q.23 Explain what is meant by the statement "A motor acts as a motor and generator at the same time." Can the same be said for a generator?

Q.24 In a toroid, is the energy density larger near the inner radius or near the outer radius ?

Q.25 Two circular loops are placed with their centres separated by a fixed distance. How would you orient the loops to have (a) the largest mutual inductance (b) the smallest mutual inductance ?

Q.26 If the resistance R in the left hand circuit of figure is increased, what is the direction of the induced current in the right hand circuit ?



ONLY ONE OPTION IS CORRECT.

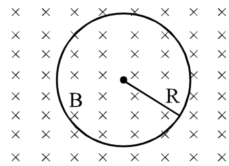
Take approx. 2 minutes for answering each question.

Q.1 An electron is moving in a circular orbit of radius R with an angular acceleration α. At the centre of the orbit is kept a conducting loop of radius r, (r <<R). The e.m.f induced in the smaller loop due to the motion of the electron is

(A) zero, since charge on electron in constant (B) $\frac{\mu_0 e r^2}{4R}\alpha$

(C) $\frac{\mu_0 e r^2}{4\pi R}\alpha$ (D) none of these

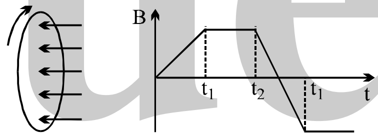
Q.2 a conducting loop of radius R is present in a uniform magnetic field B perpendicular the plane of the ring. If radius R varies as a function of time 't', as $R = R_0 + t$. The e.m.f induced in the loop is

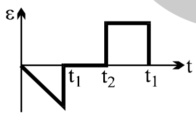


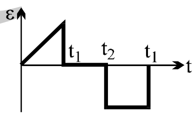
(A) $2\pi(R_0 + t)B$ clockwise (B) $\pi(R_0 + t)B$ clockwise

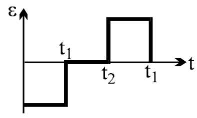
(C) $2\pi(R_0 + t)B$ anticlockwise (D) zero

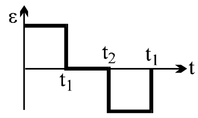
Q.3 A wire loop is placed in a region of time varying magnetic field which is oriented orthogonally to the plane of the loop as shown in the figure. The graph shows the magnetic field variation as the function of time. Assume the positive emf is the one which drives a current in the clockwise direction and seen by the observer in the direction of B. Which of the following graphs best represents the induced emf as a function of time.



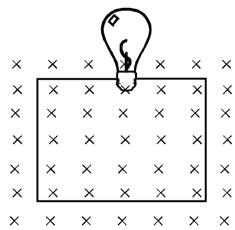
(A) 

(B) 

(C) 

(D) 

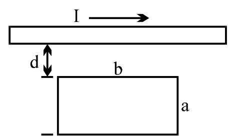
Q.4 A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt. The bulb will shine with full brightness if Δt is equal to



(A) 20 ms (B) 0.02 ms

(C) 2 ms (D) 0.2 ms

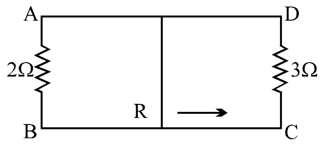
Q.5 A long straight wire is parallel to one edge as in fig. If the current in the long wire is varies in time as $I = I_0 e^{-t/\tau}$, what will be the induced emf in the loop?



(A) $\frac{\mu_0 b I}{\pi\tau} ln\left(\frac{d+a}{d}\right)$ (B) $\frac{\mu_0 b I}{2\pi\tau} ln\left(\frac{d+a}{d}\right)$

(C) $\frac{2\mu_0 b I}{\pi\tau} ln\left(\frac{d+a}{d}\right)$ (D) $\frac{\mu_0 b I}{\pi\tau} ln\left(\frac{d}{d+a}\right)$

Q.6 A rectangular loop with a sliding connector of length 10 cm is situated in uniform magnetic field perpendicular to plane of loop. The magnetic induction is 0.1 tesla and resistance of connector (R) is 1 ohm. The sides AB and CD have resistances 2 ohm and 3 ohm respectively. Find the current in the connector during its motion with constant velocity one metre/sec.



(A) $\frac{1}{110}$ A (B) $\frac{1}{220}$ A (C) $\frac{1}{55}$ A (D) $\frac{1}{440}$ A

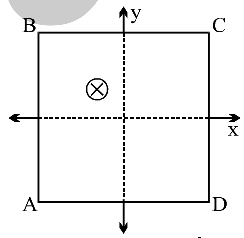
Q.7 The magnetic flux through a stationary loop with resistance R varies during interval of time T as $\phi = at(T - t)$. The heat generated during this time neglecting the inductance of loop will be

(A) $\frac{a^2T^3}{3R}$ (B) $\frac{a^2T^2}{3R}$ (C) $\frac{a^2T}{3R}$ (D) $\frac{a^2T^3}{R}$

Q.8 The dimensions of permeability of free space can be given by
(A) $[MLT^{-2}A^{-2}]$ (B) $[MLA^{-2}]$ (C) $[ML^{-3}T^2A^2]$ (D) $[MLA^{-1}]$

Q.9 A closed planar wire loop of area A and arbitrary shape is placed in a uniform magnetic field of magnitude B, with its plane perpendicular to magnetic field. The resistance of the wire loop is R. The loop is now turned upside down by 180° so that its plane again becomes perpendicular to the magnetic field. The total charge that must have flowed through the wire ring in the process is
(A) < AB/R (B) = AB/R (C) = 2AB/R (D) None

Q.10 A square coil ABCD is placed in x-y plane with its centre at origin. A long straight wire, passing through origin, carries a current in negative z-direction. Current in this wire increases with time. The induced current in the coil is :



(A) clockwise (B) anticlockwise
(C) zero (D) alternating

Q.11 A vertical bar magnet is dropped from position on the axis of a fixed metallic coil as shown in fig - I. In fig - II the magnet is fixed and horizontal coil is dropped. The acceleration of the magnet and coil are a_1 and a_2 respectively then

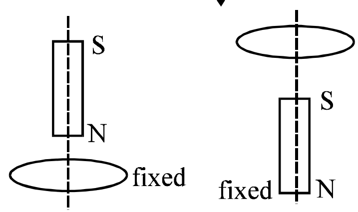
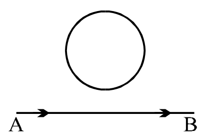


fig - I fig-II

(A) $a_1 > g$, $a_2 > g$ (B) $a_1 > g$, $a_2 < g$
(C) $a_1 < g$, $a_2 < g$ (D) $a_1 < g$, $a_2 > g$

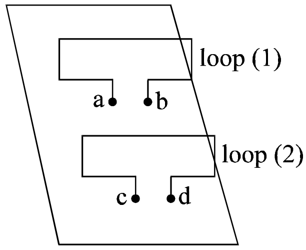
Q.12 Two identical coaxial circular loops carry a current *i* each circulating in the same direction. If the loops approach each other
(A) the current in each will decrease
(B) the current in each will increase
(C) the current in each will remain the same
(D) the current in one will increase and in other will decrease

Q.13 In the arrangement shown in given figure current from A to B is increasing in magnitude. Induced current in the loop will

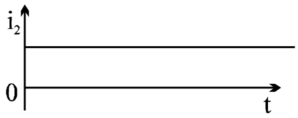


(A) have clockwise direction
(B) have anticlockwise direction
(C) be zero
(D) oscillate between clockwise and anticlockwise

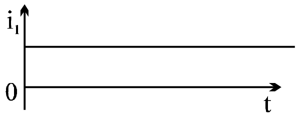
Q.14 An electric current i_1 can flow either direction through loop (1) and induced current i_2 in loop (2). Positive i_1 is when current is from 'a' to 'b' in loop (1) and positive i_2 is when the current is from 'c' to 'd' in loop (2)

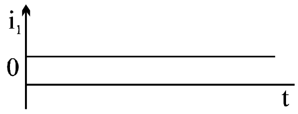


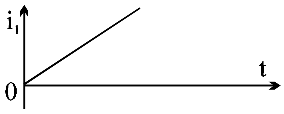
In an experiment, the graph of i_2 against time 't' is as shown below

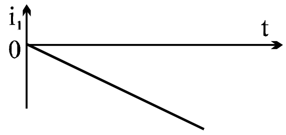


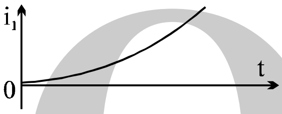
Which one(s) of the following graphs could have caused i_2 to behave as give above.

(A) 

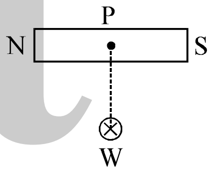
(B) 

(C) 

(D) 

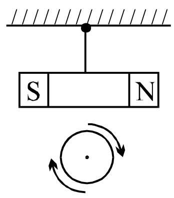
(E) 

Q.15 Figure shows a bar magnet and a long straight wire W, carrying current into the plane of paper. Point P is the point of intersection of axis of magnet and the line of shortest distance between magnet and the wire. If P is the midpoint of the magnet, then which of the following statements is correct ?



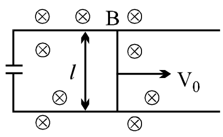
(A) magnet experiences a torque in clockwise direction
(B) magnet experiences a torque in anticlockwise direction
(C) magnet experiences a force, normal to the line of shortest distance
(D) magnet experiences a force along the line of shortest distance

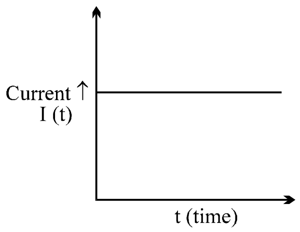
Q.16 A negative charge is given to a nonconducting loop and the loop is rotated in the plane of paper about its centre as shown in figure. The magnetic field produced by the ring affects a small magnet placed above the ring in the same plane:

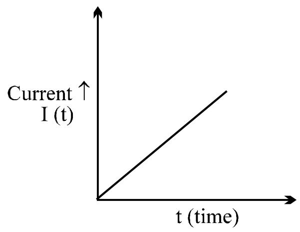


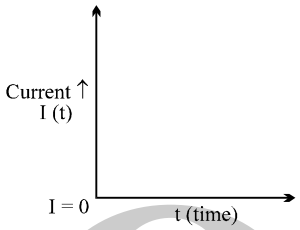
(A) the magnet does not rotate
(B) the magnet rotates clockwise as seen from below.
(C) the magnet rotates anticlockwise as seen from below
(D) no effect on magnet is there.

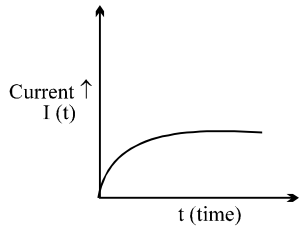
Q.17 Two infinitely long conducting parallel rails are connected through a capacitor C as shown in the figure. A conductor of length *l* is moved with constant speed v_0. Which of the following graph truly depicts the variation of current through the conductor with time ?



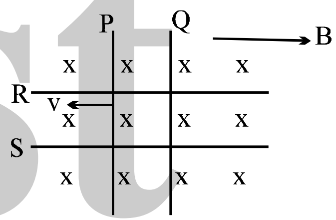
(A)


(B)


(C)


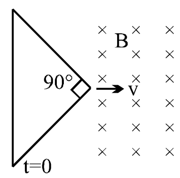
(D)


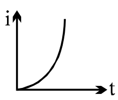
Q.18 Two identical conductors P and Q are placed on two frictionless rails R and S in a uniform magnetic field directed into the plane. If P is moved in the direction shown in figure with a constant speed then rod Q

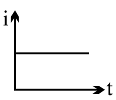


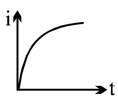
(A) will be attracted towards P
(B) will be repelled away from P
(C) will remain stationary
(D) may be repelled or attracted towards P

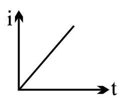
Q.19 The figure shows an isosceles triangle wire frame with apex angle equal to π/2. The frame starts entering into the region of uniform magnetic field B with constant velocity v at t= 0. The longest side of the frame is perpendicular to the direction of velocity. If i is the instantaneous current through the frame then choose the alternative showing the correct variation of i with time.



(A)


(B)


(C)


(D)


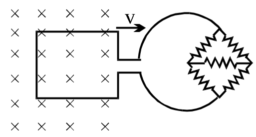
Q.20 A thin wire of length 2m is perpendicular to the xy plane. It is moved with velocity $\vec{v} = (2\hat{i} + 3\hat{j} + \hat{k})\ m/s$ through a region of magnetic induction $\vec{B} = (\hat{i} + 2\hat{j})\ Wb/m^2$. Then potential difference induced between the ends of the wire :

(A) 2 volts (B) 4 volts (C) 0 volts (D) none of these

Q.21 A long metal bar of 30 cm length is aligned along a north south line and moves eastward at a speed of 10 ms^{-1}. A uniform magnetic field of 4.0 T points vertically downwards. If the south end of the bar has a potential of 0 V, the induced potential at the north end of the bar is

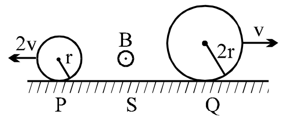
(A) + 12 V (B) – 12 V
(C) 0 V (D) cannot be determined since there is not closed circuit

Q.22 A square metal loop of side 10 cm and resistance 1 Ω is moved with a constant velocity partly inside a magnetic field of 2 Wbm^{-2}, directed into the paper, as shown in the figure. This loop is connected to a network of five resistors each of value 3 Ω. If a steady current of 1 mA flows in the loop, then the speed of the loop is



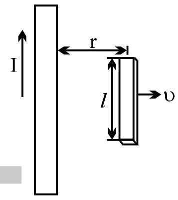
(A) 0.5 cms^{-1} (B) 1 cms^{-1} (C) 2 cms^{-1} (D) 4 cms^{-1}

Q.23 Two conducting rings P and Q of radii r and 2r rotate uniformly in opposite directions with centre of mass velocities 2v and v respectively on a conducting surface S. There is a uniform magnetic field of magnitude B perpendicular to the plane of the rings. The potential difference between the highest points of the two rings is



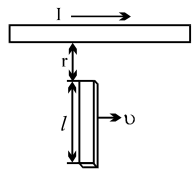
(A) zero (B) 4 Bvr
(C) 8 Bvr (D) 16 Bvr

Q.24 A conducting rod moves with constant velocity υ perpendicular to the long, straight wire carrying a current I as shown compute that the emf generated between the ends of the rod.



(A) $\dfrac{\mu_0 \upsilon I l}{\pi r}$ (B) $\dfrac{\mu_0 \upsilon I l}{2\pi r}$ (C) $\dfrac{2\mu_0 \upsilon I l}{\pi r}$ (D) $\dfrac{\mu_0 \upsilon I l}{4\pi r}$

Q.25 A conducting rod of length l moves with velocity υ a direction parallel to a long wire carrying a steady current I. The axis of the rod is maintained perpendicular to the wire with near end a distance r away as shown in the fig. Find the emf induced in the rod.



(A) $\dfrac{\mu_0 I\upsilon}{\pi} ln\left(\dfrac{r+l}{r}\right)$ (B) $\dfrac{2\mu_0 I\upsilon}{\pi} ln\left(\dfrac{r+l}{r}\right)$

(C) $\dfrac{\mu_0 I\upsilon}{\pi} ln\left(\dfrac{l}{r+l}\right)$ (D) $\dfrac{\mu_0 I\upsilon}{2\pi} ln\left(\dfrac{r+l}{r}\right)$

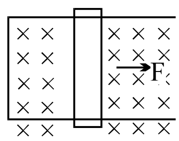
Q.26 A square loop of side a and resistance R is moved in the region of uniform magnetic field B(loop remaining completely insidefield) ,with a velocity v through a distance x . The work done is :

(A) $\dfrac{B\ell^2 vx}{R}$ (B) $\dfrac{2B^2\ell^2 vx}{R}$ (C) $\dfrac{4B^2\ell^2 vx}{R}$ (D) none

Q.27 A metallic rod of length L and mass M is moving under the action of two unequal forces F_1 and F_2 (directed opposite to each other) acting at its ends along its length. Ignore gravity and any external magnetic field. If specific charge of electrons is (e/m), then the potential difference between the ends of the rod is steady state must be

(A) $|F_1 - F_2|\, mL/eM$ (B) $(F_1 - F_2)\, mL/eM$ (C) $[\, mL/eM\,]\, ln\, [F_1/F_2]$ (D) None

Q.28 A rod closing the circuit shown in figure moves along a U shaped wire at a constant speed v under the action of the force F. The circuit is in a uniform magnetic field perpendicular to the plane. Calculate F if the rate of heat generation in the circuti is Q.

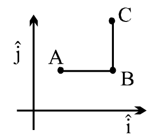


(A) F = Qv (B) $F = \dfrac{Q}{v}$ (C) $F = \dfrac{v}{Q}$ (D) $F = \sqrt{Qv}$

Q.29 Two parallel long straight conductors lie on a smooth surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists at right angles to the plane containing the conductors. They all start moving out with a constant velocity v. If r is the resistance per unit length of the wire the current in the circuit will be

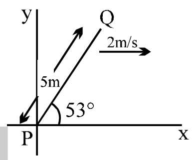
(A) $\frac{Bv}{r}$ (B) $\frac{Br}{v}$ (C) Bvr (D) Bv

Q.30 There is a uniform magnetic field B normal to the xy plane. A conductor ABC has length AB = l_1, parallel to the x-axis, and length BC = l_2, parallel to the y-axis. ABC moves in the xy plane with velocity $v_x\hat{i} + v_y\hat{j}$. The potential difference between A and C is proportional to



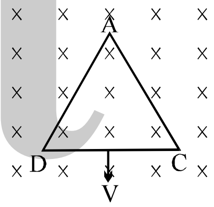
(A) $v_x l_1 + v_y l_2$ (B) $v_x l_2 + v_y l_1$
(C) $v_x l_2 - v_y l_1$ (D) $v_x l_1 - v_y l_2$

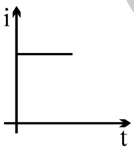
Q.31 A conducting rod PQ of length 5 m oriented as shown in figure is moving with velocity (2 m/s) $\hat{i}$ without any rotation in a uniform magnetic field $(3\hat{j}+4\hat{k})$ Tesla. Emf induced in the rod is

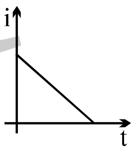


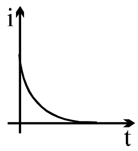
(A) 32 Volts (B) 40 Volt (C) 50 Volt (D) none

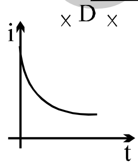
Q.32 An equilateral triangular loop ADC of some finite magnetic field $\vec{B}$ as shown in the figure. At time t = 0, side DC of loop is at edge of the magnetic field. Magnetic field is perpendicular to the paper inwards (or perpendicular to the plane of the coil). The induced current versus time graph will be as



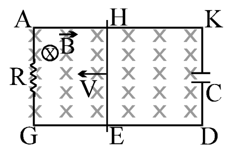
(A) 

(B) 

(C) 

(D) 

Q.33 In the circuit shown in figure, a conducting wire HE is moved with a constant speed V towards left. The complete circuit is placed in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the circuit directed in inward direction. The current in HKDE is

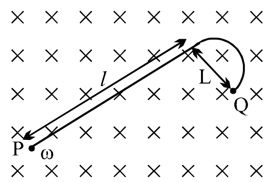


(A) clockwise (B) anticlockwise
(C) alternating (D) zero

Q.34 The magnetic field in a region is given by $\vec{B} = B_0\left(1+\frac{x}{a}\right)\hat{k}$. A square loop of edge - length d is placed with its edge along x & y axis. The loop is moved with constant velocity $\vec{V} = V_0\hat{i}$. The emf induced in the loop is

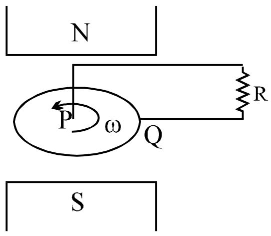
(A) $\frac{V_0B_0d^2}{a}$ (B) $\frac{V_0B_0d^2}{2a}$ (C) $\frac{V_0B_0a^2}{d}$ (D) None

Q.35 When a 'J' shaped conducting rod is rotating in its own plane with constant angular velocity w, about one of its end P, in a uniform magnetic field $\vec{B}$ directed normally into the plane of paper) then magnitude of emf induced across it will be



(A) $B\omega\sqrt{L^2+l^2}$
(B) $\frac{1}{2}B\omega L^2$
(C) $\frac{1}{2}B\omega(L^2+l^2)$
(D) $\frac{1}{2}B\omega l^2$

Q.36 A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle. What current, if any, flows through R?



(A) a current from P to Q
(B) a current from Q to P
(C) no current, because the emf in the disc is opposed by the back emf
(D) no current, because the emf induced in one side of the disc is opposed by the emf induced in the other side.
(E) no current, because no radial emf is induced in the disc

Q.37 For L-R circuit, the time constant is equal to
(A) twice the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(B) ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(C) half the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

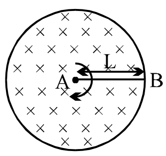
Q.38 A rectangular coil of single turn, having area A, rotates in a uniform magnetic field B an angular velocity ω about an axis perpendicular to the field. If initially the plane of coil is perpendicular to the field, then the average induced e.m.f. when it has rotated through 90° is

(A) $\frac{\omega BA}{\pi}$
(B) $\frac{\omega BA}{2\pi}$
(C) $\frac{\omega BA}{4\pi}$
(D) $\frac{2\omega BA}{\pi}$

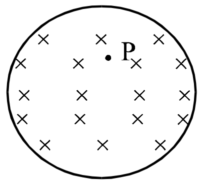
Q.39 A ring of resistance 10Ω, radius 10cm and 100 turns is rotated at a rate 100 revolutions per second about a fixed axis which is perpendicular to a uniform magnetic field of induction 10mT. The amplitude of the current in the loop will be nearly (Take : $\pi^2 = 10$)
(A) 200A
(B) 2A
(C) 0.002A
(D) none of these

Q.40 A copper rod AB of length L, pivoted at one end A, rotates at constant angular velocity ω, at right angles to a uniform magnetic field of induction B. The e.m.f developed between the mid point C of the rod and end B is

(A) $\frac{B\omega l^2}{4}$
(B) $\frac{B\omega l^2}{2}$
(C) $\frac{3B\omega l^2}{4}$
(D) $\frac{3B\omega l^2}{8}$

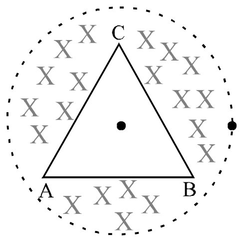


Q.41 Figure shows a uniform magnetic field B confined to a cylindrical volume and is increasing at a constant rate. The instantaneous acceleration experienced by an electron placed at P is



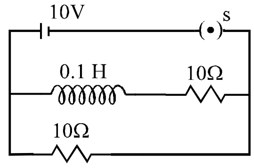
(A) zero (B) towards right
(C) towards left (D) upwards

Q.42 A triangular wire frame (each side = 2m) is placed in a region of time variant magnetic field having $dB/dt = \sqrt{3}$ T/s. The magnetic field is perpendicular to the plane of the triangle. The base of the triangle AB has a resistance 1 Ω while the other two sides have resistance 2Ω each. The magnitude of potential difference between the points A and B will be



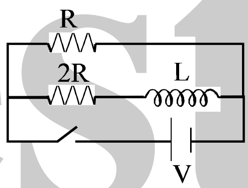
(A) 0.4 V (B) 0.6 V (C) 1.2 V (D) None

Q.43 In the adjoining circuit, initially the switch S is open. The switch 'S' is closed at t = 0. The difference between the maximum and minimum current that can flow in the circuit is



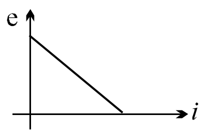
(A) 2 Amp (B) 3 Amp
(C) 1 Amp (D) nothing can be concluded

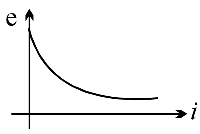
Q.44 The ratio of time constant in charging and discharging in the circuit shown in figure is

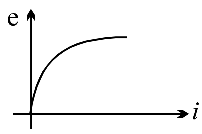


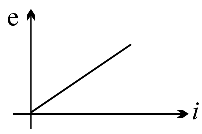
(A) 1 : 1 (B) 3 : 2
(C) 2 : 3 (D) 1 : 3

Q.45 In an L-R circuit connected to a battery of constant e.m.f. E switch S is closed at time t = 0. If e denotes the magnitude of induced e.m.f. across inductor and *i* the current in the circuite at any time t. Then which of the following graphs shows the variation of e with *i* ?

(A) 

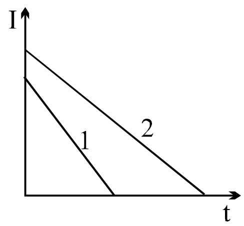
(B) 

(C) 

(D) 

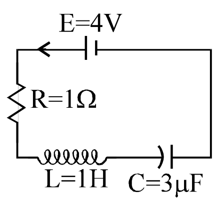
Q.46 A curren of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time is
(A) 2 J/s (B) 1 J/s (C) 16 J/s (D) 4 J/s

Q.47 Two identical inductance carry currents that vary with time according to linear laws (as shown in figure). In which of two inductance is the self induction emf greater?



(A) 1
(B) 2
(C) same
(D) data are insufficient to decide

Q.48 The current in the given circuit is increasing with a rate a = 4 amp/s. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be :



(A) 4μC (B) 5μC
(C) 6μC (D) none of these

Q.49 L, C and R represent physical quantities inductance, capacitance and resistance. The combination which has the dimensions of frequency is

(A) $\frac{1}{RC}$ and $\frac{R}{L}$ (B) $\frac{1}{\sqrt{RC}}$ and $\sqrt{\frac{R}{L}}$ (C) $\sqrt{LC}$ (D) $\frac{C}{L}$

Q.50 A coil of inductance 5H is joined to a cell of emf 6V through a resistance 10Ω at time t = 0. The emf across the coil at time $t = ln\sqrt{2}$ s is:

(A) 3V (B) 1.5 V (C) 0.75 V (D) 4.5 V

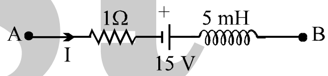
Q.51 A long solenoid of N turns has a self inductance L and area of cross section A. When a current i flows through the solenoid, the magnetic field inside it has magnitude B. The current i is equal to:

(A) BAN/L (B) BANL (C) BN/AL (D) B/ANL

Q.52 A long straight wire of circular cross-section is made of a non-magnetic material. The wire is of radius a. The wire carries a current I which is uniformly ditributed over its cross-section. The energy stored per unit lenght in the magnetic field contained within the wire is

(A) $U = \frac{\mu_0 I^2}{8\pi}$ (B) $U = \frac{\mu_0 I^2}{16\pi}$ (C) $U = \frac{\mu_0 I^2}{4\pi}$ (D) $U = \frac{\mu_0 I^2}{2\pi}$

Q.53 The network shown in the figure is part of a complete circuit. If at a certain instant, the current I is 5A and it is decreasing at a rate of 10^3 As^{-1} then $V_B - V_A$ equals

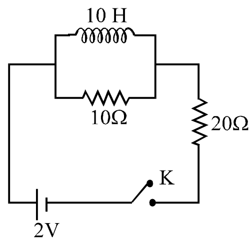


(A) 20 V (B) 15 V (C) 10 V (D) 5 V

Q.54 In **Problem 53**, if I is reversed in direction, then $V_B - V_A$ equals

(A) 5 V (B) 10 V (C) 15 V (D) 20 V

Q.55 Two resistors of 10 Ω and 20 Ω and an ideal inductor of 10 H are connected to a 2 V battery as shown. The key K is inserted at time t = 0. The initial (t = 0) and final (t →∞) currents through battery are



(A) $\frac{1}{15}$ A, $\frac{1}{10}$ A (B) $\frac{1}{10}$ A, $\frac{1}{15}$ A

(C) $\frac{2}{15}$ A, $\frac{1}{10}$ A (D) $\frac{1}{15}$ A, $\frac{2}{25}$ A

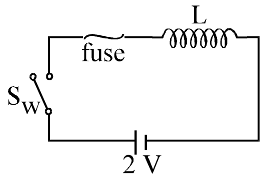
Q.56 A small coil of radius r is placed at the centre of a large coil of radius R, where R >> r. The coils are coplanar. The coefficient of mutual inductance between the coils is

(A) $\frac{\mu_0 \pi r}{2R}$ (B) $\frac{\mu_0 \pi r^2}{2R}$ (C) $\frac{\mu_0 \pi r^2}{2R^2}$ (D) $\frac{\mu_0 \pi r}{2R^2}$

Q.57 Two long parallel wires whose centres are a distance d apart carry equal currents in opposite directions. If the flux within wires is neglected, the inductance of such arrangement of wire of length l and radius a will be

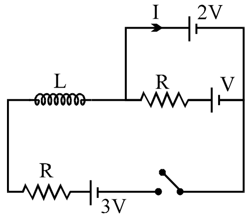
(A) $L = \frac{\mu_0 l}{\pi}\log_e\frac{d-a}{a}$ (B) $L = \frac{\mu_0 l}{\pi}\log_e\frac{d}{a}$ (C) $L = \frac{\mu_0 l}{\pi}\log_e\frac{a}{d}$ (D) none

Q.58 In the circuit shown, the cell is ideal. The coil has an inductance of 4H and zero resistance. F is a fuse of zero resistance and will blow when the current through it reaches 5A. The switch is closed at t = 0. The fuse will blow :



(A) just after t=0 (B) after 2s
(C) after 5s (D) after 10s

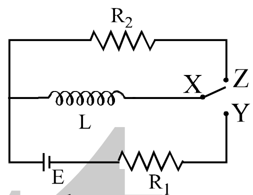
Q.59 In the LR circuit shown, what is the variation of the current I as a function of time? The switch is closed at time t = 0 sec.



(A) $\frac{V}{R}\left(1-e^{-\frac{Rt}{L}}\right)$ (B) $\frac{V}{R}e^{-\frac{Rt}{L}}$

(C) $-\frac{V}{R}e^{-\frac{Rt}{L}}$ (D) None

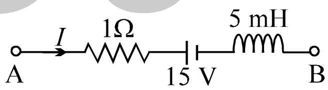
Q.60 In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :



(A) $\frac{LE^2}{2R_1^2}$ (B) $\frac{LE^2}{2R_2^2}$ (C) $\frac{LE^2}{2R_1R_2}$ (D) $\frac{LE^2R_2}{2R_1^2}$

Q.61 An induction coil stores 32 joules of magnetic energy and dissipates energy as heat at the rate of 320 watts. When a current of 4 amperes is passed through it. Find the time constant of the circuit when the coil is joined across a battery.
(A) 0.2 s (B) 0.1 s (C) 0.3 s (D) 0.4 s

Q.62 The figure shows a part of a complete circuit. The potential difference $V_B - V_A$ when the current I is 5A and is decreasing at a rate of 10^3 As^{-1} is given by



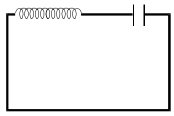
(A) 15 V (B) 10 V (C) –15 V (D) 20 V

Q.63 In a L–R decay circuit, the initial current at t = 0 is I. The total charge that has flown through the resistor till the energy in the inductor has reduced to one–fourth its initial value, is

(A) LI/R (B) $LI/2R$ (C) $LI\sqrt{2}/R$ (D) None

Q.64 A capacitor of capacitance 2 μF is charged to a potential difference of 12 V. It is then connected across an inductor of inductance 0.6 mH. The current in the circuit when the potential difference across the capacitor is 6 V is :
(A) 3.6 A (B) 2.4 A (C) 1.2 A (D) 0.6 A

Q.65 In an LC circuit, the capacitor has maximum charge q_0. The value of $\left.\frac{di}{dt}\right|_{max}$ is

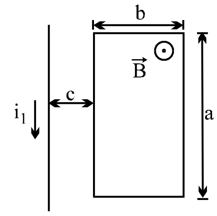


(A) $\frac{q_0}{LC}$ (B) $\frac{q_0}{\sqrt{LC}}$ (C) $\frac{q_0}{LC}-1$ (D) none of these

Q.66 An inductor coil stores U energy when i current is passed through it and dissipates energy at the rate of P. The time constant of the circuit, when this coil is connected across a battery of zero internal resistance is

(A) $\frac{4U}{P}$ (B) $\frac{U}{P}$ (C) $\frac{2U}{P}$ (D) $\frac{2P}{U}$

Q.67 The mutual inductance between the rectangular loop and the long straight wire as shown in figure is M.



(A) M = Zero

(B) $M = \frac{\mu_0 a}{2\pi} \ln\left(1 + \frac{c}{b}\right)$

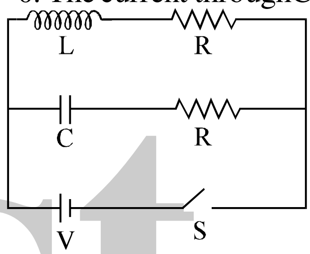
(C) $M = \frac{\mu_0 b}{2\pi} \ln\left(\frac{a+c}{b}\right)$

(D) $M = \frac{\mu_0 a}{2\pi} \ln\left(1 + \frac{b}{c}\right)$

Q.68 A long straight wire is placed along the axis of a circular ring of radius R. The mutual inductance of this system is

(A) $\frac{\mu_0 R}{2}$ (B) $\frac{\mu_0 \pi R}{2}$ (C) $\frac{\mu_0}{2}$ (D) 0

Q.69 In the circuti shown in the figure, $R = \sqrt{\frac{L}{C}}$. Switch S is closed at time t = 0. The current throughC and L would be equal after a time t equal to:



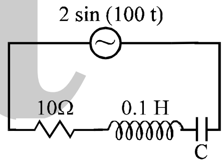
(A) CR

(B) CR ln (2)

(C) $\frac{L}{R \ln(2)}$

(D) LR

Q.70 The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to



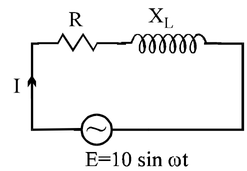
(A) 400 μF

(B) 300 μF

(C) 500 μF

(D) 200 μF

Q.71 An ac-circuit having supply voltage E consists of a resistor of resistance 3Ω and an inductor of reactance 4Ω as shown in the figure. The voltage across the inductor at $t = \pi/\omega$ is



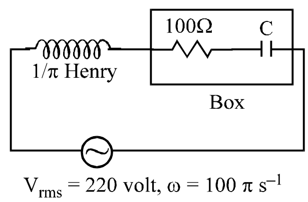
(A) 2 volts

(B) 10 volts

(C) zero

(D) 4.8 volts

Q.72 In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is



(A) $\frac{1}{\sqrt{2}}$

(B) 1

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{2}$

Q.73 When 100 V DC is applied across a solenoid a current of 1 A flows in it. When 100 V AC is applied across the same coil, the current drops to 0.5 A. If the frequency of the AC source is 50 Hz, the impedance and inductance of the solenoid are:

(A) 100Ω, 0.93 H (B) 200Ω, 1.0 H (C) 10Ω, 0.86H (D) 200Ω, 0.55 H

Q.74 An inductive circuit contains resistance of 10 Ω and an inductance of 2.0 H. If an AC voltage of 120 V and frequency 60 Hz is applied to this circuit, the current would be nearly:

(A) 0.8 A (B) 0.48 A (C) 0.16 A (D) 0.32 A

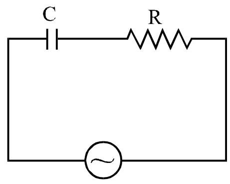
Q.75 The power in ac circuit is given by $P = E_{rms}I_{rms}\cos\phi$. The vale of $\cos\phi$ in series LCR circuit at resonance is:

(A) zero (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

Q.76 In ac circuit when ac ammeter is connected it reads i current if a student uses dc ammeter in place of ac ammeter the reading in the dc ammeter will be:

(A) $\frac{i}{\sqrt{2}}$ (B) $\sqrt{2}\,i$ (C) 0.637 i (D) zero

Q.77 In the circuit shown if the emf of source at an instant is 5 V, the potential difference across capacitor at the same instant is 4 V. The potential difference across R at that instant may be



(A) 3V (B) 9V (C) $\frac{3}{\sqrt{2}}$ V (D) none

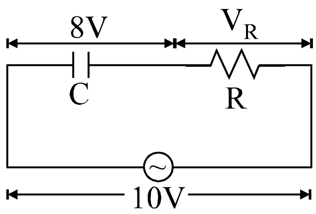
Q.78 An AC current is given by $I = I_0 + I_1 \sin wt$ then its rms value will be

(A) $\sqrt{I_0^2 + 0.5 I_1^2}$ (B) $\sqrt{I_0^2 + 0.5 I_0^2}$ (C) 0 (D) $I_0/\sqrt{2}$

Q.79 Let f = 50 Hz, and C = 100 μF in an AC circuit containing a capicator only. If the peak value of the current in the circuit is 1.57 A at t = 0. The expression for the instantaneous voltage across the capacitor will be

(A) $E = 50 \sin(100\pi t - \pi/2)$ (B) $E = 100 \sin(50\pi t)$
(C) $E = 50 \sin 100\pi t$ (D) $E = 50 \sin(100\pi t + \pi/2)$

Q.80 In a series CR circuit shown in figure, the applied voltage is 10 V and the voltage across capacitor is found to be 8V. Then the voltage across R, and the phase difference between current and the applied voltage will respectively be



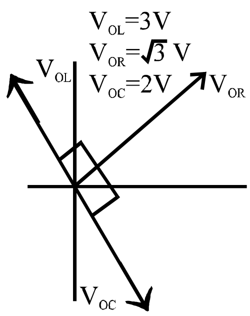
(A) 6V, $\tan^{-1}\left(\frac{4}{3}\right)$ (B) 3V, $\tan^{-1}\left(\frac{3}{4}\right)$

(C) 6V, $\tan^{-1}\left(\frac{5}{3}\right)$ (D) none

Q.81 The phase difference between current and voltage in an AC circuit is π/4 radian. If the frequency of AC is 50 Hz, then the phase difference is equivalent to the time difference :

(A) 0.78 s (B) 15.7 ms (C) 0.25 s (D) 2.5 ms

Q.82 The given figure represents the phasor diagram of a series LCR circuit connected to an ac source. At the instant t′ when the source voltage is given by $V = V_0\cos\omega t'$, the current in the circuit will be

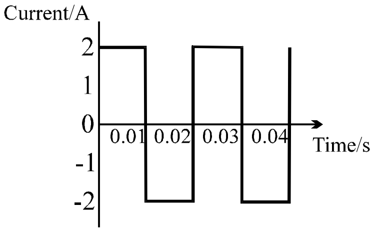


(A) $I = I_0 \cos(\omega t' + \pi/6)$ (B) $I = I_0 \cos(\omega t' - \pi/6)$
(C) $I = I_0 \cos(\omega t' + \pi/3)$ (D) $I = I_0 \cos(\omega t' - \pi/3)$

Q.83 Power factor of an L-R series circuit is 0.6 and that of a C–R series circuit is 0.5. If the element (L, C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance in the L-R circuit to the resistance in the C–R circuit is

(A) 6/5 (B) 5/6 (C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{3\sqrt{3}}{4}$

Q.84 The direct current which would give the same heating effect in an equal constant resistance as the current shown in figure, i.e. the r.m.s. current, is



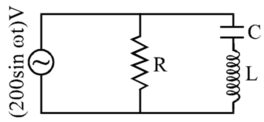
(A) zero (B) $\sqrt{2}$ A

(C) 2A (D) $2\sqrt{2}$ A

Q.85 The effective value of current i = 2 sin 100 π t + 2 sin(100 π t + 30°) is :

(A) $\sqrt{2}$ A (B) $2\sqrt{2+\sqrt{3}}$ (C) 4 (D) None

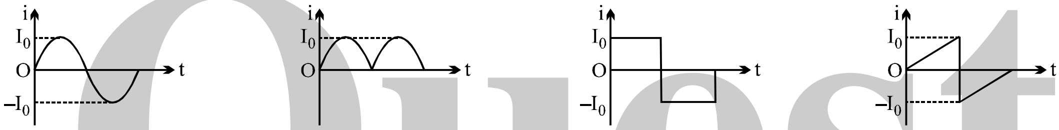
Q.86 In the circuit diagram shown, X_C = 100 Ω, X_L = 200 Ω and R = 100 Ω. The effective current through the source is



(A) 2 A (B) $\sqrt{2}$ A

(C) 0.5 A (D) $2\sqrt{2}$ A

Q.87 If I_1, I_2, I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.

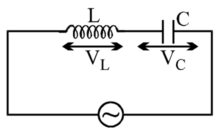


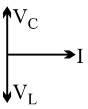
(A) $I_1 = I_2 = I_3 = I_4$ (B) $I_3 > I_1 = I_2 > I_4$ (C) $I_3 > I_4 > I_2 = I_1$ (D) $I_3 > I_2 > I_1 > I_4$

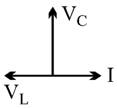
Q.88 In series LR circuit X_L = 3R. Now a capacitor with X_C = R is added in series. Ratio of new to old power factor is

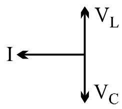
(A) 1 (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

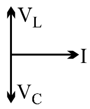
Q.89 The current I, potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown in the figure are best represented vectorially as



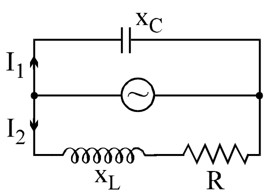
(A) 

(B) 

(C) 

(D) 

Q.90 In the shown AC circuit phase different between currents I_1 and I_2 is



(A) $\frac{\pi}{2} - \tan^{-1}\frac{x_L}{R}$ (B) $\tan^{-1}\frac{x_L - x_C}{R}$ (C) $\frac{\pi}{2} + \tan^{-1}\frac{x_L}{R}$ (D) $\tan^{-1}\frac{x_L - x_C}{R} + \frac{\pi}{2}$

Q.91 A capacitor C = 2μF and an inductor with L = 10 H and coil resistance 5 Ω are in series in a circuit. When an alternating current of r.m.s. value 2A flows in the circuit, the average power in watts in the circuit is

(A) 100 (B) 50 (C) 20 (D) 10

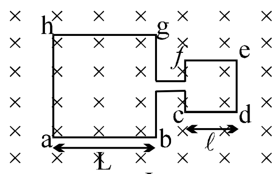
ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

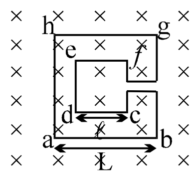
Q.1 The dimension of the ratio of magnetic flux and the resistance is equal to that of :
(A) induced emf (B) charge (C) inductance (D) current

Question No. 2 to 5 (4 questions)

The adjoining figure shows two different arrangements in which two square wire frames are placed in a uniform constantly decreasing magnetic field B.



I



II

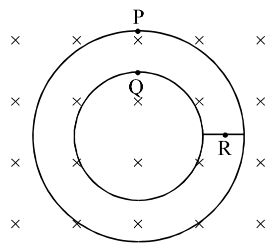
Q.2 The value of magnetic flux in each case is given by
(A) Case I: $\Phi = \pi(L^2 + \ell^2)B$; Case II: $\Phi = \pi(L^2 - \ell^2)B$
(B) Case I: $\Phi = \pi(L^2 + \ell^2)B$; Case II: $\Phi = \pi(L^2 + \ell^2)B$
(C) Case I: $\Phi = (L^2 + \ell^2)B$; Case II: $\Phi = (L^2 - \ell^2)B$
(D) Case I: $\Phi = (L + \ell)^2B$; Case II: $\Phi = \pi(L - \ell)^2B$

Q.3 The direction of induced current in the case I is
(A) from a to b and from c to d (B) from a to b and from f to e
(C) from b to a and from d to c (D) from b to a and from e to f

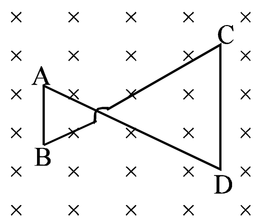
Q.4 The direction of induced current in the case II is
(A) from a to b and from c to d (B) from b to a and from f to e
(C) from b to a and from c to d (D) from a to b and from d to c

Q.5 If I_1 and I_2 are the magnitudes of induced current in the cases I and II, respectively, then
(A) $I_1 = I_2$ (B) $I_1 > I_2$ (C) $I_1 < I_2$ (D) nothing can be said

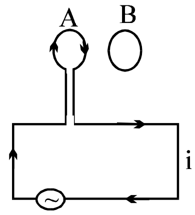
Q.6 Figure shown plane figure made of a conductor located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current
(A) at point P is clockwise
(B) at point Q is anticlockwise
(C) at point Q is clockwise
(D) at point R is zero



Q.7 A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are
(A) B to A and D to C (B) A to B and C to D
(C) A to B and D to C (D) B to A and C to D



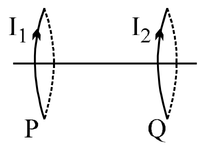
Q.8 Two circular coils A and B are facing each other as shown in figure. The current i through A can be altered
(A) there will be repulsion between A and B if i is increased
(B) there will be attraction between A and B if i is increased
(C) there will be neither attraction nor repulsion when i is changed
(D) attraction or repulsion between A and B depends on the direction of current. It does not depend whether the current is increased or decreased.



Q.9 A bar magnet is moved along the axis of copper ring placed far away from the magnet. Looking from the side of the magnet, an anticlockwise current is found to be induced in the ring. Which of the following may be true?

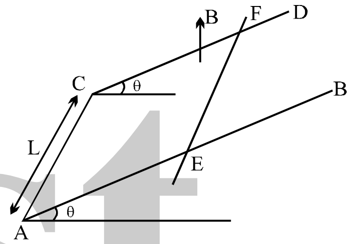
(A) The south pole faces the ring and the magnet moves towards it.
(B) The north pole faces the ring and the magnet moves towards it.
(C) The south pole faces the ring and the magnet moves away from it.
(D) The north pole faces the ring and the magnet moves away from it.

Q.10 Two circular coils P & Q are fixed coaxially & carry currents I_1 and I_2 respectively



(A) if $I_2 = 0$ & P moves towards Q, a current in the samedirection as I_1 is induced in Q
(B) if $I_1 = 0$ & Q moves towards P, a current in the opposite direction to that of I_2 isinduced in P.
(C) when $I_1 \neq 0$ and $I_2 \neq 0$ are in the same direction then the two coils tend to move apart .
(D) when $I_1 \neq 0$ and $I_2 \neq 0$ are in opposite directions then the coils tends to move apart.

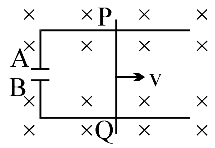
Q.11 AB and CD are smooth parallel rails, separated by a distance l, and inclined to the horizontal at an angle θ. A uniform magnetic field of magnitude B, directed vertically upwards, exists in the region. EF is a conductor of mass m, carrying a current i. For EF to be in equilibrium,



(A) i must flow from E to F
(B) $Bil = mg \tan\theta$
(C) $Bil = mg \sin\theta$
(D) $Bil = mg$

Q.12 In the **previous question**, if B is normal to the plane of the rails
(A) $Bil = mg \tan\theta$
(B) $Bil = mg \sin\theta$
(C) $Bil = mg \cos\theta$
(D) equilibrium cannot be reached

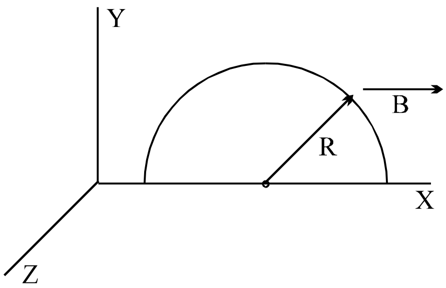
Q.13 A conducting rod PQ of length L = 1.0 m is moving with a uniform speed v = 20 m/s in a uniform magnetic field B = 4.0 T directed into the paper. A capacitor of capacity C = 10 μF is connected as shown in figure. Then



(A) $q_A = +\,800\mu C$ and $q_B = -\,800\mu C$
(B) $q_A = -\,800\mu C$ and $q_B = +\,800\mu C$
(C) $q_A = 0 = q_B$
(D) charged stored in the capacitor increases exponentially with time

Q.14 The e.m.f. induced in a coil of wire, which is rotating in a magnetic field, does not depend on
(A) the angular speed of rotation
(B) the area of the coil
(C) the number of turns on the coil
(D) the resistance of the coil

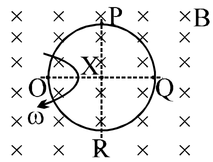
Q.15 A semicircle conducting ring of radius R is placed in the xy plane, as shown in the figure. A uniform magnetic field is set up along the x–axis. No emf, will be induced in the ring. if



(A) it moves along the x–axis
(B) it moves along the y–axis
(C) it moves along the z-axis
(D) it remains stationary

Question No. 16 to 18 (3 questions)

A conducting ring of radius a is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field B which exists everywhere. The rotational velocity is ω.



Q.16 Choose the correct statement(s) related to the potential of the points P, Q and R

(A) $V_P - V_O > 0$ and $V_R - V_O < 0$ (B) $V_P = V_R > V_O$

(C) $V_O > V_P = V_Q$ (D) $V_Q - V_P = V_P - V_O$

Q.17 Choose the correct statement(s) related to the magnitude of potential differences

(A) $V_P - V_O = \frac{1}{2}B\omega a^2$ (B) $V_P - V_Q = \frac{1}{2}B\omega a^2$

(C) $V_Q - V_O = 2B\omega a^2$ (D) $V_P - V_R = 2B\omega a^2$

Q.18 Choose the correct statement(s) related to the induced current in the ring

(A) Current flows from $Q \longrightarrow P \longrightarrow O \longrightarrow R \longrightarrow Q$

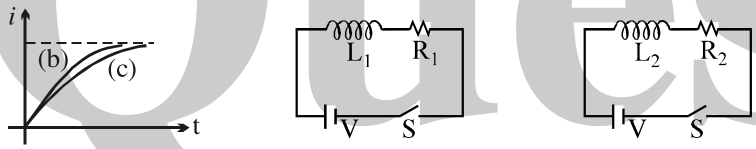
(B) Current flows from $Q \longrightarrow R \longrightarrow O \longrightarrow P \longrightarrow Q$

(C) Current flows from $Q \longrightarrow P \longrightarrow O$ and from $Q \longrightarrow R \longrightarrow O$

(D) No current flows

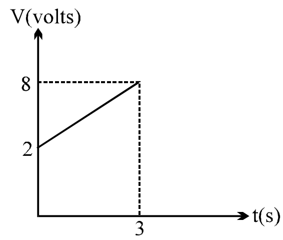
Q.19 Current growth in two L-R circuits (b) and (c) as shown in figure (a). Let L_1, L_2, R_1 and R_2 be the corresponding values in two circuits. Then

(A) $R_1 > R_2$ (B) $R_1 = R_2$ (C) $L_1 > L_2$ (D) $L_1 < L_2$



(a) (b) (c)

Q.20 A circuit element is placed in a closed box. At time t=0, constant current generator supplying a current of 1 amp, is connected across the box. Potential difference across the box varies according to graph shown in figure. The element in the box is :



(A) resistance of 2Ω (B) battery of emf 6V

(C) inductance of 2H (D) capacitance of 0.5F

Q.21 The symbols L, C, R represent inductance, capacitance and resistance respectively. Dimension of frequency are given by the combination

(A) $1/RC$ (B) R/L (C) $\frac{1}{\sqrt{LC}}$ (D) C/L

Q.22 An LR circuit with a battery is connected at $t = 0$. Which of the following quantities is not zero just after the circuit

(A) current in the circuit (B) magnetic field energy in the inductor

(C) power delivered by the battery (D) emf induced in the inductor

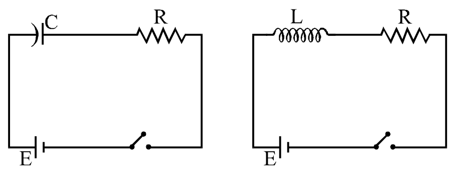
Q.23 The switches in figures (a) and (b) are closed at $t = 0$

(A) The charge on C just after $t = 0$ is EC.

(B) The charge on C long after $t = 0$ is EC.

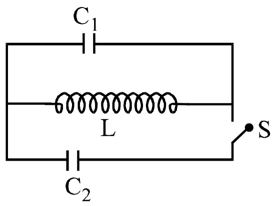
(C) The current in L just after $t = 0$ is E/R.

(D) The current in L long after $t = 0$ is E/R.



(a) (b)

Q.24 At a moment (t = 0) when charge on capacitor C_1 is zero, the switch is closed. If I_0 be the current through inductor at that instant, for t > 0,



(A) maximum current through inductor equals $I_0/2$.

(B) maximum current through inductor equals $\dfrac{C_1 I_0}{C_1 + C_2}$

(C) maximum charge on $C_1 = \dfrac{C_1 I_0 \sqrt{LC_1}}{C_1 + C_2}$

(D) maximum charge on $C_1 = I_0 C_1 \sqrt{\dfrac{L}{C_1 + C_2}}$

Q.25 For L – R circuit, the time constant is equal to

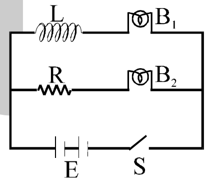
(A) twice the ratio of the energy stored in the magnetic field to the rate of the dissipation of energy in the resistance.

(B) the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance.

(C) half of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance.

(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation energy in the resistance.

Q.26 An inductor L, a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R. Which of the following statement gives the correct description of the happenings when the switch S is closed?



(A) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.

(B) B_1 light up earlier and finally both the bulbs acquire equal brightness.

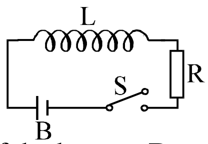
(C) B_2 lights up earlier and finally B_1 shines brighter than B_2.

(D) B_1 and B_2 light up together with equal brightness all the time.

Q.27 Which of the following quantities can be written in SI units in $Kgm^2A^{-2}s^{-3}$?

(A) Resistance (B) Inductance (C) Capacitance (D) Magnetic flux

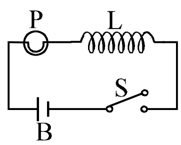
Q.28 In figure, the switch S is closed so that a current flows in the iron-core inductor which has inductance L and the resistance R. When the switch is opened, a spark is obtained in it at the contacts. The spark is due to



(A) a slow flux change in L (B) a sudden increase in the emf of the battery B

(C) a rapid flux change in L (D) a rapid flux change in R

Q.29 In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to



(A) the low resistance of P (B) the induced-emf in L

(C) the low resistance of L (D) the high voltage of the battery B

Q.30 Two coil A and B have coefficient of mutual inductance M = 2H. The magnetic flux passing through coil A changes by 4 Weber in 10 seconds due to the change in current in B. Then
(A) change in current in B in this time interval is 0.5 A
(B) the change in current in B in this time interval is 2A
(C) the change in current in B in this time interval is 8A
(D) a change in current of 1A in coil A will produce a change in flux passing through B by 4 Weber.

Q.31 A circuit has three elements, a resistance of 11Ω, a coil of inductive reactance 120Ω and a capacitive reactance of 120 Ω in series and connected to an A.C. source of 110 V, 60 Hz. Which of the three elements have minimum potential difference?
(A) Resistance (B) Capacitance
(C) Inductor (D) All will have equal potential difference

Q.32 The reactance of a circuit is zero. It is possible that the circuit contains :
(A) an inductor and a capacitor (B) an inductor but no capacitor
(C) a capacitor but no inductor (D) neigher an inductor nor a capacitor.

Q.33 In a series R-L-C circuit, the frequency of the source is half of the resonance frequency. The nature of the circuit will be
(A) capacitive (B) inductive (C) purely resistive (D) data insufficient

Q.34 An a.c. source of voltage V and of frequency 50 Hz is connected to an inductor of 2H and negligible resistance. A current of r.m.s. value I flows in the coil. When the frequency of the voltage is changed to 400 Hz keeping the magnitude of V the same, the current is now
(A) $8I$ in phase with V (B) $4I$ and leading by 90° from V
(C) $I/4$ and lagging by 90° from V (D) $I/8$ and lagging by 90° from V

ANSWER KEY

ONLY ONE OPTION IS CORRECT

Q.1	B	Q.2	C	Q.3	C	Q.4	A	Q.5	B	Q.6	B	Q.7	A
Q.8	A	Q.9	C	Q.10	C	Q.11	C	Q.12	A	Q.13	A	Q.14	D
Q.15	D	Q.16	B	Q.17	C	Q.18	A	Q.19	D	Q.20	A	Q.21	A
Q.22	C	Q.23	C	Q.24	B	Q.25	D	Q.26	D	Q.27	A	Q.28	B
Q.29	A	Q.30	C	Q.31	A	Q.32	B	Q.33	D	Q.34	A	Q.35	C
Q.36	A	Q.37	A	Q.38	D	Q.39	B	Q.40	D	Q.41	B	Q.42	A
Q.43	C	Q.44	B	Q.45	A	Q.46	C	Q.47	A	Q.48	C	Q.49	A
Q.50	A	Q.51	A	Q.52	B	Q.53	B	Q.54	C	Q.55	A	Q.56	B
Q.57	A	Q.58	D	Q.59	C	Q.60	A	Q.61	A	Q.62	C	Q.63	B
Q.64	D	Q.65	A	Q.66	C	Q.67	D	Q.68	D	Q.69	B	Q.70	C
Q.71	D	Q.72	A	Q.73	D	Q.74	C	Q.75	B	Q.76	D	Q.77	B
Q.78	A	Q.79	C	Q.80	A	Q.81	D	Q.82	B	Q.83	D	Q.84	C
Q.85	D	Q.86	A	Q.87	B	Q.88	D	Q.89	D	Q.90	C	Q.91	C

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	B	Q.2	C	Q.3	C	Q.4	B
Q.5	B	Q.6	A,C,D	Q.7	A	Q.8	A
Q.9	B,C	Q.10	B,D	Q.11	A,B	Q.12	B
Q.13	A	Q.14	D	Q.15	A,B,C,D	Q.16	B,D
Q.17	C	Q.18	D	Q.19	B,D	Q.20	D
Q.21	A,B,C	Q.22	D	Q.23	B,D	Q.24	D
Q.25	A	Q.26	A	Q.27	A	Q.28	C
Q.29	B	Q.30	B	Q.31	A	Q.32	A,D
Q.33	A	Q.34	D				

Power Coaching

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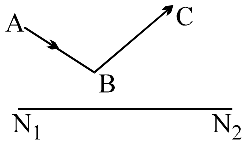
PHYSICS

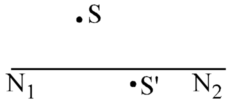
GEOMETRICAL OPTICS

SHORT QUESTIONS

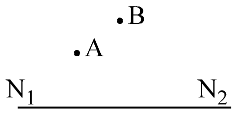
Q.1 The position of the optical axis N_1N_2, the path of ray AB incident upon a lens and the refracted ray BC are known (figure). Find by construction the position of the main foci of the lens.



Q.2 Point S' is the image of a point source of light S in a spherical mirror whose optical axis is N_1N_2 (figure). Find by construction the position of the centre of curvature and its focus.

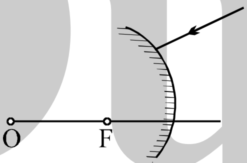


Q.3 The positions of optical axis N_1N_2 of a spherical mirror, the source and the image are known (figure). Find by construction the positions of the centre of the curvature, its focus and the pole for the cases:
(a) A– source, B – image; (b) B – source, A – image



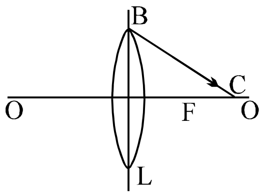
Q.4 The layered lens shown in figure is made of two kinds of glass. What image will be produced by this lens with a point source arranged on the optical axis? Disregard the reflection of light on the boundary between layers.

Q.5 A ray of light falls on a convex mirror, as shown in figure. Trace the path of the ray further.



Q.6 A double convex lens of focal length f lies between a source of light and a screen. The distance between the source of light and the screen is less than $4f$. It is known that in these conditions it is not possible to obtain an image of the source on the screen, whatever the position of the lens. How can an image of the source be obtained on the screen with quite simple means and without moving either lens or screen?

Q.7 In figure is depicted the path of a ray of light BC after refraction in a double convex lens L of principal focus F and of principal axis OO. Find by construction the path of this ray before reaching the lens.



Q.8 Where should a point source of light lie along the principal axis of a converging lens so that it is impossible to see the source and its image simultaneously from any point?

Q.9 A disk whose plane surface are parallel is cut as shown in figure (i), then the lenses so obtained are moved apart. What will happen to a beam of parallel rays falling on to the resulting system:

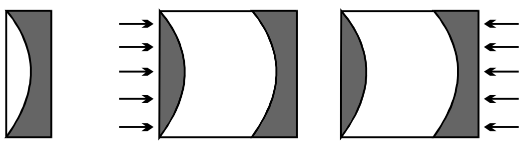


figure (i) figure (ii) figure (iii)

(a) from the side of the converging lens (figure ii),
(b) form the side of the diverging lens (figure iii)?
Consider the cases when the distance between the lenses is less than the focal length and when it is greater than the focal length.

Q.10 What will happen if a plane mirror is placed in the path of a converging beam ?

Q.11 Can a prism transmit rays at all angles of incidence?

Q.12 Why is it difficult to shoot a fish swimming in water ?

Q.13 A light ray passes through the interface between two transparent media . Under what condition will the angle of refraction be equal to the angle of incidence ?

Q.14 In what case will a bi-convex lens be diverging ?

Q.15 Is the width of a beam increased or decreased in going from air to water?

Q.16 Ordinary paper becomes transparent when it is oiled. Explain.

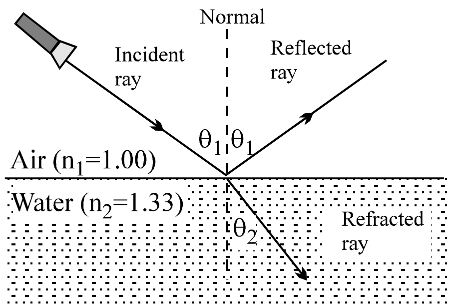
Q.17 If there are scratches on the lens of a camera, they do not appear on a photograph taken with the camera. Explain. Do the scratches affect the photograph at all?

Q.18 Concave mirrors are used as shaving mirrors. Why? Should such mirrors have short or long focal lengths?

Q.19 A sign painted on a store window is reversed when viewed from inside the store. If a person inside the store views the reversed sign in a plane mirror, does the sign appear as it would when viewed from outside the store? (Try it by writing some letters on a transparent sheet of paper and then holding the back side of the paper up to a mirror.) Explain.

Q.20 If you stand between two parallel plane mirrors, you see an infinite number of images of yourself. This occurs because an image in one mirror is reflected in the other mirror to produce another image, which is then re-reflected, and so forth. The multiple images are equally spaced. Suppose that you are facing a convex mirror, with a plane mirror behind you. Describe what you would see and comment about the spacing between any multiple images. Explain your reasoning.

Q.21 In the figure, suppose that a layer of oil were added on top of the water. The angle θ_1 at which the incident light travels through the air remains the same. Assuming that light still enters the water, does the angle of refraction at which it does so change because of the presence of the oil? Explain.



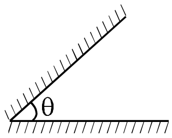
Q.22 A beam of blue light is propagating in glass. When the light reaches the boundary between the glass and the surrounding air, the beam is totally reflected back into the glass. However, red light with the same angle of incidence is not totally reflected, and some of the light is refracted into the air. Why do these two colors behave differently?

Q.23 To a swimmer under water, objects look blurred and out of focus. However, when the swimmer wears goggles that keep the water away from the eyes, the objects appear sharp and in focus. Why do goggles improve a swimmer's underwater vision?

ONLY ONE OPTION IS CORRECT.

Take approx. 2 minutes for answering each question.

Q.1 Two mirrors are inclined at an angle θ as shown in the figure. Light ray is incident parallel to one of the mirrors. The way will start retracting its path after third reflection if:

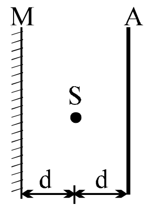


(A) θ = 45° (B) θ = 30° (C) θ = 60° (D) all three

Q.2 There are two plane mirror with reflecting surfaces facing each other. The mirrors are moving with speed v away from each other. A point object is placed between the mirrors. The velocity of the n-th image will be

(A) nv (B) 2nv (C) 3nv (D) 4nv

Q.3 A point source of light 'S' at a distance d from the screen A produces light intensity I_0 at the centre of the screen. If a completely reflecting mirror M is placed at a distance d behind the source as shown in the figure, treating object and its image as incoherent with each other, find the intensity at the centre of the screen



(A) $\frac{9}{10}I_0$ (B) $\frac{10}{9}I_0$ (C) $\frac{8}{9}I_0$ (D) $\frac{9}{8}I_0$

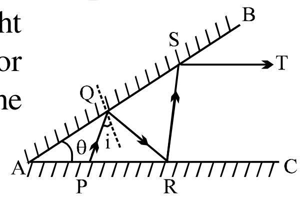
Q.4 A clock hung on a wall has marks instead of numbers on its dial. On the opposite wall there is a mirror, and the image of the clock in the mirror if read, indicates the time as 8:20. What is the time in the clock.

(A) 3:40 (B) 4:40 (C) 5:20 (D) 4:20

Q.5 A man of height 'h' is walking away from a street lamp with a constant speed 'v'. The height of the street lamp is 3h. The rate at which of the length of the man's shadow is increasing when he is at a distance 10h from the base of the street lamp is :

(A) v/2 (B) v/3 (C) 2v (D) v/6

Q.6 Two plane mirror AB and AC are inclined at an angle θ = 20°. A ray of light starting from point P is incident at point Q on the mirroe AB, then at R on mirror AC and again on S on AB finally the ray ST goes parallel to mirror Ac. The angle iwhich the ray makes with the normal at point Q on mirror AB is



(A) 20° (B) 30°
(C) 40° (D) 60°

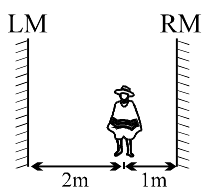
Q.7 Two plane mirrors are inclined at 70°. A ray incident on one mirror at angle θ after reflection falls on the second mirror and is reflected from there parallel to the first mirror, θ is :

(A) 50° (B) 45° (C) 30° (D) 55°

Q.8 A boy of height 1.5 m with his eye level at 1.4 m stands before a plane mirror of length 0.75 m fixed on the wall. The height of the lower edge of the mirror above the floor is 0.8 m. Then :

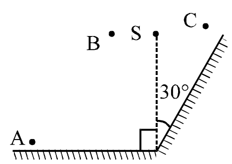
(A) the boy will see his full image (B) the boy cannot see his hair
(C) the boy cannot see his feet (D) the boy cannot see neither his hair nor his feet.

Q.9 Two mirrors, labeled LM for left mirror and RM for right mirror in the adjacent figure, are parallel to each other and 3.0 m apart. A person standing 1.0 m from the right mirror (RM) looks into this mirror and sees a series of images. How far from the person is the second closest image seen in the right mirror (RM)?



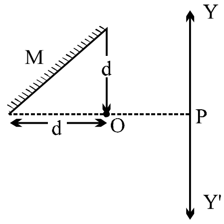
(A) 10.0 m (B) 4.0 m
(C) 6.0 m (D) 8.0 m

Q.10 A point source of light S is placed in front of two large mirrors as shown. Which of the following observers will see only one image of S?



(A) only A (B) only C
(C) Both A and C (D) Both B and C

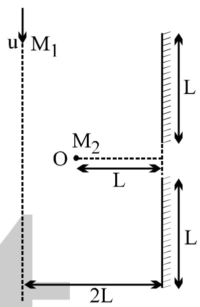
Q.11 As shown in the figure a particle is placed at O in front of a plane mirror M. A man at P can move along path PY and PY' then which of the following is true



(A) For all point on PY man can see the image of O
(B) For all point on PY' man can see the image, but for no point on PY he can see the image of O
(C) For all point on PY' he can see the image but on PY he can see the image only upto distance d.
(D) He can see the image only upto a distance d on either side of P.

Q.12 Two plane mirrors are placed parallel to each other at a distance L apart. A point object O is placed between them, at a distance L / 3 from one mirror. Both mirrors form multiple images. The distance between any two images cannot be
(A) 3L / 2 (B) 2L / 3 (C) 2L (D) None

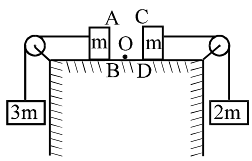
Q.13 Two plane mirrors of length L are separated by distance L and a man M_2 is standing at distance L from the connecting line of mirrors as shown in figure. A man M_1 is walking is a straight line at distance 2 L parallel to mirrors at speed u, then man M_2 at O will be able to see image of M_1 for total time:



(A) $\frac{4L}{u}$ (B) $\frac{3L}{u}$ (C) $\frac{6L}{u}$ (D) $\frac{9L}{u}$

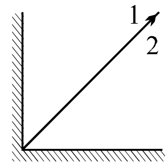
Q.14 The reflection surface of a plane mirror is vertical. A particle is projected in a vertical plane which is also perpendicular to the mirror. The initial velocity of the particle is 10 m/s and the angle of projection is 60°. The point of projection is at a distance 5 m from the mirror. The particle moves towards the mirror. Just before the particle touches the mirror the velocity of approach of the particle and its image is :
(A) 10 m/s (B) 5 m/s (C) $10\sqrt{3}$ m/s (D) $5\sqrt{3}$ m/s

Q.15 Two blocks each of mass m lie on a smooth table. They are attached to two other masses as shown in the figure. The pulleys and strings are light. An object O is kept at rest on the table. The sides AB & CD of the two blocks are made reflecting. The acceleration of two images formed in those two reflecting surfaces w.r.t. each other is:



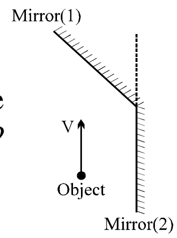
(A) 5g / 6 (B) 5g / 3 (C) g / 3 (D) 17g / 6

Q.16 A two eyed man is looking at the junction of two mutually perpendicular mirrors from a far off distance. Assume no reflection to occur from the edge. Then if both the eyes are open



(A) The eye 1 of man can see image of both eye 1 and eye 2.
(B) The eye 1 can see image of eye 1 only and eye 2 see image of eye 2 only.
(C) The eye 1 can see image of eye 2 only and eye 2 can see image of eye one only.
(D) All the above statements are false.

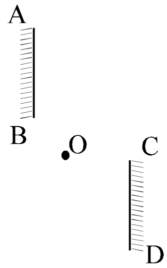
Q.17 In the diagram shown, all the velocities are given with respect to earth. What is the relative velocity of the image in mirror (1) with respect to the image in the mirror (2)? The mirror (1) forms an angle β with the vertical.



(A) 2Vsin2β (B) 2Vsinβ (C) 2V / sin2β (D) none

Q.18 A point object is kept in front of a plane mirror. The plane mirror is doing SHM of amplitude 2 cm. The plane mirror moves along the x-axis and x-axis is normal to the mirror. The amplitude of the mirror is such that the object is always infront of the mirror. The amplitude of SHM of the image is
(A) zero (B) 2 cm (C) 4 cm (D) 1 cm

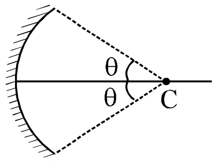
Q.19 Two mirrors AB and CD are arranged along two parallel lines. The maximum number of images of object O that can be seen by any observer is



(A) One (B) Two (C) Four (D) Infinite

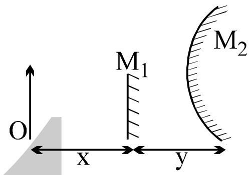
Q.20 The distance of an object from a spherical mirror is equal to the focal length of the mirror. Then the image:
(A) must be at infinity (B) may be at infinity (C) may be at the focus (D) none

Q.21 The circular boundary of the concave mirror subtends a cone of half angle θ at its centre of curvature. The minimum value of θ for which ray incident on this mirror parallel to the principle axis suffers reflection more than one is



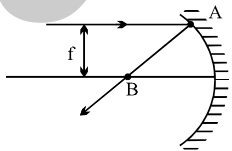
(A) 30° (B) 45° (C) 60° (D) 75°

Q.22 An object O is placed in front of a small plane mirror M_1 and a large convex mirror M_2 of focal length f. The distance between O and M_1 is x, and the distance between M_1 and M_2 is y. The images of O formed by M_1 and M_2 coincide. The magnitude of f is



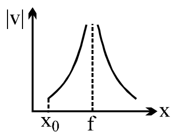
(A) $\dfrac{x^2 - y^2}{2y}$ (B) $\dfrac{x^2 + y^2}{2y}$ (C) $x - y$ (D) $\dfrac{x^2 + y^2}{x - y}$

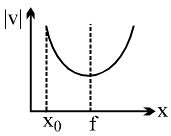
Q.23 A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray)

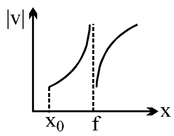


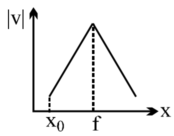
(A) $\dfrac{2}{\sqrt{3}}$ (B) $\dfrac{\sqrt{3}}{2}$ (C) $\dfrac{2}{3}$ (D) $\dfrac{1}{2}$

Q.24 A point source is situated at a distance $x < f$ from the pole of the concave mirror of focal length f. At time $t = 0$, the point source starts moving away from the mirror with constant velocity. Which of the graphs below represents best, variation of image distance $|v|$ with the distance x between the pole of mirror and the source.

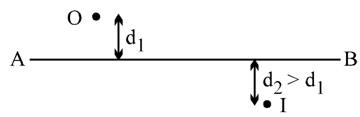
(A) 

(B) 

(C) 

(D) 

Q.25 In the figure shown, the image of a real object is formed at point I. AB is the principal axis of the mirror. The mirror must be :



(A) concave & placed towards right I
(B) concave & placed towards left of I
(C) convex and placed towards right of I
(D) convex & placed towards left of I.

Q.26 All of the following statements are correct except (for real object):
(A) the magnification produced by a convex mirror is always less thenor equal to one
(B) a virtual, erect, same sized image can be obtained using a plane mirror
(C) a virtual, erect, magnified image can be formed using a concave mirror
(D) a real, inverted, same sized image can be formed using a convex mirror.

Q.27 The distance of an object from the pole of a concave mirror is equal to its radius of curvature . The image must be :
(A) real (B) inverted (C) same sized (D) erect

Q.28 A straight line joining the object point and image point is always perpendicular to the mirror
(A) if mirror is plane only (B) if mirror is concave only
(C) if mirror is convex only (D) irrespective of the type of mirror.

Q.29 A concave mirror forms a real image three times larger than the object on a screen. Object and screen are moved until the image becomes twice the size of object. If the shift of object is 6 cm. The shift of the screen & focal length of mirror are
(A) 36 cm, 36cm (B) 36cm, 16cm (C) 72cm, 36cm (D) none of these

Q.30 The distance of a real object from the focus of a convex mirror of radius of curvature 'a' is 'b'. Then the distance of the image from the focus is
(A) $\frac{b^2}{4a}$ (B) $\frac{a}{b^2}$ (C) $\frac{a^2}{4b}$ (D) none of these

Q.31 A concave mirror is used to form image of the Sun on a white screen. If the lower half of the mirror were covered with an opaque card, the effect on the image on the screen would be
(A) negligible
(B) to make the image less bright than before
(C) to make the upper half of the image disappear
(D) to make the lower half of the image disappear

Q.32 When an object is placed at a distance of 25 cm from a concave mirror, the magnification is m_1. The object is moved 15 cm farhter away with respect to the earlier position, and the magnification becomes m_2. If $m_1/m_2 = 4$ the focal length of the mirror is (Assume image is real m_1, m_2 are numerical values)
(A) 10 cm (B) 30 cm (C) 15 cm (D) 20 cm

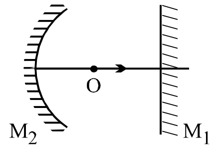
Q.33 An object is placed in front of a convex mirror at a distance of 50 cm. A plane mirror is introduced covering the lower half of the convex mirror. If the distance between the object and the plane mirror is 30 cm, it is found that there is no gap between the images formed by the two mirrors. The radius of the convex mirror is :
(A) 12.5 cm (B) 25 cm (C) 50 cm (D) 100 cm

Q.34 An infinitely long rod lies along the axis of a concave mirror of focal length f. The near end of the rod is at a distance u > f from the mirror. Its image will have a length
(A) $\frac{f^2}{u-f}$ (B) $\frac{uf}{u-f}$ (C) $\frac{f^2}{u+f}$ (D) $\frac{uf}{u+f}$

Q.35 A point object is between the Pole and Focus of a concave mirror, and moving away from the mirror with a constant speed. Then, the velocity of the image is :
(A) away from mirror and increasing in magnitude
(B) towards mirror and increasing in magnitude
(C) away from mirror and decreasing in magnitude
(D) towards mirror and decreasing in magnitude

Q.36 A luminous point object is moving along the principal axis of a concave mirror of focal length 12 cm towards it. When its distance from mirror is 20 cm its velocity is 4 cm/s. The velocity of the image in cm/s at that instant is :
(A) 6 towards the mirror (B) 6 away from the mirror
(C) 9 away from the mirror (D) 9 towards the mirror

Q.37 In the figure shown if the object 'O' moves towards the plane mirror, then the image I (which is formed after successive reflections from M_1 & M_2 respectively) will move:



(A) towards right (B) towards left (C) with zero velocity (D) cannot be determined

Q.38 The origin of x and y coordinates is the pole of a concave mirror of focal length 20 cm. The x-axis is the optical axis with x > 0 being the real side of mirror. A point object at the point (25 cm, 1 cm) is moving with a velocity 10 cm/s in positive x-direction. The velocity of the image in cm/s is approximately

(A) – 80 i + 8 j (B) 160 i + 8 j (C) – 160 i + 8 j (D) 160 i – 4 j

Q.39 A point source of light is 60 cm from a screen and is kept at the focus of a concave mirror which reflects light on the screen. The focal length of the mirror is 20 cm. The ratio of average intensities of the illumination on the screen when the mirror is present and when the mirror is removed is :

(A) 36 : 1 (B) 37 : 1 (C) 49 : 1 (D) 10:1

Q.40 Choose the correct statement(s) related to the motion of object and its image in the case of mirrors

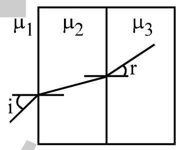
(A) Object and its image always move along normal w.r.t. mirror in opposite directions

(B) Only in the case of convex mirror, it may happen that the object and its image move in the same direction

(C) Only in the case of concave mirror, it may happen that the object and its image move in the same direction

(D) Only in case of plane mirrors, object and its image move in opposite directions

Q.41 In the figure shown $\frac{\sin i}{\sin r}$ is equal to :



(A) $\frac{\mu_2^2}{\mu_3\mu_1}$ (B) $\frac{\mu_3}{\mu_1}$ (C) $\frac{\mu_3\mu_1}{\mu_2^2}$ (D) none

Q.42 The x-z plane separates two media A and B with refractive indices μ_1 and μ_2 respectively. A ray of light travels from A to B. Its directions in the two media are given by the unit vectors, $\vec{r}_A = a\hat{i} + b\hat{j}$ & $\vec{r}_B = \alpha\hat{i} + \beta\hat{j}$ respectively where $\hat{i}$ & $\hat{j}$ are unit vectors in the x and y directions. Then

(A) $\mu_1 a = \mu_2\alpha$ (B) $\mu_1\alpha = \mu_2 a$ (C) $\mu_1 b = \mu_2\beta$ (D) $\mu_1\beta = \mu_2 b$

Q.43 A ray of light moving along the unit vector $(-\mathbf{i} - 2\mathbf{j})$ undergoes refraction at an interface of two media, which is the x-z plane. The refractive index for y > 0 is 2 while for y < 0, it is $\sqrt{5}/2$. The unit vector along which the refracted ray moves is :

(A) $\frac{(-3\hat{i}-5\hat{j})}{\sqrt{34}}$ (B) $\frac{(-4\hat{i}-3\hat{j})}{5}$ (C) $\frac{(-3\hat{i}-4\hat{j})}{5}$ (D) None of these

Q.44 A ray of light is incident at an angle of $75°$ into a medium having refractive index μ. The reflected and the refracted rays are found to suffer equal deviations in opposite direction μ equals

(A) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (B) $\frac{\sqrt{3}+1}{2}$ (C) $\frac{2\sqrt{2}}{\sqrt{3}+1}$ (D) None of these

Q.45 A ray of light is incident on a parallel slab of thickness t and refractive index n. If the angle of incidence θ is small than the displacement in the incident and emergent ray will be :

(A) $\frac{t\theta(n-1)}{n}$ (B) $\frac{t\theta}{n}$ (C) $\frac{t\theta n}{n-1}$ (D) none

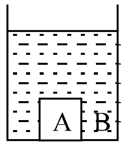
Q.46 A light ray is incident on a transparent sphere of index = $\sqrt{2}$, at an angle of incidence = 45°. What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections, and then refracts out into air ?

(A) 270° (B) 240° (C) 120° (D) 180°

Q.47 A tiny air bubble in a glass slab ($\mu = 1.5$) appears from one side to be 6 cm from the glass surface and from other side, 4 cm. The thickness of the glass slab is
(A) 10 cm (B) 6.67 cm (C) 15 cm (D) None of these

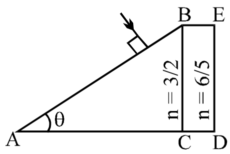
Q.48 A microscope is focused on a point object and then its objective is raised through a height of 2cm. If a glass slab of refractive index 1.5 is placed over this point object such that it is focused again, the thickness of the glass slab is :
(A) 6 cm (B) 3 cm (C) 2 cm (D) 1.5 cm

Q.49 A parallel sided block of glass of refractive index 1.5 which is 36 mm thick rests on the floor of a tank which is filled with water (refractive index = 4/3). The difference between apparent depth of floor at A & B when seen from vertically above is equal to



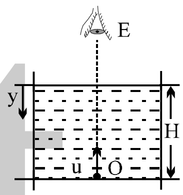
(A) 2 mm (B) 3 mm (C) 4 mm (D) none of these

Q.50 In the figure ABC is the cross section of a right angled prism and BCDE is the cross section of a glass slab. The value of θ so that light incident normally on the face AB does not cross the face BC is (given $\sin^{-1}(3/5) = 37°$)



(A) $\theta \le 37°$ (B) $\theta > 37°$ (C) $\theta \le 53°$ (D) $\theta < 53°$

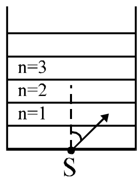
Q.51 An insect starts moving up in a liquid from point O of variable refractive index $\mu = \mu_0(1 + ay)$ where y is depth of liquid from the surface. If u is the speed of insect, its apparent speed to the observer E is



(A) $u\ln(1 + aH)$ (B) $\dfrac{u}{(1+aH)\mu_0}$ (C) $\dfrac{u}{\ln(1+aH)}$ (D) none

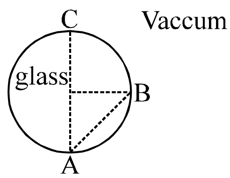
Q.52 A fish rising vertically at the rate of 3 m/s, to the surface of water observed a bird above the water, diving vertically towards at the rate of 9 m/s. The ref. index of water is 4/3. Calculate the actual velocity (velocity w.r.t ground) of bird
(A) 1.5 m/s (B) 6 m/s (C) 12 m/s (D) 4.5 m/s

Q.53 A point source S is placed at the bottom of different layers as shown in the figure. The refractive index of bottom most layer is μ_0. The refractive index of any other upper layer is $\mu(n) = \mu_0 - \dfrac{\mu_0}{4n - 18}$ where n = 1, 2, ... A ray of light with angle i slightly more than 30° starts from the source S. Total internal reflection takes place at the upper surface of a layer having n equal to



(A) 3 (B) 5 (C) 4 (D) 6

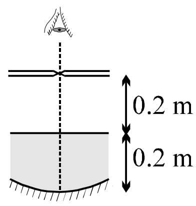
Q.54 It is found that electromagnetic signals sent inside glass sphere from A towards B reach point C. The speed of electromagnetic signals in glass cannot be:



(A) 1.0×10^8 m/s (B) 2.4×10^8 m/s
(C) 2×10^7 m/s (D) 4×10^7 m/s

Q.55 A ray of light is incident on one face of a transparent slab of thickness 15 cm. The angle of incidence is 60°. If the lateral displacement of the ray on emerging from the parallel plane is $5\sqrt{3}$ cm, the refractive index of the material of the slab is
(A) 1.414 (B) 1.532 (C) 1.732 (D) none

Q.56 When a pin is moved along the principal axis of a small concave mirror, the image position coincides with the object at a point 0.5 m from the mirror, refer figure. If the mirror is placed at a depth of 0.2 m in a transparent liquid, the same phenomenon occurs when the pin is placed 0.4 m from the mirror. The refractive index of the liquid is



(A) 6/5 (B) 5/4 (C) 4/3 (D) 3/2

Q.57 A concave mirror is placed on a horizontal table, with its axis directed vertically upwards. Let O be the pole of the mirror and C its centre of curvature. A point object is placed at C. It has a real image, also located at C (a condition called auto-collimation). If the mirror is now filled with water, the image will be:
(A) real, and will remain at C
(B) real, and located at a point between C and ∞
(C) virtual, and located at a point between C and O.
(D) real, and located at a point between C and O.

Q.58 A paraxial beam of light is converging towards a point P on the screen. A plane parallel sheet of glass of thickness t and refractive index μ is introduced in the path of beam. The convergence point is shifted by :
(A) $t(1 - 1/\mu)$ away (B) $t(1 + 1/\mu)$ away (C) $t(1 - 1/\mu)$ nearer (D) $t(1 + 1/\mu)$ nearer

Q.59 A flat glass slab of thickness 6 cm and index 1.5 is placed in front of a plane mirror. An observer is standing behind the glass slab and looking at the mirror. The actual distance of the observer from the mirror is 50 cm. The distance of his image from himself, as seen by the observer is :
(A) 94 cm (B) 96 cm (C) 98 cm (D) 100 cm

Q.60 The flat bottom of cylinder tank is silvered and water ($\mu = 4/3$) is filled in the tank upto a height h. A small bird is hovering at a height 3h from the bottom of the tank. When a small hole is opened near the bottom of the tank, the water level falls at the rate of 1 cm/s. The bird will perceive that his image's velocity is :
(A) 0.5 cm/s upward (B) 1 cm/s downwards
(C) 0.5 cm/s downwards (D) none of these

Q.61 An object is placed 20 cm in front of a 4 cm thick plane mirror. The image of the object finally is formed at 45 cm from the object itself . The refractive index of the material of the unpolished side of the mirror is (considering near normal incidence)
(A) 1.5 (B) 1.6 (C) 1.4 (D) none of these

Q.62 A ray of light from a denser medium strike a rarer medium. The angle of reflection is r and that of refraction is r'. The reflected and refracted rays make an angle of 90° with each other. The critical angle will be :
(A) $\sin^{-1}(\tan r)$ (B) $\tan^{-1}(\sin r)$
(C) $\sin^{-1}(\tan r')$ (D) $\tan^{-1}(\sin r')$

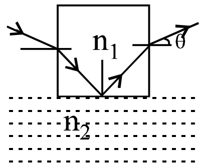
Q.63 A bird is flying 3 m above the surface of water. If the bird is diving vertically down with speed = 6 m/s, his apparent velocity as seen by a stationary fish underwater is :
(A) 8 m/s (B) 6 m/s (C) 12 m/s (D) 4 m/s

Q.64 A ray of light travels from an optical denser medium to rarer medium. The critical angle for the two media is C. The maximum possible deviation of the refracted light ray can be :
(A) $\pi - C$ (B) $2C$ (C) $\pi - 2C$ (D) $\frac{\pi}{2} - C$

Q.65 A small source of light is 4m below the surface of a liquid of refractive index 5/3. In order to cut off all the light coming out of liquid surface, minimum diameter of the disc placed on the surface of liquid is :
(A) 3m (B) 4m (C) 6m (D) ∞

Q.66 The critical angle of light going from medium A to medium B is θ . The speed of light in medium A is v . The speed of light in medium B is :
(A) $\frac{v}{\sin\theta}$ (B) $v\sin\theta$ (C) $v\cot\theta$ (D) $v\tan\theta$

Q.67 A cubical block of glass of refractive index n_1 is in contact with the surface of water of refractive index n_2. A beam of light is incident on vertical face of the block (see figure). After refraction, a total internal reflection at the base and refraction at the opposite vertical face, the ray emerges out at an angle θ. The value of θ is given by :



(A) $\sin\theta < \sqrt{n_1^2 - n_2^2}$ (B) $\tan\theta < \sqrt{n_1^2 - n_2^2}$

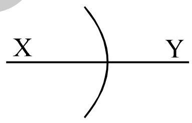
(C) $\sin\theta < \dfrac{1}{\sqrt{n_1^2 - n_2^2}}$ (D) $\tan\theta < \dfrac{1}{\sqrt{n_1^2 - n_2^2}}$

Q.68 A vertical pencil of rays comes from bottom of a tank filled with a liquid. When it is accelerated with an acceleration of 7.5 m/s^2, the ray is seen to be totally reflected by liquid surface. What is minimum possible refractive index of liquid?
(A) slightly greater than 4/3 (B) slightly greater than 5/3
(C) slightly greater than 1.5 (D) slightly greater than 1.75

Q.69 A ray of sunlight enters a spherical water droplet ($n = 4/3$) at an angle of incidence 53° measured with respect to the normal to the surface. It is reflected from the back surface of the droplet and re-enters into air. The angle between the incoming and outgoing ray is [Take sin 53° = 0.8]
(A) 15° (B) 34° (C) 138° (D) 30°

Q.70 A point source of light is placed at a distance h below the surface of a large deep lake. What is the percentage of light energy that escapes directly from the water surface is μ of the water = 4/3? (neglect partial reflection)
(A) 50% (B) 25% (C) 20% (D) 17%

Q.71 A concave spherical surface of radius of curvature 10cm separates two medium x & y of refractive index 4/3 & 3/2 respectively. If the object is placed along principal axis in medium X then



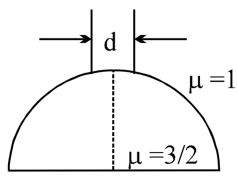
(A) image is always real
(B) image is real if the object distance is greater than 90cm
(C) image is always virtual
(D) image is virtual if the object distance is less than 90cm

Q.72 A fish is near the centre of a spherical water filled ($\mu = 4/3$) fish bowl. A child stands in air at a distance 2R (R is the radius of curvature of the sphere) from the centre of the bowl. At what distance from the centre would the child nose appear to the fish situated at the centre :
(A) 4R (B) 2R (C) 3R (D) 4R

Q.73 A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in the glass. A point object P placed in air is found to have a real image Q in the glass. The lime PQ cuts the surface at the point O, and PO = OQ. The distance PO is equal to :
(A) 5R (B) 3 R (C) 2 R (D) 1.5 R

Q.74 A spherical surface of radius of curvature 10 cm separates two media X and Y of refractive indices 3/2 and 4/3 respectively. Centre of the spherical surface lies in denser medium. An object is placed in medium X. For image to be real, the object distance must be
(A) greater than 90 cm (B) less than 90 cm.
(C) greater than 80 cm (D) less than 80 cm.

Q.75 A beam of diameter 'd' is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be:

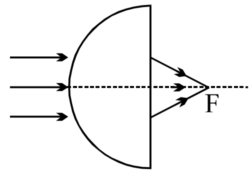


(A) 3/4 d (B) d (C) d/3 (D) 2/3d

Q.76 A concave spherical refracting surface separates two media glass and air ($\mu_{glass} = 1.5$). If the image is to be real at what minimum distance u should the object be placed in glass if R is the radius of curvature?

(A) u > 3R (B) u > 2R (C) u < 2R (D) u < R

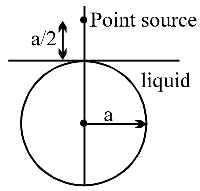
Q.77 A paraxial beam is incident on a glass (n = 1.5) hemisphere of radius R = 6 cm in air as shown. The distance of point of convergence F from the plane surface of hemisphere is



(A) 12 cm (B) 5.4 cm

(C) 18 cm (D) 8 cm

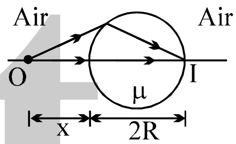
Q.78 An opaque sphere of radius a is just immersed in a transparent liquid as shown in figure. A point source is placed on the vertical diameter of the sphere at a distance a/2 from the top of the sphere. One ray originating from the point source after refraction from the air liquid interface forms tangent to the sphere. The angle of refraction for that particular ray is 30°. The refractive index of the liquid is



(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{3}{\sqrt{5}}$ (C) $\frac{4}{\sqrt{5}}$ (D) $\frac{4}{\sqrt{7}}$

Question No. 79 to 82(4 questions)

The figure, shows a transparent sphere of radius R and refractive index μ. An object O is placed at a distance x from the pole of the first surface so that a real image is formed at the pole of the exactly opposite surface.



Q.79 If x = 2R, then the value of μ is

(A) 1.5 (B) 2 (C) 3 (D) none of these

Q.80 If x = ∞, then the value of μ is

(A) 1.5 (B) 2 (C) 3 (D) none of these

Q.81 If an object is placed at a distance R from the pole of first surface, then the real image is formed at a distance R from the pole of the second surface. The refractive index μ of the sphere is given by

(A) 1.5 (B) 2 (C) $\sqrt{2}$ (D) none of these

Q.82 In **previous problem**, if the refractive index of the sphere is varied, then the position x of the object and its image from the respective poles will also vary. Identify the correct statement.

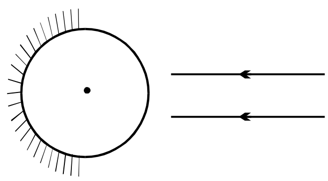
(A) If the value of μ increases the value of x decreases

(B) If the value of μ becomes equal to unity, then x tends to infinity

(C) The value of μ must not be less than 1

(D) All the above

Q.83 A glass sphere of index 1.5 and radius 40 cm has half its hemispherical surface silvered. The point where a parallel beam of light, coming along a diameter, will focus (or appear to) after coming out of sphere, will be:



(A) 10 cm to the left of centre (B) 30 cm to the left of centre

(C) 50 cm to the left of centre (D) 60 cm to the left of centre

Q.84 A lens behaves as a converging lens in air but a diverging lens in water, then the refractive index(μ) of its material is

(A) μ > 4/3 (B) μ > 3/2 (C) μ < 4/3 (D) μ < 3/2

Q.85 A bi-concave glass lens having refractive index 1.5 has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a

(A) convergent lens of focal length 3.5 R (B) convergent lens of focal length 3.0 R

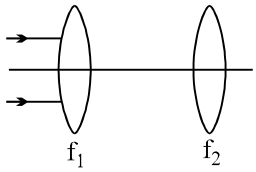
(C) divergent lens of focal length 3.5 R (D) divergent lens of focal length 3.0 R

Q.86 A converging lens forms an image of an object on a screen. The image is real and twice the size of the object. If the positions of the screen and the object are interchanged, leaving the lens in the original position, the new image size on the screen is
(A) twice the object size (B) same as the object size
(C) half the object size (D) can't say as it depends on the focal length of the lens.

Q.87 When the object is at distances u_1 and u_2 the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is :

(A) $\frac{1}{2}\sqrt{u_1 u_2}$ (B) $\frac{1}{2}(u_1 + u_2)$ (C) $\sqrt{u_1 u_2}$ (D) $2\,(u_1 + u_2)$

Q.88 Parallel beam of light is incident on a system of two convex lenses of focal lengths $f_1 = 20$ cm and $f_2 = 10$ cm. What should be the distance between the two lenses so that rays after refraction from both the lenses pass undeviated :

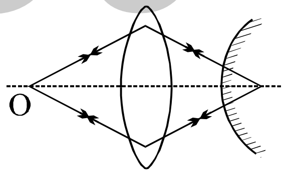


(A) 60 cm (B) 30 cm
(C) 90 cm (D) 40 cm

Q.89 A bi-concave symmetric lens made of glass has refractive index 1.5. It has both surfaces of same radius of curvature R. On immersion in a liquid of refractive index 1.25, it will behave as a
(A) Converging lens of focal length 2.5 R (B) Converging lens of focal length 2.0 R
(C) Diverging lens of focal length 4.5 R (D) None of these

Q.90 A lateral object of height 0.5 cm is placed on the optical axis of bi-convex lens of focal length 80 cm, at an object distance = 60 cm. The image formed is :
(A) virtual, erect and 4 cm high (B) virtual, inverted and 2 cm high
(C) virtual, erect and 2 cm high (D) real, inverted and 2 cm high.

Q.91 A object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed at its focus such that the image formed by the combination coincides with the object itself. The focal length of the convex mirror is



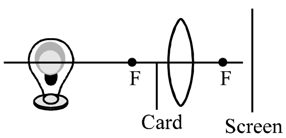
(A) 20 cm (B) 10 cm (C) 15 cm (D) 30 cm

Q.92 A thin lens of focal length f and its aperture has a diameter d. It forms an image of intensity I. Now the central part of the aperture upto diameter (d/2) is blocked by an opaque paper. The focal length and image intensity would change to
(A) f/2, I/2 (B) f, I/4 (C) 3f/4, I/2 (D) f, 3I/4

Q.93 An object is placed in front of a thin convex lens of focal length 30 cm and a plane mirror is placed 15 cm behind the lens. If the final image of the object coincides with the object, the distance of the object from the lens is
(A) 60 cm (B) 30 cm (C) 15 cm (D) 25 cm

Q.94 f_v and f_r are the focal length of a convex lens for violet and red lights respectively and F_v and F_r are the magnitude of focal lengths of a concave lens for violet and red lights respectively. Then
(A) $f_v < f_r$ and $F_v > F_r$ (B) $f_v < f_r$ and $F_v < F_r$ (C) $f_v > f_r$ and $F_v > F_r$ (D) $f_v > f_r$ and $F_v < F_r$

Q.95 A opaque card is held over the lower half of a converging lens as shown in figure. Which picture is best shows the image that appears on the screen.

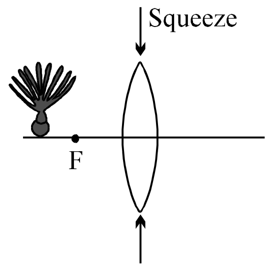


(A)  (B) 

(C)  (D) 

Question No.96 to 98(3 questions)

A turnip sits before a thin converging lens, outside the focal point of the lens. The lens is filled with a transparent gel so that it is flexible; by squeezing its ends toward its center [as indicated in figure(a)], you can change the curvature of its front and rear sides.



Figure(a)

Q.96 When you squeeze the lens, the image.
(A) moves towards the lens (B) moves away from the lens
(C) shifts up (D) remains as it is

Q.97 The lateral height of image.
(A) increases (B) decreases (C) remains same (D) data insufficient

Q.98 Suppose that the image must be formed on a card which is at a certain distance behind the lens [figure(b)], while you move the turnip away from the lens, then you should
(A) decrease the squeeze of the lens
(B) increase the squeeze of the lens
(C) keep the card and lens as it is.
(D) move the card away from the lens

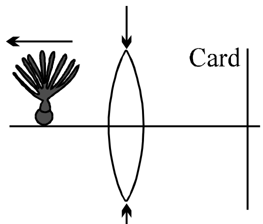
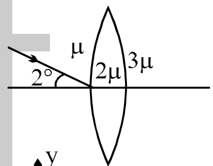


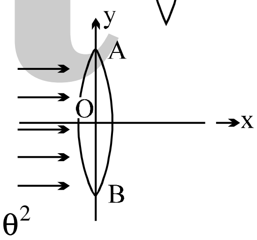
Figure (b)

Q.99 A light ray hits the pole of a thin biconvex lens as shown in figure. The angle made by the emergent ray with the optic axis will be



(A) 0° (B) (1/3)° (C) (2/3)° (D) 2°

Q.100 Monochromatic light rays parallel to x-axis strike a convex lens AB. If the lens oscillates such that AB tilts upto a small angle θ (in radian) on either side of y-axis, then the amplitude of oscillation of image will be (f = focal length of the lens):



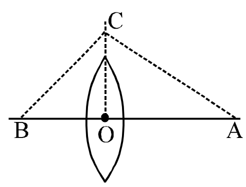
(A) $f \sec\theta$ (B) $f \sec^2\theta$ (C) $\frac{f\theta^2}{2}$ (D) $\frac{f\theta^2}{4}$

Q.101 Two point sources P and Q are 24 cm apart. Where should a convex lens of focal length 9 cm be placed in between them so that the images of both sources are formed at the same place?
(A) 3 cm from P (B) 15 cm from Q (C) 9 cm from Q (D) 18 cm from P

Q.102 If a concave lens is placed in path of converging rays real image will be produced if the distance of the pole from the point of convergence of incident rays lies between (f = magnitude of focal length of lens)
(A) 0 and f (B) f and 2f (C) 2f and infinity (D) f and infinity

Q.103 You are given two lenses, a converging lens with focal length +10 cm and a diverging lens with focal length – 20 cm. Which of the following would produce a virtual image that is larger than the object?
(A) Placing the object 5cm from the converging lens.
(B) Placing the object 15cm from the converging lens.
(C) Placing the object 25cm from the converging lens.
(D) Placing the object 15cm from the diverging lens.

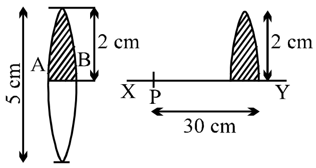
Q.104 If an object is placed at A (OA>f); Where f is the focal length of the lens the image is found to be formed at B. A perpendicular is erected at o and C is chosen on it such that the angle ∠BCA is a right angle. Then the value of f will be



(A) AB/OC^2 (B) (AC)(BC)/OC
(C) OC^2/AB (D) (OC)(AB)/AC+BC

Q.105 The height of the image formed by a converging lens on a screen is 8cm. For the same position of the object and screen again an image of size 12.5cm is formed on the screen by shifting the lens. The height of the object :
(A) 625/32cm (B) 64/12.5cm (C) 10cm (D) none

Q.106 A converging lens of focal length 20 cm and diameter 5 cm is cut along the line AB. The part of the lens shown shaded in the diagram is now used to form an image of a point P placed 30 cm away from it on the line XY. Which is perpendicular to the plane of the lens. The image of P will be formed.

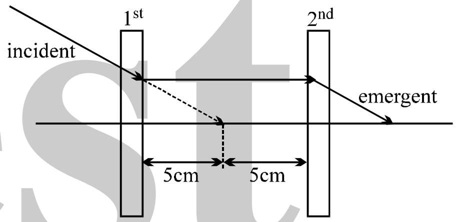


(A) 0.5 cm above XY (B) 1 cm below XY
(C) on XY (D) 1.5 cm below XY

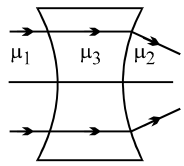
Q.107 A screen is placed 90 cm from a object. The image of an object on the screen is formed by a convex lens at two different locations separated by 20 cm. The focal length of the lens is
(A) 18 cm (B) 21.4 cm (C) 60 cm (D) 85.6 cm

Q.108 In the **above problem**, if the size of the image formed at the positions are 6 cm and 3 cm, then the highest of the object is
(A) 4.2 cm (B) 4.5 cm (C) 5 cm (D) none of these

Q.109 Look at the ray diagram shown, what will be the focal length of the 1^{st} and the 2^{nd} lens, if the incident light ray passes without any deviation?
(A) –5cm and –10cm
(B) +5cm and +10cm
(C) –5cm and +5cm
(D) +5cm and +5cm



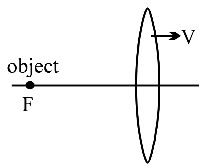
Q.110 From the figure shown establish a relation between, μ_1, μ_2, μ_3.



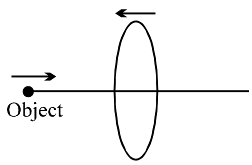
(A) $\mu_1 < \mu_2 < \mu_3$ (B) $\mu_3 < \mu_2$; $\mu_3 = \mu_1$
(C) $\mu_3 > \mu_2$; $\mu_3 = \mu_1$ (D) None of these

Q.111 An object is moving towards a converging lens on its axis. The image is also found to be moving towards the lens. Then, the object distance 'u' must satify
(A) 2f < u < 4f (B) f < u < 2f (C) u > 4f (D) u < f

Q.112 A point object is kept at the first focus of a convex lens. If the lens starts moving towards right with a constant velocity, the image will
(A) always move towards right
(B) always move towards left
(C) first move towards right & then towards left.
(D) first move towards left & then towards right.

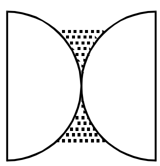


Q.113 In the diagram shown, the lens is moving towards the object with a velocity V m/s and the object is also moving towards the lens with the same speed. What speed of the image with respect to earth when the object is at a distance 2f from the lens? (f is the focal length.)



(A) 2V (B) 4V (C) 3V (D) V

Q.114 Two planoconvex lenses each of focal length 10 cm & refractive index 3/2 are placed as shown. In the space left, water (R.I = 4/3) is filled. The whole arrangement is in air. The optical power of the system is (in diopters) :



(A) 6.67 (B) – 6.67
(C) 33.3 (D) 20

Q.115 The curvature radii of a concavo-convex glass lens are 20 cm and 60 cm. The convex surface of the lens is silvered. With the lens horizontal, the concave surface is filled with water. The focal length of the effective mirror is (μ of glass = 1.5, μ of water = 4/3)

(A) 90/13 cm (B) 80/13 cm (C) 20/3 cm (D) 45/8 cm

Q.116 An object is placed in front of a symmetrical convex lens with refractive index 1.5 and radius of curvature 40 cm. The surface of the lens further away from the object is silvered. Under auto-collimation condition, the object distance is

(A) 20 cm (B) 10 cm (C) 40 cm (D) 5 cm

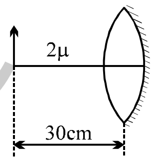
Q.117 A planoconvex lens, when silvered at its plane surface is equivalent to a concave mirror of focal length 28cm. When its curved surface is silvered and the plane surface not silvered, it is equivalent to a concave mirror of focal length 10cm, then the refractive index of the material of the lens is :

(A) 9/14 (B) 14/9 (C) 17/9 (D) none

Q.118 A concave mirror is placed on a horizontal surface and two thin uniform layers of different transparent liquids (which do not mix or interact) are formed on the reflecting surface. The refractive indices of the upper and lower liquids are μ_1 and μ_2 respectively. The bright point source at a height 'd' (d is very large in comparison to the thickness of the film) above the mirror coincides with its own final image. The radius of curvature of the reflecting surface therefore is

(A) $\frac{\mu_1 d}{\mu_2}$ (B) $\mu_1\mu_2 d$ (C) $\mu_1 d$ (D) $\mu_2 d$

Q.119 The diagram shows a silvered equiconvex lens. An object of length 1 cm has been placed in the front of the lens. What will be the final image properties? The refractive index of the lens is μ and the refractive index of the medium in which the lens has been placed is 2μ. Both the surface have the radius R.



(A) Half size, erect and virtual (B) same size, erect and real

(C) same size, erect and virtual (D) none

Q.120 On an equilateral prism, it is observed that a ray strikes grazingly at one face and if refractive index of the prism is 2 then the angle of deviation is

(A) 60° (B) 120° (C) 30° (D) 90°

Q.121 One of the refractive surfaces of a prism of angle 30° is silvered. A ray of light incident at an angle of 60° retraces it path. The refractive index of the material of prism is :

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $3/2$ (D) 2

Q.122 The refracting angle of prism is 60° and the index of refraction is 1/2 relative to surrounding. The limiting angle of incidence of a ray that the will be transmitted through the prism is :

(A) 30° (B) 45° (C) 15° (D) 50°

Q.123 One face of a prism with a refracting angle of 30° is coated with silver. A ray incident on other face at an angle of 45° is refracted and reflected from the silvered coated face and retraces its path. The refractive index of the prism is :

(A) 2 (B) $\sqrt{3}$ (C) $\sqrt{3/2}$ (D) $\sqrt{2}$

Q.124 A beam of monochromatic light is incident at i = 50° on one face of an equilateral prism, the angle of emergence is 40°, then the angle of minimum deviation is :

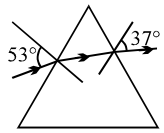
(A) 30° (B) < 30° (C) ≤ 30° (D) ≥ 30°

Q.125 An equilateral prism deviates a ray through 40° for two angles of incidence differing by 20°. The possible angles of incidences are :

(A) 40°, 60° (B) 50°, 30° (C) 45°, 55° (D) 30°, 60°

Q.126 For a prism of apex angle 45°, it is found that the angle of emergence is 45° for grazing incidence. Calculate the refractive index of the prism.
(A) $(2)^{1/2}$ (B) $(3)^{1/2}$ (C) 2 (D) $(5)^{1/2}$

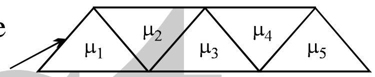
Q.127 A ray incident at an angle 53° on a prism emerges at an angle at 37° as shown. If the angle of incidence is made 50°, which of the following is a possible value of the angle of emergence.
(A) 35° (B) 42° (C) 40° (D) 38°



Q.128 A prism has a refractive index $\sqrt{\frac{3}{2}}$ and refracting angle 90°. Find the minimum deviation produced by prism.
(A) 40° (B) 45° (C) 30° (D) 49°

Q.129 R.I. of a prism is $\sqrt{\frac{7}{3}}$ and the angle of prism is 60º . The limiting angle of incidence of a ray that will be tansmitted through the prism is :
(A) 30º (B) 45º (C) 15º (D) 50º

Q.130 The diagram shows five isosceles right angled prisms. A light ray incident at 90° at the first face emerges at same angle with the normal from the last face. Which of the following relations will hold regarding the refractive indices?



(A) $\mu_1^2+\mu_3^2+\mu_5^2=\mu_2^2+\mu_4^2$ (B) $\mu_1^2+\mu_3^2+\mu_5^2=1+\mu_2^2+\mu_4^2$
(C) $\mu_1^2+\mu_3^2+\mu_5^2=2+\mu_2^2+\mu_4^2$ (D) none

Q.131 A certain prism is found to produce a minimum deviation of 38°. It produces a deviation of 44° when the angle of incidence is either 42° or 62°. What is the angle of incidencewhen it is undergoing minimum deviation?
(A) 45° (B) 49° (C) 40° (D) 55°

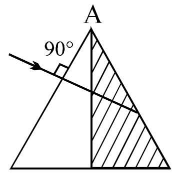
Q.132 A ray of light is incident normally on the first refracting face of the prism of refracting angle A. The ray of light comes out at grazing emergence. If one half of the prism (shaded position) is knocked off, the same ray will

(A) emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec A/2\right)$

(B) not emerge out of the prism

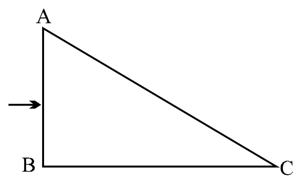
(C) emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec A/4\right)$

(D) None of these

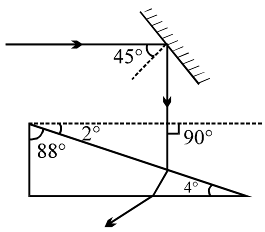


Q.133 A ray of light is incident normally on a prism of refractive index 1.5, as shown. The prism is immersed in a liquid of refractive index 'μ'. The largest value of the angle ACB, so that the ray is totally reflected at the face AC, is 30°. Then the value of μ must be :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{5}{3}$ (C) $\frac{4}{3}$ (D) $\frac{3\sqrt{3}}{4}$



Q.134 A ray of light strikes a plane mirror at an angle of incidence 45º as shown in the figure. After reflection, the ray passes through a prism of refractive index 1.5, whose apex angle is 4º. The angle through which the mirror should be rotated if the total deviation of the ray is to be 90º is
(A) 1° clockwise (B) 1° anticlockwise
(C) 2° clockwise (D) 2° anticlockwise



Q.135 A thin prism of angle 5° is placed at a distance of 10 cm from object. What is the distance of the image from object? (Given μ of prism = 1.5)

(A) $\frac{\pi}{8}$ cm (B) $\frac{\pi}{12}$ cm (C) $\frac{5\pi}{36}$ cm (D) $\frac{\pi}{7}$ cm

Q.136 Light ray is incident on a prism of angle A = 60° and refractive index $\mu = \sqrt{2}$. The angle of incidence at which the emergent ray grazes the surface is given by

(A) $\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$ (B) $\sin^{-1}\left(\frac{1-\sqrt{3}}{2}\right)$ (C) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (D) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

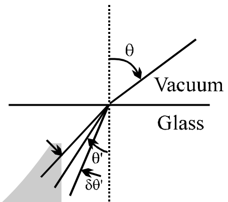
Q.137 A beam of light has a small wavelength spread $\delta\lambda$ about a central wavelength λ. The beam travels in vaccum until it enters a glass plate at an angle θ relative to the normal to the plate, as shown in figure. The index of refraction of the glass is given by $n(\lambda)$. The angular spread $\delta\theta'$ of the refracted beam is given by

(A) $\delta\theta' = \left|\frac{1}{n}\delta\lambda\right|$

(B) $\delta\theta' = \left|\frac{dn(\lambda)}{d\lambda}\delta\lambda\right|$

(C) $\delta\theta' = \left|\frac{\tan\theta'}{n}\frac{dn(\lambda)}{d\lambda}\delta\lambda\right|$

(D) $\delta\theta' = \left|\frac{\sin\theta}{\sin\theta'}\frac{\delta\lambda}{\lambda}\right|$



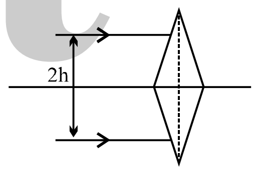
Q.138 Two identical thin isosceles prisms of refracting angle 'A' and refractive index μ are placed with their bases touching each other. Two parallel rays of light are incident on this system as shown. The distance of the point where the rays converge from the prism is :

(A) $\frac{h}{\mu A}$

(B) $\frac{h}{A}$

(C) $\frac{h}{(\mu-1)A}$

(D) $\frac{\mu h}{(\mu-1)A}$



Q.139 A parallel beam of white light falls on a convex lens. Images of blue, red and green light are formed on other side of the lens at distances x, y and z respectively from the pole of the lens. Then :

(A) $x > y > z$ (B) $x > z > y$ (C) $y > z > x$ (D) None

Q.140 The focal length of a lens is greatest for which colour?

(A) violet (B) red (C) yellow (D) green

Q.141 The power (in diopters) of an equiconvex lens with radii of curvature of 10 cm and refractive index of 1.6 is :

(A) -12 (B) $+12$ (C) $+1.2$ (D) -1.2

Q.142 Two lenses in contact made of materials with dispersive powers in the ratio 2 : 1, behaves as an achromatic lens of focal length 10 cm. The individual focal lengths of the lenses are:

(A) 5 cm, –10 cm (B) – 5 cm, 10 cm (C) 10 cm, – 20 cm (D) – 20 cm, 10 cm

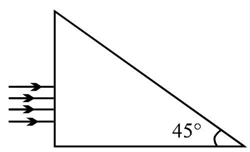
Q.143 A beam of light consisting of red, green and blue and is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will :

(A) separate part of the red color from the green and blue colors.

(B) separate part of the blue color from the red and green colours.

(C) separate all the three colors from the other two colors.

(D) not separate even partially any color from the other two colors.



Q.144 It is desired to make an achromatic combination of two lenses (L_1 & L_2) made of materials having dispersive powers ω_1 and ω_2 ($<\omega_1$). If the combination of lenses is converging then

(A) L_1 is converging
(B) L_2 is converging
(C) Power of L_1 is greater than the power of L_2
(D) None of these

Q.145 An achromatic convergent doublet of two lens in contact has a power of + 2 D. The convex lens is power + 5 D. What is the ratio of the dispersive powers of the convergent and divergent lenses?

(A) 2 : 5
(B) 3 : 5
(C) 5 : 2
(D) 5 : 3

Q.146 Two incident monochromatic waves whose wavelengths differ by a small amount $d\lambda$ are separated angularly at θ and $\theta + d\theta$ w.r.t. the incidence ray. The dispersive power is given by

(A) $d\theta/d\lambda$
(B) $d\theta/\theta$
(C) $d\lambda/\lambda$
(D) $\lambda(d\lambda/d\theta)$

Q.147 A ray of light is incident upon an air/water interface (it passes from air into water) at an angle of 45°. Which of the following quantities change as the light enters the water?

(I) wavelength
(II) frequency
(III) speed of propagation
(IV) direction of propagation

(A) I, III only
(B) III, IV only
(C) I, II, IV only
(D) I, III, IV only

Q.148 The dispersive powers of two lenses are 0.01 and 0.02. If focal length of one lens is + 10 cm, then what should the focal length of the second lens, so that they form an achromatic combination?

(A) Diverging lens having focal length 20 cm.
(B) Converging lens having focal length 20 cm
(C) Diverging lens having focal length 10 cm.
(D) Converging lens having focal length 10 cm

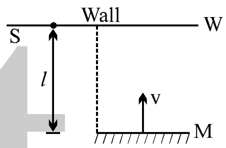
ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 A man of height 170 cm wants to see his complete image in a plane mirror (while standing). His eyes are at a height of 160 cm from the ground.
(A) Minimum length of the mirror = 80 cm
(B) Minimum length of the mirror = 85 cm.
(C) Bottom of the mirror should be at a height 80 cm.
(D) Bottom of the mirror should be at a height 85 cm.

Q.2 Two plane mirrors at an angle such that a ray incident on a mirror undergoes a total deviation of 240° after two reflections.
(A) the angle between the mirror is 60°
(B) the number of images formed by this system will be 5, if an object is placed symmetrically between the mirrors.
(C) the no. of images will be 5 if an object is kept unsymmetrically between the mirrors.
(D) a ray will retrace its path after 2 successive reflections, if the angle of incidence on one mirror is 60°.

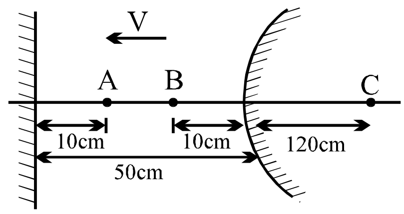
Q.3 A flat mirror M is arranged parallel to a wall W at a distance l from it. The light produced by a point source S kept on the wall is reflected by the mirror and produces a light spot on the wall. The mirror moves with velocity v towards the wall.



(A) The spot of light will move with the speed v on the wall.
(B) The spot of light will not move on the wall.
(C) As the mirror comes closer the spot of light will become larger and shift away from the wall with speed larger then v.
(D) The size of the light spot on the wall remains the same.

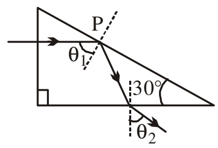
Q.4 A concave mirror cannot form
(A) virtual image of virtual object
(B) virtual image of a real object
(C) real image of a real object
(D) real image of a virtual object.

Q.5 In the figure shown consider the first reflection at the plane mirror and second at the convex mirror. AB is object.



(A) the second image is real , inverted of 1/5 th magnification
(B) the second image is virtual and erect with magnification1/5
(C) the second image moves towards the convex mirror
(D) the second image moves away from the convex mirror.

Q.6 A ray of light is incident normally on one face of 30° – 60° – 90° prism of refractive index 5/3 immersed in water of refractive index 4/3 as shown in figure.

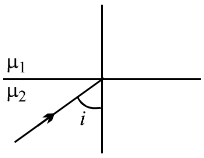


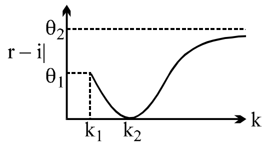
(A) The exit angle θ_2 of the ray is $\sin^{-1}(5/8)$
(B) The exit angle θ_2 of the ray is $\sin^{-1}(5/4\sqrt{3})$
(C) Total internal reflection at point P ceases if the refractive index of water is increased to $5/2\sqrt{3}$ by dissolving some substance.
(D) Total internal reflection at point P ceases if the refractive index of water is increased to 5/6 by dissolving some substance.

Q.7 A ray of light in a liquid of refractive index 1.4, approaches the boundary surface between the liquid and air at an angle of incidence whose sine is 0.8. Which of the following statements is correct about the behaviour of the light

(A) It is impossible to predict the behavior of the light ray on the basis of the information supplied.
(B) The sine of the angle of refraction of the emergent ray will less than 0.8.
(C) The ray will be internally reflected
(D) The sine of the angle of refraction of the emergent ray will be greater than 0.8.

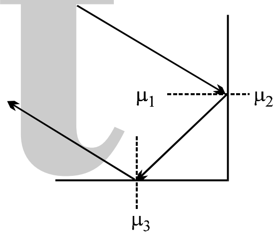
Q.8 The figure shows a ray incident at an angle $i = \pi/3$. If the plot drawn shown the variation of $|r - i|$ versus $\frac{\mu_1}{\mu_2} = k$, (r = angle of refraction)





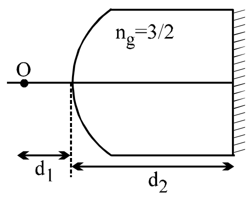
(A) the value of k_1 is $\frac{2}{\sqrt{3}}$ (B) the value of $\theta_1 = \pi/6$
(C) the value of $\theta_2 = \pi/3$ (D) the value of k_2 is 1

Q.9 In the diagram shown, a ray of light is incident on the interface between 1 and 2 at angle slightly greater than critical angle. The light suffers total internal reflection at this interface. After that the light ray falls at the interface of 1 and 3, and again it suffers total internal reflection. Which of the following relations should hold true?



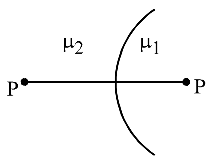
(A) $\mu_1 < \mu_2 < \mu_3$ (B) $\mu_1^2 - \mu_2^2 > \mu_3^2$
(C) $\mu_1^2 - \mu_3^2 > \mu_2^2$ (D) $\mu_1^2 + \mu_2^2 > \mu_3^2$

Q.10 In the figure shown a point object O is placed in air on the principal axis. The radius of curvature of the spherical surface is 60 cm. I_f is the final image formed after all the refractions and reflections.



(A) If d_1 = 120 cm, then the 'I_f' is formed on 'O' for any value of d_2.
(B) If d_1 = 240 cm, then the 'I_f' is formed on 'O' only if d_2= 360 cm.
(C) If d_1 = 240 cm, then the 'I_f' is formed on 'O' for all values of d_2.
(D) If d_1 = 240 cm, then the 'I_f' cannot be formed on 'O'.

Q.11 Two refracting media are separated by a spherical interface as shown in the figure. PP' is the principal axis, μ_1 and μ_2 are the refractive indices of medium of incidence and medium of refraction respectively. Then :



(A) if $\mu_2 > \mu_1$, then there cannot be a real image of real object.
(B) if $\mu_2 > \mu_1$, then there cannot be a real image of virtual object.
(C) if $\mu_1 > \mu_2$, then there cannot be a virtual image of virtual object.
(D) if $\mu_1 > \mu_2$, then there cannot be a real image of real object.

Question No. 12 to 14(3 questions)

A curved surface of radius R separates two medium of refractive indices μ_1 and μ_2 as shown in figures A and B

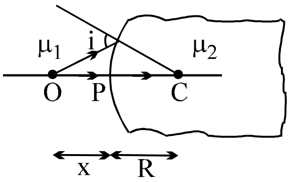


Fig. A

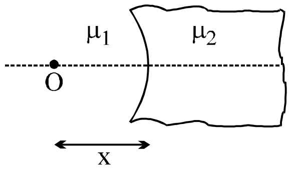


Fig. B

Q.12 Choose the correct statement(s) related to the real image formed by the object O placed at a distance x, as shown in figure A

(A) Real image is always formed irrespective of the position of object if $\mu_2 > \mu_1$

(B) Real image is formed only when $x > R$

(C) Real image is formed due to the convex nature of the interface irrespective of μ_1 and μ_2

(D) None of these

Q.13 Choose the correct statement(s) related to the virtual image formed by object O placed at a distance x, as shown in figure A

(A) Virtual image is formed for any position of O if $\mu_2 < \mu_1$

(B) Virtual image can be formed if $x > R$ and $\mu_2 < \mu_1$

(C) Virtual image is formed if $x < R$ and $\mu_2 > \mu_1$

(D) None of these

Q.14 Identify the correct statement(s) related to the formation of images of a real object O placed at x from the pole of the concave surface, as shown in figure B

(A) If $\mu_2 > \mu_1$, then virtual image is formed for any value of x

(B) If $\mu_2 < \mu_1$, then virtual image is formed if $x < \dfrac{\mu_1 R}{\mu_1 - \mu_2}$

(C) If $\mu_2 < \mu_1$, then real image is formed for any value of x

(D) none of these

Q.15 Which of the following can form diminished, virtual and erect image of your face.

(A) Converging mirror
(B) Diverging mirror
(C) Converging lens
(D) Diverging lens

Q.16 A convex lens forms an image of an object on a screen. The height of the image is 9 cm. The lens is now displaced until an image is again obtained on the screen. The height of this image is 4 cm. The distance between the object and the screen is 90cm.

(A) The distance between the two positions of the lens is 30cm.

(B) The distance of the object from the lens in its first position is 36cm.

(C) The height of the object is 6cm.

(D) The focal length of the lens is 21.6 cm.

Q.17 A diminished image of an object is to be obtained on a large screen 1 m from it. This can be achieved by

(A) using a convex mirror of focal length less than 0.25 m

(B) using a concave mirror of focal length less than 0.25 m

(C) using a convex lens of focal length less than 0.25 m

(D) using a concave lens of focal length less than 0.25 m

Q.18 Which of the following quantities related to a lens depend on the wavelength of the incident light ?

(A) Refractive index
(B) Focal length
(C) Power
(D) Radii of curvature

Q.19 A thin lens with focal length f to be used as a magnifying glass. Which of the following statements regarding the situation is true?
(A) A converging lens may be used, and the object be placed at a distance greater than 2f from the lens.
(B) A diverging lens may be used, and the object be placed between f and 2f from the lens.
(C) A converging lens may be used, and the object be placed at a distance less than f from the lens.
(D) A diverging lens may be used, and the object be placed at any point other than the focal point.

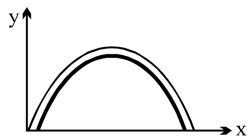
Q.20 A man wishing to get a picture of a Zebra photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.
(A) the image will look like a white donkey on the photograph.
(B) the image will look like a Zebra on the photograph.
(C) the image will be more intense compared to the case in which no such glass is used.
(D) the image will be less intense compared to the case in which no such glass is used.

Q.21 For refraction through a small angled prism, the angel of deviation :
(A) increases with the increase in R.I. of prism.
(B) will decrease with the increase in R.I. of prism.
(C) is directly proportional to the angle of prism.
(D) will be 2D for a ray of R.I.=2.4 if it is D for a ray of R.I.=1.2

Q.22 For the refraction of light through a prism
(A) For every angle of deviation there are two angles of incidence.
(B) The light travelling inside an equilateral prism is necessarily parallel to the base when prism is set for minimum deviation.
(C) There are two angles of incidence for maximum deviation.
(D) Angle of minimum deviation will increase if refractive index of prism is increased keeping the outside medium unchanged if $\mu_P > \mu_S$.

Q.23 A reflecting surface is represented by the equation $Y = \frac{2L}{\pi}\sin\left(\frac{\pi x}{L}\right)$, $0 \le x \le L$. A ray travelling horizontally becomes vertical after reflection. The coordinates of the point (s) where this ray is incident is

(A) $\left(\frac{L}{4}, \frac{\sqrt{2}L}{\pi}\right)$ (B) $\left(\frac{L}{3}, \frac{\sqrt{3}L}{\pi}\right)$ (C) $\left(\frac{3L}{4}, \frac{\sqrt{2}L}{\pi}\right)$ (D) $\left(\frac{2L}{3}, \frac{\sqrt{3}L}{\pi}\right)$



Q.24 Column-II shows the optical phenomenon that can be associated with optical components given in column-I. Note that column-I may have more than one matching options in column-II.

	Column-I		**Column-II**
(i)	Convex mirror	(A)	Dispersion
(ii)	Converging lens	(B)	Deviation
(iii)	Thin prism	(C)	Real image of real object
(iv)	Glass slab	(D)	Virtual images of real object

Answer Key

ONLY ONE OPTION IS CORRECT.

Q.1	B	Q.2	B	Q.3	B	Q.4	A	Q.5	A
Q.6	B	Q.7	A	Q.8	C	Q.9	C	Q.10	B
Q.11	C	Q.12	A	Q.13	C	Q.14	A	Q.15	D
Q.16	C	Q.17	B	Q.18	C	Q.19	A	Q.20	B
Q.21	B	Q.22	A	Q.23	A	Q.24	A	Q.25	B
Q.26	D	Q.27	A	Q.28	D	Q.29	A	Q.30	C
Q.31	B	Q.32	D	Q.33	B	Q.34	A	Q.35	A
Q.36	C	Q.37	A	Q.38	C	Q.39	D	Q.40	A
Q.41	B	Q.42	A	Q.43	B	Q.44	B	Q.45	A
Q.46	A	Q.47	C	Q.48	A	Q.49	B	Q.50	A
Q.51	B	Q.52	D	Q.53	C	Q.54	B	Q.55	C
Q.56	D	Q.57	D	Q.58	A	Q.59	B	Q.60	C
Q.61	B	Q.62	A	Q.63	A	Q.64	D	Q.65	C
Q.66	A	Q.67	A	Q.68	B	Q.69	C	Q.70	D
Q.71	C	Q.72	C	Q.73	A	Q.74	A	Q.75	D
Q.76	A	Q.77	D	Q.78	D	Q.79	C	Q.80	B
Q.81	B	Q.82	D	Q.83	D	Q.84	C	Q.85	A
Q.86	C	Q.87	B	Q.88	B	Q.89	D	Q.90	C
Q.91	B	Q.92	D	Q.93	B	Q.94	B	Q.95	D
Q.96	A	Q.97	B	Q.98	A	Q.99	C	Q.100	D
Q.101	D	Q.102	A	Q.103	A	Q.104	C	Q.105	C
Q.106	D	Q.107	B	Q.108	A	Q.109	C	Q.110	B
Q.111	D	Q.112	D	Q.113	D	Q.114	A	Q.115	A
Q.116	A	Q.117	B	Q.118	D	Q.119	C	Q.120	B
Q.121	B	Q.122	A	Q.123	D	Q.124	B	Q.125	A
Q.126	D	Q.127	D	Q.128	C	Q.129	A	Q.130	C
Q.131	B	Q.132	A	Q.133	D	Q.134	B	Q.135	C
Q.136	A	Q.137	C	Q.138	C	Q.139	C	Q.140	B
Q.141	B	Q.142	A	Q.143	A	Q.144	B	Q.145	B
Q.146	B	Q.147	D	Q.148	A				

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	B,C	Q.2	A,B,C,D	Q.3	B,D	Q.4	A
Q.5	B,C	Q.6	A,C	Q.7	C	Q.8	B,C,D
Q.9	B,C,D	Q.10	A,B	Q.11	A,C	Q.12	D
Q.13	A,B	Q.14	A,B	Q.15	B,D	Q.16	B,C,D
Q.17	C	Q.18	A,B,C	Q.19	C	Q.20	A,D
Q.21	A,C	Q.22	B,C,D	Q.23	B,D		

Q.24 (i) B,D; (ii) A,B,C,D; (iii) A,B,D; (iv) D

Power Coaching

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PHYSICS

WAVE OPTICS

SHORT QUESTIONS

Q.1 A person wets his eyeglass to clean them. As the water evaporates he notices that for a short time the glass become markedly more non reflecting. Explain.

Q.2 A lens is coated to reduce reflection. What happens to the energy that had previously been reflected? Is it absorbed by the coating?

Q.3 If interference between light waves of different frequencies is possible, one should observe light beats, just as one obtains sound beats from two sources of sound with slightly different frequencies. Discuss how one might experimentally look for this possibility.

Q.4 What is the shape of interference fringes as seen on a screen perpendicular to the line joining the sources in Young's interference experiment if the source are (a) pinholes, (b) slits?

Q.5 In Young's double slit experiment why must the slits be close and of same width?

Q.6 In Young's double slit experiment why do we use monochromatic light ? If white light is used, how would the pattern change?

Q.7 Will interference be observed in Young's double-slit experiment if the light from a source falls directly on the two slits?

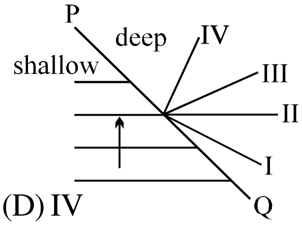
Q.8 In what direction will the fringe system shift if a glass plate is interposed in the path of one of the interfering beams?

Q.9 Suppose that a radio station broadcasts simultaneously from two transmitting antennas at two different locations. Is it clear that your radio will have better reception with two transmitting antennas rather than one? Justify your answer.

<u>***ONLY ONE OPTION IS CORRECT.***</u>

Take approx. 2 minutes for answering each question.

Q.1 Figure, shows wave fronts in still water, moving in the direction of the arrow towards the interface PQ between a shallow region and a deep(denser) region. Which of the lines shown may represent one of the wave fronts in the deep region?

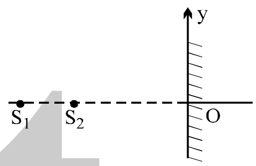


(A) I (B) II (C) III (D) IV

Q.2 Two coherent monochromatic light beams of intensities I and 4I are superposed. The maximum and minimum possible intensities in the resulting beam are :

(A) 5I and I (B) 5I and 3I (C) 9I and I (D) 9I and 3I

Q.3 Two point monochromatic and coherent sources of light of wavelength l are placed on the dotted line in front of an large screen. The source emit waves in phase with each other. The distance between S_1 and S_2 is 'd' while their distance from the screen is much larger. Then,



(1) $\rightarrow$ If $d = 7\lambda/2$, O will be a minima

(2) $\rightarrow$ If $d = 4.3\lambda$, there will be a total of 8 minima on y axis.

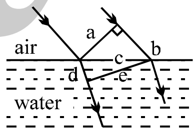
(3) $\rightarrow$ If $d = 7\lambda$, O will be a maxima.

(4) $\rightarrow$ If $d = \lambda$, there will be only one maxima on the screen.

Which is the set of correct statement :

(A) 1, 2 & 3 (B) 2, 3 & 4 (C) 1, 2, 3 & 4 (D) 1, 3 & 4

Q.4 Figure shown plane waves refracted for air to water using Huygen's principle a, b, c, d, e are lengths on the diagram. The refractive index of water wrt air is the ratio.



(A) a/e (B) b/e (C) b/d (D) d/b

Q.5 When light is refracted into a denser medium,

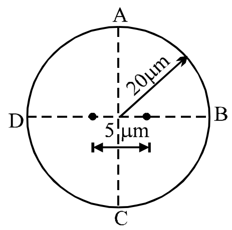
(A) its wavelength and frequeny both increases

(B) its wavelength increase but freqnency remains unchanged

(C) its wavelength decrease but freqnency remains unchanged

(D) its wavelength and freqnency both decrease.

Q.6 Two point source separated by $d = 5\ \mu m$ emit light of wavelength $\lambda = 2\ \mu m$ in phase. A circular wire of radius 20 μm is placed around the source as shown in figure.



(A) Point A and B are dark and points C and D are bright.

(B) Points A and B are bright and point C and D are dark.

(C) Points A and C are dark and points B and D are bright.

(D) Points A and C are bright and points B and D are dark.

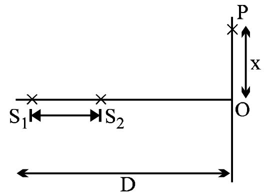
Q.7 Plane microwaves from a transmitter are directed normally towards a plane reflector. A detector moves along the normal to the reflector. Between positions of 14 successive maxima, the detector travels a distance 0.13 m. If the velocity of light is 3×10^8 m/s, find the frequency of the transmitter.

(A) 1.5×10^{10} Hz (B) 10^{10} Hz (C) 3×10^{10} Hz (D) 6×10^{10} Hz

Q.8 Two monochromatic (wavelength = a/5) and coherent sources of electromagnetic waves are placed on the x-axis at the points (2a, 0) and (–a, 0). A detector moves in a circle of radius R(>>2a) whose centre is at the origin. The number of maximas detected during one circular revolution by the detector are

(A) 60 (B) 15 (C) 64 (D) None

Q.9 Two coherent narrow slits emitting light of wavelength λ in the same phase are placed parallel to each other at a small separation of 3λ. The light is collected on a screen S which is placed at a distance D ($>> \lambda$) from the slits. The smallest distance x such that the P is a maxima



(A) $\sqrt{3D}$ (B) $\sqrt{8}\,D$

(C) $\sqrt{5}\,D$ (D) $\sqrt{5}\,\frac{D}{2}$

Q.10 Two coherent sources of light are placed at points $(-\frac{5a}{2}, 0)$ and $(+\frac{5a}{2}, 0)$. Wavelength of the light is $\lambda = \frac{4a}{3}$. How many maximas will be obtained on a CD planar circle of large radius with centre at origin.

(A) 12 (B) 15 (C) 16 (D) 14

Q.11 In YDSE how many maxima can be obtained on the screen if wavelength of light used is 200nm and d = 700 nm:

(A) 12 (B) 7 (C) 18 (D) none of these

Q.12 In a YDSE, the central bright fringe can be identified :

(A) as it has greater intensity than the other bright fringes.
(B) as it is wider than the other bright fringes.
(C) as it is narrower than the other bright fringes.
(D) by using white light instead of single wavelength light.

Q.13 In Young's double slit experiment, the wavelength of red light is 7800 Å and that of blue light is 5200 Å. The value of n for which n^{th} bright band due to red light coincides with $(n + 1)^{th}$ bright band due to blue light, is :

(A) 1 (B) 2 (C) 3 (D) 4

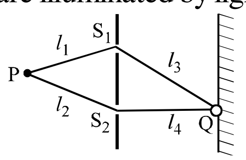
Q.14 If the Young's double slit experiment is performed with white light, then which of the following is not true.

(A) the central maximum will be white (B) there will not be a completely dark fringe
(C) the fringe next to the central will be red (D) the fringe next to the central will be violet

Q.15 Imagine a Young's double slit interference experiment performed with waves associated with fast moving electrons produced from an electron gun. The distance between successive maxima will decrease maximum if

(A) the accelerating voltage in the electron gun is decreased
(B) the accelerating voltage is increased and the distance of the screen from the slits is decreased
(C) the distance of the screen from the slits is increased.
(D) the distance between the slits is decreased.

Q.16 Two identical narrow slits S_1 and S_2 are illuminated by light of wavelength λ from a point source P.



If, as shown in the diagram above the light is then allowed to fall on a screen, and if n is a positive integer, the condition for destructive interference at Q is that

(A) $(l_1 - l_2) = (2n + 1)\lambda/2$ (B) $(l_3 - l_4) = (2n + 1)\lambda/2$
(C) $(l_1 + l_2) - (l_2 + l_4) = n\lambda$ (D) $(l_1 + l_3) - (l_2 + l_4) = (2n + 1)\lambda/2$

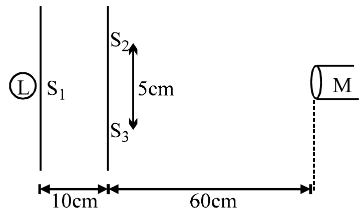
Q.17 In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same setup the two slits are sources of equal amplitude A and wavelength λ but are incoherent. The ratio of the intensity of light at the midpoint of the screen in the first case to that in the second case is

(A) 1 : 1 (B) 2 : 1 (C) 4 : 1 (D) none of these

Q.18 In a Young's double slit experiment, a small detector measures an intensity of illumination of I units at the centre of the fringe pattern. If one of the two (identical) slits is now covered, the measured intensity will be

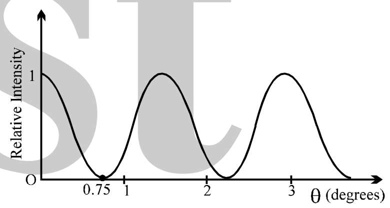
(A) 2I (B) I (C) I/4 (D) I/2

Q.19 A student is asked to measure the wavelength of monochromatic light. He sets up the apparatus sketched below. S_1, S_2, S_3 are narrow parallel slits, L is a sodium lamp and M is a microscope eyepiece. The student fails to observe interference fringes. Your first advice to him will be



(A) increase the width of S_1
(B) decrease the distance between S_2 and S_3
(C) replace L with a white light source
(D) replace M with a telescope
(E) make S_2 and S_3 wider.

Q.20 Light of wavelength 520 nm passing through a double slit, produces interference pattern of relative intensity versus deflection angle θ as shown in the figure. The separation d between the slits is



(A) 2×10^{-2} mm (B) 5×10^{-2} mm
(C) 4.5×10^{-2} mm (D) 1.1×10^{-2} mm

Q.21 In Young's double slit experiment the slits are 0.5 mm apart and the interference is observed on a screen at a distance of 100 cm from the slit. It is found that the 9th bright fringe is at a distance of 7.5 mm from the second dark fringe from the centre of the fringe pattern. The wavelength of the light used is

(A) $\frac{2500}{7}$ Å (B) 2500 Å (C) 5000 Å (D) $\frac{5000}{7}$ Å

Q.22 In a YDSE apparatus, two identical slits are separated by 1 mm and distance between slits and screen is 1 m. The wavelength of light used is 6000 Å. The minimum distance between two points on the screen having 75% intensity of the maximum intensity is :

(A) 0.45 mm (B) 0.40 mm (C) 0.30 mm (D) 0.20 mm

Q.23 In a young double slit experiment D equals the distance of screen and d is the separation between the slit. The distance of the nearest point to the central maximum where the intensity is same as that due to a single slit, is equal to

(A) $\frac{D\lambda}{d}$ (B) $\frac{D\lambda}{2d}$ (C) $\frac{D\lambda}{3d}$ (D) $\frac{2D\lambda}{d}$

Q.24 A beam of light consisting of two wavelength 6300 Å and λ Å is used to obtain interference fringes in a Young's double slit experiment. If 4^{th} bright fringe of 6300 Å coincides with 5^{th} dark fringe of λ Å, the value of λ (in Å) is

(A) 5200 (B) 4800 (C) 6200 (D) 5600

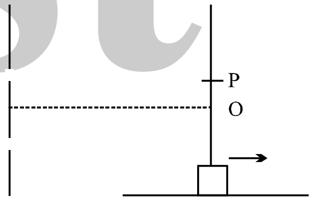
Q.25 A beam of light consisting of two wavelengths 6500Å and 5200Å is used to obtain interference fringes in Young's double slit experiment. The distance between slits is 2 mm and the distance of screen from slits is 120 cm. What is the least distance from central maximum where the bright due to both wavelength coincide?
(A) 0.156 cm (B) 0.312 cm (C) 0.078 cm (D) 0.468 cm

Q.26 In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} m. If separation between the slits is 10^{-3}m, the wavelength of light used is:
(A) 6000 Å (B) 5000 Å (C) 3000 Å (D) 4500 Å

Q.27 The ratio of the intensity at the centre of a bright fringe to the intensity at a point one–quarter of the fringwidth from the centre is
(A) 2 (B) 1/2 (C) 4 (D) 16

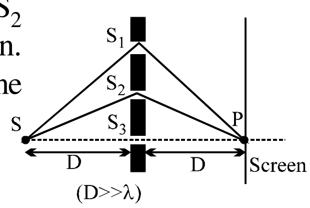
Q.28 In YDSE, let S_1 and S_2 be the two slits, and C be the centre of the screen. If θ is the angle S_1CS_2 and λ is the wavelength, the fringe width will be :
(A) $\frac{\lambda}{\theta}$ (B) $\lambda\theta$ (C) $\frac{2\lambda}{\theta}$ (D) $\frac{\lambda}{2\theta}$

Q.29 In a Young's Double slit experiment, first maxima is observed at a fixed point P on the screen. Now the screen is continuously moved away from the plane of slits. The ratio of intensity at point P to the intensity at point O (centre of the screen)
(A) remains constant
(B) keeps on decreasing
(C) first decreases and then increases
(D) First decreases and then becomes constant



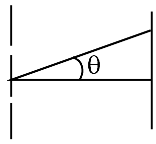
Q.30 In a double slit experiment, the separation between the slits is d = 0.25 cm and the distance of the screen D = 100 cm from the slits. If the wavelength of light used is $\lambda = 6000$Å and I_0 is the intensity of the central bright fringe, the intensity at a distance $x = 4 \times 10^{-5}$ m from the central maximum is
(A) I_0 (B) $I_0/2$ (C) $3I_0/4$ (D) $I_0/3$

Q.31 A monochromatic light source of wavelength λ is placed at S. Three slits S_1, S_2 and S_3 are equidistant from the source S and the point P on the screen. $S_1P - S_2P = \lambda/6$ and $S_1P - S_3P = 2\lambda/3$. If I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is
(A) 3 I (B) 5 I
(C) 8 I (D) zero

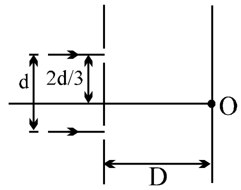


Q.32 In young's double slit experiment, the value of λ = 500 nm. The value of d = 1 mm, D = 1 m. Then the minimum distance from central maximum for which the intensity is half the maximum intensity will be
(A) 2.5×10^{-4} m (B) 2×10^{-4} m (C) 1.25×10^{-4} m (D) 10^{-4} m

Q.33 Two slits are separated by 0.3 mm. A beam of 500 nm light strikes the slits producing an interference pattern. The number of maxima observed in the angular range $-30° < \theta < 30°$.
(A) 300 (B) 150
(C) 599 (D) 149



Q.34 In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the white spot on the screen from O is [Assume $d << D$, $\lambda << d$]



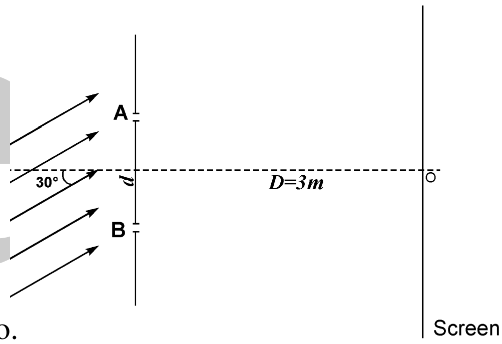
(A) 0
(B) d/2
(C) d/3
(D) d/6

Q.35 In the above question if the light incident is monochromatic and point O is a maxima, then the wavelength of the light incident cannot be

(A) $d^2/3D$
(B) $d^2/6D$
(C) $d^2/12D$
(D) $d^2/18D$

Q.36 In Young's double slit arrangement, water is filled in the space between screen and slits. Then :
(A) fringe pattern shifts upwards but fringe width remains unchanged.
(B) fringe width decreases and central bright fringe shifts upwards.
(C) fringe width increases and central bright fringe does not shift.
(D) fringe width decreases and central bright fringe does not shift.

Q.37 A parallel beam of light 500nm is incident at an angle 30° with the normal to the slit plane in a young's double slit experiment. The intensity due to each slit is Io. Point O is equidistant from S_1 and S_2. The distance between slits is 1mm.

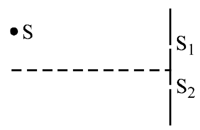


(A) the intensity at O is 4Io
(B) the intensity at O is zero.
(C) the intensity at a point on the screen 4mm from O is 4Io
(D) the intensity at a point on the screen 4mm from O is zero.

Q.38 Light of wavelength λ in air enters a medium of refractive index μ. Two points in this medium, lying along the path of this light, are at a distance x apart. The phase difference between these points is :

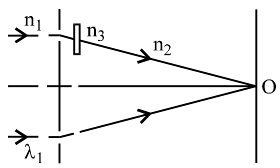
(A) $\dfrac{2\pi\mu x}{\lambda}$
(B) $\dfrac{2\pi x}{\mu\lambda}$
(C) $\dfrac{2\pi(\mu-1)x}{\lambda}$
(D) $\dfrac{2\pi x}{(\mu-1)\lambda}$

Q.39 In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits so that it is closer to the upper slit, as shown. Then,



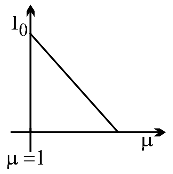
(A) the fringe width will increase and fringe pattern will shift down.
(B) the fringe width will remain same but fringe pattern will shift up.
(C) the fringe width will decrease and fringe pattern will shift down.
(D) the fringe width will remain same but fringe pattern will shift down.

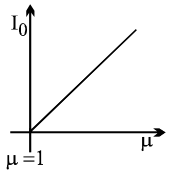
Q.40 In the figure shown in YDSE, a parallel beam of light is incident on the slit from a medium of refractive index n_1. The wavelength of light in this medium is λ_1. A transparent slab of thickness 't' and refractive index n_3 is put infront of one slit. The medium between the screen and the plane of the slits is n_2. The phase difference between the light waves reaching point 'O' (symmetrical, relative to the slits) is :

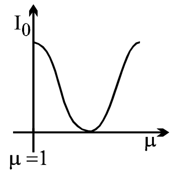


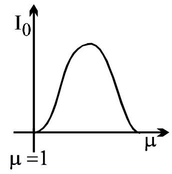
(A) $\dfrac{2\pi}{n_1\lambda_1}(n_3-n_2)\,t$
(B) $\dfrac{2\pi}{\lambda_1}(n_3-n_2)\,t$
(C) $\dfrac{2\pi n_1}{n_2\lambda_1}\left(\dfrac{n_3}{n_2}-1\right)t$
(D) $\dfrac{2\pi n_1}{\lambda_1}(n_3-n_1)\,t$

Q.41 In a YDSE experiment if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with 'μ' will be best represented by ($\mu \geq 1$).[Assume slits of equal width and there is no absorption by slab]

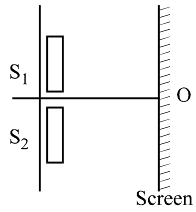
(A) 

(B) 

(C) 

(D) 

Q.42 Young's double slit experiment is carried with two thin sheets of thickness 10.4 μm each and refractive index $\mu_1 = 1.52$ and $\mu_2 = 1.40$ covering the slits S_1 and S_2, respectively. If white light of range 400 nm to 780 nm is used then which wavelength will form maxima exactly at point O, the centre of the screen ?

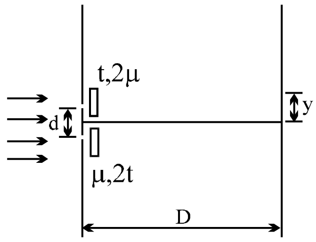


(A) 416 nm only
(B) 624 nm only
(C) 416 nm and 624 nm only
(D) none of these

Q.43 A light of wavelength 6300Å shine on a two narrow slits separated by a distance 1.0 mm and illuminates a screen at a distance 1.5 m away. When one slit is covered by a thin glass of refractive index 1.8 and other slit by a thin glass plate of refractive index μ, the central maxima shifts by 6°. Both plates have same thickness of 0.5 mm. The value of refractive index μ of the plate is
(A) 1.6 (B) 1.7 (C) 1.5 (D) 1.4

Q.44 Minimum thickness of a mica sheet having $\mu = \frac{3}{2}$ which shoule be placed in front of one of the slits in YDSE is required to reduce the intensity at the centre of screen to half of maximum intensity is
(A) $\lambda/4$ (B) $\lambda/8$ (C) $\lambda/2$ (D) $\lambda/3$

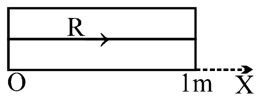
Q.45 In the YDSE shown the two slits are covered with thin sheets having thickness t & 2t and refractive index 2μ and μ. Find the position (y) of central maxima



(A) zero
(B) $\frac{tD}{d}$
(C) $-\frac{tD}{d}$
(D) None

Q.46 In a YDSE with two identical slits, when the upper slits is covered with a thin, perfectly transparent sheet of mica, the intensity at the centre of screen reduces to 75% of the initial value. Second minima is observed to be above this point and third maxima below it. Which of the following can not be a possible value of phase difference caused by the mica sheet
(A) $\frac{\pi}{3}$ (B) $\frac{13\pi}{3}$ (C) $\frac{17\pi}{3}$ (D) $\frac{11\pi}{3}$

Q.47 The figure shows a transparent slab of length 1m placed in air whose refractive index in x direction varies as $\mu = 1 + x^2 (0 \leq x \leq 1)$. The optical path length of ray R will be



(A) 1 m
(B) $\frac{2}{3}$ m
(C) $\frac{4}{3}$ m
(D) $\sqrt{2}$ m

Q.48 Two monochromatic and coherent point sources of light are placed at a certain distance from each other in the horizontal plane. The locus of all those points in the horizontal plane which have construct interference will be
(A) a hyperbola
(B) family of hyperbolas
(C) family of straight lines
(D) family of parabolas

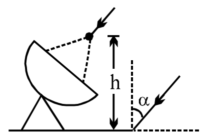
Q.49 A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate with the curved surface downwards. Monochromatic light is incident normally from the top. The observed interference fringes from this combination do not follow one of the following statements.
(A) the fringes are straight and parallel to the length of the piece.
(B) the line of contact of the cylindrical glass piece and the glass plate appears dark.
(C) the fringe spacing increases as we go outwards.
(D) the fringes are formed due to the interference of light rays reflected from the curved surface of the cylindrical piece and the top surface of the glass plate.

Q.50 A circular planar wire loop is dipped in a soap solution and after taking it out, held with its plane vertical in air. Assuming thickness of film at the top very small, as sunlight falls on the soap film, & observer receive reflected light
(A) the top portion appears dark while the first colour to be observed as one moves down is red.
(B) the top portion appears violet while the first colour to be observed as one moves down is indigo.
(C) the top portion appears dark while the first colour to be observed as one move down is violet.
(D) the top portion appears dark while the first colour to be observed as one move down depends on the refractive index of the soap solution.

Q.51 A thin film of thickness *t* and index of refraction 1.33 coats a glass with index of refraction 1.50. What is the least thickness *t* that will strongly reflect light with wavelength 600 nm incident normally?
(A) 225 nm (B) 300 nm (C) 400 nm (D) 450 nm

Q.52 It is necessary to coat a glass lens with a non-reflecting layer. If the wavelength of the light in the coating is λ, the best choice is a layer of material having an index of refraction between those of glass and air and a thickness of
(A) $\frac{\lambda}{4}$ (B) $\frac{\lambda}{2}$ (C) $\frac{3\lambda}{8}$ (D) λ

Q.53 Radio waves coming at $\angle\alpha$ to vertical are recieved by a radar after reflection from a nearby water surface & directly. What should be height of antenna from water surface so that it records a maximum intensity. (wavelength = λ).



(A) $\frac{\lambda}{2\cos\alpha}$ (B) $\frac{\lambda}{2\sin\alpha}$
(C) $\frac{\lambda}{4\sin\alpha}$ (D) $\frac{\lambda}{4\cos\alpha}$

Q.54 In a biprism experiment the distance of source from biprism is 1 m and the distance of screen from biprism is 4 metres. The angle of refraction of biprism is 2×10^{-3} radians. μ of biprism is 1.5 and the wavelength of light used is 6000Å. How many fringes will be seen on the screen?
(A) 4 (B) 5 (C) 3 (D) 6

Q.55 In a biprism experiment using sodium light λ = 6000 Å an interference pattern is obtained in which 20 fringes occupy 2 cm. On replacing sodium light by another source of wavelength λ_2 without making any other change 30 fringes occupy 2.7 cm on the screen. What is the value of λ_2?
(A) 4500 Å (B) 5400 Å (C) 5600 Å (D) 4200 Å

Q.56 A parallel coherent beam of light falls on fresnel biprism of refractive index μ and angle α. The fringe width on a screen at a distance D from biprism will be (wavelength = λ)
(A) $\frac{\lambda}{2(\mu-1)\alpha}$ (B) $\frac{\lambda D}{2(\mu-1)\alpha}$ (C) $\frac{D}{2(\mu-1)\alpha}$ (D) none

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

Q.1 To observe a stationary interference pattern formed by two light waves, it is not necessary that they must have :

(A) the same frequency
(B) same amplitude
(C) a constant phase difference
(D) the same intensity

Q.2 A light of wavelength 600nm in air enters a medium of refractive index 1.5. Inside the medium :

(A) its frequency is 5×10^{14} Hz
(B) its frequency is 7.5×10^{14} Hz
(C) its wavelength is 400nm
(D) its wavelength is 900nm

Q.3 Four monochromatic and coherent sources of light, emitting waves in phase of wavelength λ, are placed at the points $\to$ x = 0, d, 2d and 3d on the x-axis. Then

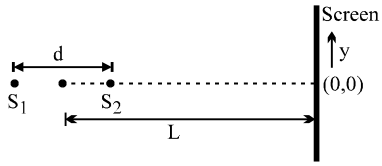
(A) points having $|x| >> d$ appear dark if $d = \lambda/4$
(B) points having $|x| >> d$ appear dark if $d = \lambda/8$
(C) points having $|x| >> d$ appear maximum bright if $d = \lambda/4$
(D) points having $|x| >> d$ appear maximum bright if $d = \lambda/8$

Q.4 In the above question, the intensity of the waves reaching a point P far away on the +x axis from each of the four sources is almost the same, and equal to I_0. Then,

(A) If $d = \lambda/4$, the intensity at P is $4I_0$.
(B) If $d = \lambda/6$, the intensity at P is $3I_0$.
(C) If $d = \lambda/2$, the intensity at P is $3I_0$.
(D) none of these is true.

Q.5 The figure shows two points source which emit light of wavelength λ in phase with each other and are at a distance $d = 5.5\,\lambda$ apart along a line which is perpendicular to a large screen at a distance L from the centre of the source. Assume that d is much less than L. Which of the following statement is (are) correct?

(A) Only five bright fringes appear on the screen
(B) Only six bright fringes appear on the screen
(C) Point y = 0 corresponds to bright fringe
(D) Point y = 0 corresponds to dark fringe.



Q.6 White light is used to illuminate two slits in a YDSE. The separation between the slits is d and the screen is at a distance D (D >> d) from the slits. At a point on the screen directly in front of one of the slits, which of the following wavelengths are missing.

(A) $\frac{d^2}{D}$
(B) $\frac{2d^2}{D}$
(C) $\frac{d^2}{3D}$
(D) $\frac{2d^2}{3D}$

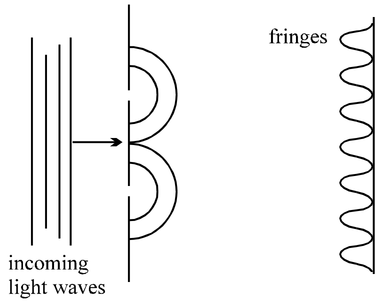
Q.7 In a YDSE apparatus, we use white light then :

(A) the fringe next to the central will be red
(B) the central fringe will be white.
(C) the fringe next to the central will be violet
(D) there will not be a completely dark fringe.

Q.8 If the source of light used in a Young's Double Slit Experiment is changed from red to blue, then

(A) the fringes will become brighter
(B) consecutive fringes will come closer
(C) the number of maxima formed on the screen increases
(D) the central bright fringe will become a dark fringe.

Q.9 In a Young's double slit experiment, green light is incident on the two slits. The interference pattern is observed on a screen. Which of the following changes would cause the observed fringes to be more closely spaced?
(A) Reducing the separation between the slits
(B) Using blue light instead of green light
(C) Used red light instead of green light
(D) Moving the light source further away from the slits.



Q.10 In a Young's double-slit experiment, let A and B be the two slits. A thin film of thickness t and refractive index μ is placed in front of A. Let $\beta =$ fringe width. The central maximum will shift :

(A) towards A (B) towards B (C) by $t(\mu - 1)\dfrac{\beta}{\lambda}$ (D) by $\mu t \dfrac{\beta}{\lambda}$

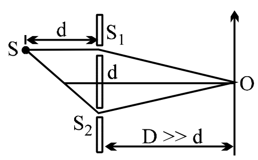
Q.11 In the **previous question**, films of thicknesses t_A and t_B and refractive indices μ_A and μ_B, are placed in front of A and B respectively. If $\mu_A t_A = \mu_B t_B$, the central maximum will :
(A) not shift
(B) shift towards A
(C) shift towards B
(D) option (B), if $t_B > t_A$; option (C) if $t_B < t_A$

Q.12 In a double slit experiment, instead of taking slits of equal widths, one slit is made twice as wide as the other. Then in the interference pattern :
(A) the intensities of both the maxima and minima increase.
(B) the intensity of the maxima increases and the minima has zero intensity.
(C) the intensity of the maxima decreases and that of minima increases.
(D) the intensity of the maxima decreases and the minima has zero intensity.

Q.13 In a YDSE, if the slits are of unequal width :
(A) fringes will not be formed
(B) the positions of minimum intensity will not be completely dark
(C) bright fringe will not be formed at the centre of the screen.
(D) distance between two consecutive bright fringes will not be equal to the distance between two consecutive dark fringes.

Q.14 If one of the slits of a standard YDSE apparatus is covered by a thin parallel sided glass slab so that it transmit only one half of the light intensity of the other, then :
(A) the fringe pattern will get shifted towards the covered slit.
(B) the fringe pattern will get shifted away from the covered slit.
(C) the bright fringes will be less bright and the dark ones will be more bright.
(D) the fringe width will remain unchanged.

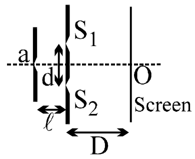
Q.15 To make the central fringe at the centre O, a mica sheet of refractive index 1.5 is introduced. Choose the correct statements (s).



(A) The thickness of sheet is $2(\sqrt{2}-1)\,d$ infront of S_1.
(B) The thickness of sheet is $(\sqrt{2}-1)\,d$ infront of S_2.
(C) The thickness of sheet is $2\sqrt{2}\,d$ infront of S_1.
(D) The thickness of sheet is $(2\sqrt{2}-1)\,d$ infront of S_1.

Question No. 16 to 19 (4 questions)

The figure shows a schematic diagram showing the arrangement of Young's Double Slit Experiment



Q.16 Choose the correct statement(s) related to the wavelength of light used
(A) Larger the wavelength of light larger the fringe width
(B) The position of central maxima depends on the wavelength of light used
(C) If white light is used in YDSE, then the violet colour forms its first maxima closest to the central maxima
(D) The central maxima of all the wavelengths coincide

Q.17 If the distance D is varied, then choose the correct statement(s)
(A) The angular fringe width does not change
(B) The fringe width changes in direct proportion
(C) The change in fringe width is same for all wavelengths
(D) The position of central maxima remains unchanged

Q.18 If the distance d is varied, then identify the correct statement
(A) The angular width does not change
(B) The fringe width changes in inverse proportion
(C) The positions of all maxima change
(D) The positions of all minima change

Q.19 Identify the correct statement(s) if the source slit S moved closer to S_1S_2, i.e. the distance ℓ decreases
(A) nothing happens to fringe pattern
(B) fringe pattern may gets less sharp
(C) fringe width remains unchanged
(D) fringe pattern may dissapear

Answer Key

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	B,D	Q.2	A,C	Q.3	A	Q.4	B
Q.5	A,D	Q.6	A,C	Q.7	B,C,D	Q.8	B,C
Q.9	B	Q.10	A,C	Q.11	D	Q.12	A
Q.13	B	Q.14	A,C,D	Q.15	A	Q.16	A,C,D
Q.17	A,B,D	Q.18	B,D	Q.19	B,C,D		

ONLY ONE OPTION IS CORRECT.

Q.1	A	Q.2	C	Q.3	C	Q.4	C	Q.5	C	Q.6	D	Q.7	A
Q.8	A	Q.9	D	Q.10	D	Q.11	B	Q.12	D	Q.13	B	Q.14	C
Q.15	B	Q.16	D	Q.17	B	Q.18	C	Q.19	B	Q.20	A	Q.21	C
Q.22	D	Q.23	C	Q.24	D	Q.25	A	Q.26	A	Q.27	A	Q.28	A
Q.29	C	Q.30	C	Q.31	A	Q.32	C	Q.33	C	Q.34	D	Q.35	A
Q.36	D	Q.37	A	Q.38	A	Q.39	D	Q.40	A	Q.41	C	Q.42	C
Q.43	A	Q.44	C	Q.45	B	Q.46	A	Q.47	C	Q.48	B	Q.49	C
Q.50	C	Q.51	A	Q.52	A	Q.53	D	Q.54	B	Q.55	B	Q.56	A